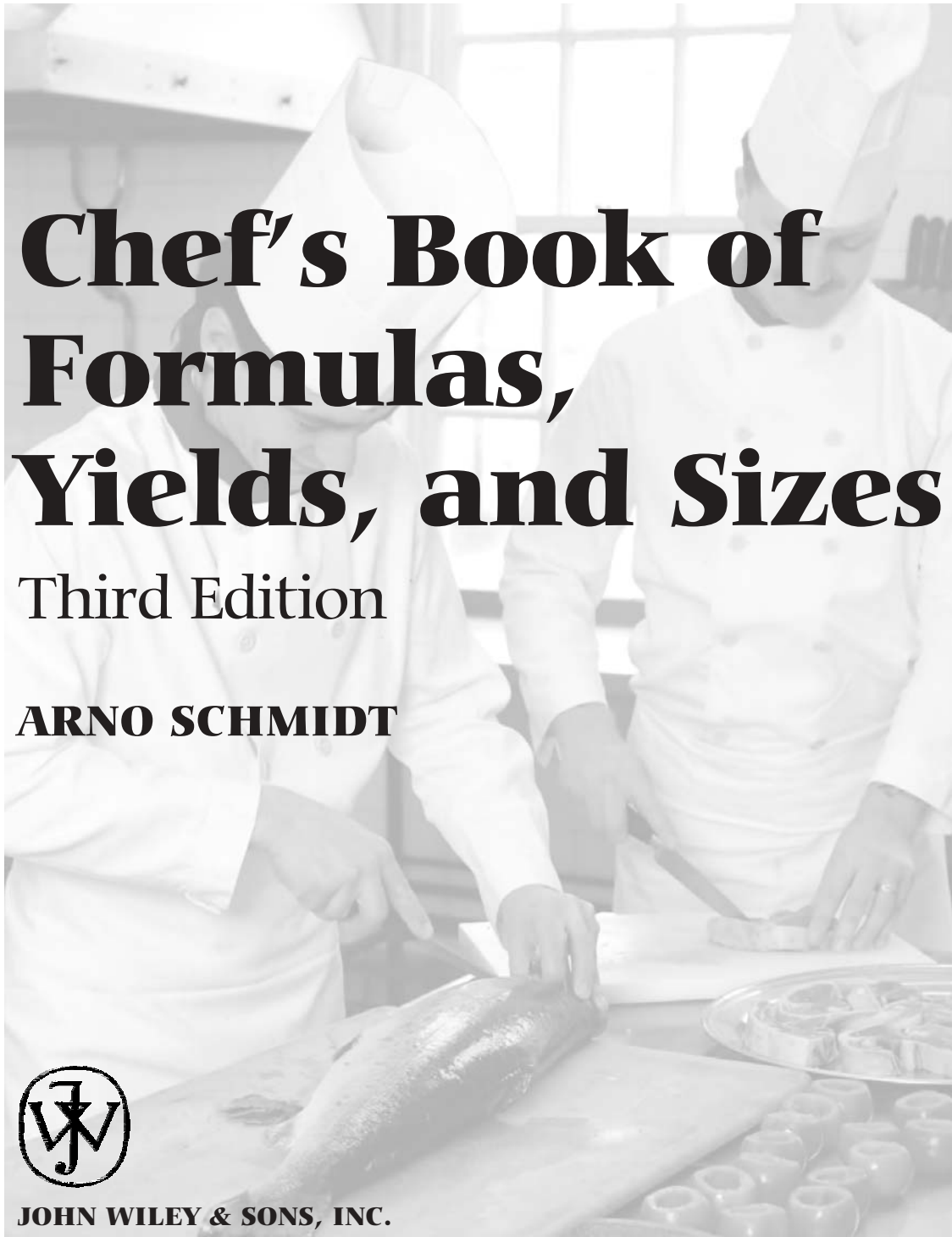


**CHEF'S BOOK
OF
FORMULAS,
YIELDS
& SIZES**

THIRD EDITION

ARNO SCHMIDT



Chef's Book of Formulas, Yields, and Sizes

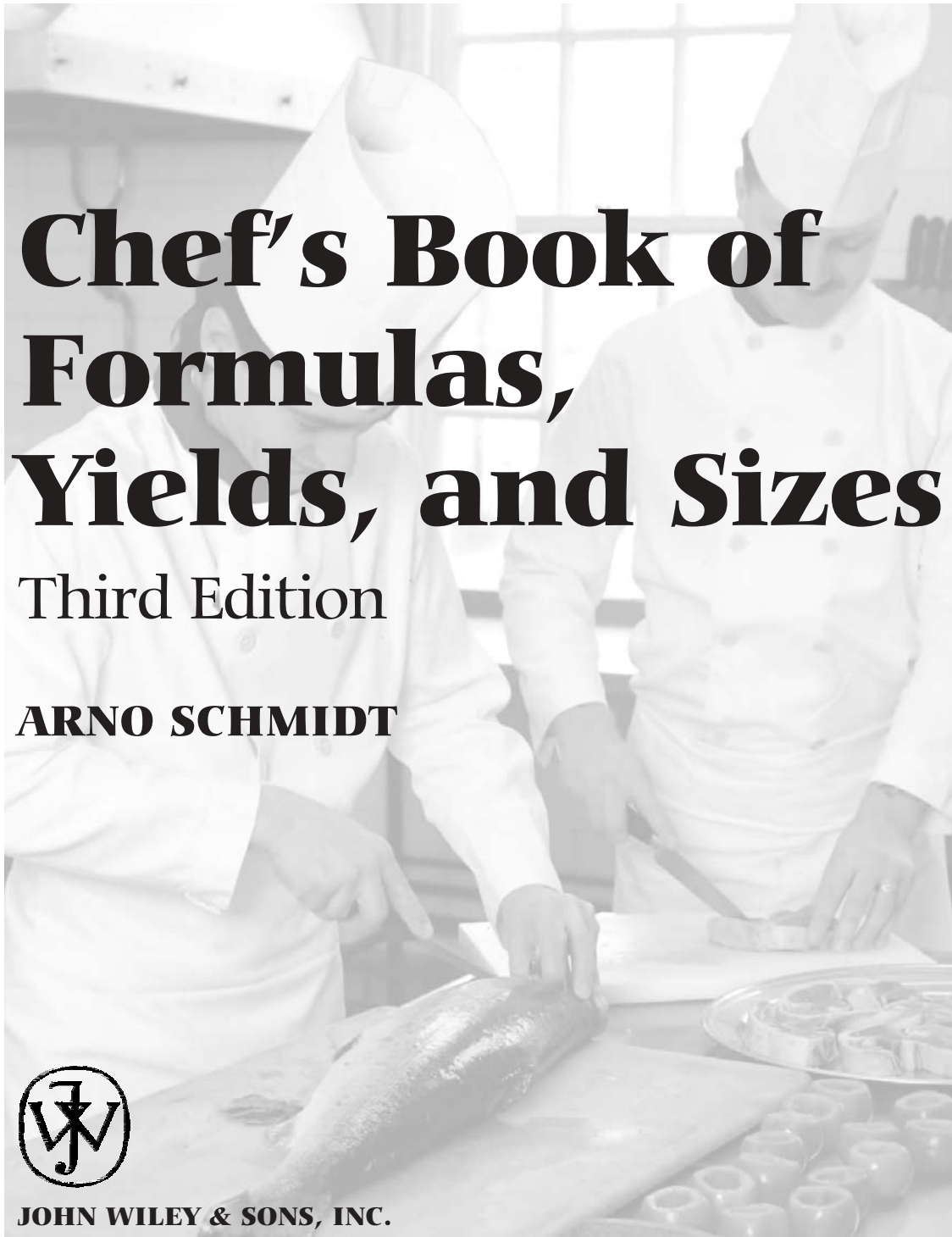
Third Edition

ARNO SCHMIDT



JOHN WILEY & SONS, INC.

Chef's Book of Formulas, Yields, and Sizes



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Third Edition

ARNO SCHMIDT



JOHN WILEY & SONS, INC.

This book is printed on acid-free paper. ∞

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Preface

The third edition of this book has been extensively updated to include new ingredients and eliminate products no longer in demand. The book is organized in broad product groups. Products normally stored in a *food storeroom* are grouped together. The *International Ingredients* section has been expanded, and tropical fruits are listed separately. Within the sections the listings are alphabetical.

The food industry has managed to produce products according to fairly precise specifications, but achieving accurate yields in commercial kitchens is vexing. Animals are not built to precise specifications like machines, but are living creatures with varying temperaments, eating habits, and needs for exercise. Therefore, meats can have varying muscle densities, hidden streaks of fat not visible during grading, and different moisture contents, all affecting yield. Two pieces of meat with the same specifications and weighing exactly the same and cooked accurately at the same temperature can yield different numbers of servings.

The same is true for vegetables and fruits. Weather conditions, as well as care in harvesting and shipping, can greatly influence yields. When the merchandise has reached the foodservice vendor, the care taken in storing it, rotating it while in storage, and issuing it also influences yields. Finally, how the kitchen crew handles the food is also of utmost importance. For

instance, one pantry person may trim strawberries just a little more than others do, and when this happens over a period of time, yields can be considerably lower than those achieved by another person. The human factor in food handling and preparation is still the most important aspect in achieving maximum yields.

The *Kitchen Yields* provided here are based on experience, rather than simple mathematical formulas that may be used elsewhere. A computer can easily calculate that 5 lb (2.2 kg) boneless ham will yield twenty 4-oz (112-g) servings. In reality, this rarely happens, because there are end pieces and scraps that cannot be served. Furthermore, not all 20 servings may be served during one meal period, and thus the leftovers diminish the yield.

The information on calories in this book has been well researched. The nutritional information provided by food manufacturers was compared with the USDA table of nutritive values, and there were some discrepancies. Therefore, samples from different manufacturers were compared to arrive at logical and reasonable calorie counts. The portion sizes given on labels were often unrealistically small, and in some instances, sizes, such as $\frac{1}{2}$ cup, were too vague and had to be verified with the actual weights.

There are two opposite trends in today's foodservice industry: many

restaurants serve huge, oversized portions to attract customers; at the same time, consumers are encouraged to cut calorie intake and eat smaller servings. The portion sizes listed in this book are generally accepted serving sizes in fine restaurants.

All quantities are listed in pounds and in metric equivalents. By necessity, metric quantities are rounded off. Cooking is not an exact science conducted in laboratories; rather, it is performed in busy kitchens, and in the real world few people will weigh ingredients to the last gram or milliliter.

Acknowledgments

My wife Margaret deserves special thanks. Although she was too busy in her own career to get directly involved in this edition of the book, she offered helpful comments and encouragement. Thanks to her, I have become much more aware of the beauty and intricacies of English. Worldwide travel and my consulting involvement in the hospitality industry has strengthened my understanding of ethnic ingredients and the need to add as many such products as practical to this book. It was a challenge to select the information most likely to be useful to readers. In the Credits section, I

mention a number of my chef colleagues who directly helped me with this book. There are many more from whom I learned and who made valuable comments about yields over the years. I am indebted to all of them. I must also give credit to the knowledgeable salespeople I met, who knew their products and freely offered advice. Unfortunately, in our computer-oriented, digital, impersonal world, they are a disappearing group. They will be sorely missed. Credit must also be given to Jim Harper, my editor, and the helpful people at John Wiley & Sons.

Credits

Abigail Kirsch Culinary Productions
Alaska Seafood Marketing Institute
Amazonas Imports, Inc.
American Agri Mushroom Business Enterprises
American Dairy Association
American Lamb Council
American Mushroom Institute
Anheuser-Busch
Anne Rutledge/Raynes
Apollo Foods, Inc.
Azar Nut Company
Batter Bake
Beatrice Hunt Wesson, Inc.
Belgian Endive Marketing Board
Berliner & Marx, Inc.
Blue Anchor Seafood
Blue Diamond Almond Growers
Bols't Indian Condiments
Bruce Church, Inc.
Bud of California
Bush Agricultural Resources
Bush Brothers and Company
Butler, Kevin, Hilton Hotels Corporation
California Apricot Advisory Board
California Artichoke Advisory Board
California Avocado Commission
California Fresh Market Tomato Advisory Board

California Iceberg Lettuce Commission
California Kiwi Fruit Commission
California Olive Industry
California Pistachio Commission
California Prune Board
California Raisin Advisory Board
California Strawberry Advisory Board
California Table Grape Commission
California Tree Fruit Agreement
C & F Foods, Inc.
Carr's Classic Game
Casa Di Lisio Products
Castle and Cooke Foods
Castroville Artichoke People
Cavalo Growers
Caviateria, Inc.
Certified Angus Beef

Chefs:

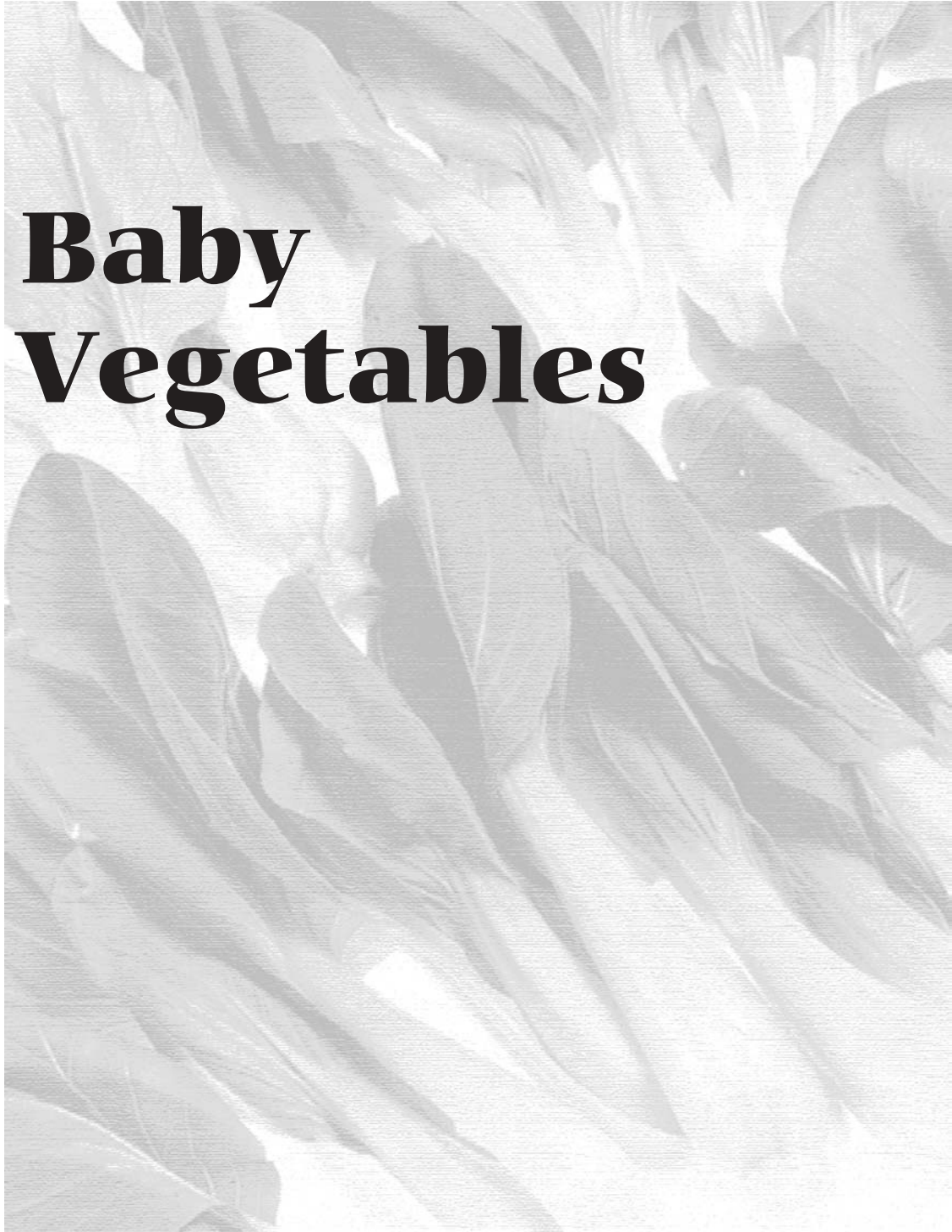
Joseph Amendola, Pastry
Anthony Bartolotta
Patricia Carlin
Rene Dionne
John Doherty
Lennart Elias
Kurt Erman
Frank Friederich
John Halligan
Demetrios Haralambatos

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Lutz Lewerenz	Eden Soy, Inc.
George Metropolis	Egg Nutrition Center
Jean Mizutani	Enfant Riant Escargots
Jean Nicolas	Fink Baking Company
Jim Padams	Fishery Products, Inc.
Tom Lo Presto	FLAV-R-PAC Frozen Foods
Herman Reiner	Florida Avocado Administrative Committee
Leslie Revsin	Florida Celery Exchange
Willi Ritz, Pastry	Florida Department of Citrus
Arno Schmidt, "The Illustrated Encyclopedia of Food" CD	Florida Lime and Avocado Committee
Arno Schmidt, <i>Notes from the Chef's Desk</i> , CBI Publishing	Florida Tomato Exchange
Arno Schmidt, <i>The Banquet Business</i> , Third Edition, C.H.I.P.S. Publishing	Food and Wines from France
Fritz Sonnenschmidt	Foods from Spain Promotion Center
Willy Spry	Freshco, Inc.
Albert Stökli	<i>Fresh Produce Food Service Directory</i> , Vance Publishing
Joseph Tarantino, Pastry	Fresh Western Marketing
Uwe Toedter	Frieda's Finest Produce
Joseph Trombetti	Frionor Norwegian Frozen Fish, Ltd.
Dutta S. Tuhin	Gamekeeper's Choice
Carlos Valle, Butcher	General Foods Corporation
Keith Way	Gilroy Farms
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Consolidated Wild Rice Company	Gourmet Fresh
Corrin Produce Sales, Inc.	Green Valley Meats
Country Skillet Catfish Co.	Harbor Banks Shrimp
Culinart, Inc.	Hazelnut Marketing Board
Culinary Institute of America, <i>The Profes- sional Chef</i> , John Wiley & Sons, Inc.	Idaho Potato Commission
Czimer Foods, Inc.	Inca's Food
Dahlen and Phillips, <i>A Popular Guide to Chi- nese Vegetables</i> , Crown Publishing	International Apple Institute
Dairyland Inc.	Ixora Restaurant
D'Artagnan, Inc.	Joanne Lewin-Jacus
De Bragga and Spitler Meats	J. R. Simplot Company
Del Monte Fresh Fruit Company	Kalustyan's Fine Foods
Divina Olives	Kaufco Sales
Dutch Valley Veal	Keller's Creamery LLC
	Kellogg's
	Diane Kennedy, <i>The Cuisines of Mexico</i> , Harper & Row
	King and Prince Seafood Corporation

Kitchen Ready Foods	National Restaurant Association, <i>Fresh Fruits Technical Bulletins</i>
Kona Coffee Council	National Turkey Federation
Kotchevar, <i>Quantity Food Purchasing</i> , John Wiley & Sons	New Zealand Farm Raised Vension Council
Hermie Kranzendorf, <i>Herbs and Spices of the World</i> , Schiffer Publishing	New Zealand Fishing Industry Board
Kronos Gyro Products	New Zealand Kiwifruit Authority
W. Wonona, Irving B. Chang, Helene W. and Austin H. Kutcher, <i>An Encyclopedia of Chinese Food and Cooking</i> , Crown Publishing	New Zealand Lamb Company
Lamb Weston	New York Times
Landreth Wild Rice Company	Paul Nicaj, Director of Banquet Service
Larsen Company	Nielsen Massey Vanillas
Albert Levie, <i>The Meat Handbook</i> , AVI Publishing	Guenther Noeth, Director of Banquet Service
Liberty Ramsey Imports	North American Blueberry Council
Lip On Trading Company	North Carolina Yam Commission
LLOYD Harbor Greens	North Star Foods, Inc.
Louisiana Crawfish Promotion Board	North West Cherries
Louisiana State Fishery Commission	Nova Scotia Food Products
Mann Packing Company	Ocean Spray Cranberries, Inc.
Maple Leaf Farms	Old El Paso Foods
Maritime Select Lobster Meat	Orange Bakery, Inc.
Peter and Joan Martin, <i>Japanese Cooking</i> , Bobbs-Merrill Company, Inc.	Oregon, Washington, and California Pear Bureau
Mash's Inc., meat company	Ore-Ida Foods, Inc.
McQuail meat company	Organic Foods Incorporated
Melissa's Produce	Organic Foods Ostrich Information
Michigan Apple Commission	Oscar Mayer Foodservice
Michigan Bean Commission	Ostrich Information
Michigan Blueberry Growers Association	Ottman Custom Processors
Michigan Marketing Association	Pacific International Rice Mills
Michigan Plum Advisory Board	Pacific Pearls Seafoods—Wakefield
Mitsui Foods, Inc.	Papaya Administrative Committee
Muscovy Grove Duck Farm	Paris Gourmet/Patisfrance
Nabisco Brands, Inc.	Patel's Indian Food Market
National Association of Meat Purveyors, Meat Buyers Guide	Pet Incorporated
National Onion Association	Pidy Gourmet
National Peach Council	Plugra Butter Products
National Pecan Marketing Council	Pollio Dairy Products
National Pork Producer Council	Potato Board of Colorado
	Quaker Oats
	RLB Food Distributors
	Salmon Institute
	S & H Organic Acres

Seald Sweet Grocers, Inc.
Singleton Seafood Company
Southeast Asia Food, Inc.
Klaus Steinke, Director of Catering
Sunkist Growers
Sun World International
Taishan City Daxin Food Co., Ltd.
Texas Sweet Citrus Advertising, Inc.
P. Prateep Thong
Uncle Ben's, Inc.
Universal Frozen Foods
U.S. Department of Agriculture *Food Buying Guide*
U.S. Department of Agriculture *Fruits in Family Meals*
U.S. Department of Agriculture *Nutrition: Food at Work for You*

U.S. Department of Agriculture *Nutritive Value of Foods*
Virga's Pizza Crust Company
Virginia Dare Extract Company
Walnut Marketing Board
Washington Asparagus Growers Association
Washington State Apple Commission
Washington State Fruit Commission
Washington State Potato Commission
Western Growers, Inc.
Wisconsin Milk Marketing Board
Yantai Xinwei Foodstuffs Co., Inc.
Yuh Ming Enterprises Co., Ltd.



Baby Vegetables

Baby vegetables have become popular in fine restaurants and for upscale banquet meals because they are colorful, dainty, and easy to prepare. For all practical purposes, they are not prepared at all but just heated. Some baby vegetables are prematurely harvested, and others are strains specially developed to stay tiny. Obviously, they have little flavor.

BABY ARTICHOKES

Season: March through May.

Pack: By weight. 10-lb (4.5-kg) carton is common.

Count: 10 to 12 pieces per pound (450 g).

Serving size: 3 to 4 pieces.

Kitchen yields: One 10-lb (4.5-kg) carton = 30 servings. Only outside leaves are peeled off.

BABY AVOCADOS/CUKES

Seedless finger or ladies' avocados, also known as *cocktail avocados*.

Season: Available year-round.

Pack: By weight. 10-lb (4.5-kg) carton is common.

Size: About 1 × 3 in (25 × 75 mm), oval-shaped.

Count: One 10-lb (4.5-kg) carton = 150 to 200.

Serving size: Varies. Baby avocados are often used as a garnish or are served as reception food, like olives.

Kitchen yields: Fruit is seedless or has soft pit; can be eaten whole.

BABY BEETS

Available in red and gold varieties.

Season: Available year-round.

Pack: Bunches; a 10-lb (4.5-kg) carton is common.

Count: 16 to 18 pieces per pound (450 g).

Serving size: 2 to 3 pieces.

Kitchen yields: One 10-lb (4.5-kg) carton = 70 servings. Discard only

the large leaves, leave the rest on, and serve whole. Do not peel.

BABY BOK CHOY

Season: Available fall and winter.

Pack: 10-lb (4.5-kg) carton.

Count: 1½ oz (40 g) each, but sizes can vary.

Serving size: 1 piece per person.

Kitchen yields: One 10-lb (4.5-kg) carton = 100 servings. Vegetable is usable as is. There is no waste.

BABY BRUSSELS SPROUTS

Season: Fall and winter.

Pack: By weight. A 10-lb (4.5-kg) carton is common.

Count: 40 to 50 per pound (450 g).

Serving size: 4 to 5 pieces.

Kitchen yields: 10-lb (4.5-kg) carton = 100 servings. Very little waste. Count varies greatly.

BABY CARROTS, UNPEELED, WITH TOPS

Long and round baby carrots are available. The color can range from red to pale white. They are very sweet.

Season: Available year-round.

Pack: 4½- to 5-lb (2- to 2.2-kg) carton.

Count: 24 bunches per case, 7 to 8 baby carrots to a bunch.

Serving size: 3 to 4 pieces.

Kitchen yields: 24 bunches = 50 servings. Some of the green top

should be left on to increase yield and eye appeal. Peeling is not necessary.

BABY CAULIFLOWER

Season: Available in fall and winter.

Pack: By weight. A 10-lb (4.5-kg) carton is common.

Count: 18 to 20 pieces per pound.

Serving size: 2 to 3 pieces.

Kitchen yields: One 10-lb (4.5-kg) carton = 70. Use as is, little waste. Do not trim off greens.

BABY CORN

Available white and yellow. For Canned Baby Corn, see the Groceries section.

Season: Summer.

Pack: By weight. A 10-lb (4.5-kg) carton is common.

Count: 12 to 13 pieces per pound (450 g), not cleaned; 37 pieces per pound, cleaned.

Kitchen yields: One 10-lb (4.5-kg) carton = 40 servings. Sizes vary widely. Whole corn can be used; no shucking is necessary.

BABY EGGPLANTS

Purple and white varieties are available. The shapes can be either round or elongated.

Pack: By weight. A 10-lb (4.5-kg) carton is common.

Count: Sizes vary greatly; 5 pieces to 1 lb (450 g) is common.

Serving size: 1 piece.

Kitchen yields: One 10-lb (4.5-kg) carton = 50 servings. The whole vegetable can be used. Baby eggplants are sometimes bitter.

BABY GREEN ASPARAGUS

Season: Available year-round.

Pack: 4-lb (1.8-kg) carton.

Count: 50 to 55 pieces per pound (450 g).

Serving size: 4 to 5 pieces.

Kitchen yields: One 4-lb (1.8-kg) carton = 45 servings. No waste; use as is.

BABY LEEKS

Season: Available year-round.

Pack: By weight. A 10-lb (4.5-kg) carton is common.

Count: 10 to 16 pieces per pound (450 g).

Serving size: 4 pieces.

Kitchen yields: One 10-lb (4.5-kg) carton = 40 servings. There is no waste.

BABY RADISHES

Season: Available year-round.

Pack: Bunches. A 10-lb (4.5-kg) carton is common.
6-oz (170-g) cello packs.

Count: One 6-oz (170-g) cello pack = 12 radishes, average.

BABY SPINACH

Season: Available year-round.

Pack: 8-oz (225-g) bags. Product is very lightweight.

Size: Leaves are about 1 to 2 in (25 to 50 mm) long.

Serving size: The tiny spinach leaves are usually served as salad or as a garnish. They are seldom cooked, because the yield would be small.

Kitchen yields: There is no waste.

BABY SQUASH

Many varieties are available, such as Baby Scallop Squash, Baby Patty Pan Squash, and Baby Zucchini. Baby squash is often sold with the flowers still intact, especially Zucchini Squash. Not to be confused with inedible ornamental squash.

Season: Usually summer, but some varieties are available year-round.

Pack: By weight. A 10-lb (2.25-kg) carton is common.

Counts:

Baby Patty Pan Squash: 20 pieces per pound (450 g)

Baby Scallop Squash: 20 pieces per pound (450 g).

Baby Zucchini: 15 pieces per pound average.

Serving size: 2 to 4 pieces.

Kitchen yields: Depend on variety. The whole vegetable can be used. There is no cleaning waste and little shrinkage.

SWEET GRAPE BABY TOMATOES

Very tiny red and yellow tomatoes have come on the market. Often the stems are still attached.

Season: Available year-round; peak occurs in summer.

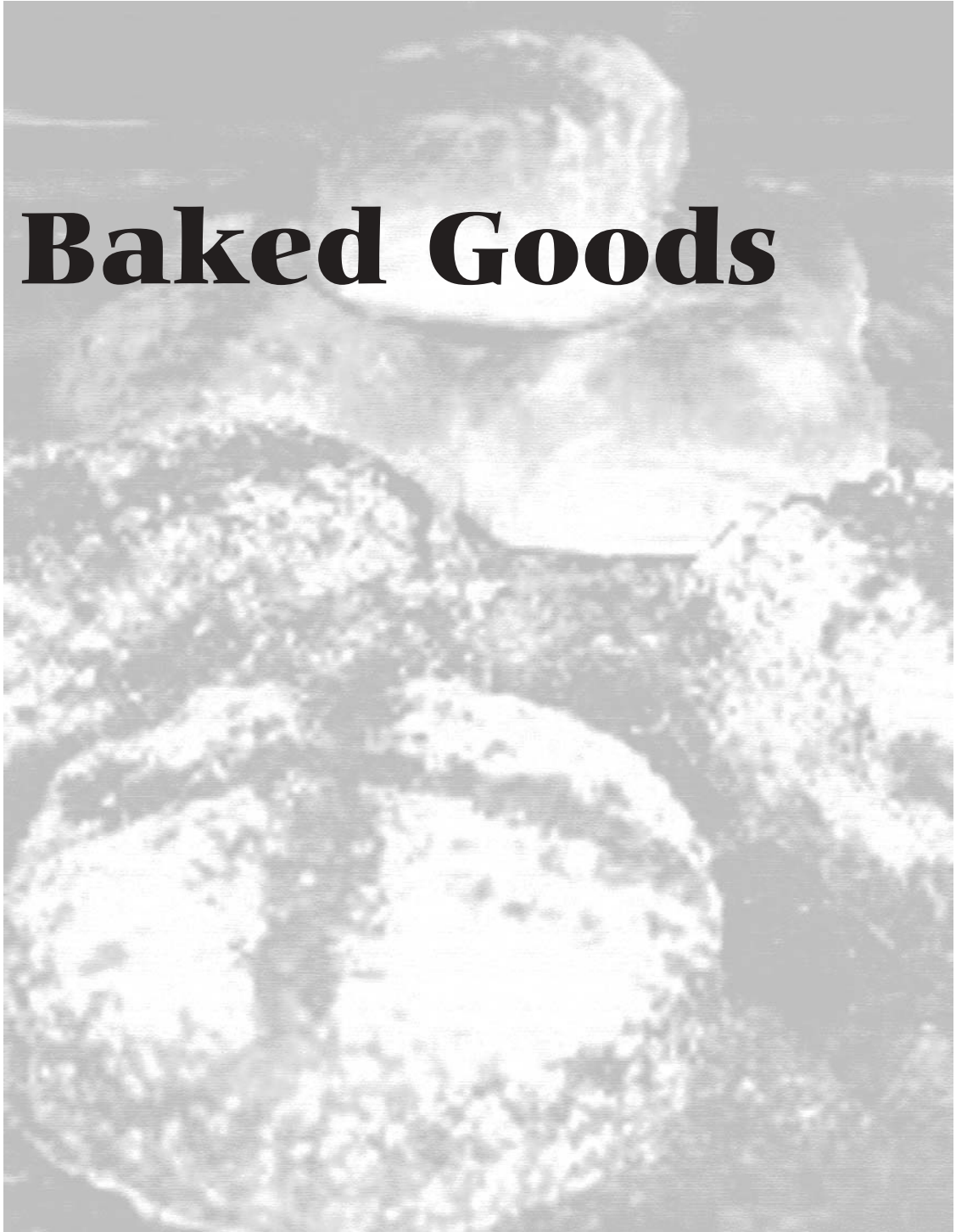
Pack: Pints (0.47 l).

Count: About 15 pieces to a pint (0.47 l), but sizes vary.

Serving size: Varies; the tomatoes are often used as a garnish.

Kitchen yields: There is no waste.

Baked Goods



BAGELS

Baked item of Jewish origin. There are two basic varieties:

- *Water Bagels:* They have a crisp crust and a chewy texture.
- *Egg Bagels:* They have a softer texture.

Both varieties are sold in different flavors and with different seed or onion toppings.

Pack: Available fresh and frozen, normally by the dozen.

Size: Common size is 3 in (85 mm) in diameter.

Weight: 4 oz (112 g).
Miniature bagels for receptions are also available fresh and frozen.

Serving size: Bagels are often served with cream cheese.
2 oz (56 g) cream cheese per serving.

Calories: One 4-oz (112-g) water bagel = 165 calories.
One 4-oz (112-g) egg bagel = 165 calories.

BREADS

Breads are available in countless varieties and sizes. Common sizes of commercial and noncommercial loaves are listed here.

BOSTON BROWN BREAD

The bread is often sold canned.

Calories: 1 slice, 1½ oz (40 g) = 95 calories.

BREAD CRUMBS

White soft, homemade crumbs. For Dry Bread Crumbs, *see* the Groceries section.

Weight: 1 cup soft white = 1.6 oz (45 g).

Kitchen yields: 1 lb (450 g) white bread, crust removed = 7 cups (315 g).

Calories: 1 cup soft white, 1.6 oz (45 g) = 120 calories.

BREAD CRUMBS, JAPANESE

See Panko under Japanese Foods in the International Ingredients section.

BREAD DISPLAYS

Preserved real breads, with unlimited shelf life for displays, are available in many sizes and shapes.

COCKTAIL LOAVES

Flavors available: Rye, pumpernickel, and marble.

Weight: 8 oz (225 g); number of slices: 30 to 32.

12 oz (400 g); number of slices: 42.

FRENCH BREAD

Pack: Loaves in many sizes.

Calories: 1-lb (450-g) loaf = 1,325 calories.

ITALIAN BREAD

Pack: Loaves in many sizes.

Calories: 1-lb (450-g) loaf = 1,250 calories.

PUMPERNICKEL

Usually made with rye flour, enriched bread flour, and rye meal.

Pack: 1-lb (450-g) loaf.

Calories: 1-lb (450-g) loaf = 1,115 calories.

RAISIN BREAD

Pack: 1-lb (450-g) loaf.

Count: 18 slices average.

Calories: 1 slice, 18 per loaf = 65 calories.

RYE BREAD

Usually made with two-thirds enriched wheat flour and one-third rye flour by weight.

Pack: 1-lb (450-g) loaf.

Calories: 1-lb (450-g) loaf = 1,100 calories.

SANDWICH LOAVES

COMMERCIAL

Commercial sandwich loaves are available sliced vertically or horizontally. Commercial loaves are classified by

weight and by the dimensions of the slice.

Flavors available: White, whole wheat, rye, and marble.

Sizes and counts:

#2 loaf weighs 2 lb (0.9 kg) and has 32 slices, $3\frac{1}{2} \times 3\frac{1}{2}$ in (90 × 90 mm).

#3 loaf weighs 3 lb (1.3 kg) and has 32 slices, $3\frac{3}{4} \times 3\frac{3}{4}$ in (95 × 95 mm).

#4 loaf weighs 4 lb (1.8 kg) and has 32 slices, $4\frac{1}{4} \times 4\frac{1}{4}$ in (107 × 107 mm).

Slice thickness, vertical slices:

Normal: $\frac{5}{16}$ in (12.7 mm), 32 slices in a 2-lb (0.9-kg) loaf.

Thin: $\frac{7}{16}$ in (11 mm), 34 slices in a 2-lb (0.9-kg) loaf.

Extra thin: $\frac{9}{16}$ in (10 mm), 40 slices in a 2-lb (0.9-kg) loaf.

Bread sliced lengthwise for canapés or finger sandwiches: 2 lb (0.9 kg) = 11 to 12 slices.

Length of loaf = 18 in (450 mm).

Kitchen yield:

One 2-lb (0.9-kg) loaf = 80 finger sandwiches, two slices.

One 2-lb (0.9-kg) loaf = 160 open-face finger sandwiches.

Commercial bread loaves:

1 loaf sliced vertically = 48 finger sandwiches.

1 loaf sliced horizontally = 80 finger sandwiches.

1 loaf sliced horizontally = 420 canapés.

NONCOMMERCIAL

Sizes:

Raisin Bread:

One 1-lb (450-g) loaf = 18 slices.

One 2-lb (900-g) loaf = 36 slices.

Rye Bread:

One 1-lb (450-g) loaf = 23 slices.

One 1½-lb (675-g) loaf = 28 slices.

One 2-lb (900-g) loaf = 33 slices.

White Bread:

One 1-lb (450-g) loaf = 16 to 18 slices.

One 1¼-lb (560-g) loaf = 19 to 22 slices.

100 slices = 5½ loaves.

One 1½-lb (450-g) loaf = 24 to 26 slices.

One 2-lb loaf (900-g) = 28 to 30 slices, $\frac{1}{2}$ in (12 mm).

One 2-lb (900-g) loaf = 36 slices, $\frac{3}{8}$ in (10 mm).

One 1-lb (450-g) loaf = 25 to 30 extra-thin slices.

Fresh cubes, white bread, weight = 1 cup = 1½ oz (30 g); #2 loaf = 12 to 13 cups.

Calories: 1-lb (450-g) loaf, white = 1,225 calories.

1 slice = 70 calories.

1 slice, 1.1 oz (32 g), enriched = 90 calories.

1 cup (0.12 l), about 1 oz (28 g) cubed = 36 calories.

1-lb (450-g) loaf, rye = 1,100 calories.

WHOLE GRAIN BREADS

There are many breads on the market with varying grains; some with added molasses.

Pack: 1-lb 8-oz (670-g) loaf.

Counts: Usually 18 slices.

Calories: 1 slice, 1.3 oz (37 g) = 100 to 110 calories.

WHOLE WHEAT BREAD (WITH BRAN AND GERM)

Pack: 1-lb (450-g) loaf.

Counts: 16 slices, soft crumb.
18 slices, hard crumb.

Calories: One slice, 1 oz (28 g) = 65 calories.

One slice, 0.7 oz (20 g) = 60 calories.

COOKIES

The typical weight for dry cookies, also known as *petits fours secs*, is $\frac{1}{2}$ oz (14 g) each.

Banquet serving size: 10 oz (228 g) for one table of 10.

CROISSANTS, FROZEN

Croissant dough, frozen croissants, and ready-to-bake preproofed croissants are

available in many sizes and with various fillings. Products are available made with 100 percent vegetable shortening, 100 percent butter, and 20 percent butter.

CROISSANT DOUGH

Pack: Six 4-lb 3-oz (1.8-kg) sheets, 15 × 23 in (375 × 575 mm).

CROISSANTS, BAKED

Available frozen and fresh from local bakeries.

Sizes: 1.35 oz (38 g) to 2.2 oz (61 g).

Serving Size: 1.35 oz (38 g) in bread baskets for breakfast.

CROISSANTS, FILLED, PREPROOFED

Available with many fillings ranging from cheese and fruit combinations to plain fruits, almond paste, and chocolate.

Packs: 108 to 56 per case.

Sizes: 1.85 oz (51 g) (mini) to 4 oz (112 g).

Plain, unbaked:

1.3 oz (34 g), for bread baskets.

2.4 oz (70 g), for single servings.

3.5 oz (100 g), for entrée sandwiches.

With sweet filling: 2.8 oz (70 g) to 4 oz (112 g).

CROISSANTS, FROZEN, PREPROOFED

Available filled and unfilled, made with 100 percent vegetable shortening or 100 percent butter.

Pack: 108 to 48 per pack, depending on size.

Sizes, unfilled: 1.2 oz (34 g) to 2.7 oz (77 g).

CROISSANTS, FROZEN, NOT BAKED

Packs: 200 each 1 oz = 12.5 lb case
60 each 3.75 oz = 14 lb case.

Sizes: 1 oz (28 g) to 3.75 oz (105 g), not filled; the most frequently used size is 2.8 oz (78 g).

CROISSANT STICKS, FROZEN, UNBAKED

Available for bread baskets, plain or with garlic.

Packs: 192 per case.

Sizes: 1.3 oz (36 g) to 1.5 oz (41 g) each.

DANISH PASTRY, FROZEN

Available items include Danish dough, frozen pastries, preproofed pastries, and fully baked products.

DANISH DOUGH

Packs: Six 4-lb 3-oz (1.8-kg) sheets, 15 × 23 in (375 × 575 mm), 4 × 4 in (125 × 125 mm) squares, or one hundred eight 1.6 oz (45 g) squares.

DANISH PASTRY, NOT BAKED

Various sizes and fillings are available.

Pack: 60 to 80 per pack, depending on size.

Sizes: From 3 oz (84 g) to 4.5 oz (125 g).

DANISH PASTRY, PREPROOFED, NOT BAKED

This is a popular product because it eliminates proofing. The product can be baked as soon as it is defrosted.

Pack: 48 pastries, 4.5 oz (125 g) each, per pack.

Calories: One 4.5-oz (125-g) pastry, baked = 475 calories, depending on filling and icing.

MATZOS

Thin crackers consisting of wheat flour and water, produced rapidly to prevent yeast cells from contaminating the dough. Matzos must be produced under rabbinical supervision; in New York City they are not manufactured between April and September.

Pack: 16-oz (450-g) packages are common.

Count: 14.

Calories: 1 matzo, 1.1 oz (32 g) = 125 calories.

PITA BREAD

Flat, round pocket bread of Middle Eastern origin.

Pack: 10 or 12 pieces, 24-oz (670-g) bags.

Size: About 7 in (177 mm) across.

Weight: 2 oz (56 g) each.

Kitchen yields: 1 piece per sandwich per person.

FILO PRODUCTS**FILO SHEETS (ALSO FILLO, PHYLLO SHEETS, OR STRUDEL LEAVES)**

Filo sheets are shipped fresh or frozen, and when frozen should be defrosted in the refrigerator for best results. Available as sheets and as kataifi, shredded dough. If not wrapped airtight, they can freeze together and become difficult to separate.

Standard Size: 14 in × 8 in (350 × 100 mm).

Thickness:

#4, thin and pliable, usually used for baklava.

#5, thicker; used to wrap shrimp and other foods.

#7, flexible; used for larger desserts and larger appetizers.

#10, thickest; used for strudel.

Pack: Cases of 12, 16, 24, and 30 packages of 16 oz (450 g) each.

Counts:

#4, thinnest = 26 sheets 12 × 17 in (300 × 430 mm).

#5, medium = 22 sheets 14 × 18 in (350 × 460 mm).

#7, medium = 20 sheets 12 × 17 in (300 × 425 mm).

#10, thickest = 18 sheets 12 × 17 in (300 × 425 mm).

Kitchen yields: The number of usable filo sheets per package is approximate, because some will break during handling.

FILO SHELLS

Fully baked filo shells are available in different sizes and flavors.

Packs: 2 boxes 45 size in master carton or 2 boxes 24 size (bigger) in master carton.

KATAIFI (SHREDDED FILO DOUGH)

Packs: 1-lb (450-g) cartons, 6 each or 12 each in case.

5-lb (2.25-kg) cartons, 4 each.

Kitchen yields: Depend on application.

There is no waste if product is properly stored.

READY-TO-USE FILLO

New product consisting of sets of 5 layers, separated by oil and margarine.

Pack: Case of 15-lb (6.75-kg) of 50 units of 14 × 18 in (350 × 450 mm) 5-sheet sets.

MUFFINS

See the Groceries section for Muffin Mixes and Batters and the Recipes section for homemade Corn Muffins.

ENGLISH MUFFINS

A number of brands are on the market. Presplit muffins are a great time saver, because splitting muffins properly with a fork is time-consuming. Muffins should be stored refrigerated.

Pack: 12-oz (340-g) packs, 24 per carton.

Count: Six 2-oz (56-g) muffins.

Calories: One 2-oz (56-g) muffin, without butter or jam = 120 calories.

PUFF PASTE PRODUCTS

Also called *pâte à choux*. Many products that were originally homemade are now available frozen from a number of manufacturers. For puff paste recipe, see *Pâte à Choux* in the Recipes section.

Eclairs, sizes: 2.6 in (65 mm), 5.5 in (140 mm), and 6.3 in (150 mm).

Cream Puffs, sizes, diameters: 1.8 in (45 mm), 3 in (75 mm) and 3.5 in (87 mm).

PUFF PASTRY PRODUCTS

Puff pastry (*pâte feuilletée* in French) is made by layering thin layers of dough with butter or fat. The process is time-consuming. Now available is ready-to-use puff pastry dough and numerous frozen puff pastry products ready to bake or already fully baked.

FROZEN PRODUCTS**BLOCKS**

Pack: 1 kg (35 oz), 4 pieces per carton.

Size: 10.6 × 7 in (270 × 180 mm); 0.7 in (18 mm) thick.

PATTY SHELLS, UNBAKED

Pack: 144 or 135 per case.

Sizes: 3 in diameter = 1.7 oz (47 g); the dough is thicker than that of the following product.

4.5 in diameter = 1.6 oz (45 g).

Other sizes are available.

SHEETS

Pack: 10-lb 12-oz (4.8-kg) carton with 15 sheets.

Size: 11.4 × 14.6 in (290 × 370 mm); 0.11 in (3 mm) thick; 11.5 oz (325 g).

Calories: 1 lb (450 g) = 1,700 calories.

1.5 oz (42 g) = 160 calories.

FULLY BAKED PRODUCTS

A large variety of fully baked hors d'oeuvre shells, dessert shells, and other items are available from a number of manufacturers. Typical of these are Patty shells, which come in various sizes and shapes:

Diameters and weights:

2.2 in (55 mm) = ½ oz (14 g).

2.75 in (70 mm) = 0.9 oz (25 g).

3 in (75 mm) = 1.15 oz (33 g).

Pack: 8-lb 12-oz (4-kg) carton.

Count: 170 pieces.

Weight: ⅔ oz (21 g).

ROLLS

Many brands of fully baked, prebaked, preproofed, and unbaked products are available on the market.

Sizes, baked rolls: 1 oz (28 g) and up for soft rolls.

1.7 oz (50 g) and up for hard rolls.

1.4 oz (40 g) for hamburger or frankfurter rolls.

4.8 oz (135 g) for hoagie or submarine rolls, 3 × 2.5 in (75 × 62 mm).

Calories:

One 1-oz (28-g) roll = 85 calories.

One 1.7-oz (50-g) roll = 155 calories.

One 1.4-oz (40-g) hamburger roll = 120 calories.

One 4.8-oz (135-g) hoagie roll, 3 in = 390 calories.

YEAST

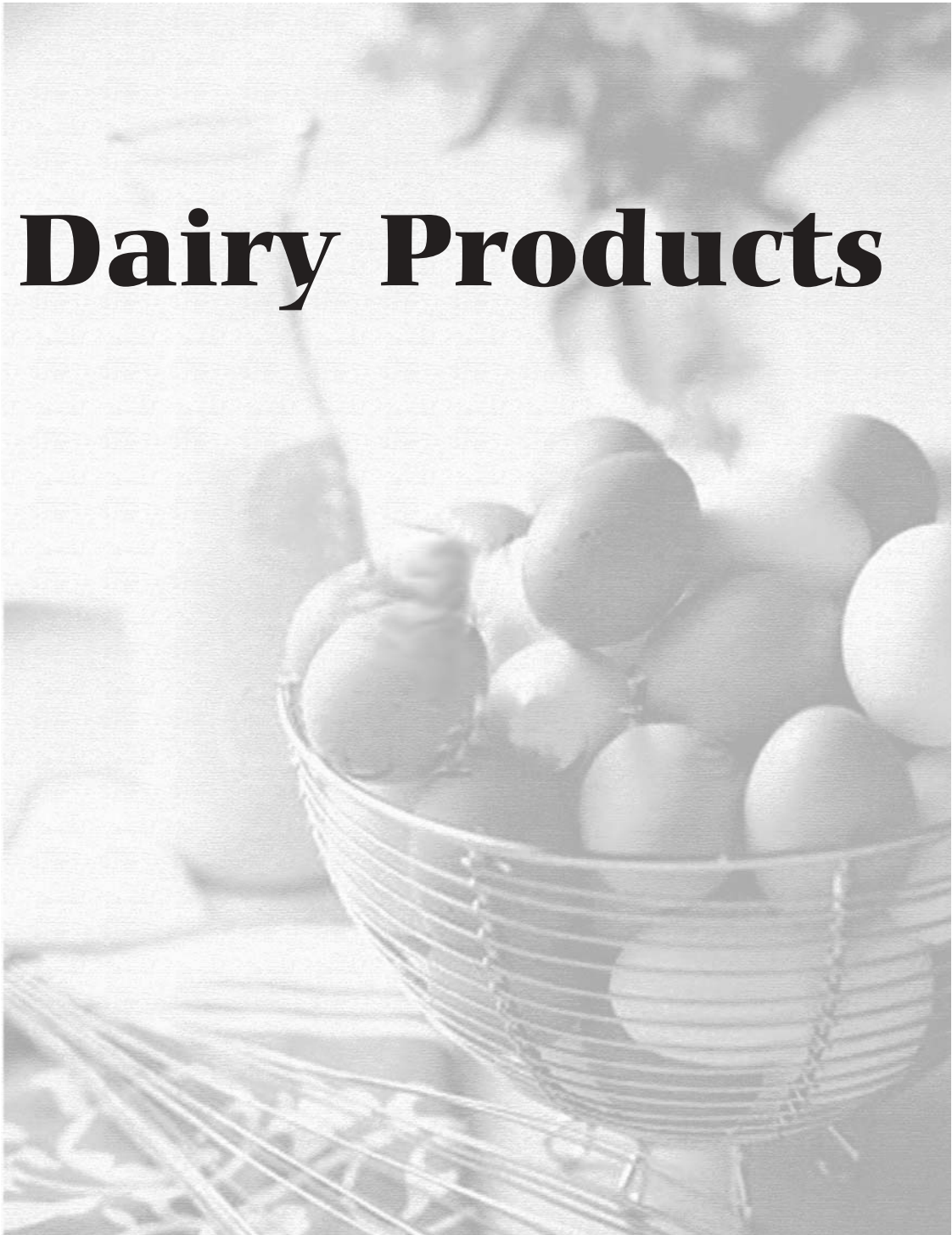
Microscopic plant cultivated in warm mash, available dry and fresh.

Kitchen yields: 1 oz (28 g) = ½ oz (14 g) dry yeast.

½ oz (14 g) fresh yeast = 1 envelope dry yeast.

Fresh yeast = 50 percent dry yeast by weight.

Dairy Products



This section lists dairy products, including eggs and cheeses.

BUTTER PRODUCTS

Butter is made from pasteurized cream. It can be purchased salted and unsalted, which is referred to as *sweet butter*. USDA AA and A grades are usually used in foodservice. Butter is also available mixed with margarine for pastry use and as a spread. It is advisable to store butter in a freezer.

Packs:

- 1-lb (450-g) prints, 36 to a case.
- ¼-lb (112-g) sticks; 36-lb (16.2-kg) cases with 144 sticks.
- 5-lb (2.2-kg), 7-lb (3.1-kg), and 24-lb (10.8-kg) tubs, whipped butter.
- 68-lb (30.6-kg), 64-lb (28.2-kg), and 50-lb (22.5-kg), bulk butter, domestic.
- 56-lb (25.2-kg) boxes, bulk butter, imported.

Packs, reddies:

- 12-lb (5.4-kg) and 15-lb (6.7-kg) cases.
- 30-lb (13.5-kg) cases of six 5-lb (2.2-kg) cartons.

Counts, reddies:

- 1-lb (450-g) = 60, 72, 90, and 108 pats.
- 8-lb (3.6-kg) cases, 90-count cups, 1.4 oz (40 g) each.

Butter chip weights:

- 60 count = ¼ oz (7.5 g).
- 72 count = ⅓ oz (5.6 g).
- 90 count = ½ oz (4.6 g).

Kitchen yields:

- 1 lb (450 g) butter = 4 sticks or 2 cups.
- 1 lb (450 g) whipped butter = 6 sticks or 3 cups.
- 1 lb (450 g) butter = 1¾ cups (340 g) melted, clear fat.
- 2½ lb (1.1 kg) butter = 2 cups (0.47 l) melted, clear fat.

BUTTER BLENDS AND SPREADS

There are many spreads on the market that resemble butter, which are made

with or without dairy products. *See* Margarine in this section.

Packs: 1 lb (450 g) and up.

Calories: 1 tb, ½ oz (14 g) = 50 calories.

BUTTER, HIGH FAT

Butter with a butterfat content of 82 percent, versus 80 percent for regular butter, is sold under various trade names and is available salted or unsalted. Rendered butter (clarified butter), with a butterfat content of 99.8 percent, is available under various brand names.

Packs:

- 1-lb (450-g) prints, 36 per case.
- 40-lb (18-kg) bulk pack.
- 1-lb (450-g) tubs, rendered butter.

Calories:

Regular butter:

- 1 lb (450 g) or 4 sticks = 3,260 calories.
- 1 stick (112 g) = 815 calories.
- 1 tb (14 g) = 100 calories.
- 1 pat, 90 count per lb = 36 calories.

High-fat butter:

- 1 lb (450 g) = 3,341 calories.

Rendered butter:

- 1 lb (450 g) = 4066 calories.

Whipped butter:

- 1 lb (450 g), or 6 sticks = 3,240 calories.
- 1 stick (75 g), or ½ cup = 540 calories.
- 1 tb (9 g), or ⅓ stick = 65 calories.
- 1 pat (120-count per lb) = 25 calories.

CHEESE VARIETIES

Cheeses can be made with cow's, goat's (often referred to as *chevre*), and ewe's (sheep's) milk. A small amount of cheese is also made with camel's milk and, in Italy, with buffalo's milk. The cheeses in this section are listed alphabetically by milk categories, except for Blue Cheeses, which are listed together. Cheeses made with blended milks from different animal species are listed under the predominant milk category. The calories listed are based on the information provided by the manufacturers. It should be noted that many cheeses are available made from skim milk or 2 percent milk, and therefore their calorie counts can vary greatly. Because most cheeses lose moisture during storage unless they are hermetically sealed, the calorie counts may be higher by weight when the cheese is used than originally indicated.

It is impossible to list all cheeses. Charles de Gaulle, former president of France, is reported to have said, "How can you govern a country that makes a thousand different cheeses?" In addition to France, practically all other countries where milk-producing animals are domesticated make cheese of some kind or other.

The following categories are listed:

- Blue cheeses, without regard to the kind of milk used
- Cow's milk cheeses
- Ewe's milk cheeses
- Goat's milk cheeses

BLUE CHEESES

Blue-veined cheeses, ranging from semi-soft and mild to crumbly and rather salty, are produced in many countries. Today these cheeses are inoculated with a penicillin-type mold.

BLEU DE BRESSE

This cheese, one of many blue cheese varieties made in France, is crumbly and hard to cut. It is a nice cheese for cheese platters because it is small and has a mild flavor and no unpleasant aroma.

Pack: 8 oz (225 g).

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: One loaf, 8 oz (225 g) = 3 servings.

Calories: N/A; refer to package information.

BLUE CHEESE SHROPSHIRE

English cheese resembling Stilton.

Pack: 16-lb (7.2-kg) wheel.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: One 16-lb wheel = 75 to 80 servings. There may be some

waste because the blue veins are not evenly distributed.

Calories: Refer to package information.

CABRALES

These cheeses are often made with goat's milk. *See* page 34.

CASHEL BLUE

Irish cheese made in County Tipperary. The cheese is firm and crumbly when young and becomes creamy when aged.

Pack: 3 lb (1.3 kg).

Needs ripening: *See* preceding information.

Serving size: 3 oz (85 g).

Kitchen yields: One 3-lb loaf = 14 servings

Calories: N/A; refer to package information.

CRUMBLED BLUE

Domestic blue cheese in pieces, often used for dressings and stuffing.

Packs: 20-lb (9-kg), 25-lb (11.2-kg) and 50-lb (22.5-kg) bags.
Other packs are available.

Needs ripening: No.

Salad dressing serving: 1 oz (28 g).

Kitchen yields: There is basically no waste.

Calories: 1 oz (28 g) = 100 calories.

DANISH BLUE

Denmark started to produce blue-veined cheeses during the First World War when imports were cut off. It is a very rich cheese with a high cream content. Its texture is semifirm to crumbly. The cheese is often marketed as Danablu.

Packs:

5-lb (2.2-kg) to 5½-lb (2.5-kg) wheels.
6-lb (2.7-kg) wheel.
Three 7½-lb (3.4-kg) wheels.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: 5-lb (2.2-kg) wheel = 25 servings.

Calories: 3 oz (85 g) = 300 calories.

FOURME D'AMBERT

Creamy cave-ripened cheese from France in cylindrical shape.

Pack: 4.5 lb (2 kg), 2 pieces per case.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: One 4.5-lb log = 22 servings.

Calories: 3 oz (85 g) = 300 calories.

GORGONZOLA

Famous Italian cheese made in a number of variations and now in a number of countries. The original Gorgonzola, known by the name *Stracchino di Gorgonzola Dolcelatte*, is sweeter, milder, and creamier than regular Gorgonzola. The cheese will melt over low heat and is often stirred into pasta.

Packs:

Regular Gorgonzola:

4-lb (1.8-kg) to 8-lb (3.6-kg) wheel.
12-lb (5.4-kg) wheel.

Gorgonzola Dolcelatte:

13 lb (5.8 kg).
Smaller packs are available.

Needs ripening: No.

Serving size:

Regular Gorgonzola: 3 oz (85 g).

Gorgonzola Dolcelatte: 3 oz (85 g) over pasta.

Kitchen yields: 4-lb (1.8-kg) regular Gorgonzola = 18 to 20 servings.
13-lb (5.8-kg) Gorgonzola Dolcelatte = 65 servings. The cheese is foil wrapped, and there is little waste.

Calories: 3 oz (85 g) = 300 calories.
Check package information for Gorgonzola Dolcelatte.

INDIVIDUALLY PACKAGED BLUE CHEESES

Available domestically and imported.

Packs: From 1¼ oz (35 g).

Needs ripening: No.

Serving size: Usually 2 oz (56 g) as snack or in fruit baskets.

Kitchen yields: There is no waste; the cheese is wrapped.

Calories: 2 oz (56 g) = 200.

MAYTAG BLUE

This is the original American blue cheese developed in Iowa. It is a whole-milk cheese with a nutty flavor.

Packs: 4-lb (1.8-kg) wheel.
12.5-lb (5.6-kg) bag, crumbled cheese.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: 4-lb (1.8-kg) wheel = 18 to 20 servings. The cheese is wrapped in foil, and there is little waste.

Calories: 3 oz (85 g) = 300 calories.

ROQUEFORT

Roquefort is a registered trademark; the texture is crumbly and the cheese can be difficult to portion. Only cheeses produced in the Roquefort area in central France can be called *Roquefort*. When Roquefort dressing is listed on the menu, only genuine Roquefort cheese can be used.

Needs ripening: No.

Pack: 5-lb (2.2-kg) wheel.

Serving size: 3 oz (85 g) as cheese course.

Kitchen yields: One 5-lb (2.2-kg) wheel = 25 restaurant servings.
One 5-lb (2.2-kg) wheel = 50 dressing servings.

Calories: 3 oz (85 g) = 270 calories.

STILTON

English blue cheese in a cylindrical shape. Older cheese can be salty and hard to cut; it is usually scooped out, which is attractive on buffets. When cheese is old, the yield is low, because the crusty rind cannot be used. Older cheeses are often

flavored with sweet port wine to soften the tangy flavor.

Packs: 8-lb (3.6-kg), 12-lb (5.4-kg), and 14-lb (6.3-kg) wheels.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: 12 lb (5.4 kg) = about 50 to 60 servings, depending on age. There is a considerable amount of rind, which cannot be used.

Calories: 3 oz (85 g) = 300 calories.

WISCONSIN BLUE

Good domestic blue cheese.

Pack: 6-lb (2.7-kg) wheel.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: 6-lb (2.7-kg) wheel = 30 servings. The cheese is foil wrapped, and there is little waste.

Calories: 3 oz (85 g) = 300 calories.

COW'S MILK CHEESES

The cheeses listed here are the most widely used and readily available varieties; many are obtainable as both regular and light cheese (low fat). The fat content of cheeses varies greatly and is also influenced by changing weights as a cheese ages and loses moisture. It is advisable to check the label information when available. The calorie contents given here are therefore approximate.

AMERICAN PASTEURIZED PROCESS CHEESE

Available unsliced, sliced vertically, and sliced horizontally. Available yellow and white, regular and light, in both colors. Used for hot and cold sandwiches.

Packs:

Six 3-lb (1.3-kg) whole loaves.

Six 5-lb (2.2-kg) whole loaves.

10-lb (4.5-kg) whole loaf.

Four 5-lb (2.2-kg) loaves, individual slices, wrapped.

Four 5-lb (2.2-kg) loaves, ribbon horizontal slices.
 40-lb (18-kg) block.
Velveeta brand: Six 5-lb (2.2-kg) whole loaves; this cheese is good for melting.

Needs ripening: No.

Serving size: 3 oz (85 g), 4 slices.

Kitchen yields:

$\frac{3}{4}$ oz (21 g) average weight of slice.
 One 3-lb (1.3-kg) loaf = 65 slices.
 One 5-lb (2.2-kg) loaf = 110 slices.
 One 3-lb (1.3-kg) loaf = 16 servings.
 One 5-lb (2.2-kg) loaf = 17 servings.
 5 cups (1.1 l) = 1 lb (450 g) grated cheese.

Calories:

Regular cheese:

1 slice, $\frac{3}{4}$ oz (21 g) = 60 calories.
 1 oz (28 g) = 80 calories.
 3-oz (85-g) serving = 240 calories.

Light cheese:

1 slice, $\frac{3}{4}$ oz (21 g) = 53 calories.
 1 oz (28 g) = 70 calories.
 3-oz (85-g) serving = 210 calories.

AMERICAN PASTEURIZED PROCESS CHEESE SPREAD

Available in different flavors, such as wine and herbs. The texture is smooth.

Packs: 1-lb (450-g) tubs. Many other packs are available.

Needs ripening: No.

Texture: Soft spread.

Kitchen yields: 1 tub (450 g) = 8 servings.

Serving size: 2 oz (56 g) as spread.

Calories: 1 oz (28 g) = 82 calories.
 2 oz (56 g) = 164 calories.

ASIAGO

Available imported from Italy; also made domestically with different flavors and textures. Originally, *Asiago* cheese was made with ewe's milk. The butterfat content is 50 percent of solids.

Packs: 4.5-lb (2-kg) wheel, aged 9 months.
 12-lb (5.4-kg) half moons.

20-lb (9-kg) wheel, aged.
 22-lb (9.9-kg) daisies.
 24-lb (10.8-kg) wheels.

Needs ripening: No.

Texture: Fresh and soft, semifirm, and aged; suitable for grating.

Serving size: 3 oz (85 g)
 1 oz (28 g), grated.

Kitchen yields: One 12-lb (5.4-kg) piece = 58 restaurant servings; 115 servings grated.

Calories: 3 oz (85 g) = 240 calories.

BAGNES CHEESE

This cheese is made in Switzerland. It is used for the melted cheese dish *Raclette*, which is made by melting the surface with an electric heater and scraping the melted cheese onto plates. The cheese is smelly, so *Raclette* should not be prepared in a dining room.

Packs: 5-lb (2.2-kg) and 6-lb (2.7-kg) loaves.

Needs ripening: No.

Texture: Semifirm.

Serving size: 6 oz (170 g) per serving as main course with potatoes.

Kitchen yields: One 5-lb loaf = 10 to 12 servings. There is some waste because not all of the loaf can be used.

Calories: N/A. Refer to label information.

BAKERS CHEESE

Cream cheese type used mostly in baking.

Packs: 3-lb (1.3-kg) loaves.
 10-lb (4.5-kg) tubs.

Needs ripening: No.

Texture: Soft, resembling cream cheese.

Serving size: Varies, depending on recipe.

Kitchen yields: There is no waste.

Calories: N/A; cheese is always used as recipe component.

BEL PAESE

Trade name of a popular Italian table cheese available from domestic sources and imported. It is a good cheese for cheese platters and banquets because it has no offensive smell. Closely related cheeses are Fleurs des Alpes, Bella Alpina, and Butter Käse, among others.

Pack: 5-lb (2.2-kg) and 6-lb (2.4-kg) wheels.

1-oz (28-g) medallions, 288 per case.

Needs ripening: No.

Texture: Semisoft.

Kitchen yields: One 5-lb wheel = 20 servings. The cheese is coated with an inedible wax, which should be removed before service.

Serving size:

Restaurant: 3½-oz (100-g) portion. 1 wheel, 5-lb (2.2-kg) = 20 servings.

Banquet service: ⅓ wheel for table of 10 covers.

Calories: 3½ oz (100 g) portion = 210 calories.

BOURSIN

Fresh, rich, delicate French cheese with high fat content; available plain and with a variety of flavor components added.

Pack: 5-oz (140-g) packages, 12 to a case.

Needs ripening: No; shelf life is about 1 month.

Texture: Soft, resembling cream cheese.

Serving size: 2.5 oz (70 g).

Kitchen yields: 1 case = 24 servings.

The cheese is foil wrapped, and there is little waste.

Calories: 2.5 oz (70 g) = 300 calories.

BRICK

Inexpensive cheese, suitable for cutting in cubes for buffets.

Packs: 5-lb (2.2-kg), 10-lb (4.5-kg), and 40-lb (18-kg) blocks.

Needs ripening: No.

Texture: Semisoft, waxy.

Serving size: 5 oz (140 g), for receptions.

Kitchen yields: One 10-lb (4.5-kg) block = 30 servings.

Calories: 5 oz (140 g) = 140 calories.

BRIE

One of the most popular soft dessert cheeses. Available as a domestic product and imported from a number of countries. The cheese may be flavored, as with herbs, peppercorns and other flavorings. Imported cheese made from unpasteurized milk is available and is considered more flavorful. Brie de Meaux, named after a small village in Normandy, is the most famous of all French Brie cheeses. The fat content varies; about 60 percent is normal.

Packs, typical sizes: 8-oz (300-g), 2-lb (900-g), 2.2-lb (1-kg), 4.4-lb (2-kg), and 6-lb (2.7-kg) wheels.

Needs ripening: Yes.

Texture: Firm when unripe, soft and runny when ripe.

Serving size:

⅛ of a 2.2-lb (950-g) imported wheel, about 4⅓ oz (125 g).

⅛ of a 2-lb (900-g) domestic wheel, about 4 oz (112 g).

2½ oz (70 g) per person for sandwiches.

Kitchen yields:

One 4.4-lb (2-kg) wheel = 20 banquet covers.

One 2-lb (900-g) wheel = 12 sandwich servings.

One 4.4-lb (2-kg) wheel = about 200 canapés.

Calories: 4⅓ oz (125 g) restaurant portion = 320 calories. Calories vary according to fat content.

CACCIOCVALLO

Cheese resembling provolone; often made with skim milk. It is a plastic-curd cheese and can be used for grating when aged.

Pack: 5-lb (2.2-kg) balls, 6 to a case.

Needs ripening: No; the cheese stores well.

Texture: Soft when young and suitable as a table cheese. The cheese will melt.

Serving size: 1 oz (28 g) for grating.

Kitchen yields: 5-lb (2.2-kg) ball = 75 servings

Calories: 1 oz (28 g) = 80 calories.

CAMEMBERT

Camembert became popular in 1836 when the French emperor Napoleon III praised it. Camembert is available as a domestic product and is also imported from many countries. The cheese originated in Normandy, France.

Packs: 8-oz (225-g) wheels, 12 to a case. Individual packs in many sizes.

Needs ripening: Yes.

Texture: Firm when unripe, soft and runny when ripe. Overripe Camembert can develop an unpleasant ammoniacal taste.

Serving size: 1 wedge, 1½ oz (37 g).

Kitchen yields: One 8-oz (225-g) wheel = 6 wedges.

Calories: 1 wedge, 1½ oz (37 g) = 115 calories. 2 oz (56 g) = 180 calories.

CANTAL

French cheese with an ancient history; a good cheese for cheese boards. Cantal, a distant relative of Cheddar, is also known as *Fourme*.

Packs: 40-lb (18-kg) to 120-lb (54-kg) wheels.

Needs ripening: No.

Texture: Hard textured.

Serving size: 3 oz (85 g).

Kitchen yields: One 40-lb (18-kg) wheel = 200 servings. The rind is inedible.

Calories: Refer to package information.

CARRÉ

Fresh cheese with about 60 percent fat content, also known as *double crème Carré*.

Packs: 4 oz (112 g). 10 oz (280 g).

Needs ripening: No.

Texture: Soft and spreadable.

Serving size: 2 oz (56 g).

Kitchen yields: 4 oz (112 g) = 2 servings.

Calories: 2 oz (56 g) = 200 calories.

CHEDDAR

Popular cheese named after the village of Cheddar in England. The cheeses are often covered with colored wax, and numerous varieties are made all over the world. This is a hard, often crumbly cheese and can be white or yellow (colored with annatto). The moisture content varies greatly, depending on age and brand, and this influences yield and slicing characteristics. The flavor can range from sharp to mild.

Packs:

Cylindrical, longhorn: 13 lb (5.8 kg).

Rectangular: 5 lb (2.2 kg), 10 lb (4.5 kg), and 40 lb (18 kg).

Wheels: 12 lb (5.4 kg), 22 lb (9.9 kg), and 75 lb (33.8 kg).

4-lb (1.8-kg) pieces, 6 to a case.

5.5 lb (2.5 kg), 2 to a case.

Shredded: 5-lb (2.2-kg) bags, 6 to a case.

Aged, sharp, New York State: 10-lb (4.5-kg) loaf.

Colored prints: 10-lb (4.5-kg) loaves, 4 to a case.

Colored: 40-lb (18.0-kg) block.

Farm House: 14-oz (400-g) pieces, 6 to a case.

Cabot white: 10-lb (4.5-kg) loaf.

Needs ripening: No.

Texture: Semisoft to hard.

Serving size: 3 oz (85 g); 1½ oz (42 g) with apple pie.

Kitchen yields: 10-lb (4.5-kg) loaf = 50 servings

Pack: 1 cup shredded = 4 oz (112 g) on average.

Calories: 3 oz (85 g) = 345 calories.
 1 cube 1 × 1 × 1 in, $\frac{5}{8}$ oz (17 g) = 70 calories.
 1 cup, shredded = 460 calories.

COLBY

Type of Cheddar cheese with softer body and therefore containing more moisture. It is sold in the same sizes as Cheddar.

Texture: Semisoft to hard.

Serving size: 3 oz (85 g).

Kitchen yields: See Cheddar, although portions may have to be slightly increased. This cheese is good for shredding.

Calories: 3 oz (85 g) = 345 calories.

COLDPACK CHEESE SPREADS

Spreads are available in a number of flavors. These are soft, spreadable, and versatile products for making canapés and sandwiches. There is no waste.

Packs: 5-lb (2.2-kg) tube.
 10-lb (4.5-kg) tube.
 30-lb (13.5-kg) tube.
 1-lb (450-g) ceramic jars.

Needs ripening: No.

Texture: Soft and spreadable.

Serving size: 2 oz (56 g) as spread.

Kitchen yields: One 5-lb (2.2-kg) tube = 40 servings.
 One 1-lb (450-g) jar = 8 servings.

Calories: 2 oz (56 g) = 164 calories.

COTTAGE CHEESE.

Fresh cheese, available creamed and with different-sized curds. Fat content also varies.

Packs: 5-lb (2.2-kg) tubs.
 8-oz (225-g) individual servings.
 Many other sizes are also available.

Needs ripening: No.

Texture: Curd sizes vary greatly, and texture can range from wet for small

curds to rather hard and dry for large curds.

Serving size: 1 scoop #6, approximately 5 oz (140 g) for fruit salad.

Kitchen yields: One 5-lb (2.2-kg) tub = 15 servings.

Calories:

Creamed: 1 cup, 8 oz (300 g),
 4 percent fat = 235 calories.
 2 percent fat = 205 calories.
 1 percent fat = 165 calories.

Dry Curd: 1 cup, $\frac{5}{8}$ oz (145 g),
 $\frac{1}{2}$ percent fat = 125 calories.

COULOMMIERS

Soft French cheese resembling Brie.

Packs: 1 lb (450 g) in wooden boxes.

Needs ripening: Should be used fresh.

Texture: The cheese is soft.

Serving size: 3 oz (85 g).

Kitchen yields: 1 lb (450 g) = 5 servings.

Calories: 3 oz (85 g) = 220 calories.
 Calorie contents vary, refer to package information.

CREAM CHEESE

Cream cheese can be made with whole milk, skim milk, or light or heavy cream.

Packs: $\frac{3}{4}$ -oz (21-g) individual packs, 100 to a case.
 1-oz (28-g) individual packs, 100 to a case.
 3-oz (85-g) individual packs.
 8-oz (225-g) individual packs.
 3-lb (1.3-kg) loaves, 6 or 10 to a case.
 18-lb (8.1-kg) carton.
 30-lb (13-kg) bulk.

Cream cheese, Philadelphia: 3-lb (1.3-kg) loaves, 6 to a case.

Cream cheese, whipped: 3-lb (1.3-kg) tubes, 6 to a case.

Needs ripening: No.

Texture: The cheese is soft.

Serving size: 2 oz (56 g) with smoked salmon.

Kitchen yields: There is no waste.

Calories: 2 oz (56 g), regular = 300 calories.

2 oz (56 g), whipped = 200 calories.

CRÈME FRAÎCHE

Usually used as a recipe component.

Packs:

10-oz (280-g) containers, 12 to a case.

3-lb (1.3-kg) tubs.

5-lb (2.2-kg) tubs.

Needs ripening: No; product should be used when delivered.

Texture: Soft, creamy.

Serving size: Varies according to recipe.

Calories: 1 oz (28 g) = 100 calories.

DOUX DE MONTAGNE

Cheese made in the Pyrenees; a yellow, firm cheese, occasionally made with ewe's milk.

Pack: 8-lb (3.6-kg) wheel.

Needs ripening: No, product should be used when delivered.

Texture: Semisoft; cheese can be cubed.

Serving size: 3 oz (85 g).

Kitchen yields: 8 lb (3.6 kg) = 40 servings.

Calories: 3 oz (85 g) = 180 calories.

EDAM CHEESE

Round or oval-shaped cheese, most often covered with red or yellow wax. The cheese is named after a Dutch town, but it is made in many countries. This is an attractive cheese for platters; it can be hollowed out and filled with cubed cheese for display.

Packs: 4-oz (112-g) individual packs.

2-lb (900-g) balls.

4-lb (1.8-kg) balls.

Needs ripening: No. This cheese has a good shelf life because it is protected with wax.

Texture: Firm.

Serving size: 3 oz (85 g).

Kitchen yields: One 4-lb (1.8-kg) ball = 20 servings.

Calories: 3 oz (85 g) = 180 calories.

EMMENTHAL

German name for genuine Swiss cheese.

See page 31.

EPOISSE

Fresh cheese made in Burgundy, France.

Pack: 8.8-oz (250-g) piece, 6 pieces to a case.

Needs ripening: Should be used fresh.

Texture: Soft, creamy.

Serving size: 3 oz (85 g).

Kitchen yields: 8.8 oz (250 g) piece = 3 servings. There is no waste.

Calories: 3 oz (85 g) = 220 calories.

Calorie contents vary, refer to package information.

EXPLORATEUR

French *Triple Crème*; soft, velvety, buttery fresh cheese.

Pack: 8-oz (225-g) pieces, 6 pieces to a case.

Needs ripening: Should be used fresh.

Texture: Soft, creamy.

Serving size: 2½ oz (70 g).

Kitchen yields: 8-oz (225-g) piece = 3 servings. There is no waste.

Calories: 2½ oz (70 g) = 250 calories.

FONTINA

Originally from the Italian Alps, this cheese is now made domestically, as well as imported from many countries. It is a good cheese for cheese boards, or for fruit baskets sent to guest rooms as an amenity, because there is little smell.

Packs: 7-lb (3.15-kg) wheel, 2 per case.

11-lb (5-kg) wheel.

13-lb (5.8-kg) wheel.

18-lb (8.1-kg) wheel.

Needs ripening: No.

Texture: Semisoft and buttery.

Serving size: 3 oz (85 g).

Kitchen yields: One 11-lb (5-kg) wheel
= 50 to 52 servings.

Calories: 3 oz (85 g) = 180 calories.

FROMAGE BLANC

French name for unripened fresh cream cheese.

Pack: 8-oz (225-g) individual packs, 8
to a case.

Serving size: 2 oz (56 g).

Needs ripening: No.

Texture: Creamy.

Kitchen yields: 8 oz (225 g) = 4
servings.

GOUDA

Cheese resembling Edam cheese. Available as a domestic product and imported, smoked and plain. The cheese is suitable for buffet displays. Baby Gouda is sometimes covered with red wax, like Edam cheese.

Packs: 8 oz (225 g), 12 per case.
2-lb (900-g) balls.
5-lb (2.2-kg) loaf.
8-lb (3.6-kg).
11-lb (4.9-kg) wheel.

Needs ripening: No.

Texture: Firm to hard.

Serving size: 3 oz (85 g).

Kitchen yields: 8 oz (225 g) = 2½
servings.

Calories: 3 oz (85 g) = 210 calories.

GRAPE CHEESE

Made in France. The outside is covered with the seeds and skins of black grapes.

Pack: 4 lb (1.8 kg).

Needs ripening: No.

Texture: White and buttery. Rind should
be served, but is not really edible.

Serving size: 3 oz (85 g).

Kitchen yields: 4 lb (1.8 kg) = 20
servings.

Calories: 3 oz (85 g) = 225 calories.

GRATED CHEESE

Grated cheese is available made from both imported and domestic cheeses. It is often a blend of various cheeses. The most common cheese varieties used are Parmesan and Pecorino Romano, because they do not melt when exposed to heat. (See also Parmesan.) *Pecorino*, listed under Goat Cheeses, can be grated. It is best to get a number of samples from different purveyors and select the best cheese suited for the operation. The least expensive cheese may not provide good yield and flavor.

Packs: 1 lb (450 g).
20 lb (9 kg) and larger.

Needs ripening: No. Fresh cheese should
be kept refrigerated because it can
get moldy.

Texture: Dry.

Serving size: 1 oz (28 g) over pasta.
Uses vary greatly.

Kitchen yields: 1 lb (450 g) = 16 servings.
1 cup = 4 oz (112 g).
1 oz (28 g) = 5½ tb.

Calories: 1 oz (28 g) = 112 calories.
1 cup = 4 oz (112 g) = 455 calories.
The calorie counts vary because
much cheese is produced with skim
milk. Refer to package information
when available.

GRUYÈRE

Version of Swiss cheese originally made in the village of Gruyère in Switzerland. Available imported from Switzerland and France; also domestically made.

Packs: 6-lb (2.7-kg) loaf without rind.
40-lb (18-kg) wheel with rind.

Needs ripening: No; the cheese has a
long shelf life.

Texture: Firm.

Serving size: 3 oz (85 g) on sandwiches.

Kitchen yields: 6-lb (2.7-kg) loaf = 30 servings. When purchased as wedge with rind on, the waste is about 5 percent. Loaves have little cutting waste.

Calories: 3 oz (85 g) = 315 calories.

HAVARTI

Mild cheese, often imported from Denmark and available with added flavor components such as caraway seeds, jalapeña peppers, dill, and others. The cheese is not smelly and is suitable for cheese platters.

Pack: 9-lb (4-kg) loaf.

Needs ripening: No.

Texture: Semifirm.

Serving size: 3 oz (85 g).

Kitchen yields: 9-lb (4-kg) loaf = 45 servings.

Calories: 3 oz (85 g) = 180 calories.

JARSLBERG

Firm cheese resembling Swiss Emmenthal cheese, originally made in Norway.

Packs: 12-lb (5.4-kg) slicing loaf without rind.
20-lb (9-kg) wheel.

Needs ripening: No; the cheese has a long shelf life.

Texture: Firm.

Serving size: 3 oz (85 g) on sandwiches.

Kitchen yields: 12 lb (5.4 kg) loaf = 60 servings. When purchased as wedge with rind on, the waste is about 5 percent. Loaves have little cutting waste.

Calories: 3 oz (85 g) = 315 calories.

KEFALOTIRI

Hard, salty Greek cheese resembling Parmesan. *See* Parmesan.

LIEDERKRANZ

Old-fashioned American cheese made originally in New York State.

Packs: 6-oz (170-g) packages.

Needs ripening: Yes.

Texture: Soft; the cheese is smelly.

Serving size: 2 oz (56 g).

Kitchen yields: One 6-oz (170-g) package = 3 servings; there is no waste.

Calories: N/A; refer to package information.

LIVAROT

Fresh cheese from Normandy.

Pack: 8.8-oz (250-g) packages, 6 per case.

Needs ripening: The cheese should be used fresh, as it will become smelly when aged too long.

Texture: Creamy.

Serving size: 2 oz (56 g).

Kitchen yields: One 8.8-oz (250-g) package = 4 servings.

Calories: N/A; refer to package information.

MASCARPONE

Very rich cream cheese of Italian origin. Available imported from Italy and domestically made. The fat content can be as high as 70 percent. The fresh cheese is often used in Italian desserts.

Packs: 8-oz (225-g) containers, 12 per case.
4-oz (112-g) cylinders, slightly aged.
1.1-lb (1-kg) tubs.

Needs ripening: No; use fresh.

Texture: Creamy and soft.

Serving size: 2 oz (56 g) for aged cheese.

Kitchen yields: There is no waste.

Calories: 2 oz (56 g) = 220 calories.

MONTEREY JACK

Cheddar-type cheese, often used in Mexican cooking. Both mild and sharp varieties are available.

Packs: 40-lb (18-kg) block.
13-lb (5.8-kg) longhorn cylinders.
10-lb (4.5-kg) prints, 4 per case.

5-lb (2.2-kg) prints.
5-lb (2.2-kg) bags shredded, 6 per case.

Needs ripening: No.

Texture: Ranges from semisoft to hard; suitable for grating.

Serving size: 3 oz (85 g) for aged cheese.

Kitchen yields: 10-lb (4.5-kg) print = 50 servings. This cheese has only a thin rind; there is little waste.

MORBIER

French cheese with a horizontal layer of decorative edible ash. This cheese looks interesting on cheese platters.

Pack: 15-lb (6.7-kg) wheel.

Needs ripening: No.

Texture: Semisoft.

Serving size: 3 oz (85 g).

Kitchen yields: One 15-lb (6.7-kg) wheel = 75 servings. The rind cannot be served.

Calories: 3 oz (85 g) = 315 calories.

MOZZARELLA

Italian cheese of the filata type. Mozzarella di bufala is made from buffalo milk. The fresh cheese is soft and becomes hard when cured.

CURED MOZZARELLA

Cheese mostly used for pizza, but also for sandwiches.

Packs: 40-lb (18-kg) block.

20-lb (9-kg) loaves.

12-lb (5.4-kg) sausage shape.

5-lb (2.2-kg) to 6 lb (2.7 kg) loaves.

5-lb (2.2-kg) bags, shredded, 6 per case.

Needs ripening: No.

Texture: Firm and elastic, suitable for grating.

Serving size: 3 oz (85 g) for sandwich.

Pizza Serving Size:

1 lb (450 g) for one 16-in pizza or 8 servings. This is a generous serving; many pizza operations use 12 oz (335 g).

10 to 12 oz (280 to 335 g) for one 12-in medium pizza.

Kitchen yields: 5-lb (2.2-kg) bag shredded = five 16-in large pizzas.
5-lb (2.2-kg) bag shredded = seven 12-in medium pizzas.
40-lb (18-kg) block, shredded = forty 16-in large pizzas.
40-lb (18-kg) block, shredded = sixty 12-in medium pizzas.

Calories: 2 oz (56 g), whole milk = 180 calories.

2 oz (56 g), part skim milk = 160 calories.

FEATHERED MOZZARELLA

Shaved mozzarella cheese, which is lighter than regular grated mozzarella cheese. As little as 5 oz (140 g) can cover one 16-in pie.

FRESH MOZZARELLA

Fresh mozzarella is usually packed in brine; it is available in individual balls, as listed here.

Packs: 3-lb (1.4-kg) tub with one hundred forty-four ⅓-oz (9-g) *ciliegini* (small balls).

3-lb (1.4-kg) tub with thirty-two 1.5-oz (42-g) *bocconcini* (medium balls).

6-lb (2.7-kg) tub with six 16-oz (450-g) balls.

6-lb (2.7-kg) pieces, fresh loaf, 8 per case.

Needs ripening: No.

Texture: Soft and elastic.

Serving size: 3 oz (56 g) for appetizer.

Kitchen yields: 3-lb (1.4-kg) tub = 10 servings, as appetizer. Use *ciliegini* for receptions.

SMOKED MOZZARELLA

Packs: 2-lb (900-g) loaves.

6-lb (2.7-kg) loaves, 2 per case.

Needs ripening: No.

Texture: Firm and elastic, suitable for grating.

Serving size: 3 oz (85 g) for sandwich.

Kitchen yields: 2-lb (900-g) loaf = 10 servings.

Calories: Whole milk: 2 oz (56 g) = 180 calories.

Part skim milk: 2 oz (56 g) = 160 calories.

MUENSTER CHEESE

Originally made in Alsace, France; imported and domestic products are available. The imported cheese has a much more pungent aroma than the domestic cheese, which is often bland.

Packs: 5-lb (2.2-kg) loaves, 6 per case, imported.

2-lb (900-g) loaves, imported.

4.3-oz (120-g) pieces, 10 loaves per case.

10- to 11-lb (4.5- to 4.9-kg) longhorn cylinder, domestic.

5-lb (2.2-kg) rectangular, domestic.

5-lb (2.2-kg) wheel, domestic.

10-lb (4.5-kg) rectangular, domestic.

40-lb (18-kg) rectangular, domestic.

Needs ripening: No.

Texture: Good cheese for sandwiches.

Slices well on machine. Rind is very thin and edible; do not remove.

Serving size: 3 oz (85 g) for sandwich.

Kitchen yields: 5-lb (2.2-kg) loaf = 25 servings.

Calories: 3 oz (85 g) = 240 calories.

OKA

Originally Canadian mild monastery cheese; no longer produced by monks.

Packs: 5-lb (2.2-kg) loaves, 6 per case.
2 lb (900 g).

Needs ripening: No.

Texture: Semisoft.

Serving size: 3 oz (85 g) for sandwich.

Kitchen yields: 5-lb (2.2-kg) loaf = 25 servings.

Calories: 3 oz (85 g) = 240 calories.

PARMESAN

Famous Italian cheese, with rather hard texture. It can be eaten as a dessert cheese, but most is used for grating. This cheese is also made domestically and in many other countries. The grated cheese labeled "imported" is not necessarily Italian. Parmigiano Reggiano is made in a specific region around the city of Parma; Parmigiano Padamo is made in other areas of Italy. Parmigiano Reggiano is usually more expensive.

Packs: 80-lb (36-kg) wheel.

40-lb (18-kg) ½ wheel.

20-lb (9-kg) ¼ wheel.

10-lb (4.5-kg) ⅛ wheel.

5-lb (2.2-kg) bag, grated.

1-lb (450-g) bag, grated.

Needs ripening: No.

Texture: Hard.

Serving size: 1 oz (28 g), grated over pasta. 2 oz (56 g), as dessert cheese.

Kitchen yields:

10-lb (4.5-kg), ⅛ wheel = 150 servings, over pasta.

10-lb (4.5-kg), ⅛ wheel = 75 servings, as cheese course.

Parmesan cheese is hard to cut. When buying cheese for grating, buy it cut in chunks.

20-lb (9-kg) ¼ wheel = 75 cups grated cheese.

1 cup = 4 oz (100 g) grated cheese.

Calories: 1 oz (28 g) = 112 calories.

1 cup = 4 oz (112 g) = 455 calories.

PONT L'EVÊQUE

Cheese with a long history, imported from France in square wooden boxes.

Packs: 10-oz (280-g) square boxes.

Needs ripening: Yes.

Texture: Creamy, soft. There is no waste.

Serving size: 2.5 oz (70 g).

Kitchen yields: 10 oz (280 g) = 5 servings.

Calories: 2.5 oz (70 g) = 250 calories.

PORT DU SALUT

Cheese originally from France, now fabricated in many countries. Good cheese for buffet and platters.

Packs: 3½ lb (1.5 kg) to 4 lb (1.8 kg).

Needs ripening: No.

Texture: Elastic, firm to semifirm.

Serving size: ⅙ wedge of one wheel, about 4 oz (112 g).

Kitchen yields: One 3½- to 4-lb (1.5- to 1.8-kg) loaf = 16 servings. The rind is thin, and there is little waste.

Calories: 4 oz (112 g) = 280 calories.

PROVOLONE

Cheese related to Caccio Cavallo, made with whole milk. Imported and domestic products are available. Good cheese to use for sandwiches and buffets, easy to slice on a machine.

Packs: 20-lb (9-kg), 40-lb (18-kg), and 50-lb (22.5-kg) rectangular blocks.
15-lb (6.7-kg) loaf.
12-lb (5.4-kg) ball.
12-lb (5.4-kg), 25-lb (11.25-kg), and 50-lb (22.5-kg) sausage-shaped pieces.
6-lb (2.7-kg) half-moon-shaped pieces.

Needs ripening: No.

Texture: Firm, dense, and smooth.

Serving size: 3 oz (85 g) for sandwich.

Kitchen yields: One 12-lb (5.4-kg) loaf = 60 sandwiches. The cheese has a thin golden-to-brownish rind that must be discarded.

Calories: 3 oz (85 g) = 300 calories.

QUESO BLANCO

Spanish name for *white cheese*. The cheeses differ greatly from country to country and can range from fresh to aged and dry. The dry version is often rather salty and is served with guayabate. See Hispanic Foods in the International Ingredients section.

REBLOCHON

Soft, ripening French cheese.

Packs: 20-oz (560-g) wheels, 6 per case.
1-lb (450-g) round loaves.

Needs ripening: Yes.

Texture: Creamy, soft. There is no waste.

Serving size: 2.5 oz (70 g).

Kitchen yields: One 20-oz (560-g) wheel = 8 servings.

Calories: 2.5 oz (70 g) = 250 calories.

RICOTTA CHEESE

Soft cheese, resembling cream cheese. Products made with skim milk, with whole milk, and from whey left over from making different cheeses, are available.

Packs: 3-lb (1.3-kg) tubes, 6 per case.
5-lb (2.2-kg) tubes.
10-lb (4.5-kg) tubes.
20-lb (9-kg) boxes.
30-lb (13.5-kg) bags.

Needs ripening: No; use as fresh as possible.

Texture: Soft. Excellent cheese for baking. Good with fruit salad.

Serving size: According to use.

Kitchen yields: There is no waste.

Calories: 1 cup [8 oz (225 g)] whole milk = 428 calories.
1 cup [8 oz (225 g)] skim milk: = 340 calories.

RICOTTA SALATA CHEESE

Ricotta, whose name means *recooked*, is made from whey left over from making various cheeses. This version, imported from Italy, is more solid than the soft ricotta sold in tubes. The aged cheese can be used for grating.

Pack: 6-lb (2.7-kg) wheel.

Needs ripening: No; use when delivered.

Texture: Dry and crumbly.

Serving size: 1 oz (28 g) grated.

Kitchen yields: 6-lb (2.7-kg) wheel = 90 servings, grated.

Calories: N/A, refer to package information.

SAINT ANDRÉ

Mild, rich triple-crème cheese with 70 percent butterfat content.

Packs: 4-lb (1.8-kg) wheel.
7-oz (200-g) pieces, 6 per case.

Needs ripening: Yes, if too hard when delivered.

Texture: Creamy, rich, and silky.

Serving size: 2.3 oz (65 g).

Kitchen yields: One 7-oz (200-g) piece = 3 servings.

4-lb (1.8-kg) wheel = 25 servings.

Calories: 2.3-oz (65-g) serving = 250 calories.

STRING CHEESE

Mozzarella cheese kneaded to form sticks and twists.

Packs: 1-oz (28-g) individually wrapped sticks.

6-oz (170-g) to 10-oz (280-g)
individually wrapped twists.

Needs ripening: No.

Texture: Stringy.

Serving size: As needed.

Kitchen yields: Individual portions are handy for cheese baskets. There is no waste.

Calories: Whole milk: 2 oz (56 g) = 180 calories. Part skim milk: 2 oz (56 g) = 160 calories.

SWISS CHEESE

DOMESTIC SWISS CHEESE

Domestic Swiss cheese is available in many sizes and shapes. Low-salt and low-calorie cheeses are on the market. The cheese will become stringy when melted.

Packs:

6-lb (2.7-kg) to 8-lb (3.6-kg), 4 × 4 in (101 × 101 mm), sandwich loaves.

9-lb (4-kg) to 15-lb (6.7-kg), 4 × 8 in (101 × 202 mm), blocks.

90-lb (40-kg) blocks.

100-lb (45-kg) blocks.

175-lb (18.7-kg) wheels with rind.

3-lb (1.3-kg) to 5-lb (2.2-kg) Baby Swiss wheels.

Needs ripening: No.

Texture: Firm; the cheese slices well on a slicing machine.

Serving size: 3 oz (85 g) for sandwich.

Kitchen yields: One 8-lb (3.6-kg) loaf, 4 in × 4 in = 40 sandwiches. Cheese purchased in loaves has no rind, and therefore very little waste. Cheese purchased in wheels has about 5 percent trimming waste.

Calories: 3 oz (85 g) = 315 calories.

IMPORTED SWISS CHEESES

Genuine Swiss cheese was originally manufactured only in the Emmenthal valley in Switzerland, but today it is also made in other parts of Switzerland as well. To distinguish Swiss-made cheese from other cheeses it is now stamped, in red, "Switzerland Swiss." Swiss cheese is also imported from many other countries, especially Austria and Finland. Battelmatt is a cheese made in the Canton (State) of Tessin, Switzerland, when there is not enough milk to make large wheels of Emmenthal cheese. The eyes of Battelmatt cheese are smaller, and the shape is cylindrical. The cheese has a 45 percent dry matter fat content.

Packs: 16-lb (7.2-kg) loaf without rind.
10- to 14-lb (4.5- to 6.3-kg).
Up to 177-lb (80-kg) wheels.

Needs ripening: No; the cheese has a long shelf life.

Texture: Firm.

Serving size: 3 oz (85 g) on sandwich.

Kitchen yields: 10-lb (4.5-kg) loaf, 4 in × 4 in = 50 sandwiches. When purchased as wedge with rind on, the waste is about 5 percent. Loaves have little cutting waste.

Calories: 3 oz (85 g) = 315 calories.

TALEGGIO

Soft, ripened cheese imported from Italy.

Pack: 5-lb (2.2-kg) loaf. Other sizes are available.

Needs ripening: Yes, but the cheese will become sharper as it ages.

Texture: Creamy, rich, and silky.

Serving size: 2 oz (56 g).

Kitchen yields: One 5-lb (2.2-kg) loaf = 35 servings. The thin rind cannot be served when the cheese is overaged.

Calories: 2 oz (56 g) = 160 calories.

TOMME DE SAVOIE

Mild, low-fat French cheese.

Pack: 4-lb (1.8-kg) wheel.

Needs ripening: No.

Texture: Firm.

Serving size: 3 oz (85 g).

Kitchen yields: One 4-lb (1.8-kg) wheel = 19 to 20 servings. The cheese has little smell and can be used on cheese boards.

Calories: 3 oz (85 g) = 240 calories.

EWE'S (SHEEP'S) MILK CHEESES

The fat content of ewe's milk cheeses varies greatly and is also influenced by changing weights as these cheeses age and lose moisture; the information on calories given here is therefore approximate. It is advisable to check the label information when available. The Italian name for ewe's milk cheeses, *picaroon*, is frequently used.

BRINATE DE MUGELLO

Soft, ripened, fresh Italian cheese resembling Brie cheese, high in butterfat.

Pack: 2-lb (950-g) pieces, 2 each per case.

Needs ripening: No; cheese should be used when delivered.

Serving size: 3 oz (85 g).

Kitchen yields: One 2-lb (950-g) piece = 10 servings.

Calories: N/A; refer to package information.

CACIO FIORE

Also called *caciotta*. This Italian cheese resembles Bel Paese (see page 22) and is flavored with saffron. Instead of an animal-based rennet, artichoke-flower-based rennet is used to curdle the milk, making this cheese potentially kosher. Note that Cacciocavallo is made with cow's milk (see page 22).

Pack: 2-lb (900-g) and 4.5-lb (2-kg) rectangular or round loaves.

Needs ripening: No; cheese should be used when delivered.

Serving size: 3 oz (85 g).

Kitchen yields: One 2-lb (900-g) loaf = 10 servings.

Calories: 3 oz (85 g) = 120 calories.

CACIO DI FOSSA

Aged sharp cheese, usable for grating.

Pack: 3.5-lb (1.5-kg) loaves.

Needs ripening: No.

Serving size: 2 oz (56 g) grated over pasta.

Kitchen yields: One 3.5-lb (1.5-kg) loaf = 25 servings.

Calories: 2 oz (56 g) = 80 calories.

FETA

Originally Greek white cheese, made with goat's or ewe's milk, sometimes blended with cow's milk. The cheese is an integral ingredient in Greek Salad and is also popular in cooking. It is slightly salty and crumbly and is shipped in brine.

Packs: 8-lb (3.6-kg) tubes, net.
28-lb (12.6-kg) tubes, net.

Needs ripening: No.

Serving size: 3 oz (85 g) for Greek Salad.

Kitchen yields: One 8-lb (3.6-kg) tube, net = 40 servings. One 28-lb (12.6-kg) tube, net = 150 servings. There is no rind and little waste.

Calories: 3 oz (85 g) = 180 calories.

LOCATELLI ROMANA

Tart cheese usable for grating.

Pack: 60-lb (27-kg) wheel.

Needs ripening: No.

Serving size: 2 oz (56 g) grated over pasta.

Kitchen yields: One 60-lb (27-kg) wheel = 460 servings.

Calories: 2 oz (56 g) = 120 calories.

MANCHEGO

Famous Spanish cheese, available from various producers and in a number of versions, including a soft cheese preserved in olive oil. The basic Manchego is a rich semisoft cheese.

Packs: 6-lb (2.7-kg) wheel.
7.5-lb (3.3-kg) wheels, 4 to a case.

Needs ripening: No.

Serving size: 3 oz (85 g).

Kitchen yields: One 6-lb (2.7-kg) wheel = 28 to 30 servings.

Calories: 3 oz (85 g) = 240 calories.

PECORINO

There are numerous versions of Pecorino; the best known is Pecorino Romano, a hard grating cheese. The cheeses range from fresh Scodelatto, aged only 20 days, to Pecorino Romano, aged 1 year. Pecorino cheeses made in Sardinia are called *Sardo*.

Packs:

Pecorino Scodelatto: 2.2-lb (1-kg) pieces, 3 pieces per case.
Pecorino Oro: 4.4-lb (2-kg) pieces, 2 pieces per case.
Pecorino Romano: 10-lb (4.5-kg) wheel.
Pecorino Romano: 15-lb (6.7-kg) wedge.

Needs ripening: No.

Serving size: 3 oz (85 g). 1 oz (28 g) for grating.

Kitchen yields:

One 2.2-lb (1-kg) piece Pecorino Scodelatto = 33 servings.
One 4.4-lb (2-kg) piece Pecorino Oro = 64 servings.
One 10-lb (4.5-kg) wheel Pecorino Romano = 150 servings, grated.
One 15-lb (6.7-kg) wedge Pecorino Romano = 225 servings, grated

Calories: Calorie count varies greatly; refer to package label.

PECORINO ROMANO, GRATED

Pack: 5-lb (2.2-kg) tub.

Needs ripening: No.

Serving size: 1 oz (28 g).

Kitchen yields: One 5-lb (2.2-kg) tub = 80 servings. There is no waste.

Calories: 1 oz (28g) = 70 calories.
1½ oz (42 g) for dressing = 105 calories.

GOAT CHEESES (CHEVRE)

The fat content of goat cheeses varies greatly and is also influenced by changing weights as the cheeses age and lose moisture. It is advisable to check the label information when available. Excellent goat cheeses are made domestically.

BANON DE CHALAISE

Small cheese from Provence aged in eau-de-vie and wrapped in chestnut leaves.

Pack: Twelve 4-oz (112-g) packages per case.

Needs ripening: Cheese should be used within 1 month.

Serving size: 2 oz (56 g).

Kitchen yields: 1 package = 2 servings.
1 case = 24 servings.

Calories: N/A; refer to package information.

BUCHERON CHEESE LOG

Aged French goat cheese with white, downy rind, soft exterior, and firm center.

Packs: 3-lb 12-oz (1.6-kg) logs.
2-lb 6-oz (1.0-kg) logs.

Needs ripening: Little; cheese should be used within 1 month.

Serving size: 2 oz (56 g).

Kitchen yields: One 3-lb 12-oz (1.6-kg) log = 30 servings.

Calories: N/A; refer to package information.

CABRALES

French cheeses made with a blend of cow's, ewe's, and goat's milk. Pasty texture with blue veins; high acidity and intensity.

Pack: 5.5-lb (2.5-kg) pieces, 2 per carton.

Needs ripening: No.

Serving size: 2 oz (56 g).

Kitchen yields: 1 piece = 40 servings.

Calories: N/A; refer to package information.

CHEVRE BLUE

Domestic goat cheese.

Packs: Many sizes, often 8 oz (225 g).

Needs ripening: Cheese should be used within 1 month.

Serving size: 2 oz (56 g).

Calories: N/A; refer to package information.

CROTTIN DE CHAVIGNOL

Well-known French cheese, available aged and fresh. The aged cheese has a darker rind.

Pack: 2.5-oz (70-g) pieces aged Crottin, 12 to a box.
2-oz (56-g) pieces fresh Crottin, 12 to a box.

Needs ripening: No; cheese should be used when delivered.

Serving size: 2 oz (56 g) or 2.5 oz (70 g).

Kitchen yields: 1 box = 12 servings.

Calories: N/A; refer to package information.

GOUDA GOAT CHEESE

Firm white cheese, usable for grating and melting.

Pack: 10-lb (4.5-kg) piece.

Needs ripening: No; cheese is already aged.

Serving size: 2 oz (56 g) for grating.

Kitchen yields: 10 lb = 75 servings.

Calories: N/A; refer to package information.

HALLOUMI

From Cyprus, semihard sheep and/or goat milk cheese flavored lightly with mint. It is served grilled and will not melt.

Pack: 9-oz (250-g) bricks.

Serving size: 3 oz (85 g).

Kitchen yields: One 9-oz (250-g) brick = 3 servings. There is no waste.

Calories: N/A; refer to package information.

MIZITHRA

Greek soft cheese resembling cottage cheese. *See* Cottage Cheese.

MONTRACHET

Ash-covered logs with soft creamy centers, originally from Burgundy.

Pack: 11-oz (300-g) log.

Needs ripening: No; cheese is already aged. Use when received.

Serving size: 2.75 oz (77 g), ¼ log.

Kitchen yields: One 11-oz (300-g) log = 4 servings.

Calories: N/A; refer to package information.

CREAM

See Milk and Milk Products in this section.

EGG PRODUCTS

Freshness, grade, and size are important considerations when using shell eggs. Eggs are graded by the U.S. Department of Agriculture (USDA) or by the states. Almost all foodservice facilities use grade A eggs. Brown eggs are not more nutritious than white eggs. It is prudent to weigh egg cases occasionally, upon delivery. Smaller eggs or even filler can inadvertently be used to fill the cases. Eggs may carry salmonella, and many foodservice operations use whole or shelled pasteurized eggs for scrambled eggs and for egg sauces. Natural hard-boiled eggs, chopped boiled eggs, and hard-boiled eggs in roll shape are convenient products. Bantam eggs are about half the size of a large egg and come from bantam (dwarf) breeds of chickens. In flavor, they cannot be distinguished from other eggs.

DEHYDRATED EGGS

Powdered whole eggs are usually of lesser importance to the average foodservice operator. The ratio is normally: 12 oz (340 g) dried sifted whole powder + 1 qt (0.94 l) water = 24 large eggs. Powdered egg whites are often used in pastry shops.

Packs: By weight.

Kitchen Yield: Use the preceding formula.

FROZEN SHELLED EGGS

Whole eggs, yolks only, egg whites only, and whole eggs with extra yolks, are available. Frozen egg yolks are mixed with sugar.

Packs: 30-lb (13.5-kg) cans.

15-lb (6.75-kg) cans.

1-qt (0.47-l) cartons, 12 per case.

Weight conversions:

Whole eggs:

One 30-lb (13.5-kg) can = 300 eggs, or 25 dozen.

One 15-lb (6.7-kg) can = 150 eggs.

Yolks only:

One 30-lb (13.5-kg) can = 675 yolks.

One 15-lb (6.7-kg) can = 338 yolks.

Egg whites only:

One 30-lb (13.5-kg) can = 540 egg whites.

One 15-lb (6.7-kg) = 270 egg whites.

Frozen egg yolks mixed with sugar:

One 30-lb can (13.5-kg) = 608 yolks and 3 lb (1.35 kg) sugar.

One 15-lb can (6.7-kg) = 304 yolks and 1.5 lb (675 g) sugar.

One cup, 8 oz (0.24-l) = 10 yolks and $\frac{2}{3}$ oz (22 g) sugar.

SHELL EGGS

Size	Weight by Dozen	Weight per Case
Jumbo	30 oz (840 g)	56 lb (25.2 kg)
Extra large	27 oz (750 g)	50½ lb (22.7 kg).
Large	24 oz (670 g)	45 lb (20.2 kg).
Medium	21 oz (590 g)	39½ lb (17.8 kg)
Small	18 oz (500 g)	34 lb (15.3 kg)
Pee Wee	15 oz (420 g)	28 lb (12.6 kg).

Packs: 1 case = 30 dozen, 360 eggs.
 One half case = 15 dozen, 180 eggs.
 One layer = 2½ dozen, 30 eggs.

Serving sizes:

3 eggs per person, scrambled eggs or omelette, individual servings.
 2½ eggs per person, banquet service.

Kitchen yields:

1 case eggs = 120 individual servings.
 2 layers broken eggs = 20 individual portions.
 1 case eggs = 144 servings for banquets.
 2 layers eggs = 24 servings for banquets.
 1 cup (0.24 l) = 4 large chopped, boiled eggs.
 1 large egg = 6 slices.
 Egg whites will about double in volume when whipped.

Volume conversions:**Large Whole Eggs:**

1 cup = 8 oz (0.24 l) = 5 eggs.
 1 pint = 16 oz (0.47 l) = 10 eggs.
 1 gal = 126 oz (3.8 l) = 80 eggs or 2 layers plus 20 eggs.

Large Egg Yolks:

1 cup = 8 oz (0.24 l) = 11 to 12 yolks.
 1 pint = 16 oz (0.47 l) = 24 yolks, or 2 dozen.

1 qt = 32 oz (0.94 l) = 46 to 48 yolks, or 4 dozen.

1 gal = 126 oz (3.8 l) = 190 yolks, or 16 dozen.

Large Egg Whites:

1 cup = 8 oz (0.24 l) = 7½ egg whites.
 1 pint = 16 oz (0.47 l) = 15 egg whites.
 1 qt = 32 oz (0.94 l) = 30 egg whites.
 1 gal = 126 oz (3.8 l) = 120 egg whites, or about 11 dozen.

Calories (based on large eggs):

1 egg, raw, 2 oz (56 g) = 80 calories.
 1 egg yolk = 65 calories.
 1 egg white = 15 calories.
 1 egg, fried in butter = 85 calories.

EGG SUBSTITUTES

Ready-to-use cholesterol-free products made with egg whites and additives are available. Products can be used for omelettes and scrambled eggs.

Packs: 2-lb (900-g) cartons, 12 per case.

1-lb (450-g) cartons, 24 per case.

½-pint (0.24-l) cartons, 48 per case.

Kitchen yields: ¼ cup, 2 oz (56 g) = 1 egg. One cup, 8 oz (225 g) = 4 eggs.

Calories: ½ cup, 4 oz (112 g) = 60.

EGGS, VEGETARIAN

Eggs produced by chickens that were fed all-vegetarian natural feed are available. Sizes and weights are the same as for other eggs.

FRESH QUAIL EGGS

For Canned Quail Eggs, see the Groceries section.

Packs: 30 eggs per flat, 18 flats per case, 540 eggs per case.

6 flats per case, 180 eggs per case.

ICE CREAM AND SHERBETS

The quality of regular ice cream is measured by butterfat content, weight (overrun), and flavor. Overrun is the amount of expansion that occurs during the freezing process; it is normal and necessary for smooth texture, but whipping additional air into the product to increase volume can decrease flavor. Ice cream with high overrun is weak in flavor and light, and has a limited shelf life. Occasionally, ice cream should be weighed when received. Ice cream will shrink during storage and handling. Sherbets are made with milk and fruit flavors. Low-calorie ice cream, ice milk, and frozen yogurt have gained in popularity.

Butterfat content is measured in percentages, based on weight:

- 16 to 18 percent, deluxe quality.
- 14 to 16 percent, high quality.
- 12 percent minimum for vanilla ice cream.
- 10 percent minimum for ice cream with added solids, such as chocolate or nuts.

Overrun is expressed in percentages in relation to weight:

- 100 percent overrun is high.
- 80 percent overrun is normal.
- 50 percent overrun means a dense product.

Packs: 2½-gal (9.5-l) tubs.

3-gal (11.4-l) tubs, standard in many locations.

Smaller packs for specialty ice cream.

Weights: 2½-gal (9.5-l) tub = 11¼-lb (5-kg) minimum weight.

2½-gal (9.5-l) tub = 15-lb (6.7-kg) high-quality ice cream.

3-gal (11.4-l) tub = 13½-lb (6-kg) minimum weight.

3-gal (11.4-l) tub = 18-lb (8.1-kg) high-quality ice cream.

1-gal (3.8-l) = 4½ lb (2 kg) minimum weight.

1-gal (3.8-l) = 6 lbs (2.7 kg) highest-quality, with 50 percent overrun.

Kitchen yields, Scooped Ice Cream:

Scoops are classified by the number of scoops in 1 qt (0.47 l). For instance, there are twelve #12 scoops in 1 qt of ice cream. Measurements are approximate, as they depend on how well the portions are dipped, and on the overrun of the product. High-overrun ice cream will shrink in storage.

Portion weights: Depend on the overrun. One #8 scoop of high-quality ice cream weighs about 3½ oz (100 g). Scoop sizes are listed in the Sizes and Miscellaneous Information section.

Banquet Serving Size: 1 qt (0.9 l) = 10 covers when served in a ring mold with fruit sauce.

Calories:

3½ oz (100 g) hardened ice cream, 11 percent butterfat = 200 calories.

3½ oz (100 g) ice cream, 16 percent butterfat = 236 calories.

One cup, 6 oz (0.24 l), soft-serve ice cream = 375 calories.

One cup, 4.6 oz (130 g), ice milk, 4.3 percent fat = 185 calories.

One cup, 6.25 oz (175 g), ice milk, soft serve, 2.6 percent = 225 calories.

One cup, 6.8 oz (193 g), sherbet, 2 percent fat = 270 calories.

MARGARINE

Margarine is available salted, sweet, whipped, and mixed with butter. Specialty margarine for baking is also available.

Packs: 1-lb (45-g) prints, 30 per case.
50-lb (22.5-kg) boxes.

4-lb (1.8-kg) tubs whipped margarine,
6 per case.

Pats are the same counts as butter; *see*
page 17.

Kitchen yields: Basically the same as for
butter.

Calories: 1 lb (450 g), regular margarine
= 3,260 calories.

4 oz (112 g), regular margarine, = 815
calories.

1 tb, $\frac{1}{8}$ stick (14 g), regular margarine
= 100 calories.

1 pat, 90-count (5-g), regular
margarine = 35 calories.

One 8-oz (225-g) container, soft spread
= 1,635 calories.

$\frac{1}{2}$ cup (76 g) whipped margarine =
545 calories.

1 tb, $\frac{1}{8}$ stick (9 g), whipped margarine
= 70 calories.

MILK AND MILK PRODUCTS

CREAM

Cream must contain at least 18 percent
butterfat to be called *cream*.

Minimum fat contents:

- *Coffee cream:* 18 to 30 percent
butterfat content.
- *Half-and-half:* 11½ percent butterfat
content.
- *Sour cream:* 18 percent or more
butterfat.
- *Whipping cream, light:* 30 to 35
percent butterfat content.
- *Whipping cream, heavy:* 36 to 40
percent butterfat content.

COFFEE OR LIGHT CREAM

Packs: Pints (0.47 l), quarts (0.94 l), and
5-gal (3.8 l) containers.

Individual packs.

Serving size: $\frac{3}{4}$ oz (22 ml) to 1 oz (29
mml) per person with coffee.

Calories: 1 cup (0.24 l) light cream, 18
percent fat = 470 calories.
1-oz (29-ml) serving = 60 calories.

CRÈME FRAÎCHE

Thick, slightly sour cream of French
origin.

Pack: 10-oz (280-g) containers, 12 per
case.

3-lb (1.3-kg) tubs.

5-lb (2.2-kg) tubs.

Kitchen yields: Vary according to recipe.

IMITATION CREAM

Products are based on vegetable fats.

Calories: 1 cup (0.24 l), frozen = 335
calories.

1 cup (0.24 l), powdered = 515
calories.

HEAVY CREAM

Heavy cream, with a butterfat content of
36 percent, is commonly used in the
industry. There is a whipping cream with
30 percent butterfat content on the
market, but it is usually too light for
making whipped cream for commercial
applications. There is also a heavy cream
with a butterfat content of 40 percent on

the market; it is used by some chefs in à la carte cooking.

Packs: ½ pints (0.24 l).

Quarts (0.47 l).

Larger packs are available.

Kitchen yields: Cream doubles in volume when whipped.

1 cup (0.24 l) = 1 pint (0.47 l) whipped.

1 qt (0.94 l) = 2 qt, ½ gal (1.9 l) whipped.

2 qt (1.9 l) = 1 gal (3.8 l) whipped.

Calories: 1 oz (28 g), heavy = 100 calories.

1 cup (0.24 l), heavy, not whipped = 820 calories.

1 oz (28 g), light = 70 calories.

1 cup (0.24 l), not whipped = 700 light.

IMITATION SOUR CREAM

Calories: 1 cup (0.24 l) = 640 calories.

SOUR CREAM

Packs: 5-qt (4.7-l) tubs.

Other packs are available.

Serving size: 3 oz (85 g), with baked potato.

Kitchen yields: One 5-qt tub = 50 servings.

Calories: 1 cup (0.24 l) = 495 calories.

One 3 oz (85 g) serving = 185 calories.

WHIPPED TOPPING

Pressurized products are available, made with either real cream or nondairy products.

Calories: 1 cup (0.24 l) = 155 calories.

1 cup (0.24 l), made with vegetable fats = 240 calories.

MILK PRODUCTS

Milk is available condensed, evaporated, dehydrated, and fresh. Milk products from animals other than cows have a limited market and are not listed here. Milk not requiring refrigeration until it is opened is listed in the Groceries section under the trade name Parmalat.

Specifications:

Buttermilk: At least 8¼ percent nonfat milk solids

Condensed milk: 8½ percent butterfat content, 28 percent milk solids, 45 percent sugar

Evaporated milk: 7.9 percent butterfat content minimum, 25 percent milk solids

Dried milk, nonfat: 1¼ percent fat content

Dried milk, whole: 2.6 percent butterfat content

Fresh milk, fat free: Less than 0.1 percent butterfat content

Fresh milk, skim: Less than 3¼ percent butterfat content

Fresh milk, whole: 8¼ percent milk solids, 3¼ percent butterfat content

Case milk conversion: 1 case of 72 half-pints (0.23 l) = 18 qt (16.9 l).

1 case of 24 qt (0.94 l) = 96 half-pints.

1 5-gal (18.8-l) container = 80 half-pints.

BUTTERMILK

Available with varying fat content.

Packs: ½ pints (0.24 l), 48 per case.

1 qt (0.47 l), 12 per case.

½ gal (1.9 l).

Calories: ½ pint (0.24 l), low fat = 100 calories.

1 cup (0.24 l), buttermilk light, 1.5 percent = 120 calories.

CHOCOLATE MILK

Packs: ½ pints (0.24 l), 48 per case.

1 qt (0.94 l), 24 per case.

Calories:

1 cup (0.24 l), regular = 210 calories.

1 cup (0.24 l), low fat, 2 percent = 180 calories.

1 cup (0.24 l), low fat, 1 percent = 160 calories.

CONCENTRATED MILK

Packs: 6-oz (0.17-l) cans.

#10 33.8 fl. oz (3-l) cans, 6 per case.
Other packs are available.

Kitchen yields: 1 #10 can + 14 cups
water = 7 qt (6.5 l) whole milk.

CONDENSED MILK

Evaporated, sweetened milk that has a higher milk solids and fat content than evaporated milk. Sugar content is equivalent to 19 to 20 lb (8.5 to 9 kg) to every 100 lb (45 kg) fresh milk.

Packs: 6-oz (0.17-l) cans.
104 fl oz (3 l) #10 cans, 6 per case.
Other packs are available.

Kitchen yields: One 6-oz (0.17-l) can +
10 oz (0.81 l) water = 2 cups (0.49 l)
milk.

Calories: 1 cup (0.24 l) = 980 calories.

DRY CRYSTAL MILK

Pack: By weight.

Weight: 1 lb (450 g) = 6½ cups.

Kitchen yields: 1½ cups powder + 2½
cups (0.85 l) water = 1 qt (0.94 l)
milk.

DRY MILK

Available as *Whole Milk* and as *Skim Milk*.

Pack: By weight.

Kitchen yields: ⅓ cup powder + ⅔
cups water = 1 cup (0.24 l).

Calories: 1 cup (0.24 l), nonfat = 325
calories.

EGGNOG

Packs: Quarts (0.47 l).

Calories: 1 cup (0.24 l), without alcohol
= 340 calories.

EVAPORATED MILK

Available as whole milk and skim milk.

Pack: #10 cans, 6 per case.
Other sizes are available.

Can Content: One #10 can = 14 cups
dry powder.

Kitchen yields: 1 cup evaporated milk +
1 cup water = 2 cups (0.47 l) milk.
One #10 can = 7 qt (6.6 l) milk.

Calories: 1 cup (0.24 l) whole milk =
340 calories. 1 cup (0.24 l) skim milk
= 200 calories.

FRESH MILK

Fresh milk is available in numerous packs and sizes, with varying fat contents, flavor, and nutritional additives. Of interest to the foodservice operator are the packs listed here:

Packs: ½ cup (0.12 l), ½ pint (0.24 l), 1
qt (0.47 l), 1 gal (3.8 l) and 5-gal
(19-l) dispenser packs.

Calories: 1 cup (0.24 l) regular, 4
percent fat = 150 calories.
1 cup (0.24 l) low fat, 2 percent fat =
120 calories.
1 cup (0.24 l) low fat, 1 percent fat =
100 calories.
1 cup (0.24 l) skim milk = 85 calories.

HALF-AND-HALF

Milk and cream blend, often served with coffee.

Packs: 1-qt (0.94-l) containers.
Individual creamers.
Other packs are available.

Calories, 12 percent fat: 1 cup (0.24 l)
= 315 calories.
1 tb = (0.015 l) = 20 calories.

MALTED MILK

See Fresh Milk for packs; 5-gal (19-l)
packs are usually not available.

MALTED MILK POWDER

Made with whole milk and malt powder.

Packs: 1-lb (450-g) containers.

Weights: 4 heaping tsp = 1 oz (28 g).

Kitchen yields: 1 cup water (0.24 l) + ⅔
oz (18 g) powder = 1 cup (0.24 l).

Calories: 1 cup (0.24 l) = 235 calories.

PARAMALAT

See the Groceries section.

SOY MILK

For Soy Products, see the Groceries section.

YOGURT

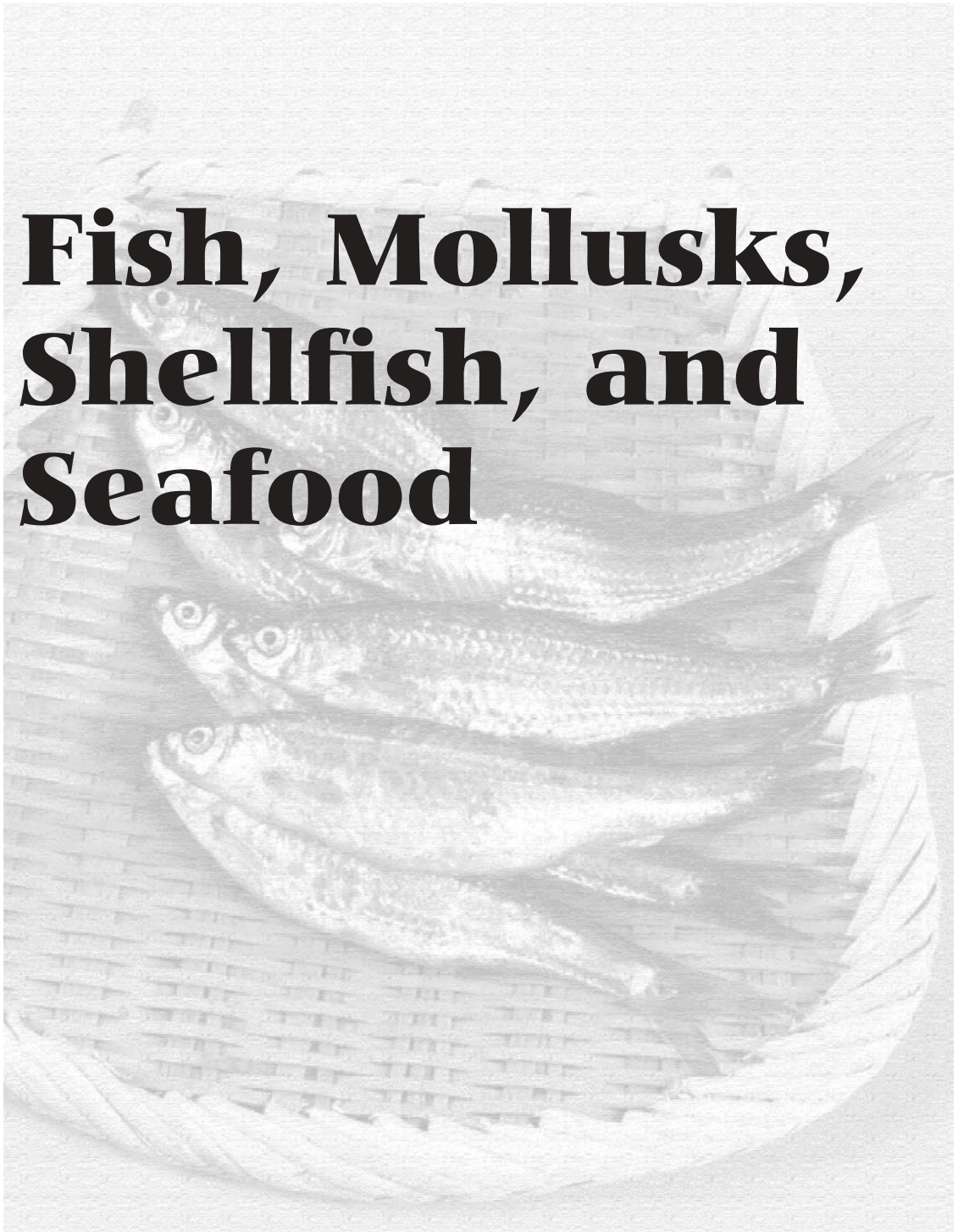
Fermented milk product, available with varying fat contents and with different fruit flavors.

Packs: 8-oz (225-g) packages.
16-oz (450-g) packages.
Other packs are available.

Serving size: 8 oz (225 g). Foodservice operators should purchase only individual servings and serve them in the original containers.

Calories: 8 oz (225 g) plain, low fat = 145 calories.
8 oz (225 g) plain, fruit flavored = 230 calories.

Fish, Mollusks, Shellfish, and Seafood



This section lists canned, fresh, frozen, and processed fish, seafood, and related products.

Fresh and frozen fish can be purchased with the following specifications:

- *Round*: Not eviscerated. Seldom marketed.
- *Drawn*: Also called *gutted*. Entrails removed.
- *Dressed*: Scaled, eviscerated, head and fins removed.
- *Steaks*: Cut across fish, including bones and skin.
- *Filletts*: Meat cut away lengthwise from spine bone of fish.

Filletts, both fresh and frozen, are marketed with skin or with skin removed. Sometimes, some bones are still attached. The buyer should inquire about how the filletts are dressed when comparing prices. Crustaceans are aquatic animals covered with a hard shell, like lobsters and shrimp. Mollusk is the generic name for water animals with a hard outer shell. They are classified as follows:

- *Univalves*, having one shell, such as snails and conch.
- *Bivalves*, having two shells, such as abalone, clams, mussels, and oysters.

ABALONE

Large mussel with beautiful shell, harvested in the Pacific Ocean. Big producers are Alaska and Mexico. California law prohibits shipping fresh or processed abalone harvested in California to other states. Available canned, dried, fresh in season, frozen whole in shell, and vacuum-packed meat.

CANNED ABALONE

Imported from a number of countries. Available as steaks and minced.

Pack: Packed in water in various-sized cans.

1-lb (450-g) can is common.

Serving size: 3 oz (85 g) drained meat for appetizer.

6 oz (170 g) for main course.

Kitchen yields: 1-lb (450-g) can = 13 oz (365 g) to 14 oz (400 g) drained weight.

1-lb (450-g) can = 4 servings for appetizer or 2½ servings for main course.

DRIED ABALONE

Available mostly in oriental markets.

Packs: By weight.

Serving size: 1 oz (28 g).

Calories: 1 oz (28 g) = 30 calories.

FRESH ABALONE

Most fresh abalone available is imported from Mexico or from the Orient.

Packs: By piece.

Serving size: 4 oz (112 g). Pound thin and sauté quickly; serve immediately.

6 oz (170 g) as main course.

Kitchen yields: Shell sizes range from 11 in (270 mm) for red abalone to about 3 to 5 in (70 to 120 mm) across for flat abalone.

Calories: N/A.

FROZEN ABALONE

Imported from Mexico and other tropical countries. Abalone in shells is occasionally available and used in Oriental cooking.

Packs:

3-lb (1.3-kg) boxes, each with twelve 4-oz (112-g) pieces.

Other packs and sizes are available.

Serving size: 4 oz (112 g) as appetizer.

6 oz (170 g) as main course.

Calories: N/A.

ALLIGATOR

See the Meat and Meat Products section.

ANCHOVIES**CANNED ANCHOVIES**

Available packed in oil as flat fillets or as rolled fillets with caper in center. Also available salted and as anchovy paste. Practically all products are imported. Anchovies packed in olive oil are more expensive than those packed in other oils.

Packs:

28-oz (785-g) flat fillets, 24 per case.

13-oz (365-g) flat fillets, 24 per case.

2-oz (56-g) flat fillets, 48 per case.

2-oz (56-g) rolled fillets, 48 per case.

Drained weight: About 20 percent of a can of anchovies is oil.

Sizes: Fillets are approximately the same size in all larger cans, slightly smaller in the smaller cans.

SALT-PACKED ANCHOVIES

Sold by weight.

Kitchen yields: Salt-packed anchovies can be rinsed and then packed in oil, thereby reducing cost.

Calories: 1 oz (28 g), 6 fillets = 25 calories.

FRESH ANCHOVIES

Small quantities of fresh anchovies are available from the Mediterranean and the Pacific. They are very perishable; their shelf

life is less than 48 hours. The fish is normally deep-fried whole.

Season: Year-round, but more likely winter because the fish spoils rapidly.

Packs: By weight.

Count: 1 lb (450 g) = 12 to 18 fish.

Kitchen yields: There is no waste.

BASS VARIETIES**BLACK SEA BASS**

Harvested along the Atlantic coast.

Season: Available year-round; peak occurs in Spring.

Average size: 24 oz (680 g).

Kitchen yields: Waste is about 50 percent. Purchase in fillets because spines are very sharp.

STRIPED BASS

Nearly extinct on the Atlantic coast, this once quite popular fish is presently available only in very limited quantities. Farm-raised striped bass has come on the market to fill the void. Louisiana is a large producer, and much is imported from Chile. *Note:* Purchase whole fish to be assured of freshness. Bones make excellent stock. A number of environmentally concerned chefs boycott Chilean bass, actually Patagonian toothfish.

Season: Available year-round; peaks occur in late fall and early spring.

Sizes:

1 lb (450 g) to 1¼ lb (560 g) dressed, farm raised.

1 lb (450 g) to 3 lb (1.3 kg) dressed, farm raised.

4 lb (1.8 kg) to 6 lb (2.7 kg).

6 lb (2.7 kg) to 8 lb (3.6 kg).

Best size for filleting: 6 lb (2.7 kg) to 8 lb (3.6 kg) whole fish.

Best size for buffet: 8 lb (3.6 kg) to 10 lb (4.5 kg).

Serving size:

3-oz (85-g) fillet for appetizer.

7-oz (200-g) fillet for main course.

Kitchen yields: Waste is about 40 percent, starting with dressed fish.

One 7-lb (3.2-kg) fish = 4-lb (1.8-kg) fillet.

One 4-lb (1.8-kg) fillet = 20 servings, appetizer = 8 servings, main course.

One 8- to 10-lb (3.6- to 4.5-kg) whole fish = 35 servings when served with other items.

WHITE BASS

Fine eating fish resembling weakfish, also called corvina. The fish is not a true bass. Available whole and as fillets.

Season: Summer.

Average sizes: 10 lb (4.5 kg).

Kitchen yields: Purchase fillets.

BLUEFISH

Popular and abundant Atlantic coast commercial and sport fish. Available whole and in fillets. Large fish can be oily and tough.

Season: Summer.

Sizes: 2 lb (900 g) to 6 lb (2.7 kg).

Serving sizes: 8 oz (225 g).

Kitchen yields: One 2-lb (900-g) dressed fish = 2 servings.

Calories: 8 oz (225 g), baked with butter = 360 calories.

BLOWFISH

Harvested in the Atlantic, this fish is also called chicken of the seas. Only the tails are used.

Season: Summer.

Average sizes: 4-oz (112-g) to 8-oz (225-g) skinned tail pieces.

Kitchen yields: Purchase skinned tail pieces only. Cook whole.

BONITO

Fish resembling tuna. Most of the catch is canned. Fresh fish should be soaked in brine and acidulated water to remove the strong taste. The fish has a bloody red flesh and is quite oily. Also known as *little tunny*.

Season: Year-round.

Sizes: 5 lb (1.2 kg) to 20 lb (9 kg).

BRILL

Imported flatfish resembling flounder; sometimes imported whole. Also called *barbue*.

Packs: 10-kg (22.2-lb) crates.

Sizes: 2-lb (0.9-kg) to 4-lb (1.8-kg) whole fish.

Kitchen yields: Purchase large fish when possible. One 4-lb (1.8-kg) fish = four 8-oz (225-g) fillets.

BUTTERFISH

Small, thin fish of little commercial distribution. Available on the Atlantic and Gulf coasts. Much production is exported to Japan.

Season: Available year-round.

Sizes: 4 oz (112 g) to 12 oz (340 g).

Kitchen yields: Use whole fish for pan-frying.

CARP

Bony freshwater fish with little commercial importance in the United States. Used in making gefilte fish, a Jewish fish dumpling popular in central Europe. Large fish can taste muddy. The flesh is dark red and bony. Most harvested is the common carp, but smaller amounts of grass carp have reached the market. Some species have scales, others do not. All have large heads and large belly cavities.

Season: Winter and early spring: common carp. Summer: grass carp.

Sizes: 2 lb (900 g) to 25 lb (11.25 kg), common carp.
6 lb (2.7 kg) to 50 lb (22.5 kg).

Kitchen yields: Purchase fillets with scales removed.

Waste factor: Whole fish = 60 percent.
Fillets = 20 percent because extensive trimming is still needed.

CATFISH AND CATFISH PRODUCTS

Catfish are freshwater fish. The most popular commercial species is the channel catfish, available wild and farmed in southern

states. Blue catfish and flathead catfish are harvested wild only. Catfish is normally sold in fillets. Other catfish varieties are caught and discarded because there is no market. The sea catfish is harvested from the ocean.

Seasons: Spring and summer for wild fish. Farmed fish is available year-round.

Sizes: 1 lb (450 g) to 5 lb (2.25 kg), farmed catfish.

1 lb (450 g) to 50 lb (22.5 kg), wild catfish.

Serving size: 5 oz (140 g) to 7 oz (200 g), breaded fillet.

7 oz (200 g) to 9 oz (250 g), blackened.

Packs:

IQF fillets: 5 oz (140 g) to 7 oz (200 g).

7 oz (200 g) to 9 oz (250 g).

9 oz (250 g) to 11 oz (310 g).

Nuggets:

3 oz (85 g) to 5 oz (140 g).

Individually quick frozen (IQF) fillet strips and steaks are available in various sizes.

Kitchen yields: Purchase fillets, nuggets, or steaks. There is no waste.

BLACKENED CATFISH

Fully seasoned blackened catfish fillets, frozen, are available ready to bake.

Packs: 10-lb (4.5-kg) cartons.

Size: 6-oz (170-g) and 8-oz (225-g) fillets.

Count: Twenty-seven 6-oz (170-g) fillets per case.

Twenty 8-oz (225-g) fillets per case.

CAVIAR VARIETIES

Only sturgeon roe can be called caviar. Unless the species from which it came is identified, fish roe cannot be labeled caviar. For example, sturgeon caviar is named after the species.

BELUGA CAVIAR

Beluga is the largest species of sturgeon, and almost all beluga are caught in the Caspian Sea, which is bordered by Russia and Iran. Beluga sturgeon produce the

largest eggs, which are considered the most expensive caviar. Overfishing has severely decimated the catch. Sturgeon caviar is now produced in the United States, China, and other countries, but is not the beluga size. The word *malossol*, often found on caviar tins, means *lightly salted* in Russian. Caviar is available fresh and pasteurized. Iranian caviar is again permitted into the United States. Caviar should be weighed each time before use to ensure exact portion sizes.

Season: Fresh and pasteurized caviar is available year-round.

Packs: Fresh caviar is imported in original tins. The product is then repacked.

Original Russian pack = 4 lb 4 oz (900 g).

Iranian pack = 3 lb 14 oz (1.7 kg).

Other packs range from 1 oz (28 g) up.

Smaller packs are often pasteurized.

Serving size: 1 oz (28 g) = 1 heaping tb.

1 oz (28 g) = 3 canapés.

Kitchen yields: 10 oz (280 g) = 30 canapés.

BOWFIN CAVIAR

The fish is harvested in Louisiana, and the flesh has little commercial importance. The roe are black and processed into an acceptable-quality caviar.

LUMPFISH CAVIAR

Less expensive caviar, often dyed black or red. It is usually sold pasteurized.

Packs: 2-oz (56-g) jars, 36 each to a case.

3½-oz (100-g) jars, 24 each to a case.

7-oz (200-g) jars, 12 each to a case.

12-oz (240-g) jars, 12 each to a case.

OSETRA CAVIAR

Produced from medium-sized osetra sturgeon. The eggs are medium sized. Pack and use are the same as for beluga caviar.

PRESSED CAVIAR

Imported sturgeon caviar of lesser quality, which has been pressed before shipment. The flavor is very concentrated and much acclaimed by connoisseurs. Seldom exported.

Packs: By weight.

SALMON CAVIAR

Salmon caviar is also known as *keta*. Available in various qualities and egg sizes, ranging in color from dark red to golden yellow. It is advisable to store the product in a refrigerator even when it is pasteurized.

Packs: 2-oz (56-g), 4-oz (112-g), 7-oz (200-g), and 12-oz (240-g) jars.

Serving size: 1 oz (56 g).

SEVRUGA CAVIAR

Caviar produced from a small sevruga sturgeon, harvested in the United States and in some other countries. The eggs are small, but can be of excellent quality. Pack varies; use is the same as for beluga caviar.

VEGETARIAN CAVIAR

A caviar substitute marketed as CAVI*ART has become available to cater to vegetarian customers. It is made from seaweed and looks like caviar.

Packs: 18-oz (500-g) jars, 4 jars to a case. The product does not need refrigeration until opened.

Kitchen yields: Use product as is. There is no waste.

WHITEFISH CAVIAR

Domestic or imported caviar consisting of small eggs.

Packs: Jars of various sizes.

CLAMS AND CLAM PRODUCTS**ATLANTIC HARD-SHELL CLAMS**

Available canned, fresh in shell, shucked fresh, and shucked frozen. Also known by the name *quahogs*. The clams are classified by size.

CANNED CLAMS

The clam variety used for canning are called *skimmer clams*, also known as *beach clams*, *giant clams*, or *sea clams*. They are normally available chopped.

Packs: 5-lb (2.2-kg) cans, 12 per case. 51-oz (1.4-kg) cans, 12 or 24 per case.

Kitchen yields: One 5-lb (2.2-kg) can = 38-oz (1.1-kg) drained clams.

One 51-oz (1.4-kg) can = 24-oz (680-g) drained clams.

Calories: 3 oz (85 g) solids and liquid = 45 calories.

CHERRYSTONES

Cherrystones are medium-sized clams, normally served raw on the half shell or baked.

Season: Available year-round. Scarce when inclement weather makes fishing difficult.

Packs: Bushels or cartons.

Count: 275 to 300 per bushel. Sometimes size and count vary. Many operators purchase clams by the piece.

Serving size: Normal portion is 6 clams on the half shell.

Kitchen yields: One bushel = 40 to 45 servings. There are always broken or open clams that cannot be used.

CHOWDER CLAMS**SHUCKED**

Available freshly shucked with juice, whole or chopped.

Packs: 1-gal (3.8-l) container.

Kitchen yields: No waste. Use 1 gal (3.8 l) for 5 gal (19 l) clam chowder.

Calories: 3 oz (85 g) raw meat = 65 calories.

WHOLE

Some operations still buy large clams for chowder.

Season: Available year-round. Scarce when winter is severe.

Packs: Bushels or cartons.

Count: 150 to 175 per bushel.

Kitchen yields: Use 1 bushel for 4 gal (15.2 l) of clam chowder.

Calories: 3 oz (85 g) raw meat = 65 calories.

CLAM JUICE

Available plain or mixed with tomato, called *clamato juice*.

Packs: 46-oz (1.3-l) cans, 12 to a carton.
16-oz (0.47-l) bottles, 12 to a carton.
8-oz (0.24-l) bottles, 12 to a carton.

Serving size: 4 oz (0.12 l).

Kitchen yields: Twelve 16-oz (0.47-l) bottles = 48 servings.

Calories: Very low.

LITTLENECKS

The smallest-sized quahogs.

Packs: Bushels or cartons.

Count: 500 to 550.

Serving size: Normally, 9 clams on the half shell.

Kitchen yields: 1 bushel = 50 to 55 servings. There are always some broken clams.

ATLANTIC SOFT-SHELL CLAMS

Also called *steamers* or *piss clams*, they are steamed and served whole with clam broth and melted butter. The shells are soft and easily crushed. The clams are often sandy.

Season: Available year-round, but can be scarce during a severe winter.

Packs: 42-lb (18.9-kg) to 46-lb (20.7-kg) bushel basket or carton.
1½-lb (0.675-kg) netted grit-free packs.

Sizes: Small, medium, and large, but seldom graded. Medium size 2 to 4 in (50 to 100 mm) in length.

Count: 550 to 600; varies according to size.

Serving size: 14 to 18 clams, about 15 oz (420 g).

Kitchen yields: About 20 percent loss because of crushed clams.
1 bushel = 35 to 38 servings.

SOFT-SHELL CLAMS, FROZEN, BREADED

Pack: 2-lb (900-g) boxes, 12 per carton.

Serving size: 8 oz (225 g), raw.

Kitchen yields: 4 servings per box.

PACIFIC HARD-SHELL CLAMS**BUTTER CLAMS**

Packs: By weight, often 80-lb (36-kg) sacks.

Kitchen yields: Clams are seldom sorted by size, although minimum sizes are established by the state.

PACIFIC HARD-SHELL GEODUCK CLAMS

Very large clams harvested along the Pacific Coast, especially in Puget Sound. The meat is often breaded and fried, or grilled. Geoduck clams are a popular selection in sushi bars. Geoduck clams are available canned, fresh, frozen, and smoked.

Season: November to May.

Packs: 40-lb (22.5-kg) box.

Size: 3 lb (1.3 kg) average, up to 5 lb (2.25 kg).

Kitchen yields: Good yield; only the fragile shell and viscera are inedible.

MANILA CLAMS

Also known as *Japanese littlenecks*.

Packs: 10-lb (4.5-kg) bags or 25-lb (11.25-kg) bushel.

Count: 15 to 20 pieces per pound (450 g).

PACIFIC HARD-SHELL LITTLENECK CLAMS

Packs: By weight; often 80-lb (36-kg) sacks.

Kitchen yields: Clams are seldom sorted by size, although minimum sizes are established by the state.

PACIFIC HARD-SHELL PISMO CLAMS

Large clams, used for chowders.

Packs: By weight; often 80-lb (36-kg) sacks.

Kitchen yields: Clams are seldom sorted by size, although minimum sizes are established by the state.

PACIFIC SOFT-SHELL CLAMS

Pacific soft-shell clams are Atlantic soft-shell clams that have been transplanted.

Packs: By weight; often 50-lb (22.5-kg) sacks.

Kitchen yields: Clams are seldom sorted by size, although minimum sizes are established by the state.

CONCH

Large tropical clams, normally sold shucked fresh, frozen raw, or frozen cooked.

Packs: 5-lb (2.2-kg), raw frozen, cartons.

Kitchen yields: About 30 percent trimming waste from raw product. Cooked product also has some cleaning waste, unless purchased chopped. 3½-lb (1.5-kg) cleaned conch = 3 gal (11.4 l) chowder.

PACIFIC SOFT-SHELL RAZOR CLAMS

Size: 3½ in to 4 in (87 to 100 mm) long.

COD VARIETIES AND PRODUCTS

Popular fish, without scales, harvested in both the Atlantic and the Pacific Oceans. Atlantic and Pacific cod are closely related species and little difference is made on the market. Most cod is processed into fish products and frozen fillets. Scrod is a small cod. Overfishing has decimated the stock, and severe fishing restrictions are in place. Fillets with skin on are better for broiling, because crisp skin is a delicacy and the fish holds together better. When buying frozen fillets, make sure they are IQF or layer-packed.

Season: Year-round; peak occurs from March to September.

Sizes: Up to 20 lb (9.1 kg).

Trade sizes:

Scrod: 1½ lb (675 g) to 2 lb (900 g), 2 lb (900 g) to 4 lb (1.8 kg).

Market: 2½ lb (1.1 kg) to 8 lb (3.6 kg).

Large: 8 kg (3.6) to 20 lb (9 kg).

Frozen fillets: 4 oz (112 g), 5 oz (140 g), 6 oz (170 g), 7 oz (200 g), and 8 oz (225 g).

Serving sizes: 5-oz (140-g) fillet as main course in institutions.
7-oz (200-g) fillet in restaurants.

Kitchen yields: 50 percent waste when dressed fish is cut in fillets.
One 5-lb (2.2-kg) market cod = 2½-lb (1.1-kg) fillet.
2½-lb (1.1-kg) fillet = 5 servings.

COD CHEEKS

New England specialty, available frozen; usually served breaded and fried.

Packs: By weight.

Size: Little nuggets, about 1 oz (28 g) each.

Kitchen yields: No waste; use as sold.

COD TONGUES

New England specialty; available frozen.

Packs: By weight.

Size: Varies, about 1½ (40 g) to 2 oz (56 g).

Kitchen yields: No waste; use as sold.

SALTED COD

Also called *bacalao* or *stockfish*, this is salted and boned dried fish, sold split or in fillets. Most is imported from Europe or Canada. Available with skin on or off.

Packs: Sold by weight.

Serving size: 3 oz (85 g), dried product.

Kitchen yields: Soak overnight. Size will about double.

SCROD

Term for immature cod, haddock, or pollock.

Season: Available year-round; peak occurs in late fall and winter.

Sizes:

Boston scrod, New York market: 1½ lb (675 g) to 2 lb (900 g).

Boston market: 2 lb (900 g) to 4 lb (1.8 kg).

Kitchen yields: Same as cod.

CRAB VARIETIES

ALASKAN CRAB MEAT SUBSTITUTES

Products based on surimi (Japanese fish cakes) technology. They are available in many shapes resembling lobster tails, crab claws, and shrimp. The most popular

products resemble king crab meat. A number of brand names are used. Some products are shipped frozen, others refrigerated.

Packs, frozen: 5-lb (2.2-kg) IQF packs, 6 per case:

Whole leg: 5 in (127 mm) length, 64 pieces.

Bite size: $\frac{3}{4}$ in (19 mm) length, 384 pieces.

Random chunks: $\frac{1}{4}$ to $\frac{1}{2}$ in (6.3 to 12.7 mm). 3-lb (1.4-kg) IQF packs, 6 per case.

Imitation shrimp: 16 pieces per pound.

Imitation lobster tail: 3 oz (85 g) each, 16 tails to a box.

Split legs: 2 oz (56 g) each, 24 to a box. 2.5-lb (1.1-kg) boxes, 12 per case.

Chunky leg and flaky body-style meat.

Packs, refrigerated: 8-oz (225-g) packages, 12 per case. *Whole leg meat, chunks (cut legs), flakes*

Serving size: 5-oz (140-g) random chunks for cocktail.

Kitchen yields: 5-lb (2.25-kg) box = 15 servings.

Calories: 5 oz (140 g) = 150 calories.

ALASKAN KING CRABS

King crabs are caught in the Bering Sea and off the coast of Alaska and are always marketed cooked. King crabs are available canned and frozen. The foodservice industry uses primarily frozen king crab meat. Only male crabs are processed, and they can weigh up to 12 lb (5.4 kg).

Season: Available year-round. The actual fishing season is from September to December.

Packs:

Whole king crab: For buffet displays, sold by weight.

Leg meat: 5-lb (2.2-kg) whole leg meat, 6 per case.

Fancy meat: 5-lb (2.2-kg) blocks, 6 per case, consisting of:
25 to 35 percent leg meat
20 to 25 percent broken leg and claw meat.

40 to 50 percent white shoulder meat.
10 percent or less shreds.

Split legs in shell: 12-oz (340-g) bags, 12 per case.

Fancy legs and claws: 25-lb (11.25-kg) carton, layer packed.

Minimum leg length: 16 in (406 mm).

Minimum leg weight: 8 oz (225 g).

Number of legs per carton: 32 to 45.

Number of claws per carton: 25 to 30.

Standard legs and claws: 20-lb (9-kg) carton, loose pack.

Minimum leg length: 16 in (406 mm).

Minimum leg weight: 7 oz (200 g).

Number of legs per carton: 24 to 36.

Number of claws per carton: 30/35.

Crab claws with arms: 25-lb (11.25-kg) carton.

Extra large: 35 to 50 per carton.

Weight: 8 oz (225 g) to 10 oz (280 g) each.

Large: 50 to 80 per carton.

Weight: 5 oz (140 g) to 8 oz (225 g) each.

Small claws: 75 to 125 per 25-lb (11.25-kg) carton.

Weight: 3 oz (85 g) to 5 oz (140 g) each.

Serving sizes: 4 oz (112 g), cocktail.

$6\frac{1}{2}$ oz (180 g), hot main course or salad.

Kitchen yields: One 5-lb (2.2-kg) frozen block = $4\frac{1}{2}$ lb (2 kg) thawed and drained.

One 5-lb (2.2-kg) block = 18 servings, cocktail.

One 5 lb (2.2 kg) block = 11 servings, main course.

Calories: 4 oz (112 g) = 60 calories.

ALASKAN SNOW CRAB

Available cooked frozen in blocks and canned. Foodservices use mostly frozen products.

Packs:

Fancy snow crab meat: 5-lb (2.2-kg) blocks consisting of:

45 percent red meat.

34 percent to 45 percent white meat.

10 percent to 20 percent shreds.

Fancy crab meat, frozen tin pack: 5-lb (2.2-kg) cans, consisting of:
30 percent whole leg meat.
70 percent broken leg and shoulder meat.

Whole leg meat, frozen tin pack: 5-lb (2.2-kg) cans, consisting of:
75 percent red meat.
25 percent white meat.

Cocktail claws with shell removed below the pincers: 2-lb (900-g) bags.

Sizes: 7 to 12 per lb (450 g).
12 to 16 per lb (450 g).
16 to 20 per lb (450 g).

Kitchen yields: 5-lb (2.2-kg) frozen block = 4½ lb (2 kg) of thawed meat.

BLUE CRABS

BLUE CRAB MEAT

Available canned, pasteurized fresh, and frozen pasteurized. The size of the pieces is important in judging quality. The meat should be reasonably free of shells and should contain little roe. Cans should be opened as they are used, and the meat should not be handled because it will break easily. Foodservices frequently buy canned pasteurized crab meat. Also available are claws and claw fingers. Fresh crab meat is very perishable.

Packs: 1-lb (450-g) tin, lump crab meat, fresh.
1-lb (450-g) tin, lump crab meat, pasteurized.
1-lb (450-g) tin, lump crab meat, frozen:
Good-quality crab meat should be wrapped in paper inside the can.

Kitchen yields: 1-lb (450-g) tin = 14 oz (400 g) drained.
1-lb (450-g) tin = 2 servings, salad.
1-lb (450-g) tin = 2½ to 3 servings.

Serving size: 7 oz (200 g) for main course salad. 5 oz (140 g) for cocktail.

SOFT-SHELL CRABS

Blue crabs, harvested in the Chesapeake Bay and along the Atlantic seacoast when molting and the shells are still soft. Fishing

for molting crabs is prohibited on the Pacific coast. Louisiana is a major producer of blue crabs. Soft-shell crabs are available live, frozen, and frozen breaded. Smaller crabs are considered superior and command a higher price. *Peelers* are crabs that did not shed their shells completely and had to be peeled by hand. Usually, they are missing legs and claws.

Season: March to November; peak occurs in April, depending on weather.

Packs: Sold by the dozen live, in flats lined with seaweed.

Sizes:

Hotel Prime: 3 to 3½ in (76 to 89 mm) across.

Large Medium: 3½ to 4 in (89 to 100 mm).

Prime: 4½ to 5 in (112 to 127 mm).

Large Prime: 5 to 5½ in (127 to 140 mm).

Jumbo: Over 5½ (140 mm).

Serving sizes: 4 small pieces, *Hotel Prime*, as main course.
3 pieces, *Prime*, as main course.
2 pieces, *Large Prime*, as main course.

Kitchen yields: Soft-shell crabs are small when the season starts.

WHOLE BLUE CRABS

Common crab harvested along the Eastern seaboard from Maine to Texas. When the crabs molt and the new shell is soft, they are marketed as soft-shell crabs. Available live in hard shell, cooked and frozen, raw, and cooked.

Season: Year-round; peak occurs May to October.

Sizes: 5-in (125-mm) to 8-in (200-mm) point-to-point carapace.
#1 crabs 6½ in (160 mm) and larger.

Packs: Live, by weight, and canned.

DUNGENESS CRABS

Dungeness crabs are fished from Alaska to southern California. Only male crabs of a specific size can be taken. Dungeness crabs are available fresh whole, frozen whole,

and as picked crab meat, frozen and canned.

Season: From May to September in Alaska and the Pacific Northwest.

Sizes: 22 lb (9.9 kg) to 24 lb (10.8 kg) per dozen, cooked whole.

Packs: Whole, average weight 2 lb (900 g) to 2¼ lb (1 kg).

Clusters, half crabs.

Single cut legs.

Cocktail claws.

Meat, 5-lb (2.2-kg) boxes.

Kitchen yields: According to application.

Calories: 3½ oz (100 g) meat = 100 calories.

OYSTER CRABS

Tiny crabs, normally served dusted with flour and deep fried. Available frozen.

Packs: 14-oz (400-g) tins.

Serving size: 5 oz (140 g) to 6 oz (170 g) for main course.

Kitchen yields: Oyster crabs are often served combined with fried whitebaits.

RED CRABS

Deep-sea crabs fished off the coast of New England. This is an underutilized species. Most of the harvest is canned.

Size: 1-lb (450-g) to 2-lb (900-g) whole crab.

STONE CRABS

Harvested off the coasts of Florida and Louisiana. Only the claws are eaten. Fresh and frozen boiled claws are available.

Seasons: February to March, September to December.

Sizes: Large: 3-in (75-mm) claws.

Medium: 2-in (50-mm) to 3-in (75-mm) claws.

CRAYFISH PRODUCTS

Crayfish, also called *crawfish*, can be farm raised or harvested wild. The industry distinguishes between red crayfish and white crayfish. Red crayfish fat is orange; white

crayfish fat is greenish. The larger the crayfish, the smaller the tail meat percentage. Available live, frozen whole, fresh tail meat, and frozen tail meat.

CRAYFISH TAILS, FRESH AND FROZEN

Available shelled, with fat on or washed. The flavor is better with fat on. Shelf life with fat on is about 2 weeks; washed, 1 month.

Packs: 8-oz (225-g) plastic bags.

1-lb (450-g) plastic bags.

Counts: 120 to 180 per lb (450 g).

Serving size: 10 tails for appetizer with pasta.

25 tails for main course with pasta.

Calories: 8 oz (225 g), washed = 105 calories.

Kitchen yields: There is no waste.

SOFT-SHELL CRAYFISH

Specialty item with little commercial distribution. The crayfish are harvested while molting.

WHOLE LIVE AND BOILED FROZEN CRAYFISH

Live crayfish are shipped from Louisiana. They easily suffocate in warm surroundings and when packed tightly together. They must have room to crawl around and should be kept refrigerated, securely covered with a moist towel. They will try to climb out of containers. Boiled Swedish pack consists of frozen whole crayfish boiled with dill weeds and seeds.

Season: From early spring to early summer.

Packs: Sold live by weight, often in 30-lb (13.6-kg) boxes.

Count: 1 lb (450 g) = about 12 to 15 crayfish.

Sizes: Large = 5 or less per lb (450 g).

Medium = 16 to 20 per lb (450 g).

Small = 21 or more per lb (450 g).

Kitchen yields: Some loss can be expected, depending on storage condition.

CROAKER

Small Atlantic coast pan fish, member of the drum–redfish family.

Season: March to October.

Sizes: 8 oz (225 g) to 2 lb (900 g).

Serving size: 8-oz (225-g) fillet as main course.

Kitchen yields: Purchase 2-lb (900-g) fish to fillet. 1 fish = 2 servings.

CUSK

Also known as *deep-sea whitefish*, this is a deep-water flatfish, harvested along the New England coast. The fish has a delicate white flesh.

Season: February to July.

Sizes: *Scrod*: 1½ lb (675 g) to 3 lb (1.3 kg).

Medium: 3 lb (1.3 kg) to 7 lb (3.15 kg).

Large: 7 lb (3.15 kg) to 15 lb (6.75 kg).

Junbo: 15 lb (6.75 kg) and up.

Kitchen yields: About 60 percent waste when cut into fillets.

DAB (PLAICE)

Flatfish, harvested in both the Atlantic and the Pacific Oceans.

Season: Available year-round.

Sizes: Small: Under 2 lb (900 g).

Large: 2 lb (900 g) and up.

Kitchen yields: Fish is used mostly for pan-frying whole.

Serving size: 1 lb (450 g) whole fish per person.

DOLPHIN

See Mahi Mahi, in this section.

DRUM VARIETIES

There are a number of species, including black drum, red drum, and the freshwater gaspergou.

BLACK DRUM

The fish can have intramuscular parasites, which have no ill effect on humans and disappear when the fish is cooked. Black

drum puppies are more expensive than the bulls.

Pack: By weight.

Sizes: 2-lb (900-g) to 6-lb (2.7-kg) puppies.

6-lb (2.7-kg) to 10-lb (4.5-kg) bulls.

Kitchen yields: Purchase fillets.

GASPERGOU

Freshwater drum with firm meat.

Season: Available year-round; peak occurs in early summer.

Sizes: Up to 12 lb (5.4 kg). Under 6 lb (2.7 kg) is the most popular size.

Kitchen yields: Fillets are sold skinless and can be grilled.

RED DRUM

Also known as *red fish*, this fish is farm raised to protect the wild population.

Sizes: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Kitchen yields: Purchase fillets.

EEL VARIETIES

Elongated, almost snakelike fatty fish that lives in fresh and salt water. Most products are smoked. Not to be confused with salt-water eels such as the moray, conger, or snake eel.

Season: Available year-round; peak occurs in November and December.

Sizes: 1 lb (450 g) to 3 lb (1.4 kg) up, skinned and head removed.

Kitchen yields: When cut for fillets, the waste is only 15 percent.

BABY EEL

Baby eel is a specialty of northern Spain and is available frozen year-round. The fish are tiny; they are fried in olive oil and eaten whole. Called *angulas* in Spanish.

Season: Early spring. Available frozen year-round.

Packs: 7-oz (200-g) individual packs.

Kitchen yields: Product is usually used in conjunction with other seafood or

as a garnish on salads. The traditional method is to stir-fry *angulas* with garlic. Used whole, there is no waste.

SMOKED EEL

Available whole, with head on or off. Skin is normally left on. Fillets are also available canned, skinless.

Sizes: 3 lb (1.3 kg) and up.

Kitchen yields: Buy 4- to 5-lb (1.8- to 2.2-kg) fish. Larger eels are fatty. Waste is about 25 percent when fish is purchased with head on. One 4-lb (1.8-kg) whole eel = 12 servings.

Serving size: 4 oz (112 g) as appetizer.

FISHBURGERS

Available as hamburger substitute and often made with salmon and tuna. Some companies claim that their products are 98 percent fat free. Available frozen, ready to use.

SALMON AND TUNA BURGERS

Sizes: 3.2 oz (90 g), 4 oz (112 g).

Packs:

3.2-oz (90-g) case pack, 50 patties, 5 to 10 patty bags.

4-oz (112-g) case pack, 40 patties, 4 to 10 patty bags.

Calories: 3.2 oz (91 g) = 100 calories.
4 oz (112 g) = 130 calories.

FISH STICKS

Fish portions, usually made with cod, had-dock, or pollock. Available breaded or batter dipped, raw or precooked, in different shapes and sizes. Fish sticks must contain at least 60 percent fish.

Packs: 2½-lb (1.1-kg) boxes.

Sizes: 1½ oz (42 g) to 5 oz (140 g).

Serving sizes: 4 oz (112 g) to 6 oz (170 g).

Kitchen yields: There is no waste.

Calories: 4 oz (112 g) = 200 calories.

FLOUNDER

See Sole and Flounder Varieties in this section.

FROGS

Usually, only the legs are sold, although oriental markets sell whole frogs live. Legs are sold fresh or frozen, fully cleaned, normally in pairs. Some shippers do not trim the spine close to the legs, thereby inflating weight and reducing yield. Most products are imported, and quality varies greatly.

Season: Year-round, except in April and May when frogs spawn.

Sizes:

Live: Up to 2½ lb (1.2 kg).

Legs: 6 to 8 pairs per lb (450 g).
8 to 10 pairs per lb (450 g).
10 to 12 pairs per lb (450 g).

Kitchen yields: Large frogs' legs can be tough and dry.

Serving size: 8 oz (225 g), untrimmed.
For receptions, buy the 12 pieces per pound size.

GAR, ALLIGATOR FISH

Fish with heavy, bony scales, sometimes used in jewelry. The flesh of large fish is tough.

Season: Year-round.

Sizes: 10 lb (4.5 kg) to 80 lb (36 kg).

GEFILTE FISH

Jewish fish dumpling, normally made with carp. Available in jars in its own jellied broth.

Packs: 1-qt (0.94-l) jars, 6 per case.

Serving size: 2 pieces as appetizer.

Kitchen yields: About 6 to 7 pieces per qt.

GROUPER VARIETIES

Groupers belong to the large family of sea bass. Groupers of various species are harvested in the Atlantic Ocean from Brazil to Virginia. All have flaky, tasty meat.

Varieties: Gag, marbled grouper, red grouper, rock hind grouper, scamp, snowy grouper, yellowedge grouper, and warshaw grouper.

Season: Available year-round; peak occurs in April to December.

Sizes: 4 lb (1.8 kg) to 100 lb (45 kg), depending on species.

Kitchen yields: Purchase fillets only from smaller fish.

HADDOCK AND HADDOCK PRODUCTS

Bottom fish with no scales, harvested in the North Atlantic Ocean. Most of the catch is filleted and frozen. A large volume is processed into frozen fish sticks and related products. Haddock and cod are related, but distinct species. However, the names are often used interchangeably.

Season: Available year-round.

Pack IQF fillets: 5-lb (2.2-kg) cartons.

Sizes:

Whole fish: 1½-lb (0.6-kg) to 6-lb (2.7-kg) whole fish.

Fillet: 6 oz (170 g), 8 oz (225 g), and 10 oz (280 g).

Serving sizes: 6 oz (170 g) to 7 oz (200 g) for broiling.

4 oz (112 g) to 5 oz (140 g) breaded for frying.

Kitchen yields: 5-lb (2.2-kg) to 6-lb (2.7 kg) whole fish.

One 5½-lb (2.5-kg) whole fish = 2-lb 7-oz (1.1-kg) fillet.

2-lb 7-oz (1.1-kg) fillet = 6 servings for broiling.

2-lb 7-oz (1.1-kg) fillet = 8 servings for frying.

Calories: 5 oz (140 g) breaded, fried = 230 calories.

FINNAN HADDIE

Smoked haddock, available in fillets fresh or frozen. Some product is only cured and colored, not smoked, and therefore has an inferior flavor. Poach in milk/water mixture to reduce saltiness.

Packs: Often sold in 15-lb (6.8-kg) cases.

Serving sizes: 4 oz (112 g) for breakfast.
6 oz (170 g) for luncheon dishes.

Kitchen yields: Trimming waste is about 5 percent or less.

HAKE

Bottom fish resembling cod, harvested in both the Atlantic and the Pacific Oceans. Most products are processed. The meat is white and lean.

Season: June to September.

Sizes: 1 lb (450 g) to 6 lb (2.7 kg).

Kitchen yields: Buy fresh fillets only.

HALIBUT

Large flatfish with lean, white meat, harvested in both the Atlantic Ocean and the Pacific Ocean. Halibut is graded by size. Whole halibut is shipped with head removed and dressed. Smoked halibut is available whole and sliced.

Season: Available year-round; peak occurs from March to August.

Sizes:

Chicken: 5 lb (2.2 kg) to 10 lb (4.5 kg).

Medium: 10 lb (4.5 kg) to 60 lb (27 kg).

Large: 60 lb (27 kg) to 80 lb (36 kg).

Whales: 80 lb (36 kg) up to 200 lb (90 kg).

Boneless fillets are called *fletches* and are available fresh or frozen, in many sizes.

Steaks: Normally sold bone in, with skin on. Trade names for steaks are:

Full moon = crosscut.

Half moon = half crosscut.

Loin cut = almost boneless cut.

Regular cut = cut from full moon cut.

Roast = tail cut, has most bones per weight.

Packs: 5 lb (22.5 kg), 10 lb (4.5 kg).
15-lb (6.8-kg) cartons.

Sizes for fillets and steaks: 4 oz (112 g) to 6 oz (170 g).

6 oz (170 g) to 8 oz (225 g).

8 oz (225 g) to 10 oz (280 g).

Large fletches by weight.

Kitchen yields: Waste factor for fresh fish = 40 percent when cut into fillets, skin removed.

Waste factor for frozen fish = 50 percent because fish is glazed with ice.

Calories: 7-oz (200-g) fletch = 200 calories.

HERRING, CANNED, CURED, AND FRESH

Herring is a small fish, abundant in both the Atlantic and the Pacific Oceans. Fresh herring is seldom marketed; almost all of a catch is canned or pickled. Herring roe is a delicacy in Japan and is exported fresh from Washington State during a limited season. Most herring on the market is canned, cured, pickled, or smoked. Herring is kosher, because the fish has scales. Principal varieties include the following:

- *Bismarck herring*: Large herring, pickled with vinegar and also with sour cream.
- *Bloater*: Hot smoked herring.
- *Brathering*: The fish is dredged in flour, pan-fried, and then pickled. Usually sold canned.
- *Bückling*: German name for smoked whole herring, ready to eat.
- *Digby chick*: Smoked fillets.
- *Herring in sour cream*: Available in fillets and bite size. Specify boneless pieces. The most popular commercial preparation of herring fillets or tidbits is marinated with onions in vinegar and sour cream.
- *Kippered herring*: Split smoked herring, served grilled, usually with scrambled eggs. Demand is usually low; purchase canned product.
- *Marinated herring*, also called *pickled herring*; the fish is marinated with spices and vinegar.
- *Matjes herring*: Dutch term for a young herring that has not yet reproduced. Cured by a natural enzyme. It is eaten fresh during the summer season.

- *Rollmops*: Herring fillets rolled up with a pickle or sauerkraut center, marinated with sliced onions.
- *Schmalz herring*: Fatty herring fillets.
- *Sill*: Swedish name for various herring species.
- *Sprat*: Small herring that can be either smoked or just pickled.

Packs: Most products are packed in 1-qt (0.94-l) or 1-gal (3.8-l) glass jars or plastic buckets.

Sizes and counts: Vary greatly according to product.

Kitchen yields: There is no waste because the products are ready to eat.

JACK VARIETIES

A number of species with usually dark and fatty flesh. Of commercial interest are amberjack, almaco jack, and crevalle jack.

Sizes: Up to 75 lb (33.7 kg) for some species.

Kitchen yields: Purchase fillets only.

JOHN DORY

Also called Saint Pierre or St. Peter Fish. See page 72.

LOBSTER PRODUCTS AND VARIETIES

The Atlantic lobster, also called *Maine lobster*, is fished in the waters off New England and the Canadian coast. Lobster farming is successful in some places, although lobsters in such situations are cannibalistic. Large amounts of live lobsters are shipped by air throughout the world. Raw lobsters, whole or split, are available frozen. *Tomalley* is the green liver, pasty when cooked, which is considered a delicacy. *Coral* is the undeveloped egg mass, which turns solid and red when cooked; it too is considered a delicacy. Lobsters are available canned, fresh (live), and frozen.

ATLANTIC LOBSTER MEAT

CANNED

Available packed in brine, often shipped frozen.

Packs: 12-oz (340-g) cans.

Drained weight: 9 oz (250 g).

Kitchen yields: Typical pack contains 2 small tails and shredded meat.

FRESH

Fresh shelled lobster meat is available by weight.

Pack: 1-gal (3.8-l) can.

Kitchen yields: 1-gallon (3.8-l) can = 8 lb (3.6 kg) net of mixed meat.

FROZEN

Available as all tail, regular, and broken meat.

Packs: 2-lb (900-g) cooked lobster meat pouches.

Serving size: 3.5 oz (100 g).

Kitchen yield: Use as is.

Calories: 3.5 oz (100 g) = 90 calories.

ATLANTIC LOBSTERS, LIVE

Live lobsters should be stored loose, refrigerated, and covered with wet material such as seaweed or paper. They should not be covered with ice.

Season: Available year-round; peak occurs in summer.

Packs: Purchased by piece and weight.

Whole lobster popular restaurant

weight: 2½ lb (1.1 kg).

Sizes:

Culls: 1 lb (450 g) and under, live but damaged, normally one claw missing.

Chicken: 1 lb (450 g).

Regular: 1½ lb (670 g).

1¾ lb (780 g) to 2 lb (900 g).

2 lb (900 g) to 2½ lb (1.1 kg).

2½ lb (1.1 kg) to 3¼ lb (1.45 kg).

Over 3½ lb (1.6 kg).

Kitchen yields: The yield depends much on freshness, because live lobsters, when stored, sustain themselves by using their own body juices.

1 lb (450 g) live after cooking = 3½ oz (100 g) to 4 oz (112 g) clean meat.

Five 1-lb (450-g) lobsters = 1¼ lb (560 g) shelled meat.

Serving sizes, shelled meat: 3½ oz (100 g), appetizer.

6 oz (170 g), main course with sauce.

5 oz (140 g) for buffet service, hot.

4½ oz (130 g) for salad.

Serving sizes, meat in shell: ½ lobster, cold, appetizer, 1½ lb (670 g).

½ lobster, cold main course, 2 lb (900 g).

Calories: 3½ oz (100 g) = 100 calories.

SPINY LOBSTER (ROCK LOBSTER)

Clawless lobster harvested in both tropical and cold waters, related to crabs and shrimp. Available in Florida live; seldom shipped whole out of state. Known by the name *langouste* in Europe, the spiny lobster makes an attractive buffet centerpiece. It is sold as lobster in Caribbean resorts and in South America.

Season: December to May.

Sizes: 2 lb (900 g) to 5 lb (2.2 kg).

SPINY LOBSTER (ROCK LOBSTER) TAILS

Most products are imported. Available as cold-water lobster tails and as warm water lobster tails. Cold-water tails are better than warm water tails. Formerly, lobster tails were a popular and inexpensive restaurant staple, but prices have risen significantly. Larger tails command a higher price. Tails are shipped in the shell.

Packs: 10-lb (4.5-kg) boxes, 5 per carton. 30-lb (13.5-kg) IQF mini-tails.

Sizes: 5 oz (140 g) to 6 oz (170 g).

8 oz (225 g) to 10 oz (280 g).

Other sizes are available.

Kitchen yields: Lobster tails are ready to use. Best use is broiled to order. When meat is boiled, it tends to be dry.

Serving sizes: Two 5-oz (140-g) to 6-oz (170-g) tails as main course.

One 8-oz (225-g) tail for Surf and Turf.

Calories: 4½-oz (127-g) meat from 8-oz tail = 130 calories.

LOTTE

French name for *Monkfish*; see facing column.

MACKEREL AND MACKEREL PRODUCTS

Fish with dark, oily flesh related to the tuna and harvested along the Atlantic and Pacific coasts. The wahoo has white flesh. Four commercial varieties are important.

Season: *Spanish mackerel*, March to November.

Chub mackerel, available year-round.

King mackerel, July and August.

Wahoo, Year-round; peak occurs in summer.

Sizes:

Chub mackerel: ½ lb (225 g) to 1½ lb (670 g).

Spanish mackerel: 1 lb (450 g) to 4 lb (1.8 kg).

King mackerel: 6 lb (2.7 kg) to 50 lb (22.5 kg).

Wahoo: 15 lb (6.75 kg) to 90 lb (40.5 kg).

Kitchen yields: About 50 percent waste when cut into fillets.

1¼-lb (560-g) fish = 2 servings for broiling or sautéing.

CANNED MACKEREL

Available whole or as fillet, smoked or plain, packed in oil.

Packs: 15-oz (425-g) cans.
Other sizes are available.

Serving size: 4 oz (112 g), as appetizer.

Kitchen yields: One 15-oz (425-g) can = 12½ oz (355 g) drained.

One 15-oz (425-g) can = 3 servings.

SMOKED MACKEREL

Smoked mackerel is available whole, in fillets, and canned.

Packs: Vary.

Sizes: 8 oz (225 g) to 10 oz (280 g) whole.
4 oz (112 g) to 5 oz (140 g), 2 fillets per pack.

Kitchen yields: Serve as is.

MAHI MAHI

The food fish *dolphin* is not related to the dolphin mammal of the same name. Most is harvested in the Pacific and sold as Hawaiian mahi mahi. Available fresh or frozen.

Season: Year-round.

Sizes: 2 lb (900 g) up to 50 lb (22.5 kg).

Pack: Sold whole or as fillet, skin on.

Kitchen yields: Fish must be skinned.

Waste is about 10 percent by weight if purchased as fillet.

MONKFISH

Also called angler fish and often included in elegant menus under the French name *lotte*. Harvested in the Atlantic.

Season: Year-round.

Sizes: 1-lb (450-g) to 2-lb (900-g) fillets.

Serving size: 6 oz (170 g), as main course.
2 lb (900 g) = 5 servings.

Kitchen yields: Trimming waste when fillets are purchased is 10 percent.

MULLET

Popular fish in the South, harvested in the Atlantic and off the southern California coast. Also known as southern kingfish, this fish is in the drum family. Available as whole fresh fish and as fresh and frozen fillets.

Season: Available year-round; peak occurs in September.

Sizes: Small: 1 lb (450 g) to 2 lb (900 g).
Medium: 2 lb (900 g) to 3 lb (1.3 kg).
Large: Over 3 lb (1.3 kg).

Kitchen yields: About 40 percent waste when cut into fillets. Purchase fillets when possible. Fillets have some bones left in.

MUSSELS

Available canned, fresh, and frozen. Fresh mussels are sold cultivated and wild. Cultivated mussels are free of grits and can be very large. New Zealand is an exporter of

cultivated *green mussels*. Live mussels should be stored refrigerated, covered with moist material but not iced.

CANNED MUSSELS

Available smoked and plain.

Packs: Vary.

FRESH MUSSELS

Season: Year-round.

Pack: Normally sold by weight.

Counts: 1 lb (450 g):

Small = 11 to 14 pieces.

Medium = 8 to 11 pieces.

Large = 5 to 8 pieces.

Extra large = 3 to 5 pieces.

Kitchen yields: Mussels must be scraped and checked for open or broken shells. The yield varies, depending on freshness and care taken during harvesting, delivery, and storage. Cultivated mussels are often sold netted and are completely clean and free of broken shells.

Serving size: 1 lb (450 g), any size, for steamed mussels.

Best size for receptions is medium.

FROZEN MUSSELS

Available shucked meat only IQF, and on the half shell.

Packs: Vary.

Kitchen yields: There is no waste.

SMOKED MUSSELS

Novelty item. The product is ready to use, and there is no waste.

Pack: By weight.

OCEAN PERCH

Atlantic Ocean food fish, available mostly in fillets. Imported frozen fillets are available. The fish is small.

Sizes: Whole fish: from ½ lb (225 g) to 2 lb (900 g). Fillets: from 3 oz (85 g) up.

Season: Early summer.

Kitchen yields: Fillets are good buys for buffets and inexpensive meals.

OCTOPUS

Raw octopus is a popular ingredient in Japanese sashimi.

Pack: 10-lb (4.5-kg) box.

Count: 3 per 10-lb (4.5-kg) box.

Kitchen yields: Fresh product is sold gutted with the eyes removed. Yield depends on application.

BABY OCTOPUS

Available fresh and frozen.

Packs: By weight.

Counts: 40 to 60 pieces per lb.

ORANGE ROUGHY

Fish with firm, white flesh, imported as fillets from the subantarctic waters around New Zealand.

Packs: 22-lb (10-kg) box of frozen fillets.
35-lb (15.7-kg) box of fresh fillets.

Sizes, fillets: 2 oz (60 g) to 4 oz (112 g).
6 oz (170 g) to 8 oz (225 g).
8 oz (225 g) and up.

Kitchen yields: Use as is; there is no waste.

Calories: 6 oz (170 g) = 270 calories.

OYSTER CRABS

See Crabs.

OYSTER PRODUCTS AND VARIETIES

Many varieties of oysters are available from both the Atlantic and Pacific Oceans. Atlantic oysters represent about 85 percent of all domestic production. Imports from Canada, Asia, and Europe are also available. Oysters are successfully cultivated. Oysters are sold live in the shell, shucked fresh and frozen. Fresh oysters should be stored in a cool, dark place, covered with wet towels. Oysters are saltwater creatures, and fresh water and ice will kill them. Oysters are also available canned,

plain and smoked. Oysters on the half shell are available frozen.

LIVE OYSTERS: BASIC INFORMATION

Packs: Oysters are packed in burlap sacks, bushels, or cartons.

Counts: Graded oysters are sold by count for a specific container. Inquire as to what container is used and verify the count.

Season: Oysters spawn during the summer and in many places are not harvested at this time. Oysters are at the peak of plumpness from November to May. Shipping live oysters during the summer when refrigeration was not possible was risky, and therefore it was common practice to avoid eating oysters in months without an *r*. For some oyster varieties, there are specific seasons when they are shipped.

Serving size: Usually 6 pieces on the half shell.

Kitchen yields: A loss of about 10 percent to 15 percent can be expected, depending on storage and season.

ATLANTIC OYSTERS

Many communities produce oysters, which are usually marketed under the site name. The following is a list of the major market varieties.

Belon oyster

European oyster, grown now in Maine.
Count: 144 per case.

Bluepoint oyster

Primarily harvested off Long Island.
Count per bushel: 220 to 230 medium.

Cape Breton oyster

Available September to January.
Count: 425 per bushel.

Cape Cod oyster

Available year-round.
Count: 150 per bushel.

Chincoteague oyster

Also called *Virginia oyster*

Available in winter.
Count: 240 to 250 per bushel.

Delaware oyster

Available in winter.
Count: 300 per bushel.

Gardiners Bay oyster

Long Island.
Available year-round.
Count: 250 per bushel.

Great South Bay oyster

Available in winter.
Count: 250 per bushel.

Lynhaven oyster

Available year-round.
Count: 125 per bushel.

Malpeque oyster

Wild Canadian oyster.
Available September to January and late spring.
Count: 225 per bushel.

Pocomoke Sound oyster

Maryland oyster.
Available in winter.
Count: 300 per bushel.

Prince Edward Island oyster

Canadian oyster.
Available September to January.
Count: 225 per bushel.

Wellfleet oyster

Available in winter.
Count: 150 per bushel.

Whitestable oyster

Imported oyster, from the Thames estuary.
Available September to February.
Count: 259 per bushel.

PACIFIC OYSTERS

Kuomomoto oyster

Count: 180 per bushel.

Olympia oyster

Also called Western Oyster. The oyster is very tiny.
Available year-round.
Count: 400 per bushel.

Pacific Golden oyster

Count: 240 per bushel.

OYSTERS, FROZEN PROCESSED

Breaded oysters are available frozen from a number of manufacturers.

Packs: 3-lb (1.4-kg) boxes.

Sizes: 1 oz (28 g) to 1½ oz (36 g).

Serving size: 8 oysters, average.

Kitchen yields: One 3-lb (1.4-kg) box = 5 servings.

OYSTERS, SHUCKED

Shucked oysters are available fresh, frozen, and processed.

Sizes: *Extra Selects:* 160 to 210 per gal (3.8 l).

Selects: 210 to 300 per gal (3.8 l).

Standards: 300 to 500 per gal (3.8 l).

Size, Olympia oysters: 500 to 600 per gal (3.8 l).

Packs: Gal (3.8 l).

Half gal (1.9 l).

Qt (0.95 l).

Other packs are available.

Frozen oysters are often sold by weight.

Kitchen yields: About 15 percent liquid can be expected.

Calories: 1 cup (0.24 l) shucked, about 16
Selects = 240.

OYSTERS, SMOKED AND CANNED

Packs: Forty-eight 3.66-oz (103-g) cans.

Count: Varies; approximately 14 to 16 pieces.

PIKE (NORTHERN PIKE OR MUSKELLUNGE)

Fine freshwater fish, but of little commercial importance.

Season: Summer.

Sizes: 1 lb (450 g) to 2 lb (900 g) and larger, whole fish.

Kitchen yields: Fish has scales and is bony. Pike is excellent in fish mousse. Buy fillets.

PLAICE

See Dab.

POLLOCK

Harvested off the coast of Alaska and in the Atlantic, pollock is also known as *Boston bluefish* or *Alaska snow cod*. The fish resembles *haddock* in taste and texture, with meatier and firmer flesh.

Season: Available year-round; peak is in late fall.

Sizes: 1 lb (450 g) to 12 lb (5.4 kg).
3 oz (85 g) to 10 oz (280 g), frozen fillets.

Calories: 3½ oz (100 g) = 78 calories.

POMPANO

Very delicate fish harvested off the Florida coast. Some imports are available. Marketed whole, as fresh fillet, and as frozen fillet. Pompano is marketed in three size classes. Fish over 2 lb (900 g) demand twice the price of smaller ones.

Sizes: Under 1½ lb (670 g).

1½ lb (670 g) to 2 lb (900 g).

Over 2 lb (900 g).

Season: Year-round; peak occurs in August to October.

Sizes: Whole from 12 oz (330 g) to 4 lb (1.8 kg).

Fillets from 6 oz (170 g) to 1¾ lb (800 g).

Serving size: 6 oz (170 g) to 7 oz (200 g).

Kitchen yields: Fish is expensive. When cut into fillets, best size is 1¾ lb (800 g). One 1¾ lb (800 g) fish = 2 fillets, 6 oz (170 g) each.

PRAWNS

See Shrimp.

ROCKFISH

Harvested along the Pacific coast in many varieties. Mostly available as fresh or frozen fillet. The most important commercial varieties are:

Bocaccio. Up to 40 in (1 meter) long.

Red rockfish. This fish resembles red snapper and is often sold as such, although not related to the genuine

Atlantic red snapper. Up to 3 ft (900 mm) long.

Orange rockfish. Up to 30 in (760 mm) long.

Yellowtail rockfish. Up to 26 in (660 mm) long.

Kitchen yields: Buy 3-lb (1.4-kg) to 4-lb (1.8-kg) fillets.

SABLEFISH

Also known by the name of *black cod*, but not related to the Atlantic cod. *Sablefish* is harvested along the Pacific Coast from Alaska to California. A large amount of sablefish is smoked.

FRESH SABLEFISH

Size: 8 lb (3.6 kg) and 30 in (0.76 m) long.

Season: Late summer and fall.

Calories: 7 oz (200 g) = 380 calories.

SMOKED SABLEFISH

Fatty Pacific fish with scales. The fish is considered kosher. Sold in fillets cured, smoked, and sprinkled with paprika.

Packs: By weight.

Size: Best size is 2-lb (900-g) to 3-lb (1.4-kg) fillets.

Serving size: 3 oz (85 g), as appetizer.

Kitchen yields: Little waste; skin only must be discarded. About 5 percent trimming waste.

SALMON AND SALMON PRODUCTS

CANNED SALMON

Most canned salmon is produced in Alaska. There are five species of Alaskan salmon, all varying in color, texture, and cost. Most canned salmon is packed in brine. There are some specialty packs of smoked or flavored salmon, packed in oil, available, but they have little commercial distribution.

Varieties:

Red sockeye salmon: Deep red firm flesh, used in salads.

Pink salmon: Pale color, less expensive than red sockeye.

King or chinook salmon: Color ranges from deep red to white; rich in oils, excellent for salads.

Chum or keta salmon: Pale-colored, coarse-textured, the least expensive variety.

Packs: 3¾-oz (105-g) cans, 48 per case.

7¾-oz (220-g) cans, 48 per case.

15½-oz (440-g) cans, 24 per case.

4-lb (1.8-kg) cans, 6 per case.

Drained weights: Good-quality salmon is solid-packed. Drained weights are about 10 percent less than net weights.

Serving sizes: For individual servings, use 3¾-oz (106-g) cans.

Salmon salad = 4 oz (112 g) fish per serving. For salads, use red sockeye salmon in the larger cans.

Kitchen yields: One 15½-oz (440-g) can = 3½ servings of salad.

One 4-lb (1.8-kg) can = 14 servings of salad.

Calories: 3¾ oz (105 g) king salmon = 188 calories.

2 oz (56 g) water-packed pink salmon = 90 calories.

FRESH OR FROZEN SALMON

Most fresh salmon is harvested in Alaska and the Pacific Northwest. Canada is a big producer of Atlantic salmon, but most of its production is smoked. Because salmon is successfully raised in ponds, it is now available year-round. Fresh pond-raised salmon is imported from Chile, Norway, and other countries. Feed and species determine flesh color. Line caught fresh salmon is available from Oregon and Washington State.

Salmon species and sizes:

King or chinook salmon: Fish with high oil content and normally very red flesh. Often smoked. White king salmon has pale flesh and is often kippered. Weight is up to 20 lb (9.1 kg).

Sockeye salmon: Fish with very red flesh. Excellent for broiling. Weight ranges from 4 lb (1.8 kg) to 12 lb (5.4 kg).

Silver or coho salmon: Fish with orange-red flesh, often smoked. Weight is about 6 lb (2.7 kg).

Pink salmon: Most abundant Alaska salmon; most production is canned. Weight ranges from about 2 lb (900 g) to 3 lb (1.4 kg).

Chum or keta salmon: Lightest in color and leanest salmon. Most production is canned. Weight ranges from 4 lb (1.8 kg) to 13 lb (5.9 kg).

Mini-salmon: Hatchery-raised fish, normally sold frozen, boneless, head on.

Season: Farm-raised salmon is available year-round. There are distinct seasons for fresh wild salmon.

King salmon: May to October.

Sockeye salmon: June and July.

Silver or coho salmon: June to September.

Pink salmon: June to November.

Chum or keta salmon: August to October.

Atlantic salmon: June to September.

Packs:

Whole salmon, headless, dressed: 6 lb (2.7 kg) to 8 lb (3.6 kg).

8 lb (3.6 kg) to 10 lb (4.5 kg).

Fillets, boneless, skin on, imported from Norway:

2½-lb (1.2-kg) sides.

22-lb (9.9-kg) to 26-lb (12-kg) cartons.

Loins, skinless, boneless pieces, individually wrapped: Sizes vary.

Steaks, crosscut, bone in and skin on:

8 oz (225 g) and other sizes.

Mini-salmon: 8 oz (225 g) to 10 oz (280 g) boneless, head and tail on:

5-lb (2.2-kg) boxes, 10 fish per box.

30-lb (13.5-kg) master with 60 fish.

Kitchen yields: About 20 percent waste when whole fish is purchased with head on.

About 10 percent waste when whole fish is purchased headless.

One 7-lb (3.1-kg) fish, head on = 5½-lb (2.4-kg) fillet, skin on.

One 10-lb (4.5-kg) fish, head off = 9-lb (4-kg) fillet, skin on.

Serving sizes: Boneless fillet, skin on, for broiling = 5 oz (140 g) to 7 oz (200 g). Boneless fillet served cold, skin removed after poaching = 7 oz (200 g). Boneless fillet to be poached and served with a sauce = 5 oz (140 g).

Calories: 5-oz (140-g) boneless fillet:

King or chinook = 270 calories.

Sockeye = 230 calories.

Silver or coho = 190 calories.

Pink = 172 calories.

Chum or keta = 171 calories.

GRAVELAX

Swedish pickled salmon, most often made on the premises. See the Recipes section for Gravelax formula. Available with different flavor combinations, presliced and whole.

Sizes: Vary greatly.

Serving size: Appetizer = 2½ oz (70 g). Buffet = 1 oz (28 g).

Kitchen yields: One 4-lb (1.8-kg) side, fully trimmed, skin on = 20 to 22 servings appetizer for 40 to 50 buffet servings. When purchased sliced, there is no waste. When purchased as whole side, refer to the yields for smoked salmon.

LOX

Cured salmon from large, fatty fish. The name *lox* is sometimes used erroneously for *smoked salmon*. Sold sliced or whole. Sliced lox is often frozen or canned.

Packs: Four 3-lb (1.4-kg) trays of sliced fish.

Size: About ½-oz (14-g) slices.

SMOKED SALMON

Smoked salmon is available in many sizes, quality levels, and packs. Some products are imported. There is a dazzling array of added flavors, derived from brandy, coriander seeds, honey, peppercorns in many

colors, and many other spices and herbs. Smoked salmon is available whole, skin on, presliced with and without skin, and whole sides, as well as individual portion packs.

Average sizes:

Canadian nova, sides: 3½ lb (1.5 kg).

Canadian nova, presliced: 2 lb (0.9 kg) to 3 lb (1.3 kg).

Gaspé nova, sides: 4 lb (1.8 kg) to 8 lb (3.6 kg).

Irish salmon, sides: 3 lb (1.3 kg) to 3½ lb (1.5 kg).

Norwegian presliced: 3 lb (1.3 kg) to 2.2 lb (1 kg).

Pacific nova, sides: 4 lb (1.8 kg) to 10 lb (4.5 kg).

Scotch salmon, sides: 2¼ (1 kg) to 3 lb (1.3 kg).

Scotch salmon, presliced: 1½ lb (675 g).

Serving sizes: 3 oz (85 g) for appetizer or sandwich.

2 oz (56 g) for buffet with other foods.

½ oz (10 g) for each canapé.

Kitchen yields: Trimming waste from whole sides is about 15 percent. Skin weight is about 5 percent, not included in trimming waste estimate.

Usable scraps, such as end pieces, are 5 percent after slicing.

Larger sides have better yield.

One 8-lb (3.6-kg) side as purchased: 30 servings appetizer.

45 servings buffet.

260 pieces canapés.

SALMON TROUT (LAKE TROUT)

Freshwater fish, with pale pink to gray-colored flesh, harvested in the Great Lakes.

Season: Peak is in summer. Small catch is available in winter.

Sizes: 2 lb (900 g) to 8 lb (3.6 kg).

Trade sizes: 2 lb (900 g) to 4 lb (1.8 kg).
4 lb (1.8 kg) to 8 lb (3.6 kg).

Kitchen yields: Delicate fish; buy whole and cut fillets. Waste is about 50

percent. Leave skin on for broiling or pan-frying. Remove skin for poaching.

SARDINES

CANNED SARDINES

Most sardines are packed in oil, but specialty packs such as sardines in tomato or mustard sauce are available. The quality of oil is important; better sardines are packed in olive oil. Distinction is made between skin-on sardines and skinless or boneless sardines. The bone structure is very soft, and all canned sardines can be completely eaten. Much product is imported from Portugal, and to a lesser degree, from other countries. There is some domestic production. Brisling, also called *silt* or *small sardines*, are normally imported from Norway and are packed in oil.

Packs: 15-oz (425-g) oval cans, mostly packed in California, 24 per case.

12-oz (340-g) cans, mostly from Maine, 48 per case.

7-oz (200-g) cans, 100 per case.

3¾-oz (100-g) cans, 100 per case.

Many other small packs are available.

Drained weights: 15 oz (425 g) = 11½ oz (326 g).

12 oz (340 g) = 10¾ oz (305 g).

Count: Varies greatly. Brisling number about 8 fish in a 3¾-oz (100-g) can.

Serving size: 2 oz, as appetizer or on sandwich. Servings size is most often by the piece, or one 3¾-oz (100-g) can as main course.

Calories: 1 oz (28 g) canned, packed in oil = 50 calories.

FRESH SARDINES

Maine sardines belong to the herring family. Relatively little production is sold fresh; most is canned.

Pack: By weight.

Sizes: 3 in (75 mm) to 4 in (100 mm) long.
6 to 9 pieces per lb (450 g).

Season: Winter.

Serving size: 4 pieces, bones removed and fried whole.

Kitchen yields: 5 lb (2.2 kg) = 9 servings.

SCALLOP VARIETIES AND PRODUCTS

Scallops are available fresh, frozen, and processed and are classified as follows:

- *Bay scallops*, which have edible muscles about ½ in (12.7 mm) across.
- *Sea scallops*, which can develop muscles up to 2 in across.
- *Calico bay scallops*, a southern variety that are larger than the genuine northern bay scallops.

Bay scallops are harvested in the estuaries along the Atlantic coast as far south as Florida. Sea scallops are produced mostly in the North Atlantic. Important sources of Sea scallops are also the waters around Alaska and along the Pacific coast as far south as Oregon. Domestic scallops are almost always fully cleaned and white. Imported scallops are occasionally shipped with the edible pink roe still attached. Scallops in their shells shipped with roe are a novelty item and seldom available. Diver scallops, supposedly hand-harvested by divers, have appeared on menus. Considering the amount sold and the limited number of divers, the designation *diver scallops* should be regarded with some skepticism.

BAY SCALLOPS, FRESH

Bay scallops are normally sold shucked.

Season: Available year-round, with peak season in the summer.

Packs: 1-gal (3.8-l) cans.

Net weight about 8 lb (3.6 kg).

Count: 450 to 480 per gal (3.8 l).

Serving sizes: 7 oz (200 g), about 1 cup, as main course.

7 oz (200 g) scallops = 4½ oz (127 g) cooked.

Kitchen yields: 1 gal (3.8 l) = 16 portions, 7-oz (200-g) servings.

Scallops must be cooked to order and served immediately because they will shed water, shrink, and become tough quickly.

BREADED SCALLOPS, FROZEN

Available in two sizes: bay scallop size and sea scallop size. The breading is about 45 percent by weight.

Packs: 3-lb (1.4-kg) trays.

4 trays to a box of 12 lb (5.4 kg).

Serving size: 6 oz uncooked.

Kitchen yields: One 3-lb (1.4-kg) tray = 8 servings.

Calories: 4¼ oz (120 g) cooked = 240 calories.

SEA SCALLOPS, FRESH

Chefs once scorned these scallops because they are large. Many restaurants have served broiled scallops, always undercooked, in many versions as appetizers and main courses. Sea scallops are almost always sold shucked.

Packs: Sold in 1-gal (3.8-l) container.

Net weight about 7 lb 12 oz (3.5 kg).

Count: Varies, about 140 to 200. Larger scallops are often preferred.

Serving size: 5 oz (140 g) for main course, or about 6 to 7 pieces.

Kitchen yields: There is no waste.

SEA SCALLOPS, FROZEN

Available loose (IQF) and as frozen block.

Packs:

Block scallops 5-lb (2.2-kg) boxes.

1 case = 50 lb (22.5 kg).

IQF Scallops 3-lb (1.4-kg) boxes.

1 case = 30 lb (13.5 kg).

Counts per pound (450 g): Under 20, 20 to 30, 30 to 40, 40 to 50.

Kitchen yields: Frozen scallops will shrink more than fresh scallops.

SMOKED SCALLOPS

Novelty item available ready to serve.

SHEEPSHEAD

Fish in the *Porgy* family with lean, white meat popular in Louisiana restaurants.

Season: Year-round; peak occurs in winter.

Sizes: 1½ lb (670 g) to 8 lb (3.6 kg).

Kitchen yields: The yield is lower than that of most other fish varieties. Purchase fillets.

SEA TROUT

Also called *weakfish*. Harvested along the coast of the Gulf states. The spotted sea trout is harvested under a strict quota system; sand sea trout is more common.

Season: Spotted sea trout available November to April; sand sea trout, September to April.

Sizes: 1 lb (450 g) to 5 lb (2.2 kg).

Kitchen yields: Lean and delicate fish; spoils rapidly. Purchase fillets or largest size available, whole fish and cut fillets. About 45 percent waste.

SEA URCHIN

Novelty item with limited commercial application. The roe, called *uni*, is a specialty item. See Japanese Foods in the International Ingredients section.

Season: Summer.

SHAD

Fatty fish remotely related to herring, harvested on both the Atlantic and Pacific coasts. The fish return to spawn to the freshwater stream where they were hatched. Shad have an erratic bone structure, which is difficult to remove, and for this reason shad is marketed as boneless fillets. The fillets are sold as two fillets wrapped in paper. Shad roe are marketed by the pair.

Season: Early spring until mid June, depending on weather.

Sizes:

Shad fillets, pairs: 16 oz (450 g) to 20 oz (570 g).

Shad roe, pairs: 6 oz (170 g) to 8 oz (225 g).
8 oz (225 g) to 10 oz (280 g).
12 oz (340 g) to 14 oz (400 g).

Serving sizes: *Shad fillet* only:
7 oz (200 g) to 8 oz (225 g).

Shad and roe: 5 oz (140 g) *Shad*, 4 oz (112 g) *Roe*.

Roe only, one pair: 8 to 10 oz (225 to 280 g). The customer is normally given a choice of ordering shad fillet only, shad and roe combination, or roe only.

Kitchen yields: It is strongly recommended to purchase boned fillets.

One pair 16-oz (450-g) to 20-oz (570-g) shad fillets = 4 servings when served with roe.

One pair roe = 2 servings, shad and roe combination.
= 1 serving when served alone without fillet.

SMOKED SHAD

Smoked shad fillets are a spring specialty.

Packs: By weight.

Sizes: About 8 oz (225 g) each fillet.

Serving size: 4 oz (112 g).

Kitchen yields: Although fillets are boneless, there is some trimming waste.

SHARK

There are many different species of shark, but only a few reach the commercial market. The fish is harvested in both the Atlantic and the Pacific Oceans. Marko shark is a well-recognized food fish. Shark flesh can retain urea and develop an off-smell, which can be removed by soaking.

Season: Year-round; peak occurs in July.

Sizes: 20 lb (9 kg) to 100 lb (45 kg).

Packs: Dressed and filleted. Dressed sharks have their heads, fins, and belly flaps removed.

Kitchen yields: Shark skin is very tough and leathery and must be removed

before cooking. Bones are very soft.
Purchase fillets only.

SHARK FINS

See Chinese Foods in the International Ingredients section.

SHRIMP VARIETIES AND PRODUCTS

Shrimp are fished in most parts of the oceans and are farmed as well. They are available in many sizes and quality levels. Also available are river shrimp, harvested in the Mississippi River. Shrimp are shipped fresh and frozen. The term *green* in regard to shrimp does not refer to color, but means that the shrimp were not processed.

Principal varieties:

Brown shrimp
Pink shrimp
Rock shrimp
Roughneck
Royal red shrimp
Tiger shrimp
White shrimp

Large shrimp are often called *prawns*. Price is dependent on size. Pink-colored shrimp are considered the most desirable in foodservice. Many species are pink-colored.

Product forms:

Fresh head on, fresh headless.
Frozen headless and head on.
Pieces large, medium, and small.
Frozen peeled, deveined and undeveined.
Frozen cooked, peeled, deveined and undeveined.
Canned; dried and breaded.

Counts per lb (450 g): Under 10 and up to 100 for head-on and headless varieties.

1 lb (450 g) shrimp, head on, will yield about 12 oz (340 g) headless.

Seasons: Spring and summer for most varieties.

Packs: 2.5-lb (1.2-kg) boxes for frozen shrimp.

BREADED SHRIMP

Breaded shrimp are classified as hand-breaded and machine-breaded. Generally,

the tails are left unbreaded on hand-breaded shrimp. Most machine-breaded shrimp have a breaded tail, but some machine-breaded shrimp brands are available with unbreaded tails. Breaded shrimp are very popular, and therefore many brands, breading flavors, and shapes are available.

Packs: 3-lb (1.4-kg) boxes, 8 per case.
8-oz (225-g) bags.
6-oz (170-g) bags.

Kitchen yields: The counts for breaded shrimp are basically the same as for green headless shrimp. The count indicated on the box is the actual count. By government regulation, weight of breading cannot exceed 50 percent. If breading exceeds 50 percent, the product must be labeled *Imitation Shrimp*.

COOKED COCKTAIL SHRIMP, FROZEN

Fully cooked and peeled shrimp are available. Most brands have the tails still attached for eye appeal. IQF shrimp should be purchased, because the glazing protects the product.

Packs: 2½-lb (1.1-kg) boxes.
Many other packs are available.

Serving size: 5 pieces 31/35-count size shrimp cocktail. [Note: If the count is, for example, 31 to 35 to a pound, the size is given as "31/35."]

Kitchen yields: 1 lb (450 g) green headless shrimp = ½ lb (225 g) peeled cooked shrimp. The number of cooked shrimp corresponds roughly with the count of green headless shrimp.

One 2½-lb (1.1-kg) box = 18 servings.

Calories: 6 oz (170 g), about 6 pieces, medium = 70 calories.

COOKED SHRIMP, CANNED

Packs: #10 cans, 6 per case.
4½-oz (125-g) cans, 12 per case.

Kitchen yields: Product is fully cooked and ready to serve. Drained weight varies.

Calories: 3 oz (85 g) = 100 calories.

COOKED TINY SHRIMP, FROZEN

Ready-to-eat small shrimp for salads and cocktails. Alaska is the largest producer.

Packs: 5-lb (2.2-kg) blocks or bags of IQF shrimp. Other packs are available.

Drained weight: One 5-lb (2.2-kg) block = 4 lb 13 oz (2.15 kg).

Counts: 350 to 500 per pound (450 g), regular.
250 to 350 per pound (450 g), medium.

Serving size: 6 oz (170 g) for shrimp cocktail.

Kitchen yields: One 5-lb (2.2-kg) block = 13 servings.

Calories: 6 oz (170 g) = 200 calories.

GREEN HEADLESS SHRIMP

Green headless shrimp are shrimp in the shell, with the head removed. They are sold on the basis of count per pound (450 g).

Pack: 5 lb (2.2 kg) per box, frozen in a solid block.

Counts:

<i>Per 1 lb (450 g)</i>	<i>Per 5-lb (2.2-kg) Box</i>
Under 8	36 to 38
Under 10	42 to 48
10 to 15	60 to 75
16 to 20	90 to 95
21 to 25	110 to 120
26 to 30	135 to 145
31 to 35	160 to 170
36 to 42	185 to 195
40 to 50	215 to 260
50 to 60	270 to 280

The most popular restaurant count is 16/20 (16 to 20) per pound.

Serving sizes: Broiled stuffed shrimp, served as main course: 4 shrimp, under-8 size.

Shrimp cocktail: 4 shrimp, 16/20 size.

Shrimp salad course: 9 to 10 each, 21/25 size.

Kitchen yields: Peeling and deveining loss is about 25 percent by weight.

Cooking loss is 25 percent by weight.

Total loss = 50 percent by weight.

5-lb (2.2-kg) under-8 size = 9 servings
Broiled Stuffed Shrimp.

5-lb (2.2-kg) 16/20 size = 22 to 24 portions Shrimp Cocktail.

5-lb (2.2-kg) 21/25 size = 13 servings.

Calories: 5 oz (142 g) = 166 calories.

P & D SHRIMP, RAW, FROZEN

P & D (peeled and deveined) shrimp are normally shipped as individually quick-frozen shrimp, also referred to as *IQF*. To prevent freezer burns, they are coated, or glazed with water. P & D shrimp are available by count per pound and as broken pieces, sold by weight. Broken pieces are usable for salads and shrimp dishes with sauce.

Packs: 3-lb (1.4-kg) boxes, 5 per case.

12½-lb (5.6-kg) bulk pack.

Other packs are available.

Count comparison between green headless and P & D shrimp:

<i>Green Headless</i>	<i>P & D</i>	<i>P & D per 3-lb (1.4-kg) Box</i>
10 to 15	18	54
16 to 20	23	69
21 to 25	28	84
26 to 30	32	96
31 to 35	38	112
36 to 42	44	132
40 to 50	53	159
50 to 60	63	190

Kitchen yields: There is no peeling loss.

Cooking loss is about 30 percent, because of glazing, which increases the weight of the product.

SMOKED SHRIMP

Novelty item, usually large sizes available, freshly smoked and frozen. Packs vary.

SKATE

Also called ray, this fish renders excellent meat. Only the wings are used.

Season: Year-round.

Sizes: 1 lb (900 g) to 25 lb (11.2 kg).

Serving size: 6 oz (170 g) to 8 oz (225 g).

Kitchen yields: Wings are available skinned and with skin. There is also a center bone that must be removed. About 30 percent waste factor for skin-on wings.

SMELTS

Small fish, related to salmon, capable of living in both fresh and salt water. Smelts are harvested in the Atlantic and Pacific Oceans and in many lakes, including the Great Lakes. They are usually served deep-fried.

Season: Late winter and early spring.

Pack: By weight.

Sizes: 1½ oz (40 g) to 1¾ oz (50 g).
7 to 8 in (175 to 200 mm) long.

Serving size: 4 to 5 pieces.

Kitchen yields: 1 lb (450 g) = 2 servings.

SNAPPER VARIETIES

Snapper is a popular fish, harvested in the Gulf of Mexico and in other tropical waters. Available whole or in fillets, fresh or frozen. Fillets are often frozen and imported. Purchase only fillets with skin on to avoid substitutions. The most popular variety is red snapper. The major species are grey snapper, lane snapper, red snapper, queen snapper, and vermillion snapper. The fish differ in size and skin color but are basically as desirable as the popular red snapper.

Season: Available year-round; peaks are in spring and fall.

Sizes:

Whole small: Under 2 lb (900 g), the most expensive size.

Whole medium: 2 lb (900 g) to 4 lb (1.8 kg).

Whole large: 4 lb (1.8 kg) to 8 lb (3.6 kg).

Fillets: 3½ oz (100 g) and larger.

Serving size: 6 oz (170 g), for broiling, skin on.

One 1¾ lb (790 g) whole fish yields 2 servings.

Kitchen yields: About 55 percent waste, because head is very large.

Buy 6-lb (2.7-kg) to 8-lb (3.6-kg) fish for cutting fillets.

One 7-lb (3.2-kg) fish = two 3-lb 3-oz (1.5-kg) fillets.

Two 3-lb 3-oz (1.5-kg) fillets = 8 servings.

Larger fish gives slightly better yield, but can be dry and tough.

SOLE AND FLOUNDER VARIETIES

Flatfish, considered the best eating fish. Sole is harvested on both the Atlantic and Pacific coasts. The names *sole* and *flounder* are often used interchangeably, but they cover a number of distinct but closely related species. Most production is sold as fresh or frozen fillets.

Atlantic sole species: Grey sole and lemon sole, also called *winter flounder* or *blackback* when weighing under 3½ lb (1.5 kg).

Pacific sole species: Petrale sole, brill sole, and English sole. English sole is often called *lemon sole*, but it is a different species than the Atlantic lemon sole. (See English or Dover sole.)

Season: Available year-round.

Sizes:

Whole fish: 1 lb (450 g) to 6 lb (to 2.7 kg).

Fillets: 4 oz (112 g) to 6 oz (170 g).

6 oz (170 g) to 8 oz (225 g).

8 oz (225 g) to 10 oz (280 g).

Serving sizes: 4-oz (112-g) appetizer for banquets.

7-oz (200-g) to 9-oz (250-g) fillet as main course.

Kitchen yields: About 50 percent waste when fillets are cut from whole fish.

Buy 5-lb (2.2-kg) to 6-lb (2.7-kg) fish when cutting into fillets.

One 5½-lb (2.5-kg) whole fish = 2-lb
14-oz (1.3-kg) fillets.
2-lb 14-oz (1.3-kg) fillets = 6 to 7
portions.

ENGLISH OR DOVER SOLE

Imported flatfish of excellent quality, available fresh or frozen. It is normally imported whole, with skin, head, and fins still attached. In Ireland Dover sole is called slip sole. Genuine Dover sole is a luxury item. Flatfish of lesser quality is harvested under the name *English sole* in the Pacific.

Sizes for whole fish:

10 oz (280 g) to 12 oz (340 g).
12 oz (340 g) to 16 oz (450 g).
16 oz (450 g) to 20 oz (570 g).
20 oz (570 g) to 24 oz (680 g).
24 oz (680 g) to 28 oz (800 g).
28 oz (800 g) and up.

Packs: 25-lb (11.25-kg) cases, fresh fish.
50-lb (22.5-kg) cases, frozen fish.

Serving sizes: 16 oz (450 g) to 20 oz (570 g) in fine restaurants.

Fish is normally served whole, one fish per serving.

20 oz (570 g) to 24 oz (680 g) for 2 servings.

Kitchen yields: Waste is about 30 percent when head, fins, and skin are removed.

Fish gives best yield when filleted after cooking.

If fish is filleted raw, buy very large fish; yield is very low, approximately only 30 percent.

FLOUNDER

Both winter flounder and summer flounder, also called *fluke*, belong to the large family of flatfish. There are a number of distinct subspecies, marketed under many local names. The fish is harvested on both the Atlantic and the Pacific coasts. The name *sole* is often used interchangeably with *flounder*. Most fish is purchased as fresh or frozen fillets under the name *fillet of sole*. There are noticeable differences in

texture and flavor between the various species, but the market often does not indicate which species is used. There is much confusion regarding names. Large amounts of fillet of sole are imported.

Atlantic coast species: Blackback or winter flounder, also called *lemon sole*; fluke or summer flounder, yellowtail flounder.

Pacific coast species: Rex, petrale sole, English sole; Dover, sand, and rock sole.

Season: Year-round, but there is a larger supply in summer.

Sizes: ½ lb (225 g) to 6 lb (2.7 kg).
3 oz (85 g) and up for fillets.

Serving sizes: ¾ lb (335 g) to 1 lb (450 g) for pan-frying whole fish.

4-oz (112-g) appetizer for banquets.

7-oz (200-g) to 9-oz (250-g) fillet per person for main course.

Kitchen yields: Buy 5-lb (2.2-kg) to 6-lb (2.7-kg) whole fish for filleting.

When cut into fillets, there is 50 percent waste from gutted fish.

STUFFED FLOUNDER

One of the most popular convenience restaurant items, available with many varieties of stuffing and in many sizes.

Packs: Vary.

Sizes: 6 oz (170 g) to 10 oz (280 g).

Kitchen yields: The amount and composition of stuffing is important when comparing prices.

ST. PETER FISH

Fish with firm meat, harvested in the Atlantic, around New Zealand, and in the Mediterranean. It is a classic ingredient in bouillabaisse. Most fish is imported. Also known as *John Dory*.

Season: Year-round.

Sizes: 20 oz (560 g) to 24 oz (670 g) gutted, whole.

Kitchen yields: The fish has a big head; thus, yield is less than 50 percent fillets.

SQUID AND SQUID PRODUCTS

SQUID, FRESH AND FROZEN

Squid is normally sold fully cleaned, with the exception of the center cartilage. Also referred to as *calamari*.

Packs: Fresh squid by weight.

5-lb (2.2-kg) boxes frozen squid.

Season: Available year-round.

Kitchen yields: Waste about 5 percent by weight.

PROCESSED SQUID

Frozen breaded squid sections are available from specialty purveyors.

Packs: 2½-lb (1.1-kg) boxes, 12 per case.

Kitchen yields: Use as is. There is no waste.

SURIMI

See Alaskan Crab Meat Substitutes.

STURGEON VARIETIES

Harvested in both the Atlantic and Pacific Oceans, the fish lives in salt water and spawns in freshwater rivers. Lake sturgeon live in fresh waters only. Sturgeon is esteemed primarily as a producer of caviar and as an excellent smoked fish. Much sturgeon is frozen and imported for smoking from caviar-producing countries and from Canada. Sturgeon species are distinguished in the caviar trade, but little distinction is made when purchasing sturgeon meat.

Season: Spring and summer.

Sizes: 5 lb (2.2 kg) to 500 lb (225 kg).

Kitchen yields: Waste is about 30 percent when buying whole fish. Sturgeon has no bones, only soft cartilage, but its skin is tough and must be removed. The spine marrow, called *vesiga*, can be used in fish soups and coulibiac. Buy 8-lb (3.6-kg) to 10-lb (4.5-kg) fish, head removed, or 8-lb (3.6-kg) to 10-lb (4.5-kg) sturgeon saddle.

SMOKED STURGEON

Elegant fish, considered kosher.

Packs: Fillets by weight.

Average Weight: 3 lb (1.4 kg) to 5 lb (2.2 kg).

Serving size: 3 oz (85 g) for appetizer.

Kitchen yields: Order #1, center cut. The pieces should be without cartilage. Skin is often left on. Some pieces have belly flap left on, thereby reducing yield. Trimming waste is about 10 percent. Slice on machine; smoked sturgeon is not available presliced in commercial quantities.

One 4-lb (1.8-kg) piece = 18 appetizer servings.

VESIGA

Gelatinous sturgeon marrow, available dried and frozen. It must be soaked overnight and then boiled. It will become a gelatinous mass.

Pack: By weight.

Kitchen yields: Use in soups and in coulibiac (baked salmon loaf).
3 oz (85 g) dried vesiga = 15 oz (425 g) cooked vesiga.

WHITE STURGEON, FARM RAISED

Also sold under the name *belusa*.

Packs: By weight; often 100-lb (45-kg) boxes.

Size: 11-lb (5-kg) to 14-lb (6.3-kg) *Bullets*, gutted, headless fish, fins and tail off.

Kitchen yields: Each fish yields two 4-lb (1.8-kg) to 5-lb (2.25-kg) fillets.

SWORDFISH

Large fish harvested in both the Atlantic and the Pacific Oceans. Available fresh in pieces and frozen in steaks. Harvesting is regulated by quotas. One period begins January 1, and another June 1.

Season: Year-round; imports available year-round.

Sizes: 30 lb (13.5 kg) to 400 lb (180 kg).

Quota sizes for headed and gutted fish:

34 lb (15.3 kg) to 49 lb (22 kg).
50 lb (22.5 kg) to 99 lb (44.5 kg).
100 lb (47 kg) and over.
Larger fish have become scarce.

Packs: Sold by weight, bone in.

Serving sizes: 3 oz (85 g) for appetizer.
8 oz (225 g) to 10 oz (280 g) for thick steak.

Kitchen yields: Purchase by weight and cut in kitchen. Specify *center cut, without belly flap*, because flap can reduce yield by as much as 25 percent. Skin is usually left on. Tail pieces have high ratio of bones to meat. Heavy pieces give better yield. One 20-lb (9-kg) center cut, no belly flaps = 17 lb (7.65 kg) usable meat.

SWORDFISH STEAKS, FROZEN

Available in many different sizes. They are usually dry.

TILAPIA

Fish with white meat, farm-raised in temperate climates. A winter-hardy tilapia has been developed. Wild tilapia are harvested in Florida and are also imported.

Season: Year-round, peak occurs in summer. Most product is shipped frozen.

Packs: Available as plain fillets, breaded fillets, and marinated fillets.

Fillet sizes: 2 oz (56 g) to 3 oz (85 g)
3 oz (85 g) to 5 oz (140 g)
4 oz (112 g) to 6 oz (170 g)
5 oz (140 g) to 7 oz (200 g)
7 oz (200 g) to 9 oz (250 g)

TILEFISH

Atlantic coast fish with firm, lean flesh and good flavor.

Season: Summer.

Sizes: 2 lb (900 g) to 25 lb (11.25 kg).

Kitchen yields: Purchase fillets. Best size is 1½ lb (675 g).

TROUT**FRESHWATER TROUT, FRESH AND FROZEN**

The best-known species is the rainbow trout. Most commercial production is hatchery-grown. Fresh trout are available in many states. The fish is normally sold gutted, with head and tail attached. It is available boned butterfly-style and as fillets. Live trout are available in selected markets, and the fish can be kept alive in a fish tank. They need clean, cold, fresh oxygenated water without chlorine.

Season: Available year-round; peak is in summer.

Frozen trout is readily available at all times.

Sizes: 5 oz (140 g) up, gutted, head on.

Serving sizes: 5 oz (140 g) stuffed, as appetizer.

7 oz (200 g) to 9 oz (250 g) whole fish, as main course.

9 oz (250 g) to 11 oz (300 g) whole fish, as main course.

20 oz (570 g), large fish for 2 servings.

Kitchen yields: The waste factor in filleting is about 45 percent.

For filleting, buy 28-oz (800-g) or larger fish.

Calories: 8-oz (225-g) whole trout, broiled = 450 calories.

LAKE TROUT

See Salmon Trout.

SEA TROUT

Listed separately under this name.

SMOKED TROUT

Available whole, with head on, and in boneless fillets. Product is sometimes frozen.

Packs: Sold by piece. 2-oz (56-g) or larger fillets. Tray pack is available.

Sizes: 6 oz (170 g) to 8 oz (225 g), whole fish, head on.

8 oz (225 g) to 10 oz (280 g), whole fish, head on.

Serving size: ½ trout, for appetizer, or two 2-oz (56-g) fillets.

Kitchen yields: Whole trout must be boned, heads and small bones removed. The fillets break easily, especially if the fish has previously been frozen.

TURTLES

FRESHWATER TURTLES

A number of freshwater turtle varieties can be legally harvested. The demand is primarily in ethnic markets. Body and leg meat are red; neck meat is white.

Varieties:

Alligator or loggerhead turtle, 20 lb (9 kg) to 150 kg (33 kg).

Red-eared turtle, 2 lb (900 g) to 5 lb (2.2 kg).

Snapping turtle, 10 lb (9 kg) to 35 lb (7.7 kg).

Available products: Live, boneless, and bone-in meat.

Packs: 1-lb (450-g) to 5-lb (2.2-kg) pouches of vacuum-packed meat.

TERRAPIN

Small saltwater turtle. The diamond back is the best species. Sold live.

Season: Summer and fall.

Sizes: 2 lb (900 g) to 4 lb (1.8 kg).

Packs: Live by the piece.

Kitchen yields: Average, 4 appetizer servings.

TUNA AND TUNA PRODUCTS

CANNED TUNA

Available packed in oil or in water. The cans are labeled *light meat*, *dark meat*, or *mixed meat*. Canned tuna is manufactured from a number of different species. Albacore tuna has the lightest meat. Bonito is not considered tuna and cannot be labeled as such. Canned tuna, usually packed in water, is imported from Asia. High-quality tuna packed in olive oil is imported from

Mediterranean countries. For salads mixed with mayonnaise, buy tuna packed in water. For Chunk Tuna Salad Plate, use only tuna packed in oil.

Product varieties:

Prime fillets, solid pack or fancy packs:

Meat from the loin and should be used for cold plates.

Chunk style: Mixture of pieces, up to 50 percent can be less than ½ in (12 mm) in size. This pack should be used for salads.

Flaked tuna: Mixture of small pieces. Can be used as salad, especially for sandwiches.

Grated tuna: Almost granular, but not pasty. Can be used as sandwich filling.

Pack: 3½-oz (100-g) cans, 48 per case.
6-oz (170-g) or 7-oz (200-g) cans, 48 per case.

8-oz (225-g) cans, 48 per case.

13-oz (370-g) cans.

42-oz (1.2-kg) cans, 12 per case, foodservice size.

66½-oz (1.9-kg) cans, 6 per case, foodservice size.

Serving sizes: One 3½-oz (100-g) can for single salad plate.

3 oz (85 g) for tuna salad sandwich, mixed with celery and mayonnaise.

Kitchen yields: One 66½-oz (1.8-kg) can = 60 oz (1.7kg) drained weight.

7 oz (200 g) = 6 oz (185 g) drained weight.

One 66½ oz (1.9 kg) can = 20 servings, sandwich.

Calories: 3½-oz (100-g) can, water pack = 105 calories.

3½-oz (100-g) can, in oil and drained = 190 calories.

1 cup, 7-oz (205-g), tuna salad made with celery and mayonnaise = 350 calories.

FRESH TUNA

Fresh tuna has become widely accepted as fresh-food fish in fine restaurants. It is

often cooked rare like a steak. Fresh tuna is one of the most important fish varieties used in sushi bars. The flesh of all species is dark and will lighten after cooking.

Varieties:

Albacore: Fish with light-colored flesh. Weight range from 30 lb (13.5 kg) to 90 lb (40.5 kg).

Bigeye tuna: Weight ranges from 60 lb (27 kg) to 160 lb (72 kg).

Blackfin: Weight range 10 lb (4.5 kg) to 35 lb (15.75 kg).

Bluefin: Weight ranges from 200 lb (90 kg) to 600 lb (270 kg). The much-sought-after fatty belly meat is called *toro* in Japanese.

Skipjack: The smallest of the four tuna species. Weights range from 4 lb (1.8 kg) to 24 lb (10.8 kg).

Yellowfin: Fish with light-colored flesh. Weights range from 40 lb (18 kg) to 150 lb (67.5 kg).

Season:

Albacore: August to December.

Bigeye: Winter.

Blackfin: Year-round.

Bluefin: December to March, some supply in summer.

Skipjack: Year-round.

Yellowfin: Year-round with peak in summer.

Serving size: 5 oz (140 g), main course cooked to order rare.

Kitchen yields: Fillets are generally the best buy. Make sure to specify loin fillets. Because tuna should be undercooked, the portion size may be small.

Calories: 5 oz (140 g) = 350 calories. The calorie count of fatty belly meat can be considerably higher.

SMOKED TUNA

Novelty item available as fillets.

TURBOT

North Sea turbot from Europe is an elegant flatfish, not to be confounded with the lesser-quality Pacific turbot. The fish is available frozen, but is occasionally freshly flown in for the luxury market. Available whole.

Packs: 22.2-lb (10-kg) to 40-lb (18-kg) cases with whole fish.

Sizes, head on:

6 lb (2.7 kg) to 9 lb (4 kg).

9 lb (4 kg) to 12 lb (5.4 kg).

12 lb (5.4 kg) to 15 lb (6.7 kg).

15 lb (6.7 kg) and larger.

Best size to buy: 9 lb (4 kg) to 12 lb (5.4 kg).

Kitchen yields: The fish is normally sold eviscerated, head on. It has a very large head and heavy bones. Waste when cut into steaks, skin on and bone in, is 40 percent. Waste when cut into boneless fillets, skin off, is 60 percent. One 10-lb (4.5-kg) fish = 4-lb (1.8-kg) fillet.

WHITEBAIT

Tiny fish, served dusted with flour and fried. Available fresh or frozen IQF or in frozen blocks.

Packs: By weight.

Serving size: 5 oz (140 g) to 6 oz (170 g).

Kitchen yields: Serve whole. About 10 to 15 percent loss caused by sorting out impurities.

WHITEFISH

Fish popular fresh around the Great Lakes and smoked in large amounts. Also known as *cisco* or *tullibee* in Canada.

CHUBS

The term *chub* is applied to small smoked whitefish. Chubs are usually sold whole, although they can also be sold fresh.

Sizes: Whole fish: 4 oz (112 g) to 6 oz (170 g).

Kitchen yield: There is considerable waste.

WHITEFISH, FRESH

Very perishable fish, and therefore seldom shipped beyond the Great Lakes area.

Sizes: 2 lb (900 g) to 3 lb (1.3 kg).

Season: Little fresh supply is available in winter.

Kitchen yields: 50 percent waste.

WHITEFISH, SMOKED

Available whole, head on. The fish is considered kosher.

Pack: By weight.

Sizes: 1½ lb (0.67kg) to 2½ lb (1.25 kg). Other sizes are available.

Serving size: 6 oz, cleaned, boneless fish for salad plate, main course.

Kitchen yields: Fish must be boned, head and skin removed. Waste is about 50 percent.

One 2-lb (900-g) fish = 2½ servings.

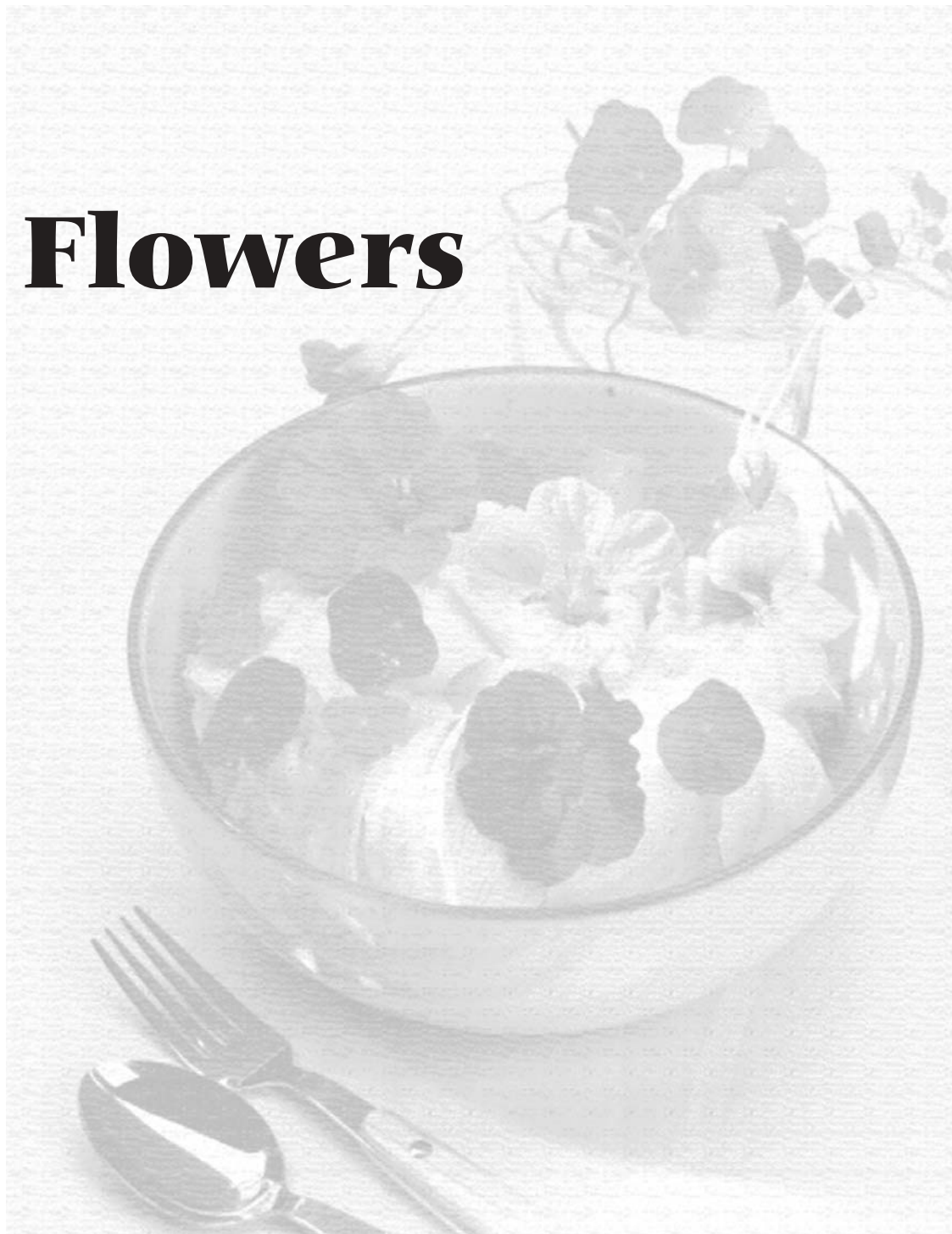
WHITING

Inexpensive fish with limited application in the hospitality industry. Harvested off the coast of New England.

Season: Summer.

Sizes: 1 lb (450 g) to 4 lb (1.8 kg).

Flowers



Flowers can be potentially toxic or may cause allergic reactions. Thus, the inclusion of this section is not to be seen as a recommendation of the culinary use of the plants and flowers listed here. Their seasons vary greatly according to location, but almost all of these flowers are now available throughout the year, shipped from many locations around the world.

ACACIA BLOSSOMS

Kitchen use: Dip acacia blossoms in batter and fry as dessert.

BANANA BLOSSOMS

See Thai/Vietnamese Foods in the International Ingredients section.

BORAGE

Tiny blue stars that taste like cucumbers.

Kitchen use: Leaves are used as bitter salad components, and flowers are candied.

CARNATIONS

Kitchen use: Buds are pickled, and blossoms are sprinkled over salads.

CHAMOMILE

Kitchen use: Fresh or dried blossoms and leaves are used as tea.

CHRYSANTHEMUM

Shungiku is the Japanese name for an Asian variety of edible chrysanthemum.

Kitchen use: Use in hot dishes as a garnish, or in salads.

CLOVER BLOSSOMS

Kitchen use: Clover blossoms can be used in salads, teas, and compound butters.

COSTMARY

Kitchen use: The pepperish lemony leaves can be used in beverages.

DAISY

Kitchen use: The flowers can be used as a garnish in beverages.

DANDELION

Kitchen use: The blossoms can be fermented for wine. *See Dandelion Greens* in the Vegetables section for salad greens.

DAY LILIES

Kitchen use: The flowers can be pickled, and the roots can be boiled as a vegetable.

ELDER

See the Fruit section.

GERANIUM

Kitchen use: Geranium leaves are used in beverages.

GLADIOLUS

Kitchen use: The petals can be added to salads.

HEATHER

Kitchen use: The blossoms are used to flavor fermented beverages.

HIBISCUS

Kitchen use: Tea made with hibiscus is popular in Hispanic countries; often referred to as *Jamaica*.

HYSSOP

Kitchen use: The flowers can be sprinkled over salads or used in beverages.

JASMINE

Kitchen use: The flower essence is used in desserts, and the blossoms are used in tea blends.

LAVENDER

Kitchen use: Lavender flowers are pickled or preserved in sugar.

LEMON BALM

Kitchen use: The leaves have a lemon scent and are used in salads and marinades.

LEMON BLOSSOMS

Kitchen use: The flowers are used as flavoring in desserts.

LEMON VERBENA

Kitchen use: The leaves can be used to substitute for lemon rind.

LILAC

Kitchen use: Lilacs are used to make candied flowers and water ice.

LIME

Kitchen use: Lime flowers are used in tea.

LINDEN

Kitchen use: Linden flowers are used in tea.

MARIGOLD

Kitchen use: The petals of the flower are used as flavoring and coloring agents in meat dishes and desserts.

NASTURTIUM

Kitchen use: The flowers and leaves have a peppery taste and can be

sprinkled over salads. The buds can be pickled.

ORANGE

Kitchen use: The essence of the orange flower may be used in ice cream and other desserts.

ORCHIDS

Kitchen use: Wanda orchids are popular in salads, desserts, and as a drink garnish.

PANSIES

Kitchen use: These flowers, of beautiful colors, can be used as a garnish or candied.

PEONY

Kitchen use: Peony flowers are used as a beverage flavoring.

ROSELLE (JELLY OKRA)

Kitchen use: These red flowers are used in jellies.

ROSE HIPS

Kitchen use: Rose fruits, called *hips*, can be made into jelly; also used as tea.

ROSES

Kitchen use: Rose petals can be candied and used for jelly and flavored water. Candied rose petals are commercially available and used as cake garnish. Rose water is used in sherbets and in almond confections.

SAGE

Kitchen use: The blue flowers can be used in cold beverages or sprinkled over meat dishes.

SQUASH BLOSSOMS

Kitchen use: Squash blossoms are used in salads and fritters.

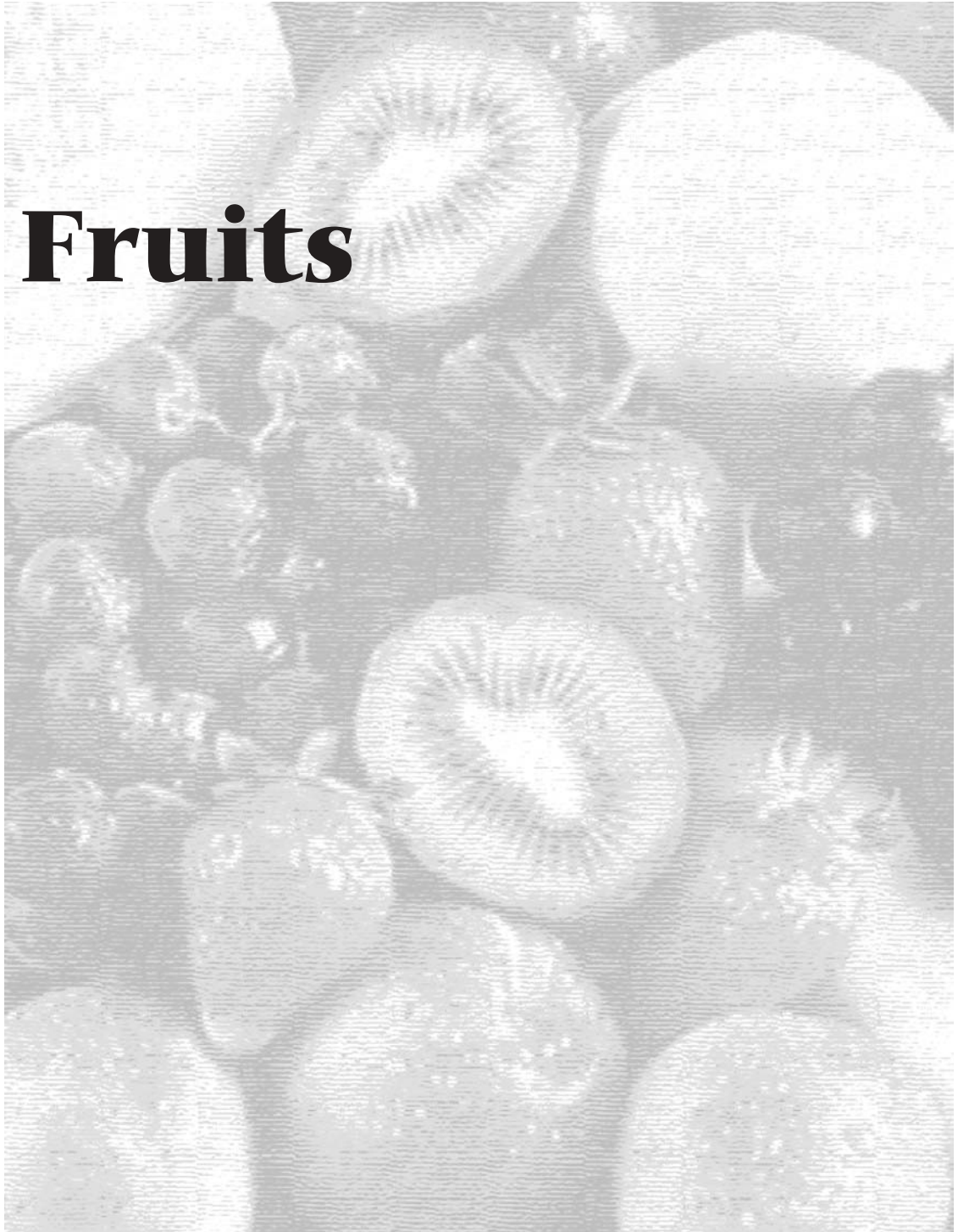
VIOLETS

Kitchen use: Violets are used in making candied petals, flavored water, water ice, and syrup.

WOODRUFF

Kitchen use: Woodruff is used as a flavoring agent in German May wine and in tea.

Fruits



In addition to whole fresh fruit, some companies also offer precut, ready-to-use fruits. Organically grown fruits and vegetables are reaching the market in increasing amounts. The packs, sizes, and yields do not always correspond with the products listed here. Canned, dehydrated, and frozen fruits and juices are listed in the Groceries section.

APPLES

Apples are generally classified as eating apples, cooking apples, and all-purpose apples.

Seasons: Apples are available year-round. Specific apple varieties have different seasons and kitchen uses. Many apple varieties are stored refrigerated, ripened with ethylene gas, and released when needed.

Varieties:

Cortland: Good for baking, salads, sauce, and freezing; September to February.

Gala: Good for eating; a New Zealand variety with rose stripings and thin skins; winter.

Golden Delicious: Good for salads, pies, baking, sauce, and freezing; year-round.

Granny Smith: Green apples, good for all uses; November to July.

Jonathan: Good for all uses except baking; September to January.

McIntosh: Good for salads and good in everything else; October to April.

Newton Pippin: An all-purpose apple, good for pie, sauce, and freezing; September to June.

Red Delicious: Bright red apple, excellent in salads, fair for all-purpose use; year-round.

Rome Beauty: Excellent for baking, fair for all-purpose use; October to June.

Winesap: Excellent in salads and for all-purpose use; October to July.

Note: Golden Delicious apples oxidize least. Cut apples can be kept in acidic water to prevent oxidation.

Packs: Loose cartons of 38 lb (17 kg) to 42 lb (18.9 kg).

Bushels of 42 lb (18.9 kg) to 52 lb (23.4 kg).

Tray pack cartons of 40 lb (18 kg) to 45 lb (20.2 kg).

Cell pack cartons of 37 lb (16.6 kg) to 43 lb (19.3 kg).

Counts, tray packs: 48, 56, 64, 72, 80, 88, 100, 113, 125, 138, 150, 163, 175, and 198.

Counts, cell packs: 64, 80, 96, 100, 120, 140, 160, 200, and 216.

Popular fancy foodservice sizes: For baking apples, 48-, 56-, or 64-count tray pack. For eating apples, 80- or 88-count tray pack.

Sizes: 1 lb (450 g) = four small, three medium-sized, or two large apples; sizes vary greatly.

Diameters for tray packs apples:

72's = 3½ in (88.9 mm).

88's = 3¼ in (82 mm).

100's = 3⅝ in (80 mm).

112's = 3 in (76 mm).

125's = 2⅞ in (73 mm).

138's = 2¾ in (70 mm).

163's = 2½ in (63 mm).

Diameters for bushel apples:

90's = 3¼ in (83 mm) to 3½ in (89 mm).

100/110's = 2¾ in (70 mm) to 3 in (76 mm).

135/145's = 2¼ in (57 mm) to 2¾ in (70 mm).

150's = 2⅝ in (67 mm).

85/195's = 2¼ in (57 to 2½ in (63 mm).

250's = 2¼ in (57 mm).

Kitchen yields: 1 medium apple, 5½ oz = 1 cup diced apple. 1 lb (450 g) apples = 3 cups diced apple.

Apple pie: 2½ lb (1.8 kg) yield = 1 pie (9 in across).

1 bushel apples yields about 10 pies.

Sauce: 1 bushel apples makes = 16 qt (15.04 l) to 20 qt (18.80 l) sauce.

Calories: 1 apple, medium, unpeeled, 5.3 oz (150 g), 2¾ in diameter = 90 calories.

1 apple, medium, unpeeled, 8 oz (225 g), 3¼ in diameter = 135 calories.

LADY APPLES

Gold-colored fruits, the size of apricots, with a distinctive flavor; often used for fruit baskets.

Season: Winter.

Pack: 20-lb (9-kg) lugs.

APRICOTS

Apricots are difficult to ship when perfectly ripe and are often sold hard and relatively flavorless. For canned, dried, and frozen apricots, *see* the Groceries section.

Season: June through August. Imports are available during the rest of the year.

Packs: 24-lb (10.8-kg) lugs.
12-lb (5.4-kg) tills.
6-lb (2.7-kg) baskets.

Counts and sizes: The size is determined by the number of fruits arranged in layers. The count on the lowest layer is usually less than that on the top layers.

6-lb (2.7-kg) baskets = 44 pieces 4 × 4.
or = 55 pieces 4 × 5.
or = 65 pieces 5 × 5.
24-lb (10.8-kg) lug = 176 pieces 4 × 4.
or = 220 pieces 4 × 5.
or = 260 pieces 5 × 5.

Sizes per lb (450 g):

Extra Extra Jumbo: 6 or 8.
Jumbo: 10.
Large: 12.
Medium: 14.

Serving sizes: ½ cup, 4 oz (112 g), sliced.

Kitchen yields: 8 to 10 large fruits = 3 cups sliced. 3 large fruits = 1 cup sliced.

Calories: 3 large apricots, 4 oz (112 g) = 55 calories.

ASIAN PEAR/APPLE

Subtropical fruit with a thin, edible skin. It is lightly sweet, and its aftertaste is perceived by some people to have an almost salty quality. There are two varieties, one resembling apples and the other pears. The fruit is also known as *aka nashi*, *apple pear*, *ottahitia apple*, *oriental apple*, *Japanese pear*, *Japanese twentieth-century pear*, *new century pear*, and *sand pear*.

Season: Fall and early winter.

Packs: 17-lb (7.6-kg) and 20-lb (9-kg) tray packs.

Size: About 8 oz (225 g).

Kitchen yields: Unlike all other pears, these fruits are tree-ripened. They remain crisp for a month when refrigerated, or for 1 week at room temperature.

BANANAS

Bananas are shipped green and are ripened with ethylene gas. US #1 bananas are normally purchased for foodservice. There are many other varieties and sizes available in ethnic food markets. For dried Banana Chips and Processed Pulp, *see* the Groceries section.

Season: Available year-round.

Packs: 40-lb (18-kg) cartons, A bunch of bananas is referred to as a "hand."

Sizes: 100 to 150 count. The average count is 115 bananas per carton. A 150-count banana is called *petites*.

Kitchen yields: 1 lb (450 g) = 12 oz (340 g) peeled. 1 medium banana = 1 cup sliced, 4½ oz (125 g).

Calories: One medium banana, 4½ oz, peeled = 100 calories.

FINGER BANANAS (MANZANITA BANANAS)

Small bananas are often called *finger bananas*.

Season: Year-round; peak is in summer.

Packs: By weight.

Weights of one hand: 12 oz (340 g) to 14 oz (400 g).

Count of one hand: 10 to 12 bananas.

RED BANANAS

A banana variety that can be eaten fresh but can also be cooked. The flesh is white.

BLACKBERRIES

Blackberries are the fruits of a small rambling bush growing wild in temperate climates; also widely cultivated. The berries have a shiny surface that becomes duller as they ripen. Boysenberries and loganberries are members of the blackberry family, but have a somewhat waxier appearance. They are actually a cross between the blackberry and raspberry. Dewberries are related, but trail on the ground and ripen earlier than blackberries. The commercial availability is limited.

Season: Late spring to fall for domestic production; imports available in winter.

Packs: 12 half-pints (0.24 l) in flats or trays.

Serving sizes: 1 cup, 5 oz (0.24 l), for breakfast or as dessert.

Kitchen yields: One flat = 12 servings for breakfast or as dessert.
One flat = 36 servings as garnish with desserts.

Calories: 1 cup, 5 oz (0.24 l) = 85 calories.

BLUEBERRIES

Most blueberries sold on the commercial market are cultivated. The juice of

blueberries is white. Huckleberries resemble blueberries, but have purple-colored juice. Blueberries are available imported during the off-season. For Canned and Frozen Blueberries, *see* the Groceries section.

Season: June to September; peak occurs in summer.

Packs: 1-pint (0.47-l) cartons, flats of 12 pints.

1-qt (0.9-l) cartons, flats of 8 quarts.
5-lb (2.2-kg) bulk cartons (rare).

Weights: 1 cup (0.24 l) blueberries = 5½ oz (150 g).

Counts: 1 cup (0.24 l) = Extra Large 90, Large 90 to 129, Medium 130 to 189, Small 190 to 250.

Serving sizes: ½ cup (0.12 l), breakfast or as dessert.

Kitchen yields: There is basically no cleaning waste.

1 pint = 4 servings.

One flat of 12 pints = 48 servings.

One flat of 8 qt = 64 servings.

For Blueberry Pie: One 9 in pie requires 3½ cups blueberries.

One flat of 12 pints = 7 pies.

For blueberry muffins: For 80 oz (2.3 l) batter, add 3½ cups blueberries.

The batter yields 48 muffins with the use of a 2-oz (#16) scoop.

Calories: 1 cup, 5 oz (0.24 l) = 90 calories.

BOYSENBERRIES

See Blackberries.

BREADFRUIT

See the Vegetables section.

CALMYRNA FIGS

See Figs.

CAPE GOOSEBERRY

Cherry-sized fruit encased in a natural papery husk, resembling a Chinese lantern. Bitter and green when unripe, it

ripens to a mustard-gold color, with an aromatic smell and medium-sweet flavor. These fruits come from South Africa, thus the name. Tomatillos are related.

Season: Winter.

Pack: Sold by weight, often in pint containers.

Sizes: 1½ in (45 mm) to 2 in (50 mm) across.

Kitchen yields: The fruit can be eaten raw and is used with the pod as exotic garnish on fruit displays.

CHAMPAGNE GRAPES

Tiny red grapes in small bunches, grown to be used for garnishes. For other information on grapes, *see* Grapes.

Packs: 5-lb (2.2-kg) lugs.

Season: Late summer; there is little supply during the rest of the year.

Serving sizes: 2-oz (56-g) bunches. The sizes of the bunches vary.

Kitchen yields: One 5-lb (2.2-kg) lug = 35 servings.

CHERRIES

See the Groceries section for Canned, Dried, and Frozen Cherries.

SOUR OR TART CHERRIES

Principal varieties are Montmorency, Early Richmond, and English Morello. Sour cherries are marketed primarily canned, frozen, or as pie filling. *See* the Groceries section for Canned and Dried Cherries.

Season: July.

Packs: By weight.

Kitchen yields: 1 lb (450 g) = 2½ cups (0.6 l).

SWEET CHERRIES

Varieties:

Bing: Dark red, large, firm, and juicy.

Lambert: Dark red, heart-shaped, slightly smaller than *Bing*.

Rainiers: Light-colored, cream to golden.

Black Republican: Very dark, medium sized.

Royal Ann: Light yellow with pink blush.

Once off the tree, cherries do not continue to ripen, but decay. Cherries should be stored in a refrigerator.

Season: May to July; imports are available in winter.

Packs: 12-lb (5.4-kg) flats.

18-lb (8.2-kg) California lugs.

20-lb (9-kg) lugs.

Counts: Depend on weather and on variety.

1 lb (450 g) *Bing* cherries = 60 to 150.

1 lb (450 g) other varieties = 280 to 290.

Serving sizes: ½ cup (0.12 l) heaping cherries, with pit.

Kitchen yields: 1 lb (450 g) = 5 servings.

3½ oz (100 g) = 15 medium cherries.

3½ oz (100 g) = 10 to 12 large cherries.

1 qt (0.9 l) whole cherries = 1 lb (450 g) pitted fruit.

Calories: 3½ oz (100 g) = 70 calories.

CHESTNUTS, FRESH

Improperly stored or old fresh chestnuts can be moldy or empty; check before accepting delivery. For Canned and Dried Chestnuts and other Chestnut Products, *see* the Groceries section.

Season: Winter.

Packs: By weight.

Kitchen yields: Peeling loss = 50 percent of raw weight.

CRAB APPLES

Crab apples are small ornamental apples, available both fresh and canned. For Canned Crab Apples, *see* the Groceries section.

Season: Fall.

Kitchen yield: Use as decoration.

CRANBERRIES

Season: Late fall and winter.

Packs: 1-lb (450-g) cello packs, 24 per case.

Bulk packs are available.

Kitchen yields: 10 lb (4.5 kg) = 10 qt (9.4 l) whole sauce.

10 lb (4.5 kg) = 7½ qt (7 l) strained sauce.

CURRENTS

There are three varieties of currants: black, red, and white. White currants are seldom available on the commercial market.

BLACK CURRENTS

Black, aromatic berries growing in clusters, available in late summer, but seldom available fresh because most of the harvest is processed. Black currants are the basic ingredient in *cassis*.

Packs: 1-pint (0.47-l) containers.

Kitchen yields: Currants are sold in small clusters, which can be used whole as a garnish on fruit plates.

RED CURRENTS

Red berries growing in clusters; tart when not completely ripe. Available in summer but rare, because most of the harvest is used in jelly.

Packs: 1-pint (0.47-l) containers.

DATES, FRESH

The date industry is centered in Indio, California. A number of varieties are grown. Medjool dates are extra-large dates. For Dried Dates, *see* the Groceries section.

Season: September, but the supply is limited.

ELDERBERRIES, ELDER BLOSSOMS

The elderberry shrub has clusters of fragrant blossoms, which mature into rather acidic purple berries.

Season: Blossoms in spring, berries in fall.

Kitchen yields: Blossoms can be dipped in batter and deep-fried. The berries can be used to make excellent stewed fruit and pie filling and can be fermented into wine. The color is strong; it will stain other fruits and is often used to enhance color in manufactured products.

FIGS

Rounded fruits, pointed at the stem end and growing on trees in temperate climates. Ripe figs vary in color from bright green to deep purple, depending on the variety. The important varieties are Calmyrna, a green fig when ripe, and Black Mission. Fresh figs are highly perishable. For Canned Figs and Dried Figs, *see* the Groceries section.

Season: June to October.

Pack: 3-lb (1.7-kg) flats, with 30 count. 5-lb (2.2-kg) flats, with 56 count.

Weight: 1½ oz (42 g) or more.

Kitchen yields: Calmyrna figs are often peeled for service. The peeling loss is about 20 percent.

Calories: 3 fruits, about 4 oz (112 g) = 230 calories.

FRAISE DES BOIS

French name for *wild strawberries*. There is very little commercial production.

GALAX LEAVES

Round, tough leaves about 2½ in (63 mm) across, green or reddish brown, used for garnishing fruit baskets. Order from a florist.

Packs: Bunches of 24 leaves each.

Kitchen yields: Leaves keep well when refrigerated.

GINGER ROOT

Brown fibrous root, used in Asian cooking, as flavoring in baking and in

beverages. Available year-round. For other Ginger Products, *see* the Groceries section.

Packs: 5-lb (2.2-kg), 10-lb (4.5-kg), and 30-lb (13.6-kg) cartons.

Sizes: Average sizes vary.

Kitchen yields: Peeling loss is about 20 percent. Because the product is fibrous, the loss can be greater.

Calories: 3½ oz (100 g) = 50 calories.

GOOSEBERRIES, FRESH

For Canned Gooseberries and Frozen Gooseberries, *see* the Groceries section.

Season: Summer.

Packs: 1-pint (0.47-l) containers. Other packs are available.

Kitchen yields: There is little cleaning necessary.

GRAPEFRUIT

The main varieties of grapefruit are Marsh Seedless White and Ruby, with pinkish flesh. Duncan grapefruit has most seeds. Florida and Texas grapefruits are generally juicier than California- and Arizona-grown fruits. Indian River and Rio counties in Texas are known for producing quality grapefruits. Grapefruits are picked when ripe and do not ripen any further once off the tree.

Season: Winter grapefruit is produced in Florida, California, and Texas.

Summer fruit is produced in Arizona and California.

Red Grapefruit is in season from October to May.

Florida:

Packs: 40-lb (18-kg) to 50-lb (22.5-kg) carton = $\frac{4}{5}$ of a bushel.

Counts: 23, 27, 32, 36, 40, 48, 56, and 64.

Texas:

Packs: 38-lb (17.1-kg) to 42-lb (18.9-kg) carton = $\frac{7}{10}$ of a bushel.

Counts: 18, 23, 27, 32, 36, 40, 48 or 56, 76, and 84.

California and Arizona:

Packs: 38-lb (17.1-kg) to 42-lb (18.9-kg) carton.

Counts: 23, 27, 32, 36, 40, 48, 56 and 64.

Sizes:

#23 = 5 in (125 mm)

#27 = 4 $\frac{9}{16}$ in (118 mm).

#32 = 4½ in (112 mm).

#36 = 4¼ in (107 mm).

#40 = 4 $\frac{5}{8}$ in (103 mm).

#48 = 3 $\frac{3}{8}$ in (98 mm).

1 fruit, medium = about 18 oz (500 g)

Common hotel sizes: 23's

Kitchen yields: 1 fruit = 10 to 12 sections.

1 fruit medium, about 18 oz (500 g) = 10 oz (280 g) sections.

1 fruit medium, about 18 oz (500 g) = 1 cup juice.

1½ medium fruits = 1 cup diced fruit.

Calories: ½ fruit, 8½ oz (240 g), pink or red = 50 calories.

FRESH GRAPEFRUIT JUICE

Blends of grapefruit juice and other fruit juices, too numerous to list, have come on the market. For Canned and Frozen Juices, *see* the Groceries section.

Packs: 1-qt (0.9-l) cartons, 24 per case. 2-qt (1.9-l) cartons, 12 per case.

Serving sizes: 6 oz (0.17 l).

Kitchen yields: One 2-qt carton = 10 servings.

Calories: 6 oz (0.17 l) = 75 calories.

FRESH GRAPEFRUIT SECTIONS

Popular banquet item. The industry distinguishes between hot-peeled and cold-peeled sections; the latter considered are better quality.

Packs: 1-gal (3.8-l) jars, 4 per case.

Drained weight: 6 lb (2.7 kg).

Serving sizes: ½ cup (0.12 l) with juice.

Kitchen yields: 1-gal (3.8-l) jar = 28 to 30 servings.

GRAPES

Grapes are classified as seedless or seeded and by color. The shades of color vary widely and can range from light to dark green and from purple to dark blue. For foodservice use, seedless grapes are preferred. Both red and green grapes can be seedless. For Canned Grapes, Frozen Grape Juice and Raisins, see the Groceries section.

Major Grape Varieties and Their Seasons

<i>Almeria, seeded, green</i>	October through February
<i>Calmeria, seeded, green</i>	October through February
<i>Cardinal, seeded, red</i>	Mid-May to mid-August
<i>Concord, seeded, blue</i>	September to November
<i>Emperor, seeded, red</i>	September through March
<i>Exotic, seeded, red</i>	June through August
<i>Flame Seedless, red</i>	Mid-June through September
<i>Golden Muscat, green</i>	September to December
<i>Italia, seeded, green</i>	August through September
<i>Perlette, seedless, green</i>	Mid-May to mid-July
<i>Queen, seeded, red</i>	August and September
<i>Ribier, seeded, red</i>	August through mid-February
<i>Ruby, seedless, red</i>	Mid-August through January
<i>Thompson, seedless, green</i>	June to November
<i>Tokay, seeded, red</i>	July through November

Packs: 23-lb (10.3-kg) lugs.
17-lb (7.6-kg) boxes.
12-lb (5.4-kg) boxes.

Sizes, preportioned bunches: 2 oz (56 g), 3 oz (85 g), 4 oz (112 g), and 8 oz (225 g).

Sizes: Vary greatly. Typically, there are 35 to 45 grapes in 1 cup.

Serving size: ½ cup, 3 oz (85 g)

Kitchen yields:

Seedless grapes: 1 lb (450 g) as purchased = 15 oz (420 g) stemmed grapes.

One 20-lb (9-kg) lug yields 50 cups (11.6 l) stemmed grapes.

Grapes with seeds: 1 lb (450 g) as purchased = 14½ oz (400 g) stemmed grapes.

One 23-lb (10.3-kg) lug = 55 cups (12.7 l) stemmed grapes.

One 23 lb lug = one hundred ten 3-oz (85-g) servings.

10 lb (4.5 kg) are needed for 100 servings.

Calories:

Seedless: 1 cup, 5¾ oz (163 g) = 107 calories.

Seeded: 1 cup, 6 oz (170 g) = 102 calories.

HUCKLEBERRIES

Huckleberries, a European variety of blueberries, have dark purple juice and are seldom available commercially. The juice of cultivated blueberries is white.

Season: Mid to late summer.

Calories: 1 cup, 5 oz (0.24 l) = 90 calories.

KIWIFRUIT

Also called *Chinese Gooseberries*, kiwifruit are green fruits with thin, fuzzy skin. The fruits are ripe when they give slightly under light pressure. To ripen kiwifruit quickly, store at room temperature in a plastic bag along with apples or bananas. Kiwiberries are grape-sized fruits with smooth skin. Some varieties are red inside rather than green.

Season: California-grown fruit is available from October until May; imported fruits are available year-round.

Packs: 7-lb (3.1-kg) flats on average.
20-lb (9-kg) cartons.

Counts: In 7 lb flats: 25, 28, 30, 33, 36, 39, 42, and 45.

Weight range: 2 oz (56 g) to 5 .5 oz (150 g).

2½ oz (70 g) to 3 oz (85 g) is average.

Kitchen yields: One 2½ oz (70 g) fruit = 5 to 6 slices.

1 cup, 6 oz (170 g) diced large fruit = 2½ fruits.

1 flat #39 = 16 cups diced fruit.

Calories: 4 oz (100 g) = 39 calories.

LEMONS

Season: Available year-round; peak occurs in May, June, and July.

Packs: 37-lb (16.6-kg) to 40-lb (18-kg) cartons.

Counts: 63, 75, 95, 115, 140, 165, 200, and 235 per case. Most common counts are from 115 to 165.

Kitchen yields: When buying for juice, the lowest price per pound is the best buy, because the yield is approximately the same regardless of price.

8 medium lemons = 1 cup juice.

1 lb (450 g) = 4 lemons, #165.

1 case #165 = 300 halves or 660 quarter wedges or 990 one-sixth wedges.

1 lemon = 7 to 10 slices when sliced on machine.

10 lemons = 1¾ oz (50 g) grated lemon rind.

Calories: 1 cup, 8.7 oz (0.24 l), fresh juice = 60 calories.

LIMES

Almost all limes sold commercially are acidic limes, often referred to as *Persian limes*. The Florida Key West lime is the only sweet lime variety with any commercial importance. See the Groceries section for Bottled Lime Juice.

Season: Available all year; peak occurs in spring.

Packs: Flats with counts of 63 and 72.

Kitchen yields: Most common hotel size is 63 count.

1 flat 63 count = 120 halves or 250 pieces.

1 lb (450 g) fruit = ¾ cup (0.17 l) juice.

Calories: 1 cup, 8.7 oz (0.24 l) = 65 calories.

KEY WEST LIMES (SWEET LIMES)

Yellow, thin-skinned fruits with a pleasant taste. They are no longer grown commercially in Key West, but some supply comes from the Caribbean Islands and Mexico.

Season: August and September, domestically grown; imports, early spring.

LOGANBERRIES

Berries growing wild, and being cultivated as well, resembling blackberries in shape and raspberries in color, but not directly related to either plant. They are very tart unless picked ripe. Commercial production is in the Pacific Northwest.

Season: June through August, generally best in July.

Packs: ½-pint (0.24-l) containers, 12 in a flat.

Kitchen yields: Used primarily for jams and jelly.

MELON VARIETIES

CANARY MELON

Also called *Juan canary melon*.

Season: Summer.

Packs: 26-lb (11.7-kg) to 27-lb (12.1-kg) cases.

29-lb (13-kg) to 30-lb (13.5-kg) cartons.

Counts: 6, 7, 8, or 10 for 26-lb (11.7-kg) case.

4, 5, or 6 for 29-lb (13-kg) carton.

Weight range: 2¼ lb (1 kg) to 4½ lb (2 kg).

CANTALOUPE

Cantaloupe melons belong to the broad group of muskmelons. They are widely available and well sized. Much of the market supply is domestically grown, and there are imports in the off-season. The melons will ripen somewhat at room temperature but lose flavor in cold storage. A clean cavity at the stem end indicates that the melon was harvested when ripe.

Season: Available year-round; peak occurs in June to August.

Packs: 75-lb (33.7-kg) to 85-lb (38.2-kg) standard crate.

53-lb (23.7-kg) to 55-lb (24.7-kg) two-thirds crate.

38-lb (17.1-kg) to 41-lb (18.4-kg) half-size crate.

Counts: 18, 23, 27, 36, or 45, standard crate.

12, 13, 18, 24, or 30, two-thirds crate.

12, 15, and 18, half-size crate.

Most common sizes: 45 or 36 pieces in 85-lb (38.2-kg) crate.

Serving sizes: One-half 45-size melon for breakfast.

One-half 36-size melon as main course.

Kitchen yields: One 45-size melon yields 25 melon balls, $\frac{7}{8}$ in (22 mm) across. 100 melon balls weigh 2 $\frac{1}{4}$ lb (1 kg).

CASABA MELON

Late-variety melon with soft, white, juicy flesh. Because the melon is normally vine ripened, it is very sweet.

Season: Late summer and early winter.

Packs: 26-lb (11.7-kg) to 27-lb (12.1-kg) case.

29-lb (13-kg) to 30-lb (13.5-kg) carton.

Counts: 6, 7, 8, or 10 for 26-lb (11.7-kg) carton.

4, 5, or 6 for 29-lb (13-kg) carton.

Weight range: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Kitchen yields: Small melons = 4 servings. Large melons = 6 servings.

CHRISTMAS MELON

See Santa Claus Melon.

CRANSHAW (CRENSHAW) MELON

This dependable melon stores well and has a rich, sweet flavor. The flesh is bright salmon color.

Season: Summer, fall, and winter.

Packs: 26-lb (11.7-kg) to 27-lb (12.1-kg) case.

29-lb (13-kg) to 30-lb (13.5-kg) carton.

Counts: 6, 7, 8, or 10 for 26-lb (11.7-kg) carton.

4, 5, or 6 for 29-lb (13-kg) carton.

Weight range: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Kitchen yields: Small melons = 4 servings. Large melons = 6 servings.

HAND MELON

A sweet, juicy, and expensive melon, cultivated at the Hand Farm in Greenwich, New York, and identifiable by a red hand decal. The season is very short, and the melons are shipped only at the peak of ripeness, primarily to the upscale New York City market.

Season: August.

Pack: 40-lb (22.5-kg) carton.

Count: 16 melons.

Average weight: 3 lb (1.3 kg) or less.

Serving sizes: $\frac{1}{3}$ melon for appetizer or dessert.

Kitchen yields: One 40-lb (22.5-kg) carton = 120 servings.

HONEYDEW MELON

A melon with light green, sweet, and delicate flesh.

Season: July to October for domestic products; winter to spring for imports.

Packs: 26-lb (11.7-kg) to 27-lb (12.1-kg) case.

29-lb (13-kg) to 30-lb (13.5-kg) carton.

Counts: 6, 7, 8, or 10 for 26-lb (11.7-kg) carton.

4, 5, or 6 for 29-lb (13-kg) carton.

Weight range: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Serving sizes: One quarter of a 6- or 8-size melon.

One sixth of a 5-size melon.

Kitchen yields: One 5-size melon = 35 melon balls.

100 melon balls = 2¼ lb (1 kg).

PERSIAN MELON

Season: Summer and fall.

Packs: 42-lb (18.9-kg) half-crates, 6 melons in each.

Average size: 7 lb (3.1 kg).

SANTA CLAUS MELON

Also called *Christmas melon*, this fruit resembles watermelon on the outside, but the interior is green.

Season: Late summer; peak occurs in winter.

Packs: 26-lb (11.7-kg) to 27-lb (12.1-kg) case.

29-lb (13-kg) to 30-lb (13.5-kg) carton.

Counts: 6, 7, 8, or 10 for 26-lb (11.7-kg) carton.

4, 5, or 6 for 29-lb (13-kg) carton.

Weight range: 4 to 6 lb (1.8 to 2.7 kg).

Kitchen yields: Small melons = 4 servings. Large melons = 6 servings.

SPANISH MELON

This melon is hard-skinned, with a green to deep yellow exterior and sweet, orange-colored flesh.

Season: Winter.

Season: Late summer; peak occurs in winter.

Packs: 26-lb (11.7-kg) to 27-lb (12.1-kg) case.

29-lb (13-kg) to 30-lb (13.5-kg) carton.

Counts: 6, 7, 8, or 10 for 26-lb (11.7-kg) carton.

4, 5, or 6 for 29-lb (13-kg) carton.

Weight range: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Kitchen yields: Small melons = 4 servings. Large melons = 6 servings.

WATERMELON

A very large melon. Also available is a smaller, seedless variety with a red or green interior.

Season: Mid-April to mid-December; peak occurs in late summer.

Packs: 70-lb (31.5-kg) to 85-lb (36-kg) carton.

Counts: 2, 3, 4, or 5 melons.

Weight range: 15 lb (6.7 kg) to 40 lb (18 kg). Larger sizes are available.

Serving size: 2-lb (0.9-kg) slice with rind.

Kitchen yields: For buffet display, purchase 30-lb (13.5-kg) melon.

One 22-lb (9.9-kg) melon = 22 cups, bite-size pieces.

One 32-lb (14.4-kg) melon = 16 servings.

Calories: 2-lb (0.9-kg) slice = 110 calories.

NECTARINES

Juicy fruit related to peaches. Each fruit has one large pit. Freestone varieties should be purchased for foodservice. There is substantial domestic production; imports are available in off-season, but are shipped unripe. Late-season fruits can show wooliness, or the flesh may have a brown discoloration.

Season: May to September; peak occurs in July and August.

Packs: 19-lb (8.5-kg) to 23-lb (10.3-kg) two-layer lugs.

25-lb (11.2-kg) and 35-lb (15.7-kg) loose-fill cartons.

Counts: 48, 56, 64, 72, 80, and 84 for two-layer lugs.

Weight: One 64-size fruit = 5 oz (140 g) average.

Kitchen yields: 1 lb (450 g) = 3 medium fruits, 2 cups (0.48 l) sliced.

One 22-lb (9.9-kg) case = 2¾ gal (10.4 l) sliced fruit.

Calories: 1 cup sliced fruit = 75 calories.

ORANGE VARIETIES

Oranges are produced in Florida, California, Texas, and Arizona. Marketing season is year-round, with a general peak from December through May. Oranges are harvested when fully ripe and do not mature further off the tree. Ripe oranges can develop a green tint.

Varieties:

Valencia: Good juice orange, good eating.

Navel: Large, seedless, sections easily.

Hamlin: Seedless, thin peel.

Pineapple: Has seeds, juicy.

Temple: Few seeds, peels easily.

Valencia is a summer orange, the *Navel* a winter orange.

Seasons:*Florida:*

Hamlin: October to January.

Navel: November to January.

Pineapple: December to February.

Temple: January to March.

Valencia: March to July.

California and Arizona:

Navel: November to June.

Valencia: March to January.

Temple: November to February.

Texas:

Valencia: February to June.

Temple: February to June.

Packs: Vary by growing areas:

Florida: 37 lb (16.6 kg) to 45 lb (20.2 kg) or $\frac{1}{2}$ bushel.

COUNTS: 64, 80, 100, or 125 and 32, 36, 40, 48, or 56 for Navels.

California and Arizona: 37-lb (16.6-kg) to 45-lb (20.2-kg) carton.

COUNTS: 48, 56, 72, 88, 112, 138, or 168.

Texas: 38-lb (17.1-kg) to 44-lb (19.8-kg) carton.

COUNTS: 56, 64, 80, 100, 125, 163, 200, 252, or 288.

Florida sizes, based on number of oranges in a 40-lb (18-kg) carton:

125 size = $2\frac{1}{2}$ -in (63-mm) diameter.

100 size = $2\frac{1}{16}$ -in (68-mm) diameter.

80 size = $2\frac{5}{8}$ -in (75-mm) diameter.

64 size = $\frac{3}{16}$ -in (81-mm) diameter.

Weights:

48's, very large, 12 to 14 oz (360 g).

88's, medium, $7\frac{1}{2}$ oz (200 g).

163's, small, 4 oz (112 g).

Common hotel sizes for fruit baskets:

56, 64, or 72 sizes, Navel or Valencia, are recommended.

Kitchen yields: One large Navel orange,

13 oz (365 g) = 8 oz (116 g) peeled fruit.

3 to 4 medium oranges = 1 cup, 8.8 oz (0.24 l), juice.

1 medium orange = 4 tsp grated peel.

Calories: One 6.5 oz (180 g) orange = 65 calories.

BLOOD ORANGES

Blood oranges are esteemed for their juice, which has a tart, sprightly flavor. The pulp ranges in color from blush, to light orange flecked with ruby, to purplish red. Now grown in southern California.

Season: Summer.

FRESH ORANGE JUICE

Canned, Frozen, and Dehydrated Juices are listed in the Groceries section.

Packs: 1-qt (0.9-l) cartons or bottles.

$\frac{1}{2}$ -gal (1.8-l) cartons or bottles.

Other packs are available.

Serving sizes: 6 oz (0.17 l).

Kitchen yields: One qt (0.9 l) = 5

servings. $\frac{1}{2}$ gal (1.8 l) = 10 servings.

Calories: 6 oz (0.17 l), unsweetened = 90 calories.

JUICE ORANGES

Florida oranges have a higher juice content than California oranges. The three best juice oranges are Valencia, Hamlin, and Pineapple. The best yield is the lowest price per carton, regardless of size.

Seasons:*Florida :*

Valencia: March to July.

Hamlin: October to January.

Pineapple: December to February.

California: Valencia: March to January.

Packs:

Florida: 125 count.

California: 112 or 138 count.

Kitchen yields: The season, climate, count, and the type of juicing machine used influence the yield. Juicing machines must be adjusted to conform to the sizes of the oranges to get the best possible yield. Higher-count oranges give slightly better yields. Storage oranges are available from July to October, but the juice yield is lower. One case #125 = 2¼ gal (8.5 l) to 2½ gal (9.5 l).

MANDARIN ORANGES

Mandarin oranges are small, sweet hybrid citrus fruits related to oranges, though with a distinctive flavor and a loose skin that usually peels away easily from the flesh. Mandarins and tangerines are considered to be similar, and the names and varieties are used interchangeably. There are a number of varieties such as Clementine, Satsuma, Wilkins/Wilkins, and Kinnow. Temple oranges are a cross between mandarins and sweet oranges, and tangelos are a cross between mandarins and grapefruit. There are other varieties on the market, with seeds and sections that are hard to separate. The boxes are covered with orange-colored netting, which makes it difficult to detect spoilage. Note that in the fall of 2001, mandarin oranges imported from Spain were taken off the shelves in western states because they could harbor fruit flies. For Canned Mandarin Sections, see the Groceries section.

Season: November to March.

Packs: 5-lb (2.3-kg) and 10-lb (4.6-kg) cartons.

Count: One 5-lb (2.3-kg) carton = 24 oranges.

Sizes: 3 in (75 mm) = 2 per pound, 2½ in (62 mm) = 3 per pound, 2¼ in (55 mm) = 3 per pound

Kitchen yields: Three 3-in (75-mm) fruits or four 2½-in (62-mm) fruits = 1 cup juice.

Calories: 1 lb (450 g) 2½-in (62-mm) fruits = 135 calories.

SEVILLE ORANGES

Bitter oranges cultivated for marmalade. One variety, grown on the island of Curaçao, is used in the production of the cordial of the same name. Bitter orange used in Mexican cooking is called *naranja agria*.

Season: February and March.

TANGELOS

A tangerine-grapefruit hybrid with thin skin, easily separated sections, and many seeds.

Season: From October to May.

Packs: 45 lb (20.2 kg) = ⅔-bushel carton, Florida packs.
27-lb (12.1-kg) to 32-lb (14.4-kg) cartons.

Counts:

Florida: 40, 48, 56, 64, 80, 100, and 125.

California: 42, 54, 72, 84, 90, 100, 105, 135, and 165.

PEACHES

Stone fruit or drupe in the rose family, with velvety skin, sweet flesh, and one large pit. There are hundreds of varieties, but peaches can be basically divided between clingstone, in which the pit clings to the flesh, and freestone, in which the pit can easily be removed. Tree-ripened peaches are much better than those that must be picked early for shipment. There are yellow and white peaches. A ripe yellow peach is of a deep, rich yellow color, and a ripe white peach is creamy gold. Peaches ripen at room temperature; refrigerating delays ripening. Early-season peaches are cling and semi-cling (the pit clinging to the flesh) varieties. All late-season peaches are

freestone. For Canned, Dehydrated, and Frozen Peaches, *see* the Groceries section.

Season: Late May until September, peaking in July and August. Imports are available during the winter.

Packs: 19-lb (8.6-kg) to 23-lb (10.3-kg) two-layer lugs.

17-lb (7.7-kg) to 18-lb (8.2-kg) boxes.

38-lb (17-kg) boxes, loose fruit.

45-lb (20.2-kg) to 48-lb (21.6-kg) bushel basket.

Counts: 50, 56, 60, 64, 72, or 80 in two-layer lugs.

40, 45, 50, 55, 60, or 65 in boxes.

Sizes: 2-in (50-mm), 2¼ in (57-mm) and 2½-in (63-mm) diameter in bushel baskets.

Kitchen yields: Common hotel size = 60-count lug.

1 lb (450 g) = 3 to 4 medium fruits.

1 lb (450 g) = 2 cups (0.47 l) sliced peaches.

Calories: 1 cup, 6 oz (170 g) sliced peaches = 65 calories.

PEARS

There are numerous varieties of pears on the market and certain seasons when they are most plentiful. Many pear varieties are available year-round. Pears do not ripen well on trees and are picked when mature but still hard. They keep well refrigerated and ripen from the inside out in a warm spot. For Canned Pears, *see* the Groceries section.

<i>Name of Variety</i>	<i>Season</i>	<i>Uses</i>
<i>Anjou</i>	October to May	Fresh and salads
<i>Bartlett</i>	August to December	Fresh and cooking
<i>Bosc</i>	September to May	Fresh and cooking
<i>Comice</i>	October to March	Fresh
<i>Forelle</i>	October to February	Fresh

Nelis October to April Fresh and cooking

Red Bartlett August to December Fresh and cooking

Seckel August to January Fresh

Packs: 44-lb (20-kg) to 46-lb (21.1-kg) layer-packed boxes.
36-lb cartons, tight fill.

Counts: 70, 80, 90, 100, 110, 120, 135, 150, or 165 in layered boxes.
56, 64, 72, 80, 90, 96, 108, 120, or 125 in cartons.

Sizes: Large pear = 90 to 110 count.

Medium pear = 135 count.

Small pear = 165 count.

Not all varieties are available in all counts.

Weights: 90's = 8 oz (225 g).

135's = 5½ oz (155 g).

165's = 4½ oz (126 g).

Kitchen yields: 1 cup (0.24 l) = 5½ oz (155 g) sliced pear without skin.
3 medium pears = 2 cups sliced pears without skins.

Calories: One Bartlett pear, 6 oz (170 g) = 100 calories.

One Bosc pear, 5 oz (140 g) = 85 calories.

One d'Anjou pear, 7 oz (200 g) = 120 calories.

PINEAPPLE

Pineapples should be harvested as ripe as possible, which can be problematic because the overripe fruit will ferment rapidly. The ripeness of the fruit is not easy to judge. Domestic products and imports come to the market at various times. Baby Pineapples, about 2½ to 3 in (63 to 76 mm) tall, are available from Hawaii. They are used primarily for decorative purposes. For Canned and Processed Pineapple, *see* the Groceries section.

Season: Available year-round; peak occurs in April and May.

Packs: 20-lb (9.1-kg) half cartons.
40-lb (18.2-kg) full cartons.

Counts: 4, 5, 6, or 7 for half cartons.
8, 9, 10, 12, 14, 15, 16, or 18 for full cartons.

Common sizes: 5's for salads.
14's for appetizer baskets.

Weights: 1 medium pineapple = 4 lb (1.8 kg). 1 cup = 5.5 oz (160 g).

Kitchen yields: 1 medium pineapple = 5½ cups diced pineapple.
One pineapple #5 = 22 oz (610 g) clean meat.
= 17 slices when sliced with machine at #34 setting.

Juice yield: One # 5 pineapple = 2½ cups (0.6 l) juice.

Calories: 1 cup, 5.5 oz (160 g), raw, diced pineapple = 80 calories.

PLUMS

There are two major varieties of plums, clingstone and freestone. Those of the clingstone variety are marketed for eating fresh, and those of the freestone variety are marketed as fresh prunes. However, there is much confusion about names. Japanese varieties are yellow to red.

Types according to color:

Purple: Italian, French Plum, Damson Plum

Red: Clayman, Climax, Hungarian, Santa Rosa, Beauty

Yellow or green: Yellow Egg, Wickson, Kelsey, Green Gage

Season: Domestic supplies peak during the summer and early fall. Italian plums, late August to October. Imports are available year-round.

Packs: 28-lb (12.7-kg) cartons or lugs.

Counts: Vary according to variety.
Yellow Egg plums can be as large as 2 in (50 mm) across, Damson plums as small as cherries.

Weights: Large plums = 4 to 5 per lb (450 g).

Small plums = 8 to 10 per lb (450 g).

Kitchen yields: 5 to 6 medium plums = 2½ cups sliced.

Calories: 4 oz (112 g) pitted plums = 50 calories.

QUINCE

Quinces resemble yellow apples. These are acidic fruits with high pectin content, used primarily to thicken jams. There is a limited market for fresh quince in foodservice.

Season: September to December.

Packs: 20-lb (9.1-kg) lugs. The fruit is often sold in units of 1 dozen.

Average weight: 10 oz (280 g) to 12 oz (340 g).

Kitchen yields: Peeling and coring loss is about 25 percent.

One lug = 15 lb = (6.7 kg) peeled fruit.

Quince jam proportion: 1 lb (450 g) peeled and cored fruit blended with 12 oz (340 g) sugar.

RASPBERRIES (BUSHBERRIES)

For Canned and Frozen Raspberries, see the Groceries section.

Season: June until November for domestically grown raspberries. Imports are available during the rest of the year.

Packs: Flats of 12 half-pints (0.24 l).

Kitchen yields: There is basically no waste when the fruit is fresh.

Serving sizes: 1 half-pint = 1⅓ servings, breakfast or dessert.

1 flat = 16 servings.

1 half-pint = 3 servings, garnish with desserts and ice cream.

1 flat = 32 servings.

1 flat = 60 to 70 servings with ice cream for a banquet.

There are about 60 to 75 berries in each half-pint.

Calories: 1 cup, 5½ oz (150 g) = 55 calories.

RHUBARB

Although technically a vegetable, rhubarb is usually referred to as fruit. For Frozen Rhubarb, *see* the Groceries section.

Season: Spring to early summer and late fall. Hothouse rhubarb is often available from January to April.

Packs: 5-lb (2.2-kg) cartons, hothouse rhubarb.

15-lb (6.7-kg) and 20-lb (9-kg) cartons, regular.

Serving size: ½ cup, 5 oz (140 g), cooked with sugar.

Sugar ratio: 1 lb (450 g) cleaned rhubarb to 8 oz (225 g) sugar.

Kitchen yields: Cleaning waste is 10 percent by weight. Hothouse-grown rhubarb may have slightly less waste. 5 lb (2.2 kg) as purchased = 17 servings.

Note: Cooked weight is the same as raw weight because of the added sugar.

Calories: ½ cup, 5 oz (140 g), cooked with sugar = 190 calories.

SHAMOUTI

Also known as *Jaffa orange*, shamouti is now grown in limited quantities in the United States.

Season: Domestic production in fall; imports in winter.

SHARON FRUIT

This fruit, which resembles a persimmon, is imported from Israel. It is seedless and can be eaten raw. *See* Persimmons in the Tropical Fruits section.

Season: Winter.

STRAWBERRIES

Domestic strawberries are available year-round. Size and shape do not determine sweetness. *Fraise des bois* is the French name for imported or domestic wild strawberries. The production is small and seldom reaches the wholesale market. For

Canned and Frozen Strawberries, *see* the Groceries section.

Season: Available year-round. Peak occurs from March through June. Worst quality is in December. Imported strawberries come on the market in the winter.

Packs: 1-pint (0.47-l) baskets, 12 per tray. 1-qt (0.9-l) baskets.

Loose packs are also on the market.

Weights: 1 pint = 14 oz (400 g).
1 tray = 10 lb (4.5 kg) of fruit.
1 qt = 1¾ lb (780 g) of fruit.

Counts: 1 pint (0.47 l) = 12 to 14 large berries or 20 to 25 smaller berries. Normally, the larger berries are packed on top.

1 tray = 6 qt (5.6 l), about 150 large or 240 to 300 small berries.

Kitchen yields: Cleaning loss is 10 percent by weight; percentage varies by season and size of berries.

1 cup = 5 oz whole berries, cleaned.

Calories: 1 cup, 5.3 oz (150 g) = 55 calories.

EXTRA-LARGE STRAWBERRIES WITH STEMS

Sometimes called *Marshall strawberries*, suitable for dipping in sugar or chocolate.

Season: Available by special order. Difficult to get in late fall and early winter.

Packs: 1-pint (0.47-l) baskets, 12 per tray.

Count: 1 pint = about 8 large berries.

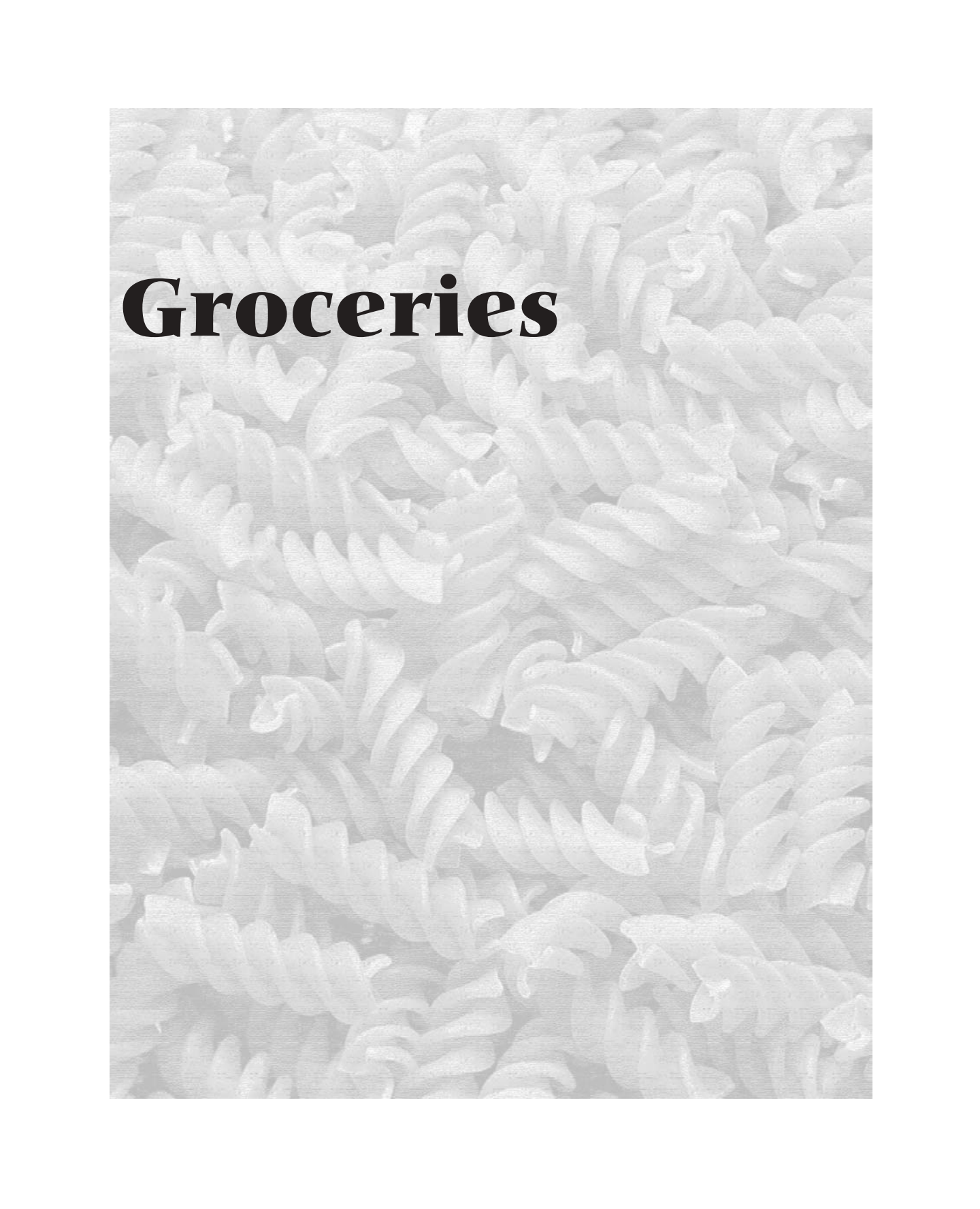
UGLY FRUIT

Juicy citrus hybrid with spotted, loose-looking skin; it should be orange-colored when mature.

Season: Winter.

Packs: By count.

Kitchen yields: This fruit is slightly smaller than grapefruit, with a thick skin.



Groceries

Listed in this section are items usually stored in the dry storeroom in a foodservice establishment. Frozen products are also listed because they too are customarily stored in and issued from the storeroom. For preserved fish, see the Fish, Mollusks, Shellfish, and Seafood section.

AGAR AGAR

Edible red seaweed in the *gelidium* family, processed into a vegetable gelatin. Also called *Japanese gelatin*, it is a natural thickening agent used in jams and ice cream. Agar agar is normally sold in thin, brittle sticks or in powder form. It will dissolve when boiled.

Packs: 1-lb (450-g) packages.
2.2-lb (1-kg) flakes.

Size: ¼-oz (7-g) sticks.

Kitchen yields: Thickening power is about 5 times greater than that of plain gelatin. It will set at room temperature and is not affected by acidity to the same extent as gelatin.

Agar agar thickening power: ½ oz (14 g) for 20 cups, or 5 qt (4.7 l).

ALMOND PRODUCTS

Almonds used in foodservice are normally purchased shelled. They are available with skin on, blanched (skin removed), and as flour. The major varieties are California, Carmel, Nonpareil, and Mission. Almonds are also available salted and with smoke flavor added.

Sizes and trade terms:

California: 18 to 40 kernels per oz (28 g)

Whole, skin on

Crosscut, skin on

Halves, skin on

Splits, skin on

Cube cuts, skin on

Diced, skin on, available in 2 sizes

Diced, buttered and roasted, skin on

Carmel: 18 to 40 kernels per oz (28 g)

Whole, skin on

Diced, skin on

Chopped pieces, skin on available in 3 sizes

Blanched, fine dice

Blanched, slivered in 2 sizes

Mission: 20 to 40 kernels per oz (28 g)

Whole, skin on

Whole and broken, skin on

Pieces, skin on

Slices, skin on in 3 sizes

Powder, all white

Nonpareil: 20 to 40 kernels per oz (28 g)

Whole, skin on

Whole, blanched

Blanched, whole and broken

Blanched splits

Blanched sliced

Blanched slivers and diced, available in 2 sizes

Packs: 5-lb (2.2-kg) cans.

10-lb (4.5-kg) cartons.

25-lb (11.2-kg) cartons of listed varieties.

30-lb (13.5-kg) case of granulated, bleached, or natural almonds.

Kitchen yields: 4½ oz (126 g) = 1 cup chopped almonds.

4 oz (112 g) = 1 cup slivered almonds.

1 lb (450 g) in shell = 2 cups nut meat.

1 lb (450 g) nut meat = 3⅔ cups.

Calories: 1 cup (0.24 l), 4.6 oz, (130 g), chopped = 775 calories.

1 cup (0.24 l) 4 oz, (112 g), slivered = 690 calories.

ALMOND PASTE

Almond paste is available with either 60 percent almond, with skin or with 60 percent blanched almonds; the rest is sugar. Almond paste made with blanched almonds is whiter than that made with unblanched almonds. Sweet almond

paste, called *marzipan* in German, is made with $\frac{2}{3}$ blanched almonds and $\frac{1}{3}$ sugar. Marzipan must be heated during fabrication, which distinguishes it from almond paste. The fat content must be at least 28 percent.

Packs: #10 cans = 104 oz (2.9 kg) each, per case.

Weight: Six #10 cans = 39 lb (17.5 kg).

Kitchen yields: Depend on use.

Calories: 1 oz (28 g) marzipan = 45 calories.

ALMONDS, SALTED

Snack item available in packs of many different sizes; plain, salted, and smoked.

Packs: Case of 200 $\frac{1}{2}$ -oz (14-g) packets. 5-lb (2.2-kg) cans.

Other packs are available.

Calories: $\frac{1}{2}$ oz (14 g) = 84 calories.

AMARANTH, GRAINS

Amaranth is a South American annual herb seed plant historically linked to ancient civilizations. It grows in temperate climates. The seeds have a slightly peppery flavor and a gelatinous consistency. Amaranth is high in protein but low in gluten. The flour can be combined with other flours and used in baking. For Amaranth Greens, *see* the Vegetables section.

Pack: By weight.

Kitchen yields: 1 cup grains simmered with 3 cups, 24 oz (0.72 l) water = 2.5 cups (0.6 l) cooked.
amaranth flour, $\frac{1}{2}$ cup (0.12 l) = 1.1 oz (30 g).

Calories: $\frac{1}{2}$ cup (0.12 l) Amaranth Flour = 110 calories.

ANISE

Bulb-shaped vegetable with fine, bushy leaves. Anise has a slight licorice flavor. It is sometimes confused with the vegetable fennel, but fennel does not form a bulb. For Fresh Anise, *see* the Vegetables

section. For Anise Seeds and Fennel Seeds, *see* the Herbs and Spices section.

Packs: 28-oz (800-g) cans, 12 cans per case.

Drained weight: 15 oz (425 g).

Serving size: One-half heart, $3\frac{1}{2}$ oz (100 g).

Kitchen yields: One 28-oz (800-g) can = 4 servings.

APPLE PRODUCTS

For Fresh Apple Varieties, *see* the Fruits section.

APPLE BUTTER

Concentrated dark brown fruit spread.

Pack: 17-oz (475-g) jars, 12 per case.

APPLE CIDER

Apple cider can be classified as follows:

Fresh, sweet apple cider: Cloudy, and with a limited shelf life.

Pasteurized apple cider: Clear or slightly cloudy, with long shelf life if unopened. It is almost undistinguishable from apple juice. Pasteurized apple cider is often substituted for apple juice.

Fermented sweet cider: Can contain up to 3 percent alcohol.

Hard cider: Fully fermented, an alcoholic beverage with 3 to 7 percent alcohol. It can be distilled into apple brandy.

Packs: Qt, 32 fl oz (0.9 l).

Half-gal, 64 fl oz (1.9 l).

Gal, 128 fl oz (3.8 l).

Serving size: 6 fl oz (0.17 l) or 8 fl oz (0.24 l).

Kitchen yields: 1 gal, 128 fl oz (3.8 l) = 21 servings of 6 fl oz each
1 gal, 128 fl oz (3.8 l) = 16 servings of 8 fl oz each

Calories: 1 cup (0.24 l) = 88 calories.

APPLE JUICE

Ready to serve; available canned, in glass bottles, and frozen concentrated.

Packs: 5½-fl oz (0.16-l) cans, 48 per case.
 10-fl oz (0.29-l) glass bottles, 24 per case.
 46-fl oz (1.3-l) cans, 12 per case.
 1-qt, 32-fl oz (0.9-l) glass bottles, 12 per case.
 64-fl oz (1.85-l) bottles, 6 per case.
 1-gal, 132-fl oz (3.8-l) glass or plastic bottles, 4 per case.

Serving size: 6 fl oz (0.17 l) is common.

Kitchen yields: One 46-fl oz (1.3-l) can = 7½ servings, 6 fl oz each.
 1-qt, 32-fl oz (0.9-l) glass bottle = 5.3 servings, 6 fl oz each.
 One 64-fl oz (1.8-l) bottle = 10½ servings, 6 fl oz each.
 1-gal, 128-fl oz (3.8-l) jug = 21 servings, 6 fl oz each.

Calories: 6 fl oz (0.17 l) = 66 calories.

APPLE JUICE, FROZEN CONCENTRATE

Packs: 32-fl oz (0.9-l) cans, 12 per case.
 46-fl oz (1.3-l) cans, 6 per case.

Serving size: 6 fl.oz (0.17 l).

Kitchen yields: Mix 1 part frozen juice with 3 parts water.
 One 32-fl oz (0.9-l) can = 1 gal (3.8 l) juice.
 1-gal (3.8-l) = 21 servings, 6 fl oz each.
 One 46-fl oz (1.3-l) can = 184 fl oz (5.3 l) juice.
 184-fl oz (5.3-l) juice = thirty 6 fl oz (0.17 l) servings.

APPLE SAUCE

Apple sauce is available sweetened or unsweetened, coarse or strained.

Packs: #10 cans, 6 to a case.
 #303 cans, 24 to a case.

Net weights: One #10 can = 104 oz (2.9 kg).

One #303 can = 15 oz (430 g).

Serving size: ½ cup, 4 oz (112 g).

Kitchen yields: One #10 can = 25 servings.
 One #303 can = 3½ servings.

Calories: ½ cup (117 g), unsweetened = 50 calories.
 ½ cup (112 g), sweetened = 115 calories.

BAKED APPLES

Canned Baked Apples

Pack: #10 cans, six to a carton.

Weight: One #10 can = 96 oz (2.6 kg).

Count: One #10 can contains 20 medium-sized apples. Other counts are available.

Kitchen yields: There is no waste.

FROZEN BAKED APPLES

Frozen baked apples are available ready to heat.

Packs: 6-lb (2.7-kg) boxes, 4 per case.

Size: 3¾ in (95 mm) across.

Weight: 8 oz (225 g) each. Other sizes are available.

Kitchen yields: 1 box, 6 lb (2.7 kg) = 12 count.
 1 case = 48 servings.

CANNED APPLE PIECES

Apple pieces are often used for pie and are also served in health care facilities. Apple pieces are available frozen as well.

Pack: #10 cans, six to a case.

Drained weights: One #10 can = 96 oz (2.6 kg) drained weight.
 1 case = 34.8 lb (15.7 kg) drained weight.

CANNED APPLE SLICES

Packs: #10 cans, 6 per case.
 #2½ cans, 24 to a case.

Drained weights: One #10 can = 96 oz (2.6 kg).
 1 case = 36 lb (16.2 kg).
 One #2½ can = 18 oz (500 g).
 1 case = 27 lb (12 kg).

CANNED SPICED APPLE RINGS

Apple rings are usually dyed dark red and will stain anything with which they

come in contact. They are popular as decorations on buffets and as a garnish with spare ribs.

Packs: #10 cans, six to a carton.

Count: One #10 can contains 85 to 90 rings.

Kitchen yields: There is no waste.

CRAB APPLES, WHOLE SPICED, CANNED

For Fresh Crab Apples, *see* the Fruits section.

Packs: #10 cans, six to a carton.

Kitchen yields: One #10 can contains 60 to 70 apples.

DEHYDRATED APPLES

Available as rings, wedges, and pie pieces.

Packs: By weight. Various packs are available.

Serving size: ½ cup (0.12 l) cooked.

Kitchen yields: 1 lb (450 g) = 4⅓ lb (1.9 kg) when cooked.

1 lb (450 g) = 14 servings cooked.

FROZEN APPLE PIECES

Peeled and sliced apples are available as Straight Pack and as Flavor Pack. Flavor Pack apples are mixed with sugar in the ratio of 7 parts apples to 1 part sugar (by weight). Frozen apples are frequently used for pie.

Packs: 40-lb (18.1-kg) can Straight Pack.

30-lb (13.5-kg) can Flavor Pack.

Serving size: ½ cup (0.12 l) stewed apples as dessert.

Kitchen yields: One 40-lb (18.1-kg) can Straight Pack = 32 pies, 9 in (225 mm) across.

One 30-lb (13.5-kg) can Flavor Pack = 110 servings as dessert.

APRICOT PRODUCTS

Listed here are Apricot Glaze, Apricot Nectar, Canned Apricots, Dried Apricots,

and Frozen Apricots. Fresh Apricots are listed in the Fruits section.

APRICOT GLAZE

Baking glaze used for Danish pastry and other products.

Packs: #10 cans, six to a carton.

Kitchen yields: One #10 can will glaze 300 large Danish pastries, brushed on.

APRICOT NECTAR

Sweetened canned apricot juice.

Packs: 5½-fl oz (0.17-l) cans, 48 per case.

46 fl-oz (1.3-l) cans, 12 per case.

Serving size: 5½ fl oz (0.17 l).

Kitchen yields: One 46-fl oz (1.3-l) can = 8 servings.

Calories: 5½ fl oz (0.17 l) = 100 calories.

CANNED APRICOTS

Available in many can sizes and counts. The most common institutional counts are listed here. Apricots are available whole (with pits) and in halves, in light and in heavy syrup.

Packs: #10 cans, 6 per case.

Drained weights: One #10 can = 62 oz (1.7 kg).

Counts per #10 can: Whole peeled, light syrup, 45 to 55.

Halves, light syrup, 86 to 130.

Halves, heavy syrup, 86 to 108.

Halves, heavy syrup, 108 to 130.

Halves heavy syrup, 150 to 175.

Serving size: 1 cup (0.24 l).

Calories: 1 cup (0.24 l) halves in heavy syrup = 220 calories.

1 cup (0.24 l) halves in light syrup = 160 calories.

DRIED APRICOTS

Apricot halves are normally purchased for foodservice; chopped pieces are available for baking. Apricot halves are often served cooked as a breakfast item for

buffets. Fancy grades are usually imported from the Middle East.

Packs: 1-lb (450-g) boxes, 24 to a case.
25-lb (11.2-kg) bags.

Sizes: #1 Jumbo, $1\frac{3}{8}$ -in diameter.
#6 Standard, $1\frac{3}{16}$ -in diameter.

Counts: 1 lb (450 g) = 75 to 90 Jumbo halves.

1 lb (450 g) = 96 Standard halves.

Serving size: 1 cup (0.24 l) cooked fruit.

Kitchen yields: One 1-lb (450-g) box = 10 servings.

Calories: 1 cup (0.24 l) cooked unsweetened fruit and liquid = 215 calories.

1 oz (28 g) not cooked = 60 calories.

FROZEN APRICOTS

Apricot halves packed in sugar are available. They are used as stewed fruits, in pie, or as Danish pastry filling.

Packs: 6½-lb (2.9-kg) cans, 6 per case.
25-lb (11.2-kg) cans.
30-lb (13.5-kg) cans.

Serving size: 1 cup (0.24 l), cooked.

Kitchen yields: One 30-lb (13.5-kg) can = 100 servings cooked fruit.

One 30-lb (13.5-kg) can = eighteen 9-in pies.

Calories: 1 cup (0.24 l), cooked = 220 calories.

ARROWROOT

Arrowroot, a starch derived from a tuber, is used as a thickener for sauces and as a baking ingredient. Sauces thickened with arrowroot are more transparent than those thickened with cornstarch.

Pack: 18-oz (500-g) boxes.

Kitchen yields: Arrowroot has slightly more thickening power than cornstarch.

ARTICHOKE PRODUCTS

Canned and frozen artichokes are available, as well as marinated bulb

artichokes, also called *globe artichokes*, which are ready to use. For Fresh Artichokes, see the Vegetables section.

ARTICHOKE HEARTS, MARINATED

Ready-to-serve product, available imported and from domestic producers.

Packs: ½-gal (1.9-l) jars, 6 jars per case.
1-gal (3.8-l) jars, 4 jars per case.

Many other packs are also available.

Count: One ½-gal (1.9-l) jar = 85 to 105 pieces.

Serving size: 4 to 5 pieces as appetizer portion.

Kitchen yields: One ½-gal (1.9 l) jar = 20 to 22 servings.

One 1-gal (3.8-l) jar = 40 to 45 servings.

CANNED ARTICHOKE

Available are bottoms, hearts, and pieces in brine.

Packs:

Bottoms: 14-oz (400-g) cans, 24 per carton.

Hearts: 14-oz (400-g) cans, 24 per carton.

Hearts: 6.6-lb (3-kg) cans, 6 per carton.

Pieces: Six #10 cans, 6 cans per carton. Other packs are available.

Counts:

Bottoms: One 14-oz (400-g) can yields 5 to 7 or 6 to 9 or 8 to 10.

Hearts: One 14-oz (400-g) can yields 8 to 10 or 14 to 16.

One 6.6-lb (3-kg) can yields 30 to 40.

One #10 can yields 90 to 100 or 110 to 120.

Pieces: One #10 can yields 180 to 200.

Serving sizes: 1 filled bottom as garnish.
4 to 5 pieces (120-count) of hearts as appetizer.

Kitchen yields: One #10 can of 120-count hearts = 24 to 26 servings.

FROZEN ARTICHOKE

Frozen artichokes are available as hearts and quarters.

Packs: 2-lb (900-g) boxes, 12 boxes per case.

Sizes: Vary; there is no count.

Serving size: 3 to 4 pieces as vegetable garnish.

Kitchen yields: Product is fully cooked, and there is no waste.

ASPARAGUS PRODUCTS

For Fresh Asparagus, see the Vegetables section.

CANNED ASPARAGUS

Green asparagus and white asparagus are available in stalks, cuts, and bottom cuts. Canned green asparagus has lost importance in foodservice because fresh and frozen asparagus are readily available. Inexpensive white asparagus is imported from Asia. Jumbo-size white asparagus, imported from Europe, is considered a luxury item.

CANNED GREEN ASPARAGUS

Packs: #303 cans, 24 per case.
#2 cans, 24 per case.
#10 cans, 6 per case.

Drained weights: #303 can = 9 oz (250 g) to 10 oz (280 g).
#2 can = 11½ oz (320 g) to 13½ oz (380 g).
#10 can = 60 oz (1.7 kg) to 64 oz (1.8 kg).

Serving size: 4 spears, 3 oz (85 g).

Kitchen yields: One #303 can = 2½ servings.

One #2 can = 3 servings.

One #10 can = 16 servings.

Calories: 4 spears, 3 oz (85 g) = 17 calories.

CANNED WHITE ASPARAGUS, INEXPENSIVE

Good product, frequently used in buffets or as garnish.

Packs: 64½-oz (1.80-kg) cans, 6 per case.

Counts: 40 to 60, or 60 to 80, or 80 to 100 counts.

Serving size: 4 pieces, 40 to 60 count as appetizer.

Other serving sizes depend on operational needs.

Kitchen yields: One 64½-oz can, 40 to 60 count = 12 to 15 appetizer servings.

Calories: 4 pieces, 40 to 60 size = 15 calories.

FROZEN ASPARAGUS

Frozen asparagus is available as a frozen block or IQF, as spears in various sizes and as cut pieces.

Packs: 2½-lb (1.1-kg) boxes, 12 per case.

30-lb (13.50-kg) bulk pack.

Other packs are also available.

Counts: One 2½-lb box contains 51 Jumbo spears.

One 2½-lb box contains 80 Medium spears, randomly packed.

One 2½-lb box contains 1 in long, randomly packed pieces; about 18 percent are tips.

Serving sizes as side order:

3 spears Jumbo size.

4 spears Medium size.

Kitchen yields: One 2½-lb (1.1-kg) box, Jumbo spears = 14 to 16 servings.

One 2½-lb (1.1-kg) box, Medium spears = 18 servings.

One 2½-lb (1.1-kg) box, pieces = 12 servings.

Calories: 4 spears, Medium size = 15 calories.

Frozen Asparagus Equivalent to Fresh

Asparagus: One 2½-lb (1.1-kg) box frozen = 5 lb (2.2 kg) to 5½ lb (2.4 kg) fresh asparagus as purchased.

AVOCADO PRODUCTS

For Fresh Avocados, see the Vegetables section. Frozen processed avocado is available in many forms and flavors, such as plain, dip, sauce, guacamole,

hot'n spicy guacamole, and Western-style guacamole.

Packs: 1-lb (450-g) cans, 12 per case.
 #10 cans, six per case.
 25-lb (11.2-kg) tubs.
 1-oz (28-g) guacamole packets, usually 100 per carton.
 ½-gal (1.8-l) pack avocado sauce, 8 per case.

Weight: One #10 can = 104 oz (2.9 kg) net weight.

Serving sizes: 2 oz (56 g) per sandwich or as topping.
 4 oz (112 g) as dip.

Kitchen yields: One 1-lb (450-g) can = 8 portions topping or 4 servings dip.
 One #10 can = 50 portions topping or 24 servings dip.

Calories: Sauce, 2 oz (56 g) = 80 calories.
 Pulp, 2 oz (56 g) = 100 calories.
 Guacamole, 4 oz (112 g) = 240 calories.

BAKING POWDER

Leavening agent consisting of bicarbonate of soda (baking soda) and an acid that activates the bicarbonate of soda and generates carbon dioxide gas, which then leavens the baked product. Kitchen yields depend on applications.

Pack: By weight.

BAKING SODA (BICARBONATE OF SODA)

White powder with the chemical formula NaHCO_3 . It neutralizes acidity and is an ingredient in baking powder. It is used as leavening agent in cakes and batters with a high acid content, such as gingerbread and fruit cakes.

Packs: 8-oz (225-g) packages.
 1-lb (450-g) packages.

Kitchen yields: Use as directed..

Calories: None.

BALSAMIC VINEGAR

See Vinegars in this section.

BAMBOO SHOOTS, CANNED

Fresh bamboo shoots are occasionally available at ethnic markets. The canned product is marketed in slices and whole. Dried bamboo shoots are available in Asian ethnic markets.

Packs: 15-oz (425-g) cans, 48 per case.
 19-oz (540-g) cans, 48 per case.
 #2½ cans, 24 per case.
 #10 cans, 6 per case.

Drained weights:

One 15-oz can = 8.5 oz (240 g).
 One 19-oz (540-g) can = 10.5 oz (300 g).
 One #2½ can = 18 oz (500 g).
 One #10 can = 65 oz (1.8 kg).

Kitchen yields: One #2½ can = about 3 cups, sliced.
 One #10 can = about 11 cups, sliced.

BANANA PRODUCTS

For Fresh Bananas, see the Fruits section.

BANANA CHIPS

Available as toasted chips with various flavors for snacks.

Packs: By weight.

Kitchen yields: There is no waste.

BANANAS, PROCESSED PULP

Available in polyester bags, cans, and drums. Banana pulp is used in baking and in ice cream.

Packs: #10 cans, 6 per case.
 5-gal (19-l) bags.
 55-gal (208-l) drums.

Kitchen yields: One #10 can = 13 cups (3 l).
 One #10 can = 6½ lb (3 kg).
 One #10 can = pulp of 24 medium-sized bananas.
 One case #10 cans = 39 lb (17.5 kg) medium bananas.
 One case #10 cans = the pulp of 144 medium bananas.

One 5-gal (19-l) bag = 40 lb (18.1 kg), or the pulp of 145 bananas.

Calories: 4 oz (112 g) = 100 calories.

BARLEY

Dried barley is used in soups and stews. It is best purchased in 1-lb (450-g) boxes.

Packs: 1-lb (450-g) boxes, 24 per case.
Larger packs are available.

Weight: 1 cup, uncooked = 7 oz (200 g).

Serving size: Varies; barley is usually a recipe component.

Kitchen yields: 1 cup, uncooked = garnish for 1¼ gal (4.7 l) soup.

Calories: 1 cup, 7 oz (200 g), uncooked = 700 calories.

BEANS

Listed here are canned beans, dried beans, and frozen beans. For Fresh Beans, see the Vegetables section.

CANNED BEANS

The calorie count of plain cooked beans of any kind, for ½ cup (0.12 l), is about 110.

ADUKI BEANS

See Japanese Foods in the International Ingredients section.

BLACK BEANS

Packs: 15-oz (420-g) cans, 24 to a case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 15-oz (420-g) can = 4 servings.

BLACK-EYED PEAS

Packs: 15-oz (420-g) cans, 24 to a case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 15-oz (420-g) can = 4 servings.

BLACK SOY BEANS

Packs: 15-oz (420-g) cans, 24 to a case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 15-oz (420-g) can = 4 servings.

CANNELLINI

Italian-type white kidney beans.

Packs: 15-oz (420-g) cans, 24 to a case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 15-oz (420-g) can = 4 servings.

FLAGEOLET BEANS

Canned flageolets are available imported from Europe; they resemble small lima beans. They are green when picked immature, and white when mature.

Packs: 15-oz (420-g) cans, 24 to a case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 15-oz (420-g) can = 4 servings.

GARBANZO BEANS

Also called *chick peas* or *ceci*. Garbanzos are often classified as peas. They are a popular component of salad bars and the main ingredient of the Middle Eastern dip hummus.

Packs: #10 cans, 6 per carton.
19-oz (540-g) cans.

Drained weights: One #10 can = 72 oz (2 kg).

Serving size: ½ cup (0.12 l).

Kitchen yields: One #10 can = 22 servings.

Calories: ½ cup = 110 calories.

GREEN BEANS

The most suitable variety for canning is Blue Lake. Canned beans are available whole, cut across, and cut lengthwise, referred to as French-style. Green bean purée is also available.

Packs: #10 cans for most varieties, 6 per case.
#303 (15.6-oz) cans, green bean purée, 24 to a carton.

Sizes: Extra Large, Large, Medium, and Small for both whole and cut beans. French-style beans are not graded by size.

Drained weights: One #10 can = 61 oz (1.7 kg) to 66 oz (1.8 kg).

Serving size: ½ cups (0.12 l).

Kitchen yields: One #10 can = 25 servings.

Calories: ½ cup (0.12 l) = 15 calories.

KIDNEY BEANS

Red kidney beans are often identified with Mexican dishes and with chili con carne. They are also used in salads. Light and dark red beans are available. A mixture of three types of beans as salad is also available. Pinto beans belong to the broad group of kidney beans.

Packs: #10 cans, 6 per carton. Other packs are available.

Drained weights: One #10 can = about 80 oz (2.2 kg).

Serving size: ½ cup (0.12 l).

Kitchen yields: One #10 can = 17 servings.

Calories: ½ cup (0.12 l) = 115 calories.

KIDNEY BEANS, REFRIED (FRIJOLES REFritos)

See Hispanic Foods in the International Ingredients section.

LIMA BEANS

Packs: #10 cans, 6 per case. Other packs available.

Sizes: Tiny, Small, Medium, and Large.

Drained weights: One #10 can = 70 oz (2 kg), drained.

Serving size: ½ cup (0.12 l).

Kitchen yields: One #10 can = 20 servings.

Calories: ½ cup = 130 calories.

NAVY BEANS

Small white beans, available “baked” with meat such as pork or frankfurters, or without meat. The flavor choices are molasses and tomato sauce. Plain navy beans, flavored only with salt, are also available. They are seldom purchased

because they can easily be cooked in-house.

Packs: #10 cans, 6 per carton. Other packs are available.

Drained weights: One #10 can = 70 oz (2 kg) drained.

Serving size: ½ cup (0.12 l).

Kitchen yields: One #10 can = 20 servings.

Calories: ½ cup (0.12 l), plain = 130 calories.

½ cup (0.12 l), baked with meat and sweet sauce = 200 calories.

½ cup, baked with pork and tomato sauce = 155 calories.

PINTO BEANS

Packs: #10 cans, 6 per carton. Other packs are available.

Serving size: ½ cup (0.12 l).

Kitchen yields: One #10 can = 20 servings.

Calories: ½ cup = 130 calories.

WAX BEANS

Wax beans are popular in salad bars because they do not discolor when blended with dressings. They are available whole and cut.

Packs: #10 cans, 6 per carton.

Drained weights: One #10 can = 62 oz (1.7 kg).

Serving size: ½ cup (0.12 l).

Kitchen yields: One #10 can = 25 servings.

Calories: ½ cup (0.12 l), without dressing = 15 calories.

DRIED BEANS

BLACK TURTLE BEANS

Black turtle beans are a staple in South America and Cuba, often served together with rice. The popular Black Bean Soup is made with black turtle beans.

Packs: 1-lb (450-g) boxes, 24 per case. 25-lb (11.2-kg) bags.

Serving size: ½ cup (0.12 l) cooked beans.

Kitchen yields: 1 lb = 2⅓ cups, dry.

1 cup, 7 oz (200 g) dried beans = 2½ cups (0.5 l) cooked.

1 lb (450 g) = 20 servings.

Calories: ½ cup (0.12 l) = 105 calories.

FAVA BEANS

Large flat, light brown, or green beans, good in salads and in stews. *See* Black Turtle Beans for average yield information.

GARBANZO BEANS

Also called *chick peas* or *ceci*. Garbanzos are often classified as peas. They are a popular component of salad bars and the main ingredient of the Middle Eastern dip hummus. *See* other Dried Beans for information.

KIDNEY BEANS

Popular in bean salads and in other dishes. In Mexican cooking, the beans are used as refried beans, called *Frijoles refritos*. *See* other Dried Beans for information.

KIDNEY BEANS, REFRIED (FRIJOLES REFritos, DRY MIX)

See Hispanic Foods in the International Ingredients section.

LIMA BEANS

Mature lima beans are white, kidney-shaped beans. *See* other Dried Beans for information.

MUNG BEANS

See Indian Foods in the International Ingredients section for information. For Fresh Bean Sprouts, *see* the Vegetables section.

NAVY BEANS

Small white beans, often called *Boston beans*. They are the traditional beans for Boston Baked Beans.

Packs: 1-lb (450-g) boxes, 24 boxes per carton.

25-lb (11.2-kg) bags.

Serving size: 1 cup (0.24 l) baked beans.

Kitchen yields: 1 lb = 2⅓ cups, dry.

1 cup, 7 oz (200 g), dried beans = 2½ cups (0.5 l) cooked.

1 lb (450 g) = 10 servings.

Calories: 1 cup cooked, drained, plain = 225 calories.

1 cup with pork and tomato sauce = 310 calories.

PINTO BEANS

Related to the pink beans of Arizona and New Mexico. Pinto beans are also used in Mexican cooking for refried beans, although red kidney beans are more common.

Packs: 1-lb (450-g) boxes, 24 per carton.

25-lb (11.2-kg) bags.

Serving size: ½ cup (0.12 l), as side dish.

Kitchen yields: 1 lb = 2⅓ cups, dry.

1 cup, 7 oz (200 g), dried beans = 2½ cups (0.5 l) cooked.

1 lb (450 g) = 20 servings.

Calories: ½ cup (0.12 l) = 105 calories.

FROZEN BEANS

BUTTER BEANS

Packs: 2½-lb (1.1-kg) boxes, 12 per case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 2½-lb box = 13 servings.

One 30-lb (13.5-kg) case = 120 servings.

Calories: ½ cup (0.12 l) = 130 calories.

GARBANZO BEANS (CHIC PEAS/CECI)

Packs: 2½-lb (1.1-kg) boxes, 12 per case.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 2½-lb box = 13 servings.

One 30-lb (13.5-kg) case = 120 servings.

Calories: ½ cup (0.12 l) = 100 calories.

GREEN BEANS (SNAP BEANS)

The Blue Lake variety is preferred for freezing. Most varieties are available frozen in block and IQF.

Frozen beans are available as:

Regular cut.
French cut (julienne).
Whole in various sizes.

Packs: 2½-lb (1.1-kg) boxes, 12 per case.
30-lb (13.5-kg) IQF, loose pack.

Serving size: ½ cup (0.12 l), about 4 oz cooked.

Kitchen yields: One 2½-lb (1.1-kg) box = 10 servings.

One 30-lb (13.5-kg) case = 120 servings.

Calories: ½ cup (0.12 l), cooked = 18 calories.

LIMA BEANS

Baby lima beans are small, have thin skins, and are bright green. Fordhook lima beans are larger and plumper, with a thicker skin, and have about one-fifth fewer calories than baby lima beans.

Packs: 2½-lb (1.1-kg) boxes, 12 per case.

Serving size: ½ cup, 3 oz (0.12 l) cup, cooked.

Kitchen yields: One 2½-lb box

Fordhooks = 13 servings.

One 2½-lb box baby limas = 12 servings.

Calories: ½ cup, 3 oz (0.12 l), cooked baby limas = 105 calories.

½ cup, 3 oz (0.12 l), cooked Fordhooks = 85 calories.

WAX BEANS (CUT)

Packs: 2½-lb (1.1-kg) boxes, 12 per case.

Serving size: ½ cup (0.12 l), about 4 oz cooked.

Kitchen yields: One 2½-lb (1.1-kg) box = 10 servings.

One 30-lb (13.5-kg) case = 120 servings.

Calories: ½ cup (0.12 l), cooked = 18 calories.

BEAN SPROUTS, CANNED

Grown from mung beans. For Fresh Sprouts, see the Vegetables section.

Packs: #10 cans, 6 to a carton.

Kitchen yields: 58 oz (1.6 kg) drained weight.

1 cup = 4 oz (112 g).

Calories: 1 cup, 4 oz (112 g) = 35 calories.

BEETS, CANNED

Canned beets are available whole, diced, sliced, and shoestring-shaped in various sizes. For Fresh Beets, see the Vegetables section.

Packs: #10 cans for all varieties, 6 per carton.

Serving size: ½ cup (0.12 l) diced or sliced beets.

4 to 5 small whole beets.

Counts, whole beets: One #10 can = 250 to 300 very tiny.

= 175 to 250 tiny.

= 174 to 125 small.

= 75 to 124 medium.

= fewer than 74 large.

Counts vary slightly.

Kitchen yields: One #10 can = 23 servings diced beets.

One #10 can = 22 servings sliced beets.

One #10 can = 30 to 35 servings whole beets.

Calories, drained: ½ cup (0.12 l) whole, small beets = 30 calories.

½ cup (0.12 l) diced or sliced beets = 35 calories.

BELGIAN ENDIVE, CANNED

The product is usually imported from Europe. The vegetable is fully cooked and can be pan-fried, served with a suitable meat sauce, or sprinkled with Parmesan cheese and baked.

Packs: 15-oz (425-g) net weight can, 24 cans per carton.

Serving size: 1 piece as hot vegetable.

Kitchen yields: 1 can = 3 to 4 servings.

BISCUIT MIXES

See Pancake Mixes.

BLUEBERRIES

Listed here are canned, dried, and IQF blueberries. For Fresh Blueberries, *see* the Fruits section.

CANNED BLUEBERRIES

Packs: #10 cans, six to a carton.
#300 cans, 13.5 fl. oz (0.4 l), 24 cans per carton.
Other packs are available.

Serving size: ½ cup (0.12 l) fruit and juice.

Kitchen yields: 1 #10 can = 8 cups drained fruit.
1 #10 can = 22 servings fruit and juice.

DRIED BLUEBERRIES

Pack: By weight.

FROZEN BLUEBERRIES

Frozen blueberries are available IQF (individually quick-frozen) and can be sprinkled over fruit salads and fruit cups.

Packs: 1-lb (450-g) polybags, 12 to a case.
3-lb (1.4-kg) polybags, 6 to a case.
25-lb (11.2-kg) can.
30-lb (13.5-kg) carton.

Serving size: ½ cup (0.12 l).

Kitchen yields: 1 lb (450 g) = 3½ servings.
25-lb (11.2-kg) can = 90 servings.

BRAZIL NUTS, SHELLED

Packs: By weight, often 5-lb (2.2-kg) cans.

Kitchen yields: 1 lb (450 g) = ¾ cups (0.75 l).

Calories: 1 oz (28 g), 6 to 8 large kernels = 185 calories.

BREAD CRUMBS

Bread crumbs can be classified as soft crumbs and dry crumbs. Soft crumbs are usually made in-house from white sandwich loaves, crusts removed. They must be stored refrigerated because they can

become moldy. Dry crumbs are made from dried white rolls or other white bread. The crust is usually not removed. Dry bread crumbs are available with different flavor components added. Their weights are basically the same as those of dry bread crumbs. The yield is according to use. For Japanese Bread Crumbs, *see* Panko under Japanese Foods in the International Ingredients section.

Packs: By weight.

Weight: 1 cup (0.24 l) = 3½ oz (100 g) dry bread crumbs.
1 cup (0.24 l) = 1.6 oz (45 g) soft bread crumbs.

Calories: 1 cup, 3½ oz (100 g), dry = 390 calories.
1 cup, 1.6 oz (45 g), soft = 120 calories.

BROCCOLI, FROZEN

Frozen broccoli is available as spears or as cut and chopped pieces. For Fresh Broccoli, *see* the Vegetables section.

Packs: 2½-lb (1.1-kg) boxes, 12 per case.
30-lb (13.5-kg) case, loose.

Serving size: ½ cup (0.12 l) cooked chopped pieces.

Kitchen yields: One 2½-lb (1.1-kg) box = 10 servings.
One case = 120 servings.

Calories: ½ cup, 2.6 oz (90 g), chopped pieces = 50 calories.
1 stalk, 1 oz (28 g) = 9 calories.

BROWN RICE

See Rice.

BRUSSELS SPROUTS

Canned brussels sprouts are seldom used in foodservice and are not listed here. Frozen brussels sprouts are graded by size. The smaller sprouts are more expensive. Fresh Brussels Sprouts are listed in the Vegetables section.

Packs: 2-lb (900-g) packages, 12 per case.
 2½-lb (1.1-kg) packages, 12 per case.
 30-lb (13.5-kg) IQF case.

Conversion: 3½ pints fresh brussels sprouts = 2 lb (900 g) frozen sprouts.

Sizes: Medium = 60 to 80 pieces per pound (450 g).
 Small = 100+ pieces per pound (450 g).

Serving size: ½ cup (0.12 l).

Kitchen yields: One 2½-lb (1.1-kg) package = 13 servings.
 One 2-lb (900-g) package = 10 servings.
 One 30-lb (13.5-kg) IQF case = 150 servings.

Calories: ½ cup: 3 oz (85 g), cooked, drained = 25 calories.

BULGUR (BULGHUR)

Wheat preserved by an ancient method, dating from the time of Assyria. The wheat is cleaned and sieved, boiled, and then dried. Bulgur is ready to eat after soaking.

Packs: 1-lb (450-g) boxes and larger packs.

Weight: 2½ oz (70 g), dry = ½ cup (0.12 l).

Serving size: ½ cup (0.12 l).

Kitchen yields: Bulgur will expand little, because it is precooked.

Calories: ½ cup (0.12 l) = 125 calories.

CABBAGE, RED, CANNED

For Fresh Red Cabbage, see the Vegetables section.

Pack: #10 cans, 6 per carton.

Serving size: ½ cup (0.12 l), 75 g.

Kitchen yields: 1 #10 can = 28 servings.

Calories: ½ cup (0.12 l), 75 g, cooked, drained, without fat = 15 calories.

CAPER BERRIES

The fruits of the *capparis* plant. When the buds blossom, the pistils develop into a berry the size and color of a small green olive. Picked with the stems on, the fruits are pickled and served at tapas bars in Spain. Like capers, they do not stay fresh long, so they must be processed as soon as they are picked and are often packed in brine before further processing.

Packs: 32-oz (900-g) jars.

Other packs are available.

Kitchen yields: There is no waste.

CAPERS

Capers are the buds of the *capparis* plant, a shrub growing in Mediterranean regions, usually wild, in full sun and poor soil. They are available pickled in vinegar or preserved with salt. Capers are graded by size; the smallest are most expensive. Nonpareils (*nonpareilles* in Italian) are small and the preferred variety. All products are imported.

Packs: 1-gal (3.8-l) jars, 4 each to a carton.

32-oz (900-g) jars.

Other packs are available.

Kitchen yields: There is no waste.

CARROTS

CANNED CARROTS

Canned carrots are available whole, diced, sliced, and shoestring-style in various sizes; they may also be mixed with peas. Carrot purée is available as well. Whole carrots are often imported.

Packs: #10 cans of whole, diced, or sliced carrots; domestic product.

5-lb 10-oz (3-kg) cans, imported whole carrots.

#303 cans, 15.6-oz (440-g), carrot purée, 24 per carton.

Counts:

Domestic whole carrots:

Small = 180 to 225.

Tiny = 260 to 320.

Extra = 350 to 425.

Imported whole carrots:

Small = 180 to 220.

Tiny = 250 to 310.

Extra = 320 to 400.

Common foodservice count: 260 to 320.

Drained weights: One #10 can = 69 oz (1.9 kg) domestic whole carrots.

One can, 5 lb 10 oz = 62 oz (1.7 kg) imported whole carrots.

One #10 can = 72 oz (2 kg) diced or sliced carrots.

Serving sizes: 2 oz (56 g), about 6 whole small carrots.

½ cup (0.12 l) diced or sliced carrots.

Kitchen yields: One #10 can = 30 servings, whole carrots.

One case = 180 servings, whole carrots.

One #10 can = 25 servings, diced or sliced carrots.

Calories: ½ cup (0.12 l) diced or sliced carrots, drained = 25 calories.

CARROT PURÉE

Pack: #303 cans, 15½ oz, or 2 cups (0.47 l).

Serving size: ½ cup (0.12 l).

Kitchen yields: One #303 can = 4 servings.

Calories: ½ cup (0.12 l) = 40 calories.

FROZEN CARROTS

Available whole, petite whole, diced, sliced, and shoestring-style in various sizes. Whole carrots are IQF.

Packs: 2-lb (900-g) packages, 12 per case. 20-lb (9.1-kg) carton.

Serving size: ½ cup (0.12 l), 2¾ oz (77 g).

Kitchen yields: 4 lb (1.8 kg) fresh carrots = 1 package frozen carrots.

One 2-lb (900-g) package = 10 servings.

Calories: ½ cup (0.12 l), 2¾ oz (77 g) = 40 calories.

CASHEW NUTS

Heart-shaped tropical fruits with a single protruding kernel beneath each fruit. The nuts should be eaten only when roasted.

Cashew trees are copious in the Brazilian rain forests.

Pack: By weight.

Calories: 1 cup (0.24 l), about 5 oz (140 g) = 785 calories.

CAULIFLOWER, FROZEN

For Fresh Cauliflower, *see* the Vegetables section. Available in block and IQF as small florets, about 1 in (25 mm) to 2½ in (63 mm) in length.

Packs: 2-lb (900-g) package blocks, 12 per carton.

2-lb (900-g) bags IQF, 12 per carton.

30-lb (13.5-kg) boxes IQF.

Serving size: ½ cup (0.12 l), 2 oz (56 g).

Kitchen yields: One 2-lb (900-g) package = 5 lb (2.2 kg) to 6 lb (2.7 kg) fresh cauliflower.

Calories: ½ cup (0.12 l), 2 oz (56 g) = 15 calories.

CELERIAC (KNOB CELERY)

Large, round roots with white flesh. Celeraic can be used raw in salads or cooked as a vegetable. The imported canned vegetable is available in slices and shoestring-style. For Fresh Celeriac, *see* the Vegetables section.

Pack: 9-lb (4-kg) cans.

Other packs are available.

Serving size: Varies.

Drained weight: One 9-lb (4-kg) can = 4 lb 13 oz (2.1 kg) fresh.

CELERY

For Fresh Celery, *see* the Vegetables section.

CANNED CELERY

Canned celery is available in stalks or as hearts.

Packs: 28-oz (800-g) cans, 12 cans per case.

Drained weight: 15 oz (425 g).

Serving size: One-half heart, 3½ oz (100 g).

Kitchen yields: One 28-oz (800-g) can = 4 servings.

FROZEN CELERY

Available diced.

Pack: 20-lb (9.1-kg) boxes diced celery.

Kitchen yields: Use like precooked fresh celery. There is no waste.

CEREAL PRODUCTS

Many brands of dry cereals are available. Most cereals are available in both self-serve bowls and regular boxes. Bulk packs are also available for dispenser and kitchen use.

Packs: Cases of 70 and 96 individual boxes each. Weights vary by product.

Cases of 4 bags. Weight varies by product.

Kitchen yields: Milk per serving = $\frac{1}{2}$ pint (0.24 l).

100 servings = 6 gal and 1 qt (23.5 l).

Calories: Bran flakes, sugar added, 1 cup, 1.25 oz (35 g) = 105 calories.

Bran flakes, raisins and sugar added, 1 cup, 2 oz (56 g) = 145 calories.

Corn flakes, sugar added, 1 cup, 1 oz (28 g) = 95 calories.

Corn flakes, sugar-coated, 1 cup, $1\frac{1}{2}$ oz (42 g) = 155 calories.

Oats, puffed, sugar added, 1 cup, 1 oz (28 g) = 100 calories.

Rice, puffed, sugar added, 1 cup, 1 oz (28 g) = 115 calories.

Wheat, puffed, sugar added, 1 cup, $1\frac{1}{2}$ oz (42 g) = 140 calories.

Note: Calorie counts vary according to product and serving size. Generally, there are between 40 and 110 calories per serving in an individual box or bowl. Refer to labels for information.

CORN FLAKE CRUMBS

Product is often used as filler in meat loaf or as breading.

Pack: Six 5-lb (2.2-kg) boxes per case.

CREAM OF WHEAT (FARINA)

Serving size: 1 cup (0.24 l), cooked.

Ratio: 1 lb (0.47 kg) requires about 4 qt, 2 cups (5 l) milk.

Kitchen yields: 1 lb (0.47 kg) = 18 servings.

Calories: 1 cup (0.24 l), enriched, cooked with milk = 120 calories.

HOMINY, CORN GRITS

Serving size: 1 cup (0.24 l), cooked.

Calories: 1 cup (0.24 l), cooked, enriched = 125 calories.

OATS, ROLLED

Serving size: 1 cup (0.24 l), cooked.

Calories: 1 cup (0.24 l), cooked = 130 calories.

CHERRY PRODUCTS AND VARIETIES

There is a distinct difference between sweet cherries and sour cherries. For Fresh Cherries, *see* the Fruits section.

CANDIED CHERRIES

Candied cherries are a component of mixed dried fruits used in baking.

Packs: 5-lb (2.3-kg) boxes.

Kitchen yields: Depending on recipe.

MARASCHINO CHERRIES

Preserved cherries, used as garnish in cocktails, available red and green, with and without stems. Maraschino cherries are always pitted. Their popularity as cake decoration has faded.

Packs: 1-gal (3.8-l) jars, 4 per case.

$\frac{1}{2}$ -gal (1.8-l) jars, 4 per case.

1-qt (0.47-l) jars, 6 per case.

Counts per gallon:

Without stems: Medium 600, Large 500, Extra Large 400.

With stems: Medium 450, Large 350, Extra Large 200.

SOUR OR TART CHERRIES

Canned and frozen sour cherries are marketed primarily as pie filling. Principal

varieties are Montmorency, Early Richmond, and English Morello.

CANNED SOUR CHERRIES

Available pitted and with pit, generally used for pie filling. Packed in syrup, in water, or as ready-made pie filling.

Packs: #10 cans, 6 per case.

#10 cans pie filling, 6 per case.

Kitchen yields: One #10 can pie filling = five 9-in (225-mm) pies.

Calories: ½ cup (0.12 l) fruit, pitted, including juice, water pack = 55 calories.

DRIED SOUR CHERRIES

Available pitted and with pits. The product is used in baking, in cereals, and as a snack.

Packs: 1-lb (450-g) boxes.

Serving size: ⅓ cup (45 g).

Kitchen yields: One 1-lb (450-g) box = 10 servings.

Calories: ⅓ cup (45 g), unpitted = 160 calories.

FROZEN SOUR CHERRIES

Available pitted, mixed with sugar at a ratio of 5 lb (2.2 kg) sugar to 25-lb (11.2-kg) cherries.

Pack: 30-lb (13.5-kg) cans.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 30-lb can = 110 servings.

One 30-lb can = 18 fillings for 9-in (225-mm) pies.

SWEET CHERRIES

CANNED SWEET CHERRIES

Canned sweet cherries are available pitted and with pits. There are two major varieties: Royal Anne cherries are light-colored, Bing cherries are dark-colored. Available in various syrup densities.

Packs: #10 cans, 6 per carton.

#2½ cans, 24 per carton.

#2 cans, 24 per carton.

Drained weights: One #10 can, 64 oz (1.8 kg) to 70 oz (2 kg).

One #2½ can, 18 oz (500 g).

One #2 can, 12½ oz (350 g).

Counts: #10 can: 210 to 235, 240 to 260, 290 to 335, 335 to 390, and 480 to 540.

#2½ can: 36 to 39, 80 to 90, 90 to 103, and 130 to 135.

#2 can: 35 to 40, 54 to 64, 64 to 75, and 90 to 95.

Serving sizes: ½ cup (0.12 l) fruit and juice as dessert.

2 oz (56 g) for dessert sauce.

Kitchen yields: One #10 can = 18 servings fruit and juice as dessert.

One #10 can = 36 servings for dessert sauce.

Calories: ½ cup (0.12 l) fruit and juice as dessert, medium syrup = 105 calories.

FROZEN SWEET CHERRIES

Available pitted, IQF, and solid-pack vacuum pack. They are often used in baking but can also be used in fruit salads or as stewed fruits.

Packs: 40-lb (18.1-kg) pitted, IQF cherries, in box.

37-lb (16.6-kg) vacuum pack, solid block.

Kitchen yields: One 40-lb (18.1-kg) box = twenty-four 9-in (225-mm) pies.

CHESTNUT PRODUCTS

For Fresh Chestnuts, see the Fruits section.

CANNED CHESTNUTS

Available whole and as purée. Chestnut Purée can be either sweetened or natural.

Pack:

Whole chestnuts in brine: 15½-oz (440-g) cans, 24 per case.

1-lb 15-oz (870-g) cans, 12 per case.

Unsweetened purée: 15½-oz (440-g) cans, 24 per case.

1-lb 15-oz (870-g) cans, 12 per case.

Sweetened purée: 17-oz (470-g) cans, 24 per case.

Roasted: 18-oz (500-g) cans.

Kitchen yields: 15½-oz (440-g) can = 10 oz (283 g) drained weight, whole.
1-lb 15-oz (880-g) can = 20½ oz (580 g) drained weight, whole.

CHESTNUT FLOUR

Excellent product for making soups.

Pack: By weight.

DRIED WHOLE CHESTNUTS

Peeled chestnuts ready to use. Soaking will increase yield.

Packs: 25-lb (11.2-kg) bags.

Kitchen yields: Dried whole chestnuts will double in weight after soaking and boiling.

GLAZED CHESTNUTS

Whole candied chestnuts, often called *marrons glacés*. Use as candy or as garnish on desserts.

Packs: 5-oz (140-g) cans.
7¾-oz (220-g) cans.
14-oz (400-g) cans.

Counts: 5-oz (140-g) can = 7 count.
7¾-oz (220-g) can = 11 count.
14-oz (400-g) can = 20 count.

Kitchen yields: There is no waste.

CHICKEN STOCK, CANNED

Canned product frequently used in foodservice, 99 percent fat free. Available regular and with low sodium.

Packs: 46-oz (1.3-l) cans.

Calories: One 46-oz (1.3-l) can = 86 calories.
1 cup (0.24 l) = 15 calories.

CHOCOLATE PRODUCTS

Basic terms:

Baking chocolate: pure unsweetened chocolate.

Chocolate liquor: the basic paste after the cocoa beans have been ground.

Cocoa butter: the natural white fat in cocoa beans.

Cocoa powder: the dry residue after the fat is removed.

Couverture: French term for *coating chocolate* with high fat content.

Dark chocolate: blend of chocolate liquor with sugar, cocoa butter, and flavoring.

Milk chocolate: blend of chocolate liquor with sugar, cocoa butter, and milk solids.

BITTER CHOCOLATE (BAKING CHOCOLATE)

Packs: 5-lb (2.2-kg) and 10-lb (4.5-kg) blocks.

Calories: 1 oz (28 g) = 140 calories.

COCOA BUTTER

Packs: 5-lb (2.2-kg) and 10-lb (4.5-kg) blocks.

Calories: 1 oz (28 g) = 180 calories.

SWEET CHOCOLATE

Packs: 5-lb (2.2-kg) and 10-lb (4.5-kg) blocks.

Calories: 1 oz (28 g) = 145 calories.

COCONUT PRODUCTS

Coconuts are the fruit of the coconut palm native to the tropics. Consistent with their name, coconuts are actually nuts, but because of their size, they are rarely classified with nuts. Processed coconut products can become rancid. For Fresh Coconuts, see the Tropical Fruits section.

COCONUT JUICE, CANNED

Homogenized coconut liquid, with sugar added for kitchen and bar use.

Packs: 11.8-oz (0.35-l) bottles, 24 per case.

15-oz (0.4-l) cans, 24 per case.

32-oz (0.9-l) bottles, 12 per case.

57.5-oz (1.6-l) cans, 12 per case.

Calories: 11.8 oz (0.35 l) bottle = 135 calories.

COCONUT MILK, CANNED

Used in dessert sauces, custards, curries, and in baking.

Pack: 13.5-oz (0.40-l) cans, 24 per case.

Calories: 2 oz (0.58 l) = 90 calories.

COCONUT, SHREDDED, DRIED

Dried coconut shreds are available in short and medium lengths and as angel hair, usually sweetened. Unsweetened shredded coconut is also available.

Packs: 10-lb (4.5-kg), 25-lb (11.2-kg), and 50-lb (22.5-kg) bags. 1-lb (450-g) boxes.

Calories: 1 oz (28 g), sweetened = 145 calories.

COFFEE

All coffees are blends, even when a specific country or region is noted on the label. The proportion of coffees harvested from different farms, and from the years when the coffee was picked, is not specified. Because the information available to the foodservice operator is incomplete, the integrity and experience of the purveyor is the only guarantee of receiving consistent quality. The expected strength and flavor of coffee varies from region to region in the country. Coffee companies adjust their products according to market preferences. The flavor differences depend on the types of beans used in the blend, the degree of roasting, and the fineness of the grind. All coffees are available in the form of whole beans or ground to different degrees of fineness. The various pieces of coffee-making equipment require particular grinds.

DECAFFEINATED COFFEE

Available naturally decaffeinated by the water wash method, also called the Swiss method, or chemically decaffeinated.

Packs: Same as regular coffee.

Kitchen yields: Same as regular coffee.

ESPRESSO COFFEE

Dark-roasted and very finely ground coffee. Many coffee companies sell

individually packaged single-portion, ready-to-use coffee pods. Some operations purchase whole beans and grind them fresh for each serving. The grinding machines can become rancid unless frequently cleaned and sharpened. Contrary to general perception, espresso coffee has less caffeine than regular brewed coffee.

Packs: 1-lb (450-g) bags.

Many other packs are available.

Serving size: 3-oz (87-ml) cup and smaller.

Kitchen yields: 1 lb (450 g) = 30 to 35 cups.

FLAVORED COFFEES

After the roasting process the coffee beans are flavored with artificial flavors such as vanilla, almond, and many others.

FROZEN COFFEE CONCENTRATE

Concentrated coffee slurry for use in special dispensers. The coffee is blended with hot water when dispensed.

Packs: Vary according to dispensers, often in 1-gal (3.8-l) bags.

INSTANT COFFEE

Brewed coffee that has been dried and will dissolve instantly. The popularity of instant coffee has diminished, as customers have become more quality conscious. Instant coffee is often used in pastry shops as a flavor enhancer. A famous brand is Sanka.

Packs: 4 trays of 100 single-service packets.
1,000-piece random packs.
Other packs are available.

REGULAR COFFEE

Ground coffee is available in vacuum-sealed packages, premeasured for use with specific equipment.

Packs: 1-lb (450-g) packs.

Many other packs are available.

Serving size: 6-oz (0.17-l) cup.

Kitchen yields: 1 lb (450 g) coffee + 2¼ gal (8.5 l) water = 2 gal (7.6 l) coffee.

2 gal = 40 cups.

Small coffeemaker capacity = 2 qt (1.8 l) = 8 to 10 cups.

Coffee needed = 2 oz (56 g) for weak coffee.

= 3 oz (85 g) for strong coffee.

SPECIALTY COFFEES

Many specialty coffees have become available, commanding a much higher price than regular coffee. Shade-grown coffee, produced on small farms with hand labor, is considered to be of better quality and to have social and environmental benefits.

BLUE MOUNTAIN COFFEE

Grown in Jamaica. Production is limited, and the price is high.

KONA COFFEE

Grown in Hawaii. Production is limited, and the price is high.

COLLARD GREENS

For Fresh Collard Greens, *see* the Vegetables section.

CANNED COLLARD GREENS

Available chopped.

Packs: #10 cans, 6 per case.

Drained weights: One #10 can = 62 oz (1.7 kg).

Serving size: ½ cup (0.12 l) drained vegetable.

Kitchen yields: One #10 can = 20 servings.

Calories: ½ cup drained vegetable, without added fat = 25 calories.

FROZEN COLLARD GREENS

Available chopped.

Packs: 3-lb (1.3-kg) boxes, 12 per case.

Serving size: ½ cup (0.12 l) drained vegetable.

Kitchen yields: One 3-lb (1.3-kg) box = 12 servings.

Calories: ½ cup drained vegetable, without added fat. = 25 calories.

COOKIE DOUGH, FROZEN

Frozen cookie dough is available from many manufacturers.

Packs: Four 6-lb (2.7-kg) buckets to a carton. Many other packs are available.

Serving size: 1 oz (28 g), uncooked.

Kitchen yields: One 6-lb (2.7-kg) bucket = 96 cookies.

CORN AND CORN PRODUCTS

Listed here are canned, frozen, and vacuum-packed corn, corn starch, corn meal, and polenta. For Fresh Corn, *see* the Vegetables section.

CANNED CORN

Available as cut corn (whole corn), creamed corn, and whole baby corn. Also available as vacuum-packed cut corn.

Packs:

Cut or creamed corn: #10 cans, 6 per case. 17-oz (480-g) cans, 24 per case.

Whole baby corn: #10 cans, 6 per case. Other packs are also available.

Weights: One #10 can cut or creamed corn = 105 oz (2.9 kg).

One #10 can whole baby corn = 105 oz (2.9 kg).

Counts: One #10 can whole baby corn = 300 pieces. Counts vary.

Serving sizes: ½ cup, 4 oz (112 g) cut corn.

½ cup, 4½ oz (130 g) creamed corn.

6 pieces = 1 oz (28 g) whole baby corn.

Kitchen yields: One #10 can cut corn = 70 oz (2 kg) drained weight.

One #10 can whole baby corn = 53 oz (1.5 kg) drained weight.

One 17-oz (480-g) can = 10½ oz (300 g) drained weight.

One #10 can cut corn = 17 servings.

Calories: ½ cup, 4 oz (112 g) cut corn = 70 calories.

½ cup creamed corn = 105 calories.

CORN KERNELS, FROZEN

Pack: 2½-lb (1.1-kg) boxes cut corn, 12 per case.

20-lb (9.1-kg) carton.

4-lb (1.8-kg) boxes cream-style corn, 12 per case.

Serving sizes: ½ cup, 3 oz (85 g) cut corn.

½ cup, 4½ oz (128 g) cream-style corn.

Kitchen yields: One 2½-lb (1.1-kg) box cut corn = 12 servings.

1 case cut corn = 150 servings.

One 4-lb (1.8-kg) box cream-style corn = 14 servings.

1 case cream-style corn = 165 servings.

Calories: ½ cup, 3 oz (85 g) cut corn = 65 calories.

½ cup, 4½ oz (127 g) cream-style corn = 110 calories.

CORN MEAL, WHITE AND YELLOW

Available coarse and fine ground. Stone-ground corn meal is considered a better product because the stone-grinding process generates less heat than steel roller grinding and retains more nutrients.

Packs: By weight.

Weights: 1 lb (450 g) = 3 cups (0.7 l).

Serving size: ½ cup (0.12 l).

Kitchen yields: 1 cup (0.28 l) corn meal boiled with 1 qt (0.9 l) liquid.

Calories: 1 cup (0.28 l) dry = 122 calories.

½ cup cooked in water = 60 calories.

CORN ON THE COB, FROZEN

Available IQF, fully cooked.

Packs: 30-lb (13.5-kg) cases or slightly smaller.

Counts: 48 whole or 96 half ears.

Kitchen yields: Product is fully cooked and ready to use.

Calories: 1 Ear 5 in (125 mm), 8 oz (255 g) = 120 calories.

CORN STARCH

Pack: 1-lb (450-g) boxes, 24 per case.

Kitchen yields: Thickening power equivalent: 1 cup corn starch = 2 cups flour by weight.

CORN, VACUUM PACKED

Pack: 1-lb (450-g) bags and larger.

Serving size: ½ cup (0.12 l), ¾ oz (100 g).

Calories: ½ cup (0.12 l), ¾ oz, (100 g) = 70 calories.

POLENTA

Ready-to-use polenta is available roll-shaped, frozen.

Pack: By weight.

Serving size: 3 oz (85 g).

Kitchen yields: There is no waste. The slices needed are cut off and pan-fried.

POPCORN

Pack: By weight; varies greatly.

Kitchen yields: ¾ oz (100 g) = 3 cups (0.7 l) popped.

Calories: 3 cups (0.7 l), popped, including fat = 120 calories.

SUCCOTASH (CORN KERNELS WITH RED AND GREEN PEPPERS), FROZEN

Pack: 2½-lb (1.1-kg) boxes, 12 per case.

Serving size: ½ cup (0.6 l), 3 oz (85 g).

Kitchen yields: One 2½-lb (1.1-kg) package = 12 servings.

COUSCOUS

See Middle East and North African Foods in the International Ingredients section.

CRANBERRY PRODUCTS

Listed here are canned, dried, and frozen cranberries and cranberry juice. For Fresh Cranberries, see the Fruits section.

CRANBERRIES, DRIED

Used as a snack and as a baking ingredient.

Pack: By weight.

CRANBERRY JUICE

Available plain and mixed with other fruit juices. White cranberry juice, made with immature cranberries, has come on the market.

Packs: 6-oz (0.17-l) individual packs, 48 per case.

1-qt (0.9-l) glass jars, 12 per case.

1-gal (3.8-l) jugs, 4 per case.

Serving size: 6 oz (0.17 l).

Kitchen yields: 1 qt (0.9 l) = 5 servings.

1 gal (3.8 l) = 21 servings.

Calories: 6 oz (0.17 l) sweetened juice = 120 calories.

CRANBERRY SAUCE, CANNED

Available strained and with whole berries, plain or with different flavors added.

Packs: #10 cans for all varieties, 6 per case. Smaller packs are available.

Serving size: 5 oz (140 g) average serving with turkey.

Kitchen yields: One #10 can = 20 to 22 servings.

Four #10 cans = 100 servings for banquets.

Calories: 5 oz (140 g), sweetened, strained = 200 calories.

CURRENT PRODUCTS

Currents are aromatic berries, available in late summer, but seldom available fresh because most of the harvest is processed. They are a basic ingredient in cassis. For Fresh Currents, *see* the Fruits section.

BAR-LE-DUC

Black currant preserves, a specialty of the Department Meuse in France.

CASSIS

Black currant syrup, often with 10 percent or more alcohol content, used as an ingredient in bar drinks.

Pack: 34-fl oz (1-l) bottles.

RED CURRANT JELLY

Pack: A variety of packs are available.

Calories: 1 tb, $\frac{2}{3}$ oz (20 g) = 50 calories.

DATE PRODUCTS

Domestic and imports are available. For Fresh Dates, *see* the Fruits section.

DATES, DRIED**Available as:**

Whole dates, pitted and unpitted.

Pieces without pits.

Macerated dates, which are ground, chopped, and broken.

Slab dates, which are ground pieces in slab form.

Packs:

15-lb (6.7-kg) carton, whole pitted.

30-lb (13.4-kg) carton, pieces.

40-lb (18.1-kg) carton, macerated dates.

50-lb (22.5-kg) carton, slab-ground dates.

Sizes with pits:

Jumbo: 16 to 19 per pound.

Large: 20 to 23 per pound.

Fancy: minor blemishes, moist, possible skin separations.

Choice: some dryness, some skin separations.

Kitchen yields:

2½ cups whole dates = 1 lb (450 g).

1 cup chopped dates = 6½ oz (175 g).

10 pitted dates = 3 oz (85 g).

Calories: 1 cup chopped dates = 178 calories.

2 oz (56 g) pitted, about 6 to 7 dates = 120 calories.

ESCARGOTS

Available mostly canned and imported. A small amount of escargots are available frozen. Much of the product is imported from Asia and is sometimes repacked in France. Large snails are of the Asian *Achatina* variety; small snails are of the

Petit Gris variety. Petit Gris are cultivated in California. Clean, empty shells are sold separately.

Packs:

Giant snails: 4½-oz (126-g) cans, with 12 snails each.

18-oz (500-g) cans, with 60 snails each.

Extra large: 4½-oz (126-g) cans, with 18 snails each.

9-oz (250-g) cans, with 36 snails each.

18-oz (500-g) cans, with 72 snails each.

Very large: 4½-oz (126-g) cans, with 24 snails each.

9-oz (250-g) cans, with 48 snails each.

18-oz (500-g) cans, with 96 snails each.

Petit Gris: 7½-oz (212-g) cans, with 36 snails each.

Shells, extra large: 2 cartons with 864 each = 1,728 case.

Shells, very large: 2 cartons with 1,152 each = 2,304 case.

Other packs are available.

FARINA

See Cereal Products in this section.

FIDDLEHEADS

For Fresh Fiddleheads, see the Vegetables section.

CANNED FIDDLEHEADS

The supply is small; available mostly in retail packs.

FROZEN FIDDLEHEADS

There is only a small supply in retail packs. There is no waste.

FIG PRODUCTS

For Fresh Figs, see the Fruits section.

CANNED FIGS

Available whole, whole and split, and broken in syrup of various densities. Kadota is the variety most often used for canning.

Packs: #10 cans, 6 per case.

#2½ cans, 24 per case.

#2 cans, 24 per case.

#303 cans, 24 per case.

Drained weights:

#10 can = 63 oz (1.8 kg).

#2½ can = 18 oz (500 g).

#2 can = 12½ oz (350 g).

#303 = 10 oz (280 g).

Counts: One #10 can = 70 to 90 whole figs.

One #2½ can = 20 to 24 whole figs.

Serving size: 4 fruits.

Kitchen yields: One #10 can = 18 to 20 servings.

Calories: 4 fruits (130 g) with syrup = 100 calories.

DRIED FIGS

A number of varieties are on the market. Adriatic, Calmyrna, and Kadota figs are light to dark brown; Mission figs are dark purple to black. Turkish figs are often of the Smyrna variety. Dried figs are frequently sold on strings tied into wreaths.

Packs: By weight; available whole, loose, and chopped for baking.

Count: 1 lb (450 g) = 30 figs, average size.

Kitchen yields: 1 lb (450 g) dried figs = 2¼ lb (1 kg) cooked.

Calories: 1 oz (28 g), about 2 figs = 70 calories.

FILBERTS (HAZELNUTS)

The name *hazelnuts* is often used interchangeably with *filberts*, although there is a botanical difference; available shelled and in the shell. Filberts are almost always purchased shelled. The nuts are available in the following forms:

Bleached (whole kernels, skins removed)

Roasted whole kernels

Chopped or diced kernels

Ground kernels

Hazelnut flour

Salted whole kernels

Packs: By weight, often in 5-lb (2.2-kg) cans.

Kitchen yields: 4 oz (112 g) chopped, or about 80 kernels = 1 cup.

1 lb (450 g) in shell = $1\frac{1}{3}$ cup nut meat.

1 lb (450 g) nut meat = $3\frac{3}{4}$ cups.

Calories: 1 cup, 4 oz (112 g) = 730 calories.

FILLO DOUGH PRODUCTS

Ready-to-bake and prebaked items are available. Typical products include: spanakopita (spinach and cheese pie), tyropitya (blended cheese pie), baklava (honey and nut dessert), kataifi (walnut and spice nest). For ready-to-use Filo Dough, *see* North African Foods in the International Ingredients section.

Packs: Usually 1-lb (450-g) boxes.

FLOUR VARIETIES

All flour is sold by weight, ranging from 1-lb (450-g) bags to 50-lb (22.5-kg) bags. For large food operations it is advantageous to purchase flour in 50-lb bags for the pastry shop and in smaller bags for kitchen use; this eliminates the need to store large amounts in a hot kitchen. There is considerable weight difference between sifted flour and unsifted flour. In baking, flour should always be used by weight.

Types of Flour:

Buckwheat: *See also* Kasha, cracked buckwheat.

Hard wheat flours:

Straight or strong flour for hard rolls and pizza.

Patent flour, also referred to as *bread flour*, for bread.

First clear flour has high gluten content, used in rye bread.

Bran flour, used in muffins and specialty breads.

Whole wheat flour, used in specialty breads.

Soft wheat flours: Cake flour for cakes and biscuits. Pastry flour.

Rye flours: Flour with low gluten, available in different shades of color.

Rye flour blends: Flour blended with high-gluten flour for making rye bread.

All-purpose flour: Blend of wheat flours.

Self-rising flour: Flour blended with baking powder.

Measure:

All-purpose flour: 1 cup, sifted = about $4\frac{1}{2}$ oz (125 g).

Buckwheat flour: 1 cup, sifted = about $3\frac{1}{2}$ oz (100 g).

Calories for 1 cup, sifted:

All-purpose flour = 420 calories.

Buckwheat flour = 340 calories.

Cake flour = 350 calories.

Self-rising flour = 440 calories.

Whole wheat flour = 400 calories.

GRAHAM FLOUR

Coarse whole wheat flour.

Pack: By weight.

Kitchen yields: 4.5 oz (126 g) = 1 cup.

Calories: 1 cup, 4.5 oz (126 g) = 470 calories.

FOIE GRAS, CANNED

Foie gras is the French name for *goose* or *duckling liver*. There are a number of products on the market with distinct quality and price differences. For Fresh Foie Gras, *see* Duck Liver in the Meats and Meat Products section.

Terminology:

Terrine: Whole liver poached in a ceramic container and studded with truffles. Available in the decorated container in which the product was cooked, or canned. Terrines are considered the top-of-the-line products. The 1.5-oz (43-g) terrine is popular as individual servings in restaurants. There is no waste, no handling, and shelf life under refrigeration is unlimited.

Whole liver: Canned whole liver in natural shape, without truffle center.

Block: Whole liver with truffle center, mostly in tunnel-shaped cans.

Parfait: Puréed liver, often with truffle center, mostly in tunnel-shaped cans.

Paté: Large chunks of liver baked in crust and filled with aspic jelly. The name is also often used for products without crust.

Purée or Mousse: Liver product to which binders, such as eggs and gelatin, have been added.

Roulade: Purée or mousse in round cans, often with truffle center.

Liver paté: Canned liver product made of pork and other livers, to which a small percentage of foie gras has been added.

Packs: There are many packs, ranging from portion sizes to foodservice packs:

Terrine: Packed in ceramic containers, in many sizes, with net weights ranging from 7 oz (200 g) to 1.5 oz (43 g).

Whole Liver: 1 lb 12 oz, or 28 oz (800 g) net.

Block: Packed in tunnel-shaped can, 10 $\frac{7}{8}$ oz (309 g) net, 7 $\frac{7}{8}$ oz (200 g) net, 15 oz (640 g) net, and 5 oz (140 g) net.

Purée: 10 $\frac{7}{8}$ oz (300 g) net.

Roulade: Packed in tall, round tin, 11 $\frac{1}{4}$ oz (320 g) net in tall, round tin.

Liver paté: 34 $\frac{1}{2}$ oz (980 g) net, 14 oz (400 g) net, or 7 oz (200 g) net.

Serving size: 1 $\frac{1}{2}$ oz (43 g) (2 slices).

Kitchen yields: 15-oz (640-g) can = 20 to 22 large slices.

5-oz (140-g) can = 6 to 7 large slices.

FRUIT POWDERS

Natural fruit powders can be used to intensify flavor. The following flavors are on the market: apple, blueberry, cherry,

lemon, lime, orange, pineapple, raspberry, and strawberry.

Pack: 1-lb 2-oz (500-g) bags.

FRUIT JAM, BAKEPROOF

Apricot, Strawberry, and other fruit jams are usually available. Prune Paste is listed separately.

Pack: 29-lb 5-oz (16.7-kg) tubs.

FRUIT PRESERVES, JAMS, AND JELLIES

There are numerous brands and package sizes on the market. The average calorie counts for the basic brands are given in the following list. Fruit jelly is made from strained fruit juice, fruit preserves contain fruit particles, and fruit jams contain fruit purée. Products can be made from a single fruit or from a blend of different fruits. Inexpensive products have low fruit content and high sugar and pectin content.

Calories: Preserves, 1 tb (20 g) = 50 calories.

Jams, 1 tb (20 g) = 50 calories.

Preserves, low sugar, 1 tb (20 g) = 20 calories.

Jams, low sugar, 1 tb (20 g) = 20 calories.

FRUIT PURÉES

Frozen fruit purées, imported from Europe, can be used to make fruit ices, sherbets, and cake fillings. Most are imported from France and are often labeled in French. Usually available without added sugar or with 10 percent sugar added. Available products include the following:

French	English
<i>Abricot</i>	Apricot
<i>Ananas</i>	Pineapple
<i>Banane</i>	Banana
<i>Cassis</i>	Black currant
<i>Citron jaune</i>	Lemon
<i>Citron vert</i>	Lime
<i>Figue</i>	Fig
<i>Fraise</i>	Strawberry

<i>Fraise des bois</i>	Wild strawberry
<i>Framboise</i>	Raspberry
<i>Fruit de la passion</i>	Passion fruit
<i>Goyave</i>	Guava
<i>Griotte</i>	Morello cherry
<i>Groseille</i>	Red currant
<i>Kiwi avec pépins</i>	Kiwi with seeds
<i>Lichi</i>	Lychee
<i>Mandarin</i>	Mandarin
<i>Mangue Alphonso</i>	Mango
<i>Mirabelle</i>	Mirabelle plum
<i>Melon</i>	Melon
<i>Mûre sauvage</i>	Wild blackberry
<i>Myrtille sauvage</i>	Wild blueberry/ huckleberry
<i>Noix de coco</i>	Coconut
<i>Orange</i>	Orange
<i>Orange sanguine</i>	Blood orange
<i>Pamplemousse</i>	Grapefruit
<i>Papaye</i>	Papaya
<i>Pêche blanche</i>	White peach
<i>Pêche jaune</i>	Yellow peach
<i>Pêche de vigne</i>	Vineyard peach
<i>Poire William</i>	William pear
<i>Rhubarb</i>	Rhubarb

Packs: 35-oz (1-kg), containers, 6 per pack.

2.1-lb (1-kg) containers, 5 per case.

44-lb (20-kg) buckets.

GARLIC, DEHYDRATED

See the Herbs and Spices section.

GARNISHING PASTE

The once-popular solid, edible paste is made of eggs, starch, and other ingredients to resemble truffles in looks, but not in taste. Available black and white.

Packs: 7¾ oz (220-g) cans.

Kitchen yields: Depend on application.

Paste is solid at room temperature and can be sliced very thin on a slicing machine.

GELATIN PRODUCTS

Plain gelatin is available granulated and in sheets. The thickening power of gelatin is influenced by acidity; therefore, more gelatin must be used when the liquid is

acid. In hot weather, gelatin amounts should be increased by about 25 percent. It is always advisable to make a small sample to check firmness of product.

Packs: 1-lb (450-g) containers of granulated gelatin.

Envelopes of ¼ oz (7 g) each.

17½-oz (500-g), gelatin sheets, often imported.

Weights: 1 cup = 5½ oz (154 g).

Kitchen yields: Unflavored gelatin thickening power: 1 oz (28 g) (4 envelopes) to 4 cups liquid. Strength is influenced by acidity; highly acidic liquids require more gelatin.

Calories: 1 oz (28 g) = 100 calories.

FRUIT-FLAVORED GELATIN

Fruit-flavored gelatin desserts are available in regular, sweetened with sugar, and diet versions. Jell-O is a registered trademark of the General Foods Corporation. Follow package instructions.

Serving size: 1 cup (0.24 l).

Calories: 1 cup (0.24 l) regular = 140 calories.

MEAT-FLAVORED GELATIN

Often referred to as *aspic powder*, meat-flavored gelatin powder is blended with hot water to produce a clear meat-flavored jelly. The package carries precise instructions. For homemade aspic, see the Recipes section.

GINGER PRODUCTS

For Fresh Ginger Root, see the Fruits section.

GINGER BEER

Alcohol-free beverage with stronger flavor than ginger ale.

Pack: 12-oz (0.3-l) bottles.

GINGER CHEWS

Chewy candy.

Pack: By weight.

GINGER, CRUSHED

Puréed ginger.

Pack: 8.4-oz (235-g) jars.

GINGER, CRYSTALLIZED

Available diced for baking, slices for garnish, and as chips.

Packs: By weight.

GINGER HICKORY SAUCE

Pack: 12.7-oz (470-g) jars.

GINGER IN SYRUP

Available as small dice, as pulp, and in nugget size.

Pack: By weight.

GINGER JUICE

Naturally pressed juice that can be added to marinades.

Packs: 12.5-oz (0.36-l) bottles.
1-gal (3.8-l) bottles.

GINGER MARMALADE

Pack: 16.9-oz (470-g) jars.

GINGER, MINCED

Pack: 8.4-oz (235-g) jars.

GINGER, PICKLED FOR SUSHI

Pack: 9.1-oz (250-g) jars.
2.2-lb (1-kg) jars.

GOOSEBERRIES, CANNED

Green or red, sometimes slightly fuzzy, often tart, berries. Frozen gooseberries are occasionally available IQF. The canned product is usually imported. Packed in light syrup or water. For Fresh Gooseberries, *see* the Fruits section.

Packs: 12½-oz (355-g) cans or jars, 12 per case.
#10 cans, 6 per case (difficult to get).

Drained weights: One 12½-oz can = 9 oz (255 g).
One #10 can = 75 oz (2.1 kg).

GRAPEFRUIT PRODUCTS

For Fresh Grapefruit and Fresh Grapefruit Juice, *see* the Fruits section.

GRAPEFRUIT JUICE

Available canned, dehydrated, and as frozen concentrate.

GRAPEFRUIT JUICE, CANNED

Canned juice is available sweetened and unsweetened, pink, red, or white.

Packs: 46-oz (1.35-l) cans, 12 per case.
6-oz (0.17-l) cans, 48 per case.
1-qt (0.47-l) glass jars.
2-qt (0.9-l) containers.

Serving size: 6 oz (0.17 l).

Kitchen yields: One 46-oz (1.3-l) can = 7½ servings.

Calories: 6 oz (0.17 l), unsweetened = 70 calories.
6 oz (0.17 l), sweetened = 94 calories.

GRAPEFRUIT JUICE, DEHYDRATED CRYSTALS

Pack: By weight.

Serving size: 6 oz (0.17 l).

Kitchen yields: 1 lb (450 g) = 1 gal (3.9 l).

Calories: 6 oz (0.17 l) = 70 calories.

GRAPEFRUIT JUICE, FROZEN

Pack: 6-oz (330-g) cans, unsweetened, 48 per case.
46-oz (1.3-l) cans, 6 per case.

Kitchen yields: Dilute with 3 parts water by volume.

Calories: 6 oz (0.17 l) = 70 calories.

GRAPEFRUIT SECTIONS, CANNED, WITH SYRUP

Packs: #3 cylinder cans, 46 fl. oz (1.3 l), 12 per carton.
#300 cans, 13.5 fl oz (0.4 l), 24, 36 or 48 per carton.

Serving size: ½ cup, 4½ oz (130 g).

Kitchen yields: One #300 can = 3 servings.
One #3 cylinder can = 10 servings with syrup.

Calories: ½ cup, 4½ oz (130 g) = 91 calories.

GRAPEFRUIT SECTIONS, FRESH

Popular banquet item. The industry distinguishes between hot-peeled and cold-peeled; the latter is considered better quality.

Packs: 1-gal (3.8-l) jars, 4 per case.

Drained weight: 6 lb (2.7 kg).

Serving size: ½ cup (0.12 l) with juice.

Kitchen yields: 1-gal (3.8-l) jar = 28 to 30 servings.

GRAPE PRODUCTS

For Fresh Grapes, *see* the Fruits section.

CANNED GRAPES

Available in light and heavy syrup.

Packs: #10 cans, 6 per case.

#2½, 29 oz (0.84 l) cans, 24 per case.

Drained weights: 1 #10 can = 8 cups (1.8 l) drained fruit.

1 #2½ can = 2 cups (0.47 l) drained fruit.

Kitchen yields: Yield depends on application; grapes are often used as a garnish on dishes.

GRAPE JUICE, CANNED

Packs: 24-oz (0.7-l) cans, 12 per case. Other packs are available.

Serving size: 6 oz (0.17 l).

Kitchen yields: One 24-oz (0.7-l) can = 4 servings.

Calories: 6 oz (0.17 l) = 90 calories.

GRAPE JUICE, FROZEN CONCENTRATED

Packs: 42-oz (1.2-l) cans concentrate, 6 per case.

Serving size: 6 oz (0.17 l).

Kitchen yields: 1 part concentrate + 3 parts water.
One 42-oz (1.2-l) can = 126 oz (3.6 l) juice = 20 servings.

GRAPE LEAVES

See Middle Eastern/North African Foods in the International Ingredients section.

HEARTS OF PALM, CANNED

For Fresh Hearts of Palm, *see* Hispanic Foods in the International Ingredients section.

Packs: 28-oz (800-g) cans, 24 per case
14-oz (400-g) cans, 24 per case

Drained weights: 28-oz (800-g) can = 17¼ oz (490 g).

14-oz (400-g) can = 7¼ oz (205 g).

Serving size: 2 stalks for salad.

Kitchen yields: One 28-oz (800-g) can = 8 to 9 stalks.

One 14-oz (400-g) can = 5 to 6 stalks.

HEMP PRODUCTS

Hemp products imported legally from Canada are available.

HEMP COFFEE

Blend of coffee beans and toasted hemp seeds.

Packs: 8-oz (225-g) and 16-oz (450-g) packs.

HEMP FLOUR

Gluten-free flour used as a baking ingredient.

Pack: 5-lb (2.2-kg) pack.

HEMP SEED OIL

Cold-pressed hemp seed oil with more than 90 percent unsaturated fats, used as salad and flavoring oil. It should not be heated over 350°F.

Pack: 17-fl.oz (0.5-l) containers.
1-gal (3.8-l) containers.

Calories: 1 oz (28 g) = 200 calories.

HEMP SEEDS, TOASTED

Used as snacks.

Packs: 2-oz (56-g) and 4-oz (112-g) packages.

Calories: 2 oz (56 g) = 285 calories.

HOMINY, CANNED

White Hominy is available fully cooked.

Packs: #10 cans, 6 per case.

Serving size: 1 cup, 8½ oz (240 g), for breakfast.

Drained weights: 1 #10 can = 75 oz (2.1 kg).

Kitchen yields: 1 #10 can = 9 servings.

Calories: 1 cup, 8½ oz (240 g) = 200 calories.

HOMINY GRITS, DRY

See Cereal Products in this section.

HONEY

The types of flowers on which bees feed influence the flavor and quality of the honey they produce. Therefore, the market offers honey in a variety of flavors. In reality, even honey labeled as gathered from specific flowers is not pure. Bees cannot be controlled while they gather the nectar. In foodservice blended honeys, without the sources identified, are normally used.

Packs: 1-kg (36-oz), 500-g (18-oz), and 200-g (7-oz) jars.

40-g (1.4-oz) portion packs.

Many other sizes are available.

Serving size: 40-g (1.4-oz) portion pack.

Kitchen yields: Honey is hard to handle; use only portion packs.

Calories: 40-g (1.4-oz) portion pack = 123 calories.

1 tb (21 g) = 65 calories.

ICE CREAM

For information on Ice Cream and Sherbets, see the Dairy Products section.

ICE PRODUCTS

The production and storage capacity of ice machines is rated in pounds.

ICE CUBES

Kitchen yields: 1 lb (450 g) of cubes will ice 3 to 4 water glasses, banquet service.

ICE FOR CARVING

Standard size block (other sizes are also available):

Weight: 300 lb (135 kg).

Size: 40 × 20 × 10 in (100 × 50 × 25 cm).

Melting rate: 1½ in (38 mm) per hour at room temperature.

ICES, ITALIAN WATER

Frozen fruit juices or fruit purées, sometimes referred to as *granité*.

ICING

See the Recipes section for Royal Icing. Fondant is a boiled icing and is usually purchased ready to use.

Packs: 5-lb (2.25-kg) tubs.

Calories: ½ cup (3.5 oz), fondant = 300 calories.

1 cup (8.7 oz), fudge = 830 calories.

1 cup (10 oz), butter cream (depending on recipe) = 1000 calories.

IQF

Abbreviation for *Individually quick frozen*. The products are frozen loose and are therefore easier to use than products frozen in solid blocks. To prevent freezer burn, the products are sprayed with water during the freezing process, which is called *glazing* in the trade.

Kitchen yields: About 10 to 15 percent lower yield per unit weight, on account of glazing, as compared with block-frozen products. The weight loss varies greatly, depending on the product.

KALE PRODUCTS

Kale is a green winter vegetable, a hardy member of the cabbage family; there are many varieties, both crinkly and smooth leaved. Collard greens are closely related. For Fresh Kale, see the Vegetables section.

CANNED KALE

Available chopped.

Packs: #10 cans, 6 per case.

Drained weight: 62 oz (1.7 kg).

Serving size: ½ cup, 2 oz (55 g).

Kitchen yields: One can = 30 servings.

Calories: ½ cup, 2 oz (55 g) = 25 calories.

FROZEN KALE

Available chopped.

Packs: 2-lb (900-g) boxes, 12 per case.

Serving size: ½ cup, 2 oz (55 g).

Kitchen yields: One 2-lb (900-g) box = 12 to 14 servings.

Calories: ½ cup, 2 oz (55 g) = 25 calories.

KASHA

Cracked buckwheat, used as cereal in Jewish and Russian cooking.

Packs: 1-lb (450-g) packages, 24 per case.

Larger packs are available.

Serving size: ½ (0.12 l) cup, cooked.

Kitchen yields: 1 lb (450 g) kasha = 2¼ cups dry.

Liquid ratio = 1 cup dry to 3 cups liquid.
1 lb (450 g) kasha = 6 servings.

KELP

See Japanese Foods in the International Ingredients section.

PRESERVED KUMQUATS

Small orange-type fruits, seldom eaten raw; preserved in light or heavy syrup. For Fresh Kumquats, see the Tropical Fruits section.

Packs: 4 each 1-gal (3.8-l) jars.

Kitchen yields: There is no waste; use as is.

LEMON JUICE

Available in bottles. For Fresh Lemons, see the Fruits section.

Packs: 10-oz (0.29-l) bottles, 24 per case.

1-qt (0.9-l) bottles, 12 per case.

Kitchen yields: 1 qt (0.47 l) = juice of 32 fresh lemons.

LIME JUICE

Bottled lime juice is available unsweetened and sweetened with sugar. Sweet Key West lime juice is available, but the supply is very limited.

Packs: 12-oz (0.35-l) bottles, 12 per case.

25-oz (0.75-l) bottles, 12 per case.

Kitchen yields: Use unsweetened juice like fresh juice. Sweetened juice is used for cocktails.

Calories: 1 cup (0.24 l), unsweetened = 65 calories.

LINGONBERRIES

A variety of cranberries harvested in Europe, smaller than the American cranberry, tart and acidic. Whortleberries are a related wild species harvested in North America.

Packs: 14-oz (400-g) cans, 12 per case.
#10 cans, 6 per case.

Other packs are available.

Serving size: 1½ oz (42 g) as condiment.

Kitchen yields: One #10 can = 50 to 55 servings.

Calories: 1½ oz (42 g) = 75 calories.

LIQUID SMOKE

A number of products are on the market. Refer to label instructions.

Diluting proportions: 1 part liquid smoke to 2 parts cold water for meat.

1 part liquid smoke to 3 parts cold water for fish.

Marinating time: 20 minutes for chops, ribs, and chicken.

2 hours for larger pieces of meat, such as briskets.

LITCHI (LYCHEE)

Tropical fruits of the soapberry tree, available canned, dried, and fresh. For

Fresh Litchi, *see* the Tropical Fruits section.

CANNED LITCHI

Packs: 20-oz (560-g) cans, 12 or 24 per case.

Drained weight: 20-oz (560-g) can = 12½ oz (350 g).

Count: 35 to 40 pieces.

Serving size: ½ cup (0.12 l) with syrup.

Kitchen yields: One 20-oz (560-g) can = 5 servings.

DRIED LITCHI (LITCHI NUTS)

The whole fruit is dried, and it separates from its shell. As it shrivels around the seed, it acquires a different flavor. The shells become paper thin and brittle. The shells and seeds are not edible.

Packs: By weight.

Count: 1 lb (450 g) = about 70 to 80 pieces.

1 oz (28 g) = 4 to 5 pieces.

MACADAMIA NUTS

Soft, round nuts, the seeds of a tropical tree. Most nuts are harvested in Hawaii; there is a small production in Guatemala. The nuts contain no cholesterol, but most are roasted in coconut oil, a saturated fat. Also known as *Queensland nuts*.

Packs: 5-lb (2.2-kg) carton, shelled.

Other packs are available. Mostly packed in jars.

Season: Nuts are harvested from August to March.

Kitchen yields: 1 oz (28 g) = ½ cup.

Calories: 1 oz (28 g) = 220 calories.

MANDARIN ORANGES

Small, sweet hybrid citrus fruits, related to oranges. Mandarins and tangerines are considered to be similar, and the names and varieties are used interchangeably. For fresh Mandarin Oranges and their varieties, *see* the Fruits section.

CANNED MANDARIN SECTIONS

Available packed in light and heavy syrup.

Packs: #10 cans, 6 per case.

Drained weights: One #10 can = 74 oz (2 kg).

Kitchen yields: For garnish.

MANGOES

Intensely flavored, succulent tropical fruits, with a slightly acidic sweet taste. Mangoes are available canned, dehydrated as powder, and as slices. Much of their production is used in the manufacture of chutney. For Fresh Mangoes, *see* the Tropical Fruits section.

DRIED MANGO POWDER

See Amchour under Indian Foods in the International Ingredients section.

MANGO CHUTNEY

See Indian Foods in the International Ingredients section.

MARRONS GLACÉS

See Chestnut Products in this section.

MARZIPAN

See Almond Products. There is a formula for Almond-Marzipan Paste for Modeling in the Recipes section.

MATZO MEAL

Finely ground baked matzo; it is used in matzo balls, in kugel, and as a soup thickener. It cannot be used as baking flour.

Packs: 1 lb (450 g) and up, in boxes and bags.

Kitchen yields: 1 cup = 4 oz (112 g).

Calories: 1 oz (28 g), without any additional ingredients = 110 calories.

MAYONNAISE

Mayonnaise is available as regular, light, and low-calorie mayonnaise. There are great differences in quality between brands of commercial mayonnaise. A

formula for Homemade Mayonnaise is included in the Recipes section.

Packs: 1-gal (3.8-l) jars, 4 per case.
5-gal (19-l) tubs.

Other packs, including portion packs, are available.

Kitchen yields: Ratios of main ingredient and mayonnaise in salads:
10 lb (4.5 kg) cooked meat, chicken, or seafood + 3 lb (1.3 kg) diced celery + 2 qt (0.9 l) mayonnaise.

Calories: 1 oz (28 g), or 2 tb, regular mayonnaise = 200 calories.
1 oz (28 g), or 2 tb, light mayonnaise = 90 calories.

MINT JELLY

Commercial food product based on apple juice, pectin, and artificial mint flavor, and dyed green, routinely served with lamb in restaurants in Britain and the United States.

Packs: 1 qt, 32 oz (920 ml) jars.

Serving size: 2 oz (56 g) = 50 servings.

Calories: 2 oz (56 g) = 50 calories.

MUFFIN MIXES AND BATTERS

Muffins can be prepared from scratch, from mix, and from ready-to-use batter.

DRY MUFFIN MIX

Packs: 5-lb bags, 6 per case.

Sizes: 1½ oz (45 g) batter, use #24 scoop.
3.2 oz (90 g) batter, use #10 scoop.

Kitchen yields: One case = 240 muffins, 3.2 oz (90 g) batter.
One case = 480 muffins, 1½ oz (45 g) batter.
Yields vary slightly according to type of muffin.

FROZEN MUFFIN BATTER

Packs: 5-lb (2.2-kg) buckets, 4 per case.
30-lb (13.5-kg) bucket.

Kitchen yields: 1 case of 5-lb (2.2-kg) buckets = 100 muffins, 3.2 oz (90 g).
1 case of 5-lb (2.2-kg) buckets = 212 muffins, 1½ oz (45 g).
One 30-lb (13.5-kg) bucket = 150 muffins, 3.2 oz (90 g).
One 30-lb (13.5-kg) bucket = 320 muffins, 1½ oz (45 g).

MUSHROOMS, CANNED, DRIED, AND FROZEN

Dried mushrooms are very light. It takes about 10 lb (4.5 kg) of fresh mushrooms to make 1 lb (450 g) of dried mushrooms. In some varieties, the flavor of the dried product is more intensified and stronger than that of the fresh variety. Fresh Mushrooms are listed in the Vegetables section.

CHAMPIGNONS

CANNED CHAMPIGNONS

Available as whole buttons, slices, and stems and pieces. Some products are imported.

Packs:

#303 cans, 24 per case.
#2 cans, 24 per case.
#10 cans, 6 per case.
Many other packs are available.

Sizes:

#1 tiny, ½-in (12-mm) to ⅝-in (15 mm) diameter.
#2 small, ⅝-in (15-mm) to ⅞-in (22 mm) diameter.
#3 medium, ⅞-in (22-mm) to 1⅛-in (28 mm) diameter.
#4 large, 1⅛-in (28-mm) to 1⅜-in (34 mm) diameter.
#5 extra large, 1⅜-in (34-mm) and larger, diameter.

Drained weights:

1 case #303 cans = 15 lb (6.75 kg).
1 case #2 cans = 18 lb (8.10 kg).
1 case #10 cans = 25½ lb (11.5 kg).

DRIED CHAMPIGNONS

Rather flavorless product; seldom used because fresh and canned mushrooms are readily available.

Packs: By weight, often 1-lb (450-g) bags.

Kitchen yields: 4 oz (112 g) = 2½-lb (1.1-kg) basket of fresh mushrooms.

CHANTERELLES**CANNED CHANTERELLES**

Canned chanterelles are mostly imported, but the mushrooms are often harvested in the United States and canned in Europe. The flavor is briny.

Packs: 1-lb 13-oz (830-g) cans, 24 per case.
14 6/10-oz (415-g) cans, 36 per case.

Drained weights: 1-lb 13-oz (830-g) can = 14.1 oz (400 g) drained.

Kitchen yields: The mushrooms are fully cooked, and there is no additional shrinkage.

DRIED CHANTERELLES

The mushrooms tend to be a little tough.

Packs: By weight, often 1-lb (450-g) bags.

Kitchen yields: 5 oz (140 g) = 2 lb (900 g) fresh mushrooms.

FROZEN CHANTERELLES

Available in Europe, but seldom imported.

Packs: 2-lb 3-oz (1-kg) bags.

Kitchen yields: There is little waste.

BLACK CHANTERELLES (TRUMPETS OF DEATH)

Despite the name, these are perfectly safe mushrooms. Dried black chanterelles are a little more flavorful than dried regular chanterelles.

Pack: By weight, often 1-lb (450-g) bags.

Kitchen yields: 5 oz (140 g) = 2 lb (900 g) fresh mushrooms.

ENOKI

Canned and dried enoki mushrooms are available but seldom used because fresh mushrooms are easily obtainable. For Fresh Enoki, see the Vegetables section.

DRIED MORELS

Both imported and domestic morels are available; the majority are imported from India and Kashmir. Drying the mushrooms over cow dung may cause the smoky smell of Indian morels. Domestic morels are normally mixed white and black varieties, but only professionals will notice the difference. Some are picked after forest fires, and these have weak walls and can be muddy. Better-quality morels are picked in clear-cut areas.

Packs: Sold by weight; 3-lb (1.3-kg) bags are common.

Kitchen yields: Dried morels have more intense flavor than fresh morels.

PORCINI**CANNED PORCINI**

Packs: 22-oz (625-g) cans, 12 per case.

Drained weight: 16 oz (450 g).

DRIED PORCINI

One of the best dried mushrooms, with a strong flavor. The quality of porcini on the market ranges widely; those of the best quality are imported from Italy. Smaller mushrooms have a stronger flavor than larger ones.

Packs: By weight, often 1-lb (450-g) bags.

Size: Some packers market very large mushrooms, which can be tough.

Kitchen yields: Because the intensity of flavor varies greatly, it is best to sample a small amount. Dried porcini can be soaked, then simmered for about ½ hour to impart maximum flavor.

PORCINI POWDER

Very useful product, packing concentrated flavor.

Pack: By weight, 1-lb (450-g) bags.

FROZEN PORCINI

Commercially flash-frozen whole and sliced mushrooms are available, mostly imported from Europe. The quality is good.

Packs: 2-lb 3-oz (1-kg) bags.

SHIITAKE

Dried shiitake mushrooms are seldom used in Western cooking. *See* Chinese Foods in the International Ingredients section.

STRAW MUSHROOMS, CANNED

See Chinese Foods in the International Ingredients section.

TRUFFLE PRODUCTS

Preserved truffles are available canned and in jars. Truffles in jars are considered better quality than canned truffles. White truffles are imported from Italy and are more expensive than black truffles. For more information on truffles, *see* Mushrooms in the Vegetables section. Canned truffle products include the following:

- Whole brushed truffles
- Peeled truffles
- Truffle juice
- Truffle oil
- Truffle paste
- Truffle peelings
- Truffle pieces
- Truffle powder

Packs: 1-oz (28-g) to 8-oz (22-g) cans or jars.

Arabic summer truffles: 1-kg (35-oz) cans, net weight 500 g (18 oz).

Kitchen yields: One 8-oz (225-g) can of truffle peelings = $\frac{1}{2}$ gal (1.8 l) sauce.
 $\frac{1}{2}$ gal sauce = 50 banquet servings.

Note: Truffles will dry quickly when not covered with liquid.

TRUFFLE OIL

Available with white and black truffle flavor. This is an economical ingredient when truffle flavor is desired, but the

flavor is often supplemented with synthetic extracts.

WOOD EARS, DRIED

Oriental mushroom, popular in Chinese cooking. *See* Chinese Foods in the International Ingredients section.

Packs: By weight.

MUSTARD GREENS

See the Vegetables section for Fresh Mustard Greens.

CANNED MUSTARD GREENS

Packs: #10 cans, chopped mustard greens, 6 per case.

Drained weights: One #10 can = 60 oz (1.7 kg).

Serving size: 1 cup, 5 oz (140 g).

Kitchen yields: One #10 can = 12 servings.

Calories: 1 cup, 5 oz (140 g) = 30 calories.

FROZEN MUSTARD GREENS

Packs: 2½-lb (1.1-kg) boxes, 12 per case. Other packs are available.

Serving size: 1 cup, 5 oz (140 g).

Kitchen yields: One 2½-lb (1.1-kg) box = 8 servings.

Calories: 1 cup, 5 oz (140 g) = 30 calories.

MUSTARDS, PREPARED

Available prepared with many different flavor combinations, and as powder, often referred to as English mustard.

Packs: 1-gal (3.8-l) jars and many other packs.
13-oz (370-g) whole grain.

Calories: Too low to list. Some manufacturers do not list any calories.

MUSTARD POWDER

Mustard seeds ground to a fine powder; it must be blended with a liquid to a

paste because it will form lumps if added as purchased. Mustard powder is very pungent and should be used sparingly.

Pack: By weight, often 1-lb (450-g) boxes.

OIL VARIETIES

Oils can be classified as flavoring oil and cooking oils. Cooking oils are derived from various plants, such as corn, cottonseed, peanut, safflower, soybean, sunflower, and others oils. Some are especially formulated to be used in deep-fat fryers.

Packs: 1-gal (3.8-l) and 5-gal (19-l) cans most often used in foodservice.

Calories: 1 cup, 7.75 oz (217 g) = 1,925 calories (for most oils, with slight variations).

FLAVORING OILS

Flavoring oils usually have a low smoking point and should be used in dressings or in marinades to add flavor, rather than as cooking oil.

ALMOND OIL

Packs: 17-fl oz (0.5-l) bottle.
8.5-fl oz (0.25-l) bottle.

DENDE OIL (*Dendê*)

See Hispanic Foods in the International Ingredients section.

FILBERT OIL

Packs: 17-fl oz (0.5-l) bottle.
8.5-fl oz (0.25-l) bottle.

HEMP OIL

See Hemp Products in this section.

PECAN OIL

Packs: 17-fl oz (0.5-l) bottle.
8.5-fl oz (0.25-l) bottle.

PISTACHIO OIL

Packs: 17-fl oz (0.5-l) bottle.
8.5-fl oz (0.25-l) bottle.

PUMPKIN OIL

Greenish, nutty-flavored oil used as flavoring oil like virgin olive oil. Much is produced in Austria.

Pack: Varies, usually 1-l (33-oz) glass bottles.

SESAME OIL

See Thai and Vietnamese Foods in the International Ingredients section.

WALNUT OIL

Packs: 17-fl oz (0.5-l) bottle.
8.5-fl oz (0.25-l) bottle.

OLIVE OILS

Quality regulations were formulated in 1986 by the Madrid-based International Olive Oil Council (IOOC), which was established under the auspices of the United Nations. Members, as of 1997, are Algeria, Cyprus, Egypt, the European Economic Community, Israel, Morocco, Tunisia, Turkey, and Yugoslavia. The United States is not a member of the council, but produces a small amount of olive oil. Prior to 1986, Italy used and exported more olive oil than it was able to produce; it imported olive oil from Spain and other countries and repacked it. The new regulations require that packaging must indicate the country of origin.

Extra virgin: From the first pressing of the olives. It has a full, fruity flavor, which requires the consumer to become accustomed to it, and the lowest acidity level. The best grades of oil are green to greenish in color; the lesser grades are yellower. Extra virgin oil is further graded.

Premium extra-virgin: Grown on small estates at very high altitudes, from 1,000 to 2,000 ft above sea level, usually on limy soil, in places such as Tuscany and Umbria in central Italy.

Virgin: Also from the first pressing of the olives. Virgin olive oil has up to 1.5 percent acidity. Virgin olive oil accounts for only about 2 percent of American consumption.

Pure olive oil: Also comes from the first pressing, but has a lighter, less fruity

taste and higher acidity level. Pure olive oil has up to 3 percent acidity and is sometimes called *olio di oliva*, *huile d'olive*, or just *olive oil*. Pure olive oil accounts for most, about 93 percent, of American consumption.

Olive pomace oil: Lower-grade oil, made from the pressed olives with the aid of solvents..

Packs: Vary, sometimes in liters rather than quarts.

Kitchen yields: The use of olive oils must be carefully evaluated because the price differences between the varieties are great.

OKRA PRODUCTS

For Fresh Okra, see the Vegetables section.

CANNED OKRA

Available cut or whole.

Packs:

#10 cans, 6 per case.

#2½ cans, 24 per case.

#2 cans, 24 per case.

Drained weights:

#10 can = 60 oz (1.7 kg).

#2½ can = 18 oz (500 g).

#2 can = 12 oz (340 g).

Serving size: 5 pods, 3¾ oz (106 g), as vegetable.

Kitchen yields: One #10 can = 15 servings.

Calories: 5 pods, 3¾ oz (106 g) = 15 calories.

FROZEN OKRA

Available whole, cut, and breaded.

Packs:

Cut okra: 3-lb (1.3-kg) boxes, 12 per case.
20-lb (9-kg) boxes.

Whole okra: 2½-lb (1.1-kg) boxes, 12 per case.

20-lb (9-kg) boxes.

Breaded Okra, IQF

20-lb (9-kg) boxes.

OLIVE VARIETIES

Olives are available green, ripe, and processed ripe. Because olives are an ancient food, many different styles are available in ethnic markets. Ripe olives are cured by several processes.

CANNED BLACK OLIVES

The process of canning olives was developed in the United States. Large-size unripe green olives are processed to become black olives. Black olives are available whole pitted and whole unpitted, and in wedges, slices, and chopped.

Packs: #10 cans, 6 per case.

Many other packs are available.

Sizes and counts for olives, #10 cans:

Size	Pitted	Unpitted
Small	578	557
Medium	486	466
Large	430	404
Extra Large	350	288
Jumbo	245	288
Colossal	199	192
Super Colossal	163	126

Drained weights: Sliced olives, one #10 can = 55 oz (1.5 kg) or 13 cups.

Olive wedges, one #10 can = 55 oz (1.5 kg) or 11¼ cups.

Chopped olives, one #10 can = 90 oz (2.2 kg) or 14½ cups.

Calories: 1 cup, sliced, 4¾ oz (135 g) = 174 calories.

1 cup, wedged, 5¼ oz (150 g) = 193 calories.

1 cup, chopped, 6⅞ oz (190 g) = 250 calories.

CANNED OR PICKLED GREEN OLIVES

The most important varieties are Sevillano, Manzanillo, Ascolana, and Mission olives. Olives are available with pits, pitted, and stuffed with various foods.

Packs: ½ pt (0.24 l).

1 pt (0.47 l).

1 qt (0.9 l).

1 gal (3.8 l).

Drained weights:

- ½ pint (0.24 l) = 5 oz (140 g).
- 1 pint (0.47 l) = 10 oz (280 g).
- 1 qt (0.9 l) = 21 oz (600 g).
- 1 gal (3.8 l) = 88 oz (2.5 kg).

Counts per pound:

- Pee wee = 181 to 220.
- Midget = 141 to 180.
- Small = 126 to 140.
- Medium = 106 to 127.
- Large = 90 to 105.
- Extra Large = 76 to 90.
- Mammoth = 65 to 75.
- Giant = 53 to 64.
- Jumbo = 42 to 52.
- Colossal = 33 to 41.
- Super Colossal = 32 and fewer.

Calories: 4 Medium, 3 Extra Large, or 2 Giant = 15 calories.

CURED SPECIALTY OLIVES

There are many varieties of specialty olives on the market, ranging from canned ripe olives in brine to cured olives in salt or with oil. Foodservices often use a small olive referred to as *niçoise*. Some shippers offer mixed olives and cracked olives.

Types:

- Kalamata*: purple.
- Niçoise*: dark.
- Mt. Athos*: green.
- Mt. Pelion*: reddish brown or green.
- Thasos*: dark black with wrinkled skin.

Packs: 10-lb (4.5-kg) cans.

Drained weights: 5 lb 12 oz (2.7 kg).

Count: 2,000 per can for small olives.

ONION PRODUCTS

For Fresh Onions, see the Vegetables section.

DEHYDRATED ONIONS

Available sliced and chopped.

Packs: #10 cans, 6 per case.

Net weight: 28 oz (800 g).

Kitchen yields: 1 lb (450 g) dry = 6¾ cups (1.4 l).

1 lb (450 g) dry = 13 cups (3 l) reconstituted onions.

FROZEN ONIONS

Available diced or sliced IQF.

Packs: 12-lb (5.4-kg), 16-lb (7.3-kg), 20-lb (9.1-kg), and 45-lb (20.4-kg). 2-lb (900-g) boxes.

Kitchen yields: The product is ready to use.

GRANULAR ONION

Pack: 20 oz (560 g) cartons.

ONION POWDER

Pack: 20 oz (560 g).

ONION SALT

Pack: 48 oz (1.3 kg).

PEARL ONIONS, CANNED**CANNED PEARL ONIONS**

Packs: #10 cans, 6 per carton.

Counts: 100 to 125, 200 to 250, 375 to 400, or 400 to 500.

Kitchen yields: Use as is; there is no waste.

Calories: 8 oz (220 g) = 80 calories.

FROZEN PEARL ONIONS

Various packs are available.

ORANGE PRODUCTS

Orange juice is available canned, dehydrated, freshly squeezed, and as frozen concentrate. For juice orange yields and for Fresh Orange Juice, see the Fruits section.

CANNED ORANGE JUICE.

Available sweetened and unsweetened.

Packs: 46-oz (1.3-l) cans, 12 per case.

Serving size: 6 oz (0.17 l).

Kitchen yields: One 46-oz (1.3-l) can = 7½ servings.

Calories: 6 oz (0.17 l), unsweetened = 90 calories.

DEHYDRATED ORANGE JUICE

Packs: 1-lb (450-g) jars.

Other packs are available.

Serving size: 6 oz (0.17 l).

Kitchen yields: 1 lb (450 g) mixed with
1 gal (3.8 l) water = 21 servings.

Calories: One 6-oz (0.17-l) serving = 75 calories.

FROZEN ORANGE JUICE CONCENTRATE

Packs: Various packs are available.
12-oz (0.35-l) can is common.

Serving size: 6 oz (0.17 l).

Kitchen yields: 1 part concentrate to 3
parts water.

One 12-oz (0.35-l) can = 48 oz (1.4 l)
juice.

One 12-oz can = 8 servings.

Calories: One 6-oz (0.17-l) serving = 90 calories.

ORANGE SECTIONS

CANNED ORANGE SECTIONS

Packs: #3 cylindrical cans, 24 per case.

#2 cans, 24 or 36 per case.

#303 cans, 24 or 36 per case.

Drained weights: One #3 cylindrical
can = 30 oz (850 g).

One #2 can = 12 oz (340 g).

One #303 = 10 oz (280 g).

FRESH ORANGE SECTIONS

Popular banquet item. The industry distinguishes between hot-peeled and cold-peeled; the latter is considered better quality.

Packs: 1-gal (3.8-l) jars, 4 per case.

Drained weight: 6 lb (2.7 kg).

Serving size: ½ cup (0.12 l) with juice.

Kitchen yields: 1-gal (3.8-l) jar = 28 to
30 servings.

PANCAKE MIX

Most operations use dry mixes for making pancakes and waffles.

Pack: 32-oz (900-g) boxes.

Other packs are available.

Serving size: 4 small pancakes.

Kitchen yields: 1 cup (0.24 l) dry mix =
10 to 12 small pancakes.

One 32-oz (900-g) box = 65 to 75
small pancakes.

Calories: 1 cup dry mix = 110 calories.

1 cup, 5 oz (140 g) dry mix, prepared
with milk and 4 eggs = 720 calories.

1 serving, without syrup = 60 calories.

PANCAKE SYRUP

Available products are genuine maple syrup, corn syrup, and maple syrup blends, and plain corn syrup. The percentage of genuine maple syrup determines price.

Packs: 1-gal (3.8-l) containers to
individual portion pack in many
sizes.

Serving size: ¼ cup (60 ml).

Calories:

¼ cup (60 ml), light = 100 calories.

¼ cup (60 ml), regular = 200 calories.

Genuine maple syrup = 200 calories.

PAPAYA PRODUCTS

For Fresh Papaya, see the Tropical Fruits section.

FROZEN PAPAYA PULP

Pack: 14-oz (400-g) packages.

Serving size: ½ cup (110 g).

Kitchen yields: About 3½ per package.

Calories: 1 serving = 40 calories.

PARMALAT

Milk not requiring refrigeration until opened. Available as whole milk and reduced-fat milk.

Pack: 1-qt (0.9-l) cartons and other
packs.

Calories: 1 cup (0.24 l) whole milk = 150
calories. Calorie counts correspond to
those of regular milk.

PASTA PRODUCTS

Pasta is made with high-gluten flour and water. Noodles are made with high-gluten

flour and eggs. For homemade Noodle Dough, *see* the Recipes section. Fresh pasta is available in most markets in many shapes and with many additives enhancing color and taste.

DRY PASTA

The quality of pasta depends on the quality of the flour used in making it, the water, and the drying process. High-quality pasta is dried more slowly than less expensive pasta. Because the base price of pasta is low, foodservice operators should use only high-quality products. There are hundreds of different pasta shapes on the market. Pastas made with flours from grains other than wheat are also available.

Packs: 1-lb (450-g), 10-lb (4.5-kg), and 20-lb (9-kg) boxes.

Serving sizes: 2 oz (56 g), dry, as main course. 1 oz (28 g) as garnish.

Kitchen yields: The relation of weight to volume of cooked pasta varies according to shape. Elbow macaroni has the largest volume. Yields depend on the degree of softness to which the product has been cooked.

1 lb (450 g) of lasagna consists of 24 sheets, $2\frac{1}{4} \times 9\frac{1}{2}$ inch (57×240 mm). Other sizes are available. Fully cooked lasagna sheets are on the market.

1 lb (450 g) of dry medium-sized macaroni produces 8 cups (1.9 l) of cooked pasta.

1 lb (450 g) of dry spaghetti produces 8 cups (1.9 l) of cooked pasta.

Yields for various pastas:

Macaroni: 1 lb (450 g) = 16 servings side order or 6 servings main course.

Lasagna: 26 oz (730 g) fill one hotel pan, 20×12 in (508×304 mm), with four layers.

Calories: 2 oz (56 g) dry pasta without sauce = 210 calories.

1 cup (4.5 oz) firm macaroni = 190 calories.

1 cup (3.6 oz) soft, cold macaroni = 120 calories.

FRESH PASTA

Fresh pasta can be purchased in many shapes and flavors. It is frequently delivered frozen.

Serving size: $3\frac{1}{2}$ oz (98 g) as main course. Serving size varies according to pasta variety.

Kitchen yields: 1 lb (450 g) fresh pasta yields approximately 5 to 6 cups (1.1 to 1.2 l) cooked pasta.

RAVIOLI PASTA, FROZEN

Little pockets filled with a wide variety of fillings.

Packs: 1-lb (450-g) packages and larger.

Counts: Counts vary greatly.

TORTELLINI PASTA, FROZEN

Little pasta pockets available with meat or cheese filling; sizes vary slightly between manufacturers. Product must be kept frozen until use.

Packs: 1-lb (450-g) packages and larger.

Counts: 1-lb (450-g) contains about 140 medium-sized pieces.

Serving sizes: 20 pieces or $2\frac{1}{4}$ oz (63 g) as appetizer.

1 cup (0.24 l), cooked, about 45 pieces as main course.

Kitchen yields: 1 lb (450 g) = 7 servings, appetizer.

1 lb (450 g) = 4 servings, main course.

Calories: 1 cup (0.24 l), cooked = 340 calories or more, depending on filling.

PEACH PRODUCTS

Available canned, dehydrated, fresh, and frozen. For Fresh Peaches, *see* the Fruits section.

CANNED PEACHES

Canned peaches are available packed in extra heavy, heavy, and light syrup, as well as in water. They are packed whole, in halves, in quarters, in slices, and diced.

Some varieties are marketed natural or spiced.

Packs: #2½ and #10 cans.

Many other packs are available.

Counts, Weights:

One #2½ can = 10 to 12 pieces, halves, average.

One #2½ can, 1 lb 13 oz (820 g) = 3½ cups (0.8 l) fruit and syrup halves or 1 lb 14 oz (840 g) slices.

#10 cans = 25 to 30, 30 to 35, 35 to 40, 40 to 50 halves, 35 to 40 pieces average.

One #10 can, 6 lb 12 oz (3 kg) = 12 cups fruit and syrup halves.

One #10 can, 6 lb 14 oz (4 kg) drained slices.

Calories: 1 cup (9 oz), syrup pack = 200 calories.

1 cup (8.7 oz), water pack = 75 calories.

DEHYDRATED PEACHES

Dehydrated peaches are available as halves, pieces, dices, and slices, and as canned low-moisture slices.

Packs: By weight, dehydrated peaches. #10 cans, low-moisture slices.

Weights: One #10 can low-moisture slices = 3 lb (1.3 kg) net weight.

1 cup = (5½ oz) dry fruit.

Kitchen yields: 1 lb (450 g) dry peaches = 5¼ cups (1.2 l) cooked fruit.

1 lb (450 g) low moisture slices = 4 cups (0.9 l) cooked fruit.

Calories: 1 cup (0.24 l) cooked, unsweetened fruit = 205 calories.

FROZEN PEACHES

Packed with approximately 25 percent sugar by weight. Available as halves and slices. The product is frequently used for peach pie.

Packs: 10-oz (280-g) cans.

1-gal (3.8-l) cans.

10-lb (4.5-kg), 25-lb (11.2-kg), and 30-lb (13.5-kg) cans.

Calories: 1 cup, or 9-oz (250-g), sliced and sweetened = 220 calories.

1 cup, or 10-oz (280-g), sliced and sweetened = 250 calories.

PEAR PRODUCTS

For Fresh Pears, *see* the Fruits section.

CANNED PEARS

Canned pears are available packed in extra heavy syrup, heavy syrup, light syrup, slightly sweetened water, slightly sweetened fruit juice, and fruit juice.

Packs: #10 cans, 6 per case.

Counts, halves: 25 to 30, 30 to 35, 35 to 40, 40 to 45, 45 to 54, 50 to 60, and 60 to 70.

Weights: #10 can, drained = 66 oz (1.85 kg), 25- to 30-size halves.

#10 can, drained = 70 oz (2 kg), slices or pieces.

#10 can, drained = 74.5 oz (2.11 kg), dices.

Serving sizes: Two #30 pears, stewed fruit.

Three #35 pears, pear salad.

Calories: 3½ oz (100 g) in fruit juice = 56 calories.

3½ oz (100 g) in water = 35 calories.

3½ oz (100 g) in light syrup = 75 calories.

3½ oz (100 g) in heavy syrup = 95 calories.

PEAS

Peas are available canned, dried, fresh, and frozen. For Fresh Peas, *see* the Vegetables section.

BLACKEYED PEAS

CANNED BLACKEYED PEAS

A number of varieties and combinations with other vegetables are on the market.

Packs: #10 cans, 6 per case.

Drained yield: One #10 can = 13 cups (3.1 l), about 85 oz (2.4 kg).

Serving size: 1 cup (5½ oz).

Kitchen yields: One #10 can = 15 servings.

Calories: 1 cup (5½ oz) = 175 calories.

DRIED BLACK-EYED PEAS

Packs: 1-lb (450-g) boxes, 24 per case.
25-lb (11.2-kg) bags.

Serving size: ½ cup, 3.5 oz (0.12 l), cooked.

Kitchen yields: 1 lb (450 g) dried product = 3 cups (0.7 l) cooked.

Calories: ½ cup, 3.5 oz (0.12 l), cooked = 95 calories.

FROZEN BLACK-EYED PEAS

The product is processed while still tender and green.

Packs: 3-lb (1.3-kg) boxes, 12 per case.
20-lb (9-kg) bulk packs.

Serving size: ½ cup, 3.5 oz (0.12 l), cooked.

Kitchen yields: One 3-lb (1.3-kg) box = 12 servings.

DRIED PEAS, WHOLE OR SPLIT

Dried peas are classified as green or mature yellow peas. In most cases, the outer skin is removed during the drying process and the peas split into two natural halves.

Packs: 1-lb (450-g) boxes, 24 per case.
25-lb (11.2-kg) bags.

Kitchen yields: 1 cup (8 oz) dried product = 2½ cups (0.6 l) cooked.
1 lb (450 g) = 5 cups (1.1 l).
10 lb (4.5 kg) = 50 cups (11.75 l).

Calories: 1 cup, 6 oz (170 g), cooked = 220 calories.

GREEN PEAS

CANNED GREEN PEAS

Canned peas are graded by size; sieve #1 is the smallest size, sieve #6 is the largest. Many packers no longer indicate sieve size. *Early* or *Fancy* does not indicate size. Pea purée is available from at least one packer in #303 cans.

Packs: #10 cans, 6 per case.

Weight: One #10 can = 70 oz (2 kg) drained weight.

Serving size: 1 cup (6 oz).

Kitchen yields: One #10 can = 12 servings.

Calories: 1 cup, 6 oz (170 g), drained = 150 calories.
1 cup, 8 oz (224 g) = 120 strained purée.

FROZEN GREEN PEAS

Peas are graded by size and tenderness. They are marketed as petit peas, small peas, and under a number of other names. Size is an important quality factor. Frozen peas are also on the market mixed with either pearl onions or cubed carrots.

Packs: 2½ lb (1.12-kg), 12 per case.
20-lb (9-kg) carton.

Serving size: 1 cup, 5.7 oz (160 g).

Kitchen yields: One 20-lb (9-kg) carton = 50 servings.

Calories: 1 cup, 5.7 oz (160 g) = 110 calories.

SNOW PEAS, FROZEN

Frozen snow peas have come on the market. They can easily be overcooked and can lose crispness unless served right away.

Packs: 2-lb (900-g) packages, 12 per case.

Serving size: 3½ oz (100 g) as purchased.

Kitchen yields: One 2-lb (900-g) package = 10 servings.

Calories: 3½ oz (100 g) = 55 calories.

SUGAR SNAP PEAS, FROZEN

Packs: 32-oz (900-g) bags IQF, 12 per carton.

Serving size: 3½ oz (100 g) as purchased.

Kitchen yields: One 32-oz (900-g) bag
= 8 to 9 servings.

Calories: 3½ oz (100 g) = 50 calories.

PEANUTS

Peanuts in the shell are occasionally purchased by bar operators. There are USDA grades for peanuts, generally referring to size and the presence of split or broken nuts. Peanut butter is made with ground, roasted peanuts. Grinds can be fine, medium, or coarse and chunky. Creamy peanut butter is emulsified.

Packs: By weight.

Kitchen yields: 1 cup = 5 oz (140 g)
roasted and salted peanuts.

Calories: 1 oz (28 g) peanut butter =
170 calories.
1 cup, 5 oz (140 g), roasted and salted
peanuts = 840 calories.

PECANS

Available shelled, in the shell, and as flour. Pecan halves are graded by size; pieces are not graded by size.

Packs: By weight.

Counts for 1 lb (450 g):

Mammoth	200 to 250.
Junior Mammoth	251 to 300.
Jumbo	301 to 350.
Extra Large	351 to 450.
Large	451 to 550.
Medium	551 to 600.

Kitchen yields: 1 cup = 4.2 oz (120 g)
chopped or pieces.

Calories: 1 cup, 4.2 oz (120 g), pieces or
120 halves = 810 calories.

PEPPERS, SWEET (BELL PEPPERS)

For Hot Peppers, *see* Mexican Foods in the International Ingredients section. For Fresh Peppers, *see* the Vegetables section. For Pepper, the spice, *see* the Herbs and Spices section.

CANNED SWEET PEPPERS

Available peeled, whole; often referred to as *pimientos*, in different sizes and qualities.

Packs: #10 cans, 6 per case.

6.6-lb (3-kg) cans, 6 per case.

12-oz (340-g) cans, 48 per case.

Other packs are available.

Drained weights: One #10 can = 70 oz
(1.9 kg).

One 6.6-lb (3-kg) can = 54 oz (1.5 kg).

One 12-oz can = 8 oz.

Calories: 1 oz (28 g), cooked = 10 calories.

FROZEN SWEET PEPPERS

Green and red peppers are available diced and in strips.

Packs: 2½-lb (1.1-kg) packages, 12 per
case.

Kitchen yields: One 2½-lb (1.1-kg)
package = 2 lb (900 g), defrosted,
drained.

PICKLED SWEET PEPPERS

Delicatessen item preserved in vinegar and spices. Available seeded and split, and as whole small peppers. The product is sometimes flavored with hot chili peppers.

Pack: 1-gal (3.8-l) jars, 4 per case.

PESTO, FROZEN

The original pesto is made with mashed basil leaves, garlic, grated cheese, pine nuts, and olive oil. Other varieties have come on the market such as cilantro pesto, sundried tomato pesto, and roasted pepper pesto.

Packs: 64-oz (1.8-kg) containers, 2 per
case.

Kitchen yields: 2 oz (56 g) sauce for 6 oz
cooked pasta.

PICKLES

Pickled cucumbers are the most common pickle products on the market, but many other vegetables are also available pickled. New York Jewish pickles are pickled in salt brine and must be kept under

refrigeration. Their shelf life is limited because they are not cured all the way through and still have a solid center. Cucumber pickles can be classified by shape:

Whole pickles
Chopped pickles
Sliced pickles, often called chips
Sticks or spears

Pickles are further classified by the methods by which they have been preserved:

Sweet brine
Regular brine
Salt brine

Pickles are also graded by size.

Packs: Typically, 1-gal (3.8-l) jars. 30-lb (13.5-kg) barrels and larger.

Sizes per gal (3.8 l):

Midgets: 445 to 545 and 330 to 440
Gherkins: 225 to 329 and 135 to 224
Small: 80 to 99, 66 to 79 and 52 to 65
Medium: 40 to 51
Large: 26 to 39 and 22 to 25
Extra Large: 16 to 21

PIE FILLINGS, CANNED

Most fruit fillings are available ready for baking. Mincemeat for pies, consisting of diced dried fruits, is also normally purchased canned.

Packs: #10 cans, 6 per case.

Kitchen yields: On average, one #10 can = five 9-in (228-mm) pies, depending on fruit variety.

PINEAPPLE PRODUCTS

For Fresh Pineapple, *see* the Fruits section.

CANNED PINEAPPLE

Available as slices, chunks, tidbits, coarse crushed, fine crushed, and broken slices. Most items are packed in a choice of heavy syrup, light syrup, or juice.

Packs: #10 cans, 6 per case.

Counts: 52, 66, and 100 to 110 slices.

Weights, one #10 can:

66 oz (1.8 kg) drained chunks.
 70 oz (2 kg) small tidbits.

90 oz (2.5 kg) coarse crushed.
 90 oz (2.5 kg) fine crushed.
 63 oz (1.75 kg) broken slices.

Serving sizes: ½ cup, 4½ oz (125 g), chunks or
 1 large slice or
 2 small slices.

Kitchen yields: One #10 can = 22 servings, chunks.

One #10 can = 50 or 60 servings, large slices.

One #10 can = 50 to 55 servings, small slices.

Calories: ½ cup, 4½ oz (125 g), chunks, heavy syrup = 100 calories.
 One large slice = 80 calories.

DRIED PINEAPPLE

Available as rings and as pieces.

Pack: By weight.

FRESH PROCESSED PINEAPPLE

Available as chilled fresh pineapple chunks.

Packs: 5-lb (2.2-kg) pouches.

Counts:

Wedges = 23 to 32.
 Spears = 10 to 12.
 Tidbits = 139 to 153.

Kitchen yields: There is no waste.

Calories: 1 cup, 5½ oz (160 g), tidbits = 90 calories.

CANNED PINEAPPLE JUICE

Available canned and fresh, sweetened and unsweetened. For Fresh Pineapple Juice, *see* the Fruits section.

Packs: 6-oz (0.17-l) cans, 48 per case.
 46-oz (1.35-l) cans, 12 per case.

Serving size: 6 oz (0.17 l) per serving.

Kitchen yields: One 46 oz (1.3 l) can = 7½ servings.

Calories: 6 oz (0.17 l), unsweetened = 110 calories.

PINE NUTS

Also called *pignoli nuts*.

Packs, shelled nuts:

#10 cans, 4 lb (1.8 kg) net.
4 lb (1.8 kg).
25-lb (11.2-kg) cartons.

Kitchen yields: 1 lb (450 g) shelled nuts
= 6½ cups.

PISTACHIOS

Available in the shell, often dyed red, and shelled. The shelled nuts can be purchased raw, roasted, salted, and unsalted. They are usually still covered with a thin skin, which can be removed from raw nuts by blanching with boiling water. Also available as flour.

Packs, shelled nuts:

#10 cans, 4 lb (1.8 kg) net.
4 lb (1.8 kg).
25-lb (11.2-kg) cartons.

Packs: By weight.

Kitchen yields: 1 lb (450 g) shelled nuts
= 6½ cups.

PIZZA COMPONENTS, PROCESSED

For homemade Pizza Dough and basic Pizza Sauce, *see* the Recipes section. Pizza Crust can be purchased fully baked, unbaked, and as dough balls.

BAKED PIZZA CRUST**Sizes and Packs:****Round:**

16 in (400 mm), 24 per case.
15 in (375 mm), 24 per case.
12 in (300 mm), 24 or 30 per case.
10 in (250 mm), 30 per case.
9 in (225 mm), 24 or 30 per case.
6½ in (184 mm), 60 per case.
5½ in (156 mm), 120 per case.

Square:

8 in × 12 in (200 × 300 mm), 24 per case.
9 in × 10 in (225 × 250 mm), 20 or 24 per case.
12 in × 16 in (300 × 400 mm), 20 each per case.

PIZZA CHEESE

Mozzarella cheese is normally used. *See* the Dairy Products section.

PIZZA DOUGH BALLS**Sizes and Packs:**

22 oz (610 g), 24 per case.
20 oz (570 g), 24 per case.
16 oz (450 g), 24 per case.
10 oz (280 g), 40 each per case.

PIZZA SAUCE

Packs: #10 cans, 6 per case.

Kitchen yields: 1½ cups (0.35 l) = one 16-in (400-mm) pie.
One #10 can = eight 16-in (400-mm) pies.

READY-TO-BAKE PIZZA

Sizes and Packs: 4 × 6 in (100 × 150 mm) squares, 48 per case.
4 × 6 in (100 × 150 mm), squares, extra cheese, 48 per case.
4 × 5 in (100 × 125 mm), squares, 48 per case.
Many other packs and sizes are available.

PLUM PRODUCTS

For Fresh Plums, *see* the Fruits section.

CANNED PLUMS

Available are purple, green gage, yellow egg, and other varieties, whole and in halves. Whole plums are usually not pitted. Available in heavy syrup, light syrup, and water pack.

Packs: #10 cans, 6 per case.

#2½ cans, 24 per case.
#2 cans, 24 per case.

Drained weights:**Whole plums:**

One #10 can = 60 oz (1.7 kg).
One #2½ can = 15½ oz (440 g).
One #2 can = 10½ oz (300 g).

Plum halves:

One #10 can = 63 oz (1.8 kg).
One #2½ can = 16¼ oz (440 g).
One #2 can = 12 oz (340 g).

Serving size: ½ cup, 4.8 oz (140 g), whole with pits and syrup.

Kitchen yields: One #10 can = 18 servings.

Calories: ½ cup, 4.8 oz (140 g), whole with pits and syrup = 110 calories.

DRIED PLUMS

See Prune Products in this section.

POMEGRANATE SEEDS, DRIED

For Fresh Pomegranate, see the Tropical Fruits section.

Pack: By weight.

POPCORN

Many brands are available. Smaller kernels make better-tasting popcorn than larger kernels. Natural blue popcorn and black popcorn have come on the market.

Packs: By weight.

Kitchen yields: Yields vary; high-quality popcorn gives better yield than cheaper popcorn.

¼ cup (0.05 l) dry measure weighs 2 oz (56 g) and makes 9 to 10 cups (2 to 2.25 l) popped corn.

1 cup (0.29 l) dry measure weighs 8 oz (224 g) and makes 2¼ to 2½ gal (8.3 to 9.2 l) popped corn.

Calories: 2 oz (56 g) with ⅔ oz (19 g) oil, 9 to 10 cups popped corn = 386 calories.

POTATO PRODUCTS

Available canned, dehydrated, and processed frozen. For fresh potatoes, see the Vegetables section.

CANNED POTATOES

Available whole, sliced, and diced.

Packs: #10 cans, 6 per case.

Counts, whole potatoes: One #10 can = 80 to 100, 100 to 120, 130 to 150, and 200 and more.

Drained weights: *Diced or Sliced Potatoes:* One #10 can = 108 oz (3 kg). Counts and weights vary according to manufacturer.

Serving size: 4 oz (112 g) each.

Kitchen yields: One #10 can = 25 servings.

DEHYDRATED POTATOES

Available as flakes, granules, slices, dices, and hash browns.

FLAKES

Pack, Idaho instant flakes:

2-lb (900-g) bags, 6 per case.

2-lb (900-g) #10 cans, 6 per case.

2½-lb (1.1-kg) and 5-lb (2.2-kg) bags.

1-lb (45-g) and 2-lb (900-g) cartons.

Serving sizes: #8 scoop, ½ cup (0.12 l).

Kitchen yields: 1 lb (450 g) + 7 cups (1.6 l) water/milk = 22 servings.

Calories: #8 scoop, ½ cup (0.12 l) = 100 calories, including butter.

GRANULES

Granules are available plain, seasoned, with milk, and with vitamin C added. They are used to produce mashed potatoes.

Pack, Idaho granules: #10 cans, 6 per case.

Kitchen yields: One #10 can = 5 gal (19.5 l).

SLICES, DICES, AND HASH BROWNS

Packs: 2½-lb (1.1-kg) cartons. 30-lb (13.5-kg) bags.

Serving size: ¼ cup (0.12 l).

Kitchen yields: One 2½-lb (1.1-kg) carton + 3 gal (11.4 l) water = 50 servings.

FROZEN BAKED POTATOES

Fully cooked potatoes, ready to heat in convection oven, steamer, or microwave oven. Also available are stuffed, ready-to-heat potatoes.

Pack: 5-lb (2.2-kg) bags 30-lb (13.5-kg) cases.

Sizes: 3 oz (85 g) = 150 per case.

4 oz (112 g) = 120 per case.

5 oz (140 g) = 96 per case.

6 oz (170 g) = 78 per case.

Calories: 3 oz (85 g) = 72 calories, or about 24 calories per oz.

FROZEN PROCESSED POTATOES

Many products are available, such as French fries in a variety of cuts, ranch or country fries, hash browns, and potato rounds.

Packs:

French fries:

5-lb (2.2-kg) bags, 6 per carton.

4-lb (1.8-kg) and 4½-lb (2-kg) bags.

Shredded hash:

2½-lb (1.12-kg) layers, 8 per case.

3-oz (85-g) portions, 96 per case.

Long shredded:

5-lb (2.2-kg) layers, 3 per carton.

4-lb (1.8-kg) layers, 8 per carton.

3-lb (1.3-kg) bags IQF, 6 per carton.

Skin-on hash: 5-lb (2.2-kg) bags, 6 per carton.

Shells: 5-lb (2.2-kg) cases, 200 shells per case, 6 per carton.

Wedges: 5-lb (2.2-kg) bags, 6 per carton.

Cubes: 4-lb (1.8-kg) bags, 6 per case.

Kitchen yields: Use as is. There is no waste.

Calories: French fried potatoes: 3 oz (85 g) = 230 calories. Hash brown potatoes: 3 oz (85 g) = 190 calories.

Note: Calorie counts are contingent on fat content.

LATKES (POTATO PANCAKES)

Available frozen and as ready-to-use mix. For Homemade Latkes (Potato Pancakes) formula, see the Recipes section.

LATKES MIX

Available with and without onions.

Pack: 1-lb (450-g) boxes. Other sizes are available.

Serving size: 2 pancakes.

Kitchen yields: About forty 3-in pancakes.

Calories: Two 3-in pancakes = 75 calories, depending on amount of fat

used. *Note:* Pancake mix can also be used for potato dumplings.

POTATO CHIPS

Available in many different shapes and flavors.

Serving size: 1 oz (28 g) or less, as garnish with a sandwich.

Kitchen yields: Packs vary greatly.

Serving size is 1 oz (28 g) average.

Calories: 1 oz (28 g) or less, as garnish = 160 calories.

POTATO SKINS

Available frozen as skins only or as cups for filling with cheese or with mashed potatoes. There are many sizes on the market. Both shells and munch skins can be filled. There are also Red Potato Shells on the market. All products are fully baked and frozen, and can be deep-fried or baked.

Packs: Skin cups, 12 to 15 count, four 6-lb (2.7-kg) bags, 24-lb (10.8-kg) case.

Munch skins, 12 to 14 count, four 4-lb (450-g) boxes, 16-lb (7.2-kg) case.

Kitchen yields: One 24-lb (10.8-kg) carton = 160 to 175 skin cups.

One 16-lb (7.2-kg) case = 52 munch skins.

POTATO STARCH

Packs: 1-lb (450-g) boxes, 24 per case.

Kitchen yields: Thickening power, about equal to corn starch.

PRUNE PRODUCTS

DRIED PRUNES

Prunes are dried plums, but not all varieties can be dried. The variety most used for drying is the California French prune. Prunes are available pitted and with pits.

Packs: 25-lb (11.2-kg) cartons.
1-lb (450-g) boxes.

Counts per pound: 43 Extra Large, 53 Large, 67 Medium, and 85 Small,

with pits. *Note:* Use Jumbo or Extra Large sizes for stuffing.

Serving sizes: Extra Large = 4 prunes,
Large = 5 prunes, Medium = 6
prunes, Small = 7 prunes.

Kitchen yields: 1 lb (450 g) =
11 servings Extra Large.
10½ servings Large.
11 servings Medium.
12 servings Small.

Calories: 4 Extra Large Prunes with pits,
uncooked = 110 calories.
4 Extra Large Prunes, cooked = 180
calories.

PRUNES, CANNED

Canned prunes are available pitted and with pits. There are three standard packs of canned prunes:

Regular: Packed in syrup.

Nectarized: Packed in water with higher fruit count.

High Moisture: No liquid added.

Packs: Many can sizes are available. For institutional use, #10 cans are normally purchased.

Counts for #10 cans, prunes without pits:

Regular: Drained weight 70 oz (1.9 kg),
140 to 160.

Nectarized: Drained weight 92 oz (2.6 kg),
190 to 210.

High Moisture: Drained weight 112 oz
(3.1 kg), 330 to 365.

Counts for #10 cans, prunes with pits:

Regular: Drained weight 70 oz (1.9 kg),
70 to 85, 115 to 135, 140 to 160.

Nectarized: Drained weight 96 oz
(2.7 kg), 155 to 175.

Moist Pack: Drained weight 7 lb (3.1 kg),
238 to 273.

Calories: 1 cup (9 oz), any size = 255
calories.

PRUNE JUICE

Canned or bottled prune juice is prepared from a water extract of dried prunes.

Packs:

Glass:

4 oz (0.12 l).

8 oz (0.24 l).

32 oz (0.9 l).

40 oz (1.15 l).

48 oz (1.4 l).

Cans:

6 oz (0.17 l).

46 oz (1.3 l).

Serving size: 6 oz (0.17 l).

Kitchen yields: 1 qt (0.9 l) = 5 servings.

46-oz can = 7½ servings.

Calories: 6 oz (0.17 l) = 130 calories.

PRUNE PASTE

Called *lekvar* on the East Coast. Prune paste is used in baking.

Packs: 25 lb (11.2 kg) to 40 lb (18.1 kg).
#10 cans, 6 per case.

Weight: One #10 can = 7 lb (3.1 kg).

Serving size: 1 oz (28 g) per piece for
large Danish pastry.

Kitchen yields: One #10 can = 110
fillings.

Calories: 1 oz (28 g) = 70 calories.

PUMPKIN, CANNED

Available as canned purée; primarily used in pies, but also in soups. For Fresh Pumpkins, see the Vegetables section.

Packs: #10 cans, 6 per case.

Kitchen yields: Depend on use.

Calories: 1 cup, 8¾ oz (245 g) = 80
calories.

QUAIL EGGS, CANNED

Packed in brine from imported or domestic sources. For Fresh Quail Eggs, see the Dairy Products section.

Packs: 15-oz (425-g) cans, 12 per case.

Count and drained weight: 8 oz (225 g),
30 eggs.

Kitchen yields: Eggs are usually peeled.
One egg = ¼ oz (7.5 g).

QUINOA

High-protein grain that originated in the Peruvian Andes. Quinoa is used as a starch garnish in place of rice or pasta.

Packs: 12 oz (340 g), 1 lb (450 g), 25 lb (11.2 kg), and 50 lb (22.5 kg).

Serving size: ½ cup (90 g).

Kitchen yields: 1 lb (450 g) = 2½ cups, cooked.

Calories: ½ cup (90 g), cooked = 360 calories. *Note:* Rinse grain before cooking.

RAISINS

Raisins are dried grapes. Seedless raisins are available, made from either seedless grapes or from seeded grapes that have their pits removed during processing. Seeded raisins, which contain their pits, are also on the market, but are of lesser importance to foodservice. The color of raisins can range from golden yellow to dark brown. Very light-colored raisins are often bleached during processing. Currants are small dark raisins produced mainly in Greece. Sultanas are large light-colored raisins. Baking raisins are moist raisins, therefore weighing more than loose raisins. The important raisin varieties are:

Natural seedless, derived from Thompson seedless

Golden seedless, derived from Thompson seedless

Sun ripened

Muscat

Zante currant

Packs: 20-lb (9.0-kg) cartons.

1-lb (450-g) boxes.

6-oz (170-g) individual packs, baking raisins.

Other packs are available.

Sizes: Seedless and seeded raisins are sized as midget, small, or select. Unseeded raisins are sized from one to four crowns, four being the largest. The sizes vary according to

variety; a select Thompson seedless is about as big as a Muscat midget.

Kitchen yields: 6½ oz (180 g) = 1 cup (0.24 l) seedless raisins.

6 oz (170 g) = 1 cup (0.24 l) baking raisins.

Calories: One cup (0.24 l) seedless = 480 calories. One cup (0.24 l) baking raisins = 240 calories.

RASPBERRIES (BUSHBERRIES)

For Fresh Raspberries, see the Fruits section.

CANNED RASPBERRIES

Packs: #10 cans, 6 per case.

#303 cans, 15.6 oz (440 g), 12, 24, or 36 per case.

Serving size: ½ cup (0.12 l) fruit and juice.

Weights: One #10 can = 6 cups fruit and juice.

Kitchen yields: One #10 can = 22 servings with juice.

One #303 can = 4 servings fruit and juice.

FROZEN RASPBERRIES

Available straight and with sugar added at the ratio of 4 parts fruit to 1 part sugar. Also available loose as IQF.

Packs: #10 cans, 6 per case.

30 lb (13.5 kg), sugar added.

28 lb (12.6 kg), straight pack.

Weight: One #10 can = 6½ lb (3 kg).

Serving size: ½ cup (0.12 l) fruit and juice.

Kitchen yields: One #10 can = 22 servings.

Calories: ½ cup (0.12 l), sweetened with juice = 130 calories.

RHUBARB, FROZEN

Frozen rhubarb is usually sweetened. For Fresh Rhubarb, see the Vegetables section.

Packs:

2½-lb (1.1-kg) packages, 12 per case.

25-lb (11.2-kg) cans.

Serving size: 1 cup (0.24 l).

Kitchen yields: One 25-lb (11.2-kg) can = 45 servings.

One 2½-lb (1.1-kg) package = 4 servings.

Calories: 1 cup, 9½ oz (270 g), with sugar = 385 calories.

RICE VARIETIES AND PRODUCTS

The basic rice varieties are long grain, medium grain, and short grain. Long grain rice varieties are commonly used in foodservice. Rice varieties are listed in the International Ingredients section under their respective geographical areas and names. Wild rice is listed separately. Formulas for Rice and Risotto are listed in the Recipes section.

Industry terms:

White rice: Polished long-grain rice.

Converted rice: Rice steam-treated to retain vitamins.

Brown rice: Whole, unpolished grain.

Instant rice: Rice that is fully cooked and then dehydrated.

Specialty varieties: Long grain and wild rice blends.

Flavored rice mixes.

Arborio rice, used for risotto.

Packs: 1 lb (450 g), 10 lb (4.5 kg), 25 lb (11.2 kg), 50 lb (22.4 kg), and 100 lb (45 kg).

Serving size: ½ cup, 3½ oz (100 g) cooked. Serving sizes vary greatly.

Kitchen yields: 1 cup (0.24 l) raw = 6.6 oz (185 g).

White rice:

1 cup (0.24 l) = 3 cups (0.7 l) cooked rice.

1 lb (450 g) = 6 cups cooked rice.

3 lb (1.3 kg) = 1 gal (3.8 l) cooked rice.

Converted rice:

1 lb (450 g) = 7½ cups cooked rice.

2 lb (900 g) = 1 gal (3.8 l) cooked rice.

Brown rice:

1 cup (0.24 l) = 3 to 4 cups cooked rice.

1 lb (450 g) = 8 cups (1.9 l) cooked rice.

Use more liquid and longer cooking time than for white rice.

Instant rice:

1 lb (450 g) = 14 cups cooked rice.

Wild rice and long grain blend:

1 lb (450 g) = 12 cups (2.2 l) cooked rice.

Rice for Stuffing: 2½ gal (9.5 l) cooked rice = 100 chicken breasts.

Calories: 1 cup, 7 oz (200 g), cooked, long grain = 225 calories.

1 cup, 6½ oz (180 g), raw, long grain = 670 calories.

RISOTTO RICE

Arborio rice, either imported or domestically produced, is preferred.

Packs: By weight, often 1 lb (450 g) and 5 lb (2.2 kg).

Serving size: 1 cup (0.24 l), cooked, as main course.

Kitchen yields: 1 cup (0.24 l) raw = 6.6 oz (185 g).

1 lb (450 g) = 2½ cups (0.6 l).

1 lb (450 g) = 8 servings.

SAGO

Granular starch made from palm trees.

Use is similar to that of tapioca.

SALAD DRESSINGS

Available ready for use or as spice blend.

See the Recipes section for Formulas for Dressings.

Packs: 1-gal (3.8-l) jars, 4 per case.

Serving size: 1 oz (28 g), amounts vary greatly.

SALSIFY, CANNED

Also called *oyster plant*, this vegetable is used most often canned. It resembles white asparagus in shape and parsnip in flavor when cooked. For Fresh Salsify, see the Vegetables section.

Packs: #10 cans, 7-lb (3.1-kg), 6 per case.

15-oz (420-g) cans, 24 per case.

Drained weights: One #10 can = 5 lb
10 oz (2.5 kg).

One 15-oz can = 12 oz (340 g).

Serving size: 4 oz (112 g).

Kitchen yields: One #10 can = 20 to 22
servings.

One 15-oz can = 3 servings.

SALT VARIETIES

Foodservice distinguishes between regular salt and kosher salt. Salt has become a boutique item, and some varieties can be pricey. Kosher salt is coarser than regular salt. Blended salts are spices blended with salt and are of lesser importance to foodservice.

Packs: By weight. Usually 1-lb (450-g) containers are convenient.

SEA SALT

Sea salt, sometimes marketed under the French name *fleur de sel*, is ocean salt produced by evaporation. To what extent it is better than regular salt is up to the user. It is much more expensive.

Packs:

14-oz (400-g) containers, 12 per case.

24-oz (670-g) containers.

55-lb (24.7-kg) barrels.

Other packs are available.

SAUCE AND SOUP BASES

Sauce and soup bases are available as refrigerated paste, dry storage paste, powder, and cubes. Most have high salt content. Flavors and yields vary greatly, and careful shopping is required to select the products best suited for the operation. Reduced natural sauces and stocks have become available in 1-lb (450-g) flash-frozen bricks.

SAUERKRAUT

Available mostly canned, although fresh sauerkraut might be available locally. Varieties include shredded kraut, chopped kraut, sweet sour kraut, and kraut with either celery seeds or caraway seeds. *Kraut* is the German name for cabbage.

Packs: #10 cans, 6 per case.

Drained weights: 80 oz (2.2 kg).

Serving sizes: 1½ oz (42 g), with hot dog.
½ cup (0.12 l), 112 g, cooked as
vegetable.

Kitchen yields: One #10 can = 50
servings with hot dogs.

One #10 can = 20 servings as
vegetable.

Calories: ½ cup (0.12 l), 112 g, solids
and liquid = 20 calories.

SEAWEED VARIETIES, DRIED

See Japanese Foods in the International
Ingredients section.

SESAME PRODUCTS

For sesame paste (tahini paste), see
Middle Eastern/ North African Foods in
the International Ingredients section.

SESAME SEEDS

The item is used in baking. Sesame seeds
can become rancid and should be pur-
chased in small quantities and stored
refrigerated. White and black seeds are
available.

Pack: By weight, often 10-lb (4.5-kg) bags.

SHALLOTS

Shallots are small bulbs, resembling
onions, with mild flavor. Used for flavor-
ing sauces and as a vegetable. For Fresh
Shallots, see the Vegetables section.
Chopped shallots have become available
dehydrated, frozen, or packed in oil.

Packs: Vary according to packer.

Kitchen yields: Use as is. There is no
waste.

SHORTENING

Solid, all-vegetable fat used in baking.

Packs: Vary from 1 lb (450 g) to 50 lb
(22.5 kg).

Calories: 1 oz (28 g) = 260 calories.

SNAILS

See Escargots.

SOY PRODUCTS

For additional Soy Products, *see* Japanese Foods in the International Ingredients section.

SOY FLOUR

Soy flour has no gluten and must be blended with at least 50 percent wheat flour for baking. There are three types of flour:

- Full-fat flour with about 20 percent fat
- Low-fat flour with about 6 percent fat
- Defatted flour with less than 1 percent fat

Packs: By weight.

SOY MILK

The product is available flavored, such as with vanilla and carob. Light Soy Milk has less than 1 percent fat; Extra Soy Milk is fortified with additional vitamins and calcium.

Packs: ½ pints (0.24 l), 24 per case.
Quarts (0.9 l).

1-l (33.8-oz) cartons, 12 per case.

Serving size: 1 cup (0.24 l).

Calories: 1 cup (0.24 l) = 100 calories.

1 cup (0.24 l), Extra = 135 calories.

1 cup (0.24 l), Light = 93 calories.

SPAETZLE

Small dumplings originally from southwestern Germany and neighboring Switzerland; available dried and ready to boil. *See* the Recipes section for homemade Spaetzle formula.

Pack: 1-lb (450-g) boxes.

SPINACH

Available canned and frozen. For Fresh Spinach, *see* the Vegetables section.

CANNED SPINACH

This product is seldom used in commercial foodservice because frozen spinach is better suited. Canned spinach is available chopped and as whole leaf.

Packs: #10 cans, 6 per case.

Drained weights: One #10 can = 60 oz (1.7 kg).

Serving size: ½ cup, 3½ oz (100 g).

Kitchen yields: One #10 can = 17 servings.

Calories: ½ cup, 3½ oz (100 g) = 25 calories.

FROZEN SPINACH

Available whole leaf, chopped, and IQF.

Packs: 3-lb (1.3-kg) boxes, chopped or whole, 12 per case.
2-lb (900-g) bags, IQF, 12 per case.

Serving size: ½ cup (100 g).

Kitchen yields: One 3-lb (1.3-kg) box = 12 servings.

Calories: ½ cup (100 g) = 23 calories.

SQUASH, FROZEN

For Fresh Squash Varieties, *see* the Vegetables section. Pumpkin is listed separately. A number of squash varieties are available frozen, including both sliced zucchini and yellow squash, as well as cooked squash purée. Summer squash has a thin skin; winter squash has a much harder rind.

Packs:

Squash purée: 4-lb (1.8-kg) boxes, 12 per case.

Zucchini or yellow squash: 2-lb (900-g) IQF boxes, 12 per case.
20-lb (9-kg) IQF boxes.

Serving size: ½ cup, 3¾ oz (105 g).

Kitchen yields: One 4-lb (1.8-kg) box squash purée = 17 servings.

One 2-lb (900-g) IQF box zucchini or yellow squash = 7 servings.

Calories:

Summer varieties: ½ cup = 3¾ oz (100 g) = 15 calories.

Winter varieties: ½ cup, 3¾ oz (105 g) = 60 calories.

STRAWBERRIES

For Fresh Strawberries, *see* the Fruits section. Canned strawberries are seldom used in commercial foodservice.

FROZEN STRAWBERRIES

Available sliced and mixed with sugar, or IQF whole. IQF berries will get very soft when defrosted.

Packs:

Sliced sweetened berries: 6½-lb (3-kg) cans.
10-lb (4.5-kg) cans.
30-lb (13.5-kg) tins.

Whole berries IQF: 30-lb (13.5-kg) cartons.

Serving sizes: ½ cup (0.12 l), sliced fruit and juice, as dessert.

¼ cup (0.06 l) as garnish with ice cream.

Kitchen yields: One 6½-lb (3-kg) can = twenty-two ½ cup servings.

One 6½-lb (3-kg) can = forty-four ¼ cup servings.

One 10-lb (4.5-kg) can = thirty-three ½ cup servings.

One 10-lb (4.5-kg) can = sixty-six ¼ cup servings.

Calories: ½ cup (0.12 l), sweetened, sliced = 80 calories.

STRUDEL DOUGH

See Filo Dough Products or *see* homemade Filo Dough or Strudel Dough in the Recipes section.

SUGAR VARIETIES

Sugar used in foodservice is normally purchased as granulated sugar and confectionary sugar. The size of the granules can range from coarse to fine. The fineness of ground sugar is designated with X's; the more X's, the finer the sugar. Brown sugar contains some molasses. Corn sugar has become an accepted sweetening agent in many food products.

BROWN SUGAR

Weight: 1 cup = 7 oz (200 g).

CONFECTIONARY (10X) SUGAR

Weight: 1 cup = 4.5 oz (125 g).

Calories: ¼ cup (30 g) = 100 calories.

GRANULATED SUGAR

Weight: 1 cup, 7 oz (200 g).

Calories: 1 tb = 15 calories.

One cup, 7 oz (200 g) = 770 calories.

PULLED SUGAR

Ready-to-use pulled sugar is now available under the trade name *Micro Magic* and needs only gentle warming to become pliable.

Pack: By kilo (35 oz).

SWEET POTATOES AND YAMS

The names *yams* and *sweet potatoes* are often used interchangeably, although there is a difference between the two products. For Fresh Sweet Potatoes and Yams, *see* the Vegetables section.

CANNED YAMS

Whole, cut, and mashed (solid-pack) yams are available. Whole and cut sweet potatoes are available in syrup pack, vacuum pack, and dry pack.

Packs: #10 cans, 6 per case.

#3 vacuum cans, 12 per case.

Whole potato counts and sizes in #10

cans: 40 to 50, 2½ oz (70 g) each.

30 to 40, 3¼ oz (90 g) each.

20 to 30, 4½ oz (125 g) each.

Serving size: One 4½-oz potato.

Kitchen yields: One #10 can syrup pack = 25 servings.

One can #3 vacuum pack = 5 servings.

Calories: 4½ (125 g) syrup pack = 175 calories.

½ cup (125 g) mashed = 140 calories.

FROZEN YAMS

Available whole, sliced, and in patties.

Packs: 2½-lb (1.1-kg) boxes, whole or sliced, 12 per case.

5-lb (2.2-kg) boxes, sliced, 6 per case.

24-lb cases of one hundred sixty-eight

2-oz (56-g) patties.

Serving size: 3 oz (85 g), or one 2-oz (56-g) patty.

Kitchen yields: 2½ lb (1.1 kg) = 13 servings. 5 lb (2.2 kg) = 26 servings.

Calories: 3 oz (85 g), candied = 130 calories.

SYRUP VARIETIES

See Pancake Syrup.

TAMARILLOS, FROZEN PULP

Tamarillos are often called *tree tomatoes*. For information about Fresh Tamarillos, see the Tropical Fruits section.

Pack: 14-oz (400-g) packages. Other sizes are available from specialty purveyors.

Serving size: ½ cup (100 g).

Calories: ½ cup (100 g) = 90 calories.

TALLOW AND RELATED PRODUCTS

SILICONE PLASTIQUE

Food-safe putty-like mixture that can be used to make molds.

Packs: 1 lb (450 g) to 50 lb (22.5 kg).

TALLOW

Tallow is used to mold or carve center-pieces for buffets. It can be purchased ready to use. For homemade Tallow, see the Recipes section. Types of tallow available include modeling tallow, supreme and super supreme modeling tallow, carving tallow, and casting tallow. Also available is a wide variety of ready-to-use tallow sculptures.

Colors: White, butter color, cheddar cheese color, and chocolate color.

Packs: 25 lb (11.2 kg) and 50 lb (22.5 kg).

TALLOW COLORING GELS

Gels that blend evenly with tallow are available in a variety of colors.

TAPIOCA

Starch made from the cassava plant, available as flour and as granules. The

flour is often used for baking, and the granules as thickener. Minute tapioca is made of ground tapioca pearls.

Packs:

Flour: 24-oz (680-g) packages.

Granules: 1-lb (450-g) packages, 24 per case.

Calories:

Flour: 1 oz (28 g) = 100 calories.

Granules: 1 oz (28 g) = 30 calories.

TEA VARIETIES

Fermented tea leaves are referred to as *black tea*, and unfermented tea or lightly fermented tea is called *green tea*. Both black and green teas come from the same shrub. For Asian teas, see Japanese Foods in the International Ingredients section.

Packs: By weight. Packages of loose or packed tea are available in a range of sizes.

8-oz (225-g) cans.

4-oz (112-g) cans.

Kitchen yields: 1 level tsp = 2 cups.

1 oz (28 g) = 12 tsp.

1 oz (28 g) = 24 cups.

TOMATO PRODUCTS

For Fresh Tomatoes, see the Vegetables section. Pizza Sauce is listed separately. There are also tomato and vegetable combinations on the market, such as tomatoes with chilies or tomatoes with okra.

BARBECUE SAUCES

There are many brands and varieties on the market.

Serving size: 1.5 oz (42 g).

Calories: 1.5 oz (42 g) = 50 calories.

CHILI SAUCE

Tomato-based sauce flavored with chili peppers. Some brands contain seeds.

Packs: #10 cans, 6 per case.

Other packs are available.

Kitchen yields: One #10 can = 13 cups (3.1 l).

COCKTAIL SAUCE

Packs: Many packs are available.

Serving size: 2 oz (56 g) for shrimp cocktail.

Calories: 2 oz (56 g) = 90 calories.

DICED TOMATOES

See Whole or Crushed Tomatoes. Specifications for diced tomatoes are basically the same.

OVEN-ROASTED TOMATOES

A new product, roasted tomato wedges packed in canola oil, is available. It has an intense roasted tomato flavor.

Pack: 1-lb (450-g) trays, packed in oil.

Calories: 3 oz = 260 calories.

SUN-DRIED TOMATOES

Available as halves, julienne, double, triple, finely chopped, and powder. Available dry or packed in oil.

Packs: Vary greatly.

Kitchen yields: Tomatoes require soaking to expand and develop flavor; the soaking liquid should be used in the preparation.

TOMATO CATSUP (KETCHUP)

Packs: #10 cans, 6 per case.

Many other packs are available.

Weight: One #10 can = 111 oz (3.1 kg).

Calories: 1 oz (28 g) = 75 calories.

TOMATO JUICE

The juice is always purchased canned. Freshly squeezed tomato juice exists only in the imagination of menu writers.

Packs: 6-oz (0.17-l) individual packs, 48 per case.
46-oz (1.3-l), cans, 12 per case.

Serving size: 6 oz (0.17 l).

Kitchen yields: One 46-oz can = 7½ servings.

Calories: 6 fl oz (0.17 l) = 35 calories.

TOMATO PASTE

Much denser product than tomato purée.

Packs: #10 cans, 6 per case.

Other packs are available.

Weight: One #10 can = 111oz (3.1 kg).

Kitchen yields: Not pertinent.

To make tomato juice with tomato paste, use one #10 can + 3 cans water = 48 cups tomato juice.

TOMATO PURÉE

Heavy purée should have about 20 to 25 percent solids; medium, about 12 percent. Lesser quality is watery and will not give the flavor needed. It should not be purchased.

Packs: #10 cans, 6 per case.

Other packs are available.

Weight: One #10 can = 104 oz (2.9 kg).

TOMATO SAUCE

Packs: #10 cans, 6 per case.

Other packs are available.

Serving size: 3 oz (85 g).

Calories: 3 oz (85 g), plain sauce = 110 calories.

WHOLE OR CRUSHED TOMATOES

Available as solid pack with no added liquid, with juice added, and with purée added. Tomatoes can be whole or crushed. Herbs such as basil are sometimes added.

Packs: #10 cans, 6 per case.

Other packs are available.

Kitchen yields: Drained weight for grade A should be 66 percent of total can contents.

Calories: 1 cup (0.24 l) solids and liquids = 50 calories.

TURNIP GREENS

For Fresh Turnip Greens, see the Vegetables section.

CANNED TURNIP GREENS

Packs: #10 cans, 6 per case.

Drained weight: One #10 can = 60 oz (1.7 kg).

Serving size: 1 cup (160 g).

Kitchen yields: One #10 can = 10 servings.

Calories: 1 cup (160 g) = 40 calories.

FROZEN TURNIP GREENS

Available chopped, with or without diced turnips.

Packs: 3-lb (1.3-kg) boxes, 12 per case.

Serving size: 1 cup (160 g).

Kitchen yields: One 3-lb (1.3-kg) box = 8 servings.

Calories: 1 cup (160 g) = 40 calories.

VANILLA PRODUCTS

Vanilla is available as whole vanilla pods, imitation vanilla concentrates, pure vanilla extracts, vanilla paste, vanilla powder, and vanilla sugar.

IMITATION VANILLA CONCENTRATES

A number of products are available in varying strengths, based on vanillin, a synthetic product.

Packs: 1-qt (0.9-l) bottles.

1-gal (3.8-l) bottles.

Other packs are available.

PURE VANILLA EXTRACTS

Packs: 1-qt (0.9-l) bottles.

1-gal (3.8-l) bottles.

Other packs are available.

Kitchen yields: 4 oz (112 g) for 5 gal (19 l) ice cream mix.

4 oz (112 g) for 50 lb (22.5 kg) dough.

VANILLA PASTE

Pure vanilla bean paste with smooth texture and still containing the vanilla seeds.

Packs: 4-oz jars.

Kitchen yields: Same strength as vanilla extracts.

VANILLA PODS

The quality depends on the location where the pods were grown and on the care exercised in curing. The best vanilla pods come from Mexico. The Bourbon grades are grown in Madagascar and on

other tropical islands. The best grades are the longest beans.

Packs: By weight. Common pack is 8 oz (225 g).

Size: Up to 10 in (254 mm) long for best grade.

Kitchen yields: Difficult to determine, because pods can be used a number of times until all flavor is extracted.

VANILLA POWDER

Available pure or mixed with vanillin, with or without ground bean specks.

Packs: By weight.

Kitchen yields: Strength varies according to product.

3 oz (85 g) for 5 gal (19 l) ice cream mix.

3 oz (85 g) for 50 lb (22.5 kg) dough.

VEGETABLE POWDERS

Natural vegetable powders are used to enhance the flavors in pasta, sauces, stocks, and soups. Available powders include beets, broccoli, carrots, corn, jalapeño, pumpkin, red bell pepper, spinach, squash, and tomato.

Packs: By weight.

Kitchen yields: Products are flavor enhancers, and yields depend on use.

VINEGARS

Plain white vinegar is used for general cooking. Flavored vinegar such as apple, grape, malt, raspberry, red wine, tarragon, white wine, and many other vinegars are used for dressings. The strength varies, and a vinegar should always be tasted before use.

Packs: 16 oz (0.47 l), 32 oz (0.94 l), and 1 gal (3.8 l). Packs are usually the same for all vinegars.

BALSAMIC VINEGAR

Special mild-flavored vinegar aged in small barrels in northern Italy, especially around the city of Modena, and usually

made in small quantities by local families. Some vinegars are very mild and pricey.

RICE VINEGAR

See Japanese Foods in the International Ingredients section.

WAFFLE PRODUCTS

FROZEN WAFFLES

A variety of packs are available, with flavor combinations such as pecans, blueberry, and others. Waffles not requiring refrigeration are also on the market. Prepared waffles can be useful for banquets.

Packs: By weight and count; packs vary greatly.

Sizes: 1.3 oz (36 g) and up.

Serving size: 2 waffles.

Calories: 2 waffles = 200 calories.

WAFFLE BATTER

Available as dry mix and as ready-to-use batter. For homemade Waffle Batter, see the Recipes section.

WALNUTS

Available in shell, shelled as granules, in halves and pieces, and as flour.

Packs: *Granules:* 4-lb (1.8-kg) cans.

Pieces: 3-lb (1.3-kg) cans.

Halves: 42-oz (1.17-kg) cans.

Kitchen yields: 1 cup halves = 60 pieces.

1 cup chopped = 4½ oz (125 g).

1 cup ground = 3½ oz (100 g).

Calories: 1 cup, 4½ oz (125 g), chopped = 790 calories.

1 cup, 3½ oz (100 g), ground = 620 calories.

WATER CHESTNUTS, CANNED

In Asian cooking, a number of different varieties of water chestnuts are clearly distinguished. The canned product is the accepted foodservice standard. For Fresh Water Chestnuts, see Chinese Foods in the International Ingredients section. Canned water chestnuts are available sliced or whole.

Pack, whole or sliced:

#10 cans, 6 per case.

29-oz (840-g) cans.

10½-oz (300-g) cans, 24 per case.

Kitchen yields: One #10 can, whole = 35 oz (1 kg) drained.

One 10½-oz (300-g) can, whole = 5 oz (140 g) drained.

One #10 can, sliced = 40 oz (1.1 kg) drained.

WILD RICE

Water grass seeds, still growing wild in Minnesota, Wisconsin, California, and Canada. The quality is determined by the number of broken kernels per pound and by the size of the kernels. Uniform kernel size is important to ensure even cooking. Precooked wild rice has become available.

Quality distinctions:

Long-grain blends = 10 percent broken kernels.

Medium-grain blends = 25 percent broken kernels.

Cracked blends = 50 percent broken kernels.

Packs:

100-lb (45-kg) bags or boxes.

25-lb (11.2-kg) pails.

5-lb (2.2-kg) boxes or cans, 6 per case.

1-lb (450-g) boxes or cans, 12 per case.

Other packs are available.

Serving size: ½ cup cooked = 1.5 oz raw.

Kitchen yields: 1.5 oz raw = ½ cup cooked.

1 lb (450 g) = 10 servings.

Long-grain blends require 3 times the liquid by volume. Cooking time is 60 minutes. Medium-grain blends and cracked blends require 2½ times the liquid amount by volume. Cooking time is 50 minutes. *Note:* Yield is increased if product is soaked overnight. Soaking water should be used in cooking to preserve nutrients.

Calories: ½ cup cooked = 170 calories.

WRAPS

Flour tortillas for filling with salads, meats, and many other fillings. Varieties available are made with white corn, rye, wheat or whole wheat flour, and flavored with basil, jalapeño peppers, spinach, tomatoes, and many other flavor combinations.

Packs: 6-in (150-mm), case of 32 packages of 12 each.
8-in (200-mm), case of 20 packages of 12 each.

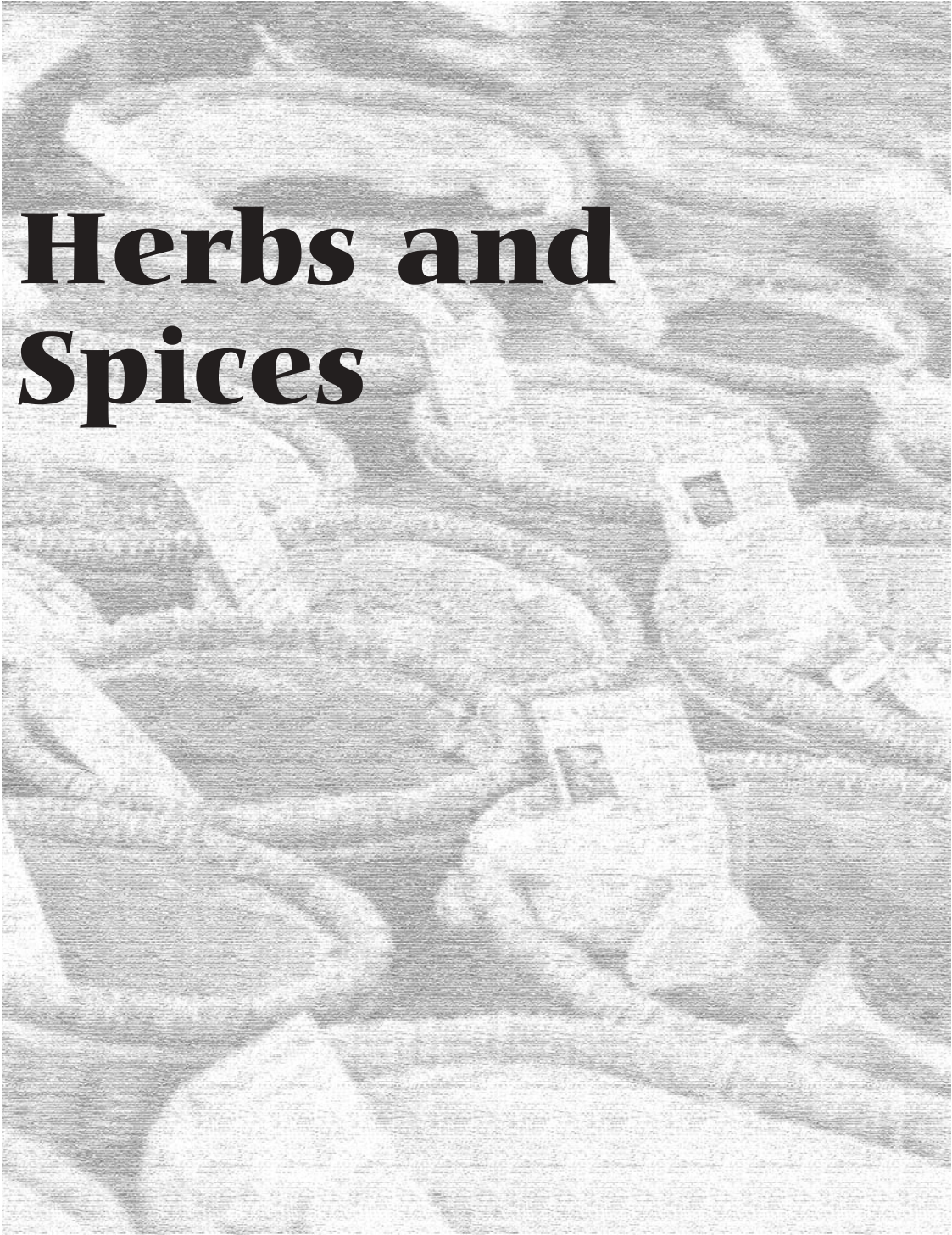
10-in (250-mm) and 12-in (300-mm), cases of 10 packages of 12 each.

Sizes: 6 in (150 mm), 8 in (200 mm), 10 in (250 mm), and 12 in (300 mm).
Most common size is 12 in.

Calories: One 2.5-oz (70-g) wrap = 180 calories.

YEAST/FRESH

See the Baked Goods section.



Herbs and Spices

Herbs and spices are listed alphabetically, including fresh, frozen, and dried herbs and spices. Kitchen yields are seldom indicated because they depend on taste and freshness of the products. The packs of dried herbs and spices vary greatly. Better-quality products are usually packed in small containers.

ACHIOTE

See Hispanic Foods in the International Ingredients section.

ALLSPICE

The dried berry of a tree of the myrtle family. It is called *allspice* because the flavor resembles a combination of clove, cinnamon, and nutmeg. Also known as *Jamaican pepper*.

Packs: 12 oz (340 g), whole.
20 oz (560 g), ground.

ANISE SEEDS

Licorice-flavored seeds frequently used in baking. The plant is native to the Middle East, a member of the parsley family. Also called *sweet cumin*.

Packs: 18 oz (500 g), ground.
16 oz (450 g), whole.

ANISE, STAR

See Chinese Foods in the International Ingredients section.

BASIL

DRIED BASIL

Pack: 6 oz (170 g), whole.

FRESH BASIL

There are many varieties on the market.

Season: Peak in summer.

Pack: Bunches or by weight.

BAY LEAVES

DRIED BAY LEAVES

Pack: 10 oz (280 g), whole.

FRESH BAY LEAVES

Season: Year-round.

Packs: Sold on stems, normally by the dozen.

Kitchen yields: Approximately 40 leaves on each stem.

BEE BALM

American herb of the mint family, genus *Monarda*. The plant has pretty scarlet, mauve, or white flowers, but it is the leaves that give flavor to tea. Varieties include bergamot or wild bergamot with an orange taste and lemon mint with a lemony-mint taste. Usually sold dried.

Pack: 6 oz (170 g) or smaller packs, whole.

BORAGE

Annual plant native to the Middle East. The fuzzy gray-green leaves have a cucumber-like taste, similar to burnet. The fresh leaves are used in fruit cups, punches, tisane, salads, and as garnishes. The flowers are deep blue, star-like, and are available fresh, dried, or candied.

Pack: 6 oz (170 g), whole.

CARAWAY SEEDS

Half-moon-shaped seeds used in breads, especially in Germany. Caraway seeds are also used in alcoholic beverages.

Packs: 18-oz (500-g) boxes.

CARDAMOM/CARDAMON

The seeds are used to flavor Indian dishes and coffee in Arabic countries; the ground seeds are used in baking. Green pods indicate that they have been dried by forced heat, yellowish pods have been dried in the sun, and white pods have been bleached with sulphur.

Packs: 12 oz (340 g), whole, green.
18 oz (450 g), ground.

CAYENNE

Very hot pepper sold as red pepper flakes or ground as cayenne pepper. Much of the production is used in manufacturing spice blends.

Pack: 16-oz (450-g) boxes.

Kitchen yields: According to taste; but should be used sparingly.

CELERY SALT

Pack: 40 oz (1.12 kg).

CELERY SEEDS**GROUND CELERY SEEDS**

Pack: 16 oz (450 g).

WHOLE CELERY SEEDS

Pack: 18 oz (500 g).

CHERVIL

Delicate herb related to parsley, having smaller, more fragile leaves, with a scent of anise. The leaves are used in sauces and salads.

DRIED CHERVIL

Pack: 4 oz (112 g).

FRESH CHERVIL

The fresh product has a limited shelf life.

Pack and size: Bunches in various sizes.

CHILI POWDER

Blend of ground red chili peppers, cumin seeds, oregano, salt, and garlic. For Chiles, *see* Hispanic Foods in the International Ingredients section.

Packs: Vary, frequently 12-oz (340-g) cans.

CHIVES**DRIED CHIVES**

Packs: 2 oz (56 g); other packs are available.

Kitchen yields: There is no waste.

FRESH CHIVES

Green, slender, round, reed-like member of the onion family. Somewhat similar to scallions in appearance, which are also known as *green onions*, but chives are thinner. Available year-round, cut or potted.

Packs: Bunches in various sizes.

Kitchen yields: Sizes of bunches vary greatly; they are smallest in early spring.

3½ average bunches = 1½ cups, cut.

1½ cups, cut = 2 oz (56 g).

FROZEN CHIVES

Packs: 4 oz (112 g). Other packs are available.

CILANTRO**DRIED CILANTRO**

Pack: 3 oz (80 g).

FRESH CILANTRO

The leaves of the coriander plant. Pungent, peppery leaves commonly used in Chinese, Southeast Asian, and Mexican cooking. Also known as *Chinese parsley*.

Packs: 30 bunches per carton.

CINNAMON

The reddish-brown inner bark of a tree growing in tropical countries. Cinnamon is used in baking and desserts, as well as in curries.

CINNAMON STICKS

Also called *quills*.

Packs: 8 oz (225 g) for 3-in sticks.

8 oz (225 g) for 6-in sticks.

1 lb (450 g) for 10-in sticks.

1 lb (450 g) for 12-in sticks.

GROUND CINNAMON

Ground cassia bark is often sold as inexpensive cinnamon.

Pack: 18 oz (450 g).

CLOVES

The unopened dried buds from an ever-green myrtle tree.

GROUND CLOVES

Pack: 18 oz (450 g).

WHOLE CLOVES

Pack: 12 oz (340 g).

CORIANDER**GROUND CORIANDER**

The seeds are dried and ground primarily for use in curry powder, desserts, breads, and custards. Pastrami is covered with crushed coriander seeds. Fresh coriander leaves are called *cilantro*.

Pack: 14 oz (400 g).

WHOLE CORIANDER

Pack: 10 oz (280 g).

CUMIN

Seeds from a plant in the parsley family. The dried seeds are used in Indian, Pakistani, and South American cooking. Cumin is one of the major seasonings in curry mixtures.

GROUND CUMIN

Pack: 20 oz (560 g).

WHOLE CUMIN

Pack: 15 oz (420 g).

CURRY

See Indian Foods in the International Ingredients section.

DILL

Delicate, feathery green annual herb.

DILL SEEDS

Pack: 16 oz (450 g).

DILL WEEDS DRIED

Pack: 5 oz (140 g).

FRESH DILL

Pack: Bunches.

EPAZOTE

See Hispanic Foods in the International Ingredients section.

FAJITA SEASONING

See Hispanic Foods in the International Ingredients section.

FENNEL SEEDS

For Fresh Fennel/Anise, see the Vegetables section.

GROUND FENNEL SEEDS

Pack: 12 oz (340 g).

WHOLE FENNEL SEEDS

Pack: 12 oz (340 g).

FENUGREEK

Ancient herb used as food and in incense. Used as a flavoring in chutney and baked goods.

FENUGREEK LEAVES

See Indian Foods in the International Ingredients section.

GROUND FENUGREEK SEEDS

Pack: 12 oz (340 g).

WHOLE FENUGREEK

Pack: 20 oz (560 g).

GARLIC, DEHYDRATED

For Fresh Garlic, see the Vegetables section. Dehydrated ground garlic is available both mixed with salt and plain.

Packs: 24 oz (1.1 kg), granular garlic.

20 oz (960 g), minced garlic in jars.

20 oz (960 g), garlic powder.

48 oz (1.3 kg), garlic salt.

Kitchen yields: 1 tb dehydrated = 1 tb minced fresh garlic.

GARAM MASALA

See Masala under Indian Foods in the International Ingredients section.

GINGER

Ginger is available in many forms. *See* the Groceries section.

GUMBO (FILÉ POWDER)

Spice and thickening agent made from the bark of the sassafras tree. It is used in Louisiana-style cooking.

Pack: By weight.

Kitchen yields: 1.6 oz (45 g) for 1 gal (3.8 l) soup.

JUNIPER BERRIES, WHOLE

Dried juniper berries are used in game dishes. They are also a major flavoring agent in gin.

Pack: 12 oz (340 g).

LAVENDER BLOSSOMS

Perennial aromatic herb, used to make a fragrant herb tea.

Pack: 4 oz (112 g).

LEMON BALM (MELISSA)

Herb with lemon-scented leaves used to make herb tea, sometimes combined with mint, lavender, rosemary, or sage.

Pack: 4 oz (112 g).

LEMON GRASS

See Thai/Vietnamese Foods in the International Ingredients section.

MACE GROUND

Mace is the ground shell or filmy red membrane of the nutmeg, the seed of a tropical tree. Mace is normally sold ground or in flakes, sometimes called *blades*.

Pack: 16 oz (450 g).

MARJORAM

Perennial herb in the mint family, closely related to oregano, available fresh and dried.

DRIED MARJORAM

Pack: 4 oz (112 g).

FRESH MARJORAM

Packs: Bunches.

MINT

Perennial herb used in herb teas or tisane, sometimes mixed with other flavors. There are many varieties.

DRIED MINT

Packs: Many packs are available, also tea blends with other herbs.

FRESH MINT

Packs: Bunches.

MUSTARD SEEDS

For Prepared Mustards and Mustard Powder, *see* the Groceries section.

Pack: 24 oz (670 g).

NUTMEG**GROUND NUTMEG**

See Mace.

Pack: 20 oz (560 g).

WHOLE NUTMEG

Pack: 20 oz (560 g).

ONIONS, PROCESSED

See the Groceries section.

OREGANO

Tender perennial herb with many culinary uses; the blossoms are also edible. Also called *sweet marjoram*. Greek oregano has a more pungent flavor than other varieties.

DRIED OREGANO

Pack: 6 oz (170 g), whole leaves.

FRESH OREGANO

Packs: Bunches.

PAPRIKA

Red spice made from ground red peppers, which originated in the Americas. The plant belongs to the chili family and was brought to Europe by Columbus. Paprika

was introduced to Hungary by the Turks, who occupied the land in the early sixteenth century, and became the national spice of Hungary. Hungarian paprika is classified as exquisite delicate, delicate, nobelsweet, semisweet, and hot. Spanish paprika is usually mild and is often used more for color than for flavor.

Packs: 18 oz (500 g). Other packs are available.

PEPPER

A spice in the shape of peppercorns. It grows on vines in the form of small berries. The berries are harvested and dried. They turn black after drying and, when ground, produce black pepper. The inside of the peppercorns is white. Ground pepper loses flavor rapidly, which is a good reason for presenting pepper in restaurants freshly ground. Green and pink peppercorns are available pickled in vinegar or dried. Unripe peppercorns are generally not harvested, as they can be slightly toxic. There are many flavored pepper blends and many grades of ground pepper on the market. For Green Peppers, see the Vegetables section.

Packs: Between 10 oz (280 g) and 20 oz (560 g).

PICKLING SPICE

Blend of spices, normally including peppercorns, mustard seed, coriander, bay leaves, and hot peppers.

Pack: 12 oz (225 g).

POPPY SEEDS

Tiny, slate-blue seeds used in baking, often sprinkled on rolls or bread. Popular in baking and in central European deserts. Poppy seeds carry oil and can become rancid and should be stored in a freezer.

Pack: 22 oz (610 g). Larger packs for baking are available.

ROSEMARY

Perennial herb with pine-needle-like leaves and tiny blue flowers. Available fresh and dehydrated, whole and ground.

DRIED ROSEMARY

Pack: 12 oz (340 g).

FRESH ROSEMARY

Packs: Bunches.

ROSEMARY SKEWERS

The rosemary branches used as skewers.

Size: Varies, up to 18 in (50 cm) long.

Pack: Sold by the piece.

SAFFRON

Saffron is imported from Spain, Greece, Turkey, and India. Imitation saffron is available from Mexico. Buy only whole saffron; ground saffron may be adulterated. Genuine saffron is very expensive. Spain is no longer the predominant producer of saffron, because the labor costs are high.

Packs: By weight. Various packs, from 1 oz (28 g) to 1 lb (450 g).

Kitchen yields: Allow flower stigmas to steep in liquid to get better flavor extraction.

$\frac{1}{8}$ oz (3.5 g) flavors 3 gal (11.4 l) stock.

SAGE

Perennial, hardy herb native to the Mediterranean.

DRIED SAGE

Pack: 6 oz (170 g), ground.
3 oz (85 g), whole leaves.

FRESH SAGE

Season: Available year-round.

Pack: Sold by weight, usually 2 oz (56 g) minimum.

SAVORY

Herb in the mint family used in game dishes, salads, and with vegetables, with a taste resembling that of thyme.

Summer savory is an annual; winter savory is a hardy perennial with a similar flavor.

DRIED SAVORY

Pack: 6 oz (170 g).

FRESH SAVORY

Season: Available year-round.

Pack: Sold by weight, usually 2 oz (56 g) minimum.

SESAME SEEDS

One of the world's oldest spices native to India, and an important oil seed crop. The seeds are white, red, brown, or black. The seeds are used in baked goods, candies, and sauces. *See* the Groceries section for Sesame Products.

Packs: 16 oz (450 g), black seeds.
22 oz (610 g), white seeds.

SORREL (SOUR GRASS)

Stems are removed and leaves are boiled in their own juices for flavoring; fresh leaves can be used in salad mix.

Season: Spring and summer.

Packs: By weight, usually 20-lb (9.1-kg) crates.

Kitchen yields: 1 lb (450 g) as purchased will flavor 1 qt (0.94 l) soup.

TARRAGON

There are two varieties, French and Russian Tarragon. The French variety

(*estragon*) has a superior, but stronger, taste and should be used sparingly. Its most common uses are in Béarnaise sauce and in tarragon vinegar.

DRIED TARRAGON

Pack: 2 oz (56 g), leaves.

FRESH TARRAGON

Pack: Bunches.

TARRAGON PRESERVED IN VINEGAR

Packs: 1-qt (0.9-l) jars.

THYME

Culinary perennial herb native to the Mediterranean and whose name is derived from the Greek word for incense, *thymon*, because it was originally used in incense.

Season: Available year-round.

Pack: Sold by weight, usually 2 oz (56 g) minimum.

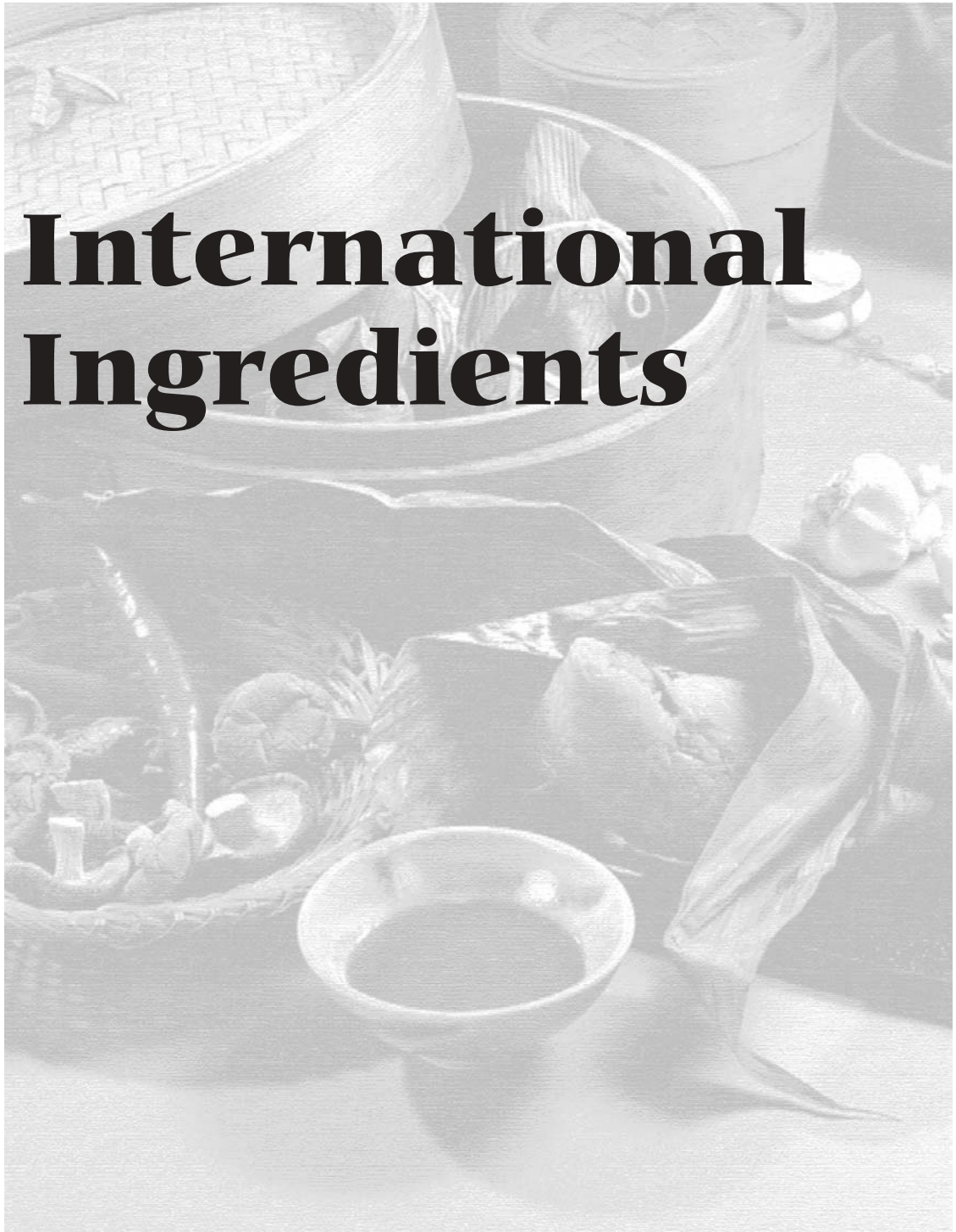
TURMERIC

Root vegetable resembling ginger. The ground root is used as a spice. Because it is yellow, it is sometimes erroneously called saffron. Turmeric is an important ingredient in curry powder and as a coloring agent in mustards.

Packs: By weight.
16-oz (450-g) packages, ground turmeric.

Size: Roots are 3 to 5 in (12 to 20 cm) long.

International Ingredients



Space restrictions and practicality made selecting international ingredients a challenge. The selection is limited to products that are usually available in specialty markets and have broad customer appeal. The kitchen yields are, by necessity, often vague because they depend on final use and taste. For best results, most ingredients should be carefully tasted and tested before use. The products are grouped as follows:

Chinese Foods
Hispanic Foods
Indian Foods
Japanese Foods
Middle Eastern/North African Foods
Thai/Vietnamese Foods

In many parts of the world processed products are readily available and used, saving preparation time. Some of these products are included here.

CHINESE FOODS

ABALONE

See the Fish, Mollusks, Shellfish, and Seafood section.

ANISE, STAR, *Ba gok/ba chio*

Star-shaped spice with licorice taste, popular in many cuisines.

ARTICHOKES, CHINESE

Small, slender tubers growing under the soil on a mint-like plant. The crisp-fleshed tubers can be eaten cooked or raw. Also known as *chorogi* or *knotroot*.

Pack: By weight

BAMBOO SHOOTS, *Chu sun*, spring, *dung sun*, winter

Bamboo shoots are the young shoots of the bamboo plant growing from an underground rootstock. Available shredded and whole, salted and pickled. For Canned Bamboo Shoots, see the Groceries section.

Packs: By weight

Sizes: There are at least 10 edible types, varying in size and season.

Kitchen yields: The products are sold fully peeled and ready to use. There is little shrinkage.

BEAN CURD (TOFU), *Dau foo*

Available fresh, canned, and dried. There are a number of varieties; the most important are:

Water bean curd: Rather solid, all-purpose.

Dry bean curd: Solid and can be shredded or stir-fried.

Fried bean curd: Brown fried squares or cubes.

Wrapped bean curd: Round curd kept in water, usually the most delicate.

Yuba: Dried curd, available in sheets or twisted ropes.

Packs: Twelve 20-oz (560-g) packages per case.

Kitchen yields: There is no waste.

Calories: 4 oz (112 g) water bean curd = 110 calories.

BEANS

BEAN FILLING, SWEET, *Dow sa*

Sweet paste that can be made with red kidney beans or with green mung beans.

Packs: Cans in many sizes.

Kitchen yields: Use as pastry filling as is.

FERMENTED BLACK BEANS, *Dow si*

Used as a vegetable and as a spice.

Packs: Bottles, cans, or plastic bags.

Kitchen yields: Use as vegetable or as condiment sauce.

LONG BEANS, *Dau gok/Dow kok/Dow gawk*

Also known as *asparagus beans*, they resemble string beans but grow up to 3 ft (0.9 m) long. The vegetable is related to southern cowpeas, growing on vines, with a taste resembling asparagus.

Size: Up to 1 yard in length, but usually sold younger and only a little more than 1 ft in length.

Packs: By weight. 35 lb (15.6 kg) and 40 lb (16.4 kg) are common.

Kitchen yields: There is some trimming to do; the challenge is to find a pot large enough.

BEAN SPROUTS

See the Vegetables section for Fresh Mung Bean Sprouts and Soy Bean Sprouts.

BÊCHE DE MER (DRIED), *Hai sum*

Sea slug in cucumber shape, available dried.

Size: 8 in (200 mm) long. Product will triple in size after soaking and cooking.

Packs: By weight.

Kitchen yields: ½ lb (225 g) soaked and softened, cut in ½-in (12-mm) chunks = 2½ cups.

BIRD'S NESTS, *Yin waw/yen wuo*

The dried saliva of birds nesting in caves and cliffs in South Asia. Available dried and cleaned, whole, and as bits. Clean whole nests are the most expensive products.

Packs: 6-oz (170-g) packages.

Kitchen yields: Soak and add to soups.

1½ oz (42 g) = 1 qt (0.47 l) soup.

2 oz (56 g) dry flakes = 1½ cups (0.36 l).

BITTER MELONS (CHINESE KARVIS)

Bitter melons are larger and less ridged than the Indian karvis.

Packs: By weight.

Sizes: 6 oz (170 g) to 10 oz (280 g).

Kitchen yields: 1 lb (450 g) = 13 oz (370 g), cleaned.

BLACK CHICKEN

Also called *silky chicken*. Chicken with black skin and dark meat, usually boiled for soup.

Size: 1.5 lb (675 g) average.

Kitchen yields: See Chicken in the Meats and Meat Products section.

CABBAGE VARIETIES

BOK CHOY, *Book choi*

White cabbage with ivory-colored wide leaf stalks. Miniature versions, served whole, have become popular with Western chefs for garnishes.

Season: Year-round.

Packs: ½ crate, 30 lb (13.6 kg) to 40 lb (18 kg).

Full crate, 65 lb (29.5 kg) to 70 lb (31.5 kg).

Sizes: 16 in (400 mm) long. Miniature versions are also available.

Kitchen yields: ½ lb = 2 cups (0.48 l), shredded. About 10 percent trimming loss. Shrinkage is about 50 percent of volume.

Calories: One cup, 2½ oz (70 g), raw, coarsely shredded = 20 calories.

½ cup, 2.6 oz (72 g) cooked drained = 15 calories.

BROCCOLI CABBAGE, *Gai lan tsoi*

Longer version of American broccoli with smaller buds.

Pack: By weight and bunches sold fresh.

Size: 12 in (300 mm) to 14 in (350 mm) long.

CELERY CABBAGE, *Wong nga baak*

Other names are *Nappa cabbage* and *Peking cabbage*. Barrel-shaped, solid, light-green crinkled leaves.

Season: Year-round.

Pack: ½ crate, 30 lb (13.6 kg) to 40 lb (18 kg).

Full crate, 65 lb (29.5 kg) to 70 lb (31.5 kg).

Size: 16 in (400 mm) long.

Kitchen yields: The cabbage can be eaten raw as salad or cooked. Trimming loss is about 15 percent of weight.

FLAT CABBAGE, *Taai goo choi*

Flat cabbage with spread-out dark green leaves.

Kitchen yields: Stems and leaves can be cooked together. There is some trimming loss.

FLOWERING CABBAGE, *Choi sum*

Green vegetable with yellow flowers.

Season: Year-round.

Pack: Bunches.

Kitchen yields: The whole vegetable, including stems, can be used.

KALE, CHINESE, *Gaai laan*

Cabbage resembling flowering cabbage, but with tough stems that must be peeled.

Kitchen yields: About 20 percent trimming loss or more by weight.

MUSTARD CABBAGE, *Daai gaai choi*

Round, stalky cabbage with strong flavor, used mostly in soups or pickled.

SLIPPERY CABBAGE, *Saan choi*

Mucilaginous soup vegetable resembling flowering cabbage, but without the flowers and green stalks.

SNOW CABBAGE, *Jiu la choi*

Strong-flavored elongated cabbage resembling celery. The stalks are soft and can be cooked together with the leaves.

Kitchen yields: Little trimming loss.

CELLOPHANE NOODLES

Transparent noodles made of mung bean flour; also known as *bean thread noodles*. The noodles puff up when fried.

Packs: 8-oz (225-g) packages and larger.

Serving size: 3 oz (84 g), as base with meat.

Kitchen yields: Noodles will not expand when boiled.

Calories: 3 oz (84 g), without any additional ingredients = 270 calories.

CHICKEN FEET

Available fresh or frozen, perfectly cleaned, to be used in soups.

Pack: By weight.

Size: 1 lb (450 g) = 12 to 14 feet, average.

CHINESE CHIVES, *Gau choi*

Tall chives with a pronounced garlic taste. They are different in appearance from other chives in that their stems are flatter. Both flowers and leaves can be used in cooking. Also known as *garlic chives*.

Packs: Bunches in various sizes.

Kitchen yields: Stalks and tips can be tough and must be discarded.

FLOWERING CHINESE CHIVES, *Gau choi fa*

Single buds on long stems, those with open flowers are considered to be too old.

Pack: Bunches in various sizes.

Kitchen yields: Stalks and tips can be tough and must be discarded.

CHINESE EGGPLANT, *Ai gwa*

The vegetable can be white or purple. See the Vegetables section for additional information.

Season: Year-round.

Size: 6 in (150 mm) to 8 in to (200 mm) long.

Packs: By weight.

CHINESE PARSLEY

See Cilantro in the Herbs and Spices section.

CHINESE TURNIPS**FRESH CHINESE TURNIPS, *Lo ba***

Long white root resembling white radish, with a strong taste. Used as a cooked vegetable.

Season: Year-round.

Packs: By weight.

Size: 8 in (200 mm) to 12 in (250 mm).

SALTED CHINESE TURNIPS, *Dy to tsoi*
Shredded and cured turnips.

Packs: By weight in packages; does not require refrigeration.

Kitchen yields: Use as is; there is no waste.

CHRYSANTHEMUM BLOSSOMS

Available dried and used for tea.

Packs: 1.4 oz (40 g).

CHRYSANTHEMUM LEAVES, *Tong Ho*

Plant closely related to chrysanthemum flowers but cultivated to produce large leaves. The leaves can be eaten raw as salad or as vegetables.

Packs: By weight.

Season: Fresh leaves are usually available in winter and early spring.

Kitchen yields: The stalks should be removed. Trimming waste is about 20 percent.

COCONUTS

See the Tropical Fruits section for Fresh Coconut and the Groceries section for Canned Coconut Juice, Coconut Milk, and Shredded Coconut. Coconut, Fresh, Trimmed, used only to extract the coconut water, is also listed in the Tropical Fruits section.

CUTTLEFISH, DRIED, *Yow yu*

Dried small squid.

Packs: By weight.

Size: 6 in (150 mm) to 8 in to (200 mm) long.

Kitchen yields: Product must be soaked before use.

DATES, DRIED, RED, *Hung jo*

Also known as *jujube*; small, hard red fruits with moist interior. Also available as sweet paste.

Packs: 14-oz (400-g) and larger packages.

Serving size: 1 oz (28 g), pitted.

Kitchen yields: Product is best used as a snack.

DUCK EGGS, SALTED

Eggs soaked in charcoal brine. The egg white should be watery and the yolk pasty and orange colored. The eggs can be eaten as purchased or hard boiled.

Packs: Sold individually or by the dozen.

DUCK FEET

Available fresh or frozen, perfectly cleaned, to be used in soups.

Pack: By weight.

Size: 1 lb (450 g) = 12 duck feet, average.

DUCK WINGS

Available fresh or frozen, perfectly cleaned, to be used in soups.

Pack: By weight.

Size: 1 lb (450 g) = 6 duck wings, average.

EGGROLL WRAPPERS, *Chwin guen pei*

Packs: 1-lb (450-g) packages.

Size: 6 in (150 mm) square. Larger sizes are also on the market.

Kitchen yields: 1 package = 14 wrappers. To seal a wrapper properly, make a paste with water and flour.

FIVE SPICE POWDER, *Ng hiong fun*

Blend of ground star anise, fennel, cinnamon, peppers, and cloves.

Pack: By weight, 4 oz (112 g) and larger packages.

Serving size: ½ oz (4.5 g) per person, or according to recipe.

GEODUCK CLAMS

Often available live in Chinese markets. See the Fish, Mollusks, Shellfish, and Seafood section.

GINGER, STEM or YOUNG, *Tsee geung*

Young ginger with pink shoots and a rather naked appearance. It has a milder flavor than other ginger. For Fresh Ginger, see the Fruits section, and for other ginger products, see the Groceries section.

GINKGO NUTS

The seeds of the ginkgo, or maidenhair, tree, popular as a park and street tree. The tree is native to Asia. The thin-shelled small nuts are surrounded by an offensive-smelling pulp, but the roasted seeds are pleasant.

Pack: By weight.

GREEN TEA

Tea varieties are listed under Japanese Foods in this section.

HOISIN SAUCE, *Hoy sin jiong*

Thick, slightly sweet condiment sauce made with sugar, water, rice vinegar, soy beans, flavors, and chili peppers.

Packs: Cans or bottles, often 8 oz (225 g).

Serving size: 1 oz (28 g).

Calories: 2 tb = 1 oz (28 g) = 70 calories.

JELLY FISH, *Hoy jit pei*

Available dried; normally whole, but sometimes shredded.

Packs: Individual fish, sold by weight.

Sizes: 2.5 in (90 mm) square and larger.

JELLYFISH, INSTANT

Fully cooked product in pouches including seasoning packages.

Pack: 5 oz (140 g).

Serving size: 1 oz (28 g) as appetizer.

Calories: 1 oz (28 g), including dressing = 10 calories.

LILY BULBS

Sliced dried bulbs.

Packs: 14-oz (400-g) and larger bags.

Serving size: 1 oz (28 g).

Kitchen yields: Products must be soaked before use and will double in size.

LILY FLOWERS, DRIED, *gum tsum*

Dried flower petals also known as *golden needles* and *tiger lily petals*. The product must be washed and boiled for 15 minutes before use.

Size: About 2 in (50 mm) long.

Packs: 6 oz (170 g) bags and larger.

Serving size: 1 oz (28 g).

Kitchen yields: One 6-oz (170-g) bag = 6 servings.

LITCHI, *Le-ee tzee*

For Canned and Dried Litchi, see the Groceries section; for Fresh Litchi, see the Tropical Fruits section.

LONGANS, *Loong gnahn*

Fruit resembling Litchi; available dried, pitted and with pits.

Packs: 4 oz (112 g) and more dried longans.

Serving size: 1 oz (28 g).

Kitchen yields: Use dried longans as a snack.

Calories: 1 oz (28 g) = 90 calories.

LOTUS ROOT, *Leen gnaw*

Available fresh, canned, or dried. The roots, when cut across, are decorative and have a lacy appearance.

Season: From July through the winter.

Packs: Fresh bulbs and dry slices are sold by weight.

14-oz (400-g) cans, water pack.

Size: 4 in (100 mm) to 8 in (200 mm) and longer, fresh bulbs.

Drained weight: 8 oz (225 g).

Kitchen yields: Fresh roots require extensive cleaning and removal of tough skin. Foodservices commonly use the canned products.

Calories: 4 oz (112 g) = 52 calories.

LOTUS SEEDS, *Leen tszee*

Available canned and dried.

Size: About ½ in (12 mm) long.

MUSHROOM VARIETIES

Dried and fresh mushrooms play an important role in Asian cooking.

CLOUD EARS, *Win yee*

Also known as *wood ears*; available dried and resembling shrunk leaves. Cloud ears are mostly sold dried, but are also sold fresh. When reconstituted or fresh, they have a slick, slippery feel and a crunchy texture.

Packs: By weight: 6-oz (170-g) and larger packages.

Serving size: 2.5 oz (70 g).

Kitchen yields: The mushrooms must be soaked.

Calories: 2.5 oz (70 g) = 50 calories.

FLOWER MUSHROOMS, *Fa gu*

Basically, lighter-colored shiitake mushrooms.

SHIITAKE (WINTER MUSHROOMS), DRIED, *Dong gwoo*

Chinese cooking normally uses dried shiitake mushrooms; for Japanese cooking fresh shiitakes are preferred.

Pack: 1-lb (450-g) bags.

Kitchen yields: The mushrooms must be soaked and the tough stems discarded. Yield is according to recipe.

STRAW MUSHROOMS

Available fresh and canned packed in water. Fresh straw mushrooms are kept in water and are sold by weight; therefore, they are heavier than other fresh mushrooms.

Packs: 1-lb (450-g) cans, 36 per case.

Drained weight: 1-lb (450-g) can = 8 oz (225 g).

Kitchen yields: The lower ends may still have to be trimmed.

WOOD EARS, DRIED

Usually available dried. The mushrooms need soaking and will stay chewy after cooking.

Packs: 1-lb (450-g) bags.

Kitchen yields: There is very little trimming needed after soaking. The mushrooms will double in size after cooking.

MUSTARD, *Gai lat, gai mo*

Mustard powder, basically the same as English mustard.

Pack: By weight.

Kitchen yields: Blend with cold water before use.

NOODLES, *Lo mein*

Wheat noodles, available with and without eggs, in different sizes. Most suppliers sell noodles fresh and dried.

FRESH SHANGHAI NOODLES, BROAD, WITHOUT EGGS

Packs: 1-lb (450-g) packages and larger.

Serving size: 4 oz (112 g) uncooked fresh noodles blended with other ingredients as main course.

Kitchen yields: Noodles cook in about 2 minutes; overcooking reduces yield.

Calories: 4 oz (112 g) = 280 calories.

FRESH WONTON THIN EGG NOODLES

Packs: 1-lb (450-g) packages and larger.

Serving size: 4 oz (112 g) uncooked fresh noodles blended with other ingredients as main course.

Kitchen yields: Noodles cook in about 2 minutes; overcooking reduces yield.

Calories: 4 oz (112 g) = 300 calories.

OKRA, *Sing kuar/Sing gwa*

Chinese okra is dark green and heavily ridged.

Size: 10 in (250 mm) to 20 in (500 mm) long.

Packs: 20-lb (9-kg) lugs.

Kitchen yields: Okra must be peeled, and because there are deep ridges, the peeling losses can be 20 percent of weight. 1 lb (450 g) = 10 oz (280 g) to 12 oz (340 g), cleaned.

100-YEAR-OLD EGGS

Preserved duck eggs, artificially aged by applying a coating of wood chips and mud. The preservation process takes about 3 months. The inside is black and pasty. The eggs are ready to eat.

Packs: Sold individually or by the dozen.

OYSTER SAUCE, *Ho yow*

Seasoning sauce made with oyster extracts, water, sugar, salt, starch, and caramel color.

Packs: Bottles and cans in various sizes.

Serving size: 1 tb, ½ oz (14 g).

Calories: 1 tb = 20 calories.

OYSTERS (DRIED), *Ho see*

Packs: 1-lb (450-g) bags or loose.

Size: About 2 in (50 mm) long.

Kitchen yields: Soak raw oysters 12 to 24 hours to remove sand. Hard uncooked, chewy when cooked.

PEA SHOOTS, *Dau miu*

The tips of peas harvested when in bloom. The vegetable has found its way onto the menus of fine Western restaurants.

Kitchen yields: The vegetable is ready to cook but will spoil rapidly in storage.

PLUM SAUCE, *Shwin mei jiong*

Popular condiment sauce, also known as *duck sauce*, available domestically and imported.

Pack: Jars or bottles in many sizes.

RADISHES

White radishes are listed as *daikon* under Japanese Foods. The Chinese green radish is basically similar, except for the color. The inside is white.

RICE, GLUTINOUS, *Noh my*

Short-grain rice available under various brand names. *See also* Japanese Rice Varieties.

Packs: 50- and 100-lb (22.5- and 45-kg) cotton bags.

Kitchen yields: 3 oz (84 g) make 1 cup (0.24 l) cooked rice.

Calories: ½ cup (0.12 l), cooked, plain = 90 calories.

RICE STICKS, *My fun*

Also known as *rice noodles*; sticks resembling spaghetti.

Packs: By weight, often 1-lb (450-g) packages.

Serving size: 2 oz (56 g).

Kitchen yields: Boil like pasta; the noodles will not expand much.

Calories: 2 oz (56 g), without any other ingredients = 200 calories.

SCALLOPS, DRIED, *Gong yu chu*

Available usually quartered, with a dull, shriveled appearance and a fishy smell.

Size: Pieces or shreds.

Packs: By weight.

SEAWEED VARIETIES

Listed together under Japanese Foods.

SESAME OIL

See Thai/Vietnamese Foods in this section.

SHARK FINS, DRIED, *Yu chi*

Dried cartilage from the fins of sharks.

Packs: By weight.

Size: Shaped like thin spaghetti. Length is indication of quality. Best are 5 in (127 mm) long.

Kitchen yields: 1 oz (28 g), dried = 1 qt (0.47 l) soup.

SHRIMP CHIPS, *Ha bang*

Popular snack items, available in many colors. Dried chips puff up when fried.

Packs: By weight.

Kitchen yields: Chips triple in size when fried.

SHRIMP, DRIED, *Ha my*

Pack: By weight; the product can be stored without refrigeration.

Size: ½ in (12 mm) to 2 in (50 mm).

SHRIMP PASTE, *Hahm ha*

See Thai/Vietnamese Foods.

SNOW PEAS, *Hoh laan dau*

See the Vegetables section for Fresh Snow Peas and the Groceries section for Frozen Snow Peas.

SOY SAUCE

A number of varieties are available:
Dark: *lo tsow*

Heavy: *jiong yow*

Light: *sang tsow*

Table sauce: *sin tsow*

Packs: Portion-sized packs and containers ranging from 20 oz (590 ml) to 1 gal (3.8 l).

SPRING ONIONS

See Green Onions/Scallions in the Vegetables section.

SQUID, DRIED, *Yow yu*

Basically the same as cuttlefish; see the Fish, Mollusks, Shellfish, and Seafood section.

STEAMED BREADS

Numerous products are on the market ready to steam heat in many sizes and shapes.

Some products are filled and can be considered dumplings.

Packs: One package consisting of 8 bread items, 2.5 oz (70 g) each.

STEM LETTUCE, *Who sun*

Long, thick stem topped with leaves. Both stem and leaves can be eaten raw; the stem is the delicacy, but the leaves can be eaten like salad.

Kitchen yields: Stems must be peeled, and there is considerable waste.

TARO ROOT, *Yu tau*

Potato-like root used in Asian and South American cooking.

Season: Year-round.

Packs: By weight; 50-lb (22.5-kg) bags are common.

Kitchen yields: About 20 percent peeling loss.

WATER CHESTNUTS, *Ma taai*

Brown bulbs with white interior. For canned Water Chestnuts, see the Groceries section.

Season: Year-round.

Packs: 10-lb (4.5-kg) lugs.

Count: 16 to 18 pieces per lb (450 g).

Size: $\frac{3}{4}$ oz (21 g) each, average.

Kitchen yields: 1 lb (450 g) = 9 oz (250 g), cleaned. The cooking loss is minimal.

WATERCRESS

See the Vegetables section.

WATER SPINACH, *Ong choi*

Vegetable resembling flat-leaf spinach, but with longer hollow stems.

Kitchen yields: The vegetable, leaves and chopped stems, will shrink by about two-thirds.

WINTER MELON, *Tung qua/Dang qua/Doong qua*

Large vegetable, round or football-shaped, with green skin and white inner meat that contains many seeds. It must be cooked. The most popular dish is winter melon soup, served in the scooped-out melon shell.

Season: Year-round; peak occurs in winter.

Packs: By weight and piece. 50-lb (22.5-kg) packs are common.

Size: Up to 20 in (500 mm) across.

Kitchen yields: The melon is 96 percent water, and this must be taken into consideration when cooking.

WONTON WRAPPERS

Wrappers are available in different sizes. Those described here are wrappers for soup-size wontons.

Packs: 1-lb (450-g) packages.

Kitchen yields: 1-lb (450-g) package = 60 skins. *Note:* To seal wrappers properly, make a paste with water and flour.

YELLOW CUCUMBER, *Wong gwa*

Vegetable that can be eaten raw like other cucumbers, but with a tough yellow skin.

Size: 10 in (250 mm) and longer, up to 2 lb (0.9 kg).

Kitchen yields: About 25 percent peeling waste, and seeds should be removed.

HISPANIC FOODS

The names of Hispanic food ingredients vary from country to country. Convenience foods that have found general acceptance are listed.

ACHIOTE/ANNATTO

Reddish/yellowish seeds from the tropical annatto tree. The very hard seeds are used as dye and as a flavoring agent. Available as paste, powder, and whole.

Packs: By weight.

Kitchen yields: Sauté in oil to release the flavor, discard the seeds. Use according to taste.

ADOBE SEASONING

All-purpose seasoning consisting of salt, ground cumin, garlic, black pepper, oregano, turmeric, and other spices.

ALMIDON

Spanish name for *starch*; Cassava starch is often used.

Pack: By weight.

Kitchen yields: Use like corn starch.

AREPA FLOUR

See Mais Products.

BATATA ESNAOLA

Sweet potato and fruit jam, available in various flavors.

Pack: 25-oz (700-g) cans, 12 per case

CACTUS FRUIT

See Tuna.

CACTUS LEAVES

See Nopales.

CALABAZA

Also called *Cuban squash*, this vegetable has a better yield than pumpkin. Buff color indicates maturity.

Season: Available all year.

Kitchen yields: 1 lb (as purchased) = 11 oz (300 g), ready to cook.

CHAYOTE

See Squash Varieties in the Vegetables section.

CHEESE SAUCE FOR NACHOS

Packs: #10 cans, 6 per case.

Weight: 106 oz (3 kg).

Serving size: 2 oz (56 g) for topping.

Kitchen yields: One #10 can = 50 servings.

Calories: 2 oz (56 g) = 95 calories.

CHICHA MORADA

Purple soft drink mix made from corn.
See Mais Products.

CHILI VARIETIES

There are numerous chili varieties on the market. Connoisseurs can distinguish them, but to the uninitiated they taste equally fiery. Those listed here are the most important varieties available in specialty markets.

AJÍ

Ají is the common term for chili peppers throughout South America; they are usually small, red berry-sized or oblong peppers growing on shrubs. The color can range from bright red to yellow.

Packs:

Canned ají: 20-oz (360-g) cans, 24 per case.

Ground dried ají: 13-oz (370-g) and 15-oz (420-g) jars, formulated for specific dishes.

Taste: Hot.

Kitchen yields: Use sparingly.

ANAHEIM CHILI

See New Mexican Chili.

ANCHO (POBLANO)

Ancho means wide in Spanish; it refers to the wide, flat shape of the dried pepper. There is much confusion about the term *poblano*; in some areas the green pepper is called *poblano* and the dried version is called *ancho*; in other areas the names are used interchangeably.

Packs: Anchos and poblanos, by weight.

Taste: Slightly hot to medium hot.

DRIED ANCHOS

Poblano chili with dark, reddish brown color. Widely used in Mexican cooking.

Pack and size: 2 oz (56 g) = 4 peppers.

Kitchen yields: Relatively mild. Soak in water or vinegar. Used shredded or ground in sauces.

FRESH ANCHOS

Resemble bell peppers and turn red when mature. Often called *poblano* on the market.

Size: About 4 in (100 mm) to 5 in (127 mm) long.

Kitchen yields: The chilis should be roasted and peeled before use. The most popular preparation is stuffed with cheese.

CANNED POBLANOS

Large green peppers, resembling deflated bell peppers. Available whole for stuffing and in strips or diced. It is important that the peppers are flame peeled.

Kitchen yields: Green poblanos are often stuffed and batter-fried.

BELL PEPPERS

See the Vegetables section.

CALIFORNIA CHILI

Probably a variety of Anaheim chili, available dried as dark red, almost mahogany-colored pods.

Taste: Mild.

Pack and size: 6 oz (170 g) = 30 peppers.

CASCABEL CHILIS

Small, rather peppery and sold dried; the seeds often rattle inside the husk.

Pack and size: 2 oz (56 g) = 10 peppers.

CAYENNE

The green pepper is available fresh. For Cayenne Spice, see the Herbs and Spices section.

Taste: Medium hot when green, fiercely hot when red.

CHERRY PEPPERS

Small red peppers resembling red cherries when ripe. Most product is pickled and used as a garnish with sandwiches.

Size: 1½ in (37 mm) across.

Taste: Medium hot.

CHIPOTLE

Jalapeño chili that has been ripened, dried, and smoked. The color is reddish brown. The chili is available dried and canned in either vinegar or adobo sauce.

Pack and size: 2 oz (56 g) = 12 to 14 peppers.

Taste: Medium hot.

DE ARBOL

Elongated hot pepper used in spice manufacturing.

FRESNO CHILIES

Bright green to reddish orange when ripe. Often referred to as *hot peppers*.

Size: About 2 in (50 mm) long.

Taste: Mildly hot to very hot.

GREEN CHILIS

Catch-all name for the popular commercially available canned diced, strips, or whole green chilis. The peppers should be peeled.

Taste: Mild.

Packs: #2½ cans, 29 oz (810 g), containing about 3½ cups (0.82 l).
#10 cans, containing 13 cups (3 l).
5-gal (19-l) cans.

Serving size: One pepper, about 1¼ oz (35 g).

Kitchen yields: One 29-oz (810-g) can = 20 to 25 whole peppers.

Calories: One pepper, about 1¼ oz (35 g) = 10 calories.

HABANERO

Flattened bell-shaped pods, ranging from yellow to orange and red.

Size: About 2½ in (62 mm) long.

Taste: Extremely hot.

JALAPEÑOS

Jalapeños are available fresh, canned, diced, in slices, and whole. The color can range from green to purple-red; the canned peppers are usually green. See also Chipotle for dried Jalapeños.

Size: 2½ in (63 mm) long.

Packs:

Canned: #2½ cans, weighing 29 oz (820 g), 12 or 24 per case.
#10 cans, 6 per case.

Fresh: By weight.

Taste: Slightly hot to very hot.

Kitchen yields: One #2½ can = 3½ cups (0.82 l)
One #10 can = 13 cups (3 l)

MULATO

Dark brown to black large pepper pods, usually sold dry.

Pack and size: 2 oz (56 g) = About 4 peppers.

Taste: Mild.

NEW MEXICAN CHILI (ANAHEIM)

Also called *Anaheim chili*. The name has been changed, because there is no logical connection to the city in California, except that canning chilis was a big industry there. These are very popular peppers, available in many sizes, canned normally while moss green, red when mature. The familiar dried chili clusters hanging from rafters and doorways in Mexico are usually New Mexican chilis.

CANNED

Available whole, diced, or in strips.

Taste: Sweet to slightly hot.

Kitchen yields: California chilis are canned already peeled. Remove seeds before use.

FRESH

Season: Year-round. Peak occurs in summer and fall.

Pack: By weight.

Size: 5 to 8 in (127 to 203 mm) long, dark green.

Taste: Sweet to slightly hot.

Kitchen yields: Anaheim chilis are canned already peeled. Remove seeds before use.

PASILLA

Long, thin, dark brown to black pods. Usually available toasted and dried, used as an ingredient in mole recipes.

Taste: Mild.

Packs and size: 2-oz (56-g) bags = about 12.

Calories: 1 oz (28 g) = 100 calories.

PIMIENTOS

Mild red chilis. *See* Sweet Peppers in the Groceries section.

POBLANO

Large green peppers, resembling deflated bell peppers. Used for stuffing. *See also* Ancho.

SERRANO

Small peppers about 3 in (75 mm) long; green when fresh and red when dried. Available fresh, dried, and pickled.

Taste: Medium hot.

Pack and size, dried: 2 oz (56 g) = 100 pods.

Kitchen yields: Ingredient in salsa cruda and in guacamole.

TABASCO

Little production is available fresh because it is cultivated for use in tabasco sauce.

Taste: Very hot.

CHORIZOS

Fatty pork sausages, available mild or hot, air-dried and fresh.

Packs: By weight.

Sizes: 1¾ oz (49 g) and larger.

Kitchen yields: 30 to 40 percent shrinkage, depending on brand, for fresh product. Little shrinkage for air-dried sausages.

Calories: One 1¾-oz (49-g) air-dried sausage = 170 calories.

CILANTRO (CORIANDER) LEAVES

See the Herbs and Spices section.

CORN HUSKS

See Mais Products.

COTIJA POLVO

Mexican grated cheese.

Packs: 16-oz (450-g) bags and larger.

Calories: 1 oz (56 g) = 200 calories.

DENDE OIL (DENDÊ)

Flavorful, heavy, dense yellowish palm oil, important to Bahian/Brazilian cooking. The oil is extracted from dendê nuts, which are harvested from African palm trees.

Pack: 1-l (34-fl. oz) bottles.

Kitchen yields: Although the oil is used for cooking and frying in Brazil,

caution is advised because the flavor is very pronounced.

EMPANADA DOUGH

Ready-to-use frozen dough for South American turnovers, usually filled with meat, cheese, or vegetables. The dough is available for frying, and a flakier version for baking.

Pack: 5-in (125-mm) disks, 10 per package.

Weight: 13 oz (370 g).

Calories: 1 disk = 110 calories.

ENCHILADA SAUCE

Red sauce made of mild chili pulp. Available canned.

Packs: #10 cans, 6 per case.

Serving size: ½ cup (0.24 l), with main course.

Kitchen yields: One #10 can = 24 servings.

EPAZOTE

Pungent herb with serrated leaves. Popular flavoring ingredient with black beans; available dry and fresh.

Pack, dry: 2 oz (56 g)

FAJITA SEASONING

Packs: By weight.

FLOURS

See Harinas.

FRIJOLES (DRIED BEANS)

There are numerous bean varieties available from specialty markets. The yields are usually comparable to those of other beans. See the Groceries section.

FRIJOLES REFritos, CANNED

Called *refried beans* in English, usually made with pinto beans. Available canned and as ready-to-use mix. See the Recipes section for a formula for homemade Refried Beans.

Packs: #10 cans, 6 per case.

20.5-oz (580-g) cans, 24 per case.

Serving size: ½ cup, 4.2 oz (120 g).

Kitchen yields: 1 can #10 = 22 servings.

Calories: ½ cup, 4.2 oz (120 g) = 160 calories.

FRIJOLES REFritos (REFRIED BEANS) DRY MIX

Packs: By weight.

Serving size: ½ cup, 3.3 oz (65 g), raw.

Kitchen yields: 1 lb (450 g) = about 5 servings.

Calories: ½ cup, 3.3 oz (65 g), low-fat mix = 230 calories.

GRAINS**ARROZ**

Long-grain, unconverted rice. See Rice in the Groceries section.

CEBADA

Peeled barley, ready to cook, used as a starch accompaniment.

Packs: By weight, usually sold in 15-oz (425-g) bags.

Calories: 1 lb (450 g), raw, without additional ingredients = about 1600 calories.

TRIGO PELADO

Hulled wheat, used as a starch component, ready to cook.

Packs: By weight, usually in 15-oz (425-g) bags.

Kitchen yields: The product needs about 3 times the amount of liquid; the ratio of product to liquid is 1 to 3.

Calories: 1 lb (450 g), raw, without additional ingredients = about 1600 calories.

GUAVA, FROZEN PULP

Fresh guava is available from specialty markets and is listed in the Tropical Fruits section.

Packs: 14-oz (400-g) packages.

Serving size: ½ cup (100 g).

Calories: ½ cup, 3.5 oz (100 g) = 90 calories

GUAYABATE

Also known as *pasta de guayaba*, the sweet paste is available in cans and in wooden boxes. It is often served with cheese.

Packs: 21-oz (600-g) cans.

3¼-lb (1.5-kg) blocks.

Serving size: 2 oz (56 g).

Calories: 2 oz (56 g) = 200 calories.

Kitchen yields: One 21-oz (600-g) can = 10 servings.

One 3¼-lb (1.5-kg) block = 25 servings.

HABAS

Dried fava beans; also available as flour.

Pack: By weight, often 15-oz (425-g) bags.

Serving size: 2 oz (56 g) dry, as garnish.

Calories: 2 oz (56 g), dry = 86 calories.

HARINAS

Numerous flour varieties (*harinas*) besides wheat flour are on the market. They have very little, if any, gluten and cannot be used for baking yeast bread unless blended with wheat flour. These flours are often used as thickeners in soups, stews, and desserts.

<i>Harina de arroz</i>	Rice flour
<i>Harina de camote</i>	Sweet potato flour
<i>Harina de cebada</i>	Barley flour
<i>Harina de chuno</i>	Potato flour
<i>Harina de garbanzo</i>	Chickpea flour
<i>Harina de haba</i>	Fava bean flour
<i>Harina de machica</i>	Barley flour
<i>Harina morada</i>	Purple corn flour

HARINA DE PLATANO

Plantain flour, used primarily in dessert puddings.

Pack: 14 oz (400 g).

Calories: 1 oz (56 g) = 90 calories.

HARINA DE YUCA

Cassava flour, used in *pan de yuca*.

Pack: 12 oz (340 g).

Calories: 1 oz (56 g) = 240 calories.

HEARTS OF PALM

Fresh hearts of palm are occasionally available from Brazilian markets. See the Groceries section for the canned product.

HUITLACOCHÉ

Corn mold available frozen and sometimes fresh. The fungus will ooze a black liquid and requires little cooking.

Pack: 8 oz (225 g), frozen.

Kitchen yields: 1 oz (56 g), as filling for quesadillas.

JICAMA

See Jicama (Mexican potato) in the Vegetables section.

LUFA

Elongated tropical squash. Tender while young, it develops a fibrous interior when mature, which is dried and used as a cleaning sponge.

MAIS PRODUCTS

AREPA FLOUR

Available as precooked flour, made from yellow corn and from white corn.

Packs: 17.6-oz (500-g) bags.

5-lb (2.25-kg) bags.

Serving size: 1 oz (28 g), average.

Kitchen yields: 17.6-oz (500-g) bag = 8 pieces arepas.

Calories: One 1-oz (28-g) arepa, without griddle oil = 110 calories.

CHICHA MORADA

Purple soft drink mix made from corn.

Packs: 16-oz (450-g) packages, 12 per case.

CHOCLO

Peruvian name for *corn*, usually large-kernel corn. Available on the cob canned, dried, and frozen.

CORN HUSKS

Available dried for making tamales.

Packs: By weight, 2-oz (56-g) pouches.

Count: 2 oz (56 g), about 50 leaves.

Kitchen yields: Depending on size; for larger tamales, 2 husks are needed.

MASA HARINA

Trade name for specially treated corn flour. Also known as *tamale flour*, *arepa flour*, and *corn tortilla flour*. See Arepa Flour.

Kitchen yields: 2 cups (16 oz, 450 g) masa + 1½ cups (250 ml) warm water = 16 tortillas, 5-in (125-mm) diameter.

POZOLE

Also known as *mote blanco* and *hominy* in English, pozole is boiled whole white corn kernels.

Packs: #10 cans, 6 lb 12 oz (3.06 kg), drained weight.

15-oz (425-g) cans, drained weight.

Other packs are available.

Serving size: ½ cup, 4.5 oz (130 g), as starch.

Calories: ½ cup, 4.5 oz (130 g) = 70 calories.

MALANGA

A tuber in the taro family. See Taro Root in Chinese Foods. The leaves, called *callaloo*, are used as a vegetable.

MOLE POBLANO

Mexican seasoning containing chili peppers, oil, starch, sugar, sesame seeds, peanuts, and cocoa, available as paste and as powder (without oil). Some brands can be fiery and should be tasted before they are purchased in large amounts. Mole verde and mole de pepitas are versions without chocolate.

Pack:

Paste: 8.35-oz (230-g) jars.

Powder: 8-oz (225-g) cans and larger.

Kitchen yields: 1 oz (28 g) per person, depending on recipe.

Calories: 1 oz (28 g), for paste = 230 calories.

NARANJA AGRIA

See Seville Oranges in the Fruits section.

NOPALES

Nopales is the Mexican term for *edible cactus leaves*, broad green pads with thorns. The flavor is mild. They can be eaten raw in salads or cooked as vegetables. Nopales reaching the market have most thorns removed. The flesh is cut into strips and boiled until tender, or used raw in salad with tomatoes, onions, oil, and vinegar.

CANNED NOPALES

Product should be drained and rinsed before use.

Packs: 15-oz (420-g) jars and larger.

Drained weight: 7 oz (200 g).

Serving size: 1 oz (28 g), sprinkled over salads.

Kitchen yields: One 15-oz (420-g) jar = 7 servings.

Calories: 1 oz (28 g) = 5 calories.

FRESH NOPALES

Season: February to November.

Packs: 8-oz (225-g) leaves, 12 per tray. Smaller sizes are available.

Sizes: Each leaf weighs about 3 to 8 oz (85 to 225 g).

Kitchen yields: About 15 percent cleaning loss.

PANELA

Hard cakes of brown raw sugar with a strong molasses taste.

Packs: 1-lb (450-g) disks.

Calories: 1 oz (28 g) = 100 calories.

PAPAS (POTATOES)

CHUÑOS NEGROS

Freeze-dried tiny black potatoes, ready to use, from Peru. The potatoes are also available dried.

Pack: 20-oz (560-g) cans.

OLLUCO

Soft yellow potato variety, available canned and frozen.

Packs: 20-oz (560-g) cans.

PAPAS AMARILLAS

Yellow potatoes, available canned and frozen.

Packs: 20-oz (560-g) cans.

PAPAS SECAS

Dried potatoes in little chunks; they must be soaked and then boiled.

Serving size: 1½ oz (42 g), dried.

Kitchen yields: Potatoes will double in size after soaking.

Calories: 1½ oz (42 g), dried = 184 calories.

PEPITAS (PUMPKIN SEEDS)

Peeled pumpkin seeds that can be toasted and sprinkled over food or eaten as a snack. Whole pumpkin seeds are called *ayote*. Ground seeds are used as a thickening agent in sauces.

Pack: By weight

Calories: ¼ cup, 1 oz (28 g) = 110 calories.

PLATANO CHIPS

Pack: 5.6-oz (160-g) packages.
There are many other packs.

Serving size: 1 oz (28 g), about 46 chips.

Calories: 1 oz (28 g) = 150 calories.

PLATANOS

Cooking bananas, called *plantains* in English. The color ranges from green to dark yellowish brown, showing maturity. Available also as flour. *See* Harinas.

Season: Available year-round.

Packs: 48-lb (21.6-kg) cartons.

Size: Average 8 oz (225 g) to 10 oz (280 g) and larger.

Serving size: 4 oz (112 g), baked.

Kitchen yields: Peeling loss is about 20 percent.

Calories: 4 oz (112 g) baked = 160 calories.

POZOLE

See Mais Products.

QUESO BLANCO

See Cow's Milk Cheese in the Dairy Products section.

SALSA

Hot sauce made of jalapeño chilis, onions, and tomatoes. Available red and green; mild, medium, and hot.

Packs: #2½ cans, 24 per case.

Size of can: 1 lb 13 oz (820 g), about 3½ cups (0.82 l).

Kitchen yields: Ready-to-use product.
There is no waste.

Calories: Calorie count is very low; one manufacturer claims no calories.

SESAME SEEDS

See the Groceries section.

SEVILLE ORANGES

Bitter oranges, also called *naranja agria*.
See Oranges in the Fruits section.

SQUASH BLOSSOMS

See the Flowers section.

TACO SAUCE

Available hot and mild.

Packs: #10 cans, 6 per case.

⅓-oz (9-g) individual packets, 500 per carton.

Kitchen yields: Serving sizes vary according to taste.

TACO SHELLS

Ready-to-fill shells made with white corn.

Packs: 200 shells per case and other packs.

Weight: ½ oz (14 g) each, average.

Serving size: 1 shell per person as snack.
2 shells per person as main course.

Calories: 2 shells without filling = 130 calories.

TAMALES

Corn husks stuffed with corn masa dough. They can be sweet or meat filled. Available canned, fresh, and frozen. For Tamale Dough, *see* the Recipes section.

TOMATILLO

Also called *tomate verde*. Sweet green tomato with a parchment-like covering; available fresh and canned.

Packs: 36-lb (16.2-kg) flat.
22-lb (9.9-kg) and 24-lb (10.8-kg) boxes.

TORTILLAS

Available as corn tortillas and flour tortillas in various sizes. *See also* Wraps in the Groceries section.

Packs: 9 oz (250 g) = 12 each, 5-in (125-mm) diameter.

Sizes: 7-in (175-mm) and 10-in (250-mm) diameter sizes are common.

Weights: 7 in (175 mm) = 1 oz (28 g).
10 in (250 mm) = 2 oz (56 g).

Calories: 7 in (175 mm), 1 oz (28 g) = 70 calories.
10 in (250 mm), 2 oz (56 g) = 160 calories.

Calories are about the same for corn and for flour tortillas.

TUNA (CACTUS FRUIT)

Tuna is the Mexican term for the fruits from certain cactus varieties. The spines are sharp, but most are removed when the fruit reaches the market. Cactus fruits are sold as *prickly pears* in English-speaking countries and as *tuna* in Mexico. The fruits are often eaten fresh, but are seedy. Much of the product is manufactured into jellies, nectar, purée, and even

a salsa ingredient. Cactus fruits should show some green skin when purchased. Other names for cactus fruit are *sabra*, *Barbary fig*, and *Indian fig*.

Season: The season in California is from August through March. Summer fruit, harvested in Mexico, can often be mealy.

Packs: By weight.

Size: 2½ to 4 in (7 to 10 cm) long.

Kitchen yields: 2 large fruits = 1 cup (0.24 l) purée.

Calories: 4 oz (112 g) edible flesh = 65 calories.

YAUTIA

Starchy root resembling yuca; *see* Yuca Root in this section.

YERBA MATE

A national beverage of Argentine, Paraguay, and Uruguay resembling green tea; it is served in a seasoned gourd and sipped with a silver straw. It contains a small amount of caffeine. It is now available with flavor components such as lemon, orange, and others.

Packs: 18-oz (500-g) packages.

Serving size: 1.7 oz (50 g). Note that the container is frequently refilled with hot water, which still produces a flavorful tea.

YUCA ROOT

Spelled *yucca root* in English, also called *cassavas*. This is a starchy root used in many tropical countries.

Season: Year-round.

Packs: By weight.

Kitchen yields: Peeling loss = 25 percent by weight.

YUCA ROOT FLOUR

See Harinas.

INDIAN FOODS

AMCHUR (AMCHOOR)

Green mango, dried, available as slivers and as powder. The powder is acidic and is used as a spice.

Packs: 3.5-oz (100-g) boxes and larger for slivers and powder.

Kitchen yields: 1 cup (0.24 l) = 2 oz (56 g) slices.

ARVI ROOTS

Starchy tubers related to cassava.

Pack: By weight

Size: 4 oz (112 g) to 8 oz (225 g).

Kitchen yields: 1 lb (450 g) = 6 oz, peeled.

ASAFETIDA, *Hing*

Spice derived from the gum of a Persian plant. The thick paste can have a nauseating smell and is used only in small quantities.

BITTER GOURDS (KARVI OR KARELI)

Available in different sizes. The gourds must be salted and allowed to marinate several hours to remove the bitter flavor. Large gourds are referred to as *Chinese karvi*. See Chinese Foods in this section.

Packs: By weight.

Sizes: 2 oz (56 g) to 3 oz (84 g).

Kitchen yields: 1 lb (450 g) = 14 oz (400 g), cleaned.

BLACK SALT

Sodium-free salt used as a flavor component, brownish black in lump form and pinkish brown when ground. It is sprinkled over fruits or other foods and gives a slightly smoky, tangy taste.

Pack: By weight.

Kitchen yields: To taste.

BOMBAY DUCK

Sun-dried fish fillets, usually deep-fried and used as a condiment with curry

dishes. The fish is a sardine-like species called *bummalao*. Most of the time, the product comes from India. Also called *bombil*.

Packs: By weight, usually 10-oz (300-g) boxes.

Kitchen yields: 1 piece per person. There are 20 to 30 pieces in each box.

CARDAMOM/CARDAMON

See the Herbs and Spices section.

CHAPATI

Flat bread made with wheat flour, available frozen in different sizes and packages. Available frozen, ready to bake or fully baked.

CHILIS, *Mirch*

See Hispanic Foods.

CHUTNEY VARIETIES

Chutneys are made with many ingredients in many variations of spiciness. The most well-known variety in Western cooking is mango chutney.

Packs: 14-oz (400-g) jars and larger.

Serving size: 1 oz (28 g), as condiment.

Kitchen yields: The product is ready to serve; it should be stored refrigerated after opening.

CURRY LEAVES (KAFFIR LIME LEAVES)

Small green leaves of the kari plant, a member of the citrus family. Curry leaves have a scent resembling citrus fruits and curry powder, but are not an ingredient in curry powder. Available fresh in specialty markets.

Pack: By weight.

Kitchen yields: Fresh leaves are sold attached to the stems. Cleaning waste is about 20 percent.

CURRY PASTE AND POWDERS

Spice blends consisting of coriander, turmeric, mustard seeds, chilies, dry ginger, cumin seeds, fenugreek, and other spices, according to brand. Available ranging from mild to very hot. Curry powders should be cooked in fat over low heat to fully develop flavor. Curry paste, called *masala paste*, is also available. The term *garam masala* indicates that the spice is hot.

Packs: 15-oz (425-g) cans and larger.

Serving size: 0.3 oz (10 g) for each pound of meat or vegetable, or according to taste.

Kitchen yields: One 15-oz (425-g) can = 45 lb (20 kg) product.

DAL VARIETIES

Generic term for *dried legumes (pulses)*. Commonly available are the following:

Ahar dal: Yellow split peas

Chana dal: Chickpea variety

Kabali: Chickpeas

Lobia: Black-eyed peas

Moong dal: Split mung beans

Udhad: Split black beans

Kitchen yields: Products are used as purée or thickeners in stews and soups.

EGGPLANTS, *Baingan*

Small purple eggplants.

Pack: By weight.

Size: 1 lb (450 g) = 6 to 8 pieces.

Kitchen yields: There is little waste; just the stems need trimming.

FENUGREEK (METHI) LEAVES

Fresh fenugreek and dried fenugreek leaves are used as seasoning and as tea. The dehydrated leaves require soaking before use. For dried whole and ground Fenugreek Seeds, see the Herbs and Spices section.

Pack: Bunches; the leaves are on long stems but are very light. 1.75-oz

(50-g) and larger packs for dried leaves.

FLOURS

Indian flours are often made with dried legumes and contain no gluten. They are used for flat breads such as roti/chappati, but cannot be used for yeast breads.

Commonly available are the following:

Ata flour: Wheat flour

Bajri flour: Millet flour

Besan flour: Split pea flour

Moong flour: Split green pea flour

Urhad flour: Black bean flour

GRAM, *Channa*

The generic term for *cereals*.

GHEE

Clarified butter or vegetarian margarine, available rendered.

Packs: 1 lb (450 g), 5 lb (2.2 kg), and larger.

Kitchen yields: Use as needed.

Calories: 1 lb (450 g) = 4,066 calories.

GREEN MANGOS, SMALL

Used as an acidic component.

Sizes: 3 oz (84 g) to 4 oz (112 g) each.

INDIAN OKRA

See Chinese Okra in this section.

MASALA

Generic term for *spices*; *garam masala* is a catch-all term for hot spice mixtures.

PAN LEAVES

Astringent leaves used raw with spices as a mouth freshener.

Pack: By count and weight.

Size: About 2 in (50 mm) wide and 3 in (75 mm) long.

PANEER (PANIR)

Curdled milk resembling cream cheese, an ingredient in many dishes. Available fresh and cubed and fried.

Pack: 8-oz (225-g), 1-lb (450-g), and larger packages.

Serving size: 1 oz (28 g).

Calories: 1 oz (28 g) of fried product = 90 calories.

PAPADOM (PAPAD)

Indian flat bread made with Udad-dal flour in a number of flavors, some of them spicy. Papadoms must be quickly fried in hot oil to puff up. Use as a garnish for curry dishes.

Packs: 8³/₄-oz (250-g) and 10-oz (280-g) packages.

Size: 7 in (177 mm) across is common.

Count: One 8³/₄-oz (250-g) package = 16 to 17 pieces.

One 10-oz (280-g) package = 25 pieces.

Kitchen yields: 1 piece per person as garnish.

RICE VARIETIES

Rice (*chawal*) is a staple in many parts of India. Specialty stores carry up to 20 varieties. The most available and best known are:

Basmati rice, also known as *patna rice*
Jasmine rice

Kitchen yields: See the Groceries section.

ROSE ESSENCE, *Ruh gulap*

Colorless liquid with strong aroma, used in desserts.

Pack: 1-pt (0.47-l) bottles.

Kitchen yields: Use sparingly.

SAFFRON, *Kesar*

India is a large producer of saffron, but the product seems to be less aromatic than the saffron from Spain or Greece.

SILVER LEAVES, *Chandik vark*

Pure silver hammered into thin leaves and used as an edible decoration on desserts.

TAMARILLO

See the Tropical Fruits section.

TAMARIND, *Imli*

Tropical fruit pods with a sour taste, native to Africa and Asia. The seeds were brought from India to the West Indies in the seventeenth century. The brown pods are sometimes sold in their semidry form, but more often as a compacted, sticky mass. When buying compacted fruit, check for telltale holes indicating insect penetration. The paste dissolves quickly and is convenient to use.

Season: Fresh pods are available in summer and autumn

Pack:

Paste: 21-oz (590-g) jars, 12 per case.

Dried and fresh pods: By weight.

Kitchen yields: Only the pulp can be used; dried pods must be soaked to extract the pulp. The paste can be used as is.

Calories: The high sugar content makes tamarind high in calories; the pulp is quite acidic.

JAPANESE FOODS

ADUKI BEANS

Small red beans, available canned and dried.

Packs: 15-oz (420-g) cans, 12 per case.
25-lb (11.25-kg) bags.
50 lb (22.5 kg).

BEAN CURD

Bean curd, commonly called *tofu* (its Japanese name), is a soybean product with a custard-like consistency. It is available in many shapes, flavors, and grades of firmness. Plain tofu is also called *Chinese-*

style tofu. It is most likely to be sold in moist squares in brine, in various sizes. *See also* Chinese Foods.

Packs: 20-oz (560-g) packages, 12 per case.

Calories: 4 oz (112 g) = 114 calories.

BEAN CURD, FRIED (TOFU)

Cubes of soft tofu, deep-fried and ready to use. The product must be kept refrigerated.

Packs: Many different packs are on the market.

Size: About 1 oz (28 g) per cube.

Kitchen yields: Product is ready to use and can be added to salads or stir-fried.

CUCUMBER

Seedless cucumber with a prickly skin.

Size: 10 in (250 mm) in length.

DAIKON

Long, white Asian radish with a flavor resembling that of an ordinary radish. It is eaten raw or cooked, used as a condiment and as a vegetable. Available fresh, dried, and pickled.

Season: Available year-round.

Packs: 3 each to a bunch, 12 bunches per case.

Size: 1 lb (450 g) to 2 lb (0.9 kg), average weight.

Calories: ¼ cup (1½ oz (42 g), shredded = 25 calories.

DASHI

Soup stock made with katsuo, dried bonito shavings, and konbu, dried seaweed. Dashi is a basic soup stock and is available as soup powder.

Packs: Both katsuo and konbu are available by weight and are very light.

Kitchen yields: The basic proportions are 6 cups (1.4 l) water + ½ oz (14 g) katsuo and ½ oz (14 g) konbu. The stock requires no lengthy boiling.

EDA MAME

Fresh soybeans, mostly sold still in their pods. The beans are served in their pods as a snack; available fresh and frozen, whole and shelled.

Season: Available year-round.

Packs: Sold by weight in bags ready to use.

Kitchen yields: There is no waste.

Calories: ¼ cup, cooked and shelled = 200 calories.

ENOKI MUSHROOMS

See the Vegetables section.

GINGER

For Fresh Ginger, *see the Fruits section*; for Pickled Ginger, *see the Groceries section*.

GOBO (BURDOCK)

Root vegetable common in most parts of Europe. A long, slender root with brown skin resembling salsify. Both the root and the leaves are eaten.

Season: Winter and spring.

Sizes: 10 oz (228 g) to 16 oz (450 g), fresh stalks without leaves.

Kitchen yields: Peeling and trimming loss is about 20 percent by weight.

HAKUSAI

Japanese name for *Chinese cabbage*.

HARUSAME

See Cellophane Noodles under Chinese Foods.

JAPANESE EGGPLANT

Small purple eggplant.

Season: Year-round.

Packs: By weight and piece.

Sizes: 4 oz (112 g) to 5 oz (140 g) each.

JAPANESE MUSHROOMS (SHIITAKE)

See Mushroom Varieties in Chinese Foods. Note that in Japanese cooking shiitake

mushrooms are usually used fresh; they are listed in the Vegetables section.

KANPYO

Dried gourd strips, which become flexible after soaking.

Pack: By weight, often 8-oz (225-g) packages.

KANTEN

Gelatin made from seaweed. *See* Agar Agar in the Groceries section.

KATSUO BUSHI (BONITO FLAKES)

Dried fish flakes, often made from skip-jack mackerel. Basic ingredient for making dashi, the basic stock.

Packs: 3½-oz (100-g), 5.3-oz (150-g) and larger bags.

Kitchen yields: ¼ cup flakes = 1½ qt (0.7 l) stock.

Calories: ¼ cup dry flakes = 25 calories.

KOBACHA SQUASH

Winter squash with green skin and orange-colored flesh.

Pack: By weight and piece.

Size: 12 in (300 mm) to 15 in (375 mm) across.

Weight: 2 lb (0.9 kg) to 6 lb (2.7 kg).

Kitchen yields: About 25 percent peeling loss.

KUZU

Granular root starch, used like arrowroot.

Pack: 11-lb (5-kg) box.

KUZU NOODLES

Transparent noodles, used like cellophane noodles.

MIKAN

Citrus fruit, often referred to as Japanese tangerine; usually available canned.

Pack: 15-oz (420-g) cans.

Calories: ½ cup (0.12 l) sections in light syrup = 70 calories.

MIRIN

Fermented sweet rice cooking liquid resembling saké with 8 percent alcohol content.

Packs:

10½-oz (0.3-l), 24.3-oz (0.7-l), 1-l (33-oz), and many other sized bottles.

5-gal (19-l) dispenser packs.

Calories: 1 oz (28 g) = 65 calories.

MISO

Fermented soy bean paste; it will dissolve in hot liquid. Available fresh in three basic flavors:

Aka: red, pungent

Chu: gold colored, mild

Shiro: white, almost sweet

Packs: Purchase by weight.

Calories: 1 oz (56 g) = 75 calories.

MISO MADE WITH OTHER CEREALS

Genmai miso: brown rice miso

Hacho miso: soybean miso

Mugi miso: barley miso

Shiro miso: sweet rice miso

Soba miso: buckwheat miso

Packs: 22 lb (9.9 kg), bulk pack case
12 oz (340 g), 12 per case.

Calories: 1 oz (28 g) = 50 calories.

MITSUBA (TREFOIL)

Plant resembling flat-leaf parsley.

Season: Available year-round.

OCTOPUS

Hoshi dako is sun-dried, and sudako is pickled-in-vinegar octopus. For Fresh Octopus, *see* the Fish, Mollusks, Shellfish, and Seafood section.

PANKO (Japanese Bread Crumbs)

Bread crumbs made from dense bread and shaved, not ground.

Packs: 12-oz (340-g) bags. Other sizes are available.

Kitchen yields: Use like other bread crumbs. Note that Japanese food is more lightly breaded than Western food.

PICKLED PLUMS (UMÉ BOSHI)

Fermented and pickled Japanese salt plums, actually made from hard green apricots. Available as paste and whole as a condiment.

Pack: 22 lb (9.9 kg) for plums and for paste, bulk pack case.

PONZU SAUCE

Bottled condiment dipping sauce.

Pack: 7¾-oz (0.2-l) bottles, 12 per case.

RICE VARIETIES

MOCHI GOMÉ

Glutinous rice used for making rice cakes.

OKOMÉ

Short grain rice, grown for Japanese cooking. Preferred varieties are California Rose, Blue Rose, or Calrose rice. Grown mostly in California, available under brand names. This rice is used for sushi and as a side dish.

Packs: 50 lb (22.5 kg), 25 lb (11.25 kg), 10 lb (4.5 kg), and 1 lb (0.45 kg).

Serving size: ½ cup (0.12 l), cooked.

Kitchen yields: 3 oz (84 g) make 1 cup (0.24 l) cooked rice.

Calories: ½ cup (0.12 l), cooked, plain = 90 calories.

RICE VINEGAR

Available as brown and as white vinegar.

Packs: 10-fl. oz (0.3-l), 12 bottles per case.
1-gal (3.8-l) containers.

Kitchen yields: Aged rice vinegars are generally less acidic than other vinegars.

SAKÉ

Fermented alcoholic rice beer beverage, usually served warm. There are

numerous versions on the market, with different alcohol contents. *Saké* is the generic Japanese term for alcohol.

Alcohol content: Between 14 percent (28 proof) to 24 percent (48 proof).

Calories: 1 cup (0.24 l) = 200 calories.

SATSUMA IMO

Sweet potatoes.

Pack: By weight.

Calories: 4 oz (112 g), baked = 140 calories.

SEA URCHIN

Called *uni* in Japanese, it is a delicacy is sushi bars.

Pack: By weight in small wooden boxes, frequently 250 g (9 oz).

Kitchen yields: There is no waste, but the product is highly perishable.

SEAWEED VARIETIES

Many varieties are available fresh and also pickled as a ready-to-use appetizer.

AGAR AGAR (KANTEN)

Gelatin derived from seaweed. *See* the Groceries section.

ARAME

Shredded, dried, precooked sea vegetable.

Pack: 2.2-lb (1-kg) bags.

Kitchen yields: Product will quadruple after soaking.

BLACK SEA MOSS

Thin, hair-like weed.

Pack: By weight.

Kitchen yields: Product is very light; 1 oz (28 g) = 1 cup (0.24 l).

DULSE

Packs: By weight.

Kitchen yields: There is no waste; just soak and rinse.

HIJIKI

Precooked, shredded wild seaweed.

Pack: 2.2-lb (1-kg) bags

Kitchen yields: Product will expand to about 5 times its volume after soaking. Cooking is required after soaking.

KOMBU (DRIED KELP)

Dashi (soup stock) ingredient, available in dried strips.

Pack: 2.2-lb (1-kg) bags.

Kitchen yields: Product will expand to 3 to 4 times its volume after cooking.

NORI/LAVAR

Available fresh and as thin, dried sheets. A lack of uniform color is an indication of good quality. Sushi nori is formulated to be used for sushi.

Season: Late summer and fall for fresh nori.

Packs: 50 sheets in 4.4-oz (120-g) packages.

Size: 6 in × 10 in (150 × 250 mm).

WAKAME

Green seaweed; available fresh, as flakes, pickled in salt, or precooked.

Pack: 2.2-lb (1-kg) bags.

SESAME OIL

See Sesame Oil in Thai/Vietnamese Foods.

SHIITAKE

See the Vegetables section for Fresh Shiitake Mushrooms; and Chinese Foods, Mushroom Varieties, for Dried Shiitake Mushrooms.

SHISO LEAVES

Seasoning leaves, usually available fresh year-round.

SOBA

Buckwheat noodles. Some manufacturers sell soba made with flour blends.

Pack: 13.2-lb (6-kg) bulk pack.

SOY SAUCE

Seasoning sauce and marinade available in regular and low-sodium varieties. The Japanese name is *shōyu*.

Packs: 10½-oz (0.3-l), 24.3-oz (0.7-l), 1-l (33-oz), and many other sized bottles.

5-gal (19-l) dispenser packs.

Calories: 1 tb (15 ml) = 15 calories.

TAMARI

Condiment sauce, a by-product of miso production with lower sodium content than soy sauce.

Packs: 10-oz (0.3-l) bottles, 12 per case. 20-oz (580-ml) bottles, 12 per case.

TEA, ASIAN VARIETIES

JAPANESE GREEN TEA (BANCHA)

The unfermented tea is available as powder and as leaf tea. The powder makes a frothy, astringent-tasting beverage, served hot or cold. When served cold, sweet sugar syrup is usually served on the side. The tea comes mostly from the Uji area of Kyoto. The famous tea ceremony traditionally used powdered tea, whipped with a bamboo whisk until frothy, and served in tiny cups. *Cha-no-yu* is the ritual tea ceremony.

Packs: 11-lb (4.5-kg) bulk pack.

Kitchen yields: 1¾ oz (50 g) = 30 cups.

HOJICHA (ROASTED GREEN TEA)

Full-flavored tea resembling fermented tea.

KUKICHA

Roasted twig tea with low caffeine content.

Packs: 11-lb (95-kg) bulk packs.

OCHA

Green tea.

TERIYAKI SAUCE

Seasoning sauce and marinade.

Calories: 1 tb (15 ml) = 15 calories.

TUNA

Fresh tuna is esteemed in many dishes, especially in sashimi and sushi. The flesh is carefully butchered and classified

according to fat content. Otoro, the fatty belly, is the most expensive. Tail meat is the least esteemed part.

UDON

Thick white wheat noodles; some manufacturers offer buckwheat and brown rice udon. Available fresh, precooked, and precooked frozen.

Packs: 13.2-lb (6-kg) bulk packs.

Serving size: 7 oz (200 g), raw.

Calories: 7 oz (200 g), raw = 570 calories.

WASABI

Powdered green horseradish. Mixed with cold water to a mustard-like consistency. Fresh wasabi is now available from specialty producers. Ready-to-use wasabi is available in tubes.

Pack: 2.2-lb (1-kg) bags.

Serving size: ½ oz (14 g) or less.

Kitchen yields: One 2.2-lb (1-kg) bag = 65 servings

MIDDLE EASTERN/NORTH AFRICAN FOODS

ALMONDS, GREEN

Green, unripe almonds are used as snacks sprinkled with salt or eaten as is. For other Almond Products, *see* the Groceries section.

Pack: By weight.

Count: 1 cup (0.24 l), 5 oz (140 g) = 30 pieces.

ANISE, STAR

See Chinese Foods.

BERBERÉ

Spice mix, usually very hot, consisting of dried, ground red peppers, salt, ginger, onions, cloves, cinnamon, garlic, nutmeg, cardamom seeds, allspice, peppercorns, fenugreek, and coriander.

Pack: By weight, often sold as paste.

BULGUR (BULGHUR)

Wheat preserved by an ancient method, dating back to Assyria. Wheat is cleaned and sieved, soaked in water, and dried. The grain is boiled, then called *danoke* or *shleepa*, dried again, and cracked. Bulgur is ready to eat after soaking. Bulgur comes in three textures: coarse, medium, and fine. It is also available as white

bulgur, made from soft wheat, and dark bulgur, made from hard wheat.

Pack: By weight.

Weight: 1 cup (0.24 l) = 4 oz.

Kitchen yields: The product is precooked and will expand little.

COFFEE

Greek or Turkish coffee is made with finely ground dark-roasted beans, blended with sugar and water and not strained.

Pack: 1-lb (450-g) packages.

Serving size: 4 oz (116 ml) or smaller.

Kitchen yields: Per person: 1 heaping tsp coffee, 1 heaping tsp sugar, and 4 oz (116 ml) water.

COUSCOUS

North African wheat product, consisting of semolina kernels. Small, medium, and large kernels are available. also available are toasted, tri-color, spinach, and tomato couscous. The Moroccan version is very fine. Many operations use instant couscous.

Packs: 17½-oz (500-g) boxes.

Larger packs are available.

12-oz (340-g) boxes, instant couscous.

Serving size: 2 oz (56 g) raw.

Kitchen yields: 1 cup = 7 oz (200 g)
raw = $3\frac{3}{4}$ (0.45 l) cups cooked.

Calories: 2 oz (56 g) raw, when cooked
with water, no fat added = 200
calories.

FILO DOUGH (PHYLO)

Filo sheets are shipped frozen and should be defrosted in the refrigerator for best results. Available as sheets and kataifi (shredded dough).

Packs: 16-oz (450-g) packages, 12, 16, 24, or 30 per case.

Sizes: 14 × 18 in (350 × 500 mm).
12 × 17 in (300 × 425 mm).
9 in × 13 in (225 × 325 mm).
Other sizes are available in bulk.

Thickness:

#4 thinnest: 26 sheets, 12 in (300 mm) × 17 in (425 mm).

#5 thin: 24 sheets, 12 × 17 in (300 × 425 mm).

#7 medium: 22 sheets, 14 × 18 in (355 × 460 mm).

#10 thickest: 18 sheets, 12 × 17 in (300 × 425 mm).

Kitchen yields: The number of filo sheets per package is approximate, because some will break during handling. It is difficult to calculate a yield for kataifi, because it is shredded fine dough.

GRAPE LEAVES

Available pickled in brine.

Packs: 35-oz (1-l) jars.

Counts: 35-oz (1-l) jar = 40 to 60 leaves.

Kitchen yields: The leaves can be salty and should be washed.

GYROS

Chopped meat cone of Middle Eastern origin, cooked on a revolving vertical spit. Usually served on pita bread.

Available fully seasoned, frozen, and ready to roast, made with lamb or beef.

Packs: 10-lb (4.5-kg) cones, 4 per case.
20-lb (9-kg) cones, 2 per case.
30-lb (13.5-kg) cone, single pack.

Sizes: 10 lb (450 g), 20 lb (9 kg), and 30 lb (13.5 kg) cones. The size is contingent on the machinery used.

Serving size: $3\frac{1}{2}$ oz (100 g) cooked meat.

Kitchen yields: One 10-lb (4.5-kg) cone = 32 to 34 servings.

HALLOUMI

Cheese from Cyprus; *see* Goat Cheese in the Dairy Products section.

ORZO

Egg-shaped pasta. *See* information in the Groceries section on Pasta Products.

ROSEWATER or ROSE ESSENCE

See Indian Foods.

SESAME PASTE

Paste made from ground seeds. Tahini is a sesame paste used in Middle Eastern cooking. The best-known dishes are baba ganoush, an eggplant dip blended with tahini paste and halvah, a sesame candy popular in Israel. Also known as tahina/tahini paste.

Packs: 15-oz (425-g) cans and larger.

Serving size: 2 oz (56 g) or more, depending on recipe.

Calories: 2 oz (56 g) = 200 calories.

SOUVLAKI

Greek pork or beef skewers, available frozen, fully seasoned.

Packs: 15-lb (6.7-kg) case.

Count: Fifty-four 4-oz (112-g) skewers.

Serving size: 2 pieces, as main course.

THAI/VIETNAMESE FOODS

BANANA BLOSSOMS

Edible purple blossoms used in stews and rice dishes in the Philippines, as a vegetable in Sri Lanka, and pickled in Chinese cooking. They must be peeled and can be eaten raw in salads. The flowers are astringent and must be soaked in lemon water.

Pack: By weight.

Size: 1 lb (450 g) and larger.

Kitchen yields: About 20 percent trimming loss.

BANANA LEAVES

Used to wrap food for steaming and as a plate substitute in some restaurants. Available from florists.

BAMBOO SHOOTS

See Chinese Foods for Fresh Bamboo Shoots, and the Groceries section for Canned Bamboo Shoots.

BASIL, THAI

Purple basil, available from produce suppliers.

BEAN PASTE/FERMENTED BLACK BEANS

See Chinese Foods.

CHILI PASTE

Very pungent seasoning paste made with red chili peppers, salt, sesame oil, soy beans, and pepper.

Packs: 8 oz (225 g) and larger.

Kitchen yields: Use sparingly because the paste is very hot.

COCONUT PRODUCTS

See the Tropical Fruits section for Fresh Coconuts, and the Groceries section for Canned Coconut Juice, Coconut Milk, and Shredded Coconut.

FRESH, TRIMMED COCONUT

Whole, wrapped in plastic, used only to extract the coconut water.

Weight: About 3 lb 10 oz (1.2 kg).

Trimmed: 3 lb 10 oz (1.2 kg) = 24 oz (670 g).

Kitchen yields: 3 lb 10 oz (1.2 kg) = 15 oz (0.43 l) coconut water.

The meat is very thin and basically worthless.

DRIED SHRIMP

Available whole, in various sizes and shades of color, and as a powder; used as a flavoring ingredient.

Pack: By weight; 4-oz (112-g) bags and larger, and loose.

Serving size: ½ oz (14 g) as flavoring ingredient.

Calories: ½ oz (14 g) = 40 calories.

EGGPLANT, THAI

See Japanese Eggplant.

FISH SAUCE/NAM PLA

Seasoning liquid made with fermented anchovies and salt; a basic ingredient in most Thai dishes.

Pack: 23-oz ((0.65-l) and larger bottles.

Kitchen yields: The sauce is very salty and should be used cautiously.

GALANGAL

These aromatic roots are also called *Siamese ginger*.

Packs: Available dried in slices, fresh, and powdered.

Kitchen yields: The dried root can be reconstituted by soaking it in water.

LEMON GRASS

A tropical grass resembling leeks. Available dried, fresh, and powdered.

Packs: Sold in bunches of 4 stalks or more.

Size: Typically sold in 12-in (300-mm) to 14-in (350-mm) lengths.

Kitchen yields: Only the white lower part is used; the flavor intensity depends on freshness. The dried form can be reconstituted by soaking it in water.

RICE

Both long grain and glutinous short grain varieties are used. For Rice Sticks, *see* Chinese Foods.

RICE PAPER

Thin white sheets, available from pastry supply houses. The paper must be moistened to become flexible.

Pack: 1-lb (450-g) packages.

Size: 8.5 in (212 mm) × 11 in (275 mm).

SESAME OIL

Rather expensive oil with a pronounced taste, usually used in combination with

milder-flavored oils. The thicker, darker versions from China and Japan have the most intense flavor. The Middle Eastern version is lighter and less aromatic. Sesame oil is also available very hot and pungent, and is used as a condiment.

Packs: From 10 oz (0.3 l) to 1 gal (3.8 l).

SESAME SEEDS

See the Groceries section.

SHRIMP PASTE

Thick paste used as a condiment, made with dried shrimp, salt, and spices. The paste is very salty and can be pungent.

Packs: 8-oz (225-g) jars and larger.

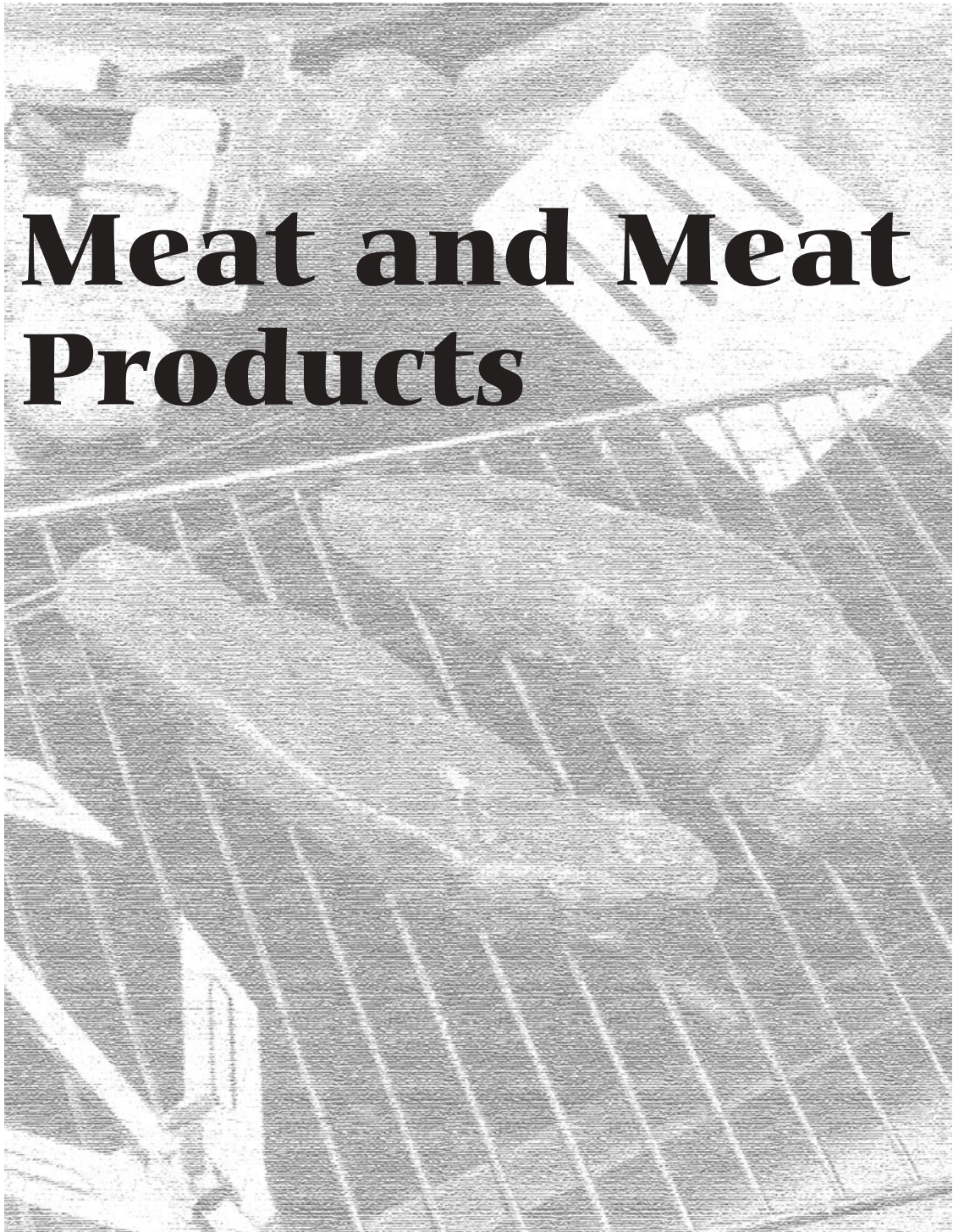
TAMARIND

See Indian Foods.

TURMERIC

See the Herbs and Spices section.

Meat and Meat Products



In this section are listed are fresh, frozen, cured, and processed meat cuts and products. Also included are the recommended cuts, portion sizes, and quantities to purchase for popular dishes. Portion sizes are based on my personal experience in New York City luxury hotels and are only guidelines. The kitchen yields are based on realistic tests, not on mathematical formulas. The potential mathematical yields are achieved only when each and every portion is served, leaving no leftovers whatsoever; when cooks carve the meat with surgical precision; and when every piece of meat is exactly uniform in fat content, density, and shape. Perhaps some lucky chefs will get one or more additional servings than indicated without cheating the customers.

Even with the best selection and grading processes, yields can vary dramatically between seemingly equal pieces of meat. Years ago meat was purchased as primal cuts, and it was the responsibility of the receiving agent to inspect the meat upon delivery and reject pieces that did not meet the specifications. Today meat is almost always delivered boxed and hardly ever inspected until it reaches the kitchen; by that time it is too late to send it back to the vendor for an exchange.

Many medical authorities recommend that meat portions should be only 3 oz (85 g) or less. Restaurants generally serve much larger portions because, certainly, most customers do not eat at a restaurant and consume the larger portions every day; moreover, the 3-oz (85-g) meat servings when presented on large plates would look puny. Customers are always right, and they can choose what to eat and the amount they want to eat. Despite publicity about the alleged dangers of eating large amounts of meat, steak houses are thriving, serving ever-larger portions. Along with the descriptions of the various cuts, the *Meat Buyers Guide* (MBG) (© North American Meat Processors Association) numbers have been used for further identification.

ALLIGATOR MEAT

Available frozen from Florida.

Packs: 5-lb (2.25-kg) cartons.

Sizes: 4-oz (112-g) to 6-oz (170-g) boneless tail pieces.

Serving size: 4 oz (112 g) to 6 oz (170 g).

Kitchen yields: 5-lb (2.25-kg) carton = 16 servings. The tail pieces are best cut into small medallions of about 2 oz (56 g) to 3 oz (85 g) each and

pounded well. Trimming waste is about 5 percent.

Calories: N/A. Meat is very lean.

BEAR

Available as frozen hind legs and loins.

Average sizes: Hind legs = 8 to 10 lb (3.6 to 4.5 kg).

Loins = 6 to 7 lb (2.7 to 3.1 kg), bone in.

Kitchen yields: Basically the same as well-done beef.

BEEF

All meat shipped interstate must be inspected by the U.S. Department of Agriculture (USDA) and reinspected every time it is processed. All states maintain a mandatory inspection service for meat slaughtered and consumed within the state. Meat grading is voluntary and is performed upon request by a USDA inspector for a fee. The grades of interest to foodservice operators are Prime, Choice, Good, and Standard. Very little meat graded *Prime* reaches the market because there is concern about the high fat content.

Weight is one criterion of the grading process, and a number of cuts are available only in Prime or Choice in specific weight ranges. Other grading criteria are marbling and configuration, which affect yield. The USDA also provides yield grading, but this is of more interest to the purchasers of whole carcasses than to restaurant buyers. There is a demand for leaner beef, and the grading criteria concerning marbling is undergoing periodic revisions. Most beef shipped is packed in Cryovac wrapping and boxed.

Much beef reaches the market that has not been graded, and it is sold under brand names. Black Angus is a brand name for a special breed of cattle; Plume de Veau is a brand name for high-quality veal.

Beef is about 10 days old when it reaches most markets and can be used without additional aging. Some purveyors dry-age sirloin strips and ribs upon request. There are losses with aging, because the meat loses moisture, and therefore weight, and much additional trimming is needed when the pieces are fabricated into portions.

Kosher meat must be slaughtered under rabbinical supervision by a licensed slaughterer; only certain cuts can be used, and the meat is used soon after slaughter. There are many additional rules concerning kosher meat that are beyond the scope of this book. In New York the term *kosher*

calf's liver is occasionally used, indicating that the liver is fresh, but not necessarily meeting the kosher requirements.

LARGER CUTS OF BEEF, OFTEN REFERRED TO AS PRIMAL CUTS

Primal cuts are purchased when meat is cut and portioned in-house rather than purchased as ready-to-use.

BLADE MEAT, MBG #109B

Also called *deckle meat* or *corner pieces*.

This boneless cut is situated over the shoulder end of the primal rib and is removed when the rib is prepared for roasting.

Packs: By weight, usually 20-lb (9-kg) cartons.

Size: 3 lb (1.3 kg) average per piece.

Best uses: Stew meat, also ground meat when fresh.

Kitchen yields: 5 percent trimming loss.

BONES, MARROW

Marrow bones are from the round (leg) and are very hard. They are purchased primarily to extract the marrow. There is little flavor in these bones.

Packs: By weight.

Kitchen yields: 50 lb (22.5 kg) of bones = 10 lb (4.5 kg) to 12 lb (5.4 kg) of marrow.

BONES, SHIN AND BACK

The best bones for making stock are back bones and neck bones. Shin bones are hard and brittle, and give little flavor to stocks.

Packs: By weight, often 50-lb (22.5-kg) cartons.

Kitchen yields: 50 lb (22.5 kg) make 5½ gal (21 l) brown stock.

50 lb (22.5 kg) make 3 gal (11.4 l) brown sauce.

50 lb (22.5 kg) make 10 gal (38 l) clear stock.

BOTTOM ROUND, GOOSE NECK, MBG #170

Muscle inside the leg, called *round*. This piece consists of the outside round and the eye round, listed separately. It also contains the heel muscle, which can only be used for stew. If this piece cannot be used, specify #170 A. The eye round is usually drier than the outside round. The following specifications are for the entire piece.

Weight range: 18 lb (8.1 kg) to 29 lb (13 kg).

Best sizes: 26 lb (11.7 kg) to 29 lb (13 kg).

Serving size: 6 oz (170 g) cooked braised steaks.

Best uses: Roast round, stew, braised steaks.

Kitchen yields for One Bottom Round, Choice, 22 lb (9.9 kg):

Waste and fat = 1 lb (450 g).

Usable trim = 3½ lb (1.5 kg).

Usable meat for roasting = 17½ lb (7.8 kg).

Roasted whole, medium rare = 13½ lb (6 kg) yield.

Steaks for braising, 7 oz (200 g) raw = 40.

BOTTOM SIRLOIN BUTT, MBG #185

Tender muscle at the end of the short loin.

Weight range: 4 lb (1.8 kg) to 10 lb (4.5 kg).

Best weight: 8 lb (3.6 kg), Choice.

Best uses: Roasted whole, steaks, cut in strips for fajitas.

Serving size: 3 oz (85 g) for fajitas raw meat.

Kitchen yields for One Bottom Sirloin Butt, Choice, 8 lb (3.6 kg):

Waste and trim = 8 oz (225 g).

Usable trim = 8 oz (225 g).

Usable meat = 7 lb (3.1 kg).

Servings fajita = 36.

BRISKET, FRESH, MBG #120

This piece has the deckel already removed. However, it is still a rather wasteful piece.

Weight range: 6 lb (2.7 kg) to 12 lb (5.4 kg).

Best sizes: 10 lb (4.5 kg) to 12 lb (5.4 kg).

Best uses: Boiling or braising.

Serving size, main course: 5 oz (140 g) cooked meat.

Kitchen yields for One Brisket, Fresh, Choice, 12 lb (5.4 kg): Cooked weight = 8½ lb (3.8 kg).

Waste fat trim after cooking = 2¼ lb (1.0 kg).

Usable cooked meat = 6¼ lb (2.8 kg).

One brisket, fresh, Choice = 19 servings.

Calories: 5 oz (140 g), cooked lean meat = 400 calories.

CHUCK, BONELESS, SQUARE CUT, MBG #115

Boneless whole shoulder, shank removed. The shoulder clod, a large outside muscle weighing between 13 lb (5.8 kg) and 18 lb (8.1 kg), can be removed and used for braising. This meat in Choice grade is considered the best ground meat. The fat content of ground meat should be approximately 20 percent of the weight. It is recommended to weigh the trimmings and lean meat before grinding to achieve the desired fat content. Meat should not be mixed or handled after it has been ground. A fat content testing machine is inexpensive and easy to use.

Weight range: 55 lb (24.7 kg) to 90 lb (40.5 kg).

Best sizes: 65 lb (29.2 kg) to 85 lb (38.2 kg), Choice.

Best uses: Ground meat, clod for roasting or braising.

Serving sizes: 4 oz (112 g), hamburger patty.
8 oz (225 g), chopped steak.

Kitchen yields for One Chuck, Boneless, Square Cut, Choice, 75 lb (33.8 kg):

Weight, after clod has been removed = 59 lb (26.5 kg).

Waste and trim = 3 lb (1.3 kg).

Usable meat, ground meat = 56 lb (25.2 kg).

56 lb (25.2 kg) ground meat yield = two hundred twenty 4 oz (112 g) hamburger patties.

56 lb (25.2 kg) ground meat yield = one hundred ten 8 oz (225 g) chopped steaks.

Calories: 4-oz (112-g) patty, 80/20 fat ratio = 324 calories.

CHUCK ROLL, MBG #116A

Boneless, netted muscle from the inside of the shoulder. Tender and juicy meat for braising; it can be fatty and fall apart after cooking. Portion size is hard to control, because muscle configuration is uneven.

Weight range: 13 lb (5.8 kg) to 25 lb (11.25 kg).

Best weights: 13 lb (5.8 kg) to 15 lb (6.7 kg).

Best uses: Braised or slow roasted.

Serving size: 5 oz (112 g) cooked meat.

Kitchen yields for One Chuck Roll, MBG #116A, Choice, 14 lb (6.3 kg):

Weight after braising = 9½ lb (2.9 kg).
One chuck roll = 30 servings.

CHUCK, TENDER, MBG #116B

Small, tender piece inbedded in the boneless, square-cut chuck. It can be cut into steaks and broiled.

Weight range: 1 lb (450 g) to 3 lb (1.3 kg).

Best size: 3 lb (1.3 kg), Choice.

Best uses: Steaks or roasted whole.

Serving size: 8 oz (225 g) or larger.

Kitchen yields for One Chuck, Tender, Choice, 3 lb (1.3 kg):

Trim and fat = 6 oz (170g).

One chuck, tender, 3 lb = Five 8-oz (225-g) steaks.

DECKEL MEAT

See Blade Meat.

EYE ROUND, MBG #171C

Muscle inside the bottom round. It is an evenly shaped piece of meat, but is generally dry. *See also* Bottom Round, Goose Neck, MBG #170.

Weight range: 3 lb (1.3 kg) to 6 lb (2.7 kg).

Best weights: 5 lb (2.2 kg) to 6 lb (2.7 kg).

Best uses: Braised whole or braised steaks.

Serving size, pot roast: 4 oz (112 g), cooked.

6 oz (170 g), raw braising steaks.

Kitchen yields for One Piece, Choice, 6 lb (2.7 kg):

Waste and fat trim = 6 oz (170 g).

Usable trim = 6 oz (170 g); can be blended into hamburger meat.

Usable meat = 5 lb 4 oz (2.25 kg).

Eighteen 4-oz (112-g) servings, pot roast.

Twelve 6-oz (170-g) raw braising steaks. There will be some usable trim left.

FILET TIPS

Also called *tails*, these are the pieces left when filet mignon is cut.

Packs: By weight, often 10-lb (4.5-kg) cartons.

Weight range: 6-oz (170-g) to 7-oz (200-g) individual pieces.

Best uses: Cut in strips or small slices and sautéed quickly to order.

Serving sizes: 4 oz (112 g) to 6 oz (170 g).

Waste: Trim, fat and gristle, about 10 percent.

Kitchen yields for One 10-lb (4.5-kg) carton: 9 lb (4 kg) usable meat.

9 lb (4 kg) usable meat = 28 servings,
5 oz (140 g) each.

FLANK STEAK, MBG #193

Skinny flat muscle from inside the flank.

Weight range: 1 lb (450 g) to 1½ lb (600 g).

2 lb (900 g) to 2¾ lb (1.2 kg).

Best sizes: 2 lb (900 g) to 2½ lb (1.2 kg).

Best uses: London broil or cut into strips for fajitas.

Serving size: 5 oz (140 g) for broiled medium rare meat as main course.
3 oz (85 g) for fajitas, raw meat.

Kitchen yields for One Piece, Purchased Weight, 2½ lb (1.2 kg):

Waste fat and trim = 3 oz (85 g).

Usable meat = 2 lb 5 oz (1.1 kg).

One piece, 2½ lb (1.2 kg) = 5 servings
London broil.

One piece, 2½ lb (1.2 kg) = 13 servings
fajitas.

GOOSENECK BOTTOM ROUND

See Bottom Round, Goose Neck, MBG #170.

HAMBURGER MEAT (CHOPPED BEEF), MBG #136

The fat content of hamburger meat can be specified when ordering. Most common is 80 percent lean and 20 percent fat, although some operations prefer an 85 percent to 15 percent blend. The meat cut can be specified as well, and buyers can select from various quality levels. Sirloin butt and chuck are considered to be the best. Ground meat will oxidize and turn brown quickly after grinding. Ground meat is available fresh and frozen. Frozen meat loses juices after cooking. Fresh hamburger meat should be handled as little as possible, as it will spoil rapidly.

Weights: Normally 10-lb (4.5-kg) bags.

Serving sizes: 4 oz (112 g) hamburger patty.

8 oz (225 g) chopped steak.

Kitchen yields:

Shrinkage varies according to fat content and how well the meat is cooked. Average shrinkage is ⅓ by weight when the meat is cooked to the well-done stage.

One 10-lb (4.5-kg) bag = Forty 4-oz (112-g) patties.

Five 10 lb (22.5 kg) bags = Two hundred 4-oz (112-g) patties.

One 10-lb (4.5-kg) bag = Twenty 8-oz (225-g) patties.

Calories: 4-oz (112-g) patty, 80/20 = 324 calories.

KNUCKLE, MBG #167

This tender piece is cut from the leg above the knee. The meat is tender and has some marbling and is still juicy after braising. The piece is odd shaped and difficult to portion.

Weight range: 8 lb (3.6 kg) to 15 lb (6.7 kg).

Best sizes: 14 lb (6.3 kg) to 15 lb (6.7 kg),
Choice.

Best uses: Pot roast, stew, tartar steak when well trimmed.

Serving size:

Pot Roast: 4 oz (112 g), cooked.

Stew: 4½ oz (125 g), cooked meat.

Kitchen yields for One Piece, Choice, 10 lb (4.5 kg):

Waste and fat trim = 1½ lb (670 g).

Usable trim = 8 oz (225 g).

Usable meat = 8 lb (3.6 kg).

1 piece, 10 lb (4.5 kg) = 23 servings
pot roast.

One piece, 10 lb (4.5 kg) = 20 servings
stew.

KNUCKLE, MBG #167A

Same piece as MGB #167, but with fat trimmed off. Referred to as *peeled knuckle*, it is more economical to purchase than knuckle, MBG #167, because there is less waste fat and the piece is easier to handle.

Weight range: 6 lb (2.7 kg) to 11 lb (4.9 kg).

Best sizes: 8 lb (3.6 kg) to 11 lb (4.9 kg),
Choice.

Best uses: Pot roast, stew, roasted whole, kebab, tartar steak.

Kitchen yields: See Knuckle, MBG #167.

OUTSIDE ROUND, MBG #171 B

Muscle inside the Bottom Round. It is an evenly shaped piece of meat inside the Bottom Round, Goose Neck MBG #170. The piece is usually marbled and is a good choice for making pot roast on account of its shape. *See also* Bottom Round, Goose Neck MBG #170, and Eye Round, MBG #171C.

Weight range: 8 lb (3.6 kg) to 16 lb (7.2 kg).

Best weights: 10 lb (4.5 kg) to 13 lb (5.8 kg).

Best uses: Braised split.

Kitchen yields: *See* Gooseneck Bottom Round, MBG #170.

OXTAILS, MBG #721

Oxtails are usually sold whole and frozen. Make sure that pieces are cut at the joints, not randomly cut with the meat saw, because the bone can be splintered and then loosened as it cooks in the sauce, becoming a potential hazard. Only the large joints can be used in stew; the tips can be used in soups. There are no grades for oxtails.

Packs: 10-lb (4.5-kg) cartons.

Weight range: 1½ lb (600 g) to 2 lb (900 g), whole tails.

Best sizes: 1½ lb (600 g) to 2 lb (900 g), the larger the better.

Best uses: Stew and soup.

Serving size: 10 oz (280 g) raw meat, or about 2 to 4 pieces.

Kitchen yields for 10 lb (4.5 kg),

Purchased Weight, Whole Tails:

Waste fat = 6 oz (170 g).

Usable meat for stewing = 6 lb 10 oz (3 kg).

Usable pieces for soup = 3 lb (1.3 kg).

One 10-lb (4.5-kg) carton = 10 servings stew.

10 lb (4.5 kg) tips = 2 gal (7.6 l) clear oxtail consommé.

RIB EYE ROLL, MBG #112

The center muscle of the rib, boneless and completely trimmed. Choice grade is often used. Catering halls frequently use rib eye rolls because they are smaller than rib roasts and occupy less oven space while yielding approximately the same number of servings. One advantage is that rib eye rolls are easy to slice. However, there is a loss of quality, and rib eye rolls are not as flavorful as rib roasts. The yield is contingent on slow roasting and on being able to serve both outside cuts.

Weight range: 5 lb (2.2 kg) to 12 lb (5.4 kg).

Best sizes: 8 lb (3.6 kg) to 10 lb (4.5 kg).

Best uses: Roasted whole or cut into rib eye steaks.

Serving size: 8 oz (225 g) roast beef.
10 oz (280 g) steaks, raw weight.

Kitchen yields for One Rib Eye Roll, 9 lb (4 kg):

Cooked weight, medium rare = 8 lb (3.6 kg).

Waste and trim after cooking = 4 oz (112 g).

One 9-lb (4-kg) rib eye roll
= Fifteen 8-oz (225-g) servings.
= Fourteen 10-oz (280-g) steaks.

RIB, OVEN-PREPARED, MBG #107

Prime rib with short ribs removed, aitchbone removed, but blade meat still attached. Note that Oven-Prepared Rib, MBG #107, has the blade bone and the muscles above and below still attached. Oven-Ready Rib, MBG #109, has the blade bone and muscles removed. These pieces weigh about 3 lb (1.3 kg) to 4 lb (1.8 kg), which is important to remember when comparing prices.

Weight range: 17 lb (7.6 kg) to 28 lb (12.6 kg).

Best sizes: 20 lb (9 kg) to 22 lb (9.9 kg) Choice.

Best uses: Prime rib, standing prime rib.

Serving size:

À la carte: 10 oz (340 g), without bones.

Banquet: 9 oz (255 g) cooked weight, medium and boneless.

Buffet: 6 oz (170 g) cooked weight, medium and boneless.

Kitchen yields for One Rib, Oven-

Prepared, 21 lb (9.45 kg): Blade bone and muscle removed before roasting = 18 lb (7.65 kg).

Cooked weight = 16 lb (7.2 kg).

Waste and trim after cooking = 2½ lb (1.1 kg).

Bones after cooking = 2¼ lb (1 kg).

Usable cooked meat = 11¼ lb (5 kg).

One rib, 21 lb (9 kg), yields:

- 16 servings *à la carte*, 10 oz (280 g), without bones.
- 20 banquet servings, 9 oz (255 g), cooked weight.
- 30 buffet servings, 6 oz (170 g), cooked weight.

Note that the yield is always lower than the mathematical potential in *à la carte* service. For optimum yield, both outside cuts must be purchased, but customers frequently ask for special cuts such as center cuts, bone-in cuts, and the like, thus lowering the yield.

RIB, OVEN READY, MBG #109

Rib bones in prime rib, but with the aitchbone removed. The piece is easier to handle than Oven-Prepared Rib, MBG #107, because it is lighter, yet it gives the same yield. The piece is often sold netted; the netting should not be removed before roasting. The yield depends on shrinkage during the roasting process. Normal roasting time is 3½ hours at 325°F for a 22-lb (10-kg) rib. Low-temperature roasting increases yield.

Weight range: 14 lb (6.3 kg) to 24 lb (9.9 kg).

Best sizes: 18 lb (8.1 kg) to 20 lb (9 kg) and 20 lb (9 kg) to 22 lb (9.9 kg).

Best uses: Prime rib, standing prime rib.

Kitchen yields for One Rib, Oven-Ready, 21 lb (9.45 kg): See yield information for Rib, Oven-Prepared, MBG #107.

RIB, PRIMAL, MBG #103

Whole rib, with short ribs still attached. Few operators purchase Primal Rib, MBG #103, because it is a heavy, unwieldy piece of meat and requires cutting off the bones with a meat saw. This piece is listed only as a matter of reference. The distance between the edge of the vertrabae and the end of the rib should be no more than 10 in, measured at the chuck end.

Weight range: 24 lb (10.8 kg) to 40 lb (18 kg).

Best sizes: 33 lb (14.8 kg) to 40 lb (18 kg).

Best use: Roast prime rib.

Kitchen yields: One 38-lb (17.1-kg) primal rib = One 22-lb (9.9-kg) #109 rib.

Waste fat, bones, and short ribs = about 45 percent by weight.

RIB, ROAST-READY AND BONELESS, MBG #110

This is a popular specification for a banquet operation, because the meat needs little trimming after roasting.

Weight range: 11 lb (4.9 kg) to 19 lb (8.5 kg).

Best weights: 13 lb (5.8 kg) to 16 lb (7.2 kg).

Best uses: Roasted whole.

Serving size:

À la carte: 10 oz (340 g).

Banquet: 8½ oz (240 g) medium, outside cuts are counted as sold.

Kitchen yields for One Rib, Purchased

Weight 16 lb (7.2 kg): Cooked weight = 12 lb (5.4 kg).

Waste and trim = 1 lb (450 g) after cooking.

Usable cooked meat = 11 lb (4.9 kg).

One rib, 16 lb (7.2 kg), yields:

- 16 à la carte servings, 10 oz (340 g) each.
- 20 banquet servings, 8½ oz (240 g) each.
- 30 buffet servings, 6 oz (170 g) each, medium.

Note: Outside cuts are included in the yield.

ROUND, CHICAGO, MBG #158

Untrimmed primal round, with aitchbone and shin attached. It is a massive chunk of meat, but useful for large buffets or in buffet brunch service. It has to be cooked very slowly. Roasting time is about 6 hours at 325°F. The temperature should be checked with a meat thermometer. The piece needs considerable butchering before it can be used.

Weight range: 25 lb (11.2 kg) to 100 lb (45 kg).

Best weights: 80 lb (36 kg) to 95 lb (42.7 kg).

Serving size: See below.

Kitchen yields for One Round, Chicago, Choice, 80 lb (36 kg):

Waste and trim fat = 8 lb (3.6 kg).

Usable scraps, raw = 6 lb (2.7 kg).

Bones = 9 lb (4 kg).

Trimmed weight, bone attached = 57 lb (25.6 kg).

150 servings as main course buffet, when other items are served.

200 servings for cocktail reception, when other items are served.

Note: There will be cooked meat trimmings that can be used for hash. The bone can still be used for making stock. Overall, the piece is very economical.

ROUND, STEAMSHIP, MBG #160

Same piece as Chicago Round, trimmed, ready for roasting.

Weight range: 47 lb (21.2 kg) to 80 lb (36 kg).

Best weights: 60 lb (27 kg) to 70 lb (31.5 kg).

Best uses: Roasted whole as Steamship Round.

Kitchen yields: Usually there is still some trimming waste before roasting. The yield is the same as for Round, Chicago, MBG #158.

SHIN MEAT

Relatively inexpensive lean muscles that can be used for stew or ground for consommé clarification meat. It makes excellent stew meat because it is very flavorful. When ordering, specify *trimmed* or *untrimmed*. When the meat is intended to be used for consommé, only the superfluous fat has to be removed.

Packs: By weight, often 10-lb (4.5-kg) cartons.

Best uses: Clarification or stew meat.

Serving size: 4 oz (112 g) cooked stew, 6 oz (170 g) raw.

Kitchen yields: Trimming loss for stew meat = 20 percent by weight.

One 50-lb (22.5-kg) carton = 40 lb (18 kg) usable meat.

40 lb (18 kg) meat = 100 servings stew, 6 oz (170 g) raw.

SHORT RIBS, THREE BONES, MBG #123

Short ribs have become popular again and are often served boneless, braised, on mashed potatoes or other vegetables.

Weight: 1 lb (450 g) each, bone in.

Best uses: Boiling and braising.

Serving size: 1 piece, 11 oz (310 g) cooked weight, bone in.

Kitchen yields: Ready to use, no further trimming required. *Note:* When bones are removed, all fat trimmed off, and the meat is braised until very soft, the portion will weigh about 5 oz (140 g).

SIRLOIN BUTT, MBG #182

The boneless end piece of the loin, with some of the fat cover, flap piece, and connecting tissues removed. When split, it is separated into Top Sirloin Butt, MBG #184, and Bottom Sirloin Butt, MBG #185.

Weight range: 11 lb (4.9 kg) to 22 lb (9.9 kg).

Best weights: 18 lb (8.1 kg) to 22 lb (9.9 kg).

Best uses: Steaks or split and roasted or cut into julienne for sauté dishes if meat is Choice grade and aged.

Kitchen yields: It is more practical to purchase the piece split as Top Butt, MBG #184, or Bottom Butt, MBG #185. See these cuts for yields.

SKIRT STEAK (BEEF PLATE), MBG #121E

Skinned diaphragm, consisting of coarse but flavorful fibers, that can be broiled and is tender when cut against the grain. The piece has been rediscovered by chefs and is often served in trendy restaurants. Skirt steaks are also available as portion cut. See MBG #1121D for portion-cut Skirt Steaks.

Weight range: 1 lb (450 g) to 3 lb (1.3 kg).

Best weights: 3 lb (1.3 kg) or smaller.

Best uses: Steaks.

Serving sizes: 4 oz (112 g) to 8 oz (225 g).

Kitchen yields: The piece is skinned and ready to be cut. One 3-lb (1.3-kg) piece = eight 6-oz (170-g) steaks.

STRIP LOIN, MBG #179

A short, cut strip loin, bone in. Few operations buy this piece of meat because boning it requires skill and an operation must be able to use all by-products. The bones are soft and make excellent stock. When the meat is aged, the bones must be parboiled (blanched) to remove unwanted flavors.

Weight range: 10 lb (4.5 kg) to 20 lb (9 kg).

Best weights: 17 lb (7.6 kg) to 20 lb (9 kg).

Best uses: Roasted whole or cut into New York shell steaks.

Kitchen yields: One 17-lb (7.6-kg) strip = 10 lb (5 kg) trimmed meat. When

buying strip loin, an operation must keep in mind that the flank width, which is basically worthless and must be trimmed off, determines the usable yield. MBG specification #175 has a 4-in flank, MBG #177 has a 3-in flank, and MBG #179 has the shortest, only a 2-in flank, all measured at the sirloin end.

Note: Prime-graded strips are fattier than Choice and can yield as much as 20 percent less usable meat than a Choice strip of same weight. Aged meat also requires more trimming, but weighs less than fresh meat because of the drying process during aging.

STRIP LOIN, MBG #180

This is the MBG #179 loin, with bones removed. It is frequently shipped in Cryovac plastic bags. There is frequently a small amount of meat juices/blood in the bag, especially if the meat was carelessly stacked, emitting a faint unpleasant smell when the bag is opened. The smell will dissipate quickly unless the meat is overaged. The piece is often referred to as shell. Make sure that the MBG #180 strip is properly trimmed. The flank should not be wider than 2 in, measured at the sirloin end.

Weight range: 8 lb (3.6 kg) to 15 lb (6.7 kg).

Best weights: 10 lb (4.5 kg) to 12 lb (5.4 kg).

Best uses: Steaks or roasted whole.

Serving sizes: 6 oz (170 g), roast, cooked medium.
10 oz (280 g) or larger, steaks.

Kitchen yields for One Strip Loin, MBG #180, Choice, 11 lb (5 kg):

Waste and trim fat = 1 lb (450 g).

Usable trimmings = 4 oz (112 g).

Usable meat = 9 lb 12 oz (4.3 kg).

One 11-lb (5-kg) strip loin = Twenty 6-oz (170-g) servings, roasted whole, medium.

One 11-lb (5-kg) strip loin = twelve 10-oz (280-g) steaks, if nerve end piece is not used. The nerve end can be used as London broil.

TENDERLOIN, MBG #189

Often referred to as *filet*, this is the full tenderloin with fat still attached. It is frequently shipped in Cryovac plastic bags. There are many confusing trim variations on the market. This is generally the most expensive beef cut, and it is important that the exact specifications, as established by the *Meat Buyers Guide*, are clearly stated when ordering. Tenderloin, MBG #189 A, is defatted but sold with the side muscle on; MBG #189B is partially defatted. Many operations buy peeled tenders, MBG #190, or skinned tenders, MBG #190A.

Weight range: 5 lb (2.2 kg) to 8½ lb (3.8 kg).

Best sizes: 7 lb (3.1 kg) to 8½ lb (3.8 kg).

Best uses: Roast tenderloin, chateaubriand, filet mignon.

Serving size: 4½ oz (125 g) cooked meat, roasted whole.

6 oz (170 g) filet mignon.

7 oz (200 g) filet mignon.

8 oz (225 g) filet mignon.

9 oz (255 g) filet mignon.

Kitchen yields: One 6½-lb (2.9-kg) tenderloin = 1¾ lb (780 g) waste and trim fat.
 = 4¼ lb (2.25 kg) defatted, side muscle on.
 = 3½ lb (1.7 kg) trimmed for roasting.
 = 8 to 9 servings, roasted whole.
 Filet mignon, side muscle on, head partially used.
 = eight 6-oz (170-g) filets mignons.
 = seven 7-oz (200-g) filets mignons.
 = six 8-oz (225-g) filets mignons.
 = five 9-oz (255-g) filets mignons.
 = 10 oz (280 g) usable tenderloin tips.

TENDERLOIN, MBG #190

Also referred to as *filet* or *tender*. This piece is defatted, with silver skin, wedge

fat, and side strip muscle still attached. It is almost always shipped in Cryovac plastic bags. Many operations buy peeled cow tenders. They are smaller than Choice-grade tenders and less flavorful, but can be used for inexpensive dishes.

Weight range: 3 lb (1.3 kg) to 6 lb (2.7 kg).

Best sizes: 5½ lb (2.2 kg) to 6 lb (2.7 kg).

Best uses: Roasting, filet mignon.

Kitchen yields: See Tenderloin, MBG #189, for yields.

TENDERLOIN, SHORT, MBG #192

Tenderloin removed from the short loin without butt and head. The piece is normally sold with natural fat on.

Weight range: 2 lb (900 g) to 5 lb (2.2 kg).

Best weights: 4 lb (1.8 kg) to 5 lb (2.2 kg).

Best use: Filet mignon.

Kitchen yields: The piece is awkward to use, because it renders only three to four 8-oz (225-g) filets mignons; the rest consists of tenderloin tips and fat.

TONGUE, FRESH

This piece is seldom used. *Also see* Tongue, Smoked, Raw, under Cured and Smoked Beef Products. The meat is lean.

Weight range: 4 lb (1.8 kg) to 5 lb (2.2 kg).

Serving size: 5 oz (140 g).

Kitchen yields: One 4½-lb (2-kg) tongue = 2 lb 9 oz (1.1 kg), cooked, peeled, and trimmed.

One tongue = 8 servings.

Note: There are usable trimmings that can be utilized.

TOP ROUND, MBG #168

Also called *inside round*. It is a large muscle, available with bone in and boneless. Specify boneless round. For roasting whole, purchase smaller rounds in the 15 lb (6.7 kg) to 18 lb (8.1 kg) range, or

purchase 21-lb (9.4-kg) to 23-lb (10.3-kg) rounds, split and netted.

Weight range: 14 lb (6.3 kg) to 23 lb (10.3 kg).

Best weights: 21 lb (9.4 kg) to 23 lb (10.3 kg).

Best uses: Roasted whole or split, pot roast, stew.

Serving size: 4-oz (112-g) sandwich.

Kitchen yields for One 17-lb (7.6-kg)

Top Round, Boneless:

Waste and trim fat, raw = 8 oz (225 g).

Cooked meat, roasted medium = 12 lb 4 oz (5.4 kg).

Scraps and trim, cooked = 12 oz (340 g).

One 17-lb (7.6-kg) top round = 45 sandwich servings, 4 oz (112 g) each.

Note: Kitchen yields are contingent on slow roasting. For pot roast and stew yields, see Bottom Round, MBG #170.

TOP SIRLOIN BUTT, MBG #184

Relatively tender muscles in Choice grade.

Weight range: 7 lb (3.1 kg) to 15 lb (6.7 kg).

Best weights: 12 lb (5.4 kg) to 15 lb (6.7 kg).

Best uses: Steaks, roast, ground meat, cut in julienne (strips) and sautéed if meat is Choice grade and aged.

Serving size: Depends on application.

Kitchen yields for One 15-lb (6.75-kg)

Top Sirloin Butt, Choice:

Waste and trim fat = 2 lb (900 g).

Usable scraps = 12 oz (340 g).

Usable meat = 12 lb (6.9 kg).

QUICK LOOK-UP FOR POPULAR BEEF DISHES

Many cuts can be purchased portion controlled, ready to use. The dishes are listed alphabetically.

BRACIOLE

Bracioles are stuffed, rolled, and then braised steaks. The same specifications

can also be used for German Beef Rouladen. This is an old-fashioned dish.

Best cuts to use: Bottom Round, MBG #170.

Top Round, MBG #168.

Eye Round, MBG #171 C.

Best weights: Bottom round, 26 lb (11.7 kg) to 29 lb (13 kg).

Top round, 21 lb (9.4 kg) to 23 lb (10.3 kg).

Eye round, 5 lb (2.2 kg) to 7 lb (3.1 kg).

Serving size: 5 oz (140 g), raw weight.

Cooked weight depends on weight of stuffing.

Kitchen yields: One 22-lb (9.9-kg) top round, not trimmed = 60 steaks.

One 27-lb (12.1-kg) bottom round = 70 steaks.

One 6-lb (2.7-kg) eye round = 16 steaks.

Note: Pieces must be pounded flat in order to achieve yield. See the listed pieces for precise yield information.

BRAISED STEAK

Often referred to as *Swiss steak*. The dish is seldom found on menus today.

Best cuts to use: Top Round, MBG #168.
Bottom Round, MBG #170.

Eye Round, MBG #171 C.

Best weights: Top round, 21 lb (9.4 kg) to 23 lb (10.3 kg).

Bottom round, 26 lb (11.7 kg) to 29 lb (13 kg).

Eye round, 5 lb (2.2 kg) to 7 lb (3.1 kg).

Serving sizes: 6 oz (170 g) to 7 oz (200 g), raw.

4½ oz (125 g) to 5½ oz (140 g), cooked.

Kitchen yields: One 22-lb (9.9-kg) top round, not trimmed = 40 servings.

One 27-lb (12.1-kg) bottom round = 50 servings.

One 6-lb (2.7-kg) eye round = 12 servings.

Calories: 5 oz (140 g), cooked = 440 calories.

BRISKET, BARBECUED OR BOILED

Best cut to use: Fresh Brisket, MBG #120.

Best sizes: 10 lb (4.5 kg) to 12 lb (5.4 kg).

Serving size: 4 oz (112 g), lean, cooked.

Kitchen yields: One 12-lb (54-kg) brisket = 6½ lb (3 kg), well trimmed, cooked.
One brisket = 22 servings.

CHATEAUBRIAND

Large tenderloin steak for 2 or more servings. Roasted beef tenderloin is often erroneously referred to as chateaubriand.

Best cut to use: Tenderloin, MBG #189 fat on.

Serving size: 24 oz (675 g) for 2 customers, including fat; 10 oz (280 g) per person after fat is trimmed off. The meat tastes better when some fat is left on during cooking and peeled away to brown the meat before service.

Kitchen yields: Typically, only one chateaubriand is cut from each tenderloin, and the rest of the meat is used for filet mignon. This is an economical way to use the tenderloin head.

CHOPPED STEAK

Freshly ground chopped meat is much better than ready-ground chopped meat, providing that the operation is able to maintain adequate safety and sanitation standards. Cutting and grinding meat in a hot kitchen can cause bacterial contamination. Once ground, the meat should be handled as little as possible.

Best cuts to use, in order of preferability:

Chuck, Boneless, Square Cut, MBG #115.

Top Sirloin Butt, MBG #184.

Blade Meat, MBG #109 B.

Top Round, MBG #168.

Best sizes: Chuck, 65 lb (29.2 kg) to 85 lb (38.2 kg).

Top sirloin butt, 12 lb (5.4 kg) to 15 lb (6.7 kg).

Blade meat, purchased by weight.

Top round, 14 lb (6.3 kg) to 15 lb (6.7 kg).

Serving sizes: 4 oz (112 g) to 10 oz (280 g) per person. Chopped steaks are seldom larger than 10 oz (280 g).

Kitchen yields: Depend on application.

KEBAB/KEBOB

Meat skewers that can be made with beef or lamb meat. Often precooked onion and pepper chunks, cherry tomatoes, and other vegetables are added.

Best cuts to use: Tenderloin tips.

Top sirloin butt, MBG #184.

Knuckle, MBG #167 A.

Strip loin, MBG #180.

Serving sizes: 5 oz (140 g), for main course plus vegetables.

1 oz (42 g), for hors d'oeuvres plus vegetables.

Kitchen yields: 50 lb (22.5 kg)

tenderloin tips, not trimmed = 130 orders, main course, or 460 orders, hors d'oeuvres.

One 14-lb (6.3-kg) top sirloin butt, not trimmed = 35 orders, main course, or 120 orders, hors d'oeuvres.

One 10-lb (4.5-kg) knuckle, not trimmed = 30 orders, main course, or 100 orders, hors d'oeuvres.

One 10-lb (4.5-kg) strip loin = 28 orders, main course, or 90 orders, hors d'oeuvres.

LONDON BROIL

The name *London broil* is freely used for just about any kind of broiled meat served sliced. Supermarkets often market top round cuts as London broil. Obviously, the meat must be tender to be broiled and should also be juicy. The sirloin nerve end cut that is left over when steaks are cut can be successfully used for London broil.

Best cuts: Flank Steaks, MBG #193.

Strip Loin, MBG #180.

Strip Loin, Nerve End Cuts.

Top Sirloin Butt, MBG #184.

Best weights: Flank steaks, 2 lb (900 g) to 2½ lb (1.2 kg).
 Strip loin, 10-lb (4.5-kg) piece.
 Strip loin, nerve end cuts, 22 oz (625 g) to 24 oz (680 g).
 Top sirloin butt, 12 lb (5.4 kg) to 15 lb (6.7 kg), not trimmed.

Serving size: 5 oz (140 g), cooked medium rare.

Kitchen yields: One 2-lb (900-g) flank steak = 4 to 4½ servings.

One 10-lb (4.5-kg) strip loin = 20 servings when cut on machine for banquets.

Sirloin nerve end cut = 3 to 12 servings.

One 14-lb (6.3-kg) top sirloin butt = 25 to 26 servings.

Calories: 5 oz (140 g) Choice grade flank steak, cooked medium rare = 290 calories.

NEW YORK-CUT SIRLOIN STEAK

Also called *strip steak* or *sirloin steak*. When bone-in steaks are required, they must be cut with a meat saw, and the meat must be very cold—almost frozen—when cut with a meat saw. Cutting steaks with a handsaw in a hot kitchen is not recommended.

Best cuts: Strip Loin, MBG #179, bone in. Strip Loin, MBG #180, boneless.

Best weights: Strip loin, MBG #179, bone in, 17 lb (7.6 kg) to 20 lb (9 kg).

Strip loin, MBG #179, bone in, 14 lb (6.3 kg) to 16 lb (7.2 kg).

Strip loin, MBG #180, boneless, 10 lb (4.5 kg) to 12 lb (5.4 kg).

Serving sizes: Bone in = 8 oz (225 g) to 16 oz (450 g).

Boneless steak = 7 oz (200 g) to 16 oz (450 g).

Kitchen yields:

One 15-lb (6.7-kg) Strip Loin, MBG #179:

- 22 steaks bone in, 10 oz (280 g) each, or

- 15 steaks, bone in, 14 oz (400 g) each, or
- 13 steaks, boneless, 10 oz (280 g) each, or
- 10 steaks boneless, 14 oz (400 g) each.

One 10-lb (4.5-kg) Strip Loin, MBG #180:

- 9 steaks, boneless, 12 oz (340 g) each.
- 17 steaks, boneless, 8 oz (225 g) each.

POT ROAST

Well-marbled meat must be used for moist and juicy pot roast. The best piece is Knuckle, MBG #167, but it is unevenly shaped and somewhat difficult to portion. Bottom and top round are fairly evenly shaped but could be dry. Chuck roll can be fatty and may fall into pieces when sliced. It is recommended to braise the pot roast for large banquets a day ahead and then slice the meat on a machine when cold.

Best cuts: Knuckle, MBG #167, 14 lb (6.3 kg) to 15 lb (6.7 kg).

Bottom Round, Goose Neck, MBG #170, 26 lb (11.7 kg) to 29 lb (13 kg).

Top Round, MBG #168, 21 lb (9.4 kg) to 23 lb (10.3 kg).

Chuck Roll, MBG #116 A, 21 lb (9.4 kg) to 23 lb (10.3 kg).

Eye Round, MBG #171 C, 5 lb (2.2 kg) to 6 lb (2.7 kg).

Serving size: 5 oz (140 g).

Kitchen yields: One 14-lb (6.3-kg) knuckle = 28 to 30 servings.

One 27-lb (12.1-kg) bottom round = 60 servings.

One 21-lb (9.4-kg) top round, boneless = 45 to 50 servings.

One 15-lb (6.7-kg) netted chuck roll = 38 servings.

One 6-lb (2.7-kg) eye round = 16 servings.

Note: Yields vary according to the fatness of the meat and the trimming necessary.

PRIME RIB OF BEEF

See Rib, Oven-Ready, MBG #109, or Rib, Roast-Ready and Boneless, MBG #110. This is a popular specification for a banquet operation, because the meat needs little trimming after roasting.

ROUND ROAST

See Top Round, MBG #168.

SIRLOIN ROAST

See Strip Loin, bone in, MBG #179, or Strip Loin, MBG #180.

TARTAR STEAK

This excellent dish is seldom served today because of apprehension about eating uncooked meat. The meat used for tartar steak must be fat free and at the same time have flavor and substance. Beef tenderloin, often recommended, is too mushy; beef strip loin is usually too fatty.

Best cuts: Knuckle, peeled, MBG #167A, 14 lb (6.3 kg).

Bottom Sirloin Ball Tip, MBG #185B, 3 lb (1.3 kg).

Tenderloin, MBG #190A, 4 lb (1.8 kg).

Serving sizes: 6 oz (170 g), as main course.

½ oz (14 g), for each canapé.

1½ oz (42 g), on buffets with other choices.

Kitchen yields: One 14-lb (45-kg) knuckle, peeled, 9½ lb (4.25 kg) clean meat

= 25 servings, main course.

= 300 canapés.

= 100 servings, buffet.

One 3-lb (13-kg) bottom sirloin ball tip, 2½ lb (1.1 kg) clean meat

= 7 servings, main course.

= 80 canapés.

= 28 servings, buffet.

One 4-lb (18-kg) tenderloin, 3 lb 10 oz (1.6 kg) clean meat

= 8 servings, main course.

= 95 canapés.

= 32 servings, buffet.

TENDERLOIN ROAST

Most operations buy trimmed Tenderloins, MBG #190, because they are basically oven ready and require little trimming. The trimmings can be used as clarification meat. However, the tenderloins purchased with fat cover, MBG #189, are considered better quality. On the downside, there is much waste to dispose of. See Tenderloins for trimming losses.

Best cuts: Tenderloin with fat, MBG #189, 7 lb (3.1 kg) to 8½ lb (3.8 kg).
Tenderloin, trimmed, MBG #190, 5½ (2.2 kg) to 6 lb (2.7 kg).

Serving sizes: 2 slices, 5 oz (140 g), total, as main course.

1½ slices, 3 oz (85 g), total, on buffets.

Kitchen yields: One 7½-lb (33-kg)

Tenderloin, MBG #189: 8 servings, main course, 12 servings, buffet.

One 5½-lb (22-kg) Tenderloin, MBG #190: 8 servings, main course, 12 servings, buffet.

PORTION CUTS OF BEEF

The following are descriptions of commonly available portion cuts. Additional cuts may be available locally. When ordering, specify grade and whether frozen or fresh. Packs vary according to products. All items are sold by both count and weight.

BRAISING STEAKS, MBG #1102

Irregularly shaped steaks with some fat cover still attached. They can be mechanically tenderized. The steaks must be braised.

Common sizes: 4 oz (112 g) to 8 oz (225 g).

Serving size: 8 oz (225 g), uncooked.

CHOPPED STEAKS AND HAMBURGERS, MBG #1336

The patties are normally machine made and can be tough because the meat is compacted. They are available with vegetable protein added.

Common sizes: 1 oz (28 g) to 10 oz (280 g).

Serving size: The typical hamburger patty weighs 4 oz (112 g).

CUBED STEAKS

Boneless steaks mechanically tenderized for broiling. They are inexpensive and can be cooked to order for sandwiches.

Common sizes: 3 oz (85 g), 4 oz (112 g), 6 oz (170 g), 8 oz (225 g).

Serving size: 8 oz (225 g).

PORTERHOUSE STEAKS, MBG #1173

Bone-in steaks, cut across the short loin including the tenderloin. Porterhouse steaks by definition must contain a piece of tenderloin. The flank trim can range from 4 in (100 mm) to 2 in (50 mm), which determines the price. It is very important to specify the precise trim.

Common sizes: 10 oz (280 g) to 24 oz (to 680 g), in 2-oz (56-g) increments.

Serving sizes: 12 oz (340 g) to 16 oz (450 g), 1 serving.
20 oz (560 g) or 24 oz (680 g), 2 servings.

Kitchen yields: When ordering steaks, make sure the trim is clearly specified.

RIB EYE STEAKS, MBG #1103

Boneless steaks cut from the eye of the prime rib, suitable for broiling. Available as bone-in with the tail, also called *lip left on*. The meat is generally fatty.

Common sizes: 8 oz (225 g) to 18 oz (500 g), bone in.
4 oz (112 g) to 12 oz (340 g), boneless.

Serving size: 12 oz (340 g).

RIB EYE ROLLED STEAKS, MBG #1112

Steaks resembling sirloin steaks but cut from the center portion of the prime rib. The steaks are fattier than sirloin steaks but are usually more flavorful. The length of the “lip” or flank attached determines the price.

Common sizes: 4 oz (112 g) to 12 oz (340 g).

Serving size: 12 oz (340 g).

ROUND STEAKS, TOP, MBG #1168, AND OTHERS

There are many choices of round steaks, depending on the trim and the primal cut from which they were fabricated. The cuts are suitable for braising.

Weight range: Typically, 3 oz (85 g) to 12 oz (340 g).

Serving size: 10 oz (280 g).

SKIRT STEAKS, MBG #1121

Tender steaks suitable for quick broiling. The market distinguishes between “inner” and “outer” skirt steaks. There is little quality difference. The steaks are thin and must be cooked to order, otherwise they will get tough.

Common sizes: 4 oz (112 g) to 8 oz (225 g).

Serving size: 8 oz (225 g).

STRIP LOIN STEAKS, BONE IN, MBG #1179, AND BONELESS, MBG #1180

Also called *sirloin steak* or *New York-cut steak*. Available with bone in and boneless. The grade and length of the flank left on the steaks should be clearly specified.

Common sizes:

Bone-in steaks: 8 oz (225 g) to 24 oz (675 g).

Boneless steaks: 6 oz (170 g) to 20 oz (560 g).

Kitchen yields: Make sure the flank length is well specified when ordering. The intermediate size has about 3 in (75 mm) flank, when measured from the eye.

T-BONE STEAKS, MBG #1174

Bone-in steaks cut across the beef loin, basically the same as Porterhouse Steaks, MBG #1173. See Porterhouse Steaks.

TENDERLOIN STEAKS, ALSO CALLED FILET MIGNON, MBG #1189 AND #1190

A number of trims are available:

- Fat on and side muscle on
- Defatted with side muscle on
- Silver skin on and side muscle removed
- Both silver skin and side muscle removed

Sizes: For most types of trim, 3 oz (85 g) to 14 oz (400 g).

Kitchen yields: Make sure trim and fat cover are clearly specified when ordering steaks.

Serving sizes:

- 10 oz (280 g), steak house size.
- 8 oz (225 g), restaurant service.
- 7 oz (200 g), banquet service.

SMOKED, COOKED, AND CURED BEEF PRODUCTS**BÜNDNERFLEISCH**

Called *viande de grison* in French, *bresaola* in Italian. Air-dried beef, available imported or domestically produced. Use like prosciutto ham.

Weights: 5 lb (2.2 kg) average.

Serving size: 1 oz (28 g).

Kitchen yields: The product is boneless, and there is little waste.

One 5-lb (2.2-kg) piece = 75 servings.

CHIPPED BEEF

Cooked corned beef, dried and shredded. It is rather salty and is served best in a cream sauce.

Packs: 4 oz (112 g), 5 oz (140 g) and 6 oz (170 g), in individual bags.
3 lb (1.3 kg), loose in boxes.

Serving size: 4 oz (112 g), in cream sauce, lunch.

Kitchen yields: There is no waste. For à la carte service, the purchase of individual packages is recommended.

Calories: 4 oz (112 g), without sauce = 232 calories.

COOKED ROUNDS

Fully cooked rounds are available fresh or frozen, normally shipped in vacuum bags. The meat is often roasted in the bag to preserve moisture and control weight loss. The pieces are often used in delis for sandwiches.

Weight range: 5 lb (2.2 kg) to 8 lb (3.6 kg), flat rounds.

5 lb (2.2 kg) to 8 lb (3.6 kg), bottom round halves.

4 (1.8 kg) to 7 lb (3.1 kg), eye round.

Portion packs are available in various sizes.

Kitchen yields: There is little trimming waste, and the meat is ready for use.

Calories: 5 oz (140 g) = 300 calories.

CORNED BEEF, CANNED

Packs: 1-lb (450-g) and 5-lb (2.25-kg) cans.

6-lb (2.7-kg) cans, 6 per case.

Other packs are available.

Calories: 4 oz (112 g), lean, 246 calories. Calories vary according to trim.

CORNED BEEF HASH

Ground cooked corned beef, mixed with diced cooked potatoes. Convenient item when there are no corned beef trimmings generated in-house.

Packs: 7.5-oz (212-g) individual serving cans.

Six #10 cans, 6 lb (2.7 kg) each, per case.

Serving sizes: 9 oz (250 g), about 1¼ cups, restaurant size.

5½ oz (150 g), about ¾ cup, buffet size.

Kitchen yields: One can #10 = 16 servings, restaurant.

One can #10 = 25 servings, buffet.

Calories: 9-oz (250-g) serving = 525 calories.

CORNERED BRISKET**COOKED CORNERED BRISKET**

Fully cooked briskets, with most fat removed. First cut is a better-trimmed piece.

Weight range: 5 lb (2.2 kg) to 8 lb (3.6 kg), whole brisket.
3 lb (1.3 kg) to 5 lb (2.2 kg), first cut.

Serving size: 5 oz (140 g), as hot deli sandwich.

5 oz (140 g), as main course, served with cabbage.

Kitchen yields: Whole cooked brisket = 5 percent trimming waste.
First cut brisket = fully trimmed, minimal waste.
One 6-lb (2.7-kg) whole brisket = 90 oz (2.5 kg) usable meat.
One 6-lb (2.7-kg) whole brisket = 18 servings.

UNCOOKED CORNERED BRISKET, DELI TRIM

Some fat has been removed from deli trim briskets. Flat-cut briskets are split along the natural seam separating the two muscles in the brisket.

Weight range: 8 lb (3.6 kg) to 13 lb (5.8 kg), deli trim.
3 lb (1.3 kg) to 4½ lb (2 kg), flat cuts.

Serving sizes: 5 oz (140 g), as hot deli sandwich.
5 oz (140 g), as main course, served with cabbage.

Kitchen yields: Cooking loss is about 40 percent, and trimming loss after cooking about 5 percent. End pieces and some small pieces can be used for hash.
One 10-lb (4.5-kg) brisket = 5½ lb (2.5 kg) cooked meat.

UNCOOKED CORNERED BRISKET, REGULAR TRIM

Raw cornered briskets are often delivered packed in brine, making it difficult to verify the real weight. The cooking and trimming losses are substantial.

Weight range: 12 (5.4 kg) to 14 lb (6.3 kg).

Serving sizes: 5 oz (140 g), as hot deli sandwich.

5 oz (140 g), as main course, served with cabbage.

Kitchen yields: About 45 percent yield.

One 13-lb (58-kg) brisket, drained but wet: = 5½ lb (1.3 kg) lean, trimmed meat and 8 oz (225 g) trimming usable for hash.

One 13-lb (58-kg) brisket, drained but wet = 16 servings, sandwich or main course.

Calories: 5 oz (140 g) = 200 calories.

CORNERED ROUNDS

Available bone in or boneless. Bottom rounds are most likely to be cornered. The meat is lean and requires little trimming, but it can be dry.

Weight range: 35 lb (15.7 kg) to 45 lb (20.2 kg).

Serving sizes: Same as for briskets.

Kitchen yields: Bone weight = about 6 percent.

Cooking loss = 35 percent average.

Trimming loss = 5 percent of cooked weight.

One 40-lb (18-kg) round = 21 lb (9.5 kg) of usable meat.

PASTRAMI

Pastrami is cured, smoked beef, rolled in crushed coriander seeds. Genuine pastrami is made from Plate, MBG #121, but pastrami made from other beef cuts is manufactured. Plate Pastrami must be steamed before use. Most other cuts are fully cooked.

Weights: 2 lb (900 g) to 4 lb (1.8 kg), plate pieces.

2 lb (900 g) to 6 lb (2.7 kg), round pieces, cooked.

Other sizes are available.

Serving size: 5 oz (140 g), for sandwiches.

Kitchen yields: Cooking loss for plate pastrami = 18 to 20 percent of weight.

Trimming loss = 5 to 10 percent, after cooking.

Trimming loss, cooked round = 5 percent.

One 3-lb (13-kg) plate pastrami = 2 lb 6 oz (1.2 kg) cooked meat = 7 servings, sandwiches.

One 5-lb (225-kg) round pastrami = 15 servings, sandwiches.

Note: The trimmings are difficult to utilize because they are covered with coriander seeds.

ROAST BEEF

Fully cooked products are normally frozen. Many ready-to-eat products are poached in pouches with seasoning and caramel color. Oven-roasted products are roasted in pouches: some are dry roasted. Some products are identified by USDA grades, normally Choice. Many different cuts are available.

Most popular cuts:

Bottom round, no heel.

Eye round.

Top round, whole.

Top round, split.

Prime rib.

Weights: Vary, depending on cut.

Serving sizes: Vary greatly, often 4 oz (112 g) for sandwiches.

Kitchen yields: The following facts should be taken into consideration when calculating yield:

Pouch weight = 2 oz (56 g).

Residual juices = up to 5 oz (140 g).

Trimming waste = 5 percent.

The trim varies according to the manufacturer and consists mostly of unusable fat.

Calories: 4 oz (112 g), lean roast round = 225 calories.

SALAMI, BEEF

Available in many flavor profiles.

Typical sizes: 5 lb (2.2 kg) to 6 lb (2.7 kg).

Serving size: 3 oz (85 g), for sandwiches.

Kitchen yields: Trimming waste, end pieces, and wrapping = 3 percent.
One 5½-lb (24-kg) salami = 28 sandwiches.

Calories: 3 oz (85 g) = 180 calories.

TONGUE, SMOKED, RAW

Sizes: 4 lb (1.8 kg) to 5 lb 2.2 kg).

Serving size: 4 oz (112 g), for sandwiches.

Kitchen yields: Cooking and trimming loss = 50 percent.

One 4-lb (18-kg) tongue = 2 lb (900 g) cooked and trimmed meat = 8 servings.

BOAR

Wild boar meat from farm-raised animals is available.

Packs: 8 lb (3.6 kg) to 12 lb (5.4 kg), fresh hams, bone in.
10 lb (4.5 kg), loins.

Kitchen yields: Difficult to estimate, because fatness of meat varies greatly.

BUFFALO

Buffalo is marketed as a leaner alternative to beef. Some purveyors break down the pieces further than mentioned in the following description, such as steaks, top sirloin, short ribs and others. Meat cutting tests should be made to determine accurate yields. The meat is not USDA graded.

Most available cuts:

Prime ribs: 18 lb (8.1 kg) to 24 lb (10.8 kg).

Rounds: 35 lb (15.7) to 50 lb (22.5 kg).

Full loins, bone in: 18 lb (8.1 kg) to 30 lb (13.5 kg).

Ground meat: 10-lb (4.5-kg) bags.

Kitchen yields: Trim varies greatly among purveyors. Use beef yield information after trim has been confirmed. *Note:* Buffalo meat is generally leaner than beef. Ribs and loins can be roasted; rounds should be braised.

CHICKEN

FRESH WHOLE CHICKENS AND VARIETIES

Whole chickens are sold by weight, including giblets. Giblets represent approximately 6 percent of weight. Fresh chickens are often delivered in ice, and it is difficult to confirm weight when receiving them. The yellow color of chicken is often achieved by mixing marigold petals in the feed before slaughter, but color is not necessarily an indication of quality. Chickens are routinely fed antibiotics to prevent the outbreak of diseases. Some concern has been raised about this practice, and major chicken producers have pledged to phase out the use of antibiotics. Free-range chickens are not raised in cages and have slightly darker, but more flavorful, meat.

Industry terms:

Dressed poultry: Picked clean, head and feet still attached, not eviscerated.

Ready-to-cook poultry: Picked clean and singed, eviscerated.

BABY CHICKEN

Also called *squab chicken*, or *poussin* in French.

Packs: By weight and by piece.

Sizes: Bone in = 14 oz (400 g) to 16 oz (450 g).

Boneless = 14 oz (400 g).

Serving size: 1 bird.

BROILER

Available in three sizes.

Packs: By weight and by piece, packed in ice.

Sizes: 2 lb (900 g), 2¼ lb (1 kg), and 2½ lb (1.12 kg).

Kitchen yields: Common size is 2½ lb (1.12 kg).

Serving size: ½ broiler per person.

CAPON

A capon is a castrated male bird. The meat is tender.

Packs: By weight and by piece.

Sizes: 3 lb (1.3 kg) to 7 lb (3.1 kg).

Serving size: 5 oz (140 g), cooked boneless meat.

Kitchen yields: One 5-lb (22-kg) bird = 3¼ lb (1.5 kg) cooked, bone in.
= 2 lb 4 oz (1 kg) cooked meat.
= 6 servings, boneless.

Note: There will be some trimmings that can be used for salad.

CORNISH HEN

Young chicken from the Cornish chicken breed. Available boned, bone in, and smoked.

Packs: By weight and by piece.

Sizes: Boned: 14 oz (400 g) to 16 oz (450 g).

Bone in: 16 oz (450 g) to 20 oz (570 g).

Serving sizes: One 14-oz (400-g) bird, usually stuffed.

One-half 20-oz (570-g) bird with bones.

FOWL

Mature chicken, used for boiling.

Packs: By weight and by the piece, packed in ice.

Sizes: 4½ lb (2 kg) to 5 lb (2.25 kg).
5 lb (2.25 kg) to 7 lb (3.1 kg).

Kitchen yields: One 5-lb (22-kg) bird = 2 lb (900 g) boneless, skinless cooked meat.

FREE-RANGE CHICKEN

Average sizes: 3 lb (1.3 kg) to 3½ lb (1.5 kg).

FRYER

Fryer is the industry term for medium-sized chicken.

Packs: By weight and by piece.

Sizes: 2¾ lb (1.2 kg), 3 lb (1.3 kg), or 3¼ lb (1.5 kg).

POULARDE

Castrated hen; the meat is tender.

Packs: By weight and by piece.

Sizes: 3 lb (1.3 kg) to 7 lb (3.1 kg).

Serving size: 5 oz (140 g), cooked boneless meat.

Kitchen yields: One 5-lb (2.2-kg) bird = 3¼ lb (1.5 kg), cooked, bone in.
= 2 lb 4 oz (1 kg), cooked meat.
= 6 servings, boneless.

Note: There will be some trimmings that can be used for salad.

ROASTER

Large chicken. The 3½-lb (1.5-kg) size is often used for cutting chicken parts.

Packs: By weight and piece.

Sizes: 3½ lb (1.5 kg), 4 lb (1.8 kg), or 4½ lb (2.0 kg).

Kitchen yields: One roaster, 3½ lb (1.5 kg) = 1 double breast, 22 oz (610 g).
= 2 legs, 9 oz (250 g) each.
= 4 oz (170 g), wings.
= 8 oz (225 g), neck and bones.
= 4 oz (112 g) giblets, wrapping paper, liquid.

FRESH CHICKEN PARTS

BREAST

Available as single and as double breast. Single breast is available with bone in and boned. Boned and skinless double breast is often marketed as chicken cutlet. The term *French-cut double breast* refers to a breast with wings and back removed.

Packs: By weight and piece.

Sizes: 10 oz (280 g), single breast, bone in, skin on.
8 oz (225 g), single breast, boneless, skin on.
16 oz (450 g) to 18 oz (510 g), double breast, French cut.

22 oz (610 g), double breast, skin on, wings attached.

CUTLETS

Boneless and skinless breast sold as double breast or single breast. Specify size when ordering. The product is practical when white chicken meat is needed, because it will cook rapidly and there is little waste.

Packs: By weight.

Sizes: Pieces range from 4 oz (112 g) to 14 oz (400 g).

Serving sizes: 7 oz (200 g), for grilled main course.

6 oz (170 g), for breaded main course.

Kitchen yields: Trimming waste is 5 percent or less, depending on the uniformity to which pieces must be cut. Trimming waste can be cooked and used as chicken salad or used in other dishes.

45 lb (20.2 kg) cutlets = 34 lb (15 kg) cooked meat. Cooking loss = 25 percent.

GIBLETS

Available as liver only or with hearts and stomachs included.

Pack: By weight. Livers are often sold in 5-lb (2.2-kg) tubes.

Kitchen yields: There is about 5 percent trimming loss for chicken livers. Hearts and stomachs have no trimming loss but must be washed well before use.

LEGS

Packs: By weight and by piece.

Sizes: 6 oz (170 g), 8 oz (225 g), and 10 oz (280 g).

Serving sizes: 8 oz (225 g), leg for broiling.

6 oz (170 g), leg for stuffing.

10 oz (280 g), leg for cutting into 6 tidbits.

Kitchen yields: Trim is important. Inexpensive legs frequently have too much backbone still attached.

NECKS AND BACKS

Sold fresh or frozen, with little difference in quality. Used for sauces and stocks.

Packs: Sold by weight, normally 5-lb (2.25-kg) bags.

Kitchen yields: 10 lb (4.5 kg) bones = 2 gal (7.6 l) chicken stock.
100 lb (45 kg) bones = 20 gal (76 l) chicken stock.

Note: Boil stock no longer than 1 hour.

TENDERS

The tender is the inside muscle of the breast, available skinless.

Packs: By weight, often 5-lb (2.25-kg) bags.

Kitchen yields: Some trimming is necessary.

WINGS

Sold by weight as whole wings or middle joints only, tips off. Precise specifications are important when ordering.

Sizes: 1.5 oz (40 g) to 2.5 oz (70 g).

Kitchen yields: 1 lb (450 g) = 10 pieces average. Trim is important; the second joint should be specified.

FROZEN CHICKEN PARTS

Most frozen chicken parts are machine cut, with bone in. Packs vary among manufacturers.

Parts, with back bone included:

Quarters: 8.7 oz (250 g), 60 pieces per pack.

Halves: 17.9 oz (508 g), 30 pieces per pack.

Pieces: 4 oz (112 g) and 4½ oz (125 g), 96 or 108 pieces per pack.

Segments:

Breast halves: 6.4 oz (180 g), 48 pieces per pack.

Thighs: 3.6 oz (100 g), 96 pieces per pack.

Drumsticks: 2.6 oz (75 g), 112 pieces per pack.

Whole legs: 6.5 oz (180 g), 50 pieces per pack.

Drumettes: 1.4 oz (40 g), 175 pieces per pack.

Wings: 2.5 oz (70 g), 96 pieces per pack.

PROCESSED CHICKEN PRODUCTS

Numerous processed chicken products are on the market. Some are fully cooked; others are ready to cook.

Breaded boneless products:

Nuggets: ½ oz (14 g) to 1¼ oz (35 g).

Patties: 2½ oz (70 g) to 4 oz (112 g).

Breast filets: 3 oz (85 g) average.

Tenders: 1.2 oz (34 g) average.

Bone-in breaded segments:

Breast halves: From 8 oz (225 g) up.

Drumsticks: From 3.9 oz (110 g) up.

Thighs: From 6.9 oz (200 g) up.

Wings: 3.3 oz (93 g).

Drumettes: From 1.6 oz (45 g) up.

Breaded random pieces: Average weight: 5 oz (140 g) per piece.

COOKED, BONELESS CHICKEN MEAT

Packs: 10-lb (4.5-kg) bags, IQF, all white meat.

10-lb (4.5-kg) bags, IQF, white and dark meat.

Kitchen yields: There is no waste, but product tends to be dry.

NONBREADED, BONE-IN, MARINATED BREASTS

Pack: IQF, ready for broiling.

Sizes: 4 oz (112 g) and 6 oz (170 g).

Kitchen yields: There is no waste.

STUFFED BREASTS

Stuffed breasts are available with a number of different stuffings. Some products are fully boneless, other products have the wing bone still attached. Available raw and prebrowned.

Packs and sizes: Twenty-four 7-oz (200-g) and 8-oz (225-g), breasts bone in. Twenty-four 6-oz (170-g) breasts, boneless.

SMOKED CHICKEN PRODUCTS

Sizes, whole birds:

Capon: 4 lb (1.8 kg) to 5 lb (2.2 kg) each.

Chicken: 1¾ lb (0.8 kg) to 2 lb (900 g) each.

Cornish hen: 18 oz (510 g) to 21 oz (600 g) each.

Chicken Breasts: 8 oz (225 g) average, boneless.

Kitchen yields: The products are fully cooked and ready to serve.

DUCK VARIETIES AND PRODUCTS

Most ducks are shipped frozen, but fresh ducks are also available in some markets. The most common breeds are Barberie, Peking, Mallard, and Moulard. Canvas-back duck is a wild duck and seldom available commercially. The name for a male duck is drake.

BARBERIE DUCKLING PRODUCTS

Bred from the Muscovy stock, this duckling is often sold as breast only, to be grilled and served medium rare. Barberie duckling is leaner than other breeds.

BARBERIE DUCKLING BREASTS

Packs and Sizes: Boned breasts, male, six 27-oz (750-g) to 38-oz (1-kg) case. Boned breast, female, twelve 19-oz (530-g) to 23-oz (640-g) case.

Weight: For double breasts.

Kitchen yields: Drake, double breasts = 5 to 6 servings, medium rare. Hen, double breasts = 2 servings, medium rare.

BARBERIE DUCKLING LEGS

Packs: 36 per case, drake or hen.

Sizes: 10 oz (280 g) to 12 oz (340 g), drake. 5 oz (140 g) to 7 oz (200 g), hen.

Serving size: 1 drake leg = 1 serving, main course. 1 hen leg = 1 serving, appetizer.

WHOLE BARBERIE DUCKLING

Packs: 4 drakes, 22 lb (9.9 kg) to 32 lb (2.8 kg), per case. 6 hens, 21 (9.4 kg) to 30 lb (3.5 kg), per case.

Sizes: Drake, average 8 lb (3.6 kg). Hen, average 4 lb (1.8 kg).

Weight: Includes gizzards and necks.

Kitchen yields:

Drake: 5 to 6 servings, using breast only, served medium rare. 2 leg servings, main course.

Hen: 2 servings, using breast only, served medium rare. 2 leg servings, appetizer, or 1 serving, main course.

MALLARD DUCKLING

Mallard ducks are farm-raised gamy birds; they are leaner than Peking ducks.

Packs: 10 to 12 ducklings per carton.

Average sizes: 1¾ (0.8 kg) to 2¾ lb (1.2 kg).

Kitchen yields: 1 duckling per serving.

MOULARD DUCK BREASTS

Moulard ducks are a cross of Muscovy or Barbary drakes and Peking duck hens. The breed was developed in Israel for the production of foie gras and is now raised in the United States. Moulard breasts are used in fine restaurants and the breasts are usually referred to as *magret*.

Pack: By weight and piece.

Size: Boneless single breast: 10 oz (280 g) to 16 oz (450 g).

Boneless double breast: 2 lb (900 g).

Kitchen yields: One boneless breast = 2 servings, cooked medium rare.

PEKING DUCK

PEKING DUCK BREASTS, BONELESS

Packs: 12 breasts per bag, 4 bags to a master carton.

Sizes: 10 oz (280 g) to 12 oz (340 g).

12 oz (340 g) to 14 oz (400 g).

16 oz (450 g) to 18 oz (500 g).

Serving sizes: One 12-oz (340-g) to 14-oz (400-g) breast, well done.

PEKING DUCK LEGS

Pack: 36 legs per case.

Sizes: 6 oz (170 g) to 8 oz (225 g), bone in.

Serving size: 2 legs per serving, main course.

Kitchen yields: 1 case = 18 portions.

PEKING DUCKS, WHOLE

Available fresh or frozen. Long Island used to be a large shipper of these ducks.

Packs: 6 ducks per case, in various weight ranges:

Sizes: 4 lb (1.8 kg) to 4½ lb (2 kg).

4½ lb (1.8 kg) to 5 lb (2.25 kg).

5 lb (2.25 kg) to 6 lb (2.7 kg).

Serving size: Half duck, boned.

Kitchen yields: 4½ lb (1.8 kg) to 5 lb (2.25 kg) when serving ½ boned duck.

6 lb (2.7 kg) when serving ¼ duck, bone in.

OTHER DUCK PRODUCTS

BREASTS, SKINLESS AND BONELESS

Raw breasts are available plain or marinated in teriyaki, vinaigrette, or green peppercorn marinade.

Packs:

4 oz (112 g) = 32 portions per case.

6 oz (170 g) = 16 portions per case.

8 oz (225 g) = 16 portions per case.

Serving sizes: 4 oz (112 g), appetizer.

6 oz (170 g) or 8 oz (225 g), as main course.

Kitchen yields: There is no waste.

DUCK FAT, RENDERED

Packs: 6-lb 11-oz (3-kg) tubs.

20-lb (9-kg) tubs.

DUCK LIVER, FRESH FOIE GRAS

Domestic foie gras of excellent quality is produced in a number of states and is readily available. The size and weight of the livers and the presence of blemishes determines the grade. For canned Goose Liver Products, see the Groceries section.

Season: Fall and winter, but often available year-round.

Packs and Grades: Grade A: 14 oz (400 g) to 22 oz (625 g).

Grade B: 11 oz (310 g) to 13 oz (360 g).

Grade C: 7 oz (200 g) to 10 oz (280 g).

Serving size: 3 oz (85 g), raw (2 slices), as appetizer.

Kitchen yields: One 18-oz (500-g) liver = 4 to 5 appetizer servings. There will be some trimmings left over. The largest livers are the best for cooking in slices. Livers should always be cooked pink. The smaller livers are more economical when making paté.

DUCK MEAT, IQF

Uncooked leg meat, good product for stir-fried dishes.

Packs: 5-lb (2.25-kg) boxes.

Serving size: 5 oz (140 g), stir-fried.

Kitchen yields: 5-lb (2.25-kg) box = 16 servings.

DUCK STRIPS, BREADED

Frozen, raw strips, available lightly breaded with or without almonds.

Packs: 5-lb (2.25-kg) boxes.

Sizes: 1 oz (28 g) to 1½ oz (42 g) strips.

Kitchen yields: About 65 pieces per box. Use as hors d'oeuvres or as appetizer.

ROASTED DUCK

Fully cooked product, available boneless and with bone in.

Packs: Boneless ducks, 12 per carton.
Bone-in ducks, 10 per carton.

Sizes: 11 oz (310 g) to 13 oz (360 g), boneless.
14 oz (400 g), boneless breasts with sauce.
15 oz (420 g) to 17 oz (480 g), bone in.

Kitchen yields: Product is ready to use.
There is no waste.

SMOKED DUCK

Available whole and as smoked breast.

Weights: 3½ lb (1.5 kg) to 5 lb (2.2 kg) for whole duck.
Breast sizes vary greatly.

Kitchen yields: Products are fully cooked.

GOAT

Available in ethnic markets.

Sizes: Whole carcass, from 30 lb (13.5 kg) to 60 lb (27 kg).

Best sizes: 30 to 40 lb (13.5 to 18 kg).

Kitchen yields: Goat is generally less fatty than lamb. Use lamb cuts and yields as guidelines.

Breasts: 2 lb (900 g) average.

Serving size: 3 oz (85 g), as appetizer.

Kitchen yields: Purchase breasts only for best yield. Trimming waste and fat are about 20 percent of weight.

GOOSE

Available fresh and frozen.

Packs: 4 geese per carton.

Sizes, whole geese:

8 lb (3.6 kg) to 10 lb (4.5 kg).
10 lb (4.5 kg) to 12 lb (5.4 kg).
12 lb (5.4 kg) to 14 lb (6.3 kg).
14 lb (6.3 kg) and larger.

Best size: 10 lb (4.5 kg). Larger geese may be tough.

Kitchen yields: One 10-lb (4.5-kg) goose = 6 servings.

GOOSE LIVER

See Duck Liver for fresh products; for Canned Foie Gras, see the Groceries section.

SMOKED GOOSE

Available whole and as boneless breasts.

Sizes: Whole birds: 7 lb (3.1 kg) to 9 lb (4 kg).

GUINEA HEN

Available fresh and frozen. Normally sold whole. Called *pintade* in French.

Average sizes: 2¼ lb (1 kg) to 2½ lb (1.1 kg).

Kitchen yields: 1 bird = 2 servings.

HARE

Available fresh and frozen. Hare has dark meat, as opposed to rabbit, which has light meat. Sold skinned and with skin on.

Season: Fall and winter.

Average size: 4 lb (1.8 kg), skinned and eviscerated.

Serving sizes: 1 leg or 1 whole saddle piece.

Kitchen yields: 1 hare = 3 servings, consisting of 2 legs and 1 saddle piece. Shoulders and racks can be boned and the meat used as an appetizer, but it is a labor-intensive process.

LAMB AND MUTTON

Lamb is less than 1 year old; mutton is over 1 year old, but should not be older than 2 years. The term *yearling* is used for animals 1 year old. The term *spring lamb* indicates that the animal was born the same year it is slaughtered. It is generally 3 months old, but can be marketed as late as October of the same year it was born. The American Lamb Council is recommending a change in some of the specifications listed here in order to make the cuts easier to use. The new specifications are listed on pages 226–227. Note that carcasses are not split.

LARGER CUTS OF DOMESTIC LAMB AND MUTTON

BABY LAMB

Sold often whole, with the pelt on, in ethnic markets. It is a delicacy, and portioning requires considerable skill. Baby lamb is not a banquet item.

Weight range: 20 lb (9 kg) to 40 lb (18 kg).

Best sizes: 26 lb (11.7 kg) to 30 lb (13.5 kg), pelt on.

Serving size: 5 oz (140 g).

Kitchen yields: One 28-lb (12.6-kg) lamb = 22 servings. The carcass is divided into legs, boneless shoulders, racks, and saddle. The parts are very small, and slices from various parts are normally included in each serving. The meat should be cooked lightly pink.

LAMB OR MUTTON CUTS

BACK, MBG #236

Whole back with rack and loin still attached.

Weights:

Lamb: 8 lb (3.6 kg) to 15 lb (6.7 kg).

Mutton: 11 lb (4.9 kg) to 26 lb (11.7 kg).

Best size:

Lamb: 15 lb (6.7 kg).

Mutton: 20 lb (9 kg).

Servings:

Lamb: 25 on buffet.

Mutton: 30 on buffet.

Kitchen yields: Good buffet piece when roasted whole.

BREAST, MBG #209

Weights: 6 lb (2.7 kg) to 11 lb (5 kg).

Kitchen yields: The piece is fatty. Best use is boiling, then removing excess fat and riblets. When the meat is cold, it can be cut into slivers, breaded, and grilled. For barbecued riblets, see Denver-Style Ribs, MBG #209 A.

CHUCK, SHOULDER, MBG #207

Whole shoulder, square cut, not split, bone in. The meat can be fatty. Use for stew or make boneless roast. Boneless roast is fatty and will not slice well. The piece requires boning, and it is advisable to purchase Boneless Shoulder, MBG #208.

Weight range, lamb: 10 lb (4.5 kg) to 18 lb (8.1 kg).

Best weight: 18 lb (8.1 kg).

Serving size, stew: 10 oz (280 g) raw, about 6 oz (170 g) cooked.

Kitchen yields: 18 lb (8.1 kg), shoulder = 10 lb (4.5 kg), boned and trimmed. 10 lb (4.5 kg), meat = 16 servings.

CHUCK, SHOULDER, BONELESS, MBG #208

The product is often sold rolled and tied, ready for roasting, but it can be cut for stew.

Weight range: 4 lb (1.8 kg) to 8 lb (3.6 kg).

Best size for roasting: 6 lb (2.7 kg).

Serving size: 5 oz (140 g), roast.

Kitchen yields: One 6-lb (2.7-kg) roast = 12 servings. Note that the shoulder must be cooked well done.

Calories: 5 oz (140 g), cooked roast = 575 calories.

DENVER-STYLE RIBS, MBG #209 A

Weights: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Serving size: 12 oz (340 g), raw.

Kitchen yields: The product has much better trim than Breast, MBG #209. The ribs are trimmed to about 3 in (75 mm).

LEGS, MBG #233

The weights listed are for pairs of legs, bone in. Note that single legs, boneless, shank off, and tied, are available in the 6 lb (2.7 kg) to 8 lb (3.6 kg) range as MBG #234B.

Sizes: 14 lb (6.3 kg) to 23 lb (8.5 to 10.3 kg), pair, not split.

Best sizes: 18 lb (8.1 kg), pair, 9 lb (4 kg), single leg.

Serving size: 6 oz (170 g), roast, medium rare.

Kitchen yields: One 9-lb (4-kg) single leg, bone in, oven ready = 7½ lb (3.3 kg).

Sliced, with bone in = 8 to 10 servings.

One 9-lb (4-kg) single leg, boneless and tied = 6¼ lb (2.8 kg).

Machine sliced = 12 servings.

Calories: 4 oz (112 g), roast = 480 calories.

LOIN, MBG #232

Full loin, with kidneys, kidney fat, and flanks attached. This piece requires extensive skilled butchering and generates much waste. It is partially trimmed.

Sizes: 5 lb (2.2 kg) to 12 lb (4.5 to 5.4 kg).

Best sizes: 8 lb (3.6 kg) to 10 lb (4.5 kg).

Serving size: 5 oz (140 g), boneless roast.
2 pieces, 3¼-oz (90-g) boneless medallions.

1 piece, 7-oz (200-g) bone-in loin chop.

Kitchen yields:

One 10-lb (4.5-kg) loin = 2 halves, boned, trimmed, and rolled.

1 half loin = 20 oz (570 g).

1 half loin = 3 servings roast.

1 loin = 6 servings, roast, or 12 medallions.

1 loin = 8 chops.

RACK, MBG #204

The rack is usually sold with 8 ribs, although in some parts of the country 9 ribs are acceptable. The rack is not split and requires skilled butchering. There is considerable waste material to dispose of. Most operations purchase racks ready to roast. The term *French lamb chop* refers to a completely trimmed chop, all fat removed.

Weight range: 4 lb (1.8 kg) to 5 lb (2.2 kg).

5 lb (2.2 kg) to 6 lb (2.7 kg).

6 lb (2.7 kg) to 8 lb (3.6 kg).

Best sizes: 6 lb (2.7 kg) to 8 lb (3.6 kg).

Serving size: 8 oz (225 g), roast, cooked with bone, 2 slices medium rare.

1 piece, 9 oz (250 g), rib chop, or 2 pieces, 4½ oz (125 g) each.

Kitchen yields: One double rack, 7 lb (3.1 kg) = 6 servings, roast or chop.

SHANK, MBG #210

Foreshank for braising whole. The item has been rediscovered by young chefs because it is flavorful and inexpensive.

Weights: 1 lb (400 g) to 2.5 lb (1.1 kg).

Best size: 1 lb (400 g).

Serving size: 1 shank.

NEW SPECIFICATIONS FOR DOMESTIC LAMB AND MUTTON**BONELESS SADDLE**

Sizes: 1½ lb (675 g) to 2 lb (900 g).

Serving size: 5 oz (140 g), roast, cooked medium rare.

Kitchen yields: 1 loin = 4 servings.

BONELESS SARATOGA ROLL

Rolled boneless shoulder, leaner and smaller than shoulder roll.

Sizes: 1½ lb (675 g) to 2 lb (900 g).

Serving size: 6 oz (170 g), roast, cooked, well done.

Kitchen yields: One 2-lb (900-g) piece = 4 servings.

BONELESS SIRLOIN

Roast whole or cut into medallions.

Sizes: 2 lb (900 g) to 3 lb (1.3 kg).

Serving sizes: 5 oz (140 g), roast, medium.

6 oz (170 g), raw, 2 medallions per serving.

Kitchen yields: One 3-lb (1.3-kg) piece = 8 servings, roast.
= 7½ servings, medallions.

BONELESS SHOULDER ROLL

Sizes: 4 lb (1.8 kg) to 6 lb (2.7 kg).

Serving size: 6 oz (170 g), roast, cooked, well done.

Kitchen yields: One 5-lb (2.2-kg) piece = 10 to 11 servings, roast.

DENVER RIB

Whole ribs, trimmed and ready to use.

Size: 12 oz (340 g).

Serving size: 12 oz (340 g).

DOUBLE BONELESS LOIN

Weight: 3½ lb (1.5 kg) to 4½ lb (2 kg).

Packs: 2 double loins per box.

Serving Size: 5 oz (140 g), roast, medium rare.

6 oz (170 g), raw, chops, 2 per person.

Kitchen yields: One 4-lb (1.8-kg) piece = 11 servings.
= 10 chops.

DOUBLE BONELESS LOIN CHOP

Lean, without tenderloin.

Size: 6 oz (170 g).

FRENCH RACK

Sizes: 1¾ lb (0.8 kg) to 2¼ lb (1 kg).

Packs: 6 racks per box.

Kitchen yields: 1 rack = 3 servings.

STEAMSHIP LEG

Ready-to-roast leg, sirloin removed, aitchbone removed, shank left attached.

Sizes: 6 lb (2.7 kg) to 8 lb (3.6 kg).

Serving size: 6 oz (170 g), cooked, medium rare.

Kitchen yields: One 7-lb (3.1-kg) leg = 12 to 14 servings on buffet.

TENDERLOIN

All fat removed, silver skin on. The item is suitable for quick roasting whole or serving on skewers.

Sizes: 4 oz (112 g) to 5 oz (140 g).

Pack: 5 lb (2.2 kg) per box.

Serving size: 1 piece.

Kitchen yields: 1 box = 16 to 18 pieces.

QUICK LOOK-UP FOR POPULAR LAMB DISHES

BONE-IN LOIN CHOP

Serving size: 1 piece, 7 oz (200 g).

Purchase: 10-lb (4.5-kg) Loin, MBG #231.

Kitchen yields: 1 loin = 8 chops.

BONELESS LOIN CHOP

Serving size: 2 pieces, 3¼ oz (90 g) each.

Purchase: 10-lb (4.5-kg) Loin, MBG #231.

Kitchen yields: 1 loin = 6 servings, 12 pieces.

BONELESS ROAST LOIN

Serving size: 5 oz (140 g), cooked, medium rare.

Purchase: 10-lb (4.5-kg) Loin, MBG #231.

Kitchen yields: 1 loin = 6 servings.

LAMB STEW

It is recommended that a blend of shoulder meat, for flavor, and leg meat, for consistency, be used. Shoulders can be fatty and may require extensive trimming.

Serving size: 10 oz (280 g), raw; about 6 oz (170 g), cooked.

Purchase: 2 Shoulders, Boneless, MBG #208, 8 lb (3.6 kg).
1 Leg, boneless, MBG #234 B, 9 lb (4 kg).

Total usable meat: 23 lb stew meat.

Kitchen yields: 35 servings.

MOUSSAKA

Baked eggplant and meat casserole, often blended with rice or potatoes.

Serving size: 4 oz (112 g) ground lamb.

Purchase: 2 Shoulders, Boneless, MBG #208, 8 lb (3.6 kg).

Total usable meat: 15 lb (6.75 kg).

Kitchen yields: 60 servings.

RIB CHOPS

Serving size: One 9-oz (250-g) piece or two 4½-oz (125-g) pieces.

Purchase: 9 Half Racks, 8 Ribs, Hotel Size, MBG #204, 7 lb (3.1 kg).

Kitchen yields: 50 servings.

SMALL CHOPS FOR RECEPTIONS

Serving size: One 3½-oz (100-g) chop.

Purchase: 7 Racks, 8 Ribs, Hotel Size, MBG #204, 5 lb (2.2 kg) each.

Kitchen yields: 100 chops.

ROAST LEG OF LAMB

Serving size: 5 oz cooked meat, medium rare.

Purchase: 2 Double Legs, MBG #233, 18 lb each, if machine sliced, or
5 single legs, 9 lb (4 kg) each, if sliced on bone, or
3 Boneless Legs, MBG #234 A, 7 lb (3.1 kg) to 8 lb (3.6 kg) each.

Kitchen yields: 50 servings.

SHISH KEBAB

Serving size: 6 oz, raw.

Purchase: 2 Double Legs, MBG #233, 18 lb (8.1 kg) each, or

50 pieces New Zealand boneless short loin, 6.2 oz (176 g) each.

Kitchen yields: 50 servings.

PORTION-CONTROLLED LAMB

LOIN CHOPS

The chops are cut across the loin, all bones in, flank still attached.

Sizes: 4 oz (112 g) to 10 oz (280 g), in increments of 1 oz (28 g).

Serving size: Two 6-oz (170-g) chops is the average dinner portion.

RIB CHOPS

The chops are cut across the rack; the blade bone and surrounding muscle are removed.

Sizes:

Regular chops: 3 oz (85 g) to 10 oz (85 to 280 g), in 1-oz (28-g) increments.

Frenched chops: 3 oz (85 g) to 8 oz (225 g), in 1-oz (28-g) increments.

Serving size: Two 6-oz (170-g) chops is the average dinner portion.

SHOULDER CHOPS

Bony chops cut across the blade and arm bone; best served braised with vegetables.

Sizes: 4 oz (112 g) to 8 oz (225 g), in 1-oz (28-g) increments.

Serving size: Two 6-oz (170-g) chops is the average dinner portion.

STEW MEAT

Packs: By weight.

Kitchen yields: Stew meat is cut from different muscles, and therefore tenderness varies. Shrinkage is about 35 to 40 percent. 10 lb (4.5 kg) = 15 servings.

SPECIFICATIONS FOR NEW ZEALAND LAMB

New Zealand lamb is generally half the size of domestic lamb. It is best purchased fully trimmed, ready for use.

BONE-IN LOIN CHOPS

Weights: 3 oz (75 g) to 3¼ oz (92 g).

Packs: 48 to 52 per 10-lb (4.5-kg) box.
4 boxes per master carton.

Serving size: 3 pieces, as main course.

Kitchen yields: 16 servings per box.

BONE-IN RACKS

Packs: 2 pieces per vacuum pack.
10 pieces per box.
40 pieces per carton.

Weights: 6 oz (175 g) to 8 oz (225 g).

Serving size: 1 piece per person.

Kitchen yields: Ready-to use-product,
little trimming is needed. Popular
banquet item.

BONELESS SHORT LOIN

Weight: 6.2 oz (175 g).

Packs: 4 pieces per vacuum pack.

20 pieces per box.

80 pieces per carton.

Serving size: 1 piece per person, roasted
whole or used as shish kebab.

BONELESS STEW MEAT

Packs: 10-lb (4.5-kg) boxes.
40-lb cartons.

SEMIBONELESS LEGS

Weight: 3 lb (1.3 kg).

Packs: 4 pieces per vacuum pack.

Serving size: 4 oz (112 g).

Kitchen yields: One 3-lb (1.3-kg) leg =
8 servings, medium rare.

SHANKS

Packs: 20 pieces per box.
80 pieces per master carton.

Weight: 9.2 oz (260 g) each.

Serving size: 2 shanks.

OSTRICH

Ostrich meat is low in fat and is a low-cholesterol alternative for other dark meats. The meat became somewhat popular in Europe in the year 2000 because of mad cow and foot-and-mouth diseases. Ostrich meat is concentrated in the thighs and legs, as the breast muscle is rather small. Because the meat is dry, it is best cooked medium rare or braised. The eggs are edible and are considered a delicacy. Most supply is domestic.

OSTRICH EGGS

About 3 lb (1.3 kg) and larger.

OSTRICH MEAT**LEG**

Inside leg Medium tender, average
weight 2 lb (900 g).

Outside leg Medium tender, average
weight 2 lb (900 g) to 3 lb (1350 g).

Mid-leg Medium tender, average weight
1.5 lb (675 g) to 2 lb (900 g).

BACK TENDERLOIN

Tender, average weight 1.5 lb (675 g).

THIGH

Top loin: Tender, average weight 1.5 lb
(675 g).

Fan: Tender, average weight 3 lb (1350
g) to 4 lb (1800 g).

Round: Medium tender, average weight
3 lb (1350 g) to 4 lb (1800 g).

Tip: Medium tender, average weight 2 lb
(900 g).

Oyster: Medium tender, average weight
1.5 lb (675 g).

Inside strip: Tender, average weight 1.5 lb
(675 g).

Outside strip: Tender, average weight 1.5 lb
(675 g).

Packs: 10 lb (4.5 kg).

Ground meat: Available in bulk and in
8-oz (225-g), 5.3-oz (160-g), and
4-oz (110-g) individual packages.

Available portion packs:

Eighty 2-oz (56-g) medallions in 10-lb (4.5-kg) cartons.

Forty 4-oz (112-g) medallions in 10-lb (4.5-kg) cartons.

Seventy 6-oz (170-g) medallions in 15-lb (6.75-kg) cartons.

Serving size: 6 oz (170 g), raw.

Kitchen yields: Meat is completely trimmed, and there is little shrinkage when cooked medium rare.

Calories: 6 oz (170 g) = 194 calories (98 percent fat free).

PARTRIDGE

Chukar and Bartavelle partridges are available.

Average sizes: 14 oz (400 g) to 16 oz (450 g).

Serving size: 1 whole bird.

PHEASANT

Available eviscerated, fresh and frozen. Male birds with feathers are also available for display. Pheasants are in the poultry family, and farm-raised birds have light-colored meat and little game flavor.

Sizes, whole birds:

15 oz (425 g) to 18 oz (510 g).

20 oz (570 g) to 32 oz (900 g).

2 lb (900 g) to 2½ lb (1.1 kg).

2½ lb (1.1 kg) to 3 lb (1.3 kg).

Serving sizes: One 15-oz (425-g) to 18-oz (510-g) bird.

One-half 2-lb (900-g) bird.

Kitchen yields: Use 3-lb (1.3-kg) birds when serving breast only.

SMOKED PHEASANT

Available as whole bird. Smoked pheasant breasts are available in some markets.

Sizes: 2¼-lb (1-kg) to 2¾-lb (1.2-kg) whole birds.

PORK AND PORK PRODUCTS

Pork products generally come from animals under 1 year of age, and for this reason all cuts are relatively tender. Pork parts are not quality graded, although the carcass is yield graded. Selection #1 is normally used for foodservice.

PORK, LARGER CUTS**BACK RIBS, MBG #422**

Rib bones with some loin meat still attached. Ribs over 3 lb (1.3 kg) can be tough.

Weight ranges: Under 1½ lb (0.6 kg).
1½ lb (600 g) to 3 lb (1.3 kg).

Over 3 lb (1.3 kg).

Serving size: 1 lb (450 g).

BELLY, MBG #408

Fresh belly is often referred to as *green belly*; the same piece is also available salted. Completely boneless piece, with skin attached; skinned belly is available as MBG #409.

Sizes:

With skin: 10 lb (4.5 kg) and up.

Skinned: 7 lb (3.15 kg) and up.

Best size: 10 lb (4.5 kg), skinned belly.

Kitchen yields: Skin is difficult to remove. Skin weighs about 20 percent of whole belly.

BONES, MBG #421

Neck bones, excellent for making stocks or sauces. Available by weight.

BUTT, SHOULDER, MBG #406

Called a *Boston butt*, this is a piece from the shoulder, with the blade bone in. Roast, stew, or use for ground pork.

Sizes: 4 lb (1.8 kg) to 8 lb (1.8 to 3.6 kg).
8 lb (3.6 kg) to 12 lb (3.6 to 5.4 kg).

Best size: 7 lb (3.1 kg).

Kitchen yields: One 7-lb (3.1-kg) piece
= 6 lb (2.7 kg), boned and trimmed.

CAUL FAT

Net-like fat layer, which keeps the intestines in place. Most caul sold to foodservice operations is from pigs. Caul is used to wrap *paté*, meat loaves, and other soft ground-meat products to provide moisture during the cooking process and to retain the shape.

Pack: By weight.

Kitchen yields: Caul is available fresh or frozen. It should be kept in ice water until used. Occasionally it is sold salted.

FRESH HAM, BONED AND NETTED, MBG #402 B

Fresh boneless ham with all skin removed. Smaller hams are easier to slice than larger pieces. Smaller hams are generally leaner than larger hams.

Sizes: 6 lb (2.7 kg) to 8 lb (3.6 kg).
8 lb (3.6 kg) to 10 lb (4.5 kg).
10 lb (4.5 kg) to 12 lb (5.4 kg).

Recommended size: 7 lb (3.1 kg).

Serving size: 5 oz cooked, roast ham, main course.
4 oz cooked, as sandwich meat.

Kitchen yields: One 7-lb (3.1-kg) ham, slow roasted = 5 lb 4 oz.
5 lb 4 oz cooked meat = 16 servings, main course.
= 20 servings, sandwich.

HAM, REGULAR, SHORT SHANK, MBG #401 A

Basic bone-in fresh ham with shank cut short, some skin left on.

Weight ranges:

10 lb (4.5 kg) to 14 lb (6.3 kg).
14 lb (6.3 kg) to 17 lb (7.6 kg).
17 lb (7.6 kg) to 20 lb (9.0 kg).

Recommended size: 14 lb (6.3 kg).

Serving size: 5 oz (140 g), roast ham.

Kitchen yields: One 14-lb (6.3-kg) ham
= 24 to 26 servings.

HAM, SKINNED, SHORT SHANK, MBG #402 A

Same specifications as for regular ham, but skin and underlying fat are partly removed.

Recommended size: 12 lb (5.4 kg).

Kitchen yields: The removed skin and fat are about 8 to 10 percent of weight. Yield is about the same as for regular ham.

LIVER

Pork liver is generally used as a *paté* ingredient.

Weights: 3 lb (1.3 kg) to 4 lb (1.8 kg).

Kitchen yields: 5 percent loss, trimming and skin.

LOIN, MBG #410

Full loin, including tenderloin and blade. The piece requires skilled butchering before use.

Weight ranges:

10 lb (4.5 kg) to 14 lb (6.3 kg).
14 lb (6.3 kg) to 17 lb (7.6 kg).
17 lb (7.6 kg) to 20 lb (9 kg).

Serving size: 4 oz (112 g), cooked roast.

Recommended size: 12 lb (5.4 kg).

Kitchen yields: 1 lb (450 g), as purchased = 8 oz (225 g), boneless, cooked.

One 12-lb (5.4-kg) loin = 20 servings, boneless roast.

One 12-lb (5.4-kg) = 8 lb (3.60 kg) raw boneless meat.

Calories: 4 oz (112 g) cooked roast = 440 calories.

LOIN, BONELESS, MBG #413

Full loin, including end pieces, but fully boned; use for roasting whole.

Weight ranges:

- 6 lb (2.7 kg) to 8 lb (3.6 kg).
- 8 lb (3.6 kg) to 10 lb (4.5 kg).
- 10 lb (4.5 kg) to 12 lb (5.4 kg).

Recommended size: 8 lb (3.6 kg).

Serving size: 4 oz (112 g), cooked.

Kitchen yields: One 8-lb (3.6-kg) loin = 19 servings.

LOIN, CENTER CUT, MBG #412A

The piece still has 8 ribs, but the chine bone and most other bones are removed. It is ready for roasting. The tenderloin is removed. Economical piece to purchase when trimmings are not needed.

Weight ranges:

- 4 lb (1.8 kg) to 6 lb (2.7 kg).
- 6 lb (2.7 kg) to 8 lb (3.6 kg).
- 8 lb (3.6 kg) to 10 lb (4.5 kg).

Recommended size: 8 lb (3.6 kg).

Kitchen yields: One 8-lb (3.6-kg) loin = 6 lb (2.7 kg) raw boned meat.

PIG ROAST

See also Suckling Pig in this section.

Recommended size: 100 lb (45 kg).

Kitchen yields: Calculate about 1½ lb (670 g) per person. However, most pig roast dinners include hamburgers, hot dogs, corn on the cob, and other items; therefore, one 100-lb (45-kg) pig is usually sufficient for 100 customers.

RIBS, COUNTRY STYLE, MBG #423

Basically a loin chop from the shoulder end, chine bone removed, cut horizontally into two pieces, one piece containing the feather bones, the other piece the ribs.

Weight ranges:

- 1 lb (450 g) to 2 lb (900 g).
- 2 lb (900 g) to 4 lb (1.8 kg).
- 4 lb (1.8 kg) and up.

Serving size: 1 lb (450 g) or more.

SHOULDER HOCKS, MBG #417

Also available are Ham Hocks, MBG #417 A. Both items are usually used cured and smoked.

Weight ranges: ½ lb (225 g) and up.

Kitchen yields: There is very little meat when the skin and bones are removed. Hocks make a very gelatinous stock.

SHOULDER, SKINNED, BONELESS, MBG #407

This piece is boneless, with most of the fat removed. It is excellent for braising or for making stew.

Weight ranges: 1½ lb (675 g) to 7 lb (3.15 kg).

Best size: 5 lb (2.25 kg).

Serving size: Stew = 8 oz (225 g), raw; 5½ oz (155 g), cooked.

Kitchen yields: One 5-lb (2.25-kg) piece = 4¾ lb (2.1 kg).

One 5-lb (2.25-kg) piece = 9 servings stew.

SPARERIBS, MBG #416

Whole sides with at least 11 ribs from belly and breast.

Weight ranges:

- 1½ lb (0.6 kg) to 3 lb (1.3 kg).
- 3 lb (1.3 kg) to 5 lb (2.2 kg).
- 5 lb (2.2 kg) and up.

Serving size: 1 lb (450 g) up.

Kitchen yields: The piece is unwieldy, and the ribs are long. It is advisable to purchase Spareribs, Breast Off, MBG #416 A. The size would be 10 oz (280 g) to 16 oz (450 g) less.

SUCKLING PIG

Weights: 8 lb (3.6 kg) to 40 lb (18 kg).

ROASTED WHOLE PIG FOR BUFFET SERVICE

Recommended sizes: 25 lb (11.25 kg) to 30 lb (13.5 kg). This size pig will still fit in a convection oven.

Servings: 15 to 20 servings, as main course.

PIG FILLED FOR GALANTINE

Size: 10-lb (4.5-kg).

Servings: 40 to 50 servings, as appetizer on buffet.

TENDERLOINS, MBG #415 A

Excellent product, completely trimmed, side muscle off; available mostly frozen.

Weight ranges:

4 oz (112 g) to 8 oz (225 g).

8 oz (225 g) to 12 oz (340 g).

12 oz (340 g) to 16 oz (450 g).

Serving sizes: Vary. Braised, whole = 8 oz (225 g) to 10 oz (280 g) raw.

Kitchen yields: Very little trimming is needed. Roast or braise whole, or cut into small medallions.

TRIMMINGS

Trimming is available 90 and 80 percent lean. Good buy when making sausages. Also available is Ground Pork, MBG #496, in 5-lb (2.25-kg) bags.

QUICK LOOK-UP FOR POPULAR PORK DISHES

BARBECUED SPARERIBS

Purchase: Spareribs, MBG #416, ribs from belly and breast.

Best weights: 1½ lb (600 g) to 3 lb (1.3 kg).

Or

Purchase: Back Ribs, MBG #422, rib bones with some loin meat still attached.

Best weight: 1½ lb (1.3 kg).

Serving size: 1½ lb (675 g), raw.
1 lb (450 g), cooked weight.

Kitchen yields: 10 lb (4.5 kg) = 7 servings.

BROILED CHOPS

Purchase: Loin, Center Cut, MBG #412.

Best weight: 8 lb (3.6 kg).

Serving size: 2 chops, 5 oz (140 g) each, only rib bones still attached.

Kitchen yields: One 8-lb (3.6-kg) loin = 10 servings.

ROAST FRESH HAM

Smaller hams are easier to slice than larger pieces and are generally leaner than larger hams.

Purchase: Ham, Boned and Netted, MBG #402B.

Recommended size: 7 lb (3.1 kg).

Serving size: 5 oz (140 g) roast ham, main course.

4 oz (142 g) sandwich meat.

Kitchen yields: One 7-lb (3.1-kg) ham = 16 servings, main course, 20 servings, sandwich. *Note:* In some operations customers expect pieces of crisp skin when ordering ham. In that case, item MBG #401A must be purchased. This piece is not boned.

ROAST FRESH SHOULDER

Purchase: Butt, MBG #406, called a *Boston butt*, with the blade bone in or Butt, MBG #406A, boneless, netted.

Best weights: 7 lb (3.1 kg), bone in.
5 lb (2.25 kg), boneless.

Serving size: 5 oz (140 g).

Kitchen yields: One 7-lb (3.1-kg) or 5-lb (2.25-kg) butt = 12 to 14 servings.

ROAST LOIN

Purchase: Loin, Boneless, MBG #413, full loin, including end pieces, but fully boned.

Best size: 8 lb (3.6 kg).

Serving size: 5 oz (140 g) cooked meat.

Kitchen yields: One 8-lb (3.6-kg) loin = 18 to 20 servings.

PORTION CUTS OF PORK

BUTT STEAKS

Available with bone in and boneless. The meat is rather fatty.

Weights: 4 oz (112 g), 5 oz (140 g), 6 oz (170 g), and 8 oz (225 g).

Serving size: 6 oz (170 g).

Kitchen yields: Use as braised steak with vegetables or with sauerkraut.

CHOPS

Available in many sizes and configurations.

BONE-IN CHOPS

Cut across the loin, with all bones still attached, starting at shoulder end. The least expensive cutlets.

Weights: 3 oz (85 g), 4 oz (112 g), 5 oz (140 g), and 6 oz (170 g).

Serving size: Two 4-oz (112-g) chops.

BONELESS CHOPS

Cut from the boned full loin, from end to end. The completely boneless chops are the most expensive cuts.

Weights: 3 oz (85 g), 4 oz (112 g), 5 oz (140 g), 6 oz (170 g), and 8 oz (225 g).

Serving size: Varies.

CENTER-CUT CHOPS

Available with or without the chine bone removed; also completely boneless.

Weights: 3 oz (85 g), 4 oz (112 g), 5 oz (140 g), 6 oz (170 g), and 8 oz (225 g).

Serving size: Varies.

CHOPS WITH POCKET

Available as rib chops or cut from the full loin. Rib chops have the incision made at the rib bone; the other chops have the incision made from the outside.

Weights: 5 oz (140 g), 6 oz (170 g), and 8 oz (225 g).

Serving size: One 6-oz (170-g) or 8-oz (225-g) chop, stuffed.

PORK PRODUCTS

BREAKFAST SAUSAGES

Available in different sizes. Sausages without pork are also available.

Pack: 5 lb (2.2 kg).

Sizes: 8 sausages per lb (450 g), 10 or 12 per lb (450 g).

1-lb (450-g), 2-lb (900-g), and 3-lb (1.3-kg) rolls.

5-lb (2.25-kg) and 8-lb (3.6-kg) bulk buckets.

Serving size: 3 sausages for breakfast, 10 or 12 per lb size.

Kitchen yields: Fat content of sausages varies greatly. Average shrinkage by weight is 35 percent. Precooked products with little shrinkage are on the market.

One 5-lb (2.2-kg) box, 8 size = 13 servings.

One 5-lb (2.2-kg) box, 12 size = 20 servings.

CANADIAN BACON

Cured and smoked boneless loin. Genuine Canadian bacon is a product in its natural shape, often tied to a thin wood plank. There is also a pressed and rolled product on the market.

Weights: Imported: 5 lb (2.2 kg) to 7 lb (3.1 kg).

Roll: 7 lb (3.1 kg) to 9 lb (4 kg).

Also available sliced in various sizes.

Kitchen yields: The imported product is packaged on top of a thin wooden board. Waste and trim loss are about 5 percent. The rolled product has very little waste.

Serving size: 2 slices, 2 oz (56 g) each, total 4 oz (112 g), for breakfast.

Kitchen yields: 6 lb (2.7 kg), imported product = 24 servings.

8 lb (3.6 kg), rolled product = 30 servings.

FATBACK

Available fresh (called *green*) or salted.

The product is also called *larding pork*. See also Belly, MBG #408.

Weight range: 12 lb (5.4 kg) to 14 lb (6.3 kg), whole sides.

Kitchen yields: Skin weighs about 20 percent of product. When it is used for paté, make sure to specify unsalted fatback (green).

Calories: 1 oz (28 g) = 200 calories.

IRISH BACON

Very similar to pancetta. See the following entry.

KASSLER, MBG #545

Kassler is the German name for smoked pork loin.

Weights: 10 lb (4.5 kg) to 14 lb (6.3 kg) and larger.

Best weight: 12 lb (5.4 kg).

Serving size: 8 oz (225 g), 2 slices cooked, usually with 1 rib per serving.

Kitchen yields: Product is normally sold as full loin, with all bones attached, and requires boning when cooked. There is much waste.

One 12-lb (5.4-kg) loin = 12 servings.

LARD

Rendered pork fat.

Packs: 1-lb (450-g) packages and larger.

Calories: 1 cup, 7.3 oz (200 g) = 1,885 calories.

PANCETTA

Pancetta is the Italian term for *rolled bacon*, used for cooking or eaten uncooked.

Weight: 4½ lb (2 kg).

SLICED BACON

Bacon is available sliced in different thicknesses; also available precooked, which reduces shrinkage and cooking time.

Packs: 10-lb (4.5-kg), 12-lb (5.4-kg), and 15-lb (6.7-kg) boxes. Other sizes are available.

Counts: 16 to 18 slices per pound (450 g).
18 to 22 slices per pound (450 g).
30 slices special lean, no shrinkage.
50 slices precooked.

Serving size: Breakfast, 3 slices, 16 to 18 size is common.

Salad garnish, 3 slices, 18 to 22 size, about 1 oz (28 g) cooked.

With pasta, such as carbonara, 4 slices, 18 to 22 size, about 1½ oz (37 g).

Kitchen yields:

16 to 18 size:

1 lb (450 g) = 5½ servings, breakfast.

9 lb (4 kg) = 50 servings.

18 lb (8.1 kg) = 100 servings.

18 to 22 size:

1 lb (450 g) = 6½ servings.

8 lb (3.6 kg) = 50 servings.

15½ lb (7 kg) = 100 servings.

Other yields: 3¾ lb (1.7 kg) uncooked bacon = 25 salad servings.

5 lb (2.2 kg) uncooked bacon = 25 pasta carbonara servings.

1 lb (450 g) = 5 oz (140 g), cooked.

1 cup crisp bacon bits, chopped fine = 4 oz (112 g).

13 oz (364 g), raw = 1 cup (0.23 l) chopped bits, loose.

5-lb (2.25-kg) bacon = 6 cups bacon bits.

Calories: 3 slices, 16 to 18 size, crisp, 1 oz (28 g) = 175 calories.

3 slices crisp, 18 to 22 size, ¾ oz (21 g) = 130 calories.

HAM VARIETIES AND HAM PRODUCTS

CANNED HAM

Available in natural shape, square shape, and rectangular shape called *Pullman ham* in many quality levels, ranging from very lean to fatty. Pullman ham is available at 4 x 4 in (100 x 100 mm), which is the standard size of sandwich bread. Some imported hams, such as Danish ham and Krakau (Polish) ham, have a reputation of being lean and mild.

Weights: 6 lb (2.7 kg) to 16 lb (7.2 kg).

Best sizes: 8 lb (3.6 kg) to 10 lb (4.5 kg).

Serving size: 3 to 4 oz (85 to 112 g), sandwich.

Kitchen yields: Basically, there is very little waste. Natural-shaped ham is harder to portion because of the shape.

Calories: 4 oz (112 g), very lean = 260 calories.

COOKED HAM

Boneless ham, fully cooked, shaped and wrapped. This ham is often used as breakfast ham. A number of manufacturers produce hams under trade names.

Sizes: 7 lb (3.15 kg) to 10 lb (4.5 kg).

Serving sizes: 3 oz (85 g) to 4 oz (112 g), breakfast ham.
6 oz (170 g) to 7 oz (200 g), ham steak.

Kitchen yields: The ham is evenly shaped and can be cut on a slicing machine. There is about 5 percent trimming waste.

One 8-lb (3.6-kg) ham = Forty 3-oz (85-g) breakfast servings.

Calories: 3 oz (112 g), lean and fat = 250 calories.

HAM STEAKS

Fully trimmed, ready to eat, center-bone-in steaks, usually individually packaged.

Packs: 15-lb (6.75-kg) or 30-lb (13.5-kg) boxes.

Weights: 8 oz (225 g), 12 oz (340 g), 14 oz (400 g), and 16 oz (450 g).

Serving size: 12 oz (340 g); serving is large, but has center bone in.

Kitchen yields: There is no waste.

PICNIC HAM, MBG #526

Shoulder ham, often fatty, irregularly shaped.

Sizes: Vary, from 4 lb (1.8 kg) to 10 lb (4.5 kg).

Kitchen yields: The ham must be boiled, and the irregular shape requires trimming.

PROSCIUTTO HAM

Available with bone in, boneless in natural shape, and boneless round-shaped.

Products available are imported ham from Parma, Italy, and domestically produced ham. Boneless hams are available fully boned and with the shank on. The amount of skin left on also varies greatly. The waste factor of bone-in ham is high, because skin, shank, and bones must be removed, which have only limited kitchen use. Boneless hams in a natural shape have high waste factor as well, because the shank and skin must be removed. The fat cover can also be substantial on both ham varieties.

Sizes: 11 lb (4.9 kg) to 14 lb (6.3 kg), bone in.
6 lb (2.7 kg) to 10 lb (4.5 kg), boneless.

Serving size: 1½ oz (42 g), about 2 large or 3 medium well-trimmed paper-thin slices.

Kitchen yields:

One 12-lb (5.4-kg) bone-in prosciutto = 4 lb 6 oz (2.1 kg) bones, fat, and trimmings.
= 7 lb 10 oz (3.4 kg) usable meat.
= 75 servings, slices of different shapes.
One 9-lb (4-kg) boneless natural shape prosciutto = 3 lb 3 oz (1.4 kg) shank, fat, and scraps.
= 5 lb 13 oz (2.6 kg) usable meat.
= 60 servings, slices of different shapes.
One 7-lb (3.1-kg) boneless round prosciutto = 12 oz (340 g) waste and trim.
= 6¼ lb (2.8 kg) usable meat.
= 60 servings, slices of smaller shapes.

SERRANO HAM

Spanish ham resembling prosciutto. Yield is comparable to that of prosciutto ham.

SMITHFIELD HAM

Another name for *Genuine Virginia Ham*.

VIRGINIA HAM

Genuine Virginia Ham is manufactured from razorback hogs, air dried, lightly smoked, and coated with coarse ground pepper. Available raw and fully cooked, with bone in and boned. Raw ham must

be soaked overnight to remove salt and then simmered. Fully cooked ham is available sliced.

Weights: 12 lb (5.4 kg) to 16 lb (7.2 kg), whole, uncooked.

10 lb (4.5 kg) to 12 lb (5.4 kg), ready to eat, bone in.

8 lb (3.6 kg) to 10 lb (4.5 kg), ready to eat, boneless.

Kitchen yields: Uncooked ham has about 50 percent waste.

Cooked ham has about 40 percent trimming and boning waste.

Ready-to-eat boneless ham has about 15 percent waste.

WESTPHALIAN HAM

Juniper wood-smoked and dried ham, served uncooked in very thin slices. Available from specialty butchers.

Weights: 3 lb (1.3 kg) to 5 lb (1.3 to 2.2 kg), boneless.

Serving size: 3 oz (100 g) for cold plate or sandwich.

Kitchen yields: Better qualities have skin and fat layer still attached, and the piece is irregularly shaped. Trimming waste is about 20 percent.

WHOLE SMOKED HAM, BONE IN, MBG #501

Available both fully cooked, referred to as *ready-to-eat*, and uncooked. Most smoked ham on the market is fully cooked. The hams are usually covered with net. The shank is short, but all other bones are still attached. There may be a considerable fat layer, and the knuckle is covered with skin. Spiral-cut hams are popular consumer products consisting of bone-in seasoned and fully cooked ham halves, machine sliced to the bone.

Weights: 12 lb (5.4 kg) to 20 lb (9 kg), ready-to-eat ham.

12 lb (5.4 kg) and more, uncooked ham.

Serving size: 4 oz (112 g), sandwich.

Kitchen yields:

One 14-lb (6.3-kg) ready-to-eat ham = 2 lb (900 g) bones and waste.

= 10 lb 8 oz (4.75 kg) usable meat for slices.

= 1 lb (0.67 kg) usable trimmings.

= 40 servings, in slices.

One 15-lb (6.75-kg) uncooked ham = 2 lb 8 oz (680 g) cooking loss.

Calories: 4 oz (112 g) = 330 calories.

POUSSIN

Poussin is the French name for *small chicken*; see Baby Chicken under Chicken.

QUAIL

Farm-bred quails are available fresh, frozen, or smoked, in the following forms: whole, bone in, whole boned, or split. For Quail Eggs, see the Dairy Products section.

BONED QUAIL

Sizes: 3½ oz (100 g).

Packs: 40 per case.

WHOLE QUAIL

Sizes: 4 oz (112 g) to 5 oz (140 g).

5 oz (140 g) to 6 oz (170 g).

4 oz (112 g), split.

Packs: 12, 24, 40, and 56 whole birds to a case, or 6 per tray, 72 per master.

Serving size: 2 birds.

Kitchen yields: Meat stuffing needed to stuff 50 birds = 3½ lb (1.5 kg).

RABBIT

Available fresh and frozen, whole and in pieces. Rabbits have white meat.

Packs:

Whole skinless, head and feet off

44-lb (19.8-kg) case, 1¾ lb (700 g) to 2½ lb (1 kg) each.

28-lb (12.6-kg) case, 1½ lb (600 g) to 3½ lb (1.5 kg) each.

26-lb (11.7-kg) case, 4 lb (1.8 kg) to 6 lb (2.7 kg) each.

Irregular pieces:

15-lb (6.7-kg) case.

Loins, boneless:

Pack is by weight and piece, in sizes 6 oz (170 g) to 9 oz (255 g).

Kitchen yields: One 2-lb (900-g) whole rabbit = two 8-oz (225-g) hind legs.

= one 8-oz (225-g) back piece.

= 8 oz (225 g) shoulder and trim.

One 2-lb (900-g) to 2½-lb (1-kg) rabbit = 3 servings.

Calories: 5 oz (142 g), cooked = 290 calories.

SQUAB

A squab is a young pigeon and should not be confused with the squab chicken, which is a baby chicken. Squab has dark

meat. Available fresh and frozen, whole with bone in and boneless. Whole birds with bone in are often shipped with the head on. Boning squab is labor-intensive.

Season: Year-round; larger supply in spring.

Pack: 12 squabs to a carton.

Sizes:

Whole:

12 oz (340 g).

14 oz (400 g).

16 oz (450 g).

20 oz (560 g).

Boneless:

13 oz (365 g).

Best serving size: 16 oz (450 g) whole or 13 oz (365 g) boneless.

Kitchen yields: One 20-oz (560-g) bird will yield 2 each 5–6-oz (140–168-g) breasts.

TURKEY, WHOLE AND PARTS

FRESH OR FROZEN WHOLE TURKEY

Whole turkey is available fresh and frozen. Although most production is frozen, there is a growing market for fresh turkey.

Sizes: 5 lb (2.25 kg) to 9 lb (4.1 kg), fryer/roaster.

8 lb (3.6 kg) to 16 lb (7.3 kg), hen.

16 lb (7.3 kg) and larger, tom (male bird).

Best size: 22-lb (10.4-kg) tom. The larger the bird, the higher the percentage of edible meat. Tom birds have a higher percentage of meat than fryers and hens.

Average weight of parts:

Breast = 41 percent.

Drumstick = 14 percent.

Thigh = 17 percent.

Wings = 11 percent.

Necks and giblets = 8 percent.

Back, juices, paper = 9 percent.

Serving size: 6 oz (170 g) cooked meat, white and dark.

Kitchen yields: One 22-lb (10.4-kg) bird = 12 lb (5.4 kg) roasted meat, including all scraps.

22 servings if sliced directly on the frame by hand.

24 servings if boned after cooking and sliced on machine; there will be a considerable amount of scraps.

40 servings on buffet when served with other meats and salads.

One 25-lb (11.25-kg) turkey = 4 rolls, 4 lb (1.8 kg) each, galantine.

Calories for roasted bird: 6 oz (170 g), white meat only without skin = 300 calories.

6 oz (170 g), dark meat only without skin = 360 calories.

6 oz (170 g) diced, mixed meat, 1 cup = 320 calories.

RAW TURKEY PARTS**BONELESS BREAST AND THIGH ROAST**

Natural or slab shape, rolled or netted, ready to roast.

Pack: 2 to 4 pieces per case.

Weights: 4 lb (1.8 kg) to 14 lb (6.3 kg).

Serving size: 4 oz (112 g).

Kitchen yields: One 12-lb (5.4-kg) breast = 7½ lb (3.4 kg).

One 12-lb (5.4 kg) breast = 28 servings.

BREAST MEAT BONE-IN CHOPS

98 percent fat free, IQF slices with breast bone.

Weights and Packs: 4-oz (112-g) chops in 36-count boxes.

6-oz (170-g) chops in 24-count boxes.

8-oz (225-g) chops in 18-count boxes.

Kitchen yields: This product has limited use because it has the bone in.

BREAST MEAT CUBES AND STRIPS

All-natural whole muscle meat, 99 percent fat free, portion controlled, IQF.

Packs: 2 bags per carton.

Weight: 5-lb bags per carton, 10 lb (4.5 kg).

Kitchen yields: This is a ready-to-use product, but there is considerable weight loss after thawing and cooking shrinkage.

BREAST ROAST

All white, boneless netted roast with adhering skin. The product may be seasoned.

Pack: 2 to 8 breasts per case.

Weights: 8 lb (3.6 kg) to 12 lb (5.4 kg).

Serving size: 5 oz (140 g).

Kitchen yields: One 8-lb (3.6-kg) breast = 20 servings.

Calories: 5 oz (140 g) = 200 calories.

BREAST SLICES (CUTLETS)

All boneless and skinless meat, sliced across the grain. Product is packed with

paper or plastic film between slices. It can be breaded or flattened for roulades or turkey scallopini. Available frozen; 99 percent fat free.

Weights and Packs: 2 oz (56 g) each, ⅛ in (3 mm) to ⅓ in (8 mm) thick in 30-lb (1.3-kg) and 40-lb (1.8-kg) cases.

3 oz (85 g) each, in 64-count packs.

4 oz (112 g) each, in 48-count packs.

6 oz (170 g) each, mechanically tenderized, in 32-count packs.

8 oz (225 g) each, in 24-count boxes.

Serving sizes: Two 2-oz (56-g) pieces, breaded, for main course; 3 pieces as scallopini.

One 2-oz (56-g) piece for buffet.

Kitchen yields: There is no waste, but shrinkage is considerable.

Calories: 4 oz (112 g) = 176 calories.

BREAST, WHOLE, BONE IN

Pack: 2 to 8 breasts per case.

Weights: 8 lb (3.6 kg) to 22 lb (9.9 kg).

Serving size: 4 oz (112 g).

Kitchen yields: One 12-lb (5.4-kg) breast = 7½ lb (3.4 kg).

One 12-lb (5.4-kg) breast = 28 servings.

DRUMSTICKS

Weights: ½ lb (225 g) to 1½ lb (675 g).

Packs: Bulk.

Serving size: 10 oz (100 g).

Calories: 10 oz = 3½ oz (100 g) cooked meat = 190 calories.

DRUMSTICK STEAKS

Dark meat, skinless, whole muscle, mechanically tenderized, IQF.

Packs: 10-lb (4.5-kg) packs, 40 steaks.

Size: 4 oz each.

GROUND TURKEY

Available as all-breast, 99 percent fat-free meat; lean ground, 7 percent fat; and

seasoned meat such as taco, Italian, or Sloppy Joe.

Packs: 4 each 3-lb (1.3-kg) tubs, in 12-lb (5.4-kg) carton.

GROUND TURKEY, MOSTLY DARK MEAT

At least 85 percent lean, but the product can contain skin. Use for patties, meat sauce, and pizza.

Packs: 10-lb (4.5-kg) rolls, 2 or 4 rolls per case.

Calories: 4 oz (112 g), cooked = 202 calories.

GROUND TURKEY PATTIES

Available as:

All-breast ground, 1 percent fat, one side scored.

Lean, 7 percent fat, one side scored.

Seasoned patty, 7 percent fat, one side scored.

Taco/nacho or Italian patty.

Pack: 40 patties, IQF, 10-lb (4.5-kg) case.

Weight: 4-oz (112-g) individual patty, ready to cook.

TENDERLOIN

Frozen whole muscle from inside center of breast, with plastic film between layers.

Packs: 20-lb (9-kg) to 40-lb (18-kg) cases.

Weights: 6 oz (170 g) to 12 oz (340 g).

Serving size: 6 oz (170 g) average, raw. There is considerable shrinkage.

Kitchen yields: The tenderloins can be cut across the grain into medallions, kebabs, and pieces for stir-fry dishes. There is some cutting waste, depending on portion size.

Calories: 6 oz (170 g), raw = 270 calories.

TENDERLOIN FILET MIGNON

Weights and packs: 1-oz (56-g) size in 160-count packs.

3-oz (85-g) size in 48-count packs.

4-oz (112-g) size in 40-count packs.

4-oz (112-g) size, butterfly cut, in 40-count packs.

Serving size: Two 3-oz (85-g) pieces.

Kitchen yields: There is no waste.

THIGHS

Bone in, skin on, dark meat.

Weights: ½ lb (225 g) to 1½ lb (675 g).

Packs: Bulk.

Serving size: 4 oz (112 g), boneless.

Kitchen yields: Purchase large thighs.

One 1½-lb (675-g) thigh = 12 oz cooked meat with skin, 3 servings.

THIGH STEAKS

Boneless and skinless raw dark meat; can be braised, broiled, or baked.

Weights: 3 oz (85 g) and 4 oz (112 g).

Packs: 40 to 54 individual steaks in 10-lb (4.5-kg) carton.

Serving size: Two 3-oz (85-g) pieces.

Kitchen yields: One 6-oz (170-g) serving weighs about 4 oz (112 g) cooked.

Calories: 4 oz (112 g), cooked = 232 calories.

WINGS

Turkey wings are available as:

Whole wings or with wing tips removed

Drumettes (first joint next to breast)

Second, flat joints

Flippers, second joints and wing tips

Packs: 30-lb (13.5-kg) cases.

Weights: ¾ lb (340 g) to 1¼ lb (560 g).

Kitchen yields: It is of the utmost importance to specify exactly what parts are needed, because each joint has different applications and yield. Drumettes are meaty and can make an acceptable main course, second joints can be precooked and grilled, and wing tips are suitable only for stock.

COOKED TURKEY PRODUCTS**BREAST AND THIGH**

Boneless combination of natural proportion white and dark meat, usable for sandwiches. Product is available flavored and browned. Economical and easy to slice. The shape of the product varies, which has an impact on yield.

Pack: 2 to 4 pieces per case.

Weight: 7 lb (3.1 kg) to 10 lb (4.5 kg).

Kitchen yields: Same as for Formed Breast.

Calories: 4 oz (112 g) = 140 calories.

FORMED BREAST

All-breast meat, made with large sections, held together with a number of extenders, including ground turkey meat and water. Product is available flavored and browned. Economical and easy to slice. The shape of the product varies, which has an impact on yield. There is little trimming waste.

Pack: 2 or 4 pieces per case.

Weights: 4 lb (1.8 kg) to 10 lb (4.5 kg).

Best size: 6 lb (2.70 kg). Large pieces are difficult to handle.

Serving size: 4 oz (112 g), for sandwich.

Kitchen yields: One 6-lb (2.70-kg) breast, skinless = 24 sandwiches.

GROUND ROLL

Ground turkey, available as all white meat or in combination with dark meat. Can include skin, extenders, binders, and water added. The least expensive turkey product for sandwiches.

Weights: 7 lb (3.1 kg) to 10 lb (4.5 kg).

Pack: 2 to 4 pieces per case.

Kitchen yields: There is little waste; the product can be machine sliced when thoroughly chilled.

Calories: 4 oz (112 g) = 170 calories.

PULLED OR DICED BONELESS MEAT

Fully cooked meat, which may contain dark and white meat combined, IQF. Only the servings required at the time need be removed from bag.

Packs: 5-lb (2.2-kg) bags, 4 to a case.

Serving size: 5 oz (140 g) frozen, in sauce or salad.

Kitchen yields: Product can be used for any kind of hot turkey dish.
One 5-lb (2.2-kg) bag = 16 servings.

THIGH LOAF

Boneless, all-dark-meat product, oval-shaped or roll.

Pack: 2 to 4 pieces per case.

Weights: 4 lb (1.8 kg) to 10 lb (4.5 kg).

Kitchen yields: Inexpensive meat for sandwiches. There is some trimming loss.

Calories: 4 oz (112 g) = 144 calories.

WHOLE-MUSCLE BREAST

Boneless, fully cooked, hand packed. Available with skin on and skinless. Up to 98 percent fat free when skinless. This is the best-quality, but also the most expensive, cooked breast. Many brands are available from different packers. The product is available colored with caramel, browned, and with low salt. One package can include up to three breast muscles. The shape is oval, and there is at least 3 percent cutting waste for skinned breast, and more for skin-on breast.

Pack: 2 to 8 pieces per carton.

Weights: 2 lb (900 g) to 10 lb (4.5 kg).

Best size: 6 lb (2.70 kg). Large pieces are difficult to handle.

Serving size: 4 oz (112 g), for sandwich.

Kitchen yields: One 6-lb (2.70-kg) breast, skinless = 22 sandwiches.

Calories: 4 oz (112 g), skinless = 125 calories.

BREADED COOKED TURKEY PRODUCTS**CUTLETS**

Manufactured from whole-muscle breast meat, breaded or batter dipped. Precooked or fried.

Pack: 50 pieces 10 lb (4.5 kg) carton.

Count: 5 cutlets per lb (450 g).

Weight: 3.2 oz (90 g).

Serving size: 2 pieces.

Kitchen yields: Ready to use.

NUGGETS

Available batter fried or breaded, fully cooked.

Packs: 8 lb (3.6 kg) to 12 lb (5.4 kg).

Counts: 215 to 390.

Weights: ½ oz (14 g) to ¾ oz (20 g).

Kitchen yields: Ready-to-use product for buffets or receptions. Nuggets taste best heated.

Calories: 4 oz (112 g) = 328 calories.

PATTIES

Ground product, shaped into patties and precooked or prebrowned. Available as all white meat, all dark meat, or blended meat.

Packs: 10-lb (4.5-kg) to 12-lb (5.4-kg) cases, 48 to 72 patties each.

Counts: 5 to 7 patties per lb.

Weight: 3 oz (85 g) average.

TURKEY DELICATESSEN ITEMS**SMOKED BREAST**

Available boneless and on frame. On some products, the type of smoke used is indicated; some are flavored with artificial liquid smoke. Whole breasts are usually vacuum packed.

Pack: 2 to 10 breasts per case.

Weights: 6 lb (2.7 kg) to 12 lb (5.4 kg), on frame.

2 lb (900 g) to 4 lb (1.8 kg), boneless.

Kitchen yields: Product may be skinless or with skin on, in natural shape or round, slab, or football shape. It is best to choose the item most suitable for the operation.

SMOKED WHOLE TURKEY

The product is available fresh or frozen. It is best used as a display piece on buffets. If sliced meat is needed, purchase boneless breasts. Check the quality; some products are flavored with artificial smoke and boiled in the bag rather than smoked. All smoked meats should be served at room temperature, providing that all sanitary precautions are taken.

Pack: 4 turkeys per case.

Weights: 8 (3.6 kg) to 10 lb (4.5 kg).

10 (4.5 kg) to 12 lb (5.4 kg).

12 (5.4 kg) to 14 lb (6.3 kg).

14 (6.3 kg) to 16 lb (7.2 kg).

Kitchen yields: One 10-lb (4.5-kg) turkey, carved on a buffet table = 20 appetizer servings.

TURKEY HAM, WHOLE MUSCLE

Dark meat product resembling ham in flavor and appearance. Available oval shaped and as a roll, referred to as *Canadian brand*. It is leaner than pork ham. Related products are chunked/chopped turkey ham and cured drumstick boneless turkey.

Weights: From 5 lb (2.2 kg) up.

Kitchen yields: Ready-to-eat product, often less expensive than similar pork products. Product should be identified on the menu as turkey.

Calories: 4 oz (112 g), whole-muscle ham = 150 calories.

TURKEY PASTRAMI

All dark, fully cooked, shaped meat, covered with crushed coriander seeds. The product is 95 percent lean and appeals to diet-conscious customers.

Packs: 12-lb (5.4-kg) to 48-lb (21.6-kg) cases.

Sizes: 3 lb (1.3 kg) to 8 lb (3.6 kg).

Kitchen yields: Although the product is fully cooked, yields vary according to shape; the scraps are covered with seeds and are difficult to use.

Calories: 4 oz (112 g) = 160 calories.

TURKEY SAUSAGES

Numerous fully cooked products are on the market, meeting the demand of diet-conscious customers because of the low

fat and calorie contents. Available products include salami, bologna, and frankfurters in different sizes.

WILD TURKEY

Available fresh or frozen.

Sizes: 10 lb (4.5 kg) to 20 lb (9 kg).

Kitchen yields: 1 lb (450 g), as purchased = 35 percent usable cooked meat.

VEAL

Animals up to 3 months of age are classified as veal, over 3 months up to 1 year as calves. For foodservice, veal is most likely to be specified, except for liver, which should come from calves because veal liver is too small for commercial use. Veal is not normally graded by the USDA. Some veal is raised and marketed with proprietary names, such as *Plume de Veau* and *Provini*. The animals are heavy, yet have light meat. They are raised in small enclosures with special diets. Some consumer groups and chefs are concerned about the treatment of the animals and the antibiotics added to the feed. *Nature veal* is an industry term for heavy veal with light meat. When purchasing, make sure to specify veal, because some cuts are available in both veal and calf weight ranges. Veal has soft, almost white meat; calf has redder and coarser meat. The *Meat Buyers Guide* specification numbers are used to further identify the cuts.

LARGER CUTS OF VEAL

BACK, MBG #341

Whole back with 9 ribs, trimmed. The piece requires extensive butchering. It makes an attractive buffet centerpiece when roasted whole. For ready-to-cut steaks, see Boneless Strip Loin, MBG #344.

Weight range: 10 lb (4.5 kg) to 30 lb (13.5 kg).

Recommended weight: 18 lb (8.1 kg).

Kitchen yields: Use for buffet display; servings on buffet = 50.

BONES

Bones from the neck and loin are soft and render flavor; knucklebones add gelatin but less flavor to stocks and sauces.

Packs: By weight, usually 50-lb (22.5-kg) boxes, frozen.

Kitchen yields: 50 lb (22.5 kg) veal bones = 10 gal (38 l) brown stock. Note that judging the strength of stock is subjective. The proportions given here are for good-quality stock.

BRAINS

Brains should only be purchased fresh, because the skin and membranes are difficult to remove from frozen brains. To remove skins and blood, soak in lukewarm water.

Packs: ½-lb (225-g) piece, trayed.
16 pieces per 10-lb (4.5-kg) box.
40-lb (22.5-kg) cartons.
Also available loose by weight.

Average size: ½ lb (225 g) per piece.

Kitchen yields: One 8-oz (225-g) piece = 7 oz (200 g) cleaned. Note that

careless handling results in broken pieces, which are difficult to portion.

BREAST, MBG #313

For best use, bone and stuff, bone and roll, or braise with bones in. The product is available with a pocket, but still with the bone in as, MBG #314.

Packs: 4 breasts per 48-lb (21.5-kg) master carton. Other packs are available.

Weight ranges:

- 3 lb (1.3 kg) to 6 lb (2.7 kg).
- 6 lb (2.7 kg) to 8 lb (3.6 kg).
- 8 lb (3.6 kg) to 10 lb (4.5 kg).
- 10 lb (4.5 kg) to 12 lb (5.4 kg).

Recommended size: 10 lb (4.5 kg).

Kitchen yields:

Rolled veal breast: One 10-lb (4.5-kg) breast

- = 24 oz (670 g) bones.
- = 8 oz (225 g) waste trim, fat.
- = 8 lb (3.6 kg) rolled breast meat.
- = 5 lb 10 oz (2.5 kg) after roasting
- = 12 servings.

Stuffed breast: One 10-lb (4.5-kg) breast

- = 24 oz (670 g) bones.
- = 8 oz (224 g) waste trim, fat.
- = 8 lb (3.6 kg) ready-to-stuff meat.
- = 3 lb (1.3 kg) ground meat stuffing.
- = 11 lb (5 kg) stuffed, ready to roast.
- = 8 lb 5 oz (3.8 kg), weight after roasting.
- = 15 to 16 servings.

Serving size: 7 oz (200 g), rolled breast.
8 oz (225 g), breast with meat stuffing.

BUTT TENDERLOINS, MBG #346A

Short tenders, fully trimmed.

Size: ½ lb (225 g) each.

Pack: Sixty-four ½-lb (225-g) tenders per 37-lb (16.6-kg) master carton.

Kitchen yields: Very tender pieces, expensive but easy to portion. Best use is for scallopini or breaded cutlets.

CHUCK, BONELESS, MBG #309B

Economical piece for cutting stew meat.

Weight range: 12 to 40 lb (5.4 to 18 kg) whole, double chuck.

Recommended size: 20 lb (9 kg).

Serving size: 6 oz (170 g), cooked meat in stew.

Kitchen yields: Trimming waste = 8 percent.

1 lb (450 g) raw meat = 10 oz (280 g) cooked meat.

One 20-lb (9-kg) chuck = 30 servings.

CHUCK ROLL, TRIMMED, MBG #311

Netted boneless piece, ready to roast.

Pack: Pack by one purveyor: five 4½-lb (2-kg) pieces per 22-lb case.

Sizes: 4½ lb (2 kg), 9 lb (4 kg) to 30 lb (13.5 kg).

Best size: 4½ lb (2 kg). Large pieces are hard to handle.

Serving size: 5 oz (140 g), roast.

Kitchen yields: One 4½-lb (2-kg) roll = 10 servings.

1 lb (450 g) as purchased = 11 oz (310 g) cooked meat.

Calories: 5 oz (140 g), roast = 380 calories.

CHUCK, SCOTCH TENDER, MBG #310C

Relatively newly available and convenient fully trimmed piece. It can be used for roasting or braising whole or cut for stew. The piece is very practical for banquet or buffet use.

Pack: Sixty 1-lb (450-g) pieces per 60-lb (27-kg) box.

Size: 1-lb (450-g) piece, vacuum packed.

Kitchen yields: 1 lb (450 g), roasted whole = 12 oz (340 g) cooked meat.

CHUCK, SHOULDER CLOD ROAST, MBG #310B

Ready-to-roast, netted, football-shaped piece; it may separate after roasting.

Weights: 2 lb (400 g) to 7 lb (3.15 kg).

Recommended size: 4 lb (1.80 kg).

Serving size: 5 oz (140 g), roast.

Kitchen yields: One 4-lb (1.80-kg) piece
= 8 to 9 servings.

HEART

Relatively lean muscles that should be boiled or braised, although in South America calf and beef hearts are cut into cubes, skewered, grilled, and served as appetizers.

Packs: By weight and piece. 25 to 30 pieces per 50-lb (22.2-kg) box.

Average sizes: 1 lb (450 g) to 2 lb (900 g) each.

Serving size: 6 oz (170 g) cooked.

Kitchen yields: About 10 percent trimming is needed after cooking.
One 1-lb (450-g) heart = 2 servings.

Calories: 4 oz (112 g) = 310 calories.

HIND SHANK, MBG #337

When purchasing, specify hock bone cut off; this piece requires a meat saw to portion properly. For osso buco, do not purchase Foreshank, MBG #312.

Pack: Fifteen 3½-lb (1.5-kg) shanks, trimmed, per 55-lb (24.7-kg) box.

Weight range: 1 lb (400 g) to 4½ lb (2 kg).

Recommended sizes: 4½ lb (2 kg) for osso buco.

3 lb (1.3 kg) for roasting whole.

Serving size: 2 pieces osso buco.

Kitchen yields: Cut shank in 4 pieces.
1 shank, roasted whole = 2 servings.

HIND SHANK, OSSO BUCO, MBG #338

Cut veal shank, available already cut. The bone size varies.

Packs: 10-lb (4.5-kg) cartons, 4 cartons per master case.

Count: 80 to 90 pieces.

Weights: 6-oz (170-g) to 8-oz (225-g) pieces.

Serving size: 2 pieces, usually.

KIDNEYS

Available with the fat on or peeled, fat removed. Most purveyors sell peeled kidneys only.

Pack: 12-oz (340-g) pieces, 14 pieces per 10-lb (4.5-kg) box.

Sizes: 8 oz (225 g) to 12 oz (340 g), peeled.

Recommended size: 10 oz (280 g) to 12 oz (340 g), peeled.

Serving size: 1 kidney.

Kitchen yields: Trimming loss from peeled kidney = 5 percent.

LEG, MBG #334

Legs are normally sold in pairs. When a number of single legs are purchased, make sure that only every other leg has the tail still attached, because the tails weigh about 8 oz (225 g) to 10 oz (280 g) and can be used only as bone. Few operations purchase whole legs because skillful butchering is needed, and there is a considerable amount of by-products to use up. Leg meat is available boned and trimmed; these pieces are described in the following entry. Common uses for veal leg include scallopini, breaded cutlets, roast veal, stew, and veal strips sauté.

Weight range: 12 lb (5.4 kg) to 45 lb (20.5 kg), whole legs.

Recommended sizes: 20 lb (9 kg), for roasting.

45 lb (20.5 kg), for boning.

Serving sizes: 6 oz (170 g), scallopini, three pieces combined.

5 oz (140 g), breaded cutlet.

Kitchen yields:

One 42-lb 12-oz (19.2-kg) leg
= 6 lb 6 oz (2.8 kg) clean, best quality meat, ready to cut, fully trimmed, 15 percent.

= 7 lb (3.1 kg) clean meat, good quality, 16 percent.

= 4 lb 5 oz (1.9 kg) stew meat, second quality, 10 percent.

- = 6 lb 10 oz (3 kg) scraps, usable for stock, some for ground meat, 16 percent.
- = 6 lb 6 oz (2.8 kg) fat, 15 percent.
- = 8 lb 9 oz (4 kg) bones, 20 percent.
- = 3 lb 8 oz (1.5 kg) shank, 8 percent.
- One 42-lb 12-oz (19.2-kg) leg
- = 37 servings scallopini or
- = 36 servings cutlets *and* 6 servings scallopini.

Calories: 6 oz (170 g) scallopini, raw = 365 calories.

LEG, BONELESS, SHANK OFF, MBG #336

Good piece for roasting whole, usually purchased netted. Because of its uneven shape, the piece is more useful as a buffet piece, sliced in the dining room, than as a banquet item.

Weight range: 7 lb (3.1 kg) to 23 lb (10.3 kg).

Recommended size: 12 lb (5.4 kg).

Serving size: 4 oz (112 g), cooked.

Kitchen yields: One 12-lb (5.4-kg) leg = 20 to 22 servings, main course.
 = 25 banquet servings, if cut on machine.
 = 30 buffet servings.
 1 lb (450 g) = 13 oz (365 g), cooked meat.

LEG PARTS, BONELESS, MBG #348

The leg is divided into four parts and, to a large extent, trimmed. Some additional trimming may be necessary.

Weights: 15 lb (6.75 kg) and up.

Best size: Purchase as small as possible.

LIVER (CALF'S LIVER)

On the New York market, the term *kosher calf's liver* is used to indicate freshness, although the liver itself may not be certified kosher. Livers should be light in color and fresh. The livers can come from heavy veal or from small calves.

Weight range: 2½ lb (1.1 kg) to 7 lb (3.1 kg).

Packs: By the piece and by weight.

7-lb (3.1-kg) whole livers, 8 pieces per case.

Best size: 3 lb (1.3 kg) to 3½ lb (1.6 kg).

Serving size: 5 oz (140 g) to 6 oz (170 g), in 2 slices.

Kitchen yields: The liver must be skinned. There is a small flap, and some trimming waste, which can be used in paté or as julienne of liver sauté.

One 3-lb (1.3-kg) liver = 7 servings.

Calories: 5 oz (140 g), raw = 210 calories.

LOIN, STRIP, BONELESS, MBG #344

Also referred to as *calf's sirloin*. This is a boneless, well-trimmed piece of meat with 1-in (25-mm) flank, skin on. It can be cut into boneless chops resembling sirloin steaks, with some additional trimming.

Packs: Sixteen 3¼-lb (1.5-kg) loins per 51-lb (23-kg) master carton.

Seven 8-lb (3.6-kg) loins per 56-lb (25.2-kg) master carton.

Weight range: 5 lb (2.25 kg) to 8 lb (3.6 kg).

Kitchen yields: The piece requires minimal trimming.

LOIN, TRIMMED, MBG #344A

This expensive piece is trimmed to the loin eye.

Weight range: 2½ lb (1.1 kg) to 3 lb (1.3 kg).

Pack: Sixteen 2¾-lb (1.2-kg) loins per 45-lb (20-kg) carton.

Kitchen yields: Completely trimmed loin with no waste.

LOIN, WHOLE, MBG #331

Whole loin, including two ribs, kidneys, and tenderloins. The same piece trimmed to 4 in flank, with kidneys removed, is called MBG #332. The piece is also available boneless and rolled. The loin is the

best part of the animal and very expensive.

Weight range: 10 lb (4.5 kg) to 14 lb (6.3 kg).

14 lb (6.3 kg) to 18 lb (8.1 kg).

18 lb (8.1 kg) to 23 lb (10.3 kg).

Recommended size: 22 lb (9.9 kg).

Serving size: 6½ oz (180 g), loin chop, boneless, 1 piece per serving.

5 oz (112 g), medallion, completely trimmed.

5½ oz (160 g), scallopini, raw.

5 oz (112 g), loin roast, cooked meat.

Kitchen yields: One 22-lb (9.9-kg) loin

= 7 lb (3.1 kg) trimmed meat.

= 16 servings loin chop, boneless.

= 16 servings medallion, 1 piece per serving.

= 17 servings scallopini, 3 pieces per serving.

= 18 servings roast loin.

Whole loin presented with bone in, cold, on buffet:

Recommended size: 14 lb (6.3 kg) to 18 lb (8.1 kg) = 30 servings.

NECK, BONELESS

Flavorful piece for cutting stew meat.

Weight range: 9 lb (4 kg) to 11 lb (4.9 kg).

Serving size: 5½ oz, cooked stew meat, main course.

3½ oz (100 g) cooked stew on buffets.

Kitchen yields: Trimming waste = 5 percent.

1 lb (450 g) stew meat = 11 oz (310 g) cooked meat.

10 lb (4.5 kg) raw stew meat = 20 servings, main course.

RACK, MBG #306

Racks with seven ribs are standard.

Weight range: 5 lb (2.2 kg) to 20 lb (9 kg).

Recommended size: 16 lb (7.2 kg) to 18 lb (8.1 kg).

Serving size: 9 oz (250 g) rib chop, bone in.

7½ oz (210 g), roast rack including rib bone.

Kitchen yields: The piece has low yield because of the massive bone structure and trimming loss. The proportion of usable meat versus bone is better in larger pieces.

However, rib chops cut from 16-lb (7.2-kg) and larger racks are very heavy because of the large rib bone.

One 19-lb 4-oz (8.6-kg) rack

= 7 lb 8 oz (3.3 kg) racks, two pieces Frenched, 39 percent.

= 3 lb 8 oz (1.5 kg) stew meat, 18 percent.

= 2 lb 8 oz (1.1 kg) scraps (some can be ground), 13 percent.

= 1 lb 8 oz (0.6 kg) fat, 8 percent.

= 4 lb 4 oz (1.9 kg) bones, 22 percent.

One 19-lb 4-oz (8.6-kg) rack = 10 rib chops and two servings scallopini, 6 oz (170 g).

= 12 servings rib roast.

RIB EYE, MBG #307

Completely trimmed and boneless muscle ready to be cut. The piece is obviously expensive.

Size: 3 lb (1.3 kg) and up.

Packs: Ten 3-lb (1.3-kg) pieces per 30-lb (13.5-kg) carton.

Kitchen yields: The piece is boneless and well trimmed. It can be cut into boneless steaks or roasted whole.

RIBLETS

One in (25 mm) or 2 in (50 mm) wide riblets cut from the rack. Back ribs are also available.

Packs: 40-lb (18-kg) cases.
15-lb (6.75-kg) cartons.

Count: Varies: about 4 pieces to the pound.

Serving size: 12 oz (340 g) to 16 oz (450 g), raw.

Kitchen yields: Short cut riblets are ready to use.

STEW MEAT, MBG #395

Purchase: By weight.

Kitchen yields: See Neck or Chuck for yield information.

SWEETBREADS, MBG #715

Veal sweetbreads are lighter in color than calf sweetbreads; specify fresh when possible. Veal sweetbreads can be tough. Sweetbreads are sold in pairs, consisting of two uneven pieces. They can also be sold separated.

Packs: By weight, usually 5-lb (2.2-kg) baskets.
40-lb (18-kg) boxes.

Sizes: 1 lb (450 g) average, per pair.

Serving size: 8 oz (225 g), raw, as main course.

Kitchen yields:

Trimming waste = 10 percent.

Cooking loss = 15 percent.

5 lb (2.2 kg) = 10 servings.

TBS PACK

One packer sells packs of fully trimmed top round, bottom round, and sirloin, vacuum packed.

Pack: 2 packs per 30-lb (13.5-kg) case.

TONGUE

Sizes: 1 lb (450 g) to 1½ lb (675 g).

Packs: By piece and weight.

Thirty-two 1½-lb (675-g) pieces per 40-lb (18-kg) case.

Kitchen yields: 1 lb (450 g) as purchased = 10 oz (280 g) cooked and skinned.

TOP ROUND, MBG #349 A

Fully trimmed, solid piece, the most economical cut to purchase for cutting cutlets and scallopini.

Pack: Eight 4½-lb (2-kg) pieces per 37-lb (16.6-kg) master carton.

PORTION CUTS OF VEAL

CUBED STEAKS, MBG #1300

Boneless slices that have been mechanically tenderized. The meat can be dry.

Sizes: 3 oz (85 g), 4 oz (112 g), 5 oz (140 g), 6 oz (170 g), and 8 oz (225 g).

Serving size: 4 oz (112 g), lunch size when breaded.

Kitchen yields: Best use is for breaded slices.

CUTLETS, MBG #1336

Boneless slices made from leg meat only. The meat can be cut against or with the grain. Pounding is needed to make the meat tender.

Packs: 2-oz (56-g) pieces: four 8-lb (3.6-kg) boxes = 256 pieces.

3-oz (85-g) pieces: four 8-lb (3.6-kg) boxes = 170 pieces.

4-oz (112-g) pieces: four 8-lb (3.6-kg) boxes = 126 pieces.

Sizes: 2 oz (56 g), 3 oz (85 g), 4 oz (112 g), 5 oz (140 g), and 6 oz (170 g).

Serving sizes: 5 oz (140 g), breaded.

4 oz (112 g), slice, cut in half for reception.

5 oz (140 g) slice for braised roulade.

Kitchen yields: Meat can be tough and dry because of the loss of juices after thawing. Best use is breaded, but the product can be used as veal marsala or for scallopini.

GROUND PATTIES, MBG #1396 A

Available in different sizes, plain and breaded. Low-cholesterol substitute for beef hamburger patties.

PLAIN PATTIES

Packs: 3-oz (85-g) patties: four 8-lb (3.6-kg) boxes = 170 pieces.

4-oz (112-g) patties: four 8-lb (3.6-kg) boxes = 126 pieces.

Weights: 3 oz (85 g), 4 oz (112 g).

BREADED PATTIES

Packs: Four 8-lb (3.6-kg) boxes = 126 pieces.

Weight: 4 oz (112 g).

LIVER (CALF'S LIVER)

Frozen calf's liver is available sliced. Not to be confounded with baby beef liver.

Weights: 3 oz (85 g) and 4-oz (112-g), slices.

Packs: 3-oz (85-g) slices: four 8-lb (3.6-kg) boxes = 170 pieces.

4-oz (112-g) slices: four 8-lb (3.6-kg) boxes = 126 pieces.

Serving size: 2 pieces, usually.

LOIN CHOPS, MBG #1332

Cut from the loin, with bones still attached. Up to 3 in (75 mm) of flank can be left on.

Sizes: 5 oz (140 g), 6 oz (170 g), 8 oz (225 g), and 10 oz (280 g).

Serving sizes: One 8-oz (225-g) or 10-oz (280-g) piece.

Two 5-oz (140-g) pieces per serving.

Kitchen yields: Item is rather expensive and is best used as a dinner item.

RIB CHOPS, MBG #1306

Cut from the rack, with all bones still attached but blade removed.

Packs: Vary according to size and packer.

Sizes: 5 oz (140 g), 6 oz (170 g), 8 oz (225 g), and 10 oz (280 g).

Kitchen yields: Item is rather expensive and is best used as a dinner item.

Serving size: 8 oz (225 g) or 10 oz (280 g).

SHOULDER CHOPS, MBG #1332

Cut from the chuck, with bones still in.

Sizes: 5 oz (140 g), 6 oz (170 g), 8 oz (225 g), and 10 oz (280 g).

Serving size: 6 oz (170 g).

Kitchen yields: Best used for braising. Good lunch item.

QUICK LOOK-UP FOR POPULAR VEAL DISHES**FRICASSEE**

Use high-quality stew meat from shoulder or purchase stew meat. Do not buy

frozen stew meat because it can make the sauce cloudy.

Serving size: 8 oz (225 g) to 10 oz (280 g), raw meat, for lunch.

LOIN ROAST

The roast can be stuffed with any stuffing, or with kidneys, and rolled. It can also be made with loin halves, flank still attached, or with the whole small loin.

Kitchen yields:

Stuffed half loin roast: 4 lb (1.8 kg) meat plus 1 lb (450 g) stuffing = 10 servings.

Stuffed full loin roast: 7 lb (3.1 kg) meat plus 1 lb (450 g) stuffing = 16 servings.

OSSO BUCO

Purchase large shanks, about 4 lb (1.8 kg) each. Cut center of shank in 4 pieces. There will be about 1½ lb (680 g) bone without meat left over. Shanks are also available ready cut.

Serving size: 2 pieces, 8 oz (225 g) to 10 oz (280 g) each.

PAILLARD

Very thin, pounded cutlet, grilled or sautéed to order. Use best-quality leg meat.

Serving size: 7 oz (200 g).

PICCATA

Veal scallopini served with lemon juice and sliced lemon.

Serving size: 3 pieces per serving, 6 oz (170 g) total weight.

See Veal Leg and Leg Parts.

ROASTED VEAL SHANKS (HAXEN)

Haxen is the German word for *whole roasted shanks*. Use 3-lb (1.3-kg) shanks.

Serving size: 2 servings per shank.

ROGNONNADE

Rognonnade is the French name for *roast loin stuffed with kidneys*. See under Loin Roast.

ROLLATINE

Rollatine is the Italian name for *stuffed, rolled cutlet*. Use 6-oz (170-g) thin cutlets.

SALTIMBOCCA ALLA ROMANA

Veal scallopini with prosciutto and sage.

Serving size: Three 2-oz (56-g) pieces, total 6 oz (170 g).
1 oz (28 g) prosciutto, sliced in 3 pieces.

SCALLOPINI

The meat can be cut from the leg, from the loin, or from butt tenderloins.

Serving size: Three 2-oz (56-g) pieces, total 6 oz (170 g).

VEAL BIRDS

Rolled, stuffed, and braised cutlets. Cubed cutlet can be used.

Serving sizes: 6 oz (170 g) meat plus 1 oz (28 g) stuffing = 1 serving, main course.

3 oz (85 g) meat plus ½ oz (14 g) stuffing, 2 pieces per person or for buffet receptions.

VEAL MARSALA

Scallopini flavored with Marsala wine. *See* under Scallopini.

VEAL OSCAR

Veal cutlet topped with asparagus, crab meat, and hollandaise sauce.

Serving size: 6 oz (170 g), boneless cutlet.
1 oz (28 g) crab meat.
3 asparagus tips.

VITELLO TONNATO

Cold veal roast, covered with cold tuna-fish-based sauce. Any lean roast can be used. *See* the Recipes section for Tonnato Sauce under Sauces.

Serving size: 4 oz (112 g), cooked roast, sliced thin, as main course.
2 oz (0.06 l) sauce.

WIENER SCHNITZEL

German name for *boneless breaded cutlet*.

Serving sizes: 4 oz (112 g), lunch.
5 oz (140 g), dinner.

VENISON

Available fresh or frozen from domestic sources and imported. Game meat is very lean and can be dry when braised. Boneless venison medallions, cut from the racks, are excellent. Available cuts include fully trimmed hind legs, racks, shoulders, boneless stew meat, sausages, and ground venison.

Packs: Hind leg, shank cut short: 16 lb (7.2 kg) to 25 lb (11.25 kg).

Racks, bone in: 10 lb (4.5 kg) to 14 lb (6.3 kg).

Boneless stew meat: 10-lb (4.5-kg) bags.

Boneless loins: sizes and packs vary.

Best sizes: 16 lb (7.2 kg) to 18 lb (8.1 kg), leg, bone in.
12 lb (5.4 kg), rack, bone in.

Serving sizes: 7 oz (200 g), leg, roasted or braised.

6 oz (170 g), steak, raw, cooked to order medium rare.

Kitchen yields: One 16-lb (7.2-kg) leg = 16 to 18 servings, roasted medium.

One 16-lb (7.2-kg) leg = 14 to 16 servings, braised.

One 12-lb (5.4-kg) rack, bone in = 20 servings, boneless filet steaks.

One 12-lb (5.4-kg) rack, bone in = 16 servings, roasted on the bone.

SMOKED VENISON LEG

Available as a specialty item. Serve sliced very thin, like prosciutto ham.

Size: Varies, usually 8 lb (3.6 kg), bone in.

VIANDE DE GRISON

See Bündnerfleisch, under Smoked and Cured Beef Products.

Chicken Breasts Braised in Tomato Port Wine Sauce

The distinct flavor of port, a wine fortified with brandy, makes this simple chicken dish far from dull.

Recipe

Vegetable cooking spray, olive-flavored	—	Garlic, minced	1 T
Chicken breast halves, skinless, trimmed of fat	7½ lb (24 5-oz pieces)	Port wine, dry	1½ pt
Onions, minced	2 lb	Rich white chicken stock (see page 7)	1½ pt
Tomatoes, peeled, coarsely chopped	2½ lb	Parsley, minced	¼ c + 2 T
		Cornstarch	2 oz
		Salt	To taste
		Pepper, freshly ground	To taste

1. Coat a nonstick skillet with cooking spray. Place over medium heat until hot. Add the chicken in batches and cook, turning once, until no longer pink, or about 8 minutes. Transfer the chicken from the skillet to a counter pan; keep warm.
2. Recoat the skillet with cooking spray. In batches, or using a saucepan, add the onions and garlic; cook until tender.
3. Add the tomatoes and next three ingredients; heat to a boil; reduce the heat to low; simmer until the sauce is pulpy, or about 10 minutes. Pour over the chicken. Bake in a 350°F (175°C) oven until tender, or about 30 minutes. Remove the chicken from the pan. Hold in a warm place.
4. In a small bowl, mix the cornstarch with cold water as needed to form a smooth paste. Whip the cornstarch slurry into the cooking sauce. Simmer until the sauce is thickened, or a few minutes. To adjust the sauce's consistency, thicken by adding more cornstarch slurry or thin by adding stock. Season with salt and pepper to taste. Serve the chicken breasts with the sauce ladled over.

Servings	Calories	Protein (g) (%)	Fat (g) (%)	Cholesterol (mg)	Carbohydrates (g) (%)	Fiber (g)	Sodium (mg)
1	185	28 (63%)	3.4 (18%)	72.3	8.4 (19%)	1.3	79.6

No alcohol is listed in the nutrition analysis. It is assumed to be cooked off during preparation.

This section contains basic recipes commonly needed in commercial kitchens.

ALMOND FILLING FOR BAKING

Formula:

2 lb (900 g) almond paste
2 lb (900 g) sugar
1 lb (450 g) butter
8 oz (225 g) flour
1 qt (0.47 l) or about 20 whole eggs
Grated lemon peel to taste

Method: Combine ingredients in mixer, use paddle.

Serving sizes: 1 oz (28 g) filling per piece for Danish pastry.
8 oz (225 g) for each coffee cake ring.

Kitchen yields: 130 pieces Danish pastry.
17 coffee rings.

ALMOND MACAROONS

Formula:

10 lb (4.5 kg) almond paste
8 lb (3.6 kg) sugar
3 lb (1.3 kg) 6X sugar
2 qt (0.94 l) egg whites

Method: Combine ingredients in mixer with paddle. Dress macaroons on kraft paper and bake. Moisten paper to remove macaroons.

Serving size: 1 oz (28 g).

Kitchen yield: Total mix = 380 pieces.

ALMOND/MARZIPAN PASTE FOR MODELING

The paste can be used for modeling marzipan fruits and figures.

Formula:

One #10 can, 6.5 lb (3.7 kg) almond paste, warmed
1 lb (450 g) glucose, warmed
3 lb (1.3 kg) confectioners sugar

Method: Combine ingredients in mixer, use paddle.

Kitchen yield: 11 lb (5 kg) paste.

ALMOND MERINGUE CRUST

Formula: 1 qt (0.94 l) egg whites

2 lb (900 g) ground blanched almonds
2 lb 3 oz (1.8 kg) superfine sugar
1½ cups, 10.5 oz (300 g) corn starch

Method: Combine almonds with cornstarch and 2 lb sugar. Beat egg whites until stiff, and add remaining sugar. Fold in almond mixture. Butter and flour pastry sheets (or use parchment paper). Dress meringue in circles or any other shape desired. Bake at 27°F, but do not brown. Dry overnight. Fill with whipped cream or ice cream to order.

Kitchen yield: Sixteen 10-in meringue circles.

ALMOND TART

Formula, Dough:

3 lb (1.3 kg) cake flour
1 lb (450 g) sugar
8 eggs
1 tsp vanilla extract
1 tb lemon juice
1 lb (450 g) butter

Method: Combine dough ingredients in food processor and make dough. Refrigerate. Line quiche molds with removable bottoms, or other suitable molds, with dough, including sides.

Formula, Filling:

1 lb (450 g) butter
1 lb 6 oz (610 g) sugar
2 lb (900 g) ground blanched almonds
12 eggs
2 tb dark rum
1½ lb (670 g) sliced blanched almonds
Apricot glaze

Method: Cream butter, sugar, and eggs. Add rum. Add ground almonds. Fill molds with mixture, sprinkle with

almonds, and bake at 400°F. Brush with apricot glaze.

Kitchen yield: 8 tarts, 9 in (228 mm) across.

ASPIC

Clear meat or fish jelly. It can also be made with ready-to-use aspic powder.

BEEF ASPIC

Formula:

- 1 gal and 1 pint (4.2 l) warm beef stock
- 2 lb (900 g) lean ground beef
- 8 oz (225 g) mirepoix
- 1 cup (0.24 l) canned stewed tomatoes
- 1 tb peppercorns
- 1 cup (0.24 l) egg whites
- 4 oz (112 g) plain gelatin

Method: Combine ingredients very thoroughly and bring slowly to boil, stirring frequently until the mixture has reached 180°F. Simmer slowly without stirring for 1 hour. Cool and drain off, strain through cheesecloth. Remove fat when cold.

Kitchen yields: 1 gal (3.8 l).

FISH ASPIC

Formula:

- 1 gal (3.8 l) fish stock
- 1 cup (0.24 l) very dry white wine
- 8 oz (225 g) mirepoix without carrots
- 1 tb peppercorns
- 2 tb white vinegar
- 1 cup (0.24 l) egg whites
- 5 oz (140 g) plain gelatin

Method: Combine ingredients very thoroughly and bring slowly to boil, stirring frequently until the mixture has reached 180°F. Simmer slowly without stirring for 1 hour. Cool and drain off, strain through cheesecloth.

Kitchen yields: 1 gal (3.8 l).

BEEF POT ROAST

For best cuts and sizes, *see* Beef, Pot Roast in the Meats and Meat Products section.

Formula: 10 lb (4.5 kg) meat

- 1 lb (450 g) mirepoix
- 2 cups (0.47 l) tomato purée
- 2 to 2¼ qt (1.9 to 2.1 l) brown stock cooking liquid
- 4 oz (112 g) corn starch

Serving size: 3 to 4 oz (85–112 g) cooked meat.

1½ fl. oz (0.04 l) to 2 fl. oz (0.06 l) sauce.

Kitchen yields: 10 lb (4.5 kg) trimmed meat = 30 servings.

Calories: 4 oz (112 g) meat and 1½ fl. oz gravy = 380 calories.

BEER BATTER

Basic batter that can be used for meat, fish and vegetables. *See also* Tempura Batter. There are many ready-to-use mixes on the market, which can be found in the Groceries section.

Formula:

- Six 12 fl. oz (2 l) cans beer
- 3 cups (0.7 l) warm water
- ½ cup (0.17 l) oil
- 8 cups (1.3 kg) flour
- 3 tb salt

Method: Combine ingredients and let rest 1 hour at kitchen temperature.

Kitchen yields: 1 gal (3.8 l).

BISCUITS

Note: Southern cooks use only soft cake flour for biscuits. For ready-to-use mixes, *see* the Groceries section.

Formula 1:

- 1½ lb (675 g) cake flour
- 1½ lb (675 g) pastry flour
- 4 oz (112 g) baking powder
- 6 oz (170 g) sugar
- 8 oz (225 g) shortening
- 1 oz (28 g) salt
- 5 eggs
- 3 cups (0.7 l) milk

Method: Combine ingredients to make soft dough.

Serving size: 1½ oz (42 g).

Kitchen yields: 55 pieces.

Formula 2:

4½ lb (2 kg) patent flour
8 oz (225 g) baking powder
8 oz (225 g) sugar
1½ lb (670 g) shortening
12 eggs
1 pt (0.47 l) milk

Serving size: 1 oz (28 g), 2 in (5 cm) diameter.

Kitchen yields: 125 pieces.

Calories: 1 oz (28 g) = 105 calories.

BLINTZES

Thin pancakes, baked on one side only and filled with cottage cheese or fruit, served with sour cream and fruit sauce. Used in Jewish cooking.

Formula:

14 oz (400 g), about 2½ cups, cake flour
10 eggs
5 cups (1.16 l) milk
¼ cup (0.06 l) oil
¼ tsp salt

Kitchen yields: Forty 6-in (150-mm) blintzes.

Formula for Cottage Cheese Filling:

5 lb (2.2 kg) cottage cheese, medium curd
8 egg yolks
1 lb (450 g) sugar
4 grated lemon rinds

Serving size: 1 oz (28 g).

Kitchen yields: 50 blintzes.

BLUE CHEESE STUFFING**Formula:**

10 lb (4.5 kg) blue cheese, crumbled
5 lb (2.25 kg) cream cheese
1 lb (0.4 kg) butter
1 pint (0.47 l) heavy cream, whipped

Kitchen yields: 2 gal (7.6 l).

BREAD BASKET DOUGH

Baskets can be made with noodle dough. See under Noodle Dough.

Formula: 2½ lb (1.1 kg) bread flour

1 egg
1 oz (28 g) shortening
1 oz (28 g) oil
2 cups (0.47 l) cold water

Note: Baskets for display are dried rather than baked.

BREAD DOUGH FOR BASIC HARD ROLLS**Formula:** 5 lb (2.25 kg) bread flour

2 oz (56 g) salt
2 oz (56 g) sugar
2 oz (56 g) shortening
2 oz (56 g) egg whites
3 lb (1.3 kg) water
3 oz (85 g) yeast

Baking instructions: 400°F, steam for the first 10 minutes.

Serving sizes: Rolls scale at 1.3 oz (35 g). French flutes scale at 5.3 oz (150 g).

Kitchen yields: 1 lb (450 g) dough = 1 dozen rolls.

Total mix = 8½ dozen rolls.

1 lb (450 g) dough = 3 French flutes.

Calories: 1 roll = 112 calories.

BREAD DOUGH SHRINKAGE

Average shrinking loss of yeast dough in baking:

Kitchen yields: 18 oz (500 g) dough = 1 lb (450 g) baked bread.

BREAD PROOFING TEMPERATURE CHART

Sweet dough = 95 to 98°F, humidity 80 percent.

Bread dough = 80 to 82°F, humidity 80 percent.

Note: Yeast dies at 138°F.

BRINE**Formula for Meat Brine:**

14 oz (400 g) salt
4 oz sugar (112 g)
1½ oz (63 g) salt peter
2 oz (56 g) pickling spice
1 gal (3.8 l) water

Formula for Fish Brine:

2½ gal (9.5 l) water

1 cup (225 g) salt

¼ cup, 2 oz (56 g) sugar

Note: Soak thin fillets 30 minutes, thick pieces overnight. Air dry in refrigerator before smoking.

CAKE SCALING GUIDE

The following information helps to calculate yields:

SPONGE CAKES

<i>Size of Cake</i>	<i>Scaling Weights</i>
6 in (150 mm) round	10 oz (280 g)
7 in (175 mm) round	14 oz (340 g)
8 in (200 mm) round	22 oz (600 g)
10 in (250 mm) round	28 oz (780 g)
12 in (300 mm) round	42 oz (1.18 kg)
14 in (350 mm) round	64 oz (1.8 kg)
18 × 24 in (450 × 600 mm) sheet	85 oz (2.3 kg)

ANGEL FOOD CAKE

<i>Size of Cake</i>	<i>Scaling Weights</i>
8 in (200 mm) round	13 oz (365 g)
9 in (224 mm) tube pan	28 oz (780 g)

CARAMEL FOR DIPPING FRUITS**Formula:**

2 lbs (900 g) sugar

10 oz (270 g) glucose

1½ cups (0.33 l) water

Method: Boil to 310°F.

Kitchen yields: Approximately 50 orange sections or 40 large strawberries.

CASINO BUTTER**Formula:**

½ cup (0.12 l) chopped garlic

½ cup (0.12 l) chopped shallots

2 cups (0.47 l) diced green pepper

2 cups (0.47 l) diced red pimentos

1 cup oil

10 lb (4.5 kg) butter

Salt and pepper to taste

Method: Sauté all ingredients except butter. Cool, carefully blend with butter in mixer, using paddle. Do not mash diced vegetables.

Kitchen yields: 1½ gal (5.7 l).

Total mix = 450 Littleneck Clams Casino.

CHEESE CAKE**Formula:**

12 lb (5.4 kg) bakers cheese

3 lb (1.4 kg) cream cheese

7¾ lb (3.5 kg) sugar

3 lb (1.4 kg) softened butter

2¼ qt (2.1 l) egg yolks

1½ lb (675 g) patent flour

12 oz (340 g) corn starch

8½ qt (8 l) sour cream

Method: Thoroughly blend all ingredients together. Whip and fold in:

2¼ qt (2.1 l) egg whites

1¼ lb (560 g) sugar

Scaling Sizes:

2 lb (900 g) for 8-in (200-mm) molds.

3 lb (1.3 kg) for 10-in (254-mm) molds.

Kitchen yields: Twenty-three 8-in cakes. Fifteen 10-in cakes.

CLARIFICATION MEAT

When butchering is done in-house, the lean trimmings from tenderloins and other cuts can be used. They should be ground through the largest-diameter plate of a meat grinder. If meat for clarification is purchased, it should be as lean as possible. Shin meat is best for this purpose. See the formula for Beef Consommé under Soups.

COCKTAIL SAUCE

See Tomato Products in the Groceries section for ready-to-use Cocktail Sauce.

One-quarter #10 can tomato purée
One #10 can chili sauce
One #10 can ketchup
1 pt (0.24 l) horseradish, grated
2 lemons, juice
2 oz salt/tabasco sauce
8 oz sugar

Kitchen yields: 8 qt (7.4 l) or about 120 servings.

Note: For European-style cocktail sauce, add 1 cup brandy and 1 qt whipped cream.

Serving size: 2 oz for shrimp cocktail.

COOKIES

The weight of butter cookies varies greatly. The count is usually 25 to 35 per pound, baked. For Frozen Cookie Dough, see the Groceries section.

Serving size: 10 oz (228 g) cookies for each table of 10 for banquets.

BUTTER AND SHORTENING COOKIES

Formula:

3 lb (1.3 kg) butter
3 lb (1.3 kg) shortening
4½ lb (2 kg) sugar
1½ qt (0.7 l) heavy cream
4½ lb (2 kg) patent flour
5 lb (2.25 kg) cake flour.
Salt, vanilla, and lemon flavor to taste.

Size: 1 oz (28 g) each.

Kitchen yields: 360 cookies.

BUTTER COOKIES

Formula:

6½ lb (3 kg) butter
4½ lb (2 kg) sugar
1½ qt (0.7 l) heavy cream
4½ lb (2 kg) patent flour
4½ lb (2 kg) cake flour
Salt, vanilla, or lemon to taste

Kitchen yields: 350 pieces.

CORN BREAD

Formula:

2½ lb (1.1 kg) corn meal
1 lb (16 oz) flour

2 oz (56 g) baking powder
2 oz (56 g) salt
2 qt (0.97) milk
1 pt (0.24 l) whole eggs
1 lb (450 g) melted butter

Method: Combine ingredients to smooth batter.

Kitchen yields: 1 sheet pan, 18 × 24 in.

CORN MUFFINS

See the Groceries section for ready-to-use Muffin Mixes and Batters.

Formula:

1 lb (450 g) sugar
8 oz (225 g) butter
3 eggs
3 cups (0.29 l) milk
1 lb (450 g) corn meal
1½ lb (680 g) cake flour
1 oz (28 g) salt
1½ oz (42 g) baking powder

Serving size: 2 oz (56 g) batter, use #16 scoop.

Kitchen yields: 48 muffins.

CORN STARCH, THICKENING POWER EQUIVALENT

1 cup corn starch = 2 cups flour.

COUSCOUS

For instant Couscous Mix, see Middle Eastern/North African Foods in the International Ingredients section.

Formula:

1¼ cup (250 g) couscous
2 oz (56 g) fat
2⅞ cups (0.55 l) boiling water or stock

Method: Sauté couscous in fat, mix with boiling liquid, let stand 10 minutes; bake in oven or steam 10 minutes.

Serving size: ½ cup (0.12 l).

Kitchen yields: 3¾ cups cooked, or about 7 servings.

CREAM PUFF PASTE

See under Pâte à Choux.

CRÊPES (THIN PANCAKES)**Formula:**

2 lb (900 g) cake flour
10 eggs
1 qt (0.9 l) milk
1 qt (0.9 l) light cream

Method: Blend ingredients, strain and let rest 15 minutes to allow air bubbles to dissolve.

Kitchen yields: 50 crêpes, 4 in (100 mm) across.

CUSTARD FOR DESSERTS**Formula:**

1 qt (0.94 l) milk
8 oz (225 g) sugar
7 or 8 whole eggs
Flavor according to taste

Method: Warm milk, add to remaining ingredients. Blend but do not whip. Strain into molds and poach in water bath at low temperature until set. The custard will be softer if light cream is used.

Serving size: Depends on mold size; frequently 4½ oz (120 g).

Kitchen yields: 10 servings.

Calories: 4½ oz (120 g) (when made with milk) = 185 calories.

CUSTARD FOR QUICHE

See under Quiche.

CUSTARD SAUCE**Formula:**

1 qt (0.9 l) milk
1 tb corn starch
6 oz (170 g) sugar
7 egg yolks
Flavoring according to taste

Method: Heat milk; combine remaining ingredients. Stir in hot milk and heat until thick. The sauce will curdle if boiled.

Serving size: 2 oz (0.05 l), sauce for dessert.

Kitchen yields: 18 to 20 servings.

DANISH PASTRY CHEESE FILLING**Formula:**

16 lb (7.2 kg.) bakers cheese
6 lb (2.7 kg) sugar
32 eggs
1 lb (0.450 g) cake flour
Lemon or vanilla flavor to taste

Serving size: 1½ oz (42 g) for large Danish.

Kitchen yields: 260 pieces large Danish.

DANISH PASTRY DOUGH**Formula:**

2 lb (900 g) sugar
4 qt (3.8 l) milk
2 lb (900 g) yeast
4 qt (3.8 l) whole eggs
4 lb (1.8 kg) butter
20 lb (9 kg) patent flour
4 lb (1.8 kg) cake flour
¼ cup (50 g) salt
Lemon flavor, nutmeg to taste
12 lb (5.4 kg) sweet butter, folded in

Method: Combine ingredients (except butter) and make yeast dough. Proof and then chill and let rest. Fold in butter.

Sizes:

Scale, 1½ oz (40 g) each before filling for miniature Danish.

Scale, 2 oz (55 g) before filling for breakfast Danish for baskets.

Scale, 3½ oz (100 g) before filling for large Danish.

Kitchen yields: 50 dozen (600 pieces) miniature Danish.

37 dozen (440 pieces) breakfast Danish for baskets.

22 dozen (260 pieces) large Danish.

DRESSINGS

Many operations purchase salad dressings ready for use or add additional ingredients to purchased dressings. For Salad Dressings, see the Groceries section.

FRENCH DRESSING**Formula:**

1 gal (3.8 l) oil, preferably olive oil
 ½ gal (1.8 l) cider vinegar
 ½ gal (1.8 l) wine vinegar
 1 cup (0.24 l) prepared mustard
 1 cup (0.24 l) dry mustard
 1 cup (0.24 l) sugar
 ½ cup (0.12 l) crushed peppercorns
 ½ cup (0.12 l) salt

Serving size: 1 oz.

Kitchen yields: 2 gal (7.6 l).
 Total mix = 250 servings.

GREEN GODDESS DRESSING**Formula:**

1 qt (0.9 l) chopped parsley
 1 qt (0.9 l) chopped, uncooked spinach
 1 pint (0.47 l) chopped chives
 1 can: 28 oz (800 g), anchovy fillets with oil
 ¼ cup chopped garlic
 1 gal (3.8 l) mayonnaise
 ½ gal (1.9 l) sour cream
 ¼ cup lemon juice
 1 cup (0.24 l) tarragon vinegar

Method: Purée in food processor: chopped parsley, chopped, uncooked spinach, chopped chives, anchovy fillets with oil, and ¼ cup chopped garlic. Blend mayonnaise, sour cream, lemon juice, and tarragon vinegar. Combine with vegetable purée as needed. Dressing will lose color when stored too long.

Serving size: 2 oz (56 g).

Kitchen yields: 1¾ gal (6.6 l).
 Total mix = 110 servings.

RUSSIAN DRESSING**Formula:**

3 qt (2.7 l) mayonnaise
 1 qt + 1 cup (1.1 l) cocktail sauce
 1 pint (0.47 l) cooked chopped eggs
 1 cup (0.24 l) lemon juice

Serving size: 1½ oz (42 g).

Kitchen yields: 1 gal (3.8 l).
 Total mix = 85 servings.

THOUSAND ISLAND DRESSING**Formula:**

1 gal (3.8 l) mayonnaise
 ½ gal (1.9 l) cocktail sauce
 ½ gal (1.9 l) whipped cream
 1 qt (0.9 l) chopped dill relish
 1 qt (0.9 l) cooked, chopped beets
 1 cup (0.24 l) lemon juice

Serving size: 1½ oz (42 g).

Kitchen yields: 2½ gal (9.5 l).
 Total mix = 210 servings.

VINAIGRETTE DRESSING**Formula:**

1 qt (0.9 l) dry white wine
 1 pint (0.47 l) wine vinegar
 2 qt (1.8 l) olive oil
 1 tb chopped garlic
 2 tb thyme
 1 cup (0.24 l) lemon juice
 1 cup (0.24 l) chopped onions
 ½ cup (0.12 l) prepared mustard
 3 tb crushed peppercorns

Serving size: 1 oz (28 g).

Kitchen yields: 1 gal (3.8 l).
 Total mix = 128 servings.

DUMPLINGS, BAKING POWDER**Formula:**

1 lb (450 g) cake flour
 4 tsp baking powder
 1 tsp salt
 14 oz (0.4 l) milk

Method: Blend ingredients with paddle in machine to a soft dough. Portion with a scoop or spoon.

Serving size: 1 oz (56 g).

Kitchen yields: 30 dumplings.

FILO (FILLO) DOUGH or STRUDEL DOUGH

Most operations purchase ready-to-use filo dough. For information, see the Groceries section.

Formula:

1½ lb (0.675 g) bread flour
 1½ oz (42 g) oil

2 eggs
1½ cups (0.35 l) warm water

Method: Combine ingredients to rather soft dough, which must be worked thoroughly to develop the gluten. Dough should rest, wrapped in plastic, before use. It can be stretched on a floured tablecloth.

FLOUR

Converting Flour Volume to Weight

Volume	Weight Unsifted	Weight Sifted
1 cup	6 oz (170 g)	4¼ oz (120 g)
1 pint	12 oz (340 g)	10 oz (285 g)
1 qt	1½ lb (675 g)	1 lb 2 oz (510 g)
1 gal	6 lb (2.7 kg)	4½ lb (2 kg)

The chart is based on bread flour; cake flour is slightly lighter.

FRUIT PARFAIT

Formula:

5 egg whites
12 oz (440 g) sugar
½ cup (0.12 l) water
20 oz (560 g) fruit purée or fruit juice
Lemon juice according to taste
1 qt (0.9 l) heavy cream (36 percent)

Method: Beat egg whites with 2 oz sugar until stiff. Combine remaining sugar with water and boil to 200°F. Add boiled sugar to egg whites in steady stream, beating continuously. Beat slowly until cool, add fruit purée. Beat heavy cream until thick and blend into cooled mixture. Put in suitable molds and freeze until solid.

Kitchen yields: 3 qt, frozen.

GELATIN STRENGTH

Note: The strength of gelatin is influenced by the acidity of the product and by the weather. Highly acidic stocks require more gelatin. Pineapple juice will not jell.

Amount needed per gal (38 l): 3 oz (56 g), very soft.

6 oz (170 g), sliceable.
10 oz (280 g), for show platters.

GRAVELAX—SCANDINAVIAN CURED SALMON

The classic gravelax recipe has now been modified to include many additional flavor components.

Formula:

2 cups salt
1 cup sugar
¼ cup crushed juniper berries
¼ cup crushed peppercorns
2 cups coarsely chopped dill, including stems

Method: Rub mixture over salmon sides, wrap in plastic and refrigerate. Turn over occasionally. The salmon should be cured in 3 days.

Kitchen yields: Mixture is for 2 fillets, skin on, 4 lb (1.8 kg) each.

HIPPENMASSE (ICE CREAM WAFERS)

Formula:

18 oz (500 g) sugar
4½ oz (125 g) almond paste
7 whole eggs
1½ oz (42 g) cream
14½ oz (400 g) all-purpose flour

Method: Combine sugar, almond paste, cream, and two eggs. Work until smooth. Add remaining ingredients, alternating between flour and eggs. Let rest 1 hour. The batter should be spreadable. Spread on waxed cookie sheet, using a stencil to get even servings. Remove while still hot and shape over suitable mold.

Kitchen yields: 120 rolled wafers, about 3 × 4 in, rolled in cigarette shape.

KASHA

Formula:

1 lb (450 g) kasha (buckwheat grits)
2 whole eggs
½ cup shortening
4 cups water or stock

Method: Break eggs and blend. Mix eggs with dry kasha until all granules are evenly coated. Melt shortening in hot water. Bring mixture to boil, add kasha, and simmer or bake about 40 minutes.

Kitchen yields: 10 servings.

KEY LIME PIE

Formula:

- 1 can, 14 oz (0.4 l), sweetened condensed milk
- 4 egg yolks
- ½ cup (0.12 l) lime juice
- 4 egg whites
- 4 oz sugar
- One 9-in prebaked pie crust

Method: Blend milk and egg yolks, stir in lime juice. Fill pie shell with mixture. Make meringue with egg whites and sugar and spread over pie. Bake at low heat until brown.

Kitchen yields: One 9-in pie = 8 servings.

LATKES (POTATO PANCAKES)

Latkes are available frozen and can also be made from ready-to-use mix. See the Groceries section for information.

Formula:

- 10 lb (4.5 kg) peeled russet potatoes
- 3 lb (900 g) peeled onion
- 1 bunch parsley, well washed
- 10 eggs
- 1 cup (0.2 l) matzo meal
- 1 cup (0.2 l) flour
- 1 tb salt

Method: Grind potatoes, onions, and parsley through fine plate of meat grinder, drain in colander until most liquid has drained off. Add remaining ingredients. Adding antioxidation powder can prevent oxidation of ground potatoes. Omit parsley if latkes are to be served with applesauce.

Serving size: 2 pieces for reception.
3 pieces as garnish with meat.

Kitchen yields: 100 pieces.

LAVASH

Formula:

- 17 lb (7.65 kg) all-purpose flour
- 8 oz (225 g) granulated sugar
- 8 oz (225 g) salt
- 1 oz (28 g) cayenne pepper
- 3 lb (1.25 kg) butter, soft
- 4 oz (112 g) fresh yeast
- 1 gal (3.8 l) warm water

Method: Combine all ingredients with dough hook. Chill. Divide dough into 8 pieces and press into desired garnish, such as poppy seeds, sesame seeds, dried herbs, spices, or parmesan cheese. Roll out each piece to about ¼ in (6 mm) thick. Place on paper-covered sheet pan and cut into serving pieces. Bake at 400°F about 10 minutes or until brown and crisp.

Kitchen yields: 600 pieces.

LIAISON, BASIC FORMULA

Liaison is added to cream soups and sauces to provide texture. It also adds calories and is seldom used today.

Basic Formula: 1 pt (0.48 l) cream
6 egg yolks

Kitchen yields: Use this mixture for 1 gal (3.8 l) cream soup.

MATZO BALLS

Formula: 1 qt (0.94 l) matzo meal
10 oz (0.3 l) whole eggs
6 oz (0.17 l) melted chicken fat
Salt and pepper to taste

Method: Blend ingredients and let mixture rest at least 1 hour to solidify. Make sample dumpling before using the whole mix. The consistency of matzo meal varies.

Serving sizes: ½ oz for soup.

Kitchen yields: 80 matzo balls.

MAYONNAISE, HOMEMADE

Few operations make mayonnaise on-site, and most customers prefer

purchased mayonnaise because it is lighter. See the Groceries section for Mayonnaise.

Formula:

12 dozen egg yolks (4 layers)
or
5 pints (2.2 l) pasteurized egg yolks
3 cups (0.81 l) vinegar
1 cup (225 g) salt
1 cup (200 g) dry mustard
5 gal (19 l) oil
1 qt (0.9 l) hot water

Method: All ingredients should be at room temperature. Combine egg yolks with spices. Slowly add oil while blending at medium speed. If mayonnaise is getting too thick or oily add warm water in small increments.

Kitchen yields: 5 gal and 3 qt (21.7 l).

MEATBALLS

10 lb (4.5 kg) ground meat
1 lb (450 g) stale rolls, soaked
1 lb (450 g) onions
2 tb chopped garlic
6 eggs
1 bunch parsley
Spices

Serving size: 1 oz (28 g) for hors d'œuvres.

2½ oz (70 g), use scoop #12, for main courses.

3 meatballs each serving for spaghetti and meat balls.

Kitchen yields: 200 meatballs, 1 oz (28 g) each, for hors d'œuvres.

75 meatballs, 2½ oz (70 g) each.

25 servings spaghetti and meat balls.

MEAT LOAF

See Hamburger Meat in the Meats and Meat Products section for information. A cheaper grade of beef or a mixture of pork and beef is often used for meat loaf.

Formula:

10 lb (4.5 kg) hamburger meat
1 lb (450 g) onions

2 lb (900 g) stale rolls, soaked and squeezed

2 tb chopped garlic

6 eggs

1 bunch parsley, well washed

Spices and herbs to taste

Method: Grind all ingredients and blend well in mixer with paddle.

Serving size: 4 oz (112 g), cooked, 2 slices.

Kitchen yields:

Four 3-lb (1.3-kg) loaves.

1 loaf = 20 slices, 10 servings.

4 loaves = 40 servings.

Calories: 4 oz (112 g), cooked, 2 slices = 270 calories.

MERINGUE

Formula: 1 lb (450 g) egg whites

2 lb (900 g) sugar

Method: Beat egg whites, add sugar and beat until sugar is well incorporated.

MIREPOIX

Mixed vegetables used for flavoring stewed meats, sauces, and gravies.

Formula:

50 percent onions, coarse cut

25 percent carrots, coarse cut

25 percent celery, coarse cut

Crushed peppercorns and bay leaves
optional

Herbs optional

Kitchen yields: Vary according to application.

MOUSSE

Mousse is the name for a wide variety of dishes, which can be served hot or cold. The common denominator is that the main ingredients are finely ground and smooth.

BASIC COLD MOUSSE (NOT SWEET)**Formula:**

5 lb (2.2 kg) cooked meat, fish, or shellfish

- 1 qt (0.94 l) cold cream sauce or mayonnaise
- 6 cups aspic jelly with the same flavor as the main ingredient
- 6 cups 36 percent heavy cream, whipped

Serving size: 3 oz (85 g), appetizer.

Kitchen yields: Total mixture = 65 servings.

BASIC COLD SWEET MOUSSE

Formula:

- 2 gal (7.6 l) milk
- 48 egg yolks
- 1½ lb (675 g) corn starch
- 3 lb (1.3 kg) sugar
- 1 qt (0.94 l) suitable concentrated fruit flavor
- 12 oz (340 g) gelatin
- 48 egg whites
- 1½ lb (675 g) sugar
- 2 gal (7.6 l) heavy cream

Method: Make pastry cream with first 6 ingredients. Cool. Beat egg whites with sugar to make meringue. Fold into pastry cream. Whip heavy cream and fold into almost cold mousse. Portion into suitable molds or glasses.

Serving size: ¾ cup (0.17 l).

Kitchen yields: Mixture yields 6 gal (22.8 l) = 125 servings.

CHOCOLATE MOUSSE

Formula 1 (Basic):

- 8 lb (3.6 kg) semisweet chocolate.
- 3 lb (1.3 kg) sugar
- 1 pint (0.47 l) water
- 1½ qt (0.7 l) egg whites
- 5 qt (2.3 l) heavy cream

Method: Melt chocolate and cool to 80°F. Boil 3 lb (1.3 kg) sugar with water to 235°F. Whip 1½ qt (0.7 l) egg whites and pour hot sugar into egg whites to make Italian meringue. Whip 5 qt (2.3 l) heavy cream. Fold melted chocolate and heavy cream into

meringue. Mousse can be served frozen.

Serving size: 5 oz (0.14 l).

Kitchen yields: Total mix = 120 servings.

Formula 2:

- 5 lb (2.25 kg) sugar
- 1 pt (0.47 l) water
- 6 oz (170 g) gelatin
- 1½ cups (0.35 l) water
- 2 qt (1.8 l) pasteurized egg yolks
- 4 lb (1.8 kg) sweet butter
- 6 lb (2.70 kg) sweet chocolate
- 2 lb (900 g) bitter chocolate
- 3 qt (1.4 l) heavy cream

Method: Combine sugar with water and boil to 235°F (155°C). Melt 6 oz (170 g) gelatin in 1½ cup (0.35 l) water. Cream egg yolks with 4 lb (1.80 kg) sweet butter, add cooled sugar, whip until foamy. Melt sweet and bitter chocolate to 86°F (30°C). Stir (temper) chocolate until cool. Whip heavy cream. Blend gelatin with egg mixture, fold in melted chocolate and cream.

Kitchen yields: 100 servings, 6 oz (170 g) each.

FISH MOUSSE, HOT

Formula: 8 lb (3.6 kg) solid fish, such as halibut, sole, pike, or salmon, completely boneless

- 20 egg whites
- 3½ qt. (3.1 l) heavy cream, 36 percent
- Salt and pepper to taste

Note: The fish should be fresh; if frozen fish is used, the amount of egg whites should be increased. It is advisable to mix frozen fish with about 30 percent fresh scallops to improve the binding of the mixture. Make sure all ingredients are very cold. **Note:** Some people are allergic to bivalves

Method: Chill, almost freeze, fish. Grind twice through fine plate of a meat grinder. Blend in egg whites and cream in a food mixer. Always make

a sample; if the mousse is too stiff, add more cream; if too loose, add egg whites. Fill molds with mousse and steam.

Serving size: 5 oz, appetizer.
7 oz, main course.

Kitchen yields: Appetizer = 50 servings.
Main course = 40 servings.

FROZEN SWEET MOUSSE

Formula:

4 lb (1.8 kg) sugar
5 cups (1.1 l) water
32 egg yolks
1 qt (0.94 l) heavy cream
8 oz (0.24 l) liquor, such as Grand Marnier, or other flavoring

Method: Boil sugar and water about 10 minutes to make syrup. Add hot to egg yolks and beat over water bath until thick and frothy, but be careful not to make the mixture too hot because it will curdle. Whip until cold in mixer. Whip cream. Fold cream and flavoring into mousse. Portion into glasses.

Kitchen yields: 1 gal (3.8 l).

NOODLE DOUGH

Formula:

5 lb (2.2 kg) bread flour
16 eggs
 $\frac{1}{2}$ cup water
1 oz (28 g) oil

Method: Make stiff dough in mixer with dough hook.

Serving sizes: Main course: $3\frac{1}{2}$ oz (98 g), which yields 1 cup (0.24 l) cooked.
Garnish: $1\frac{1}{2}$ oz (42 g), which yields $\frac{1}{3}$ cup cooked.

Kitchen yields: 7 lb (3.1 kg) fresh noodles.
Mixture yields 30 servings, main course.
Mixture yields 70 servings, garnish.

PASTILLAGE

Inedible sugar dough for decorations. The dough can be tinted and must be dried.

Formula:

$1\frac{1}{2}$ oz (42 g) plain gelatin
6 lb (2.7 kg) confectioners sugar
4 oz (112 g) cornstarch
1 pt (0.47 l) warm water

Kitchen yields: 7 lb (3.1 kg), enough to make 4 simple cookie stands.

PASTRY CREAM

Formula:

$7\frac{1}{2}$ qt. (7 l) milk
 $3\frac{1}{2}$ lbs (2.5 kg), or 7 cups, sugar
36 whole eggs (1 qt and 3 cups)
1 lb (450 g) corn starch

Kitchen yields: 11 qt (10 l) = 44 cups.

PÂTE À CHOUX (Cream Puff Paste)

Formula:

1 qt (0.94 l) milk
1 lb (450 g) butter
 $1\frac{1}{4}$ lb (560 g) patent flour
 $\frac{1}{4}$ oz (7 g) salt
1 qt (32 oz or 0.9 l) or 20 eggs

Kitchen yields: $3\frac{1}{2}$ qt (3.2 l).

PÂTÉ DOUGH

Formula:

3 lb 5 oz (1.5 kg) bread flour.
1 lb (450 g) shortening.
1 pint (0.47 l) warm water.
1 oz (28 g) salt.

Kitchen yields: 5 lb 5 oz (2.4 kg) dough.

PIZZA, HOMEMADE INGREDIENTS

For purchased Pizza Components, see the Groceries section.

PIZZA DOUGH

Formula:

50 lb (22.5 kg) high-gluten flour
2 gal and 3 pt (9 l) water
1 cup, 8 oz (225 g) oil
1 cup, 8 oz (225 g) salt
 $\frac{1}{2}$ cup (112 g) sugar
8 oz (225 g) fresh yeast

Method: Combine ingredients in mixer with dough hook. Do not proof, but scale immediately. After scaling, the dough must be tightly rolled into firm balls.

Kitchen yields: 70 large 16-in (40-cm) pies, scaled at 16 oz (450 g).
110 medium 9-in (23-cm) pies, scaled at 10 oz (284 g).
400 hors d'oeuvre pies, scaled at 1¾ oz (78 g).

PIZZA SAUCE

Formula:

Five #10 cans tomato sauce
Three #10 cans crushed tomatoes
One #10 can tomato ketchup
½ cup (0.12 l) oregano

Kitchen yields: 75 pies, 16 in (400 mm) across.

POLENTA

Formula:

Polenta is yellow corn meal mush, sometimes flavored with grated Parmesan cheese. There is a basic distinction between soft polenta, with a consistency of soft mashed potatoes, and stiff polenta, which is spread on sheet pans and can be cut into pieces when cold. Ready-to-use stiff polenta is available. See Corn and Corn Products in the Groceries section for information.

Formula for Stiff Polenta:

3 cups (0.7 l) coarse corn meal
8 cups (2 l) chicken stock
4 oz (112 g) butter
¼ cup (0.05 l) heavy cream
Salt, nutmeg, and grated cheese to taste

Serving size: ½ cup (0.12 l).

Kitchen yields: Total mixture = 12 cups or 24 servings.

Formula for Soft Polenta:

2 cups (0.48 l) coarse corn meal
8 cups (2 l) chicken stock
4 oz (112 g) butter

¼ cup (0.05 l) heavy cream
Salt, nutmeg, and grated cheese to taste

Serving size: ½ cup (0.12 l).

Kitchen yields: Total mixture = 12 cups or 24 servings.

POPOVERS

Formula:

24 whole eggs
3 lb (1.3 kg) bread flour
2½ qt (2.25 l) milk
1 oz (56 g) salt
Nutmeg to taste

Method: Combine ingredients to make a batter resembling pancake batter. Fill muffin tins with rendered beef drippings or oil, about ¼ full. Place muffin tins on sheet pan and heat until smoking hot. Ladle batter into tins and bake at 425°F. Popovers will puff up.

Kitchen yields: 60 pieces.

POTATO GNOCCHI, BASIC

Formula:

5 lb (4.5 kg) russet potatoes, peeled and steamed
2 lb (900 g) bread flour
2 eggs
Salt, pepper, and nutmeg to taste

Method: Put steamed potatoes through ricer while still hot. When potatoes have cooled, work in remaining ingredients. Make small sample to adjust amount of flour if necessary.
Note: Potato dough can be flavored with grated cheese, herbs, or spinach purée. Ricotta cheese is also used in some recipes.

Kitchen yields: According to use.

QUICHE

The basic quiche is filled with cheese, ham or cooked bacon, chives, and custard. Many new versions have been made by varying the fillings.

Dough Formula:

3 lb (1.4 kg) pastry flour
 2 lb (900 g) shortening
 14 oz (0.40 l) cold water
 2 oz sugar (56 g) sugar
 1 oz salt (28 g)
 Prebake shells at 375°F 15 minutes.

Kitchen yields: 12 crusts, 9 in (228 mm) across.

Garnish Formula:

¼ cup chopped chives
 8 cups (0.6 kg) diced Swiss cheese
 4 lb (1.8 kg) bacon, cooked and diced,
 or 2 lb (900 g) cooked ham

Kitchen yields: Twelve 9-in (228-mm) quiches.

Custard Formula:

6 qt (5.6 l) light cream, slightly warmed
 36 whole eggs
 3 tb salt
 1 tb white pepper
 1 tsp nutmeg

Note: Do not overmix; there should be no air bubbles in the mix. Bake at 350°F until set.

Kitchen yields: Twelve 9-in (228-mm) quiches.

REFRIED BEANS**Formula:**

1 lb (450 g) pinto beans or red kidney beans
 5 cups (1.25 l) water
 1 cup (0.24 l) diced onions
 ½ cup (0.12 l) bacon drippings

Serving size: ¾ cup (0.19 l).

Kitchen yields: 5½ cups (1.3 l) = 7 servings.

RICE, LONG GRAIN**Formula:**

Rice	Water	Fat	Salt	Yield
1 lb	5 cups	2 oz	1 tb	6 cups
5 lb	6½ qt	10 oz	½ cup	30 cups
10 lb	13 qt	1 lb 4 oz	1 cup	3¾ gal

RISOTTO**Formula:**

1 pt (0.47 l) Arborio rice
 1 cup (0.24 l) chopped onion
 1 cup (0.24 l) melted butter
 1½ (1.35 l) qt chicken stock or other stock
 1 cup (0.24 l) Parmesan cheese
 4 oz (112 g) fresh butter

Serving size: 1 cup (0.24 l.), as appetizer.

Kitchen yields: Total recipe = 6 servings.

ROUX

Roux is little used in kitchens today because most sauces are based on reductions and thickened when needed with corn starch. Roux is classified by color as white, blond, or brown. Brown roux is seldom used today, except in Cajun dishes.

Basic Formula:

1 lb (450 g) fat, such as oil or clarified butter
 18 oz (500 g) bread flour

Kitchen yields:

White roux: 1 gal (3.8 l) liquid = 12 oz (335 g) for soup.

= 14 oz (400 g) for medium-thick cream sauces.

= 18 oz (500 g) for very thick sauces.

Brown roux: Roux loses thickening power through roasting. Brown roux has approximately 25 percent less thickening power than white roux.

ROYAL ICING**Formula:**

4 lb (1.8 kg) confectioners sugar
 ¾ lb (340 g) egg whites
 ½ tsp cream of tartar

Note: Icing gets hard and brittle when dry. Use for decorative cakes only.

Kitchen yields: Recipe is for one 20-in wedding cake dummy.

SALT DOUGH

Inedible salt dough is used for decoration and can be painted. When dry and stored in a dry environment, it will not spoil.

Formula 1:

- 2 cups (140 g) flour
- 1 cup (225 g) salt
- 1 cup (235 l) water

Dough can be dried or baked at a low temperature.

Formula 2:

- 1 pint (0.47 l) water
- 12 oz (340 g) corn starch
- ½ oz (14 g) gelatin
- 1½ oz (3 tb) water
- 2 lb (900 g) extra-fine salt

Method: Boil starch with 1 pint water, add dissolved gelatin and remaining ingredients. Dough will dry well.

SANDWICH FILLING QUANTITIES

Butter or margarine: 1 lb (450 g) softened product is sufficient to spread 70 slices of bread #2.

Cheeses, sliced: 2 to 3 oz (56 to 85 g) per serving.

Meats: 2 to 4 oz (56 to 112 g) per serving.

Kitchen yields: 1 lb (450 g) canned ham = 5 to 6 sandwiches.

1 lb (450 g) roast beef, cold = 5 sandwiches.

Note: Meat should be sliced very thin. Weight of meat varies according to moisture content.

Salad fillings:

For 2 oz (56 g), use #16 scoop.

For 2½ (70 g), use #12 scoop.

For 3½ oz (100 g), use #10 scoop.

For 4 oz (112 g), use #8 scoop.

For 30 tuna salad sandwiches, 2½ oz (70 g): 1 can tuna: 55 oz (1.5 kg), drained.

1 pint (0.47 l) mayonnaise.

1 pint (0.47 l) diced celery.

Use #12 scoop.

SAUCES

The following are basic sauce formulas for banquet service. For Sauce Bases, see the Groceries section.

BÉARNAISE SAUCE

Use the formula for Hollandaise Sauce and add ¼ cup (0.06 l) chopped tarragon and ¼ cup (0.06 l) chopped parsley. The vinegar listed in the recipe can be replaced by tarragon vinegar.

BONNE FEMME SAUCE FOR FISH**Formula:**

One 3-lb (1.3-kg) basket sliced mushrooms

1 cup (0.24 l) chopped shallots

½ lb (225 g) butter

1 pint (0.47 l) dry white wine

1 cup (0.24 l) cut chives

1 gal (3.8 l) heavy fish velouté

1 qt (0.9 l) hollandaise sauce

1 qt (0.9 l) heavy cream, whipped

Note: The hollandaise sauce and whipped cream are folded into the sauce at the last moment.

Kitchen yields: 1½ gal (5.7 l) = sauce for 100 portions of fish.

BROWN MUSHROOM SAUCE**Formula:**

One 3-lb (1.3-kg) basket sliced mushrooms

1 cup: 8 oz (0.24 l) chopped shallots

½ gal (1.9 l) white wine

1 gal (3.8 l) brown sauce

Kitchen yields: 1¼ gal (4.7 l).

CHICKEN VELOUTÉ**Formula:**

4 gal (15.2 l) strong chicken stock

1 gal (3.8 l) light cream

3 lb (1.4 kg) mushroom trimmings

3 qt (2.7 l) white roux

1 lb (450 g) cold, fresh butter

Note: The butter should be added at the last moment.

Kitchen yields: 5 gal (19 l).

CUMBERLAND SAUCE**Formula:**

Three #10 cans red currant jelly
3 qt (2.7 l) sweet port wine
3 cups (0.7 l) orange juice concentrate
1½ cups (0.3 l) lemon juice
1 cup (0.24 l) chopped shallots
3 tb dry mustard
¼ cup (0.06 l) chopped ginger
1 pint (0.47 l) water
Add for garnish: 3 cups (0.7 l) blanched
orange zest and 2 cups (0.47 l)
blanched lemon zest.

Kitchen yields: 5 gal (19 l).

FISH VELOUTÉ**Formula:**

3 gal (11.4 l) fresh fish stock
1 gal (3.8 l) chicken stock
1 bottle (0.750 l) dry white wine
3 lb (1.4 kg) mushroom trimmings
3 qt (2.7 l) white roux
1 gal (3.8 l) light cream
1 lb (450 g) fresh butter

Kitchen yields: 5 gal (19 l).

HOLLANDAISE SAUCE**Formula:**

9 lb (4 kg) butter = 3½ qt (3.2 l),
melted
24 egg yolks
1 cup water
¼ cup (0.06 l) vinegar
¼ cup (0.06 l) lemon juice
Salt and cayenne pepper to taste

Kitchen yields: 1 gal (3.8 l).

LEMON SAUCE FOR DUCKLING**Formula:**

3 cups (0.7 l) vinegar
3 cups (0.7 l) sugar
3 cups (0.7 l) frozen lemonade
concentrate
2 cups (0.46 l) red currant jelly
8 lemons, cut in quarters
2 qt (1.8 l) brown duckling stock
¾ cup (0.18 l) corn starch
1 cup (0.24 l) dry sherry wine

For garnish: 2 cups (0.47) lemon peel
julienne

Kitchen yields: 1 gal (3.8 l).

MINT SAUCE**Formula:**

1 gal (3.8 l) white vinegar
1½ qt (1.4 l) sugar
1½ qt (1.4 l) water
1 pint (0.4 l) mint jelly
2 qt (1.8 l) chopped mint leaves

Method: Combine all ingredients except
mint leaves. Bring to boil and pour
over mint leaves.

Kitchen yields: 1½ gal (5.7 l).

NEWBURGH SAUCE**Formula:**

1 pint (0.47 l) chopped onions
1¼ cup (0.3 l) Spanish paprika
2 lb (900 g) butter
½ gal (1.9 l) dry sherry
1 gal (3.8 l) light cream
3½ gal (13.3 l) heavy cream sauce

Kitchen yields: 5 gal (19 l).

ORANGE SAUCE FOR DUCKLING**Formula:**

3 cups (0.7 l) vinegar
3 cups (0.7 l) sugar
3 cups (0.7 l) orange juice concentrate
2 cups (0.46 l) red currant jelly
4 large oranges, cut in quarters
4 lemons, cut in quarters
2 qt (1.8 l) brown duckling stock
¾ cups (0.17 l) corn starch
1 cup (0.24 l) dry sherry wine
For garnish: 2 cups (0.47) blanched
julienne orange peel

Kitchen yields: 1 gal (3.8 l).

SAUCE VERTE (GREEN SAUCE)**Formula:**

Purée in food processor:
1 bunch watercress, stems discarded
4 bunches chives
½ bunch parsley, stems discarded
1 bunch chervil

1 cup (0.24 l) cooked spinach
 3 pieces anchovy fillets
 1 tb chopped garlic
 4 cups (0.9 l) mayonnaise

Kitchen yields: 3 pints (1.4 l).

TARTAR SAUCE GARNISH

Formula:

1 gal (3.8 l) coarse-cut onions
 1½ gal (5.7 l) dill pickle, drained
 ½ bunch parsley
 1 bunch watercress
 1 pint (0.47 l) capers, drained
 1½ tb chopped garlic
 Grind all ingredients through medium plate of food chopper.

Kitchen yields: Blend 1 pint (0.47 l) garnish with 3 qt (2.7 l) mayonnaise.

TONNATO SAUCE

Formula:

84 oz (2.25 kg) = two 42-oz cans tuna in oil
 1 cup (0.24 l) capers, drained
 24 pieces anchovy fillets
 2 tb grated lemon peel
 2 cups (0.47 l) olive oil
 3 pints (1.4 l) mayonnaise

Method: Combine all ingredients with food chopper.

Kitchen yields: 1 gal (3.8 l).

SAUCE QUANTITIES

Restaurant service: Serving size:

Brown sauces: 1½ oz (0.04 l).

Banquet service: Serving size:

Brown sauces:

100 covers = 5 qt (4.7 l).

200 covers = 2 gal (7.6 l).

Cream sauces:

100 covers = 1½ gal (5.7 l).

200 covers = 3 gal (11.4 l).

Jus for roast beef:

100 covers = 3 qt (2.7 l).

200 covers = 1½ gal (5.7 l).

SOUFFLÉ

Basic Dry Mix Formula:

2 lb (900 g) flour
 1 lb 12 oz (790 g) butter
 8 oz (225 g) sugar

Method: Combine ingredients like cookie dough, store under refrigeration.

Formula for 2 Servings:

2½ oz basic dry mix
 ½ pint (0.24 l) boiling milk
 4 egg yolks
 3 egg whites whipped with 2 oz (56 g) sugar
 ½ oz (0.14 l) cordial or other flavoring

Method: Add dry mix to boiling milk; boil until thick. Add egg yolks, whipped egg whites, and flavoring.

Kitchen yields: 32 single servings or 16 double servings.

Formula for 40 Servings:

6 qt (5.6 l) milk
 70 egg yolks
 54 egg whites
 2 lb 8 oz (1.1 kg) sugar
 10 oz (0.30 l) cordial or other flavoring

Kitchen yields: 40 servings.

Method: Proceed according to method for 2 servings.

SOUPS, BASIC INGREDIENTS

BEAN SOUP WITH TOMATO

20 lb (9 kg) navy beans
 12 gal (45.6 l) chicken stock
 One #10 can chopped tomatoes
 1½ gal (5.7 l) light cream

Kitchen yields: 10 gal (38 l).

BEEF CONSOMMÉ

10 lb (4.5 kg) clarification meat
 30 egg whites (1 qt = 0.9 l)
 One #10 can peeled tomatoes
 1 gal mirepoix (diced soup vegetables)
 12 gal (45.6 l) beef stock

Kitchen yields: 10 gal (38 l).

BLACK BEAN SOUP

15 lb (6.7 kg) black turtle beans

5 lb (2.2 kg) navy beans
12 gal (45.6 l) chicken stock
1½ gal (5.7 l) light cream
½ gal (1.9 l) dry sherry

Kitchen yields: 10 gal (38 l).

CHILLED CHERRY SOUP

6 gal (22.8 l) frozen pitted sour cherries
4 gal (15.2 l) water
2 lb (900 g) corn starch
2 lb (900 g) sugar
4 cinnamon sticks
Peel from 8 lemons
1 tb cayenne pepper

Kitchen yields: 10 gal (38 l).

CRAYFISH BISQUE

10 doz. crayfish, live
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream
½ gal (1.9 l) white wine
1 qt (0.9 l) cream sherry
1 qt (0.9 l) brandy

Kitchen yields: 10 gal (38 l).

CREAM OF ASPARAGUS

10 lb (4.5 kg) = 6 boxes frozen
asparagus bits and pieces
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

CREAM OF BROCCOLI

7½ lb (3.3 kg) = 3 boxes frozen
broccoli bits and stems
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

CREAM OF CAULIFLOWER

7½ lb (3.3 kg) = 3 boxes frozen
cauliflower
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

CREAM OF CHICKEN

¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream
1 gal (3.8 l) cooked, diced chicken as
garnish

Kitchen yields: 10 gal (38 l).

CREAM OF CORN

Two #10 cans creamed corn
One #10 can kernel corn, for garnish
½ gal (1.9 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

CREAM OF CURRY

½ lb (225 g) curry powder
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream
1 gal (3.8 l) diced fresh apples, added at
the last moment as garnish

Kitchen yields: 10 gal (38 l).

CREAM OF MUSHROOM

9 lb (4 kg) fresh mushrooms (three
baskets)
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

CREAM OF TOMATO

Three #10 cans tomato purée
¾ gal (2.8 l) roux
12 gal (45 l) strong chicken stock
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

CUCUMBER SOUP, CHILLED

25 lb (11.25 kg) russet potatoes
12 gal (45 l) strong chicken stock
20 lb (9 kg) cucumbers
2.5 gal (9.5 l) dry white wine
2.5 gal (9.5 l) heavy cream, 36 percent
butterfat

Note: Reserve some diced cucumbers for
garnish.

Kitchen yields: 10 gal (38 l).

FENNEL OR ANISE SOUP

2 gal (4.6 l) chopped fennel, no greens
 ½ gal (1.9 l) chopped fennel with greens
 ½ cup, 2 oz (56 g), fennel seed
 ¾ gal (2.8 l) roux
 12 gal (45 l) strong chicken stock
 1 gal (3.8 l) light cream
 ½ gal (1.9 l) diced fennel for garnish
 1 qt (0.9 l) chopped fennel green for garnish

Kitchen yields: 10 gal (38 l).

GAME CONSOMMÉ

15 lb (6.7 kg) browned venison or other game bones
 5 lb (2.2 kg) clarification meat
 15 egg whites (1 pint, 0.47 l)
 One #10 can peeled tomatoes
 1 gal mirepoix
 ¼ cup, 2 oz (56 g), crushed juniper berries
 12 gal beef stock
 1 qt (0.9 l) dry sherry
 1 qt (0.9 l) cream sherry

Kitchen yields: 10 gal (38 l).

GAZPACHO

10 lb (4.5 kg) onions
 10 lb (4.5 kg) seeded green peppers
 15 lb (6.7 kg) peeled, seeded cucumbers
 One #10 can peeled whole tomatoes
 5 lb (2.2 kg) celery stalks
 One #10 can tomato purée
 2 cans (46 oz, 2.6 l) tomato juice
 2 qt (1.8 l) vinegar
 1 qt (0.9 l) olive oil
 ½ cup crushed garlic
 24 whole eggs
 ½ cup salt
 ¼ cup ground pepper
 ½ cup sugar
 1 oz (28 g) dried tarragon

Note: Grind all ingredients through medium-fine plate of meat grinder. Store refrigerated at least 1 day before service. The traditional garnish consists of croutons (not toasted, in Spain), diced green

peppers, diced cucumbers, and diced peeled, seeded tomatoes.

Kitchen yields: 10 gal (38 l).

GIBLET AND BARLEY SOUP

2.5 gal (9.5 l) ground raw giblets
 ½ gal (1.9 l) oil
 5 lb (2.2 kg) pearl barley
 ½ gal (1.9 l) diced onions
 ½ gal (1.9 l) diced celery
 ½ gal (1.9 l) diced carrots
 ½ gal (1.9 l) diced turnips
 1 qt (0.9 l) brown roux
 12 gal (45.6 l) beef stock

Kitchen yields: 10 gal (38 l).

GUMBO

3½ gal (11.4 l) chicken stock
 3½ gal (11.4 l) beef stock
 2 qt (1.9 l) brown roux
 ½ gal (1.9 l) diced green peppers
 ½ gal (1.9 l) diced onions
 ½ gal (1.9 l) diced celery
 ½ gal (1.9 l) diced cabbage
 One #10 can chopped tomatoes
 10 lb (4.5 kg) = 2 boxes frozen diced okra

or

Two #10 cans diced okra
 1 gal (3.8 l) cooked rice
 5 lb (2.2 kg) diced ham
 ½ gal (1.9 kg) cooked chicken
 1 lb (450 g) filé powder

Kitchen yields: 10 gal (38 l).

JELLIED MADRILENE CONSOMMÉ

4 cans 46 oz (2.4 l) tomato juice
 Two #10 cans peeled tomatoes
 1 lb (450 g) plain gelatin
 ½ gal (1.9 l) chopped celery
 10 lb clarification meat
 30 egg whites (1 qt, 0.9 l)
 One #10 can peeled tomatoes
 8 gal (30.4 l) beef stock

Kitchen yields: 10 gal (38 l).

LOBSTER SOUP

50 lobster heads and carcasses, from
 1¼ lb (560 g) lobsters
 One #10 can tomato purée

¾ gal (2.8 l) white roux
8 gal (30.4 l) strong chicken stock
4 gal (15.2 l) fresh fish stock
1 gal (3.8 l) light cream
½ gal (1.9 l) white wine
1 qt (0.9 l) cream sherry
1 qt (0.9 l) brandy

Kitchen yields: 10 gal (38 l).

MINESTRONE

1 cup (8 oz) chopped fresh garlic
½ gal (1.9 l) olive oil
5 lb (2.2 kg) ham hocks
3 lb (1.3 kg) navy beans
½ gal (1.9 l) diced green peppers
½ gal (1.9 l) diced onions
½ gal (1.9 l) diced celery
½ gal (1.9 l) diced carrots
½ gal (1.9 l) diced cabbage
One #10 can chopped tomatoes
One #10 can tomato purée
3 lb (1.3 kg) broken spaghetti
7 gal (26.6 l) beef stock

Kitchen yields: 10 gal (38 l).

ONION SOUP

4 gal (15.2 l) sliced onions
½ gal (1.9 l) oil
12 gal (45.6 l) strong beef stock
¼ cup dried thyme
1 qt (0.9 l) applejack

Kitchen yields: 10 gal (38 l).

PEA SOUP

20 lb (9 kg) green split or yellow peas
12 gal (45.6 l) beef or chicken stock
5 lb (2.25 kg) ham hocks
1½ gal (5.7 l) light cream

Kitchen yields: 10 gal (38 l).

PHILADELPHIA PEPPER POT

1 gal (3.8 l) diced green peppers
1 gal (3.8 l) diced onions
1 gal (3.8 l) diced potatoes
1 gal (3.8 l) diced celery
1 gal (3.8 l) cooked spaetzle
1 gal cooked, diced tripe
8 gal (30.4 l) beef stock

Kitchen yields: 10 gal (38 l).

PUMPKIN SOUP

½ gal (1.9 l) roux
Three #10 cans pumpkin purée, plain
12 gal (45 l) strong chicken stock
¼ cup, 2 oz (56 g) nutmeg
½ cup, 4 oz (112 g) lemon juice
1 gal (3.8 l) light cream

Kitchen yields: 10 gal (38 l).

VEGETABLE SOUP

½ gal (1.9 l) diced green peppers
½ gal (1.9 l) diced onions
½ gal (1.9 l) diced potatoes
½ gal (1.9 l) diced celery
½ gal (1.9 l) diced carrots
½ gal (1.9 l) diced cabbage
One #10 can chopped tomatoes
7 gal (26.6 l) beef stock

Kitchen yields: 10 gal (38 l).

VICHYSOISE (POTATO SOUP)

25 lb (11.25 kg) russet potatoes
8 lb (3.6 kg) onions
8 lb (3.6 kg) white of leeks
8 gal (30.4 l) strong chicken stock
2 gal (7.6 l) light cream
2.5 gal (9.5 l) heavy cream, 36 percent
butterfat

Note: Make base with all ingredients except cream. Store refrigerated. Add cream to base as needed. Garnish with snipped chives and season with Worcestershire sauce.

Kitchen yields: 10 gal (38 l).

SOUP QUANTITIES

Kitchen yields:

1 gal (3.8 l) = 20 servings.
5 gal (19 l) = 100 servings.
10 gal (38 l) = 200 servings.
25 gal (95 l) = 500 servings.
38 gal (145 l) = 750 servings.

Note: Always make about 10 percent more than needed in case of spillage.

SPAETZLE

Small dumplings from southwestern Germany and neighboring Switzerland.

Spaetzle are available ready-to-use as convenience product. See the Groceries section.

Formula:

3 lb (1.3 kg) bread flour
12 eggs
1 qt (0.9 l) milk
½ cup (0.12 l) oil
1 tb salt
1 tb baking powder

Serving size: ½ cup (0.12 l).

Kitchen yields: 40 servings.

Note: To shape spaetzle, use a food mill with very large holes. Some chefs shape spaetzle on a wet wooden board and scrape them directly into the boiling water. There are specific spaetzle-shaping devices available in ethnic stores.

SPONGE CAKE (BASIC)**Formula:**

1 qt (0.47 l) egg yolks
3 qt (1.41 l) whole eggs
4 lb (1.8 kg) sugar
3 lb (1.3 kg) cake flour
1 lb (450 g) patent flour
8 oz (225 g) corn starch
1 lb (450 g) melted butter
Salt and lemon to taste

Kitchen yields: Twenty 8-in (200-mm) layer cakes.

SUGAR, PULLED**Formula:**

5 lb (2.25 kg) granulated sugar
1 pt. water (0.47 l).
¼ oz (6.75 g), or 2 level tsp, cream of tartar.

Note: Boil sugar to 318°F or more, depending on use and weather. Pulled sugar is also available ready for use. See the Groceries section.

Kitchen yields: 1 medium-sized sugar basket.

SUGAR, SPUN**Formula:**

5 lb (2.25 kg) sugar

1 lb (0.45 kg) glucose
1 pint (0.47 l) water

Method: Boil to 320°F.

Kitchen yields: Garnish for 50 desserts.

TALLOW FOR CARVING

Tallow is available ready to use. See the Groceries section.

Formula:

5 lb (2.25 kg) beeswax
5 lb (2.25 kg) rendered beef fat
5 lb (2.25 kg) paraffin

TALLOW, PLIABLE**Formula:**

5 lb (2.25 kg) shortening
5 lb (2.25 kg) rendered beef fat
4 lb (1.8 kg) paraffin

TAMALE DOUGH**Formula:**

2 lb (900 g) lard
4½ lb (2 kg) masa harina
6 tb salt
2 qt (1.9 l) warm water or chicken broth

Kitchen yields: Total recipe = 4½ qt (4 l).

Note: Some chefs add a small amount of baking powder.

TEMPURA BATTER**Formula:**

1 gal (3.9 l) ice water
4 egg yolks
2½ cups flour

Method: Combine ingredients. Japanese chefs use chopsticks and do not seem to mind if the batter is lumpy. The batter will be very thin and transparent.

TORTILLAS, CORN

Tortillas are usually purchased ready to use. See Hispanic Foods in the International Ingredients section.

Formula:

2¼ lb (1 kg) masa harina

4½ cups (1 l) warm water
1 cup (8 oz) lard

Kitchen yields: 36 tortillas, about 7 in (175 mm) across.

WAFFLE BATTER

Waffles are usually prepared from a mix or from ready-to-use batter. See the Groceries section.

Formula:

30 eggs
1½ lb (675 g) sugar
8 lb (3.6 kg) cake flour
2½ lb (1.1 kg) melted shortening

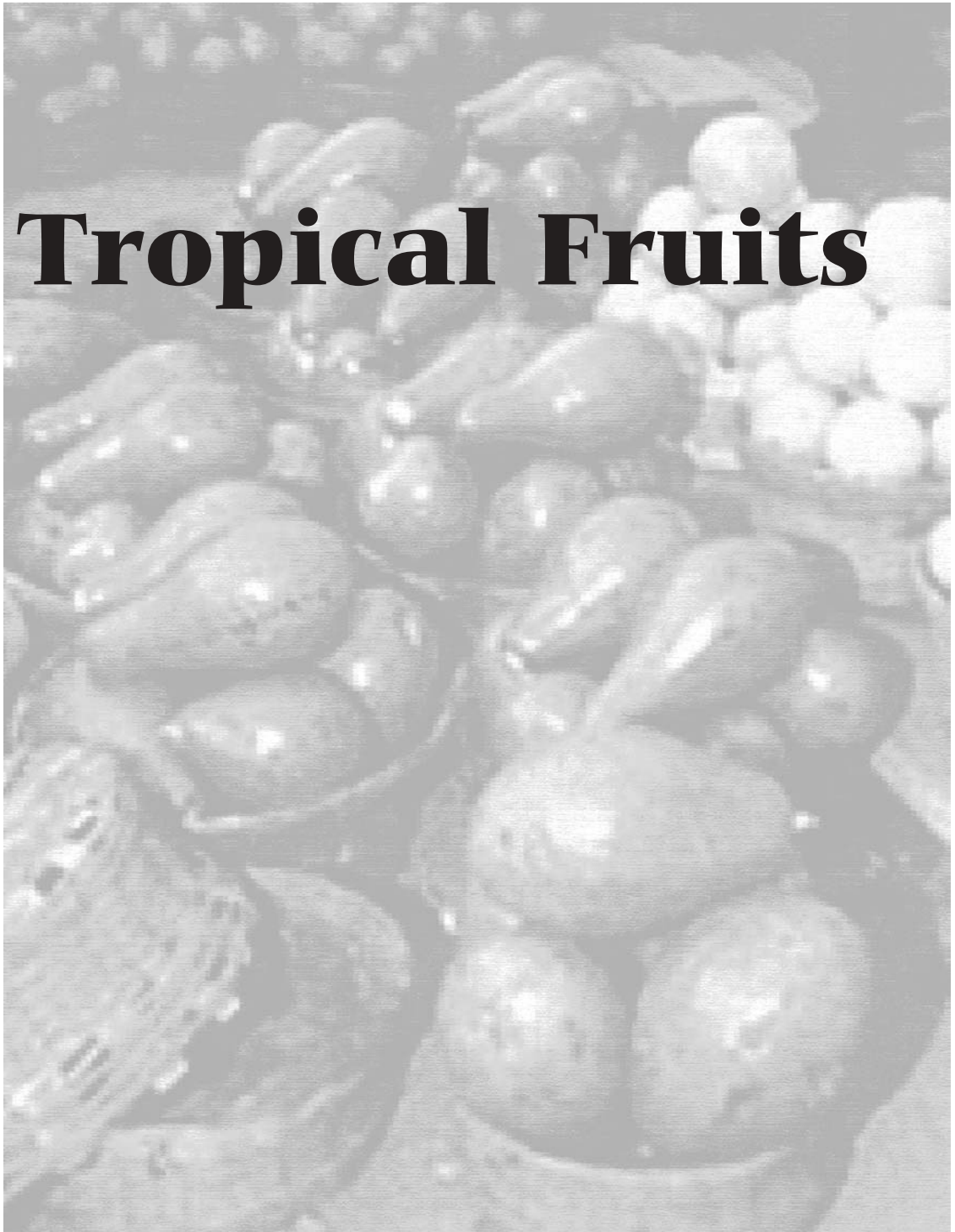
5 qt (4.7 l) milk
2 oz (56 g) salt
8 oz (225 g) baking powder
Vanilla flavor to taste

Serving size: 7-in (177-mm) waffle = 4 oz (112 g) batter.

Kitchen yields: Total recipe = 80 servings.

Note: Batter will get stiff when stored in the refrigerator. It can be diluted with milk. Additional baking powder must be added if diluted and stored longer than 24 hours.

Tropical Fruits



ACKEE

Tropical fruit about the size of a large plum. Only the kernels are edible.

Note: The fruit is poisonous until it is ripe and has opened naturally, exposing the kernels.

Season: Year-round.

Packs: By count.

Weight: About 4 oz (112 g).

Serving size: About 9 double kernels as appetizer garnish. Sauté quickly and serve hot.

Kitchen yields: 9 double kernels requires 3 fruits. There are 3 double kernels to each fruit.

Calories: N/A.

ATEMOYA

Tropical American custard-like dessert fruit, of the family Annonaceae; a hybrid cross between cherimoya and sugar apple. Its bumpy gray-green exterior is somewhat like a pinecone in shape. The flesh is studded with heavy black seeds. The fruit is very sweet and creamy textured. It is said to taste like a melange of strawberries, pineapple, and coconut. The fruit is not widely available outside its tropical growing areas because it does not transport well, although it is being increasingly grown in Florida. Rich in potassium.

CACTUS FRUIT (TUNAS)

See Hispanic Foods in the International Ingredients section.

CAPE GOOSEBERRIES

See the Fruits section.

CARAMBOLA

See Starfruit.

CAROB

Also known as St. John's bread. Pods, normally available dried, each containing 12 hard seeds. The pods are very hard when dried.

Pack: Sold by weight, often in 10-lb (4.5-kg) cartons.

Size: About 8 in (20 cm) long.

CHERIMOYA

This fruit resembles a large pinecone. It has large black seeds and custard-like flesh.

Season: November to May.

Pack: By piece.

Size: 3½ in (7 cm) to 5 in (12 cm) across.

Kitchen yields: Used in fruit displays, and sorbet can be made with the fruit pulp.

Calories: 4 oz (112 g) edible fruit pulp = 130 calories.

COCONUT

Fruit of the coconut palm native to the tropics. The outer coating is a thick greenish to yellowish fibrous husk. The husk is often removed before shipping to save weight and to make sure that the inside nut is not damaged or rancid. Coconuts with the husk removed are referred to as *balds*. Green coconuts have become increasingly available in ethnic markets and are often tapped for fresh juice. For Processed Coconut Products, see the Groceries section.

Season: Available year-round, with peak supply during winter.

Packs: 50-lb (22.5-kg) burlap bags, or loose by the piece.

Sizes:

Balds: 24's per 50-lb (22.5-kg) bag.

Balds: 30's per 50-lb (22.5-kg) bag.

With husks: 12's per 50-lb (22.5-kg) bag.

Weights: Balds: from 26 oz (750 g) to 33 oz to (900 g).

Kitchen yields: Balds have waste that is about 50 percent of weight. Amount of usable meat depends on amount of liquid left in nut.

Calories: Fresh, shredded, 1 cup, 3 oz (85 g) = 280 calories.

FRESH COCONUT, TRIMMED

Whole, wrapped in plastic, used only to extract the coconut water. *See* Thai/Vietnamese Foods in the International Ingredients section.

DATES

Fresh dates are available in fall, but are seldom available in most markets. For Dried Dates and other Date Products, *see* the Groceries section.

Season: Fall.

FELJOA

Oval, greenish fruit with shriveled skin. Also known as *pineapple guava*.

Season: Summer.

Pack: By piece.

Kitchen yields: Use raw in fruit salads. Peeling waste is about 20 percent.

GUAVA

Tropical fruit native to South America. The shape ranges from apple-shaped to pear-shaped, the flesh color ranges from white-yellow to pink to red. Best eaten raw. In South America and the West Indies, guava is often made into a paste-like preserve called *pasta de guayaba*, not to be confused with the fruit of the same name. For Frozen Pulp Guava and Pasta de Guayaba, *see* the Groceries section.

Packs: 10-lb (4.5-kg) cartons.

Count: 30 to 32 pieces.

Sizes: 4½ in (12 cm) long, 5 oz (150 g) each.

Serving size: About 2½ oz (70 g).

Season: Spring and summer. Small supplies are available year-round.

Note: The fruit is often picked underripe, and if picked too green will not ripen.

Calories: 4 oz (112 g) usable pulp = 70 calories.

JUJUBE

Fruit native to Southeast Asia with olive green color and about the size of a hen's egg. It has a single stone imbedded in the firm flesh, which resembles that of an unripe pear. Also known as *jujula*, and *june plum* in the Caribbean. *See* Dates under Chinese Foods in the International Ingredients section.

Season: Late summer until spring.

Size: 1 in (2.5 cm) to 2 in to (5 cm) long.

Weight: 2½ (70 g) to 3 oz (85 g)

KIWANO

The kiwano fruit is a member of the gourd family, christened *kiwano* in New Zealand. Brilliant yellow-orange on the outside when ripe, with a bright lime green jelly-like pulp on the inside, and numerous edible seeds. Also known as African horned melon, African horned cucumber, or jelly melon because it has distinctive spikes.

Season: Spring and early summer.

Pack: By weight.

Size: 4 in (100 mm) to 6 in (125 mm), about 12 oz (340 g).

KUMQUATS

Small, orange-type fruits. The fruits are seldom eaten raw but are used as decoration, as candied fruits, or in syrup. *Nagami* is a Japanese kumquat. For Kumquats Preserved (in syrup), *see* the Groceries section.

Season: Fall and winter.

Packs: 10-lb (4.5-kg) cartons, usually in pints.

Kitchen yields: Stems and leaves are often still attached.

Calories: 3.5 oz (100 g) = 65 calories.

LITCHI

Tropical fruits of the soapberry tree. Litchi fruits are grown in Florida and are usually of a red variety. Imports can be green, yellow, or pink. The flesh is white

and fragrant. Shelf life is limited, but the fruit is available preserved in cans and freezes well. For canned and dried products, *see* the Groceries section. Dried litchi are referred to as *litchi nuts*.

Season: June until late July. The fruits have short season in July and August.

Packs: By weight. The whole fruit clusters, including the tips of the branches, are normally shipped. One cluster can have a few berries or as many as 30.

Size: About 1½ in (38 mm). The pit is not edible, but it slips out easily; it is about ¾ in (20 mm) long.

Kitchen yields: 10 fruits = 4 oz (112 g) fresh pulp.

Calories: 10 fruits = 66 calories.

LOQUATS

Seedy fruits, yellow to orange, with white to pink flesh, from an evergreen tree called *Japanese medlar*, often used as an ornamental tree in backyards. Loquats resemble apricots. The fruit will blemish easily and is normally sold in clusters.

Season: Spring.

Packs: 10-lb (4.5-kg) carton.

Size: About 3 in (76 mm) long.

Calories: 3.5 oz (100 g) edible flesh = 50 calories.

MANGOES

Intensely flavored, succulent tropical fruits with a slightly acidic sweet taste, although inferior ones have a hint of turpentine in the taste. The fruit is elongated, with a thin skin that is green when unripe and yellowish red when ripe. It will ripen somewhat at room temperature, but will not achieve the sweetness associated with ripe fruits. Unripe mangoes are often sold sprinkled with salt as a snack. For canned and dehydrated Mango Products, *see* the Groceries section; for other mango products,

see Indian Foods in the International Ingredients section.

Season: May to September; peak occurs in August; imports available year-round.

Packs: 10-lb (4.5-kg) to 12-lb (5.4-kg) cartons.

Counts per carton: 8, 9, 10, 12, 14, 15, 16, 18, 20, 24, 28, or 30.

Weight: 15 or 16 size = 10 oz (280 g) to 12 oz (340 g).

Kitchen yields: Pits can be hard to remove; sometimes the fruit flesh is woody and stringy. One 12-oz (340-g) mango = ¾ cup flesh.

MANGOSTEEN

Reddish brown fruit with a thick rind and white flesh. The flavor can be compared to a blend of pineapples and apricots.

Size: 3 oz (85 g) to 4 oz (112 g) each.

Calories: 3.5 oz (100 g) of flesh = 65 calories.

PAPAYA

Also called *tree melon*, this oval to pear-shaped fruit varies greatly in size. The flesh is hard when the fruit is not ripe, indicated by green skin, but soft and very perishable when ripe, indicated by yellow skin. Papaya can reach 10 lb (4.5 kg) in some tropical countries, but Hawaiian papaya commonly used in foodservice is smaller and more uniform in size. For Frozen Papaya Purée, *see* the Groceries section.

Season: Year-round; peak occurs in spring and summer.

Packs: 10-lb (4.5-kg) containers.

Size: 14 oz (400 g) to 17 oz (475 g) each.

Count: 8 to 12 fruits per container.

Serving size: ½ fruit as appetizer or as main course component.

Kitchen yields: ½ fruit peeled, seedless = 5½ oz (150 g) to 6½ oz (180 g).

Calories: 1 cup cubed fruit, 5 oz (140 g) = 55 calories.

PASSION FRUIT

A purple- or red-skinned fruit with yellow flesh and many edible seeds. In Australia and New Zealand the name *grenadilla* is used. Passion fruits are cultivated in many hot regions, a red-skinned variety is grown in Florida, and a purple-skinned variety is grown in California. The trade distinguishes between sweet *grenadilla* and giant *grenadilla*.

Packs: By weight and by the piece.

Season: September and October and again in February to July. As the fruit ripens, the skin begins to shrivel. Underripe fruit will ripen at room temperature and will last refrigerated about a week.

Size:

Sweet grenadilla: 3 oz (85 g) average, oval, about 3 in diameter, occasionally twice as big.

Giant grenadilla: 10 oz (280 g).

Kitchen yields: The fruit is a gelatinous mass of yellow-orange pulp and small seeds. The seeds are usually eaten with the pulp, but are sometimes strained out when the pulp is used in beverages and smooth sauces.

Calories: 3½ oz (100 g) = 75 calories.

PAWPAWS

Oblong, yellowish fruit, with many seeds in a sweet banana-like pulp.

Season: August through winter.

Size: 10 oz (280 g) to 12 oz (340 g).

Calories: 3.5 oz (100 g) pulp = 150 calories.

PERSIMMONS

Two varieties of persimmons are marketed. Fuyu persimmons have orange-colored skin and flesh and should be eaten while still firm. Those of the Hachiya variety are more pointed and

should be eaten when soft. Both varieties are used in salads and as a fruit garnish. The fruits are tart when unripe and extremely sweet when ripe. They can be stored under refrigeration when ripe.

Season: Fall and early winter.

Packs: Flats with 24 and 30 counts.

Kitchen yields: 1 fruit = 4 oz (112 g) edible flesh.

Calories: 4 oz (112 g) edible flesh = 140 calories.

PLANTAINS

For Plantains (platanos), *see* Hispanic Foods in the International Ingredients section.

PLUMCOTS

Hybrid cross between plum and apricot, with little supply reaching the market.

POMEGRANATES

Red-skinned fruit, used mostly in the manufacture of grenadine syrup. *See* the Groceries section for Dried Pomegranate Seeds. The word *pomegranate* means “apple with many seeds.”

Season: August to December.

Packs: 22-lb (9.9-kg) lug.

Size: 3 in (80 mm) to 4 in (120 mm) across.

Kitchen yields: Primary use is for flavoring sauces and syrups, because numerous seeds are imbedded in the flesh.

Calories: 4 oz (112 g) edible flesh = 90 calories.

PRICKLY PEARS (CACTUS FRUIT)

See Tuna under Hispanic Foods in the International Ingredients section.

RHUBARB

Although rhubarb is technically a vegetable, *see* the Fruits section.

SAPOTA

Tropical fruit from South America and the Caribbean; the sapota is the national fruit of Cuba and Mexico. Looking like a small football, it has a rough brown skin and dense orange-pink pulp. It has creamy, pudding-like flesh, with a honey-sweet flavor. Little supply reaches the market. There is considerable confusion between sapota and sapodilla.

SAPPODILLA

Also known as *dilly fruit* in Florida. Round to oval tropical fruit with thin brown or orange skin. Some varieties are seedless, but most have 2 or more inedible seeds. Fruit is sweet when ripe. Also known as *naseberry*.

Season: All year.

Packs: 10-lb (4.5-kg) flats.

Sizes: Large, medium, and small.

SPANISH LIME

Florida fruit not related to lime at all, though it has a similar flavor. The fruit resembles litchi nuts. It has a tough brown skin. The fruit is eaten by sucking the milky flesh from the seeds. Also known as *mamoncillo*.

STARFRUIT

There are two varieties on the market: Green Star Fruit, also called *Cucumber Tree Fruit*, and a yellow variety called *Star Apple*. Both are in the shape of a five-pointed star when cut across. The Green Star Fruit is sour and cannot be eaten raw. The Star Apple is slightly sweet. Also known as *carambola*.

Season: Year-round, peak in fall and winter.

Packs: By weight and by piece.

Sizes:

Green Star Fruit: 3 in (8 cm) long.

Star Apple: 4 in (12 cm) long.

SUGAR CANE

Available as sticks and whole.

Season: Summer.

Sizes: The supply reaching the market is cut into 4 ft (1 m) to 6 ft (1.5 m) pieces. 6 in (150 mm) skewers.

Kitchen yields: Pieces of young sugar cane can be used as a garnish or as skewers, but are basically inedible. In some ethnic markets the fresh cane is crushed and the brownish juice sold fresh.

TAMARILLO

Tropical fruits, glossy purplish-red, red, or golden, egg-shaped, with a slightly pointed end. Because of their strong acidity, they are good combined with meats, in making sauces or preserves, and commonly used in chutney. Tamarillos are often called *tree tomatoes*. For Frozen Whole Tamarillos, see the Groceries section.

Pack: 10-lb (4.5-kg) cartons.

Size: About 2½ oz (70 g) to 3 oz (85 g) each.

Calories: One fruit = about 80 calories.

TAMARIND

Acidic large brown bean pods; the fruit is used for making syrups. See Indian Foods in the International Ingredients section for tamarind-based seasoning.

Season: Year-round.

Packs: 10-lb (4.5-kg) carton.

WHITE SAPOTE

Also known as *custard apple*, this tropical fruit can be eaten raw or cooked. The skin is edible and should be light green-yellow when mature. White sapote is the most common commercial variety. Other principal varieties are black sapote and yellow sapote.

Season: April to November.

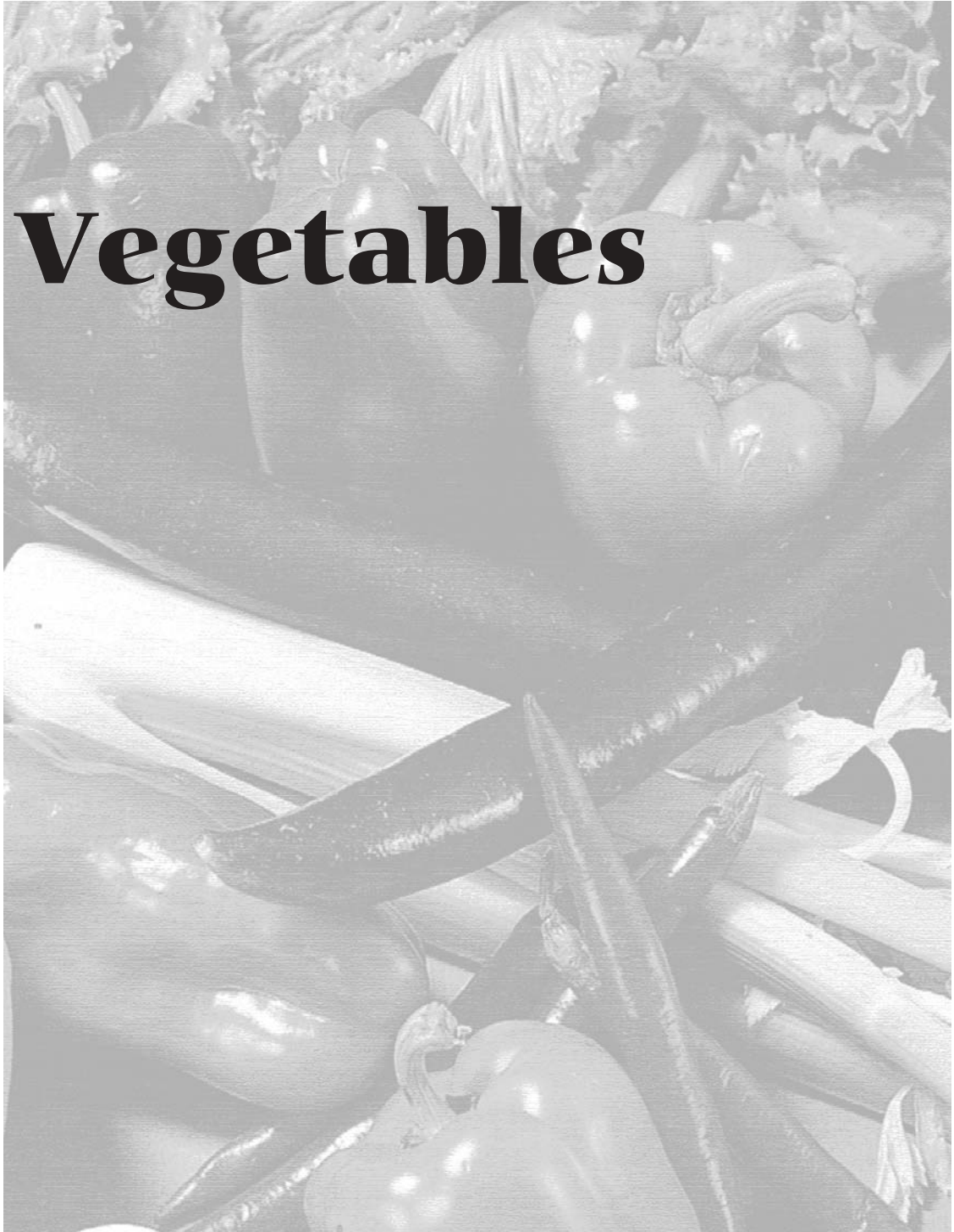
Packs: 10-lb (4.5-kg) flats.

Count: Varies. Average fruit is the size of a large orange.

UGLY FRUIT

See the Fruits section.

Vegetables



Many vegetables are now available custom cut and are usually sold by weight. The products are perishable and should be kept refrigerated and used promptly. Some companies also offer hand-shaped vegetables, referred to as tournés. Organically grown fruits and vegetables are reaching the market in increasing amounts. The packs, sizes, and yields do not always correspond with those listed here.

ALFALFA SPROUTS

See under Sprouts.

AMARANTH GREENS

Spinach-like vegetable with dark green leaves, used as vegetable and as salad. Also known as *callaloo*, *Chinese spinach*, *dasheen*, *elephant's ears*, *sag*, *sagaloo*, and *surinam spinach*. For Amaranth Grains, see the Groceries section.

Season: Available year-round.

Packs: By weight.

Kitchen yields: Depend on use. See Spinach.

ANISE

Bulb-shaped vegetable with fine, bushy leaves. The flavor is slightly like licorice. The vegetable is sometimes confused with the vegetable fennel, but fennel does not form a bulb. For Canned Anise, see the Groceries section.

Season: Domestic anise is available fall to spring; imports, the rest of the year.

Packs: 40-lb (18-kg) to 50-lb (22.5-kg) boxes.
22-lb (10-kg) boxes, mostly imports.

Counts: 20 to 24 bunches to a box.

Size: 1¾ lb (780 g) to 2 lb (900 g) with greens still attached. Imported anise is shipped trimmed.

Kitchen yields: Average size of bulb with most greens removed = 10 oz (280 g).

ARTICHOKES

Fresh artichokes are sized by number packed in a carton and by diameter. For Canned and Frozen Artichokes, see the Groceries section.

Season: Peaks occur in March through May and November to February. Imports are available year-round.

Packs: 20-lb (9-kg) and 25-lb (11.2-kg) boxes.

Counts: 18, 24, 36, 48, and 60 per box. The counts are applicable to boxes of both sizes and should be verified when the product is purchased. Small, loose artichokes without counts are available by weight.

Sizes

Small: Less than 2 in (50 mm) across.

Medium: 36 size, weighing about 10 oz (280 g) each.
48 size, weighing about 8 oz (225 g) each.

Large: 18 size, weighing about 20 oz (560 g) each.
24 size, weighing about 15 oz (420 g) each.

Serving sizes: Whole artichoke served as appetizer, use 18 or 24 size. Bottom for stuffing, use 24 size. It will be about 2½ in (63 mm) in diameter when peeled and cleaned.

Calories: One 18 size, 20 oz (560 g), artichoke, without sauce = 20 calories.

ARUGULA

A dark green, narrow-lobed salad green, pleasantly bitter, peppery, and pungent.

The best are the young leaves about 3 in long. Older leaves develop a tough stem and are best used in stir-fried dishes. Also known as *rocket* and *rocket lettuce*.

Season: Available year-round.

Packs: 6-lb (2.7-kg) to 7-lb (3.1-kg) flats, 24-count bunches.

Kitchen yields: Use 1 part arugula with 3 parts milder salad greens.

ASPARAGUS

For Canned and Frozen Asparagus, see the Groceries section.

GREEN ASPARAGUS

Season: Peaks are March to June and October to November. Green asparagus is now available year-round, often imported.

Packs: 30-lb (13.5-kg) 12-bunch pyramid wooden crates.
15-lb (6.7-kg) 6-bunch crates.
27-lb (12.1-kg) loose-pack pyramid crates.
13½-lb (6-kg) loose-pack half-pyramid crates.
24-lb (10.8-kg) to 25-lb (11.2-kg) cartons.
Other packs are also available.

USDA diameter sizing:

Small: ¼ in (6 mm) to ⅜ in (9 mm).

Standard: ⅜ in (9 mm).

Large: ⅞ in (11 mm).

Jumbo: 1⅜ in (20 mm).

Trade names: Trade names vary by growing regions. The most well-recognized names are Colossal, Jumbo, Large, Medium, Small, and Pencil (or Grass).

Lengths: 7 in (175 mm), 8 in (200 mm), and 9 in (225 mm).

Weights of bunches: 1 lb (450 g), or 2 lb (900 g) or 2½ lb (1.1 kg).

Note: The serving sizes vary because at the beginning of the season less trimming of the ends of the stalks is needed.

Serving sizes:

Appetizer portion:

Jumbo or Large: 5 to 6 spears at the beginning of the season; 6 to 8 spears at the end of the season.

Medium or Small: 6 to 7 spears at the beginning of the season; 7 to 9 spears at the end of the season.

Grass: 12 spears per person regardless of season.

Side orders:

Jumbo or Large: 2 spears, cut in half (4 pieces) at the beginning of the season; 3 spears per serving at the end of the season.

Medium or Small: 2½ spears cut in half (5 pieces) at the beginning of the season; 4 spears at the end of the season.

Grass: 4 to 5 pieces per serving.

Kitchen yields: The yield varies greatly, because the stem ends of the stalks get tougher as the season progresses. At the beginning of the season, almost the whole stems can be eaten; toward the end, about half must be discarded.

Appetizer Servings:

Beginning of Season:

Jumbo or Large: One 30-lb (13.5-kg) case = 50 servings.

One half case, 15 lb (6.7 kg) = 25 servings.

Medium or Small: One 30-lb (13.5-kg) case = 45 servings.

One half case, 15 lb (6.7 kg) = 22 servings.

Appetizer Servings:

End of Season:

Jumbo or Large: One 30-lb (13.5-kg) case = 40 servings.

One half case, 15 lb (6.7 kg) = 20 servings.

Medium or Small: One 30-lb (13.5-kg) case = 45 servings.

One half case, 15 lb (6.7 kg) = 22 servings.

Side Order Servings:

Beginning of Season:

Jumbo or Large: One 30-lb (13.5-kg) case = 140 servings.

One half case, 15 lb (6.7 kg) = 70 servings.

Medium or Small: One 30-lb (13.5-kg) case = 150 servings.

One half case, 15 lb (6.7 kg) = 75 servings.

Side Order Servings:

End of Season:

Jumbo or Large: One 30-lb (13.5-kg) case = 90 servings.

One half case, 15 lb (6.7 kg) = 45 servings.

Medium or Small: One 30-lb (13.5-kg) case = 95 servings.

One half case, 15 lb (6.7 kg) = 45 servings.

Grass: One 30-lb (13.5-kg) case = 100 servings.

One half case, 15 lb (6.7 kg) = 55 servings.

Note: The quantities given here are approximate and depend on the amount of peel removed and the length of stalk discarded.

Calories: 4 medium spears, 4 oz (112 g) = 20 calories.

WHITE ASPARAGUS

White asparagus is a specialty item; some is grown in California and some is imported. Sizes and packs vary greatly. See the Baby Vegetables section for very tiny white asparagus.

Packs: Bunches.

AVOCADOS

Domestically grown avocados are produced in California and in Florida. There are two principal California varieties: Hass, with a dark green to black, rough and leathery skin (Alligator Pear), and Fuerte, with a thin, pliable skin. Florida varieties are Booth, Lula, Hill, and Hickson. When ordering, specify ripe or

green, but this is no guarantee that the fruits will be perfectly ripe. The fruit will continue to ripen at kitchen temperature and should never be stored in the refrigerator.

Seasons:

California: Year-round, with Hass peaking in February to October; Fuerte, in fall and winter.

Florida: July through February, peaking in October.

Packs: 12½-lb (5.6-kg) flats with a single layer of fruit.
25-lb (11.2-kg) lugs with 2 layers of fruit.

Counts: 12½ lb flats: 9, 12, 14, 16, 18, 20, 24, 30, and 35.
25-lb lugs: 18, 20, 24, 28, 32, 36, 40, 48, 60, 70, 84, and 96.

Most common sizes used in hotels: 12-, 14-, and 16-count flats. Larger avocados, some weighing 1 lb (450 g) or more, come to the market frequently.

Average pulp yield:

Count per Carton	Weight per Fruit	Pulp Yield per Fruit	Pulp Yield per Carton
84	4½ oz (125 g)	3½ oz (100 g)	18¼ lb (8.2 kg)
50	6¾ oz (190 g)	5 oz (140 g)	18¾ lb (8.4 kg)
48	8½ oz (240 g)	6⅓ oz (180 g)	18¾ lb (8.4 kg)
40	10½ oz (300 g)	7¾ oz (220 g)	19½ lb (8.7 kg)

Kitchen yields: One and a half #36 avocado or one #14 avocado = 1 cup pulp.

Two #48 avocados or one #12 avocado = 1 cup diced.

Calories: 3½ oz (100 g) pulp = 153 calories.

BEANS

Fresh beans are referred to as immature beans. Immature beans are available

canned, dehydrated, fresh, and frozen. Mature beans are available canned and dried. Some varieties are available in many different forms. For Canned, Dried, and Frozen Beans, *see* the Groceries section.

FAVA BEANS

Season: Spring and summer.

Packs: By weight.

Kitchen yields: Yields varies because quality can be very uneven. Buying fresh fava beans is not recommended.

GARBANZO BEANS

Also called *chick peas* or *gandules*; they are occasionally available shelled.

Season: Late summer and fall.

Packs: 11-oz (310-g) bag, shelled, 12 per carton.

Kitchen yields: There is no cleaning waste.

GREEN BEANS

Fresh green beans are also called *bush* or *snap beans*. They can dry and become stale rapidly. A good indication of freshness is that they snap when broken. The term *string beans* is old-fashioned because beans no longer have “strings” that must be removed.

Season: Year-round, peaks occur in summer and fall.

Packs: 26-lb (11.7-kg) to 31-lb (14-kg) bushel baskets or cartons.

Counts: Vary. Length of beans vary and is no indication of quality.

Serving size: 3 oz, cooked.

Kitchen yields: Trimming loss is about 10 percent of weight or less. Many chefs trim off only the stem part.

1 lb (450 g) = 15 oz (400 g), cleaned.

1 lb (450 g) = 2½ cups, cooked, cut.

1 lb (450 g) = 5 servings.

1 30-lb (13.5-kg) bushel = 150 servings.

Calories: 3 oz, cooked = 22 calories.

HARICOTS VERTS

Haricots verts is the French name for very tiny beans, usually imported.

Packs: 5-lb (2.25-kg) cartons.

Serving size: 4 oz (112 g).

Kitchen yields: One 5-lb (2.25-kg) carton = 20 to 22 servings. When product is fresh, there is little cleaning waste.

LIMA BEANS

Fresh lima beans are seldom used in foodservice.

Season: July to December.

Packs: 26-lb (11.7-kg) to 31-lb (14-kg) hampers, whole.
11-oz (300-g) bags, shelled, 12 bags per carton.

Serving size: ½ cup (0.12 l).

Kitchen yields:

In pod:

1 lb = 6 oz (170 g) shelled, about 1 cup (0.24 l).

1 lb (450 g) = 2 servings.

30 lb = 11½ lb (5 kg) shelled.

30 lb = 30 servings.

Shelled:

1 lb (450 g) = 5 servings.

10 lb (4.5 kg) = 50 servings.

Calories: ½ cup (0.12 l) = 85 calories.

PINTO BEANS

Available already shelled.

Season: July to December.

Packs: 11-oz (300-g) bags, 12 per carton.

Serving size: ½ cup (0.12 l).

Kitchen yields: One 11-oz (300-g) bag = 3½ servings.

Calories: ½ cup (0.12 l) = 85 calories.

WAX BEANS

Yellow beans with a distinct flavor. They will not discolor when heated or when blended with dressings.

Season: Year-round, peaks occur in summer and fall.

Packs: 26-lb (11.7-kg) to 31-lb (13.9-kg) bushels.

Serving size: 3 oz (85 g).

Kitchen yields: 1 lb (450 g) as purchased = 15 oz (420 g) cleaned.
14 oz (400 g) cleaned = 11 oz (300 g) cooked.
1 lb (450 g) as purchased = 5 servings.
One 30-lb (13.5-kg) bushel = 150 portions.

Calories: 3 oz (85 g) = 20 calories.

BEAN SPROUTS

See Sprouts. For Canned Bean Sprouts, see the Groceries section.

BEETS

Beets are sold with and without leaves; whether or not leaves are included is important to verify when comparing prices. The greens can be used as a vegetable. Beets are often roasted rather than boiled. They should not be peeled raw because the color will leech out. For Canned Beets, see the Groceries section.

Season: Available year-round, peak occurs in winter.

Packs: 25-lb (11.2-kg) and 50-lb (22.5-kg) bags, without leaves.
43-lb (19.3-kg) and 47-lb (21.1-kg) cartons of 24 bunches, with leaves still attached.

Counts: None established; sizes vary.

Serving size: 3 oz (84 g), sliced.

Kitchen yields: 1 lb (450 g) beets, no leaves = 14 oz (400 g) cooked, sliced.
14 oz (400 g), cooked, sliced = 4½ servings.

Calories: ½ cup (0.12 l) = 25 calories.

BELL PEPPERS

See Sweet Peppers under Peppers.

BIBB LETTUCE

See under Lettuce.

BREADFRUIT

Large roundish fruit, belonging to the same family as figs. There are seedless and seeded breadfruit varieties. The unripe fruit has a slightly unpleasant aroma, which disappears when the fruit is ripe.

Season: May until February.

Packs: By count and weight.

Weight range: 2 lb (900 g) to 15 lb (6.7 kg).

Best sizes are 3 lb (1.3 kg) to 3½ lb (1.5 kg).

Sizes: Vary greatly, but normally 8 in (20 cm) to 12 in (30 cm) long.

Serving size: 1/6 breadfruit.

Kitchen yields: One 3-lb (1.3-kg) fruit = 6 servings as vegetable garnish.

Calories: 3 oz (112 g) = 115 calories.

BROCCOLI

Broccoli is available in a broad range of different trims. The stem length of the spears is crucial when calculating yield. The edible part is about 6 in down from the top; the rest is stem, which can be used in soups or purees. *Broccolino* is a cross between broccoli and cauliflower. Broccoli is traditionally shipped with ice, and this makes checking weights problematic. Some shippers have switched to a plastic packaging, called *Intellipac*. See the Groceries section for Frozen Broccoli.

Season: Available year-round, with no discernible peak months. The supply is lower in summer.

Packs: Carton, 42-lb (19-kg), 18 bunches, 2¼ lb (1 kg) to 2½ lb (1.1 kg) each.

½ carton, 20 (9 kg) to 23 lb (10.3 kg), 14 to 18 bunches, 1¼ lb (560 g) to 1½ lb (670 g) each.

Carton, 21 lb (9.45 kg), 57 loose-pack spears, 6 in (150 mm) each.

Intellipac, 18 lb (8 kg), cut crowns.

Intellipac, 18 lb (8 kg), Asian cut crowns (heads only).

Intellipac, 18 lb (8 kg), loose florets.
Carton, 15 lb (7.75 kg), loose-pack spears, no count, 3 in (75 mm) to 4 in (100 mm).

Carton, 9 lb (4.1 kg), loose-pack florets, no count, 1 in (25 mm) to 2¾ in (70 mm).

Carton, 9 lb (4.1 kg), 80 buds, 3 pieces per lb (450 g).

Intellipac, 14-count bunches of broccoli.

Sizes of spears: 2¼ lb (1 kg) to 2½ lb (1.1 kg) each, 8 in (200 mm) to 9 in (230 mm).

1¼ (560 g) to 1½ lb (670 g) each, 8 in (200 mm) to 9 in (230 mm).

Serving size: 3½ oz (100 g) cleaned vegetable, raw.

Kitchen yields: One 1¼-lb bunch = 13 oz (360 g) or less usable product.
One 1½-lb bunch = about 4 servings.
One 42-lb (19-kg) carton = 130 servings.

Calories: 3½ oz (100 g) cleaned vegetable, cooked = 25 calories.

BROCCOLI RABE (BROCCOLI DI RAPE)

Also known as *rapini* or *rappini*, a leafy bitter vegetable common in Italian and Chinese cooking. A member of the turnip green family, it looks like a thin version of budded broccoli, and, like broccoli, it is an excellent source of nutrients. The leaves should be crisp and green on small, firm stems. The yellow flowers are edible.

Season: Specifically winter, although available year-round.

Packs: Bunches in 25-lb (11.2-kg) cartons.

Kitchen yields: There is about 30 percent trimming loss, and the vegetable will shrink.

BRUSSELS SPROUTS

For Frozen Brussels Sprouts, see the Groceries section.

Season: Available year-round; peak occurs from August to March. Low supply May to July.

Packs: 1-pint, 16-oz (0.47-l), containers, 12 per flat.
10-oz (280 g) containers, 12 per flat.
25-lb (11.1-kg) cartons, loose.

Counts: Vary greatly. 1 pint (0.47 l) = 25 to 30 medium sprouts.

Weights: 1 flat pint container weighs 9 lb (4.1 kg).
1 flat 10-oz (280-g) containers weighs 7½ lb (3.4 kg).

Serving size: 4 oz (112 g) raw.

Kitchen yields: Trimming waste by weight is about 20 to 25 percent.
10 oz (280 g) as purchased = 8 oz (225 g), ready to cook.
10 oz (280 g) as purchased = 2½ servings.
1 pint (0.47 l) as purchased = 4 servings.

Calories: 4 oz (112 g), cooked = 40 calories.

CABBAGE

BOK CHOY (CHINESE CABBAGE)

See Chinese Foods in the International Ingredients section.

CELERY CABBAGE

See Chinese Foods in the International Ingredients section.

GREEN CABBAGE

Green cabbage is available whole, sliced, and as ready-to-use coleslaw mix. For Sauerkraut, see the Groceries section.

Varieties: Domestic is a spring variety. Danish is a firm winter cabbage.

Seasons: Available year-round. Early cabbage is available from April to May.

Packs: 40-lb (18-kg), 50-lb (22.5-kg), and 60-lb (27.2-kg) cartons.
10-lb (4.5-kg) bags coleslaw mix, 2 per case.

Sizes:**Early cabbage:**

Small, under 1½ lb (0.6 kg).

Medium, 1½ lb (600 g) to 3 lb to 1.3 kg).

Large, over 3 lb (1.3 kg).

Danish:

Small, under 2 lb (900 g).

Medium, 2 lb (900 g) to 5 lb (2.2 kg).

Large, over 5 lb (2.2 kg).

Serving size: 1 cup, 2½ oz (70 g), raw cabbage for coleslaw.

½ cup, 2.6 oz (72 g), cooked chopped cabbage.

Kitchen yields: 1 lb (450 g) as purchased = 12 oz (340g), ready to cook.

10 lb (4.5 kg) as purchased = 50 servings coleslaw.

10 lb (4.5 kg) as purchased = 40 servings.

Calories: 1 cup, 2½ oz (70 g), raw, coarsely shredded = 20 calories.

½ cup, 2.6 oz (72 g), drained = 15 calories.

RED CABBAGE

For Canned Red Cabbage, *see* the Groceries section.

Season: Available year-round.

Packs: 40-lb (18-kg), 50-lb (22.5-kg), and 60-lb (27.2-kg) cartons.

10-lb (4.5-kg) bags coleslaw mix, 2 per carton.

Kitchen yields: Same yields as for green cabbage.

Calories: 1 cup, 2½ oz (70 g), raw, shredded = 20 calories.

SAVOY CABBAGE

Curly, soft cabbage, edible only when cooked.

Season: Late summer until spring.

Packs: 40-lb (18 kg), 50-lb (22.5-kg), and 60-lb (27.2-kg) cartons.

Serving size: ½ cup (0.12 l) cooked.

Kitchen yields: 1 cup shredded = 2½ oz.

1 lb (450 g) as purchased = 12 oz (340 g), shredded.

1 lb (450 g) as purchased = 6 servings.

Calories: ½ cup (0.12 l), cooked, plain = 15 calories.

CACTUS LEAVES (NOPALES)

See Hispanic Foods in the International Ingredients section.

CARDONI

The name is also spelled *cardoon*. This vegetable looks like celery, but is part of the thistle family. It must be cooked before use.

Season: Fall and winter.

Packs: By weight.

Size: Up to 2 ft (500 mm) in length.

Kitchen yields: Trimming loss = 30 percent by weight.

CARROTS

There are many carrot varieties. Important to foodservice operators are the size and shape. Evenly shaped carrots, regardless of size, are easier to peel and have less waste. The shapes can range from long and slender to round. Smaller carrots are more tender than large carrots. Baby carrots can be either immature carrots of standard varieties or mature miniature varieties. Belgian carrots are a special variety of miniatures. The color of all carrots can range from light yellow to dark orange. Carrots can be purchased with the green leaves on or off. Large carrots are seldom marketed with the greens on. Carrots store well. For Canned and Frozen Carrots, *see* the Groceries section.

Season: Year-round. Peak occurs from late fall to spring.

Packs:

Carrots without tops:

50-lb (22.5-kg) bags or boxes

1-lb (450-g), 2-lb (900-g), and 3-lb (1.4-kg) bags.

Carrots with tops:

23-lb (10.3-kg) to 27-lb (12.1-kg)
crates, 24 bunches.

1-lb (450-g) cello pack, cleaned small
carrots.

Shredded carrots:

Four 5-lb (2.2-kg) bags.

Sticks, random pack:

Four 5-lb (2.2-kg) bags.

Sticks, precision pack:

Two 9-lb (4.1-kg) bags.

Sizes: Large or Jumbo: 1½ in (38 mm)
to 2½ in (60 mm) diameter; 7 in
(175 mm) to 12 in (300 mm) long.
Medium: ¾ in (18 mm) to 1½ in
(38 mm) diameter; 7 in (175 mm) to
9 in (225 mm) long.

Serving sizes: ½ cup, 2 oz (56 g),
shredded, raw.

½ cup, 2½ oz (70 g), sliced, cooked.

Kitchen yields: 1 lb (450 g), no tops =
14 oz (400 g), cleaned and peeled.
1 lb (450 g) as purchased = 12 oz
(340 g), cooked.
1 lb cleaned carrots, grated = 3 cups
(0.7 l).

Calories: ½ cup, 2 oz (56 g), shredded,
raw = 22 calories.

½ cup sliced, 2½ oz (70 g), cooked =
25 calories.

CATTAIL SHOOTS

Wetland plant with edible shoots, avail-
able in some locations during the season.

Season: March to May.

CAULIFLOWER

Cauliflower is available as whole heads
and cut into florets. For Frozen Cauli-
flower, *see* the Groceries section.

Season: Year-round; slightly larger
supplies October through January.

Packs:

Whole heads: 16-lb (7.2-kg) to 23-lb
(10.3-kg) cartons.

Loose florets: 18-lb (8-kg) Intellipacs.

6-lb (2.7-kg) cartons, two 3-lb (1.35-
kg) bags each.

12-lb (5.4-kg) cartons, four 3-lb (1.35-
kg) bags each.

Counts: 9, 12, or 16 heads per carton.

Serving size: 4 oz raw or ½ cup, 2½ oz
(70 g), cooked.

Kitchen yields: Florets are ready to use.

One 2-lb (900-g) to 2½-lb (1.1-kg)
head = 21 oz (600 g) florets.

One 2-lb (900-g) to 2½-lb (1.1-kg)
head = 8 servings, cooked.

One 21-lb (9-kg) carton with 9 heads
= 60 to 70 servings.

Calories: 4 oz (112 g), raw = 31
calories.

½ cup, 2½ oz (70 g), cooked = 15
calories.

CELERIAC (KNOB CELERY)

Large, round roots with white flesh.
Celeriac can be used raw in salads or
cooked as a vegetable. Available with or
without green tops; the yield can change
significantly if it is not clearly specified
whether greens are to be included. For
Canned Celeriac, *see* the Groceries section.

Season: Fall and winter.

Packs: ½ bushel, trimmed, or in
bunches with tops.

Size: 2½ in (63 mm) to 4 in (100 mm)
across.

Kitchen yields: About 25 percent
trimming waste after greens have
been removed.

CELERY

The Pascal variety is popular because of
its lack of stringiness. Golden celery is
yellowish white because it has been cov-
ered with soil during the growing pro-
cess. It is a specialty item, and there is
very little on the market. For Canned and
Frozen Celery, *see* the Groceries section.

Season: There is no particular season.

Packs: 55-lb (24.7-kg) or 56-lb
(25.2-kg) cartons or crates, whole.

Hearts:

25-lb (11.2-kg) to 28-lb (12.6-kg)
cartons.

Bag counts of 12, 18, or 24.

Sticks:

Four 5-lb (2.2-kg) bags, cut and
trimmed.

Diced:

Four 5-lb (2.2-kg) bags.

Counts: 18, 24, 30, 36, and 48.

Serving size: ½ cup, 2 oz (0.12 l),
chopped or diced.

Kitchen yields: 1 lb (450 g) = 12 oz
(340 g) ready to use, about 6
servings.

Calories: ½ cup, 2 oz (0.12 l), chopped
or diced = 10 calories.

CHARD

See Swiss Chard.

CHAYOTE

Also called *Mirliton squash*, chayote is
melon that grows on vines and is used in
cooking. The squash is round to pear-
shaped and varies in color from light to
dark green.

Season: Available year-round.

Packs: 10-lb (4.5-kg) and 15-lb (6.8-kg)
boxes.

Counts: Vary according to pack.

Size: 1 lb (450 g) average.

Kitchen yields: Cleaning waste = 30
percent.

CIPOLLINE

Small, bitter bulb-shaped vegetable in the
onion family, used as seasoning
ingredient.

COLLARD GREENS

For Canned and Frozen Collard Greens,
see the Groceries section. Grown mainly
in the Southern states. There are four
basic varieties:

Georgia: Large, crumpled blue-green
leaves

Louisiana Sweet: Large, thick leaves
with short stems

Morries Heading: Broad, medium leaves
with short stems

Vates: Thick green leaves with short
stems

Season: Available year-round; peak
supply in winter and low production
June to August.

Packs: By weight.

Serving size: ½ cup, 3 oz (0.12 l).

Kitchen yields: 1 lb (450 g) as
purchased = 14 oz (320 g) ready to
cook.

1 lb (450 g) as purchased = 4½
servings.

Calories: ½ cup, 3 oz (0.12 l) = 30
calories.

Note: Collard greens are often cooked
with ham or bacon, which increases the
calorie count.

CORN

For Canned and Frozen corn, see the Gro-
ceries section.

CORN ON THE COB

There are many local varieties of corn on
the market. In foodservice, the basic dis-
tinction is between yellow and white
sweet hybrid corn. A well-developed ear
of corn has 12 to 14 even rows of ker-
nels. Corn will lose quality rapidly,
although chilling helps to prevent flavor
loss.

Season: Year-round. Peaks occur in
Florida from May to August, from
July to September in other parts of
the country.

Packs: 45-lb (20.2-kg) to 50-lb
(22.5-kg) cartons.

35-lb (15.7-kg) bushel with 42 to 48
ears.

Counts: 54, 60, and 66 in the 50-lb
(22.5-kg) carton.

Kitchen yields: 2 ears, 60 size = 1 cup
(0.24 l) fresh corn kernels.

Calories: 1 ear, 5 in (125 mm) x 1¾ in (44 mm), 8 oz (225 g) = 120 calories.
1 cup, 6 oz (170 g) = 130 calories.

COURGETTES

See under Zucchini.

CUCUMBERS

Cucumbers are classified as Slicing Cucumbers, Greenhouse, and Pickling (or Kirby) Cucumbers. Greenhouse Cucumbers have almost no seeds and are burpless. They are often grown hydroponically. The stems of cucumbers can be bitter. Cucumbers are often covered with edible wax to retain moisture.

HOTHOUSE CUCUMBERS

Hydroponically grown and virtually seedless.

Season: Available year-round.

Packs: By weight or by piece; usually 12 cucumbers per case.

Average weight: 1 lb (450 g).

Kitchen yields: 1 cucumber = 26 slices cut on machine, set at #30.

PICKLING (KIRBY) CUCUMBERS

Packs: By weight.

Size: 4 oz (225 g) average.

SLICING CUCUMBERS

Season: Year-round; peaks occur in May and June.

Packs: 26-lb (11.7-kg) to 28-lb (12.6-kg) cartons, with 24 count.
30-lb (13.5-kg) to 32-lb (14.4-kg) cartons, with 35 and 40 counts.
50-lb (22.5-kg) to 55-lb (24.7-kg) cartons, with 60, 70, 80, and 90 counts.

Serving size: ½ cup unpared, sliced.

Kitchen yields: 1 lb (450 g) as purchased = 15 oz (425 g) pared.
1 lb (450 g) as purchased = 11 oz (310 g) peeled and cored.

One medium-sized cucumber = 1¾ cup (0.4 l).

Calories: 6 to 8 slices, with peel, 1 oz (28 g) = 5 calories.

DANDELION GREENS

Season: Year-round. Peak supply in April and May.

Packs: By weight, often packed in bunches.

Serving size: ½ cup, 3 oz (84 g) cooked, drained.

Kitchen yields: When used for salads, cleaning waste is about 20 percent. Shrinkage is about 45 percent after cooking.

Calories: 3 oz (84 g) cooked, drained = 27 calories.

EGGPLANT

Eggplants are grown in many states and are always in good supply, because the seasons overlap in the various growing areas.

Season: Year-round.

Packs: 20-lb (9-kg) to 22-lb (9.9-kg) lugs, with 24 counts.
30-lb (13.5-kg) to 34-lb (15.3-kg) bushels, with 30 to 35 counts.

Weights: 15 oz (420 g) average, but weights vary greatly.

Best size: 3 in (75 mm) diameter and 6 in (155 mm) long.

Kitchen yields: Size and shape are important for best yield.
1 eggplant = 2 cups, diced.
1 eggplant = 22 cross slices, ¼ in (6 mm) thick.

Calories: 4 oz (112 g) = 28 calories.

FIDDLEHEADS

The curly tops of ferns. Not all ferns are edible, and only the tops of ostrich ferns should be used. Some people have an allergic reaction when eating fiddleheads.

For Canned and Frozen Fiddleheads, *see* the Groceries section.

Season: Middle of April until June.

Packs: By weight.

Serving size: 3 oz (84 g).

Kitchen yields: 1 lb (450 g) = 5 servings. There is basically no waste.

GARLIC

Common white garlic is available from domestic sources and imported. Domestic harvest starts in July in California. The community of Gilroy, California, is famous for growing garlic. For Dehydrated Garlic, *see* the Herbs and Spices section. Many operations purchase peeled or chopped garlic.

Season: Year-round.

Packs: By weight.

Kitchen yields: Peeling waste is about 12 percent of weight.

ELEPHANT GARLIC

Large garlic with milder flavor than ordinary garlic. Can be served raw or cooked as a vegetable.

Season: May to December.

Packs: By weight.

Size: 8 oz (225 g) to 16 oz (450 g) each head.

Kitchen yields: There is little peeling waste.

ITALIAN GARLIC

Slightly pink garlic with strong flavor and consisting of smaller cloves.

Packs: By weight.

PROCESSED GARLIC

Processed garlic is available peeled only as well as peeled and chopped.

Packs: 1-gal (3.8-l) containers, peeled. 1-qt (0.9-l) jars, peeled, 12 per case. Smaller packs are also available.

TAHITI GARLIC

Large garlic with good yield.

Size: 2-in (50-mm) to 3-in (75-mm) diameter bulbs.

HORSERADISH

FRESH HORSERADISH

Season: Available year-round.

Packs: By weight.

Kitchen yields: Peeling waste is about 20 percent of weight. There is also waste when horseradish is grated because it can be stringy. Keep peeled roots in ice water to avoid discoloration.

GRATED HORSERADISH

Available plain or flavored with red beet juice.

Packs: 1-qt (0.9-l) jars and smaller.

JERUSALEM ARTICHOKE

See Sunchoke.

JICAMA (MEXICAN POTATO)

White tuber resembling a large turnip. Can be used raw in salads, or steamed, fried, or boiled. The tuber peels easily, and there is little waste.

Season: September to June.

Packs: Sold by weight.

Average weight: 6 oz (170 g) to 8 oz (225 g) and larger.

Kitchen yields: About 25 percent peeling loss.

KALE

A green winter vegetable, a hardy member of the cabbage family; there are many varieties of kale, both crinkly and smooth-leaf. Collard greens are closely related. Scotch Kale and Blue Kale are important varieties with curly leaves. Smooth-leaf varieties are also available, especially in the spring. Kale can be purchased whole in bunches or stripped. It is often shipped iced, which increases the weight. For Canned and Frozen Kale, *see* the Groceries section.

Season: December to April.

Packs: 18-lb (8.1-kg) to 25-lb (11.2-kg) bushels.

Serving size: 2 oz (0.12 l), cooked and drained leaves without stems.

Kitchen yields: Very difficult to estimate because of the amount of stripping waste, which is about 30 percent by weight.

Calories: 2 oz (0.12 l) cooked and drained leaves without stems = 25 calories.

KOHLRABI

Also called *cabbage turnip*, kohlrabi is a vegetable with a swollen stem that grows aboveground. Available with green or purple skin. Kohlrabi for foodservice should not be larger than 2 in (50 mm) across. Larger kohlrabi is often woody and cannot be used.

Season: Available year-round, with peaks in July and August.

Packs: By weight or by bunches. When purchasing by weight, ascertain whether the leaves are trimmed off.

Kitchen yields: Waste is at least 40 percent after trimming leaves.

LEEKs

Vegetables with large flat leaves and white stems. The white part is primarily used in cooking, and it should extend at least 3 in (75 mm) to 4 in (100 mm) from the roots.

Packs: 24-lb (10.8-kg) and 30-lb (13.5-kg) crates.

Counts: 12 bunches to a crate.

Size: White stems should be at least $\frac{3}{4}$ in (1.9 cm) in diameter, but not larger than $1\frac{1}{4}$ in (3 cm), because large leek centers are often woody.

Kitchen yields: There are about 12 to 25 leeks to a bunch, but sizes vary greatly. For this reason, leeks should be purchased by weight. If only the

white part is used, trimming loss is about 50 percent. Leeks are woody in early spring, and the core cannot be used. This can reduce the usable yield an additional 50 percent by weight after trimming.

LETTUCE

Lettuce can be classified as head lettuce and loose-leaf lettuce. Many varieties are on the market, and new varieties are being developed. Baby lettuce, with one head used as one serving, is becoming popular. Many operations purchase washed, cleaned, ready-to-use salad blends to save labor and storage space. Packs vary.

Season: Lettuce is available year-round, because growing seasons overlap, but price, quality, and variety can vary greatly because of the weather. Some lettuce is grown hydroponically and is shipped with the roots still attached. This increases the keeping quality.

BELGIAN ENDIVE

Rather expensive vegetable, most often used as salad, but it can also be braised, grilled, or sautéed. A purple variety has come on the market. For Canned Endive, see the Groceries section.

Season: Fall until spring.

Packs: 10-lb (4.5-kg) boxes.

Counts: Approximately 50 pieces per box.

Serving size: 4 oz (112 g) for salad, cut up.

Kitchen yields: Very little waste. One 10-lb (4.5-kg) box = 40 servings.

BIBB LETTUCE

Small lettuce, related to Boston lettuce, with soft leaves and a firm head. Available green as well as with slightly red leaves. Kentucky Bibb Limestone lettuce has firm, crisp heads and is considered a delicacy.

Packs: Baskets with 18 to 22 heads.

Serving size: When heads are firm, ½ head per serving.

Kitchen yields: One basket = 40 servings.

BOSTON LETTUCE

Also called *butterhead lettuce*. There are a number of varieties available, which range from lettuce with firm heads and crisp, yet tender leaves, to rather loose-headed lettuce.

Pack: Carton with 24 heads.

Kitchen yields: Heads are larger than those of bibb lettuce, but less firm.

Serving size: ½ head.

Calories: ½ head, 4 oz (112 g) = 15 calories.

CHICORY (ENDIVE)

Also referred to as *endive* or *frisée*, chicory has a loose head with curly, edge-indented leaves, which can be dark green at the edges and almost white at the center. Slightly bitter.

Season: Summer.

Packs: 35-lb (15.7-kg) to 40-lb (18-kg) cartons with 24 heads.
18-lb (8.1-kg) cartons with 12 to 18 heads.

Kitchen yields: Waste factor is about 30 percent.

Calories: 1 cup, 2 oz (56 g), cut = 10 calories.

CORN SALAD

See under Mache.

COS

See under Romaine.

ESCAROLE

Lettuce variety closely related to curly endive or chicory. It has a looser head and broader, less ruffled leaves. The lettuce should be crisp. The Batavian variety is the most common on the market.

Season: Year-round; peak occurs in December.

Packs: 35-lb (15.7-kg) to 40-lb (18-kg) cartons with 24 heads.
18-lb (8.1-kg) cartons with 12 to 18 heads.

Kitchen yields: There can be considerable waste if most dark green leaves are disposed of. Waste is about 45 percent. Note that repeated washing is necessary to remove all sand.

FRISÉE

See Chicory.

GREEN OAK LEAF

Baby lettuce with lobed green leaves. Sharper flavored than Red Oak Leaf lettuce.

Season: Summer.

Packs: Individual packs, 24 per case.

Serving size: 1 head.

Kitchen yields: 24 servings.

ICEBERG LETTUCE

Firm lettuce, will keep well under refrigeration. The common practice of smashing the heads on a hard surface to remove the stem damages the outer leaves.

Season: Year-round, but weather may influence supply and quality.

Available products: Chopped or shredded, whole cored and trimmed, and whole uncut.

CHOPPED OR SHREDDED

Packs: 20-lb (9-kg) net cartons containing either four 5-lb (2.2-kg) or two 10-lb (4.5-kg) clear poly bags.

Kitchen yields: There is no waste if product is fresh.

WHOLE, CORED AND TRIMMED

Packs: 30-lb (13.5-kg) carton, 24 or 30 heads packed in poly bags of 6 heads each.

Kitchen yields: Good product to use for wedges. There is little waste.

WHOLE, UNCUT

Packs: 50-lb (22.5-kg) cartons.

Sizes: 24 heads per carton is common; 30 and 18 heads per carton are also available.

Weight: Average head trimmed: 20 oz (720 g).

Serving size: 5-oz (140-g) wedge. Small tossed salad, 1 cup, 2 oz (56 g).

Kitchen yields: One 24-head carton = 96 wedges.
= 120 to 144 large leaves.
= 240 small tossed salads.
= 15 gal (5.7 l) torn or shredded.
= 10 gal (38 l) chunks.

Calories: 1 cup, 2 oz (56 g), chopped = 5 calories.

5-oz (140-g) wedge = 12 calories.

LEAF LETTUCE

Also called *garden lettuce*, leaf lettuce is a leafy variety, not forming a head.

Season: Year-round; peak occurs in summer.

Packs: 6-lb (2.7-kg) baskets.

Kitchen yields: Waste is 40 percent.

LAMB'S LETTUCE

Also called *corn salad*. See under Mache.

LOLLO BIONDO

Ruffled, crinkly baby leaf lettuce with white tips. Mostly grown hydroponically.

Packs: Individual heads.

LOLLO ROSA

Ruffled, crinkly baby leaf lettuce with rosy tips. Mostly grown hydroponically.

Packs: Individual heads.

MACHE

Mache, which has delicate, teaspoon-shaped leaves, is also known as *corn salad* or *lamb's lettuce*.

Packs: 11-lb (5-kg) case.

8-oz (225-g) plastic trays, hydroponically grown.

Serving size: 1½ oz (42 g).

Kitchen yields: The product is very light and is often used as a garnish or blended with other salad greens.

Note that it will wilt quickly as soon as dressing is applied. The product requires much labor if not absolutely fresh, because wilted leaves and stems must be removed.

8 oz (225 g) hydro grown with roots = 5 oz (140 g) clean salad.

MESCLUN

Blend of different baby lettuce varieties ready to use.

Packs: By weight, often 3-lb (1.35-kg) boxes.

Kitchen yields: There is no waste.

PERELLA

Red and green baby lettuce grown hydroponically.

Packs: Individual heads.

Kitchen yields: Basically no waste because the lettuce is grit free.

RADICCHIO

Red leaf lettuce in the chicory family with compact heads and a bitter flavor; used as salad and as a vegetable. Much product is imported from Italy. Look for small, compact heads.

Pack: 5-lb (2.5-kg) and 10-lb (4.5-kg) boxes.

Weight of head: 3 oz (84 g) to 6 oz (170 g).

Kitchen yields: There is very little waste when product is fresh.

RED OAK LEAF

Baby lettuce with deeply cut, finely divided leaves with burgundy red centers. About eight leaves per head.

RED ROMAINE

Also called Rouge d'Hiver, this is a red-tipped baby romaine lettuce, grown hydroponically.

Packs: Individual plastic packs, 12 or 24 per case.

Kitchen yields: Basically no waste, because product is delivered clean and grit free.

ROMAINE (COS)

Head lettuce in loaf shape with broad, white-stemmed leaves. Leaves change from dark green on the exterior to light yellow in the interior.

Season: Year-round.

Packs: 40-lb (18-kg) cartons with 24 heads.
18-lb (8.1-kg) cartons 12 to 18 heads.

Kitchen yields: 35 percent cleaning loss, depending on the amount of green leaves used.

Calories: 1 cup (0.24 l) chopped = 55 calories.

ROMAINE HEARTS.

Hand-trimmed inner leaves.

Packs: 14-lb (6.3-kg) cases.

Counts: 48 each.

Serving size: 1 heart per serving.

Kitchen yields: There is no waste.

TREVISE

Red lettuce, hothouse grown.

Packs: 11-lb (5-kg) cases.

STEM LETTUCE, WHO SUN

See Chinese Foods in the International Ingredients section.

MUSHROOMS, FRESH

A large quantity of wild mushrooms is harvested in the United States and exported. Some of these mushrooms are processed abroad and then reimported canned or dried. The location where the mushrooms are grown is the most important quality factor. The midwestern and

eastern states grow the best-tasting mushrooms. Nova Scotia also produces excellent wild mushrooms. However, the Pacific Northwest is the most prolific producer. Professional mushroom pickers are very fastidious, and some put every mushroom in a separate bag to avoid infestation by worms, especially when picking Porcini. Wild mushrooms are inspected a number of times before they are shipped and are very safe; there has never been a reported case of mushroom poisoning attributed to commercially picked wild mushrooms. For Canned, Dried, and Frozen Mushrooms, see the Groceries section. See Chinese Foods in the International Ingredients section for Asian Mushroom varieties.

CHAMPIGNONS OR COMMON MUSHROOMS

Marketed as white, off-white, and brown mushrooms; the most common variety is white. Champignons are sized by the number of mushrooms per lb (450 g). Mushrooms are perishable and store best refrigerated and in darkness.

Season: Available year-round; low season is middle of summer.

Packs: 3-lb (1.4-kg), 5-lb (2.2-kg), and 10-lb (4.5-kg) cartons.

Sizes:

	<i>Small</i>	<i>Medium</i>	<i>Large</i>	<i>Extra Large</i>
<i>Diameter (inches):</i>	3/4 - 1 1/4	1 1/4 - 1 3/4	1 3/4 - 3	3 +
<i>Count per pound:</i>	30-40	18-20	12-14	7-10

Medium-size mushrooms are often marketed as "silver dollar" size. Nonselected mushrooms are also on the market, usable for chopping or slicing.

Kitchen yields: 1 lb (450 g) = 1 qt (0.47 l) whole, raw.
= 5 cups sliced, raw.
= 6 cups chopped, raw.
= 2 cups cooked, sliced.
= 2 cups diced, cooked.

Waste: Trimming waste is about 5 percent by weight.

Calories: 1 cup, 2½ oz (70 g), raw, chopped = 20 calories.

CHANTERELLES

Available from domestic sources and imported. American chanterelles are related to the French girolles, but are not exactly the same species. Chanterelles should be dry when picked, because they can easily become worm infested. They should smell like apricots.

Season: From June until almost February.

Seasons start:

June: Morocco, North Africa.

July, August: Pennsylvania, followed by Nantucket and Nova Scotia.

Late August: California, Oregon, Washington State, and Canada.

Packs: By weight.

Size: There is no count. Early chanterelles are small, and weigh between ¼ oz (7 g) and ½ oz (14 g) each. Small chanterelles have a better flavor than later picked mushrooms. Late mushrooms can be as large as 1 lb (450 g) each.

Kitchen yields: There is little shrinkage.

Note: To preserve mushrooms, store in a dry place or sauté quickly and then freeze.

CORALS

Corals look like hens-of-the-woods. They are golden tinged with white at the base of the florets. However, these mushrooms do not taste as good as the hen-of-the-woods variety; some, especially the cultivated ones, can be bitter. These are best just for show.

Season: Summer.

Packs: Sold by weight.

CRAB MUSHROOMS

Crab mushrooms look like a lady's powder puff; they look much better than

they taste. They have an unpleasantly bitter taste, and though safe to eat, are not recommended.

CREMINI

Cremini mushrooms are the forerunners of the common cultivated mushrooms. They are more like those our grandparents would have known. Shelf life is about 1 week.

Season: Available year-round.

Packs: Sold by weight, often in 5-lb (2.25-kg) to 6-lb (2.7-kg) boxes.

Kitchen yields: The same as for champignons.

ENOKI

Enoki mushrooms look like enlarged pins. Each has a long skinny stem, about 3 in (75 mm) or 4 in (100 mm) in length, and a tiny round head or cap. They have almost no taste, but make a great appearance. They are common in Japanese cuisine.

Season: Available year-round.

Packs: 3½-oz (100-g), 10-oz (228-g), 1-lb (450-g), and 2-lb (900-g) packages.

Kitchen yields: There is some trimming loss because the mushrooms are sold attached at the root end. They are usually purchased in small quantities. They can be used raw, sprinkled over dishes as a garnish.

Calories: 3½ oz (100 g) = 45 calories.

GOAT'S BEARD

Rare mushrooms that look like goats' beards.

Season: Fall.

HEDGEHOG MUSHROOMS

The caps are beige/orange colored, the underside is white.

Season: Winter and very early spring.

HEN-OF-THE-WOODS (CAULIFLOWER)

Excellent mushrooms, perhaps the best of all, but very rare and expensive. They are

delicious raw. Only domestically harvested mushrooms are available. They resemble the head of a cauliflower, with a brownish-gray tinge.

Season: Spring and fall.

Packs: 3-lb (1.3-kg) flats or by weight.

Kitchen yields: There is no trimming loss. The mushrooms should be served cut into florets, but it is impressive to show one whole.

HON SHIMEJI MUSHROOMS

Season: Summer.

Packs: By weight.

LOBSTER MUSHROOMS

Orange to red in color.

Season: July and August.

Packs: By weight.

MORELS

Nearly all fresh morels in the U.S. markets are domestic. The United States exports large quantities to Europe for canning. Cultivation of morels is not yet possible. These are underground plants; the visible “hat” is the fruit. Morels picked in the United States are normally not as sandy as their imported counterparts. There are more than 22 varieties of morels found in the United States, but basically only black and white morels are of commercial importance. White morels are of superior quality, because they are lighter, more delicate than black morels. The best come from Michigan and Wisconsin. Although white morels are rarer than black morels, they are not necessarily more expensive. False morels, called *snow morels*, reach the market occasionally. They are less expensive, but should never be used. All morels, although not toxic, can cause stomach cramps and indigestion if eaten in large quantities. Many European cookbooks recommend parboiling all morels to remove any trace amounts of toxin.

Season: End of March until end of June in the Northern Hemisphere.

Packs: By weight. When morels are shipped, they generate heat, which can reach up to 100°F. Heated morels develop an ammonia smell and should not be accepted.

Size: Morels are not graded by size. A packer may pick large morels for stuffing upon request for a premium price.

Kitchen yields: Morels do not shrink much. There is little cleaning loss. Morels are often sandy and must be well washed.

PIEDS DE MOUTON

Called *lambs’ feet mushrooms*, they look like chanterelles but have more gills, which come off in cooking and make the product look muddy.

Season: November and December.

Packs: By weight.

Size: Small, about 1¼ in (32 mm) across.

PLEUROTES (OYSTER MUSHROOMS)

Pleurotes are grown in straw or sawdust. There are four key varieties in commercial cultivation; some are grown domestically, and others are imported. When purchasing, look for mushrooms with short stalks.

AMERICAN GOLDEN PLEUROTES

Also known as *golden trumpets* because of their shape and color. American varieties come trimmed, with or without stalks. The American pleurotes are usually not as large as the Italian ones.

Packs: 3-lb (1.4-kg) and 5-lb (2.25-kg) boxes.

AMERICAN WHITE PLEUROTES

Packs: 1-lb (450-g) boxes.

ITALIAN GOLDEN PLEUROTES

Imported yellowish to white mushrooms.

Packs: 6-lb 9-oz (3-kg), 8-lb 12-oz (4-kg), or 11-lb (5-kg) wooden flats.

ITALIAN WHITE PLEUROTTE

Also known as *white trumpets*, these imported mushrooms are very difficult to grow and are usually quite small in size. White pleurottes have a short shelf life.

Packs: 1-lb (450-g) boxes.

Kitchen yields: The mushrooms are shipped untrimmed.

PORCINI

Porcini are considered the best eating wild species of mushrooms except for truffles. The German name is *Steinpilze*; the French name is *cèpes*; *porcini* is the Italian name. Place of origin does not influence quality. The whole mushroom can be used. These mushrooms are very perishable and subject to infestation.

Season: Imports from Europe during summer and early fall. Short season in spring and a longer season in fall for domestic mushrooms.

Packs: By weight.

Size: Size is no indication of quality. The large mushrooms are as tasty as the small mushrooms. The large caps can be up to 15 in (375 mm) across and are used for grilling. The whole mushroom can weigh as much as 3 lb (1.35 kg), and a small one can be the size of, and as tight as, a baby's fist.

Kitchen yields: Very large caps have dark brown gills, which are often mushy and should be removed. Infestation by snails can also be a problem, and sometimes the trim loss is more than 20 percent.

PORTOBELLO OR ROMAN MUSHROOMS

Actually the same mushroom as cremini, but left to mature longer in the field. As these mushrooms mature, they become flatter and the flavor intensifies. When cooking them, it is advisable to use an

acid such as lemon juice to avoid having them turn very dark.

Season: Year-round.

Packs: Sold by weight, often in 5-lb (2.2-kg) boxes.

Size: About 6 large mushrooms = 1 lb (450 g).

Kitchen yields: The stems are usually discarded. Waste is about 25 percent by weight.

PUFF BALLS

Not commonly available, although not rare.

Season: Late spring to fall.

Size: Average is about ½ lb (450 g) each, up to 3 lb (1.3 kg).

SHIITAKE

Shiitake mushrooms are the most commonly cultivated mushrooms in the world. Their American production is expected to surpass the production of cremini mushrooms in the next 20 years, because of the influences of Asian cuisines and because shiitake mushrooms taste delicious. Commercially cultivated shiitake mushrooms taste better than those harvested in the wild. Shiitake mushrooms are marketed in two basic varieties: indoor grown and outdoor grown. The outdoor grown mushrooms are usually denser, woodier, and heavier. Fresh shiitakes have a long shelf life.

Grades: There are two grades for both indoor- and outdoor-grown shiitake mushrooms:

Grade A: Uniform in size with clear, neatly structured gills.

Grade B: Larger and nonconforming in shape and size.

Packs: 3-lb (1.3-kg), 5-lb (2.2-kg), 7-lb (3.1-kg), and 10-lb (4.5-kg) flats.

Kitchen yields: Use grade A when using the cap whole.

Use grade B for cutting up, or select the large caps for grilling.

STRAW MUSHROOMS

See Straw Mushrooms under Chinese Foods in the International Ingredients section.

TRUFFLES

Fresh truffles are available domestically and imported. By far the largest amounts, and the best qualities, are imported.

DOMESTIC TRUFFLES

Two varieties of domestic truffles come to the market. They are different from imported varieties in both size and smell. Like imported truffles, they are available both black and white, the white being more plentiful and slightly cheaper than the black. Most come from the West Coast, especially Oregon. Some domestic truffles have a camphor-like smell, which is not acceptable. The supply is very limited.

IMPORTED TRUFFLES

Imported truffles come mostly from France and Italy. Both black and white varieties are available, the black variety being slightly more abundant than the white. The flavor is carried in the volatile oils, and is easily imparted to other foods. Truffles can be successfully flash-frozen.

Seasons: White truffles: October to December.

Black truffles: November to December.

Packs: By weight.

Size: Varies from 1 oz (28 g) to 4 oz (112 g); larger truffles, up to 12 oz (400 g), are occasionally found.

Kitchen yields: Truffles are very expensive and maximum use must be made of the product. Truffles should be sliced thin and to order. White truffles are excellent sliced raw over pasta dishes.

SUMMER TRUFFLE

Summer truffles, black truffles with less flavor than winter truffles, are imported

from France, Italy, and North Africa. Summer truffles are also much less expensive than winter truffles.

Season: May to September.

Packs: Sold by weight.

Sizes: 2 oz (56 g) to 8 oz (225 g) and larger.

WINE CAPS

Mushrooms with lavender-colored caps with gills.

Season: Spring and fall.

WOOD EARS

See Cloud Ears (*win yee*) under Chinese Foods in the International Ingredients section.

YELLOW FOOT CHANTERELLES

These mushrooms are very different from other chanterelles, because they are very light and tiny and look like miniature yellow carnations.

Season: Spring and fall.

Size: 1 oz (28 g) = 20 to 30 mushrooms.

MUSTARD GREENS

Mustard greens are grown extensively in the South, but supplies also come from other states. For Canned and Frozen Mustard Greens, see the Groceries section.

Varieties: Broad Leaf, Southern Giant Curled, Fordhook Fancy, Smooth Leaf, and Tendergreen.

Season: Winter.

Packs: 18-lb (8.1-kg) to 25-lb (11.2-kg) bushels.

Serving size: ½ cup, 2.5 oz (70 g), cooked, leaves only.

Kitchen yields: 1 lb (450 g) raw = 4 servings.

Calories: ½ cup, 2.5 oz (70 g), cooked, leaves only = 15 calories.

OKRA, FRESH

Also called *gumbo*; long and short varieties are on the market in both green and yellow. Green okra is the most common variety. *See* the Groceries section for Canned and Frozen Okra.

Season: Available year-round; peak occurs July to October.

Packs: By weight in hampers, bushels, or baskets.

Best size: 2 in (50 mm) to 3½ in (90 mm) long.

Kitchen yields: 1 lb (450 g) = about 35 pods.

Calories: 5 pods, 2 oz (56 g), cooked = 15 calories.

OLIVES

Fresh California-grown olives are available on a limited basis; green and black varieties are available. For Canned and Cured Olives, *see* the Groceries section.

Season: September to December.

Pack: By weight.

ONIONS

For Frozen and Dehydrated Onions, *see* the Groceries section.

CREAMER ONIONS

See Pearl Onions in this section.

GREEN ONIONS (SCALLIONS)

Season: Available year-round; peak in spring.

Packs: 15-lb (6.8-kg) to 25-lb (11.2-kg) cartons, with 48 bunches. 24- and 12-bunch cartons.

Kitchen yields: Waste depends on use of green part versus white part. If whole vegetable is used, cleaning waste is about 10 percent by weight.

Calories: 6 stalks, 1½ oz (42 g) = 20 calories.

MATURE ONIONS

Onions are classified as storage onions and fresh onions. Storage onions can be

recognized by the darker-colored outer skin, which is thicker than that of fresh onions. Storage onions are known for their firmness. Fresh onions can be recognized by their lighter color and thinner skin. They have higher water and sugar contents than storage onions and are more susceptible to bruising and decay. Both varieties are available most of the time in three colors: red, white, and yellow. Bermuda onions have a mild flavor and are early ripening. Sweet Spanish onions are large yellow storage onions from Idaho and eastern Oregon. Sweet onions are sweeter than normal onions and do not irritate the eyes when being processed. They come from various growing regions. Red onions are strong-flavored creole-variety onions. Vidalia onions are mild and are grown in a designated area in Georgia. Walla Walla Sweets are in season during the summer. Many operations purchase onions already peeled, chopped, or sliced.

Season:

Fresh onions: Available year-round.

Storage onions: From July until April of next year.

Packs: 50-lb (22.5-kg), 25-lb (11.2-kg), 10-lb (4.5-kg), and 5-lb (2.25-kg) bags. 50-lb (22.5-kg) and 40-lb (18-kg) cartons.

Sizes:

Name	Diameter Size	Weight
Colossal	4+ in (100+ mm)	14–18 oz (400–500 g)
Jumbo	3¼–4½ in (80–112 mm)	8–10 oz (225–280 g)
Large	2¼–3¼ in (56–80 mm)	3–5 oz (84–140 g)
Medium	1¾–2¼ in (45–57 mm)	1½–3 oz (84–45 g)
Prepack	½–1 in (12–25 mm)	
Boiler, creamer	⅞ in (12 mm)	

Counts:

50 lb: 45 to 50 Colossal, 80 to 100 Jumbo, 180 Large, 350 Medium.
 25 lb: 22 Colossal, 45 Jumbo, 90 Large, 170 Medium.
 10 lb: 12 Colossal, 23 Jumbo, 45 Large, 85 Medium.
 5 lb: 6 Colossal, 11 Jumbo, 22 Large, 44 Medium.

Kitchen yields: Peeling loss is about 15 percent by weight.
 One large, 1-lb (450-g), onion = 4 to 5 cups onion rings.
 1 cup chopped = 6 oz (170 g).
 50-lb (22.5-kg) bag = 110 cups chopped.

Calories: 1 cup, 6 oz (170 g), raw, chopped = 65 calories.
 1 cup, 4 oz (112 g), raw, sliced = 45 calories.

PEARL ONIONS

Also called *creamer onions*, small white onions used in stews and in vegetable preparations. See Canned and Frozen Pearl Onions in the Groceries section.

Season: Available year-round.

Packs: 28-lb (12.6-kg) lugs.

Counts: Vary.

Kitchen yields: 15 percent peeling loss.

PROCESSED FRESH ONIONS

Available peeled, sliced, or diced. Product must be kept refrigerated.

Packs: 20-lb (9-kg) boxes with four 5-lb (2.2-kg) bags each.
 20-lb (9-kg) boxes with two 10-lb (4.5-kg) bags each.

Kitchen yields: There is no waste.

SCALLIONS

See Green Onions.

OYSTER PLANT

See Salsify in this section.

PARSLEY, FRESH

There are two basic varieties, curly leaf and Italian parsley. Curly leaf is used

mostly for garnishing, Italian (flat-leaf) parsley for flavoring.

Available: Year-round.

Packs: Sold by bunches. Size of bunches varies.

Kitchen yields: Average weight of bunch is 2 oz (56 g). One tablespoon chopped weighs about 4 g.

PARSLEY ROOT

Also called *Hamburg parsley*. The vegetable looks like a yellow carrot and is used as a garnish in soups and stews.

Season: Available year-round.

Packs: By weight.

Sizes and Counts: Vary.

Kitchen yields: Peeling loss is 15 percent.

PARSNIPS

Parsnips are a root vegetable; they are best when left in the ground until after the ground freezes.

Season: October to January. Small amount available until spring.

Packs: 25-lb (11.2-kg) and 50-lb (22.5-kg) bags.

Twelve 20-oz (560-g) bags per carton.

Kitchen yields: About 4 parsnips = 1 lb (450 g).

Large parsnips can have a woody core, which will reduce yields.

1 lb (450 g) = 2 cups cooked, diced.

Calories: 1 cup, 6 oz (170 g), cooked, diced = 100 calories.

PEAS

For Canned, Frozen, and Dried (Mature) Peas, see the Groceries section.

BLACK-EYED PEAS

Available shelled.

Season: Mid to late summer.

Packs: By weight.

Serving size: 1 cup, 5.8 oz (165 g).

Calories: 1 cup, 5.8 oz (165 g), cooked and drained = 180 calories.

GREEN PEAS

Fresh green peas are almost never purchased in pods because shelling is very labor-intensive and the kernel sizes vary greatly. Frozen peas are processed as quickly as harvested and are available in many sizes. For Canned, Frozen, and Dried (Mature) Peas, *see* the Groceries section.

Season: Available all year; peak occurs in early summer.

Packs: 10-lb (4.5-kg) and 30-lb (13.5-kg) lugs.

Serving size: ½ cup, 3 oz (84 g), shelled.

Kitchen yields: 1 lb (450 g) = 1 cup, 5½ oz (155 g), shelled peas.
30-lb (13.5-kg) carton = 10 lb (4.5 kg) shelled peas.
30-lb (13.5-kg) carton = 26 servings.
Note: Yields vary greatly.

Calories: ½ cup, 3 oz (84 g), cooked, drained = 75 calories.

SNOW PEAS

Snow peas are a cross between green peas and sugar snap peas.

Season: Available year-round; peak occurs in January/February and July/August.

Packs: 10-lb (4.5-kg) and 30-lb (13.5-kg) cartons.

Serving size: 1¼ oz (35 g).

Kitchen yields: Cleaning loss by weight = 10 percent or less.
1 lb (450 g) = 11 servings.

SUGAR SNAP PEAS

Sugar snap peas can be used whole; only the ends must be snapped off, and the string along the back removed.

Season: February to September.

Packs: 10-lb (4.5-kg) and 30-lb (13.5-kg) cartons.

Serving size: 2½ oz (70 g).

Kitchen yields: 1 lb (450 g) = 120 pieces, approximately 6 servings.

Calories: 2½ oz (70 g) = 35 calories.

PEPPERS**HOT PEPPERS**

See Hispanic Foods in the International Ingredients section.

SWEET PEPPERS

Most sweet peppers are sold in the immature stage, which is green. All peppers eventually turn red or yellow when they mature. Red, yellow, purple, brown, and white varieties have been developed. Red peppers keep their color when cooked; the color of other varieties fades. For Canned and Frozen Sweet Peppers, *see* the Groceries section. The market differentiates between Choice and Fancy grades. The prices of colored peppers vary greatly according to season.

Season: Year-round; peak occurs May to August.

Packs: 26-lb (11.7-kg) to 30-lb (13.5-kg) bushels.
10-lb (4.5-kg) cartons.
Specialty colored peppers should be purchased by weight.

Counts: Vary greatly. About 5 pieces per pound (450 g).
Some red peppers can weigh 7 oz (200 g) and up.

Kitchen yields: For chopping, purchase Choice. For other applications, purchase Fancy.
Cleaning loss = 20 percent by weight.
Cleaning loss, seedless varieties = 5 percent by weight.
10 medium peppers = 6 cups (1.6 l) diced.
20 lb (9 kg) = 40 cups diced and cooked.

Calories: 1 cup, 4 oz (112 g), cooked, diced = 20 calories.

POTATOES

For Canned, Dehydrated, and Frozen Processed Potatoes, *see* the Groceries section. *See also* Specialty Potatoes in this section.

Potatoes are available peeled and custom cut.

Potatoes are classified into four broad groups:

- *Long Whites*: boiling and salads
- *Round Red*: boiling and salads
- *Round Whites (Irish Potatoes)*: boiling, salads, and frying
- *Russet*: baking and frying

In each group are a number of varieties. Potatoes are grown in many states. They are available year-round, although fall is the harvesting season in most states. The main producing areas are Idaho, Washington State, Colorado, and Maine. Washington State potatoes have a slightly higher nutrient content than the national average because they receive more sunshine.

LONG WHITE POTATOES

Good variety for salads, hash browns, and home-fried potatoes.

Season: Available year-round.

Packs: 50-lb (22.5-kg) bags.

Serving size: Varies greatly, about 4 oz (112 g).

Kitchen yields: 1 lb (450 g) raw = 14 oz (400 g) peeled when cooked.

1 lb (450 g) raw = 3 portions hash browns or home fried.

Calories: 4 oz (112 g), without fat = 80 calories.

PEELED POTATOES

Ready-to-use potatoes, treated with preservative to retain color, are available in various sizes. The potatoes must be stored under refrigeration.

Packs: 25-lb (11.1-kg) bags.

Size: About 110 potatoes per bag.

ROUND RED (RED BLISS) POTATOES

Available small and large. In many markets, grade A indicates a larger potato than grade B. The small variety is often served whole as boiled potatoes, with only part of the peel removed before cooking.

Season: Year-round.

Packs: 50-lb (22.5-kg) bags.
25-lb (11.2-kg) boxes.

Counts: 1 lb (450 g) = 6 to 7 small potatoes, grade B.

Serving size: 3 small potatoes, restaurant serving.
2 small potatoes, banquet serving.

Kitchen yields: 50-lb (22.5-kg) bag, small size = 110 restaurant servings.

Calories: 2 potatoes, about 5 oz (140 g) = 105 calories.

ROUND WHITE POTATOES (IRISH POTATOES)

Also referred to as *chef's potatoes*.

Season: Year-round.

Packs: 50-lb (22.5-kg) bags.

Counts: 4 to 5 potatoes per 1 lb (450 g).

Kitchen yields: 1 lb (450 g) = 3 to 4 portions boiled potatoes.

1 lb (450 g) = 3 portions potato salad.
50 lb (22.5 kg) = 150 portions potato salad.

Calories: 2 potatoes, about 5 oz (140 g) = 105 calories.

RUSSET POTATOES

Most russet potatoes are grown in Idaho, Washington State, and Colorado.

Season: Fresh harvest in October, available year long.

Packs: 50-lb (22.5-kg), net, cartons.

Counts per case:

Count	Average Weight
40	20 oz (560 g).
50	14 oz (400 g) to 17 oz (475 g)
60	12 oz (335 g) to 13 oz (365 g)
70	11 oz (300 g) to 13 oz (320 g)
80	9 oz (250 g) to 12 oz (335 g)
90	8 oz (225 g) to 11 oz (300 g)
100	7 oz (200 g) to 9 oz (250 g)
110	6 oz (170 g) to 8 oz (225 g)
120	5 oz (140 g) to 7 oz (200 g)
140	3 oz (85 g) to 4 oz (110 g)

Most common restaurant sizes are 80 or 100 count.

Shrinkage: 25 percent in weight when baked without foil.

Serving sizes: 40, 50, and 60 size for complete entrée or steak house size.

80 size common hotel size.

90 size for double baked (stuffed) potatoes.

100 and 110 size for commercial operations and cafeterias.

120 size elementary schools, as side dish.

140 size = 3 oz (84 g) cooked, meets $\frac{2}{3}$ cup vegetable requirement for school foodservice.

Kitchen Yields for Homemade French Fries:

4 lb (1.8 kg) russet potatoes yields:

$\frac{1}{4}$ -in fries, skin on = 44 oz (1.2 kg) or about 11 servings.

$\frac{1}{4}$ -inch fries, peeled = 20 oz (560 g) or about 5 servings.

Shoestring fries, skin on = 48 oz (1.3 kg) or about 12 servings.

Shoestring fries, peeled = 28 oz (780 g) or about 7 servings.

Serving size: 4 oz (112 g) cooked.

Kitchen Yields for Homemade Mashed

Potatoes: 6 lb (2.7 kg) unpeeled + 3 to 4 cups (0.7 to 0.9 l) milk + 4 oz (112 g) butter = 25 servings.

Serving size: $\frac{1}{2}$ cup (0.24 l), #8-scoop.

Calories: $\frac{1}{2}$ cup (0.24 l) = 120 calories.

SPECIALTY POTATOES

A number of novelty potatoes have come on the market. Most are used as boiled potatoes, except for Yukon Gold, which is a fairly starchy potato and can be mashed.

Varieties:

Baby Red

Blue Fingerling

German Yellow Fingerling

Peanut Potato

Red Creamer

Ruby Crescent Fingerling

Russian Banana Fingerling

Yukon Gold

Packs: By weight.

Counts: Vary greatly; most potatoes are small.

YUKON GOLD POTATOES

Average size: 5.5 oz (150 g).

Calories: One 5.5-oz (150-g) potato = 100 calories.

PUMPKINS

Season: Late fall.

Weights: Vary greatly, up to 200 lb (90 kg). Product of more than 20 lb (9 kg) is hard to handle.

Serving size: $\frac{1}{2}$ (0.24 l) cup, mashed.

Kitchen yields: Trimming loss is about 30 percent by weight.

50 lb (22.5 kg) as purchased = 3 gal (11.4 l) mashed.

Calories: $\frac{1}{2}$ cup (0.24 l), mashed, without addition of syrup and butter = 45 calories.

RADISHES

BLACK RADISHES

Black radishes have black roots and white flesh; sharp and pungent when eaten raw, they are popular in Germany with beer. Black radishes can also be cooked as a vegetable.

Packs: Sold by weight.

DAIKON (JAPANESE RADISH)

See Japanese Foods in the International Ingredients section.

RED RADISHES

Available with the green leaves still attached or trimmed or sliced (cello pack). Radishes are classified by shape, such as globular, oval, turnip-shaped, oblong, and long. Most production is globular.

Season: Year-round.

Packs: 36-lb (16.2-kg) cartons of 48 bunches each, with leaves.

30-lb (13.5-kg) cartons of 12 cello packs each.

5-lb (2.2-kg) bags, sliced.

Weights: 1 fresh bunch, including leaves = 12 oz (340 g) to 14 oz (400 g).

1 fresh bunch, without leaves = 8 oz (225 g).

Kitchen yields: 1 fresh bunch = 12 to 13 pieces average.

1 cup, sliced = 4½ oz (120 g).

10 radishes, trimmed = 1.6 oz (45 g).

Calories: 4 radishes, trimmed, 0.6 oz (18 g) = 5 calories.

RAMPS

Wild leeks with a strong garlic flavor. Ramps are a specialty item, with little supply reaching the market.

Season: Spring.

RAPINI (RAPPINI)

Bitter broccoli. *See* under Broccoli Rabe.

RHUBARB

Although technically a vegetable, rhubarb is usually referred to as fruit. For Frozen Rhubarb, *see* the Groceries section; for Fresh Rhubarb, *see* the Fruits section.

RUTABAGAS

Large yellow root vegetable.

Season: Fall and winter.

Packs: 50-lb (22.5-kg) bags or cartons.

Serving size: ½ cup (0.12 l) cubed, cooked.

Kitchen yields: Waste = 20 percent by weight.

50 lb (22.5 kg) = 6 gal. (22.8 l) cubed, cooked.

SALSIFY

Also called *oyster plant*, salsify resembles white asparagus in shape and parsnip in flavor when cooked. Available with white or black skin. The inside flesh is always

white. This vegetable is typically used canned. *See* the Groceries section.

Season: Fall and winter.

Packs: By weight.

Kitchen yields: Peeling loss = 25 percent.

SAVOY CABBAGE

See Cabbage.

SEAWEEDS

For Dehydrated Seaweed, *see* Japanese Foods in the International Ingredients section.

DULSE

Packs: By weight.

Kitchen yields: There is no waste; just soak and rinse.

KELP

Season: April, May, and June for fresh kelp.

SEA BEANS

Seaweed, available fresh year-round.

Packs: By weight.

Kitchen yields: Large stems are tough and must be discarded.

Cleaning loss by weight = 20 percent.

SEA LETTUCE

Seaweed, available fresh. It has the texture of romaine lettuce and a spicy taste.

Season: April to August.

Packs: By weight.

SEA PALM

Seaweed harvested on the Northwest coast, available fresh.

Season: April to August.

Packs: By weight.

WAKAME

See Japanese Foods in the International Ingredients section.

SHALLOTS

Small bulbs resembling onions, with mild flavor, used for flavoring sauces and also as a vegetable. Peeled and chopped shallots are now available. For Processed Shallots, *see* the Groceries section.

Season: Available year-round.

Packs: 1-pint (0.47-l) cartons.
5-lb (2.2-kg) bags.

Sizes and counts: 1 lb (450 g) Large = 40 to 50 pieces.
1 lb (450 g) Jumbo = 20 to 25 pieces.

Kitchen yields: 1 cup chopped = 4 oz (112 g).
1 lb as purchased = 2¼ cups.

SPINACH

There are three basic varieties of spinach: curly leaf Savoy spinach, semi-Savoy spinach, and flat-leaf spinach, also called *broadleaf spinach*. Curly leaf Savoy spinach ships better than flat-leaf spinach. Curly spinach has thicker rib stems and requires careful washing because grit adheres to the leaves. Loose spinach is shipped both clipped, with some stems removed, and in bunches. For Canned and Frozen Spinach, *see* the Groceries section.

Season: Year-round, with peaks in spring.

Packs: 20-lb (9-kg) to 22-lb (9-kg) bushels.
10-oz (280-g) cello packs, 8 per 5-lb (2.25-kg) carton.

Serving sizes: 2 cups, 4 oz (112 g), as main course salad.
1 cup, 2 oz (56 g), as side order salad.
½ cup, 3½ oz (100 g), cooked, drained.

Kitchen yields: Bushel spinach trimming waste is about 30 percent by weight.
Cello pack spinach trimming waste is 10 percent or less by weight.
One 20-lb (9-kg) bushel = 13 to 14 lb (5.8 to 6.3 kg) cleaned spinach.

One 20-lb (9-kg) bushel = 54 main course salads or 100 side order salads.
1 5-lb (2.25-kg) cello pack carton = 72 oz (2 kg) cleaned spinach.
1 cello pack = 2 main course salads or 4 servings side order salad.

Cooking losses: One 20-lb (9-kg) bushel = 14 lb (6.3 kg), well-drained cooked spinach.

One 20-lb (9-kg) bushel = 60 servings.

Calories: Raw, chopped, 1 cup, 2 oz (56 g) = 15 calories.
Cooked, drained, ½ cup, 4 oz (0.12 l) = 20 calories.

SPROUTS

Some sprout varieties are available both cut and live in planters' boxes. For Canned Sprouts, *see* the Groceries section. Fresh sprouts can be contaminated with salmonella bacteria. Sprouts have become important ingredients in some markets. One purveyor lists following varieties:

Adzuki beans
Blackeyed peas
Broccoli
Clover
Daikon radish
Garbanzo
Green pea
Onion alfalfa
Onion
Pea leaves
Pea shoots
Sunflower
Wheat berry
Wheat grass

Packs: 3-oz (84-g) to 6-oz (170-g) containers.

Kitchen yields: Sprouts are very perishable, and some packs include the seeds. It is important to ascertain the exact pack size needed before ordering.

ALFALFA SPROUTS

Packs: 4-oz (112-g) or 6-oz (170-g) containers, 12 per flat.
4-lb (1.8-kg) flats.

Serving size: 1 oz (28 g) as salad garnish.

Kitchen yields: One 4-oz (112-oz) flat = 45 servings. There is little waste.

Calories: 1 oz (28 g), as salad garnish = 5 calories.

BEAN SPROUTS

Usually mung bean sprouts. For Canned Bean Sprouts, *see* the Groceries section.

Packs: 5-lb (2.2-kg) and 10-lb (5.5-kg) bags.

Serving size: 1 cup, 3½ oz (100 g).

Kitchen yields: 1 lb (450 g) = 6 cups. There is no waste; use as is.

Calories: 1 cup, 3½ oz (100 g) = 33 calories.

SOY BEAN SPROUTS, *DAAI DAU NGA CHOI*

Large sprouts with the seeds still attached. Raw sprouts are slightly poisonous and must be cooked.

Packs: Usually in bunches.

Kitchen yields: Lower portion must be trimmed off; about 20 percent waste. 1 lb (450 g) as purchased = 4 cups cleaned sprouts.

SQUASH VARIETIES

Squash is classified as:

- Soft shell, also called summer squash
- Mature small winter squash
- Hard-shelled varieties, also called winter squash

Summer squash and mature winter squash are highly perishable and can be used with the skin left on. Therefore, it is important that the product is unblemished and has a smooth skin. Winter squash has a hard shell and should feel heavy; light squash may be dried out and stringy. For Canned and Frozen Squash, *see* the Groceries section.

Season: Most varieties are available year-round.

Packs: 25-lb (11.2-kg) carton for most varieties.

Sizes: Vary greatly.

Kitchen yields: For summer squash, the smallest pieces are the best, give the most yield, are firm, and shrink little. For the following major varieties, season, pack, and yields are about the same.

Summer squash: Crookneck, patty pan, spaghetti, yellow straightneck, and zucchini. Within each group are many varieties. Chayote is listed separately in this section. *Courgette* is the French name for small zucchini. The term is frequently used in England.

Winter squash: Acorn, banana, buttercup, hubbard, and turban. Pumpkin is listed separately. Some of these varieties are available as mature winter squash, harvested while still soft.

SUNCHOKES (JERUSALEM ARTICHOKEs)

Cultivated tubers, a native American vegetable known as *topinambours* in French. The plant is related to the sunflower. Jerusalem artichokes look like large ginger roots, but the flesh is crisp white, and resembles that of water chestnuts. They can be eaten raw as salad, but are normally used like potatoes in soups, stews, and purées.

Packs: 12-lb (5.4-kg) carton.

Season: Late autumn and winter.

Kitchen yields: Waste = 20 percent by weight.

SWEET POTATOES AND YAMS

The names *yam* and *sweet potato* are often used interchangeably, although there is a difference between the two. Yams are considered to be moister than sweet potatoes. The market distinguishes between dry-meated and moist-meated varieties. Boniato is a sweet potato with red skin and white flesh. For Canned and

Frozen Sweet Potatoes and Yams, *see* the Groceries section.

Season: Available year-round; peak occurs in the winter months.

Packs: Cartons with varying counts and weights; 50 lb (22.5 kg) is common.

Weights: 5 oz (140 g) to 6½ (185 g), best average size.

Serving size: ½ medium potato, 2½ oz (70 g), baked.
½ cup (0.12 l), mashed.
4 oz (112 g), candied.

Kitchen yields: Shape is important for good yield. Look for clearly identifiable pieces.

1 lb (450 g) = 14 oz (310 g) baked.
1 lb (450 g) = 12 oz (340 g) mashed.
1 lb (450 g) = 3 servings candied.

Calories: 4 oz (112 g), baked in skin = 160 calories.
4 oz (112 g), candied = 190 calories.

SWISS CHARD

Vegetable in the beet family with green leaves and red stems.

Season: Fall and winter; some supplies are available in early summer.

Packs: 20-lb (9-kg) to 25-lb (12.5-kg) bushel baskets.
30-lb (13.5-kg) to 35-lb (15.7-kg) cartons.

Kitchen yields: Trimming loss is about 30 percent. Yield largely depends on how the vegetable is used. Some chefs braise the stems and cream the chopped leaves; others use the stems only.

TARO ROOT

Also called *poi* or *Hawaiian potato*.

Packs: 10-lb (4.5-kg) cartons.

Kitchen yields: Waste = 25 percent by weight.

TOMATILLO

See Hispanic Foods in the International Ingredients section.

TOMATOES

Tomatoes are available year-round. They are harvested green or turning and then ripened by being exposed to carefully regulated temperatures and ethylene gas. Tomatoes are best when vine-ripened, but because they are very perishable when fully ripe, only locally grown fruits are marketed ripe. There is no comparison in flavor between vine-ripened tomatoes and warehouse-ripened tomatoes. Refrigerated storage retards ripening; tomatoes ripen at room temperature. There are many varieties on the market, including plum-shaped tomatoes, often referred to as Italian tomatoes. Yellow tomatoes have come on the market as a specialty item. Tomatoes are shipped according to types. Mature, but still green tomatoes are shipped in cartons with volume-fill packs and locally repacked when red. Pink and vine-ripened tomatoes are shipped in one- or two-layer flats, lugs, loose-packed cartons, or baskets. Imports are available from a number of countries. Heirloom tomatoes are available in limited quantities according to season. For Canned and Processed Tomato Products, *see* the Groceries section.

Industry terms concerning ripeness:

- **Green:** Surface is completely green.
- **Breakers:** 10 percent of surface is showing pink or red.
- **Turning:** 10 percent or more, but less than 30 percent, of surface is showing pink or red.
- **Pink:** 30 percent to 60 percent of surface is red.
- **Light Red:** 60 percent to 90 percent of surface is red.
- **Red:** 90 percent or more of surface shows red.

BEEFSTEAK TOMATO

Large, fleshy tomato served sliced as salad or stuffed as a main course. The name refers to the meatiness of the tomato. Excellent beefsteak tomatoes are grown in New Jersey.

Season: July to November.

Packs: By weight.

Size: Up to 1 lb (450 g) each.

CHERRY TOMATOES

Season: Year-round.

Packs: 1-pint (0.47-l) containers in 12-unit flats.

Counts: 1-pint = 23 to 26 or 35 to 45 tomatoes.

GRAPE TOMATOES

Very sweet, teardrop-shaped small tomatoes.

Season: Year-round.

Packs: 1-pint (0.47-l) containers in 12-unit flats.

Counts: 1 pint (0.47 l) = 35 to 45 tomatoes.

REGULAR TOMATOES

Packs: 18-lb (8.1-kg) to 20-lb (9-kg) flats.
28-lb (12.6-kg) to 30-lb (13.5-kg) lugs.
25-lb (11.2-kg) cartons.
10-lb (4.5-kg) baskets.

Weights:

Maximum Large: 8 oz (225 g) to 10 oz (280 g).

Extra Large: 5 oz (140 g) to 7 oz (200 g).

Large: 3 oz (84 g) to 5 oz (140 g).

Small: under 3 oz (84 g).

Counts: 40 Maximum Large to 60 Extra Large in flats.

108 Large, 126 Medium, or 147 Small in three-layer lugs.

Cartons and baskets have varying counts.

Size Classifications: Size classifications are determined by the number of tomatoes packed in a carton or a lug in rows. For example, a straight-pack 6 × 6 lug will have 108 tomatoes. However, additional rows of the next size are permitted in lower layers, thus increasing the number of tomatoes in a carton. A

lug of 5 × 5 tomatoes can contain 85 tomatoes.

Kitchen yields: 1 lb (450 g) = 3 to 4 medium tomatoes.

1 lb (450 g) = 1½ cups peeled, diced, and seeded tomatoes.

Calories: 1 tomato, raw, medium size, 5 oz (140 g) = 27 calories.

TURNIP GREENS

Available with and without roots still attached. Sometimes broccoli greens are shipped for turnip greens. Roots should be no larger than 1½ in (38 mm) across. For Canned and Frozen Turnip Greens, see the Groceries section.

Packs: By weight.

Kitchen yields: Trimming loss is about 40 percent.

Calories: 1 cup, 5 oz (140 g), chopped, with stems = 30 calories.

TURNIPS

Season: Available year-round; smaller supply in summer.

Packs: 25-lb (11.2-kg) and 50-lb (22.5-kg) bags.
43-lb (19.3-kg) to 47-lb (21.1 kg) cartons with 24 bunches each, greens still attached.

Best size: 2 in (50 mm) across.

Kitchen yields: Smaller turnips are better than larger turnips. If vegetable is not fresh, regardless of size, it can be spongy. This reduces yield considerably. Peeling loss = 20 percent by weight for turnips without greens.

Calories: 1 cup, 5½ oz (154 g), cooked, diced = 35 calories.

WATERCRESS

Season: Year-round; peak occurs May to July.

Packs: By the bunch.

Weight of bunch: Varies, but usually
4 oz (112 g).

Kitchen yields: Trimming waste is
about 10 percent by weight if stems
are cut 2 in (50 mm) from bottom.
1 lb (450 g) = 10 cups (2.25 l)
chopped.

WINTER MELON

See Chinese Foods in the International
Ingredients section.

YUCCA ROOT (*Yuca in Spanish*)

For Yuca Flour and Fresh Yuca, *see* His-
panic Foods in the International Ingredi-
ents section. *See also* Tapioca in the
Groceries section.

Sizes and Miscellaneous Information



BEER

Keg Sizes:

Barrel = 31 gal (112.8 l).
Standard keg, ½ barrel = 15½ gal (58.9 l).
Standard can or bottle = 12 fl. oz cans (0.355 l).

Yields:

One 31-gal (112.8-l) barrel = 330 cans, 12 oz (0.355 l) each.
1 standard keg, 15½ gal (58.9 l) = 165 cans, 12 fl. oz (0.355 l) each.
Note: The yield varies, based on the thickness of the head (foam) and the shape of the glass.

Glasses per standard keg, 15½ gal (58.9 l):

Pitchers:

60 fl. oz (1.7 l) with 1-in head = 41
40 fl. oz (1.1 l) with 1-in head = 62

Schooners:

12 fl. oz (0.355 l) with 1-in head = 256
12 fl. oz (0.355 l) with ¾-in head = 214
10 fl. oz (0.29 l) with 1-in head = 330
10 fl. oz (0.29 l) with ¾-in head = 293

Sham pilsner:

10 fl. oz (0.29 l) with 1-in head = 264
10 fl. oz (0.29 l) with ¾-in head = 248
8 fl. oz (0.24 l) with 1-in head = 345
8 fl. oz (0.24 l) with 1-in head = 317

Footed pilsner:

12 fl. oz (0.355 l) with 1-in head = 208
10 fl. oz (0.29 l) with 1-in head = 248

Stein:

14 fl. oz (0.40 l) with 1-in head = 170
14 fl. oz (0.40 l) with ¾-in head = 165
10 fl. oz (0.29 l) with 1-in head = 248

Paper cup:

14 fl. oz (0.40 l) with 1-in head = 195
14 fl. oz (0.40 l) with ¾-in head = 180
12 fl. oz (0.355 l) with 1-in head = 245
12 fl. oz (0.355 l) with ¾-in head = 220
10 fl. oz (0.29 l) with 1-in head = 265
10 fl. oz (0.29 l) with ¾-in head = 245

Calories:

One 12-fl. oz (0.355-l) can or bottle = 150 calories, average.

Note: Calorie counts vary greatly between brands. Imported beer frequently has a higher calorie count than domestic beer.

BOTTLE SIZES

Champagne Bottle Sizes

Split: 0.187 l = 6.4 fl. oz
Half Bottle: 0.375 l = 12.8 fl. oz
Fifth: 0.75 l = 25.35 fl. oz
Magnum: 1.5 l = 2 Fifths, 50.7 fl. oz
Jeroboam: 3 l = 4 Fifths, 102.4 fl. oz
Rehoboam: 4.5 l = 6 Fifths, 152.1 fl. oz
Salmanazar: 9 l = 12 Fifths, 304.2 fl. oz
Balthazar: 12 l = 16 Fifths, 405.6 fl. oz
Nebuchadnezzar: 15 l = 20 Fifths, 507 fl. oz
Sovereign: 26 l = 34.6 Fifths, 879 fl. oz

Liquor Bottle Sizes

Half-pint = 8 fl. oz (0.23 l)
Pint = 16 fl. oz (0.47 l)
Fifth = 25.35 fl. oz (0.75 l)
Quart = 32 fl. oz (0.94 l)
Liter = 33.8 fl. oz
Half-gallon = 64 fl. oz (1.9 l)
Gallon = 128 fl. oz (3.8 l)

Wine Bottle Sizes

Half-bottle: 0.375 l = 12.8 fl. oz
Bottle: 0.75 l = 25.35 fl. oz (one fifth)
Liter = 33.8 fl. oz
One and a half liters = 50.7 fl. oz
One-half gallon = 59 fl. oz (1.75 l)
Gallon = 128 fl. oz (3.785 l)
Magnum: 1.5 l = 2 fifths, 50.7 fl. oz
Double magnum: 3 l = 4 fifths, 101.4 fl. oz
Jeroboam: 4.5 l = 6 fifths, 152.1 fl. oz
Imperial: 6 l = 8 fifths, 202.8 fl. oz
Note: The 0.75-l bottle is the most common wine bottle size. Some producers have switched to 1-l bottles.

Drink yields of liquor bottles:

Approximate number of drinks per bottle, allowing for spillage and overpouring:

One-liter (33.8-fl. oz) bottle:

Drink size	No. of Drinks
$\frac{7}{8}$ fl. oz (0.025 l)	38
1 fl. oz (0.029 l)	32
$1\frac{1}{8}$ fl. oz (0.032 l)	30
$1\frac{1}{4}$ fl. oz (0.036 l)	27
$1\frac{1}{2}$ fl. oz (0.043 l)	23

Half-gallon (1.9-l) bottle:

Drink size	No. of Drinks
$\frac{7}{8}$ fl. oz (0.025 l)	75
1 fl. oz (0.029 l)	64
$1\frac{1}{8}$ fl. oz (0.032 l)	58
$1\frac{1}{4}$ fl. oz (0.036 l)	52
$1\frac{1}{2}$ fl. oz (0.043 l)	43

Gallon (3.8-l) bottle:

Drink size	No. of Drinks
$\frac{7}{8}$ fl. oz (0.025 l)	150
1 fl. oz (0.029 l)	130
$1\frac{1}{8}$ fl. oz (0.032 l)	115
$1\frac{1}{4}$ fl. oz (0.036 l)	104
$1\frac{1}{2}$ fl. oz (0.043 l)	87

Common serving sizes for wine:

Champagne and sparkling wines = 4 fl. oz (0.116 l)
 Dessert wines, often sold in half-bottles = 3 fl. oz (0.087 l)
 White wines = 5 fl. oz (0.145 l)
 Red wines = 6 fl. oz (0.174 l)

Glasses per wine bottle:**Half-bottle (3.75 l):**

4-fl. oz (0.116-l) serving = 3.2 glasses.
 5-fl. oz (0.145-l) serving = 2.5 glasses.
 6-fl. oz (0.174-l) serving = 2 glasses.

Bottle (0.750 l):

4-fl. oz (0.116-l) serving = 6.4 glasses.
 5-fl. oz (0.145-l) serving = 5 glasses.
 6-fl. oz (0.174-l) serving = 4.3 glasses.

Liter (33.8 fl. oz):

4-fl. oz (0.116-l) serving = 8.5 glasses.
 5-fl. oz (0.145-l) serving = 6.8 glasses.
 6-fl. oz (0.174-l) serving = 5.7 glasses.

One and one-half liters (50.7 fl. oz):

4-fl. oz (0.116-l) serving = 13 glasses.
 5-fl. oz (0.145-l) serving = 10 glasses.
 6-fl. oz (0.174-l) serving = 8½ glasses.

Two liters (67.6 fl. oz):

4-fl. oz (0.116-l) serving = 17 glasses.

5-fl. oz (0.145-l) serving = 13 glasses.

6-fl. oz (0.174-l) serving = 11½ glasses.

Banquet wine serving sizes:

Champagne toast = 6 glasses per bottle.

White wine = 2 bottles for a table of 10 covers.

Red wine = 2 bottles for a table of 10 covers.

Calories, wine:

4 oz (0.11 l) dry wine = 100 calories.

4 oz (0.11 l) sweet wine = 160 calories.

Calories, gin, rum, vodka, and whisky:

For 1½-fl. oz (0.043-l) servings, 80 proof = about 80 calories.

CAN SIZES

There are many different can sizes on the market. The following list includes the most common can sizes in institutional foodservice, their approximate sizes in fluid ounces and cups, and the number of cans per case. The weight of cans varies greatly, depending on the density of the contents.

Can Name	Volume in Cups	Content in Fluid Ounces	Content in Milliliters	Number per Case
#1 picnic	10.3	10.5	311	24 or 48
#2	2.5	20	590	12 or 24
#211 cylinder	1.5	12	355	24, 36, or 48
#2½	3.6	29	843	12 or 24
#300	1.7	13.5	400	24, 36, or 48
#303	2	15.6	440	12 or 24
#3 cylinder	5¾	46	1360	12
#5	7	56	1650	12
#10	13	103.5	3070	6
Gallon	16	126	3800	4 or 6

Transposing Can Sizes

This table does not take count (number of pieces in each can) into consideration.

To fill one can #10, it will take approximately:

Three #3 cans, or five #2 cans, or two and one-half #3 cylinder cans.

To make up a case of:

Twenty-four #303 cans = nineteen #2 cans are needed.

Twenty-four #2 cans = fourteen #3 cans are needed.

Twenty-four #3 cans = three #10 cans are needed.

Twelve #10 cans = thirty-six #3 cans are needed.

<i>Scoop Number</i>	<i>Tablespoons</i>	<i>Fluid Ounces</i>	<i>Liters</i>
30	2	1	0.032
24	2¾	1.33	0.039
20	3	1.6	0.047
16	4	2	0.059
12	5	2.6	0.073
10	6	3.2	0.095
8	8	4	0.118
6	10	5.3	0.160
4	12	6.4	0.190
2	16	8	0.240

LADLE SIZES

<i>Size</i>	<i>Part of Cup</i>	<i>Number to Quart</i>	<i>Number to Liter</i>
1 oz	⅛	32	34
2 oz	¼	16	17
2⅔ oz	⅓	12	13
4 oz	½	8	8.6
6 oz	¾	5⅓	5.7
8 oz	1	4	4.3
12 oz	1½	2.6	2.8

LITER, QUART, AND CUP CONVERSIONS

All conversions are rounded off.

1 gal (3.8 l) = 4 qt (0.94 l)

1 qt (0.94 l), 32 fl. oz = 2 pints

1 pint, 16 fl. oz = 2 cups

1 cup, 8 fl. oz = (0.24 l)

1 l = 33.8 fl. oz

2 tb = 1 fl. oz

1 tb = 3 tsp

ROASTING PAN SIZE

G.I. Pan: 20 in (500 mm) × 17 in (425 mm) × 6.5 in (160 mm)

SCOOP SIZES

Scoops are rated by the approximate number of scoops contained in 1-qt (0.46-l) level fill.

SHEET PAN SIZES

Standard pan: 18 in (450 mm) × 24 in (600 mm) × 1 in (25 mm) deep

Half pan: 9 in (225 mm) × 12 in (300 mm) × 1 in (25 mm) deep

STEAM TABLE PAN SIZES AND CAPACITIES

Note: The capacities shown are measured to the rim of the pan. The effective capacities are considerably less when pans are filled and transported.

<i>Depth</i>	<i>Capacities</i>	<i>4-oz Portions</i>
Full Size 12¾ × 20¾ in		
1 in (25 mm)	3½ qt (3.3 l)	28
2 in (50 mm)	7 qt (6.5 l)	56
2.5 in (63 mm)	9 qt (8.4 l)	72
3 in (76 mm)	11 qt (10.3 l)	88
4 in (101 mm)	15 qt (14.1 l)	120
6 in (152 mm)	22 qt (20.6 l)	176
8 in (203 mm)	31½ qt (29.6 l)	252
Two-Thirds 13¾ × 12¾ in		
1 in (25 mm)	2¼ qt (2.1 l)	18
2 in (50 mm)	4½ qt (4.2 l)	36
2.5 in (63 mm)	6 qt (5.6 l)	48
3 in (76 mm)	7¼ qt (6.8 l)	58
4 in (101 mm)	10 qt (9.4 l)	80
6 in (152 mm)	14¼ qt (13.3 l)	112
8 in (203 mm)	21⅞ qt (20.0 l)	171

One-Half 10³/₈ × 12³/₄ in

1 in (25 mm)	1½ qt (1.4 l)	12
2 in (50 mm)	3½ qt (3.3 l)	28
2.5 in (63 mm)	4 qt (3.8 l)	32
3 in (76 mm)	5 qt (4.7 l)	40
4 in (101 mm)	7 qt (6.6 l)	56
6 in (152 mm)	10 qt (9.4 l)	80
8 in (203 mm)	15 qt (14.1 l)	120

One-Third 6⁷/₈ × 12³/₄ in

1 in (25 mm)	1 qt (0.9 l)	8
2 in (50 mm)	2 qt (1.9 l)	16
2.5 in (63 mm)	2 ⁵ / ₈ qt (2.5 l)	21
3 in (76 mm)	3¼ qt (3.0 l)	26
4 in (101 mm)	4½ qt (4.2 l)	36
6 in (152 mm)	6½ qt (6.1 l)	52

One-Quarter 10⁵/₁₆ × 6⁵/₁₆ in

1 in (25 mm)	⁵ / ₈ qt (0.6 l)	5
2 in (50 mm)	1¼ qt (1.2 l)	10
2.5 in (63 mm)	1 ⁵ / ₈ qt (1.5 l)	13
3 in (76 mm)	2 qt (1.9 l)	16
4 in (101 mm)	3 qt (2.8 l)	24
6 in (152 mm)	4¾ qt (4.5 l)	38

One-Sixth 6⁷/₈ × 6⁵/₁₆ in

1 in (25 mm)	½ qt (0.47 l)	4
2 in (50 mm)	1 qt (0.94 l)	8
3 in (76 mm)	1½ qt (1.4 l)	12
4 in (101 mm)	2 qt (1.9 l)	16

One-Ninth 6³/₄ × 4¹/₄ in

1 in (25 mm)	³ / ₈ qt (0.355 l)	3
2 in (50 mm)	³ / ₅ qt (0.70 l)	6

TABLE SIZES (BANQUET FOLDING TABLES)

<i>Size</i>	<i>Seating Capac- ity</i>	<i>Tablecloth Size</i>
Round Tables		
2 ft (0.6 m)	2	72 × 72 in (1.82 m)
2½ ft (0.76 m)	3	72 × 72 in (1.82 m)
3 ft (0.91 m)	4	72 × 72 in (1.82 m)
4½ ft (1.36 m)	6	90 × 90 in (1.82 m)
5 ft (1.52 m)	8	90 × 90 in (1.82 m)
5½ ft (1.67 m)	10	108 × 108 in (2.73 m)
6 ft (1.82 m)	12	108 × 108 in (2.73 m)

Rectangular Tables

18 in × 6 ft (0.45 × 1.82 m)	4	Two 62 × 62 in (1.57 m)
30 in × 4 ft (0.76 × 1.21 m)	6	72 × 72 in (1.82 m)
30 in × 6 ft (0.76 × 1.82 m)	10	Two 72 × 72 in (1.82 m)
3 × 6 ft (0.9 × 1.82 m)	8	Two 72 × 72 in (1.82 m)
3 × 8 ft (0.9 × 2.4 m)	12	Two 72 × 72 in (1.82 m)

Crescent Tables

6 × 3 ft (1.82 × 0.9 m)	4	Two 90 × 90 in (1.82 m)
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TABLE SKIRTING SIZES

To calculate skirting length required for round tables:

$$\frac{\text{Diameter} \times 1.14}{12} = \text{length in feet}$$

Round off to the next 6-in increment.

WEIGHT AND VOLUME EQUIVALENTS**Weights:**

1 lb (16 oz) = 453.6 g
1 oz = 28.35 g
3½ oz = 100.00 g
1 kilogram (1000 gram) = 35.27 oz

Volume:

1 gal (3.886 l) = 4 qt (128 oz)
4 qt (128 oz) = 8 pints
1 quart (0.946 l) = 4 cups
1 pint (0.473 l) = 6 fluid ounces
1 pint (0.473 l) = 2 cups
1 cup (0.237 l) = 8 fl. oz
1 cup (0.237 l) = 16 tb
2 tb (0.029 l) = 1 fl. oz
1 tb (0.015 l) = 3 tsp
1 l = 1 qt and 1.9 fl. oz
1 l = 4 cups and 2 tb

Note: For the sake of practicality, measurements in this book have been rounded off.

Useful Charts

Sizes and Counts of Fresh Green Asparagus

Grade or Name	No. of Spears per lb (450 g)	No. of Spears per Case
Colossal	7 or 8	30 lb (13.5 kg) = 225 to 235 15 lb (6.75 kg) = 110
Jumbo	8 to 9	30 lb (13.5 kg) = 260 12 lb (5.4 kg) = 100
Large	9 to 10	30 lb (13.5 kg) = 300 12 lb (5.4 kg) = 112
Medium	10 to 12	30 lb (13.5 kg) = 350 15 lb (6.75 kg) = 175
Small	12 to 14	30 lb (13.5 kg) = 390 15 lb (6.75 kg) = 190
Pencil or Grass	14 to 16	30 lb (13.5 kg) = 450 12 lb (5.4 kg) = 180
Grass	26 to 28	30 lb (13.5 kg) = 800

Conversion Amounts for Dehydrated Eggs,
Water, and Fresh Eggs

Large Eggs	Dried Whole Sifted	Water
6	3 oz (84 g): 1 cup	1 cup (0.23 l)
12	6 oz (170 g): 2 cups	2 cups (0.47 l)
24	12 oz (340 g): 1 qt	1 qt (0.94 l)
50	1 lb 9 oz (700 g) 2 qt + ⅓ cup	2 qt + ⅓ cup (1.95 l)
100	3 lb 3 oz (1.43 kg) 1 gal + ⅓ cup	1 gal + ⅓ cup (3.95 l)

Egg Size Substitution Chart

Jumbo	Extra Large	Medium	Small
2	2	2	3
5	6	7	8
10	12	13	15
21	24	27	28
44	50	56	62

Converting Flour Volume
to Weight

Volume	Weight Unsifted	Weight Sifted
1 cup	6 oz (170 g)	4¼ oz (120 g)
1 pint	12 oz (340 g)	10 oz (285 g)
1 qt	1½ lb (675 g)	1 lb 2 oz (510 g)
1 gal	6 lb (2.7 kg)	4½ lb (2 kg)

Liquid Measurements

	Cups	Fluid Ounces	Liters
1 tsp		⅓	0.005
3 tsp	1 tb	⅓	0.015
4 tb	¼	2	0.060
8 tb	½	4	0.120
16 tb	1	8	0.240
1 pint	2	16	0.480
1 qt	4	32	0.960
1 gal	16	128	3.800

Counts and Weights for Canned Pineapple

Product Name	Can Weight	Drained Weight	Count
Sliced	108 oz (3.06 kg)		52
Sliced	108 oz (3.06 kg)		66
Sliced	108 oz (3.06 kg)		100 to 110
Chunks	109 oz (3.09 kg)	66 oz (1.87 kg)	
Small tidbits	108 oz (3.06 kg)	70 oz (2 kg)	
Coarse crushed	107 oz (3.03 kg)	90 oz (2.5 kg)	
Fine crushed	107 oz (3.03 kg)	90 oz (2.5 kg)	
Broken slices	107 oz (3.03 kg)	63 oz (1.8 kg)	

Ingredients and Yields for Idaho Instant Flakes

Ingredients	25/20	50/40	100/80
Water	1¾ qt (1.6 l)	3½ qt (3.2 l)	1¾ gal (6.4 l)
Margarine or butter	⅓ cup (74 g)	¾ cup (170 g)	1½ cups (340 g)
Milk	3 cups (0.69 l)	1½ qt (1.4 l)	3 qt (2.7 l)
Salt	1½ tsp	1 tb	2 tb
Potato flakes	1 lb (450 g)	2 lb (0.9 kg)	4 lb (1.8 kg)

Available Forms of Idaho Instant Mashed Potatoes

Granule Type	Packing Size	Yield per Can
Plain	Six #10 cans 6-lb (2.7-kg) can	5 gal (19.5 l)
With vitamin C	Six #10 cans 6-lb (2.7-kg) can	5 gal (19.5 l)
Complete	Six #10 cans 87-oz (2.4-kg) can	4½ gal (17.1 l)
Complete with vitamin C	Six #10 cans	4½ gal (17.1 l)
Seasoned, complete	Six #10 cans 5 lb 8 oz(2.4 kg) 3½ lb (1.5 kg)	4 gal (15.2 l)

Yields for Unpeeled Russett Potatoes, 80 to 120 Size

Form of Potato	25 Servings	50 Servings	100 Servings
¼-in fries, unpeeled	10¼ lb (4.6 kg)	21 lb (9.4 kg)	41½ lb (18.6 kg)
Jacket fries, skin on	9½ lb (4.2 kg)	18½ lb (8.3 kg)	37½ lb (16.8 kg)
Shoestring fries, unpeeled	9½ lb (4.2 kg)	19 lb (8.5 kg)	37 lb (16.6 kg)
¼-in fries, peeled	20 lb (9 kg)	40 lb (18 kg)	80 lb (36 kg)
Round fries	15¼ lb (6.8 kg)	30½ lb (13.7 kg)	61 lb (27.4 kg)
Shoestring fries, peeled	14¼ lb (6.4 kg)	28½ lb (12.8 kg)	57 lb (25.6 kg)

Rehydration and Yields for Slices, Dices, and Hash Browns

Ingredient	25 Servings	50 Servings	100 Servings
Potato product	22 oz (616 g)	42 oz (1.1 kg)	5 lb 5 oz (2.25 kg)
Water	3 qt (2.7 l)	6 qt (5.5 l)	3 gal (11.4 l)

Weight Yields for 4 lb (1.87 kg) Fresh Idaho Potatoes

Form of Potato	Cooked Weight	Number of Servings
¼-in fries, unpeeled	44 oz (1.2 kg)	11
Jacket fries, skin on	42 oz (1.1 kg)	10
Shoestring fries, unpeeled	48 oz (1.35 kg)	12
¼-in fries, peeled	20 oz (560 g)	5
Round fries	26 oz (728 g)	6
Shoestring fries, peeled	28 oz (785 g)	7

Ingredients Needed for Different Quantities of ½-cup (#8 Scoop) Servings Mashed Potatoes

Ingredient	25 Servings	50 Servings	100 Servings
Whole unpeeled russet potatoes	6 lb (2.70 kg)	12 lb (5.4 kg)	24 lb (10.8 kg)
Whole peeled russet potatoes	5 lb (2.25 kg)	10 lb (4.5 kg)	20 lb (9 kg)
Milk, hot	3 to 4 cups (0.7 to 0.9 l)	1½ to 2 qt (1.4 to 1.9 l)	3 to 4 qt (2.8 to 3.8 l)
Fresh butter	¼ lb (112 g)	½ lb (225 g)	1 lb (450 g)
Salt	Salt to taste	Salt to taste	Salt to taste

Sizes and Counts for 25-lb (11.25-kg) Cartons of Pitted Dried Prunes

Count per 1 lb (450 g)	Average Number per Case	Servings per 1 lb (450 g)	Servings per Case
Jumbo, 39 and larger	950	8 (5 prunes)	200
Extra large, 40 to 49	1125	7½ (6 prunes)	150
Large, 50 to 61	1375	9 (6 prunes)	229
Medium, 62 to 79	1750	10 (7 prunes)	250
Small, 80 and smaller	2000	10 (8 prunes)	250

Sizes and Counts for 25- and 30-lb (11.25- and 13.5-kg) Cartons of Unpitted Dried Prunes

Carton Weight	Count per 1 lb (450 g)	Average Number per Case	Servings per 1 lb (450 g)	Servings per Case
25 lb (11.25 kg)	20 to 30	625	6½ (4 prunes)	160
	25 to 35	800	8 (4 prunes)	200
	30 to 40	900	9 (4 prunes)	225
	40 to 50	1125	9 (5 prunes)	225
30 lb (13.5 kg)	25 to 35	850	7 (4 prunes)	212
	30 to 40	1080	9 (4 prunes)	270
	40 to 50	1350	9 (5 prunes)	270

Ingredients for White Rice

Rice	Water	Fat	Salt	Yields	Portions
1 lb (450 g)	5 cups (1.17 l)	2 oz (56 g)	1 tb	6 cups (1.4 l)	10
5 lb (2.25 kg)	6½ qt (6.17 l)	10 oz (280 g)	½ cup	30 cups (7 l)	50
10 lb (4.5 kg)	13 qt (12.35 l)	1 lb 4 oz (560 g)	1 cup	3¾ gal (14 l)	100

Size Comparisons Between Cooked Cocktail and Green Headless Shrimp

Name	Tail Style	Cooked Count per 1 lb	Raw Count (Green Headless) per 1 lb	Cooked Count per 2½-lb (1.1-kg) Box
Super colossal	Tail on or off	18 to 25	10 to 12	52 pieces
Colossal	Tail on or off	21 to 30	10 to 15	65 pieces
Super jumbo	Tail on or off	31 to 40	16 to 20	90 pieces
Jumbo	Tail on or off	41 to 50	21 to 25	115 pieces
Extra large	Tail on or off	51 to 60	26 to 30	140 pieces
Large	Tail on or off	61 to 70	31 to 35	165 pieces
Salad	Tail off only	70 to 90, 110 to 130, 130 to 200, 200 to 300 or 300 to 500	36 to 42 or 50 to 60	200 pieces or more

Table Skirting Sizes

Skirt Length	Round Tables	Head Tables	Buffet Tables
12 ft (3.64 m)		2½ × 6 ft (0.76 × 1.82 m) <i>or</i> 3 × 6 ft (0.9 × 1.82 m)	
13 ft (3.9 m)	4 ft diameter (1.2 m)	2½ × 8 ft (0.76 × 2.4 m)	
16 ft (4.8 m)	5 ft diameter (1.5 m)		
17½ ft (5.25 m)	5½ ft diameter (1.65 m)	two 2½ × 6 ft (0.76 × 1.82 m) side-by-side	2½ × 6 ft (0.76 × 1.82 m) freestanding
19 ft (5.7 m)	6 ft diameter (1.8 m)		3 × 6 ft (0.9 × 1.82 m) freestanding
21½ ft (6.3 m)		two 2½ × 8 ft (0.76 × 2.4 m) side-by-side	2½ × 8 ft (0.76 × 2.4 m)

Temperature Ranges for Baking

Heavy fruit cakes	325 to 350°F
Custards, quiche	325 to 350°F
Angel and sponge cakes	350 to 375°F
Cookies	350 to 375°F
Puff pastry	350 to 375°F
Eclairs, cream puffs	375 to 400°F
Danish Pastry	375 to 400°F
Breads, rolls	400 to 425°F
Pizza	525 to 550°F

Fahrenheit to Celsius Conversion

Fahrenheit	Celsius
32°F	0°C
50°F	10°C
68°F	20°C
86°F	30°C
100°F	40°C
115°F	45°C
120°F	50°C
130°F	55°C
140°F	60°C
160°F	70°C
170°F	75°C
180°F	80°C
185°F	85°C
195°F	90°C
200°F	95°C
212°F	100°C
230°F	110°C
250°F	120°C
265°F	130°C
285°F	140°C
300°F	150°C
325°F	165°C
350°F	175°C
360°F	180°C
375°F	190°C
400°F	200°C
425°F	220°C
450°F	230°C
485°F	250°C
500°F	260°C
575°F	300°C

Tomato Size Classifications and Diameters

Size Classification	Minimum Diameter	Maximum Diameter
7 × 7	2 ⁵ / ₃₂ in (54 mm)	2 ⁵ / ₁₆ in (58 mm)
6 × 7	2 ¹ / ₄ in (56 mm)	2 ⁹ / ₁₆ in (64 mm)
6 × 6	2 ¹ / ₂ in (62 mm)	2 ¹³ / ₁₆ in (71 mm)
5 × 6	2 ³ / ₄ in (69 mm)	3 ¹ / ₁₆ in (80 mm)
5 × 5	2 ⁷ / ₈ in (72 mm)	3 ³ / ₈ in (84 mm)
4 × 5	3 in (76 mm)	3 ⁵ / ₈ in (91 mm)

Uniform Sizes for Male Personnel

Height	Weight	Chest	Coat Length	Waist	Trouser Inseam
5 ft 4 in (1.6 m)	120 lb (54 kg)	34 in (0.86 m)	27½ in (0.69 m)	36¾ in (0.93 m)	27 in (0.68 m)
	150 lb (67.5 kg)	38 in (0.96 m)	28 in (0.71 m)	37 in (0.93 m)	27 in (0.68 m)
	180 lb (81 kg)	42 in (1.06 m)	28 in (0.71 m)	37½ in (0.95 m)	27 in (0.68 m)
5 ft 6 in (1.67 m)	165 lb (74.2 kg)	40 in (1.01 m)	28½ in (0.72 m)	38 in (0.96 m)	27½ in (0.67 m)
	200 lb (90.0 kg)	44 in (1.11 m)	29 in (0.73 m)	39 in (0.99 m)	27½ in (0.69 m)
5 ft 8 in (1.72 m)	170 lb (76.5 kg)	40 in (1.01 m)	29½ in (0.75 m)	40 in (1.01 m)	29 in (0.73 m)
	185 lb (83.2 kg)	42 in (1.06 m)	30 in (0.76 m)	40½ in (1.02 m)	29 in (0.73 m)
5 ft 10 in (1.77 m)	175 lb (78.7 kg)	40 in (1.01 m)	30½ in (0.77 m)	41 in (1.04 m)	29½ in (0.75 m)
	190 lb (85.5 kg)	42 in (1.06 m)	31 in (0.78 m)	41 in (1.04 m)	29 in (0.73 m)
6 ft (1.82 m)	180 lb (81 kg)	40 in (1.01 m)	31½ in (0.80 m)	42 in (1.06 m)	30½ in (0.77 m)
	195 lb (87.7 kg)	42 in (1.06 m)	32 in (0.81 m)	42 in (1.07 m)	30 in (0.76 m)
6 ft 2 in (1.87 m)	198 lb (89.1 kg)	42 in (1.06 m)	33 in (0.84 m)	43½ in (1.10 m)	31½ in (0.80 m)

Uniform Sizes for Female Personnel

Dress Size	Height	Weight (Regular)	Bust Girth	Waist Girth	Hip Girth
4	5 ft 2½ in (1.58 m)	85 to 90 lb (38.3 to 40.5 kg)	30 in (0.76 m)	23½ in (0.60 m)	33 to 34 in (0.84 to 0.86 m)
6	5 ft 3 in (1.6 m)	95 to 100 lb (42.7 to 45 kg)	31 in (0.78 m)	25 in (0.63 m)	34 to 36 in (0.86 to 0.91 m)
8	5 ft 3½ in (1.61 m)	105 to 110 lb (47.2 to 49.5 kg)	32 in (0.81 m)	25½ in (0.67 m)	35 to 37 in (0.89 to 0.94 m)
10	5 ft 4 in (1.62 m)	115 to 120 lb (51.7 to 54 kg)	33 in (0.84 m)	26½ in (0.67 m)	36 to 38 in (0.91 to 0.96 m)
12	5 ft 4½ in (1.63 m)	125 to 130 lb (56.2 to 58.5 kg)	34½ in (0.88 m)	28 in (0.71 m)	38 to 39 in (0.96 to 0.99 m)
14	5 ft 5 in (1.65 m)	135 to 140 lb (60.7 to 63 kg)	36 in (0.91 m)	29 in (0.73 m)	39 to 41 in (0.99 to 1.04 m)
16	5 ft 5½ in (1.66 m)	150 to 155 lb (67.5 to 69.7 kg)	37½ in (0.95 m)	31 in (0.78 m)	41 to 42 in (1.04 to 1.06 m)
18	5 ft 6 in (1.67 m)	165 to 170 lb (74.2 to 76.5 kg)	39½ in (1.0 m)	32 to 33 in (0.81 to 0.84 m)	43 to 44 in (1.09 to 1.11 m)

Weight Equivalents

Avoirdupois Weight	Metric Equivalent	Avoirdupois Weight	Metric Equivalent	Avoirdupois Weight	Metric Equivalent	Avoirdupois Weight	Metric Equivalent
22½ lb	10.00 kg	4 lb	1.80 kg	14½ oz	400 g	4½ oz	125 g
20 lb 1½ oz	9.00 kg	3 lb 14½ oz	1.75 kg	131/3 oz	375 g	4½ oz	120 g
18 lb 4¾ oz	8.20 kg	3 lb 9 oz	1.60 kg	13 oz	360 g	3½ oz	100 g
17 lb 4 oz	8.00 kg	3 lb 6 oz	1.50 kg	12½ oz	350 g	3½ oz	90 g
15 lb 10 oz	7.00 kg	3 lb 2 oz	1.40 kg	11½ oz	320 g	3 oz	80 g
13 lb 6½ oz	6.00 kg	2 lb 14½ oz	1.30 kg	10¾ oz	300 g	2¾ oz	75 g
11 lb 2½ oz	5.00 kg	2 lb 11 oz	1.20 kg	11¾ oz	275 g	2½ oz	70 g
10 lb 2 oz	4.50 kg	2 lb 7½ oz	1.10 kg	9 oz	250 g	2⅓ oz	65 g
9 lb	4.00 kg	2 lb 3¾ oz	1.00 kg	8⅓ oz	240 g	2 oz	60 g
7 lb 13½ oz	3.60 kg	2 lb	900 g	8 oz	225 g	1¾ oz	50 g
7 lb 13 oz	3.50 kg	1 lb 14⅓ oz	850 g	7 oz	200 g	1½ oz	40 g
7 lb 1 oz	3.20 kg	1 lb 12½ oz	800 g	6½ oz	180 g	1⅓ oz	35 g
6 lb 11 oz	3.00 kg	1 lb 10¾ oz	750 g	6½ oz	175 g	1 oz	28 g
5 lb 13½ oz	2.60 kg	1 lb 9 oz	700 g	6 oz	170 g	¾ oz	20 g
5 lb 10 oz	2.50 kg	1 lb 7½ oz	650 g	5¾ oz	160 g	½ oz	15 g
5 lb 6½ oz	2.40 kg	1 lb 5½ oz	600 g	5⅓ oz	150 g	⅓ oz	10 g
5 lb	2.25 kg	1 lb 2 oz	500 g	5 oz	140 g	¼ oz	7 g
4 lb 7½ oz	2.00 kg	1 lb	450 g	4½ oz	130 g	⅛ oz	3½ g

Conversion from Liters to Fluid Ounces

Liter	Fluid Ounces	Liter	Fluid Ounces	Liter	Fluid Ounces
0.01 l	$\frac{1}{3}$ oz	0.35 l	11 $\frac{1}{2}$ oz	2.50 l	2 qt 22 oz
0.02 l	$\frac{3}{4}$ oz	0.40 l	13 $\frac{3}{4}$ oz	3.00 l	3 qt 7 $\frac{1}{2}$ oz
0.025 l	1 oz	0.45 l	1 pint	3.50 l	3 qt 24 $\frac{1}{2}$ oz
0.03 l	1 oz	0.50 l	1 pint 1 $\frac{1}{2}$ oz	4.00 l	1 gal 10 oz
0.04 l	1 $\frac{1}{2}$ oz	0.55 l	1 pint 3 oz	4.50 l	1 gal 27 oz
0.05 l	1 $\frac{3}{4}$ oz	0.60 l	1 pint 4 oz	5.00 l	1 gal 44 $\frac{1}{2}$ oz
0.06 l	2 oz	0.65 l	1 pint 5 $\frac{3}{4}$ oz	6.00 l	1 gal 78 oz
0.075 l	2 $\frac{1}{2}$ oz	0.70 l	1 pint 8 oz	6.50 l	1 gal 3 qt
0.08 l	2 $\frac{3}{4}$ oz	0.75 l	1 pint 9 $\frac{3}{4}$ oz	7.00 l	1 gal 113 $\frac{1}{3}$ oz
0.10 l	3 $\frac{1}{3}$ oz	0.80 l	1 pint 11 $\frac{1}{2}$ oz	8.00 l	2 gal 20 oz
0.12 l	4 $\frac{1}{2}$ oz	0.85 l	1 pint 13 $\frac{1}{3}$ oz	9.00 l	2 gal 54 $\frac{2}{3}$ oz
0.15 l	5 oz	0.90 l	1 pint 15 oz	10.00 l	2 gal 90 oz
0.175 l	6 oz	1.00 l	1 qt 2 $\frac{1}{2}$ oz	12.00 l	3 gal 30 oz
0.20 l	6 $\frac{3}{4}$ oz	1.20 l	1 qt 91/3 oz	14.00 l	3 gal 100 oz
0.225 l	7 $\frac{3}{4}$ oz	1.50 l	1 qt 20 oz	15.00 l	4 gal 5 $\frac{1}{2}$ oz
0.25 l	8 $\frac{1}{2}$ oz	1.80 l	62 oz	20.00 l	5 gal 50 oz
0.30 l	10 oz	2.00 l	2 qt 5 oz	24.00 l	6 gal 60 oz

Common Volume Equivalents

Level Measure	Equivalent
1 gal (3.886 l)	4 qt (8 pints)
1 qt (0.946 l)	4 cups (2 pints)
1 pint (0.473 l)	2 cups (16 fl oz)
1 cup (0.237 l)	8 fl oz ($\frac{1}{2}$ pint or 16 tb)
2 tb (28.35 ml)	1 fl oz
1 tb (14.17 ml)	3 tsp
1 l	1.05 qt (1 qt + 1.9 fl oz)
1 l	4 cups + 2 tb

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